

GENERAL THETA FOUNDATIONS I: STOCHASTIC REALIZATION AND FINITE-USE MEASUREMENT GEOMETRY

QIAN QI

ABSTRACT. We develop stochastic realization and finite-use description geometry for controlled systems. Same-width stationarization, physical-equivariant purification and finite-quotient criteria give a structural theory of classical realizations. Orbit geometry and spectral estimates yield quantitative width and numerical-streaming laws, with constructive rational realizations. Positive instrument coordinates extend these methods to descriptions stable under arbitrary common adaptive quantum testers, including entangled references and bounded stopping. The quantitative development treats strictly positive instruments, singular preparations, coherent tensor families and observable readouts. For the full body of ordered binary quantum measurements in each fixed input dimension d , a regularized midpoint Sylvester form gives a uniform finite-use comparison. Horizontal dilations and compressed nonadaptive tests supply the two bounds at every rank and multiplicity. Actual operational-ball volumes and a nested endpoint integral yield covering order $N^{d^2/2}[\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}$ for small error. A common learner of an unknown biased qubit effect attains the matching worst-case training order $N\delta^{-2} \log(1/\eta)$ under future- N adaptive loss. Simultaneous endpoint calibration and controlled two-dimensional compressions extend this learning law to every fixed matrix dimension, with an explicit upper bound $Cd^4N\delta^{-2} \log(d/\eta)$. A deterministic exact construction attains the matrix covering order with a finite rational codebook, with computation charged separately. For fixed $d \geq 2$, a cap $b \leq N$ on independent training blocks has minimax cost $\Theta_d((N^2/b)\delta^{-2} \log(1/\eta))$, including classical feedback and collective output processing. A conditional information bound gives the growing-dimensional lower $\Omega(d^2N\delta^{-2})$ under arbitrary coherent training. On the fixed interior $I/4 \preceq E \preceq 3I/4$, a dimension-free comparison gives the joint fixed-confidence law $\Theta(d^2N\delta^{-2})$, achieved by one-call Choi acquisition. Its full confidence refinement is $\Theta(N\delta^{-2}[d^2 + d\log(1/\eta)])$, with absolute constants jointly in all four parameters. On the same interior, the description length is $(d^2/2) \log_2 N + d^2 \log_2(1/\delta) + O(d^2)$ uniformly in dimension; randomization and failure multiply its leading optimal expected prefix length by $1 - \eta$. A finite rational dictionary and an algebraic collective readout computed by real quantifier elimination attain the joint query and description orders. Finite trusted control specifications preserve these and the original common learning guarantees. A second construction uses bounded-denominator rational readouts and finite parameter-net certificates. Their exact inequalities certify risk for all real effects and attain the same joint query and description orders without a quantifier-elimination step. For each fixed finite outcome count, balanced interiors of ordered noncommuting POVMs admit dimension-independent horizon comparison, sharp joint dimension–horizon–confidence learning, and public optimal-order rational descriptions. A normalization-preserving horizontal lift and a coupled inward grid prove these statements; finite collective-readout inequalities transfer to the real tuple body. The exact qubit code charges both spectral endpoints and direction; the unbiased slice retains its weighted covering criterion and boundary-contact trichotomy. This complete edition contains the proofs and mathematical refinements of the structural and quantitative articles, including the arithmetic, profinite, causal-process and rational-compilation developments. A normalized-tuple covariance now extends the adaptive upper geometry to the complete finite-outcome body, with rank changes and zero effects. Its regularized energy, binary restriction, and noise crossover are proved in full. An exact affine map gives polynomial balanced legalization without altering the inherited learning orders.

Date: 5 October 2026.

2020 Mathematics Subject Classification. Primary 93B15, 68Q45; Secondary 22C05, 60J10, 11J13.

Key words and phrases. stochastic realization, finite quotient, Ramsey extraction, compact group, spectral gap, nonuniform width.

Complete research edition. The focused quantitative and structural articles are independently complete; this edition retains the whole active mathematical development.

The principal additional conclusions include Theorem 52.1 (all-dimensional binary geometry), Theorem 53.1 (matrix description entropy), Theorem 54.1 (common qubit learning), Theorem 55.1 (common matrix learning), Theorem 56.1 (deterministic matrix descriptions), Theorem 57.2 (independent-block minimax law), Theorem 58.2 (joint dimension lower bound), Theorem 58.5 (sharp interior learning), Theorems 59.1–59.5 (joint interior descriptions and learning), Theorem 60.1 (joint confidence-optimal learning), Theorem 62.2 (effective optimal collective readout), Theorem 63.1 (finite uniform-risk certificates), Theorem 63.3 (rational optimal collective readout), Theorems 64.1–65.2 (finite-outcome geometry, learning, exact codes and certificates), and Theorem 61.2 (finite trusted preparation), together with Theorem 35.2 (rational instrument rounding), Theorem 35.4 (adaptive reference stability), Theorem 36.3 (variable-description streaming), and Proposition 35.5 (stopped posterior control), as well as Theorem 30.1 (uniform numerical streaming space), Corollary 30.6 (rational orthogonal actions), Theorem 29.5 (exponential-accuracy crossover), Theorem 16.2 (return-free occupation), Theorem 16.5 (joint accuracy law), Theorem 3.4 (full-error process minimum), Theorem 28.1 (six-command numerical example), Theorem 15.5 (synchronized occupation), Theorem 27.2 (effective spectral example), Theorem 24.1 (all-direction orbit caps), Theorem 25.2 (explicit rational robust separation), Theorem 26.2 (constructed finite tables), Theorem 2.2 (repeatable-observation classification), and Theorem 70.1 (adaptive path-law construction).

1. INTRODUCTION

A finite stochastic register has a continuum of probability distributions. Its number of labels is therefore not determined by the cardinality of the numerical orbit it realizes. Allowing an external clock creates a further difficulty: the command rows may change at every position, and the entire realization may be redesigned at each horizon. The central question is whether these freedoms change the least width needed to preserve a fixed numerical experiment with a prescribed uniform error.

Our first theorem identifies the limiting clocked width with the stationary minimum, without a loss of labels or accuracy. The synchronization hypothesis is eventual approximation of the physical identity at every sufficiently large length. Under it, arbitrarily many cuts of width at most k force a stationary realization on k labels, even when the intervening registers are unrestricted. The proof selects approximately homogeneous composite transition tables by finite Ramsey theory. It applies to irreversible topological monoids as well as groups. It does not assume that a physical return acts trivially on the hidden register.

For compact groups a second reduction makes the stationary realization physically equivariant. Transition randomness can be removed, and hidden permutations above the physical identity can be averaged out, without increasing width or uniform error. The stationary minimum is therefore intrinsic to a finite continuous group action with randomized initialization. Combining the two reductions gives an equality-inclusive finite-quotient criterion for clocked boundedness, including positive critical radii on groups with infinitely many components. A moving-frame construction gives the exact limiting minimum in each return-length phase, again without additional labels.

1.1. The interface and its two resources. Unless a section explicitly specifies a different physical space, H is a compact Hausdorff group and $\mathcal{A}$ is a finite nonempty alphabet whose letters $u_a \in H$ generate a dense subgroup. The execution convention is

$$u_{a_1 \cdots a_n} = u_{a_n} \cdots u_{a_1}.$$

There are finite nonempty seed and query sets $\mathcal{S}, \mathcal{J}$ and continuous maps $F_{s,j} : H \rightarrow Y_j$. In the structural results Y_j is a nonempty norm-compact convex subset of a real Banach space. The

localization and algebraic sections specify finite-dimensional output spaces. For a categorical law, $Y_j = \Delta_{M_j}$ and the norm on differences is $\frac{1}{2}\|\cdot\|_1$, namely total variation. A prescribed joint law is one query; incompatible measurements are not assigned an artificial joint law.

Definition 1.1 (Stationary realization). A realization on k labels consists of initialization rows $\alpha_s \in \Delta_k$, row-stochastic matrices T_a of order k , and decoder values $D_j(i) \in Y_j$, all independent of time and word length. Its error is

$$\sup_{s,j,w \in \mathcal{A}^*} \|\alpha_s T_{a_1} \cdots T_{a_n} D_j - F_{s,j}(u_w)\|_j.$$

The empty word is included. Let M_ε be the least k at error at most ε , or ∞ when none exists. A permutation realization has permutation command matrices; its initialization need not be deterministic.

The clocked resource $W_{N,\varepsilon}$ instead minimizes the maximum size of the registers $S_0, \dots, S_N$ at a fixed horizon, with arbitrary row-stochastic transitions depending on time and horizon, and a terminal query decoder. Definition 7.1 states the model formally. Every advertised persistent label counts, including unused padding. Private information retained for future commands is part of the current label. The original width model does not charge the external clock, real table descriptions, their construction, exact arithmetic, or exact atomic sampling. Section 21 gives separate finite-description and fair-random-bit bounds; it does not reinterpret those resources as free computational memory. A numerical decoder vector is not an additional sampled-answer register.

For an open normal subgroup $U \leq H$ put

$$(1.1) \quad r(U) = \max_{s,j,h \in H} \text{rad}_{Y_j}(F_{s,j}(hU)), \quad \text{rad}_Y(A) = \min_{y \in Y} \sup_{x \in A} \|x - y\|.$$

The coset images are compact by continuity. The radius depends continuously on the translate, and compactness gives the displayed extrema. The restriction of the center to the legal output set is essential. The component radius R_0 uses $U = H^\circ$, whether or not this subgroup is open; it is the infimum of the finite-quotient radii. At the critical error, the relevant issue is attainment of that infimum, not a limit from larger errors.

1.2. The principal conclusions. Theorem 4.2 proves $W_{N,\varepsilon} \rightarrow M_\varepsilon$ under eventual approximate returns. Finite limits are eventually equal, and every $k < M_\varepsilon$ has a uniform narrow-cut occupation bound. Theorems 6.1 and 6.3 then give the intrinsic group-action minimum and its finite-quotient existence criterion. Theorem 6.4 treats each length phase by a time-dependent change of physical frame. Table 1 separates these quantifiers and hypotheses.

The quantitative consequences have additional hypotheses. For a specified Diophantine planar alphabet the full-subcritical matching polynomial law is retained. A fixed 2-adic interface has linear exact width and bounded width at every positive error. For a spherical interface in dimension d with a spectral-gap alphabet, Theorem 13.1 gives a block estimate; Theorem 15.5 strengthens it to the matching order $N^{(d-1)/2}$. A rational alphabet generating $\text{SO}(3)$ supplies a genuinely noncommuting example. Its spectral gap is imported from Bourgain–Gamburd, not inferred from a finite numerical test. Theorem 21.1 gives an $\exists\mathbb{R}$ -complete finite input problem, while Theorem 21.3 prices dyadic descriptions and fair random bits.

The quantitative theory also extends beyond spheres. Theorem 14.3 separates a small-ball exponent uniform over all unit feature directions from the dimension of the target orbit. For conjugation on traceless Hermitian matrices a spectral split gives the uniform exponent $2(d-1)$, including directions with repeated eigenvalues. The pure-state orbit has the same dimension. Theorem 17.3 therefore gives nearly matching width exponents $d-1$ for full legal density-matrix

outputs in every dimension. Its exact upper construction stays inside the positive cone, rather than replacing the legal output set by an ambient Euclidean ball.

At an extreme target, the exact answer can be very different. Theorem 18.1 identifies the entire optimal exact width profile with the reachable orbit cardinalities. Theorem A.2 applies it to one fixed five-letter algebraic dense free alphabet in $SU(d)$: exact pure-state width is $2 \cdot 3^N - 1$, while each fixed positive subcritical error has the polynomial bounds just described. The specified seed is transcendental, not a rational-input instance. These assertions concern a full matrix-valued terminal interface, not adaptive observations or the zero-error endpoint of a single scalar readout.

The compact nonnegative-matrix group structure used in purification is classical, and is separated from the external numerical error constraints throughout Section 23. Neutral-letter collapse for advice automata is also a relevant predecessor to stationarization. The additional assertions proved here concern the same stochastic label bound and the same closed numerical error balls, arbitrary intervening widths, and approximate rather than exact returns. Neither classical theorem is relabeled as a new result.

TABLE 1. Resources, hypotheses, and conclusions. All widths count persistent labels; all errors are worst-word numerical errors.

Object	Hypotheses	Conclusion
M_ε	Compact H ; finite dense alphabet; compact convex outputs; no returns needed	Least finite continuous H -action with randomized initialization (Theorem 6.1)
$r(U)$	U open normal; arbitrary number of components	$M_\varepsilon < \infty$ iff some $r(U) \leq \varepsilon$, equality included (Theorem 5.5)
$W_{N,\varepsilon}$	Eventual approximate returns; even an irreversible topological monoid	$W_{N,\varepsilon} \rightarrow M_\varepsilon$; finite limits eventually equal; all-profile occupation (Theorem 4.2)
R_0	Compact group and eventual approximate returns	$R_0 = \inf_U r(U)$; critical boundedness iff the infimum is attained (Theorem 6.3)
$M_{r,\varepsilon}$	Block length p , phase subgroup J ; block approximate returns	$W_{N,\varepsilon} \rightarrow M_{r,\varepsilon}$ for $N \equiv r$; no extra phase labels (Theorem 6.4)
ε_r	Finitely many components; nonempty exact-return language	Component radius in the reachable phase; earlier localized occupation proof (Theorem 9.3)
Exact boundary	Cofinite exact returns; finite-dimensional outputs	Finite translate rank yields a finite observable quotient (Theorem 10.5)
Quantitative width	Separate arithmetic, profinite, or uniform-orbit spectral-gap hypotheses	Planar and dyadic laws; sharp spherical and matrix-orbit power laws (Theorems 15.5, 17.3)
Extreme outputs	Affine group action; target orbit consists of legal extreme points	Exact profile equals orbit cardinalities; free example $2 \cdot 3^N - 1$ (Theorems 18.1, A.2)
Finite encodings	Explicit finite group and rational categorical tables	$\exists \mathbb{R}$ -complete stationary decision; dyadic precision and random-bit budgets (Theorems 21.1, 21.3)

2. REPEATABLE OBSERVATIONS AND CAUSAL REALIZATION

Terminal numerical accuracy and accuracy of a generated observation process are different requirements. Repeating a nondisturbing observation can separate arbitrarily close, but distinct, numerical responses. We now formulate this distinction within the same finite-label resource model.

Let H be a compact Hausdorff group, let $\mathcal{A}$ be a finite nonempty command alphabet generating a dense subgroup, and let $\mathcal{S}$ and $\mathcal{J}$ be finite nonempty sets. For each s, j fix a continuous law

$$p_{s,j} : H \longrightarrow \Delta_{M_j}.$$

The physical state starts at $g = e$. A command a replaces g by $u_a g$ and emits a fixed null symbol. A probe j leaves g unchanged and emits a fresh draw from $p_{s,j}(g)$, conditionally on the public past and current physical state. At each step the next command or probe may be chosen from the entire observed transcript by a randomized policy. The seed is specified at initialization. Probes are repeatable statistical observations; when $p_{s,j}$ is obtained from a density matrix, this is an oracle model, not repeated nondisturbing measurement of one unknown quantum system.

Definition 2.1 (Causal finite-label machine). At horizon N a machine has registers $S_0, \dots, S_N$, initialization rows E_s , and, for each control c and possible output y , a nonnegative matrix $K_{t,c,y} : S_{t-1} \rightarrow S_t$ such that $\sum_y K_{t,c,y}$ is row-stochastic. Its entries jointly specify the emission and next label. The matrices may depend on t, N but on the past only through the retained label and current control. Its error is

$$\sup_{s,\pi} \text{TV}(\mathbb{P}_{s,N}^\pi, \mathbb{Q}_{s,N}^\pi),$$

where the laws include the complete control–output transcript under the same policy π . Write $\text{PW}_{N,\varepsilon}$ for the minimum peak register size at error at most ε .

Every advertised label and all retained private randomness are counted. The horizon, external clock and arbitrary real tables remain free. The policy is an external experimenter, not an uncharged private memory for the machine. Outputs may influence future controls, but are not supplied again as a free history input. A sampled output register, when charged, adds the separate fixed output-alphabet cost. The error above is one budget for the entire transcript, not a per-step error budget.

For (s, g) define its future response function

$$(2.1) \quad \Theta_{s,g}(x, j) = p_{s,j}(xg), \quad x \in H, \quad \mathcal{X} = \{\Theta_{s,g} : s \in \mathcal{S}, g \in H\}.$$

Equip the finite product of continuous-function spaces with the uniform norm. Put $n = |\mathcal{X}|$ when this set is finite, and $n = \infty$ otherwise. This is a behavioral quotient, not the dimension of a linear predictive representation. In particular, infinitely many distinct predictive rows do not by themselves imply infinite nonnegative realization size for a general hidden Markov process [26, Theorem 2 and Section 4]. Repeatable observations are the additional input in the theorem below.

Theorem 2.2 (Uniform process error and finite physical actions). *For every fixed $0 \leq \varepsilon < 1$ the following are equivalent:*

- (1) $\sup_N \text{PW}_{N,\varepsilon} < \infty$;
- (2) $n < \infty$;
- (3) *some open normal subgroup $U \triangleleft H$ makes every $p_{s,j}$ constant on each coset of U ;*
- (4) *a stationary exact causal realization exists with permutation command updates and state-preserving probe emissions.*

If $n < \infty$ and $0 \leq \varepsilon < 1/2$, then

$$(2.2) \quad \text{PW}_{N,\varepsilon} = n \quad \text{for all sufficiently large } N.$$

If $n = \infty$, then, for every $\varepsilon < 1$ and every $k \geq 1$, there is $L_{k,\varepsilon} < \infty$ such that every admissible realization satisfies

$$(2.3) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_{k,\varepsilon}.$$

The same occupation conclusion holds for finite n , $\varepsilon < 1/2$ and $k < n$. Intervening registers are unrestricted. No return condition is imposed on the command alphabet: the probes themselves leave the physical state unchanged and supply the padding used in the proof.

The threshold 1 in the boundedness criterion is sharp: total variation never exceeds one, so at tolerance one a single label always suffices. The elementary one-continuation proof of (2.2) uses $\varepsilon < 1/2$. Theorem 3.4 below proves the full minimum for every $\varepsilon < 1$ by a different tail-event argument; the present proof and its stronger one-cut test remain useful in their stated range.

Lemma 2.3 (A common separating itinerary). *Choose $m \geq 2$ different functions in (2.1), represented by executable seed-prefix pairs. For every $\delta > 0$ there is one finite deterministic control itinerary with disjoint output events $A_1, \dots, A_m$ such that its target law from the i th state assigns probability at least $1 - \delta$ to A_i .*

Proof. Equality of two future response functions is preserved and reflected by left multiplication of their physical states by any $u \in H$: replace x in (2.1) by xu , a bijection of H . Positive executable products are dense in H . Indeed positive powers approximate the inverse of every generator in a compact group, so the closed positive semigroup contains the dense generated group.

List the $\binom{m}{2}$ pairs. At each stage the current two states for that pair still have different future response functions. By continuity and positive-word density, a finite executable word followed by one probe separates some coordinate of their output laws. Append that word and record the coordinate, then proceed to the next pair. This constructs one itinerary, not a family of incompatible pair-dependent suffixes. No reset or inverse word is required. Let $J = \binom{m}{2}$, and let $\mu_{i,r}$ be the selected coordinate probability at the r th probe location when the initial state is the i th representative. There is $\Delta > 0$ such that every pair of rows of this finite matrix differs by at least Δ in some coordinate.

At each location repeat the chosen probe L times. These repetitions leave the physical state fixed. The empirical coordinate frequencies $\hat{\mu}_r$ obey

$$\mathbb{P}_i\{\max_r |\hat{\mu}_r - \mu_{i,r}| \geq \Delta/3\} \leq 2J \exp(-2L\Delta^2/9).$$

For completeness, the scalar bound follows from $\mathbb{E}e^{z(B-\mathbb{E}B)} \leq e^{z^2/8}$ for a Bernoulli variable, exponential Markov inequality, and optimization in z ; a union bound gives the display. The moment bound follows by differentiating the logarithmic moment generating function twice: its second derivative is the variance of a tilted Bernoulli variable and is at most $1/4$. Choose $L \geq 1$ with the displayed upper bound at most δ . The events $A_i = \{\max_r |\hat{\mu}_r - \mu_{i,r}| < \Delta/3\}$ are disjoint, because $2\Delta/3 < \Delta$, and have the required probabilities. $\square$

Lemma 2.4 (Testing across a finite register). *Suppose m equal-length seed-prefix pairs at cut t admit an itinerary of length ℓ as in Lemma 2.3. For every $N \geq t + \ell$, an error- ε causal realization satisfies*

$$(2.4) \quad |S_t| \geq m(1 - \varepsilon - \delta).$$

If $\varepsilon + \delta < 1/2$, the stronger bound $|S_t| \geq m$ holds.

Proof. Use the same deterministic suffix for all representatives and pad its end with probes if necessary. Marginalize all prefix outputs. Write $\alpha_i(z)$ for the resulting cut-label distribution and

Q_z for the suffix-output law conditional on label z . Causality makes Q_z independent of the chosen prefix. Total variation contracts under marginalization, so correctness gives

$$1 - \delta - \varepsilon \leq \sum_z \alpha_i(z) Q_z(A_i).$$

This uses no conditional correctness assertion after a rare observed prefix. Summing in i , using $\alpha_i(z) \leq 1$, and then disjointness, yields

$$m(1 - \delta - \varepsilon) \leq \sum_z \sum_i Q_z(A_i) \leq |S_t|.$$

If $1 - \delta - \varepsilon > 1/2$, each i has a positive-mass label z_i with $Q_{z_i}(A_i) > 1/2$. Disjointness forces the labels z_i to be different. This proves the stronger assertion. $\square$

Proof of Theorem 2.2. The map $g \mapsto \Theta_{s,g}$ is continuous in the uniform norm. This follows from continuity on the compact product $H \times H$ and a finite-cover argument, uniformly in the first coordinate. Thus each seed's image is compact. Executable images are dense in it. If $\mathcal{X}$ is infinite, arbitrarily many distinct types therefore have executable representatives; if it is finite, every type has one.

Choose finitely many such representatives and pad their prefixes to a common length t_0 by any fixed probe. Their physical states are unchanged. Apply Lemma 2.3. For every $t_0 \leq t \leq N - \ell$, pad further with the same probe, marginalize its outputs, and apply Lemma 2.4. The itinerary and ℓ do not depend on t, N or on the machine.

If $n = \infty$, choose m with $m(1 - \varepsilon) > k$ and then choose $\delta > 0$ with $m(1 - \varepsilon - \delta) > k$. Every interior cut has width greater than k . At most $t_0 + \ell$ positive cuts lie outside this interval, including when it is empty. This proves (2.3), divergence, and the implication (i) $\Rightarrow$ (ii). For finite n and $\varepsilon < 1/2$, use $m = n$ and $0 < \delta < 1/2 - \varepsilon$ when $n \geq 2$. All sufficiently interior cuts require n labels. The case $n = 1$ follows from nonempty registers.

For finite $\mathcal{X}$, the maps

$$\Theta(x, j) \mapsto \Theta(xu, j), \quad u \in H,$$

define a continuous permutation action of H on $\mathcal{X}$. Continuity follows from uniform continuity of each of the finitely many functions. Its kernel U is open and normal. Evaluating at $x = e$ shows that every $p_{s,j}$ is constant on U -cosets, proving (ii) $\Rightarrow$ (iii). Conversely (iii) gives at most $|\mathcal{S}||[H : U]$ response types. More economically, use the n distinct types as labels, initialize at $\Theta_{s,e}$, update by the permutation above, and emit $\Theta(e, j)$ on probe j without changing the label. This produces the exact conditional response at each step. Induction on transcript length proves equality for every adaptive policy, hence (iv) and (i). It also supplies the upper bound n in (2.2). The occupation assertion for $k < n$ follows from the stronger testing bound just established. $\square$

Corollary 2.5 (Connected groups). *If H is connected, uniformly bounded causal width at any fixed total transcript error below one is possible exactly when every $p_{s,j}$ is constant on H . If at least one is nonconstant, every fixed width has the occupation bound (2.3).*

Proof. A continuous action of a connected group on a finite discrete set is trivial. The assertion follows from the theorem. $\square$

Remark 2.6 (Why terminal purification is insufficient). A two-label variable chosen once with equal masses and then repeatedly reported has the correct Bernoulli(1/2) marginal at every time, but its length- L law is supported on the two constant strings. Its total variation distance from L independent fair bits is $1 - 2^{1-L}$. Thus even exact numerical marginals do not supply a path-law realization. The causal theorem concerns an actual nonnegative emission instrument, not a list of separately correct decoders. It neither classifies arbitrary irreversible hidden-state systems nor asserts a measurement procedure for noncommuting quantum observables.

3. A STRONG CONVERSE FOR REPEATABLE OBSERVATION PROCESSES

The one-continuation testing bound distinguishes the full number of response types only when the error is below one half. Repeating a common diagnostic cycle gives a stronger obstruction: a finite hidden register cannot support more positive-probability observation-tail types than it has labels. The underlying tail-event principle is classical [30, 31, 32]. We give the short argument needed here, including the hidden-emission and nonhomogeneous conventions, before applying compactness to separately redesigned finite-horizon machines.

Lemma 3.1 (Tail capacity at narrow cuts). *Let $(S_t, Y_t)_{t \geq 0}$ be a hidden emission process with finite registers and finite output alphabets, whose joint next-state and emission law depends on the entire past only through S_t and the external time. Its transition kernels may be nonhomogeneous and irreversible. Suppose $|S_t| \leq k$ at infinitely many deterministic times. Then there cannot be more than k pairwise disjoint observation tail events having positive probability. Here an observation tail event belongs to $\bigcap_{t \geq 0} \sigma(Y_{t+1}, Y_{t+2}, \dots)$.*

Proof. Let $B_1, \dots, B_r$ be such events. Include the initialization variable, all past hidden labels, and all past emissions in $\mathcal{F}_t$. The Markov property for future emissions gives a version

$$\mathbb{P}(B_i \mid \mathcal{F}_t) = b_{i,t}(S_t), \quad b_{i,t}(s) = \mathbb{P}(B_i \mid S_t = s).$$

Only labels of positive probability are relevant. Since each B_i is measurable with respect to $\mathcal{F}_\infty = \sigma(\bigcup_t \mathcal{F}_t)$, upward martingale convergence gives

$$b_{i,t}(S_t) \longrightarrow \mathbf{1}_{B_i} \quad \text{almost surely and in } L^1.$$

Consequently, for every sufficiently large t and each i , some positive-probability label s_i satisfies $b_{i,t}(s_i) > 1/2$. There is one common sufficiently large time for the finite collection of events. Disjointness implies $\sum_i b_{i,t}(s_i) \leq 1$, so these labels are distinct. At an arbitrarily late narrow cut, $r \leq k$. This is the conditional-probability zero-one mechanism of [32, Theorem 1 and Section 5.1], applied to the emitted tail; no stationarity, mixing rate, or lower bound on state masses is used. $\square$

Fix pairwise distinct laws $P_1, \dots, P_n$ on one finite alphabet Ω . An R -step k -label instrument has one initialization row α_i for each i and common, possibly time-dependent, nonnegative emission matrices $K_{t,y}$ of size $k \times k$ with $\sum_y K_{t,y}$ row-stochastic. Let $Q_{i,R}$ be its output law, and set

$$(3.1) \quad e_{k,R} = \min_{\alpha_i, K_{t,y}} \max_{1 \leq i \leq n} \text{TV}(Q_{i,R}, P_i^{\otimes R}).$$

The minimum is attained in a finite product of compact simplexes. Initialization may be randomized and may depend on i ; subsequent kernels may not depend on an unrecorded value of i .

Proposition 3.2 (Uniform finite witnesses for product laws). *If $k < n$, then $e_{k,R}$ is nondecreasing in R and*

$$(3.2) \quad \lim_{R \rightarrow \infty} e_{k,R} = 1.$$

In particular, for every $\varepsilon < 1$ there is an integer $R(k, \varepsilon, P_1, \dots, P_n)$ for which no such instrument has error at most ε . When the laws and the tolerance are rational, one can find such an integer by a terminating sequence of finite real-algebraic feasibility decisions. No effective bound on the number of decisions, or polynomial-time claim, is made.

Proof. Truncation and contraction of total variation under marginalization show that $e_{k,R} \leq e_{k,R+1}$. Suppose its limit were some $e < 1$. Consider the compact countable product consisting of the initialization rows and all kernels $(K_{t,y})_{t \geq 1}$. For every R , impose all closed constraints $\text{TV}(Q_{i,r}, P_i^{\otimes r}) \leq e$ for $i \leq n, r \leq R$. Each finite system is feasible by attainment in (3.1). Nested compactness gives one infinite hidden process satisfying every finite constraint, with common kernels and k labels at every time.

Let Q_i denote its infinite output law and P_i^∞ the product law. The finite constraints imply $\text{TV}(Q_i, P_i^\infty) \leq e$. Indeed, for any event, approximate its indicator in L^1 under the finite measure $Q_i + P_i^\infty$ by indicators of cylinder events; these generate the product sigma-field. The bound on event-probability differences passes to the limit. Equivalently, one can apply martingale convergence to conditional probabilities under this finite measure.

For each i , let B_i be the event that empirical frequencies of all symbols converge to P_i . These are disjoint observation tail events, and the strong law gives $P_i^\infty(B_i) = 1$. Hence $Q_i(B_i) \geq 1 - e > 0$. Mix the initialization index uniformly, retaining the same common transition kernels. The resulting hidden process has output law $Q = n^{-1} \sum_i Q_i$, so $Q(B_i) > 0$ for every i . Lemma 3.1 contradicts $n > k$. This proves the limit.

For the last assertion, introduce the finitely many stochastic rows as real variables. Every word probability is a polynomial in their entries; the targets are rational. Absolute-value slack variables express each finite total-variation inequality by polynomial equalities and inequalities. Real quantifier elimination decides feasibility. Test $R = 1, 2, \dots$ until it is infeasible. The limit just proved guarantees termination. This is a finite-witness search, not an implementation of a general minimum-width algorithm. $\square$

Lemma 3.3 (A diagnostic cycle). *Suppose the response-type set $\mathcal{X}$ in (2.1) is finite, of cardinality $n \geq 2$. There is a finite deterministic control word B of positive length ℓ such that its action on $\mathcal{X}$ is the identity and its output laws $P_1, \dots, P_n$ from the n types are pairwise distinct. Successive executions of B give independent, identically distributed P_i -blocks from initial type i . Any intervening fixed probe controls may be inserted if their outputs are discarded.*

Proof. Every type has an executable seed-prefix representative, as proved in Theorem 2.2. Apply Lemma 2.3 to all n types with, for example, $\delta = 1/4$. The resulting finite deterministic itinerary T has pairwise distinct output laws: its disjoint diagnostic events have probability at least $3/4$ under the corresponding laws. Its command product induces a permutation π of the finite set $\mathcal{X}$. Let $d \geq 1$ be the order of this permutation and take $B = T^d$. Its action on types is the identity, and its output laws remain distinct because their first- T marginals are distinct.

The output law of any fixed continuation depends only on the response type, not on its chosen representative. Probes emit fresh conditionally independent observations and have no physical effect. Since B returns the type deterministically, its conditional output law after any number of previous blocks is again P_i . Induction gives the product law. Intervening probes also preserve the type; their ignored outcomes do not change the deterministic controls or the conditional block laws. The cycle is a return on the *finite response quotient*, not an assumed identity-product word in the physical group. $\square$

Theorem 3.4 (The full-error causal minimum). *In the repeatable-probe model of Definition 2.1, let $n = |\mathcal{X}| \in \mathbb{N} \cup \{\infty\}$. For every fixed $0 \leq \varepsilon < 1$,*

$$(3.3) \quad \lim_{N \rightarrow \infty} \text{PW}_{N, \varepsilon} = n.$$

When $n < \infty$, equality with n holds at every sufficiently large horizon. For every integer $k < n$ there is $L_{k, \varepsilon} < \infty$ such that every error- ε realization has

$$(3.4) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq L_{k, \varepsilon},$$

with no restriction on intervening widths. Thus the finite-response cardinality is the eventual minimum throughout the whole interval below the maximal total-variation error, not only below one half. The full-transcript and fresh nondisturbing probe assumptions are unchanged. No command return condition is imposed.

Proof. For $n = \infty$, Theorem 2.2 already proves (3.4) for every k and $\varepsilon < 1$. For $n = 1$, nonempty registers and the exact one-type machine prove the assertion. Suppose $2 \leq n < \infty$ and $k < n$.

Choose executable representatives of all types, and pad their prefixes with a fixed probe to one common length $b \geq 0$. Let B , ℓ and P_i be as in Lemma 3.3. Fix an integer R for which $e_{k,R} > \varepsilon$, supplied by Proposition 3.2.

Consider an advertised profile. Discard narrow cuts with $t < b$, and greedily select from the others $t_1 < \dots < t_J$ with gaps at least ℓ . If there are $R + 1$ selected cuts, start from each of the equal-length representatives, pad to t_1 with the same probe, and put one copy of B immediately after each of $t_1, \dots, t_R$. Fill the remaining positions before the next selected cut with that probe. Complete the horizon with any fixed controls. For each representative the retained R block outputs have target law $P_i^{\otimes R}$.

Marginalize all prefix, filler, and post-block outputs. In the proposed machine, the actual transition from one selected cut to the next, jointly with the retained block output, is a nonnegative instrument between at most k labels. Summing over the discarded emissions gives its stochastic normalization. All its entries are common to the representatives: the schedule is deterministic and the retained label is the only private past information available to a row. Intervening registers may be arbitrarily wide. The cut- t_1 distributions are allowed to be the arbitrary initialization rows α_i in (3.1). After padding unused labels, these composite kernels therefore define an R -step k -label instrument.

Whole-transcript correctness under these deterministic policies and contraction of TV under marginalization give its error at most ε . This contradicts $e_{k,R} > \varepsilon$. Thus $J \leq R$. The discarded cuts number at most b , and each selected cut accounts for at most ℓ remaining narrow cuts. Consequently one may take

$$L_{k,\varepsilon} = b + \ell R.$$

This count includes empty eligible intervals and excludes cut zero. The prefix, cycle, and witness integer depend only on the interface, k and ε , not on the horizon or machine. No correctness conditional on a rare observed prefix was used.

An exact stationary n -type machine works at every horizon by Theorem 2.2. Applying the occupation bound to $k = n - 1$ shows that a smaller peak is impossible once $N > L_{n-1,\varepsilon}$. This proves eventual equality and the theorem. $\square$

Corollary 3.5 (Strong converse below the response cardinality). *If $k < n$, the least worst-policy whole-transcript error achievable by peak width at most k tends to one as the horizon tends to infinity. At error one, one label always suffices. When the finite group action and all probe tables are explicitly rational, the proof supplies a terminating search for a valid occupation constant via the finite product-law feasibility problems.*

Proof. For each $\varepsilon < 1$, the preceding occupation bound excludes a k -peak realization for every $N > L_{k,\varepsilon}$. TV is at most one, which proves the limit and the endpoint. For explicit finite data, response types and their command permutations can be enumerated. The common finite itinerary and its permutation order can be found by finite search, its cycle laws are rational, and Proposition 3.2 finds R . The claimed bound $b + \ell R$ then follows. No efficiency conclusion follows from this termination proof. $\square$

Proposition 3.6 (Why one half is not the process boundary). *For $P_1 = \delta_0$, $P_2 = \delta_1$ and one hidden label,*

$$(3.5) \quad e_{1,R} = 1 - 2^{-R}.$$

Nevertheless, a single width-one cut, followed by two-label storage, can have worst error one half for these two product laws at arbitrarily large remaining horizons.

Proof. A one-label instrument has independent time-dependent emissions with probabilities p_t of zero. Its two target-string probabilities are $u = \prod_t p_t$ and $v = \prod_t (1 - p_t)$. The two TV errors are

$1 - u$ and $1 - v$. Since $uv \leq 4^{-R}$, their maximum is at least $1 - 2^{-R}$, attained by $p_t = 1/2$ for every t . For the last assertion, after the one-label cut choose a fair bit, store it in two labels, and emit it repeatedly. Its law is the equal mixture of the two deterministic strings and is at distance one half from each. Thus for errors above one half the theorem needs control of *repeated* narrow cuts; it does not assert the false pointwise lower bound at every interior cut. $\square$

The tail-capacity lemma is a necessary condition for arbitrary hidden emission processes, including irreversible ones. The diagnostic-cycle application, however, still uses the stated repeatable-probe model. It is neither a general HMM/POMDP realization classification nor a claim about nondisturbing repeated quantum measurements. The new conclusion is the all-error finite-horizon strong converse and occupation bound obtained from the classical tail principle, finite-product compactness, and one common physical diagnostic cycle.

4. STATIONARIZATION FROM RECURRENT NARROW CUTS

The first reduction does not require reversible physical dynamics. In this section only, replace H by a topological monoid G with identity 1, retain a finite command alphabet, and let $F_{s,j} : G \rightarrow Y_j$ be continuous. The output sets are norm-compact convex subsets of real Banach spaces. Define M_ε and $W_{N,\varepsilon}$ by the same stochastic models. Compactness of the physical monoid is not required.

Definition 4.1 (Eventual approximate returns). The alphabet has eventual approximate returns if, for every neighborhood V of 1, there is an integer $g(V) \geq 0$ such that every integer $n \geq g(V)$ is the length of a word with product in V . Cofinite exact returns mean that a single g works with product exactly 1.

The statement concerns physical products only. The hidden rows on a return word need not be close to an identity matrix. Approximation here is eventual at every length, not merely along a return subsequence.

Theorem 4.2 (Width-preserving stationarization). *Assume eventual approximate returns. For every $\varepsilon \geq 0$ and integer $k \geq 1$, the following are equivalent:*

- (1) *a stationary realization with at most k labels has error at most ε ;*
- (2) *among error- ε clocked realizations, the number of positive cuts having width at most k is unbounded.*

Consequently, if $M_\varepsilon > k$, there is $L_{k,\varepsilon} < \infty$, independent of the horizon, such that every error- ε realization satisfies

$$(4.1) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_{k,\varepsilon}.$$

All other registers may be arbitrarily large. Moreover,

$$(4.2) \quad \lim_{N \rightarrow \infty} W_{N,\varepsilon} = M_\varepsilon \quad \text{in } \mathbb{N} \cup \{\infty\}.$$

If the right side is finite, equality holds for every sufficiently large N .

The theorem preserves the proposed width, including randomized initialization. It does not assert that the microscopic rows of the clocked machine become stationary. They can be entirely unrelated at different times.

4.1. Finite witnesses and homogeneous transition tables. For a matrix A write $\|A\|_r = \max_i \sum_j |A_{ij}|$. Put $a = |\mathcal{A}|$ and $D_* = \max(1, \max_j \text{diam } Y_j)$. Let $\mathcal{X}_k$ be the compact parameter space of stationary k -label machines. For $m \geq 0$ define

$$e_{k,m} = \min_{X \in \mathcal{X}_k} \max_{s,j, |w| \leq m} \|X_{s,j}(w) - F_{s,j}(u_w)\|_j.$$

The finite maximum is continuous, so the minimum is attained. If no stationary error- ε machine exists, compactness of $\mathcal{X}_k$ and the finite intersection property give an m with

$$(4.3) \quad e_{k,m} = \varepsilon + \gamma, \quad \gamma > 0.$$

Indeed the nested closed sets defined by the constraints $|w| \leq m$ otherwise have a common point. This argument also applies when the full infinite-word error is not a continuous function of the parameters.

We shall use the finite edge-coloring theorem of Ramsey [25]. Denote by $R_q(h)$ an integer such that every coloring of the edges of a complete graph on $R_q(h)$ vertices with at most q colors has a monochromatic h -vertex subgraph. Only finiteness is essential.

Remark 4.3 (An explicit but coarse Ramsey bound). For definiteness one may take

$$(4.4) \quad R_q(h) \leq q^{qh+1} \quad (q \geq 2, h \geq 2).$$

To see this elementary bound, successively select a vertex and retain a largest color class among its remaining neighbors. With q^{qh+1} initial vertices this produces at least $q(h-1) + 1$ selected vertices, each of whose edges to later selected vertices has one color. At least h of their associated colors agree; these vertices form the required subgraph. The deliberately excessive exponent avoids any endpoint rounding issue.

Partition $[0, 1]$ into at most $\lceil k/\delta \rceil + 1$ intervals of length at most δ/k . Two tuples in the same entrywise cell of

$$\text{Stoch}(k)^{\{0\} \cup \mathcal{A}}$$

have corresponding matrices within δ in $\|\cdot\|_r$. The number of occupied cells is at most

$$(4.5) \quad q = \max(2, (\lceil k/\delta \rceil + 1)^{(a+1)k^2}).$$

Each representative used below is chosen from an occupied cell and is therefore stochastic; an entrywise rounded matrix need not be stochastic and is not used as a representative.

Proof of Theorem 4.2. The forward implication follows by using one stationary machine at every horizon. For the converse, suppose (4.3). Choose $0 < \eta < \gamma/4$. Continuity of multiplication and of the finitely many $F_{s,j}$ gives a neighborhood V of 1 with the following property. In any word of length at most m , inserting at most $m+3$ factors from V , including before and after the word, changes every output by less than η . There are only finitely many relevant command words and insertion patterns; continuity at the tuple of identity factors, followed by a finite intersection of neighborhoods, proves the assertion. This uses neither an invariant metric nor commutativity.

Let $g = g(V)$, put $b = g + 1$, and choose

$$0 < \delta \leq \min(1, \gamma/(4D_*(m+1))).$$

Take a clocked realization with so many narrow cuts that, after discarding cuts $t < g$ or $t > N - g$ and greedily retaining cuts separated by at least b , at least $R_q(m+2)$ remain. Such a choice is possible by hypothesis; at most $2g$ cuts are discarded and each retained cut accounts for at most b of the others.

Identify each retained register with a subset of $\{1, \dots, k\}$. Extend rows from dummy labels arbitrarily to stochastic rows without altering any row on the actual register. For retained cuts $t_i < t_j$ write $\ell = t_j - t_i \geq g + 1$. Choose a length- ℓ return word with product in V , and let A_{ij} be its actual composite kernel between these cuts. For each letter $c \in \mathcal{A}$, choose the word consisting of c followed by a length- $(\ell - 1)$ return word with product in V , and let $B_{ij,c}$ be its composite kernel. Thus the physical products of these words are respectively in V and of the form vu_c with $v \in V$. All the matrices are of order k , however large the intermediate registers may be.

Color the edge $\{i, j\}$, oriented from the earlier cut to the later cut, by the cell of $(A_{ij}, (B_{ij,c})_c)$. Ramsey's theorem gives $m + 2$ ordered cuts

$$t_0 < t_1 < \cdots < t_{m+1}$$

whose edges have one color. Fix a stochastic representative $(E, (B_c)_c)$ from that color. Choose fixed return words for the prefix of length t_0 and the suffix of length $N - t_{m+1}$, both with products in V . Let α_s be the actual state distribution after the prefix. Let $D_j(i)$ be the conditional expected original decoder after the suffix, given label i at its start, with arbitrary legal values on dummy labels. Convexity gives $D_j(i) \in Y_j$.

Consider the stationary machine

$$(4.6) \quad \alpha_s, \quad T_c = B_c, \quad d_j = ED_j.$$

Its decoder is legal. To test $w = c_1 \cdots c_\ell$, $0 \leq \ell \leq m$, use the first ℓ consecutive clique edges with kernels B_{i-1, i, c_i} and then the single edge from t_ℓ to t_{m+1} with kernel $A_{\ell, m+1}$. For $\ell = 0$ there is only this last edge. Add the fixed prefix and suffix. This is a genuine word of length N ; its physical product is obtained from u_w by at most $\ell + 3$ insertions from V . Its target output is consequently within η of $F_{s,j}(u_w)$.

Each of its $\ell + 1$ edge matrices differs from the corresponding matrix in (4.6) by at most δ . Stochastic multiplication contracts the ℓ^1 distance between probability rows. A telescoping expansion, and subtraction of any fixed point of Y_j from the decoder, therefore bound the difference between the actual output and $\alpha_s B_{c_1} \cdots B_{c_\ell} ED_j$ by

$$D_*(\ell + 1)\delta.$$

Wordwise correctness of the original machine implies $e_{k,m} \leq \varepsilon + \eta + D_*(m + 1)\delta < \varepsilon + \gamma$, a contradiction. The representative E is used only as part of the terminal decoder. No idempotence of E , consistency between different filler words, or hidden-state identity on a physical return was assumed.

The same argument shows that one may take

$$(4.7) \quad L_{k,\varepsilon} = 2g + (g + 1)R_q(m + 2).$$

For exact cofinite returns, use exact fillers and omit η and the neighborhood choice; g is the given exact-return bound. In general g depends on the finite witness and the continuity neighborhood. The remaining parameters depend only on the interface, the alphabet, k , and ε , never on a machine or horizon.

If $M_\varepsilon = \infty$, (4.1) for each k gives $W_{N,\varepsilon} \rightarrow \infty$. If $M_\varepsilon = K < \infty$, the stationary machine gives $W_{N,\varepsilon} \leq K$ for every N , and the bound for $k = K - 1$ gives eventual equality when $K > 1$. When $K = 1$ equality holds for every N because registers are nonempty. $\square$

Proposition 4.4 (Approximate returns without exact returns). *Let the two circle commands have angles θ, ϕ , where $1, \theta/(2\pi), \phi/(2\pi)$ are rationally independent. They have no nonempty exact-return word, but satisfy Definition 4.1.*

Proof. A length- n product has angle $n\phi + j(\theta - \phi)$, $0 \leq j \leq n$. The positive orbit of the irrational rotation $\theta - \phi$ is dense. Compactness supplies, for every accuracy, a finite initial orbit segment that is a net of the circle. All longer initial segments remain nets; rotation by $n\phi$ preserves their accuracy. Thus a product is arbitrarily close to the identity at every sufficiently large length. An exact return would be a nontrivial integer relation among $1, \theta/(2\pi), \phi/(2\pi)$. $\square$

A single irrational letter fails the eventual condition: there is only one word at each horizon. Its target can be placed in a one-label horizon-dependent decoder even when no finite stationary approximation at the same tolerance exists. Thus mere recurrence along a subsequence is not a substitute for Definition 4.1.

Remark 4.5 (Quantitative interpretation). The Ramsey estimate gives a finite occupation constant, not a useful asymptotic width rate. The homogeneous neutral representative remains inside the terminal decoder in the extraction proof. It is neither an identity transition nor an assumed idempotent.

5. STATIONARY STOCHASTIC PURIFICATION

Write $\text{Stoch}(k)$ for the compact semigroup of row-stochastic matrices of order k . The identity of a group inside this semigroup can be a singular idempotent; it must not be confused with the identity matrix.

Lemma 5.1 (An idempotent over the physical identity). *Let $\mathcal{T}$ be a compact subsemigroup of $H \times \text{Stoch}(k)$ whose projection onto H is surjective. There is an element $e = (1, E) \in \mathcal{T}$ with $E^2 = E$ and rank equal to the minimum matrix rank in $\mathcal{T}$. The monoid $e\mathcal{T}e$ is a compact group with identity e and still projects onto H .*

Proof. Only the integer ranks $1, \dots, k$ occur, so their nonempty set has a minimum which is attained. Choose (h, T) of that minimum rank r . The powers of T are bounded, since they are stochastic. Thus its complex eigenvalues have modulus at most one and all Jordan blocks on the unit circle have size one: a larger peripheral Jordan block would produce polynomially growing powers. The closure of positive powers of an element of a compact group is a group: arbitrarily large positive powers approach its identity, and the preceding powers approach its inverse. Apply this observation to the compact monothetic group generated by $(h, \lambda_1, \dots, \lambda_b)$ in $H \times (S^1)^b$, where the λ_i are the peripheral eigenvalues of T . There is a net of integers tending to infinity for which $h^n \rightarrow 1$ and all peripheral eigenvalues raised to the n th power tend to one. Along it T^n tends to the spectral projection E onto the peripheral subspace. This is a real stochastic idempotent. Its rank is at most r and at least r by minimality, so it is r . Nets avoid any metrizability assumption on H .

Every matrix B in $e\mathcal{T}e$ satisfies $B = EBE$ and has rank r . Its action on the row space of E is consequently invertible. Restriction embeds $e\mathcal{T}e$ as a compact submonoid of $H \times \text{GL}(r, \mathbb{R})$, with identity $(1, I_r)$; injectivity follows because the restriction determines $(xE)B = xB$ for every row x , by $B = EBE$, and hence determines every row of B . A compact submonoid of a topological group is a subgroup, by the same positive-power return argument. The inverse images of its inverses are still stochastic. Finally, sandwiching an element with physical coordinate h between e 's leaves that coordinate unchanged. Surjectivity follows. $\square$

Lemma 5.2 (The recurrent simplex). *If E is a stochastic idempotent of rank r , then*

$$\Delta_k E = \{p \in \Delta_k : pE = p\}$$

is a simplex with exactly r vertices. Every group of stochastic matrices with identity E permutes these vertices.

Proof. Regard E as a transition matrix. Its closed communicating classes each have a unique stationary probability row supported on that class. Every stationary row is a convex combination of these rows: mass on a transient class is zero, and stationarity within a closed irreducible class determines a one-dimensional space. For completeness, transient mass vanishes because each transient state has a positive probability of reaching a closed class in a bounded number of steps; a common bound over the finitely many states makes transient mass decrease by a fixed factor unless it is zero. The closed-class rows have disjoint supports, hence are linearly independent.

Each row of E is stationary because $E^2 = E$. Conversely, every stationary row equals its product with E . The span of the closed-class stationary rows is therefore precisely the row space of E , so there are r classes and r simplex vertices. If B has a stochastic inverse relative to E ,

multiplication by B and by this inverse are inverse affine maps of this simplex. Affine bijections permute its vertices. $\square$

Theorem 5.3 (Purification without a width or error loss). *For every compact-group interface in Definition 1.1, a stationary k -label realization of error at most ε admits a permutation realization of error at most ε on at most k labels. There is no return-word hypothesis and no restriction on the number of connected components of H .*

Proof. For the given machine form

$$\mathcal{T} = \overline{\{(u_w^{-1}, T_w) : w \in \mathcal{A}^*\}} \subset H \times \text{Stoch}(k).$$

The inverse on the physical coordinate is important: concatenation satisfies $u_{vw}^{-1} = u_v^{-1}u_w^{-1}$ and $T_{vw} = T_v T_w$. Thus $\mathcal{T}$ is a compact semigroup. Positive-word density makes its projection onto H surjective. By continuity, its every element (h, B) satisfies

$$(5.1) \quad \|\alpha_s B D_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon \quad (s \in \mathcal{S}, j \in \mathcal{J}).$$

Apply Lemma 5.1 and denote the vertices of $\Delta_k E$ by $\nu_1, \dots, \nu_r$. For each letter the element

$$e(u_a^{-1}, T_a)e = (u_a^{-1}, E T_a E)$$

belongs to the group $e\mathcal{T}e$. By Lemma 5.2, $B_a = E T_a E$ permutes the ν_i . Use that permutation as the new command. If V has rows ν_i and the induced row permutation is Π_a , the precise intertwining identity is $V B_a = \Pi_a V$. Write uniquely

$$\alpha_s E = \sum_{i=1}^r \beta_s(i) \nu_i, \quad \tilde{D}_j(i) = \nu_i D_j \in Y_j.$$

The last inclusion uses convexity of the legal output set. For a nonempty word the output of the resulting r -label machine is $\alpha_s B_{a_1} \cdots B_{a_n} D_j$. For the empty word it is $\alpha_s E D_j$. In both cases the corresponding joint element lies in $\mathcal{T}$ and has physical coordinate u_w^{-1} . Equation (5.1) proves the error bound. Since $r \leq k$, the width has not increased. $\square$

The new machine need not reproduce the old machine's erroneous output word by word; it remains in the same closed error balls about the target. Nor do we assert $T_w = I$ on a physical return, invertibility of the original T_a , or deterministic initialization. These three incorrect shortcuts would invalidate the reduction.

5.1. Finite quotients and the unrestricted stationary minimum. For a tuple of permutation matrices $\sigma = (\Pi_a)_{a \in \mathcal{A}}$ of order k , set

$$\mathcal{L}_\sigma = \overline{\{(u_w^{-1}, \Pi_w) : w \in \mathcal{A}^*\}} \subset H \times \mathfrak{S}_k.$$

The topology is the product of the compact topology on H and the discrete topology on the finite group $\mathfrak{S}_k$. This is a compact subgroup. Its definition includes the possible discrepancy between physical relations and hidden-state relations.

Theorem 5.4 (An exact stationary minimum). *The value M_ε is the least positive integer k for which some tuple σ of permutations of order k , some $\beta_s \in \Delta_k$, and some $d_j(i) \in Y_j$ satisfy*

$$(5.2) \quad \|\beta_s \Pi d_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon \quad \text{for all } (h, \Pi) \in \mathcal{L}_\sigma \text{ and all } s, j.$$

For fixed k there are at most $(k!)^{|\mathcal{A}|}$ discrete choices (a crude bound before simultaneous conjugacy reduction), and the remaining feasible set is compact. The optimal stationary error at any fixed width is attained.

Proof. Necessity follows from Theorem 5.3, padding by unused labels if its output has fewer than k states, and then taking the closure of word products. Sufficiency is the restriction of (5.2) to words. The probabilities and legal decoders lie in a finite product of compact sets. Each displayed inequality defines a closed subset of that product. The worst error is continuous in these parameters, uniformly in the compact group variable, so its minimum exists. There are finitely many permutation tuples. $\square$

Theorem 5.5 (Finite observable quotient criterion). *For arbitrary compact H and the stated continuous interface,*

$$M_\varepsilon < \infty \iff r(U) \leq \varepsilon \text{ for some open normal subgroup } U \leq H.$$

If such U exists, $|\mathcal{S}[[H : U]]$ deterministic-initialized permutation labels suffice. Conversely, a stationary k -label realization supplies such a U with $[H : U] \leq k!$.

Proof. Purify a stationary realization and use its joint group $\mathcal{L}_\sigma$. Put

$$U = \{h \in H : (h, I) \in \mathcal{L}_\sigma\}.$$

This is a closed normal subgroup of H : conjugate (h, I) by an element above any $g \in H$, using surjectivity of the first projection. For a fixed permutation Π , a nonempty fiber in $\mathcal{L}_\sigma$ is a coset hU . These at most $k!$ fibers cover H , so U has finite index. A closed subgroup of finite index in a compact group is open, since its complement is a finite union of closed cosets.

For each s, j and each fiber, the single legal vector $\beta_s \Pi d_j$ is within ε of every $F_{s,j}(h^{-1})$ in that fiber. Explicitly, $(hU)^{-1} = Uh^{-1} = h^{-1}U$ by normality, so inversion preserves the ordinary U -coset form. Hence $r(U) \leq \varepsilon$. Conversely, store (s, gU) , update the coset by left multiplication by u_a , and decode a minimizing center of $F_{s,j}(gU)$. Compactness supplies the centers. This gives the asserted finite realization. $\square$

For finitely many connected components, every open subgroup contains $K = H^\circ$, and K itself is open normal. Therefore $M_\varepsilon < \infty$ holds exactly when

$$\varepsilon \geq \max_{s,j,c \in H/K} \text{rad}_{Y_j}(F_{s,j}(c)),$$

without a return assumption. Executable returns enter the stronger clocked converse, not this stationary statement. The index bound $k!$ is an existence bound on a deterministic quotient, not a claim that the stationary minimum equals its index.

Proposition 5.6 (Random initialization strictly saves states). *Let $H = \mathbb{Z}/6\mathbb{Z}$, with an identity letter and a generator, one seed, and binary success probability*

$$f(n) = \frac{1}{2} + \frac{1}{4}(-1)^n + \frac{1}{4}\cos(2\pi n/3).$$

At zero error the least deterministic component-quotient width is six, whereas the unrestricted stationary stochastic minimum is five.

Proof. The values at $n = 0, \dots, 5$ are

$$(1, \frac{1}{8}, \frac{5}{8}, \frac{1}{2}, \frac{5}{8}, \frac{1}{8}).$$

This sequence has least period six. A deterministic quotient of the cyclic component action preserving these outputs must therefore have six states. For a five-state realization take the disjoint union of a two-cycle and a three-cycle, both updating the index by $i \mapsto i + 1$ modulo their respective lengths, initialize with mass $1/2$ at the zeroth vertex of each, and use decoder rows $(1, 0)$ on the two-cycle and $(1, 1/4, 1/4)$ on the three-cycle. The identity command fixes all five labels. Direct averaging gives exactly $f(n)$, with legal binary probabilities.

If a stationary realization on at most four labels existed, purification would give one whose generator is a permutation of at most four letters. Its order is one of $1, 2, 3, 4$, as follows by listing

the possible cycle lengths. Every resulting numerical sequence has a period dividing that order, contradicting the least period six. Thus five is the unrestricted minimum. $\square$

6. PHYSICAL EQUIVARIANCE AND THE COMPLETE BOUNDARY

Return now to a compact Hausdorff group H with a finite dense generating alphabet. The output sets may still be norm-compact convex subsets of real Banach spaces. Purification removes transition randomness; a further averaging removes hidden relations that have no physical counterpart.

Theorem 6.1 (Physical-equivariant purification). *Every stationary k -label realization of error at most ε admits one on $r \leq k$ labels of the form*

$$(6.1) \quad \alpha_s \rho(h^{-1}) d_j, \quad h \in H, \quad \rho : H \longrightarrow S_r \text{ a continuous homomorphism,}$$

with the same uniform error. Here permutations are represented by row-action matrices. In particular the command row is $\rho(u_a^{-1})$, all physical group relations hold on the new register, and initialization may remain random.

Proof. Apply Theorem 5.3, whose proof uses only finite convex combinations of decoder values. Denote the resulting initialization by β_s , its decoder by D_j , and its compact joint group by $\mathcal{L} \subset H \times S_k$, with the product topology and discrete permutation factor. Unused labels may be adjoined when necessary. Let P be the second projection and

$$N = \{\pi \in P : (1, \pi) \in \mathcal{L}\}.$$

Then N is normal in P . The averaging matrix $Q = |N|^{-1} \sum_{n \in N} n$ is stochastic, satisfies $Q^2 = Q$, and commutes with P . For $(h, \pi) \in \mathcal{L}$, every $(h, n\pi)$ belongs to $\mathcal{L}$. Averaging their error inequalities and using convexity of the norm ball gives

$$(6.2) \quad \|\beta_s Q \pi D_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon.$$

Let $O_1, \dots, O_r$ be the N -orbits in the label set, and let v_i be the uniform probability row on O_i . The rows of Q are precisely these uniform orbit rows. Write V for the matrix with rows v_i . Every $\pi \in P$ maps N -orbits bijectively onto N -orbits and therefore induces a permutation $\bar{\pi}$ with

$$(6.3) \quad V\pi = \bar{\pi}V.$$

The induced action of each $n \in N$ is trivial. Two elements of $\mathcal{L}$ with the same first coordinate differ by an element of $\{1\} \times N$. Hence $(h, \pi) \mapsto \bar{\pi}$ descends to a homomorphism $\rho : H \rightarrow S_r$. It is continuous: the surjection from the compact group $\mathcal{L}$ onto the Hausdorff group H is a quotient map, and the original orbit action is continuous into a finite discrete group.

Set $\alpha_s(i) = \sum_{x \in O_i} \beta_s(x)$ and $d_j(i) = v_i D_j$. Then $\alpha_s V = \beta_s Q$ and $d_j = V D_j$. Equation (6.3) gives

$$\alpha_s \rho(h) d_j = \beta_s Q \pi D_j \quad ((h, \pi) \in \mathcal{L}).$$

Replace h by h^{-1} in (6.2). This proves (6.1); all decoder values are legal by convexity, and $r \leq k$. $\square$

Corollary 6.2 (Intrinsic stationary minimum). *The number M_ε is the least cardinality of a finite continuous H -set for which some probability initialization for each seed and legal decoder for each query approximate the interface within ε . It is independent of the chosen finite dense alphabet. It is unchanged by replacing every target with*

$$h \longmapsto F_{s,j}(\varphi(h)q),$$

where φ is a continuous group automorphism and $q \in H$.

Proof. Theorem 6.1 proves necessity, and restriction to command words proves sufficiency. For the last assertion, replace ρ by $\rho \circ \varphi$ and α_s by $\alpha_s \rho(q^{-1})$ in (6.1). Applying the inverse change of variables proves equality of the minima. $\square$

The finite quotient in Theorem 5.5 has a more precise description than its factorial bound. With $\mathcal{L}, P, N$ as above and $U = \{h : (h, I) \in \mathcal{L}\}$, the correspondence $(h, \pi) \mapsto (hU, \pi N)$ induces $H/U \simeq P/N$. Thus $[H : U] = |P/N| \leq k!$. The orbit action in Theorem 6.1 can have an additional kernel; no equality between this index and the minimal label count is asserted.

Theorem 6.3 (Complete clocked finite-quotient classification). *Assume eventual approximate returns on the compact group H . At every $\varepsilon \geq 0$, including a positive critical radius, the following are equivalent:*

- (1) $\sup_N W_{N,\varepsilon} < \infty$;
- (2) $M_\varepsilon < \infty$;
- (3) $r(U) \leq \varepsilon$ for an open normal subgroup U of H .

For each $k < M_\varepsilon$ the all-profile occupation bound (4.1) holds, and $W_{N,\varepsilon}$ converges to M_ε . Put

$$R_0 = \max_{s,j,h \in H} \text{rad}_{Y_j}(F_{s,j}(hH^\circ)).$$

Then $R_0 = \inf_{U \triangleleft H \text{ open}} r(U)$. Above R_0 bounded realizations exist; below it every fixed width has bounded occupation. At $\varepsilon = R_0$ bounded clocked realization exists exactly when the infimum is attained by an open normal subgroup.

Proof. The first two conditions are equivalent by Theorem 4.2. The second and third are equivalent by Theorem 5.5. That proof, including constrained center attainment, uses compactness and convexity of Y_j , not finite dimensionality.

For completeness, every open subgroup contains H° : the image of the connected identity component in a finite discrete quotient is a singleton. Thus $r(U) \geq R_0$. The quotient H/H° is profinite, so its open normal subgroups form a neighborhood basis at the identity. The inverse images U have intersection H° . Given a neighborhood O of H° in H , compactness supplies one such $U \subset O$: otherwise their intersections with the closed set $H \setminus O$ have the finite intersection property. Uniform continuity of the finite family of targets gives, for every $\eta > 0$, a neighborhood V of 1 such that $\|F_{s,j}(hkv) - F_{s,j}(hk)\|_j < \eta$ for all $h \in H, k \in H^\circ$, and $v \in V$. Choose $U \subset H^\circ V$. Every $F_{s,j}(hU)$ is then within η of $F_{s,j}(hH^\circ)$, so $r(U) \leq R_0 + \eta$. This proves the infimum formula and all remaining assertions. $\square$

This result includes the formerly separate zero and positive boundary cases. The finite-rank proof of Theorem 10.5 is retained below because it identifies an additional exact translate-space obstruction. No passage from an error strictly larger than R_0 to equality is used in Theorem 6.3.

6.1. Length phases without a width overhead. Let $p \geq 1$ be an integer and fix a command with product a . Let J be the closure of all products of lengths divisible by p . The group argument of Lemma 9.2, which uses only the block length, shows that J is open normal, H/J is cyclic of order dividing p , and all letters belong to Ja . Regard $\mathcal{A}^p$ as a finite alphabet on J . Assume that this block alphabet has eventual approximate returns. In particular this holds when the original nonempty exact-return language has greatest common divisor p .

For $0 \leq r < p$ let $M_{r,\varepsilon}$ be the intrinsic stationary minimum on J for

$$f_{r,s,j}(h) = F_{s,j}(ha^r).$$

Theorem 6.4 (Exact limiting width in each phase). *Under the preceding assumptions, for every r and $\varepsilon \geq 0$,*

$$(6.4) \quad W_{N,\varepsilon} \leq M_{r,\varepsilon} \quad (N \equiv r \pmod{p}), \quad \lim_{\substack{N \rightarrow \infty \\ N \equiv r \pmod{p}}} W_{N,\varepsilon} = M_{r,\varepsilon}.$$

Finite limits are eventually attained. If $k < M_{r,\varepsilon}$, the number of positive cuts of width at most k is bounded uniformly over that horizon phase, with no restriction on other widths. Boundedness in the phase is equivalent to the existence of an open normal subgroup $U \leq J$ such that

$$(6.5) \quad \max_{s,j,h \in J} \text{rad}_{Y_j}(F_{s,j}(hUa^r)) \leq \varepsilon.$$

Proof. Suppose first that $M_{r,\varepsilon} = K < \infty$ and choose a physical-equivariant stationary realization of f_r on K labels, with representation $\rho : J \rightarrow S_K$. For a fixed $N \equiv r \pmod{p}$, let g_t be the physical product after t commands and set

$$(6.6) \quad z_t = a^{N-t} g_t a^{-r} \in J, \quad c_{t,b} = a^{N-t} u_b a^{-(N-t+1)} \in J.$$

Normality of J and $u_b a^{-1} \in J$ justify the membership assertions. We have $z_0 = a^{N-r}$, $z_t = c_{t,b} z_{t-1}$ when the t th letter is b , and $z_N = g_N a^{-r}$. Initialize with $\alpha_s \rho(z_0^{-1})$, use the clock-dependent permutation $\rho(c_{t,b}^{-1})$, and keep decoder d_j . The row product is $\alpha_s \rho(z_N^{-1})$, so its error from $F_{s,j}(g_N) = f_{r,s,j}(z_N)$ is at most ε . This uses the free external clock but no phase register or partial-block storage. It proves the upper bound with exactly K labels.

Fix $k < M_{r,\varepsilon}$ and a cut residue $0 \leq \ell < p$. Put $b = (r - \ell) \bmod p$, chosen in $\{0, \dots, p-1\}$. Restrict the input to the fixed prefix a^ℓ , arbitrary p -letter blocks, and fixed suffix a^b . Integrate the prefix into initialization and the suffix into the decoder. The block target on J is

$$G_{\ell,s,j}(h) = F_{s,j}(a^b h a^\ell) = f_{r,s,j}(\text{Ad}_{a^b}(h) a^{\ell+b-r}).$$

Here $a^{\ell+b-r} \in J$ and conjugation by a^b is an automorphism of J . Corollary 6.2 therefore identifies its stationary minimum with $M_{r,\varepsilon}$. Theorem 4.2 on the block alphabet bounds its positive narrow cuts by a constant L_ℓ . These cuts are precisely the original cuts in residue ℓ between the prefix and suffix, apart from the initial block cut. At most $\ell + b + 1 \leq 2p$ further cuts need be charged for this residue, including small horizons shorter than the chosen prefix and suffix. Summing gives the uniform bound $\sum_{\ell=0}^{p-1} L_\ell + 2p^2$. As before, a peak at most k would force N below this bound. This proves the limit and occupation assertions. Finally apply Theorem 5.5 to f_r on J to obtain (6.5). $\square$

If the order of H/J is a proper divisor q of p , the minima and quotient-radius criteria repeat with period q : multiplication by $a^q \in J$ is a right translation of the target. The earlier phase radii have the same repetition. Neither the physical quotient order nor the return-length greatest common divisor is substituted silently for the other.

6.2. Compact outputs and finite hidden state. Theorems 4.2, 6.1, 6.3, and 6.4 hold for norm-compact convex Banach-valued outputs. A single-machine purification even needs only convexity: its new decoder values lie in the convex hull of finitely many old values. Compact output sets are used instead in finite-witness extraction, optimal-error attainment, and constrained-center existence. The representative-function and algebraic sections below retain their explicitly finite-dimensional hypotheses.

Finiteness on the hidden side has a different role. On countably many labels the full simplex is not compact in total variation; mass can escape to infinity, matrix ranks need not admit a useful finite minimum, and a recurrent simplex may have infinitely many vertices. On a general compact convex hidden state space affine automorphisms need not permute a finite vertex set: rotation of the disk already acts continuously on infinitely many extreme points. Thus these proofs give neither a finite-label purification theorem for arbitrary operator kernels nor an online observation theorem. Their terminal-output assertion is strictly the one stated in the models.

Remark 6.5 (Action and counting conventions). Probability vectors act on the right of transition matrices. Chronological row multiplication consequently corresponds to the inverse physical coordinate h^{-1} used above. The coarse deterministic quotient construction stores both the seed

and the coset; its $|\mathcal{S}||[H : U]$ bound is not the unrestricted stochastic minimum. In the phase theorem the chosen block period p , the physical quotient order $q \mid p$, and the greatest common divisor of actual return lengths retain their separate meanings.

7. THE CLOCKED COMPONENT-RADIUS THRESHOLD

A finite stochastic register can carry infinitely many probability distributions. Thus an infinite numerical orbit is not, by itself, an obstruction to finite positive realization. A further difficulty arises when the realization may be redesigned at every horizon and its transitions may depend on a free external clock. The obstruction established here survives both freedoms. It is measured by the radius of an observable image inside its allowed output set.

For the clocked theorem in this section, let H be a compact Hausdorff group with finitely many connected components. Put $K = H^\circ$ and $C = H/K$. Then K is an open normal subgroup and C is a finite group. Fix a finite alphabet $\mathcal{A}$ with elements $u_a \in H$ whose generated group is dense in H . Our convention is

$$u_{a_1 \cdots a_n} = u_{a_n} \cdots u_{a_1}.$$

The identity e need not be an alphabet element. Instead assume the following executable return condition:

(7.1)

there is an integer $g \geq 0$ such that, for every $n \geq g$, some word of length n has product e .

An identity letter gives $g = 0$. The condition concerns physical products; the stochastic rows on a return word may perform arbitrary processing.

For nonempty finite seed and query sets $\mathcal{S}, \mathcal{J}$, let

$$F_{s,j} : H \longrightarrow Y_j$$

be continuous, where Y_j is a nonempty compact convex subset of a finite-dimensional real normed space $(V_j, \|\cdot\|_j)$. A query is selected only after the command word. A prescribed joint law of finitely many outputs is treated as one query, not as a collection of marginal constraints. In particular, a categorical interface has

$$Y_j = \Delta_{M_j} = \{p \in \mathbb{R}^{M_j} : p_i \geq 0, \sum_i p_i = 1\}, \quad \|x\|_j = \frac{1}{2} \sum_i |x_i|.$$

Then $\|p - q\|_j$ is total variation. Binary means correspond to $Y = [-1, 1]$ with $\|x\| = |x|/2$.

Definition 7.1 (Clocked realization). At horizon N choose finite nonempty registers S_t , $0 \leq t \leq N$, initialization rows E_s , row-stochastic command matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal decoder maps $D_j : S_N \rightarrow Y_j$. The realized output vector on $w = a_1 \cdots a_N$ is

$$E_s T_{1,a_1} \cdots T_{N,a_N} D_j.$$

Its error is the maximum, over all seeds, length- N words, and queries, of its distance from $F_{s,j}(u_w)$. The command width is $\max_t |S_t|$, counting all declared labels, including unused padding labels. Write $W_{N,\varepsilon}$ for its least possible value at error at most ε .

All retained private information, including randomness kept for future use, is charged to the current label. The rows may depend on t and N , but not on an unrecorded input history or on the future query. The external clock and horizon, arbitrary real table descriptions, their construction and lookup, real arithmetic, and exact sampling are uncharged. This is a numerical realization invariant, not uniform workspace or program size. For a finite categorical interface, charging the terminal answer register changes the peak to $\max(W_{N,\varepsilon}, \max_j M_j)$. Thus all boundedness and growth results are unchanged; the convention that also charges a sampled binary-answer register has a minimum of two labels. A decoder vector by itself is not an additional sampled register.

For $A \subset Y_j$ define its constrained radius by

$$\text{rad}_{Y_j}(A) = \min_{y \in Y_j} \sup_{x \in A} \|x - y\|_j.$$

The minimum is attained when A is compact. The critical error is

$$(7.2) \quad \varepsilon_c = \max_{s,j,c \in C} \text{rad}_{Y_j}(F_{s,j}(c)).$$

Here a component is identified with its coset in H ; its image is compact by continuity. The requirement that the center belong to Y_j is essential for categorical outputs.

Theorem 7.2 (Intrinsic radius classification). *Under (7.1), for every $\varepsilon \geq 0$ the following are equivalent:*

- (1) $\sup_N W_{N,\varepsilon} < \infty$;
- (2) *one stationary finite-state realization with permutation command updates has error at most ε at every horizon*;
- (3) $\varepsilon \geq \varepsilon_c$.

At the boundary, at most $|\mathcal{S}||C|$ command labels suffice. If $\varepsilon < \varepsilon_c$, then for every integer $k \geq 1$ there is an integer $L_k < \infty$ such that every error- ε realization satisfies

$$(7.3) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_k.$$

All intervening widths are unrestricted. In particular, $W_{N,\varepsilon} \rightarrow \infty$.

Neither a positive lower bound on state masses nor a reachable-rank, observability, or conditioning assumption occurs in the theorem. The upper realization is deterministic until its terminal decoder. It need not have the minimum width among all stochastic realizations. Theorem 5.4, rather than a deterministic component partition, characterizes that stationary minimum.

For binary means, (7.2) reduces to one quarter of the largest componentwise oscillation. For a genuine categorical law it is generally not half its diameter; an analytic three-outcome example below has critical total-variation error $2/3$, whereas every individual event has binary threshold $1/2$. The proof uses finitely many localized vector tests rather than a separate midpoint for each coordinate.

The finite-component assumption does not assert that H is a Lie group. For example, take $H = \prod_{n \geq 1} \mathbb{R}/\mathbb{Z}$, choose an element whose every finite coordinate tuple together with 1 is rationally independent, and include this element and e as commands. The positive orbit is dense and H is connected but has no faithful finite-dimensional representation. The theorem applies to every finite continuous interface on this group. With infinitely many components the component construction need not be finite; Section 10 gives the finite-observable-quotient criterion and the strict-side clocked theorem in that setting.

There is also a quantitative counterpart. For planar rotations with a dual badly approximable s -tuple of angles and signal amplitude $0 < \rho < 1$, Theorem 12.1 proves

$$W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}) \quad \text{for every fixed } 0 \leq \varepsilon < \rho/2.$$

The upper realization is exact even when $\varepsilon > 0$. The new lower estimate reaches the entire sub-critical interval, by combining harmonic localization with the arithmetic orbit-packing mechanism. This quantitative theorem and the radius classification have distinct hypotheses: algebraic matrix entries alone do not imply the badly approximable angle condition.

7.1. Executable tests. Let $\mathcal{P} = \{u_w : w \in \mathcal{A}^*\}$. We use standard Haar averaging and the density of matrix coefficients of finite-dimensional unitary representations, in realified form [13]. No quantitative mixing assumption for a preassigned random walk is imposed.

Lemma 7.3 (Positive words and common lengths). *The semigroup $\mathcal{P}$ is dense in H , and $\mathcal{P} \cap K$ is dense in K . Every finite subset of $\mathcal{P} \cap K$ has executable representatives of a common positive length.*

Proof. For an element u of a compact group, positive powers return arbitrarily close to e with exponent at least two. Indeed cover the compact closure of its powers by finitely many translates of a sufficiently small neighborhood. Among sufficiently many powers, three lie in one member of the cover; the first and last exponents differ by at least two, and their quotient is arbitrarily close to e . This argument also covers finite-order elements by taking a sufficiently large multiple of the order. If u^r approaches e with $r \geq 2$, then $u^{r-1} = u^r u^{-1}$ approaches u^{-1} and is still a positive power. Thus u^{-1} is in the closure of positive powers. The closed semigroup $\overline{\mathcal{P}}$ therefore contains the group generated by the letters, and hence equals H . Since H has finitely many components, K is open, and a relatively open subset of K is open in H ; density gives the second assertion. For finitely many representative lengths ℓ_i , choose $B \geq \max_i \ell_i + g$, $B \geq 1$, and append return words of lengths $B - \ell_i$. The products are unchanged. $\square$

Lemma 7.4 (Finite executable gap). *Let $R : K \rightarrow O(E)$ be a continuous representation on a nonzero finite-dimensional Euclidean space, with no nonzero fixed vector. For every $k \geq 1$ there are $B \geq 1$, a probability law ν on finitely many executable length- B words with products in K , and $d \in (0, 1)$ such that*

$$(7.4) \quad \mathbb{E}_{w \sim \nu} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - d$$

for all unit vectors $u, v_1, \dots, v_k \in E$.

Proof. On the compact product of unit spheres consider the minimum

$$\gamma = \min_{u, v_1, \dots, v_k} \int_K \left(1 - \max_i \langle v_i, R(h)u \rangle \right) d\mu(h).$$

Repeated centers are permitted. The integrand is nonnegative. If a minimizing tuple had integral zero, full support of Haar measure would give $R(h)u \in \{v_1, \dots, v_k\}$ for every h . The orbit of u is connected, hence a singleton, contradicting the fixed-vector hypothesis. Thus $\gamma > 0$.

Partition the compact image $R(K)$ into finitely many measurable sets of small diameter in the Euclidean operator norm, and replace their Haar masses by atoms at nearby executable products in K . Cells of zero Haar mass can be discarded. The function $A \mapsto \max_i \langle v_i, Au \rangle$ is uniformly 1-Lipschitz in that norm. An approximation radius less than $\gamma/2$ therefore preserves a positive defect uniformly in the unit vectors. Lemma 7.3 gives a common length. Decreasing the defect to a number in $(0, 1)$ proves the assertion. $\square$

Lemma 7.5 (Conditional-centroid contraction). *Let $Z \in E$ be a bounded random physical vector and S the current register. Apply an independent word $w \sim \nu$ from Lemma 7.4, and let the actual block kernel generate a terminal state S' having at most k labels. Then*

$$(7.5) \quad \mathbb{E} \|\mathbb{E}[R(u_w)Z \mid S']\| \leq (1 - d) \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

A fixed intervening return word cannot increase the same conditional norm, regardless of its intermediate or terminal widths.

Proof. Put $z_s = \mathbb{E}[Z \mid S = s]$ and write $T_w(s, i)$ for the actual composite row of the block. Since this row depends on the past only through S ,

$$\Pr(S' = i) \mathbb{E}[R(u_w)Z \mid S' = i] = \sum_s \Pr(S = s) \mathbb{E}_w T_w(s, i) R(u_w) z_s.$$

Choose v_i in the direction of each nonzero terminal centroid, arbitrarily otherwise. Pair, sum in i , and use stochasticity to bound the sum by

$$\sum_s \Pr(S = s) \mathbb{E}_w \max_i \langle v_i, R(u_w) z_s \rangle \leq (1 - d) \sum_s \Pr(S = s) \|z_s\|.$$

Zero-probability states contribute zero. No restriction was placed on registers inside the block. On a fixed return word the physical vector is unchanged at its endpoints; the new state is a stochastic image of the old one, so conditional Jensen gives nonincrease. Independence is used only at complete block boundaries. $\square$

8. REPRESENTATIVE FUNCTIONS AND LOCALIZED RADII

A real representative function on K is a matrix coefficient of a finite-dimensional continuous orthogonal representation, or a finite sum of such coefficients. Their algebra contains the constants, is closed under multiplication by tensor products, and is uniformly dense in $C(K, \mathbb{R})$ by the Peter–Weyl approximation theorem [13]. A finite collection of representative functions can be written on one space

$$\mathcal{V} = \bigoplus_{i=1}^a \text{End}(E_i), \quad \Phi(h) = (R_i(h))_{i=1}^a,$$

where the scalar product is the sum of Frobenius products. Left multiplication defines an orthogonal representation R on $\mathcal{V}$ and satisfies $\Phi(gh) = R(g)\Phi(h)$. Adjoin a trivial summand when necessary. Let

$$P = \int_K R(h) d\mu(h), \quad E = (I - P)\mathcal{V}, \quad Z(h) = (I - P)\Phi(h), \quad Q_0 = \|Z(e)\|.$$

Haar averaging is the orthogonal projection onto fixed vectors. Consequently $Z(gh) = R(g)Z(h)$, the representation on E has no fixed vectors, and $P\Phi(h) = P\Phi(e)$ for $h \in K$.

Choose and fix a coefficient vector a_p for each scalar representative function p , and a linear coefficient map A_G for each vector-valued representative function G . Their norms need not be intrinsic; fixing any such choices suffices. For an arbitrary joint distribution of $h \in K$ and a finite state S , put $Q = \mathbb{E}\|\mathbb{E}[Z(h) \mid S]\|$ and $\bar{p} = \int p d\mu$. The fixed/moving decomposition gives

$$(8.1) \quad \mathbb{E}[p(h) \mid S] = \bar{p} + \langle a_p, \mathbb{E}[Z(h) \mid S] \rangle,$$

$$(8.2) \quad \|\mathbb{E}p - \bar{p}\| \leq \|a_p\|Q, \quad \|\mathbb{E}G - \bar{G}\| \leq \|A_G\|Q.$$

Here the norm of A_G is the operator norm from the Euclidean feature space to the indicated output space. If $D : S \rightarrow Y$ and $M_Y = \max_{y \in Y} \|y\|$, then

$$(8.3) \quad \|\mathbb{E}[p(h)D(S)] - \bar{p}\mathbb{E}D(S)\| \leq M_Y \|a_p\|Q.$$

Indeed condition on S in (8.1), multiply by $D(S)$, and apply the triangle inequality. These are moment estimates, not total-variation convergence of a hidden physical distribution.

Lemma 8.1 (Finite localization of a radius). *Let $f : K \rightarrow Y$ be continuous, with Y compact and convex in a finite-dimensional normed space. Suppose $\text{rad}_Y(f(K)) > \varepsilon$. There are a vector-valued representative function G , a number $\eta > 0$, and nonnegative representative functions $p_1, \dots, p_a$ of Haar mean one such that*

$$\|G - f\|_\infty < \eta, \quad r_* := \min_{y \in Y} \max_i \|p_i \bar{G} - y\| > \delta := \varepsilon + \eta.$$

Proof. Choose $\varepsilon < t < \text{rad}_Y(f(K))$. For each $y \in Y$ some $h \in K$ has $\|f(h) - y\| > t$. The corresponding open subsets of Y cover Y , so a finite subcover yields $h_1, \dots, h_a$ with

$$\min_{y \in Y} \max_i \|f(h_i) - y\| > t.$$

The strict inequality follows also at a minimizing y , since the maximum is everywhere greater than t and Y is compact.

Fix a sufficiently small positive η . Approximate the coordinates of f uniformly by representative functions to obtain $\|G - f\|_\infty < \eta$. For each i , choose a nonzero nonnegative continuous bump supported in a neighborhood where f differs from $f(h_i)$ by less than η . Such a bump exists by normality of the compact Hausdorff space; it has positive Haar integral by full support. Approximate its square root uniformly by a representative function, square the approximation, and normalize its Haar integral. The resulting nonnegative representative function p_i has mean one, and its weighted f -mean can be made arbitrarily close to that of the bump. Thus we may ensure

$$\|p_i \bar{G} - f(h_i)\| < 3\eta.$$

The constrained radius changes by at most the maximum perturbation of the finitely many points. Taking $4\eta < t - \varepsilon$ gives $r_* > \varepsilon + \eta$. $\square$

Proposition 8.2 (Residual moment from a vector interface). *Fix the data of Lemma 8.1. Choose one feature space representing all p_i and $p_i G$, and fix their coefficient choices before defining*

$$L = \max_i (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|), \quad \Delta = r_* - \delta > 0.$$

Suppose a terminal decoder $D(S) \in Y$ obeys

$$\|\mathbb{E}[D(S) \mid w] - G(h(w))\| \leq \delta$$

for every complete deterministic word in a test law. Then $L > 0$, $E \neq \{0\}$, $Q_0 > 0$, and

$$(8.4) \quad Q \geq \Delta/L.$$

Proof. For $p = p_i \geq 0$, wordwise correctness and the triangle inequality imply

$$\|\mathbb{E}[p(h)D(S)] - \mathbb{E}[p(h)G(h)]\| \leq \delta \mathbb{E}p(h).$$

This is a conditional mean error, not a pointwise bound on the random decoder. Since $\bar{p}_i = 1$, calibration gives explicitly

$$\mathbb{E}p_i \leq 1 + \|a_{p_i}\|Q.$$

Write $y = \mathbb{E}D(S)$, which belongs to Y by convexity. Equations (8.2)–(8.3), together with $\bar{p} = 1$, give

$$\|y - p_i \bar{G}\| \leq \delta + (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|)Q.$$

Maximizing in i yields $r_* \leq \delta + LQ$. Hence $LQ \geq \Delta > 0$. In particular the moving feature and its initial norm cannot vanish: by equivariance, $Q_0 = 0$ would imply $Z(h) = 0$ for all h and therefore $Q = 0$. $\square$

8.1. Proof of Theorem 7.2. Choose (s, j, c) with component radius greater than ε . By density there is an executable prefix v with $u_v \in c$. Put $b = |v|$ and $f(h) = F_{s,j}(hu_v)$ on K . Since $Ku_v = c$, this map has the same constrained radius. Apply Lemma 8.1 and Proposition 8.2.

The moving feature is nonzero independently of any particular test law. Otherwise all represented functions would be constant on K ; since each p_i has mean one, this would force $p_i = 1$ and G constant. Then all $p_i \bar{G}$ would equal a point at distance less than η from Y , contradicting $r_* > \delta \geq \eta$.

For a fixed width k , apply Lemma 7.4 to the resulting moving feature. Let B, d be its block parameters and set

$$\chi = -\log(1 - d) > 0, \quad A = \max(1, LQ_0/\Delta).$$

Start a test word with v ; between independent length- B test blocks use fixed return words. The accumulated physical product is hu_v . Initially $h = e$, so the conditional feature norm after the

prefix equals Q_0 , regardless of the hidden-state distribution. If J block endpoints have width at most k , Lemma 7.5 gives

$$Q_N \leq Q_0(1 - d)^J.$$

The machine is correct on every deterministic word of length N . Thus any external law supported on such words inherits the conditional mean bound, with error $\delta = \varepsilon + \eta$ for G . Proposition 8.2 implies

$$(8.5) \quad J\chi \leq \log A.$$

The law depends on the advertised width profile, not on the machine's private state. When $A = 1$, (8.5) excludes even one selected narrow block; no logarithmic loss has occurred.

For clarity we give the spacing count with the full return hypothesis. Discard narrow cuts before $b + B + g$ and after $N - g$. From the remaining cuts greedily choose the first and then the next at distance at least $B + g$. Put an independent block immediately before each selected cut. The initial gap, all intervening gaps, and the final suffix have lengths at least g , so (7.1) fills them with executable returns. At most $b + B + 2g - 1$ cuts were discarded. Each chosen endpoint accounts for at most $B + g$ eligible cuts. Therefore one may take

$$(8.6) \quad L_k = \left\lceil b + B + 2g - 1 + (B + g) \frac{\log A}{\chi} \right\rceil.$$

The constants B, d depend on the fixed moving representation and k ; $\chi = -\log(1 - d)$ depends on that defect. The localization, prefix length b , and A depend only on the alphabet, interface, chosen component, and error. Consequently L_k depends on these fixed data, the return certificate g , and k , but not on N or the machine. When the eligible interval is empty the same estimate follows directly by counting. This proves occupation without restricting any register inside a block or gap. If the entire peak is at most k , then $N \leq L_k$, which proves divergence directly. Monotonicity in the horizon is not needed when an identity letter is absent.

For the upper bound, use labels $(s, c) \in \mathcal{S} \times C$, including any labels not reached at a given horizon. Initialize at (s, K) and update by left multiplication

$$(s, c) \mapsto (s, [u_a]c).$$

Choose a minimizing center of $F_{s,j}(c)$ in Y_j as decoder at (s, c) for query j . These are legal decoder values, and the error is at most (7.2). All command updates are permutations of one finite state set, independent of clock and horizon. This proves boundary attainment and the equivalence. $\square$

Proposition 8.3 (A finite return certificate). *Condition (7.1) follows if there are finitely many executable identity-product words whose positive lengths have greatest common divisor one.*

Proof. Let the lengths be $\ell_1, \dots, \ell_a$. Their nonnegative sums modulo ℓ_1 form the subgroup generated by their residues: a subsemigroup of a finite group is a subgroup. The gcd hypothesis makes this the whole residue group. Choose one nonnegative sum n_r for each residue r modulo ℓ_1 , and put $g = \max_r n_r$. Every $n \geq g$ equals its representative n_r plus a nonnegative multiple of ℓ_1 . Concatenating the corresponding return words proves the claim. $\square$

For example, the alphabet $\{R_\theta, R_{-\theta}, R_{2\pi/3}\}$, with $\theta/2\pi$ irrational, contains no identity letter but has return lengths two and three. It satisfies (7.1) with $g = 2$. A single irrational rotation letter does not: there is only one word at each horizon, and a horizon-specific constant decoder can store its numerical answer with one command label. Thus synchronization cannot be silently omitted. Nor is a bounded irreversible semigroup enough: for the identity command $I = 1$ and contraction command $a = \lambda \in (0, 1)$, the mean $\rho\lambda^{\#a(w)}$ has an exact alive/absorbed two-state realization. The group generated with inverses is not bounded. The present theorem is not a compact-semigroup classification.

9. EXACT-RETURN LANGUAGES AND LENGTH PHASES

The existence of cofinite returns is not the only useful synchronization regime. In fact an exact-return language, when nonempty, has a rigid eventual form. Here and in this section a return has positive length.

Lemma 9.1 (The return period). *Suppose*

$$\mathcal{R} = \{|w| > 0 : u_w = 1\} \neq \emptyset, \quad p = \gcd \mathcal{R}.$$

There is an integer g_p such that every multiple dn with $n \geq g_p$ belongs to $\mathcal{R}$, and no length outside $p\mathbb{N}$ belongs to $\mathcal{R}$. In particular, every nonempty exact-return language is eventually one arithmetic progression.

Proof. Concatenation makes $\mathcal{R}$ additive. A finite subset of its elements already has $\gcd p$: start with one length and successively decrease the $\gcd$ when necessary; a strictly decreasing chain of its positive divisors is finite. Divide these finitely many lengths by p . Their $\gcd$ is one, so the residue argument of Proposition 8.3 shows that their nonnegative sums contain every sufficiently large integer. Multiplying by p proves the assertion. The exclusion of other lengths is the definition of the $\gcd$. $\square$

Fix a letter with product $a \in H$, and let J be the closure of the products of words whose lengths are multiples of p , including the empty word.

Lemma 9.2 (The phase subgroup). *The group J is an open normal subgroup of H . The quotient H/J is cyclic of order dividing p , generated by aJ , and*

$$(9.1) \quad \overline{\{u_w : |w| \equiv r \pmod{p}\}} = a^r J \quad (0 \leq r < p).$$

If H/H° is finite, then $J^\circ = H^\circ$.

Proof. The compact closure defining J is a group, by the positive-power argument, and is generated densely by the block alphabet $\mathcal{A}^p$. Also $a^p \in J$. If $x = u_w$ for a word w of length p , then the executable word consisting of $p-1$ copies of the chosen letter, followed by w , followed by one more copy, has product axa^{p-1} and length $2p$. Thus $axa^{p-1} \in J$, whence $axa^{-1} \in J$. It follows that $aJa^{-1} \subset J$. Applying this inclusion p times and using $a^p \in J$ shows equality. Every letter b has $ba^{p-1} \in J$, so $ba^{-1} \in J$. Thus all letters lie in Ja , normalize J , and their products of length n lie in Ja^n . The finite union $\bigcup_{r=0}^{p-1} Ja^r$ is a closed group containing the letters, hence is H . This proves normality, finite index, and openness. Conversely, words consisting of r copies of a followed by arbitrary p -blocks have products dense in Ja^r , proving (9.1). An open subgroup contains the identity component, and its own identity component is therefore the same one. $\square$

The quotient order may be a proper divisor of p ; the phase radii repeat whenever the physical cosets repeat. The phase index records executable length, not merely the physical component quotient.

Assume now that H has finitely many components and put $K = H^\circ$. For $r \in \{0, \dots, p-1\}$ define

$$(9.2) \quad \varepsilon_r = \max_{s,j, c \in H/K: c \subset a^r J} \text{rad}_{Y_j}(F_{s,j}(c)).$$

Theorem 9.3 (The phasewise clocked threshold). *Under the sole synchronization assumption $\mathcal{R} \neq \emptyset$, for every r and $\varepsilon \geq 0$,*

$$\sup_{N \equiv r \pmod{p}} W_{N,\varepsilon} < \infty \iff \varepsilon \geq \varepsilon_r.$$

Equality is attained by a finite component-permutation realization on the indicated horizons. If $\varepsilon < \varepsilon_r$, then for each k there is a constant $L_{k,r}$, independent of N , such that every error- ε realization at $N \equiv r \pmod{p}$ satisfies

$$\#\{1 \leq t \leq N : |S_t| \leq k\} \leq L_{k,r}.$$

The intervening widths are unrestricted. Consequently the full family is bounded exactly at and above $\max_r \varepsilon_r = \varepsilon_c$; below it, divergence is asserted phase by phase, not necessarily along every horizon.

Proof. Every product of length $N \equiv r$ lies in $a^r J$. Store its component and the seed and decode component centers as before. This proves the upper bound, including equality, with at most $|\mathcal{S}||H/K|$ labels. Centers on other components may be chosen arbitrarily when only this phase is under consideration.

For the converse fix a component $c \subset a^r J$ and s, j with radius greater than ε . We must count narrow cuts in every residue class, not only at block boundaries of one phase. For each $\ell \in \{0, \dots, p-1\}$ choose $b_\ell \in \{0, \dots, p-1\}$ congruent to $r - \ell$ modulo p . By (9.1) and the openness of components there is a fixed word v_ℓ , of length $p_\ell \equiv \ell \pmod{p}$, with product in $a^{-b_\ell} c$. Indeed this component is contained in $a^\ell J$. Consider the block experiment on J with alphabet $\mathcal{A}^p$ and target

$$G_\ell(h) = F_{s,j}(a^{b_\ell} h u_{v_\ell}), \quad h \in J.$$

It is a legal compact-convex interface. Its image on $K = J^\circ$ is $F_{s,j}(c)$, since K is normal. Lemma 9.1 gives cofinite exact returns in block length for this alphabet.

Restrict the original machine to words that begin with v_ℓ , end with b_ℓ copies of the chosen letter, and have $M = (N - p_\ell - b_\ell)/p$ arbitrary p -blocks in between. When $M \geq 0$, absorb the fixed prefix into the initialization and the fixed suffix into the terminal decoder; stochastic averaging keeps that decoder in Y_j . This produces an error- ε clocked block machine for G_ℓ , with registers at original cuts $p_\ell + dm$, $0 \leq m \leq M$. Theorem 7.2 bounds its narrow cuts $1 \leq m \leq M$ by a constant depending only on k and the fixed data. The omitted cuts of residue ℓ number at most $p_\ell + b_\ell + 1$: this also charges the block cut zero, while all counted cuts remain in $1, \dots, N$. If $M < 0$, the whole horizon is bounded by $p_\ell + b_\ell$. Sum these bounds over the p residue classes. This counts all cuts $1, \dots, N$, independently of all other register widths. $\square$

The constants in this theorem depend on the fixed alphabet, return period and certificate, interface, error, width k , and chosen finite prefixes. They do not depend on the horizon or on the machine. The original constants B, d, χ, L_k in Section 8 use d for a contraction defect. The return period here is denoted p to keep the two parameters distinct.

Proposition 9.4 (Different phases can have different thresholds). *There is an alphabet with return period two for which $W_{N,0} = 1$ on every even horizon, but $W_{N,\varepsilon} \rightarrow \infty$ along odd horizons for every $0 \leq \varepsilon < \rho/2$.*

Proof. Take $H = (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z}$ and letters $(\theta, 1), (-\theta, 1)$, with $\theta/2\pi$ irrational. The generated group is dense in H . An exact return must have zero signed count and hence even length; the two-letter return gives every positive even length. Here $J = (\mathbb{R}/2\pi\mathbb{Z}) \times \{0\}$. For one binary-mean query put $F(\phi, 0) = 0$ and $F(\phi, 1) = \rho \cos \phi$, where $0 < \rho \leq 1$. The even-phase target is identically zero and needs one label. The odd-phase radius in the norm $|\cdot|/2$ is $\rho/2$. Apply Theorem 9.3. $\square$

10. INFINITE COMPONENT GROUPS AND BOUNDARY ATTAINMENT

This section retains independent finite-dimensional proofs of the strict-radius and exact translate-rank conclusions. Its output spaces are finite-dimensional. Let H again be an arbitrary compact

Hausdorff group, and set $K = H^\circ$. The quotient H/K is profinite. We use two standard compact-group consequences of this fact and of Peter–Weyl theory [13]: open normal subgroups form a neighborhood basis in a profinite group, and, for a continuous surjection $\pi : H \rightarrow G$ onto a compact Lie group, one has $\pi(K) = G^\circ$. To see the latter assertion, $\pi(K)$ is connected and normal, while $G/\pi(K)$ is a continuous group quotient of the profinite group H/K and is therefore totally disconnected. This forces $G^\circ \subset \pi(K)$; the reverse inclusion is immediate. A continuous finite-dimensional representation of a profinite group has finite image, by the closed subgroup theorem and the no-small-subgroups property of a Lie group [14]: an open normal subgroup can be placed inside a no-small-subgroups neighborhood, so its image is trivial.

Define the component radius

$$(10.1) \quad R_0 = \max_{s,j,g \in H} \text{rad}_{Y_j}(F_{s,j}(gK)).$$

Unlike $r(U)$, this quantity need not itself be realized by any finite quotient.

Lemma 10.1 (Approximation by finite quotient radii). *With $r(U)$ as in (1.1),*

$$\inf_{U \text{ open normal in } H} r(U) = R_0.$$

Proof. Every open subgroup contains K , so $r(U) \geq R_0$. Fix $\eta > 0$. Uniform continuity, for the finite family of interfaces, gives an identity neighborhood V such that

$$\|F_{s,j}(xv) - F_{s,j}(x)\|_j < \eta \quad (x \in H, v \in V, s, j).$$

There is an open normal subgroup U containing K and contained in KV , by the profinite quotient basis. Write each $u \in U$ as kv with $k \in K, v \in V$. Then $F_{s,j}(gu)$ lies within η of $F_{s,j}(gk)$. A center for the latter component image consequently covers the former coset image with an additional error at most η . Thus $r(U) \leq R_0 + \eta$. $\square$

Lemma 10.2 (A legal finite-dimensional interface approximation). *For every $\eta > 0$ there is a continuous homomorphism $\pi : H \rightarrow G \subset O(D)$ onto a compact Lie group and continuous maps $\tilde{F}_{s,j} : G \rightarrow Y_j$ such that*

$$\max_{s,j} \|F_{s,j} - \tilde{F}_{s,j} \circ \pi\|_\infty < \eta.$$

Moreover, for every $g \in H$,

$$\left| \text{rad}_{Y_j}(F_{s,j}(gK)) - \text{rad}_{Y_j}(\tilde{F}_{s,j}(\pi(g)G^\circ)) \right| < \eta.$$

Proof. Approximate all coordinates of all $F_{s,j}$ by finitely many real representative functions. The direct sum of their orthogonal representations is π , and the approximants are continuous functions on its compact image G . Choose Euclidean inner products on the finitely many output spaces. Project each approximant to the closed convex set Y_j by nearest-point projection. This projection is continuous and nonexpansive in the chosen Euclidean norm. Choose constants $a_j, b_j > 0$ with $a_j\|v\|_j \leq |v|_2 \leq b_j\|v\|_j$. An initial error less than $a_j\eta/b_j$ in $\|\cdot\|_j$ then gives a projected error less than η , by Euclidean nonexpansiveness. These constants are fixed before choosing the approximants. The closed subgroup $G \subset O(D)$ is a compact Lie group. Since $\pi(K) = G^\circ$, the two indicated images have Hausdorff distance less than η . Their constrained radii therefore differ by less than η . $\square$

Theorem 10.3 (Arbitrary compact groups on both strict sides). *For any compact-group interface, $\varepsilon > R_0$ implies a finite stationary permutation realization. At $\varepsilon = R_0$, such a realization exists exactly when $r(U) \leq R_0$ for some open normal U .*

Suppose in addition that executable exact returns have all sufficiently large lengths. If $\varepsilon < R_0$, then for each k there is a horizon-independent upper bound on the number of cuts of width at

most k , with all intervening widths unrestricted. In particular $W_{N,\varepsilon} \rightarrow \infty$. No finite-component assumption is made in either assertion.

Proof. The stationary assertions follow from Lemma 10.1 and Theorem 5.5. For the lower bound choose a component image of radius greater than ε and then $\eta > 0$ with 2η smaller than that strict difference. Apply Lemma 10.2. A machine correct for the original interface at error ε is correct for the approximating interface at error at most $\varepsilon + \eta$. The chosen component radius of that interface is greater than $\varepsilon + \eta$. The projected commands still have cofinite exact returns and generate G densely. Theorem 7.2 therefore gives the desired occupation bound for the very same registers. This argument never assumes that K is open or that executable products lying exactly in K are dense. $\square$

Theorem 6.3 now identifies this stationary boundary criterion with clocked boundedness, including $R_0 > 0$, and under the weaker approximate-return hypothesis. The zero-error boundary also admits the following independent translate-rank argument.

Lemma 10.4 (Evaluation rank and translate dimension). *For a real function f on a group, the supremum of the ranks of finite matrices $(f(y_l x_i))_{i,l}$ equals the dimension, possibly infinite, of the span of the functions $x \mapsto f(yx)$. These are left translates with the convention $(L_g f)(x) = f(g^{-1}x)$, after replacing y by g^{-1} .*

Proof. A finite-dimensional span bounds every evaluation rank. Conversely, if $f_1, \dots, f_m$ are independent functions, their evaluation vectors $(f_1(x), \dots, f_m(x))$ span $\mathbb{R}^m$: a proper span would have a nonzero annihilator, giving a linear dependence of the functions. Select m evaluation vectors forming a basis. This gives an invertible evaluation matrix and proves the reverse inequality for every finite m . $\square$

Theorem 10.5 (The exact clocked boundary). *Assume cofinite executable exact returns. The following are equivalent:*

- (1) $\sup_N W_{N,0} < \infty$;
- (2) all $F_{s,j}$ factor through one finite continuous quotient H/U , with U open normal;
- (3) there is an exact stationary finite permutation realization.

This includes groups with infinitely many connected components.

Proof. The implications from the second assertion to the third and from the third to the first are immediate. Suppose the first holds with bound k . Theorem 10.3 implies $R_0 = 0$, so every interface is constant on each K -coset.

Fix a real coordinate f of one $F_{s,j}$. For any finite families of executable prefix products x_i and suffix products y_l , use cofinite returns to pad all selected prefix words to one common length A and all suffix words to one common length B . Choose an exact realization at horizon $A + B$ of width at most k . At its cut A the matrix

$$(f(y_l x_i))_{i,l}$$

factors through the k hidden labels: the left factors are the actual state distributions after the padded prefixes and the right factors are the actual conditional terminal expectations on the padded suffixes. Hence this matrix has rank at most k . Padding does not need to preserve hidden states; it only preserves the physical products used in the displayed entries. By density and continuity the same rank bound holds for arbitrary $x_i, y_l \in H$.

It follows that the span of the left translates of f in $C(H, \mathbb{R})$ has dimension at most k . Otherwise $k + 1$ linearly independent translates would have an invertible evaluation matrix at some $k + 1$ points, contrary to the preceding rank bound. Lemma 10.4 formalizes this evaluation assertion. Equivalently it follows inductively: a nonzero function is nonzero somewhere, and a function outside the span of the preceding ones cannot have all its evaluations forced by theirs.

Take the sum V of these translate spaces over all seed–query coordinates. It is finite-dimensional and invariant under left translation. Translation is continuous in the uniform norm and preserves that norm, so it gives a continuous representation of H on V with compact image; Haar averaging makes it orthogonal. Because K is normal and every original coordinate is constant on K -cosets, K acts trivially on every translate and thus on V . The representation factors through H/K . Its image is finite by the profinite finite-dimensional representation fact above. Its kernel U is open normal. Since every original coordinate belongs to V , all interfaces are U -invariant and descend to H/U . $\square$

Proposition 10.6 (A nonattained zero-radius boundary). *On a compact group with infinitely many components, every positive error can have a bounded stationary realization while the zero-error clocked width is unbounded.*

Proof. Let $H = \mathbb{Z}_2$ be the additive group of 2-adic integers, with commands 0 and 1. Their positive products are dense, and the zero command supplies exact returns of every length. If $x = \sum_{n \geq 0} x_n 2^n$, $x_n \in \{0, 1\}$, prescribe the binary success probability

$$f(x) = \sum_{n \geq 0} \frac{2x_n}{3^{n+1}}.$$

Uniform convergence makes f continuous. It is injective: at the first differing digit the contribution $2/3^{n+1}$ exceeds the sum $1/3^{n+1}$ of all possible later differences. Since H is totally disconnected, $K = \{0\}$ and $R_0 = 0$. Truncation after m digits factors through $\mathbb{Z}/2^m\mathbb{Z}$ and has uniform error at most $\sum_{n=m}^{\infty} 2/3^{n+1} = 3^{-m}$. Thus every positive tolerance admits a finite quotient machine. But the injective function f cannot factor through any finite quotient. Theorem 10.5 excludes bounded exact clocked width, and Theorem 5.5 excludes a finite exact stationary realization. $\square$

Theorem 10.7 (Linear exact width at a profinite boundary). *For the interface of Proposition 10.6,*

$$W_{N,0} = \Theta(N).$$

More precisely, if n is a power of two with $2n - 1 \leq N$, then $W_{N,0} \geq n$, while $W_{N,0} \leq N + 1$. Every exact realization, with unrestricted intermediate widths, has at most $4k$ cuts in $\{1, \dots, N\}$ of width at most k . Nevertheless, for each $\varepsilon > 0$, $\sup_N W_{N,\varepsilon} < \infty$.

Proof. For nonnegative integers z write $v_2(z+1)$ for the exponent of two in $z+1$. Incrementing the binary expansion, with $v_2(z+1)$ trailing ones, gives

$$(10.2) \quad b(z) := f(z+1) - f(z) = \frac{5}{3} 3^{-v_2(z+1)} - 1.$$

Let $n = 2^m$, and form $B_n = (b(i+j))_{0 \leq i, j < n}$. Define a function on $\mathbb{Z}/n\mathbb{Z}$ by

$$\psi_m(x) = 3^{-v_2(x)} \quad (x \neq 0), \quad \psi_m(0) = 3^{-m}.$$

Here the valuation of a nonzero residue is independent of its representative. Since $1 \leq i+j+1 \leq 2n-1$, its valuation equals that used in $\psi_m(i+j+1 \bmod n)$, including the case $i+j+1 = n$. Thus replacing row i by the old row $(-i-1) \bmod n$ changes its entry at column j to the kernel value at $j-i \bmod n$. This is the circulant with kernel $(5/3)\psi_m - 1$.

On $\mathbb{Z}/n\mathbb{Z}$ one has the elementary identity

$$\psi_m = 1 - 2 \sum_{\ell=1}^m 3^{-\ell} \mathbf{1}_{2^\ell | x}.$$

Use the eigenvector with entries $\exp(2\pi i q j / n)$; its eigenvalue is the kernel sum against $\exp(2\pi i q x / n)$. The sign is immaterial for subgroup indicators, whose transforms below are real. The Fourier

eigenvalue at frequency zero is

$$\mu_0 = \frac{2}{3}n6^{-m} > 0.$$

At frequency $q \neq 0$ it is

$$\mu_q = -\frac{10n}{3} \sum_{\substack{1 \leq \ell \leq m \\ n/2^\ell | q}} 6^{-\ell} < 0.$$

Indeed a subgroup indicator has Fourier sum $n/2^\ell$ exactly when $n/2^\ell$ divides q , and zero otherwise. For $q \neq 0$ the summand $\ell = m$ is always present. At $m = 0$ the sole eigenvalue is $2/3$. Consequently B_n is invertible for every dyadic n .

At an arbitrary command cut $t < N$, take a dyadic $n \leq \min(t+1, N-t)$. For each $i < n$ choose a prefix of length t with i increment commands. For each $j < n$ choose two suffixes of length $N-t$ with respectively $j+1$ and j increment commands. Fill all remaining positions with zero commands. The difference of their exact outputs is $b(i+j)$. Hence B_n factors through the actual register S_t : one factor consists of the prefix state rows and the other of the differences of the two conditional suffix decoder columns. Invertibility gives $|S_t| \geq n$. The difference column need not be a legal decoder; it is used only for this rank bound on two legal experiments.

Taking $t = n-1$ proves $W_{N,0} \geq n$ whenever $2n-1 \leq N$. The largest dyadic $n \leq (N+1)/2$ is greater than $(N+1)/4$, proving the linear lower order. Storing the number of increment commands uses at most $N+1$ labels and gives the exact upper bound.

For occupation, if $t < N$ and $|S_t| \leq k$, the largest dyadic integer not exceeding $\min(t+1, N-t)$ is at most k , so $\min(t+1, N-t) < 2k$. Such cuts lie within fewer than $2k$ positions of one of the two ends. Including the terminal cut gives at most $4k$ positive cuts. No width at another cut was restricted. The positive-error assertion is the finite-quotient truncation in Proposition 10.6. $\square$

This supplies a sharp quantitative law on a genuinely infinite profinite group. Its linear exact-width obstruction and bounded positive-error realizations concern one fixed continuous interface; they are not a uniform approximation statement over all continuous outputs on that group.

11. BINARY, CATEGORICAL, AND OBSERVABLE INTERFACES

Corollary 11.1 (Component oscillation for binary means). *For a finite continuous binary-mean interface $f_{s,j} : H \rightarrow [-1, 1]$, with the return hypothesis of Theorem 7.2, the exact bounded-width threshold in total variation is*

$$(11.1) \quad \varepsilon_c = \frac{1}{4} \max_{s,j,c} \left(\max_{h \in c} f_{s,j}(h) - \min_{h \in c} f_{s,j}(h) \right).$$

It is attained by a stationary permutation realization. Below it every fixed width has the occupation bound (7.3).

Proof. In the norm $|\cdot|/2$, the radius of an interval is one quarter of its length. Its midpoint lies in $[-1, 1]$. Apply Theorem 7.2. $\square$

This recovers the scalar component theorem, including all of its nonuniform quantifiers. The vector theorem does not follow by applying this corollary separately to coordinates, because the coordinate centers need not belong to the output simplex.

Proposition 11.2 (A three-outcome threshold). *On the circle let $\omega = e^{2\pi i/3}$ and, for $j = 0, 1, 2$, set*

$$p_j(\phi) = \frac{1}{9} \left| 1 + e^{i\phi} \omega^{-j} + e^{2i\phi} \omega^{-2j} \right|^2.$$

For any dense circle command alphabet satisfying the return condition, the categorical law $p = (p_0, p_1, p_2)$ has critical total-variation error $2/3$. Each separately requested binary event $\{j\}$ has threshold $1/2$.

Proof. Orthogonality of the three characters gives $\sum_j p_j = 1$; each entry is nonnegative. At $\phi = 2\pi j/3$ the vector is the simplex vertex e_j . Hence every center $q \in \Delta_3$ has maximum distance at least

$$\max_j \text{TV}(e_j, q) = 1 - \min_j q_j \geq 2/3.$$

The uniform center has distance at most $2/3$ from every point of Δ_3 , by convexity and the values at its vertices. Thus the constrained radius is exactly $2/3$. On the other hand each p_j has range $[0, 1]$, so its binary mean $2p_j - 1$ has oscillation two. Apply Theorem 7.2 and Corollary 11.1. In particular the categorical radius exceeds half the diameter, which here equals $1/2$. $\square$

Corollary 11.3 (Joint measure-once unitary laws). *Let unitary commands have compact closure H and satisfy the return condition. For a fixed unit seed ψ_s and a finite measurement $E_{j,l} \geq 0$, $\sum_l E_{j,l} = I$, prescribe the law*

$$F_{s,j}(h)_l = \langle h\psi_s, E_{j,l}h\psi_s \rangle.$$

The critical error is its componentwise simplex radius in total variation. For closure $U(d)$ and one orthonormal-basis measurement, this radius is $1 - 1/d$.

Proof. The displayed map is a continuous simplex-valued polynomial in realified matrix entries. A closed unitary matrix group has finitely many components, so Theorem 7.2 applies. For $U(d)$, every unit vector is reachable from the seed, and its squared coordinate magnitudes range over the whole simplex. The simplex vertices force radius at least $1 - 1/d$, attained by the uniform distribution. $\square$

A query selects one complete measurement, not simultaneous answers to incompatible measurements. Several compatible finite outputs, with specified correlations, instead form one joint categorical law. Language recognition at a cutpoint is a different requirement.

For linear binary interfaces on a compact matrix group, write

$$f_{s,j}(h) = q_j^T h x_s \in [-1, 1], \quad V_r = \text{span}\{h x_s : h \in H, s \in \mathcal{S}\},$$

$$V_n = \{x \in V_r : q_j^T h x = 0 \text{ for all } h \in H, j \in \mathcal{J}\}.$$

Both spaces are invariant. The observable space $V_o = V_r \cap V_n^\perp$ is the orthogonal complement of V_n inside V_r , canonically realizing the observable quotient.

Theorem 11.4 (Intrinsic zero-error criterion). *For these linear means, bounded exact clocked width is equivalent to finiteness of the generated group's image in $O(V_o)$, and to trivial action of K on V_o . An exact stationary permutation realization then exists.*

Proof. Corollary 11.1 identifies bounded exact width with constancy of all interface means on every component. Under that condition, for $k \in K$ and $g, h \in H$, normality gives

$$q_j^T h(k - I)g x_s = q_j^T (h k h^{-1}) h g x_s - q_j^T h g x_s = 0.$$

Thus $(k - I)V_r \subset V_n$. On the invariant orthogonal complement V_o the difference also lies in V_o , hence vanishes. The observable action factors through the finite component group. Conversely, if K acts trivially on V_o , the same difference is invisible and all means are componentwise constant. A finite image of the dense generated group has the same finite image closure from H , whose connected subgroup K acts trivially. The zero-dimensional quotient is included. $\square$

Corollary 11.5 (The planar threshold). *Let an orthogonal planar alphabet contain I and an irrational rotation, with any additional orthogonal letters. For signed coordinate seeds $\pm \rho e_1, \pm \rho e_2$, coordinate queries, and $0 < \rho \leq 1$, the threshold is $\varepsilon_c = \rho/2$. At and above it one command label with a fair binary decoder suffices. Below it every fixed width has a finite occupation budget.*

Proof. The identity component is $SO(2)$, and every specified mean has range $[-\rho, \rho]$ on each component of $H = SO(2)$ or $O(2)$. All midpoint decoders equal zero, so the seed/component register collapses to one label. $\square$

For the noncommuting rational alphabet

$$I, \quad R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

one has $FRF = R^{-1} \neq R$. The rotation has infinite order: otherwise $(3 + 4i)/5$ and its inverse would be algebraic integers, whereas their rational sum $6/5$ is not an integer. This example has the sharp threshold above. We do not infer a badly approximable angle from rational matrix entries.

Proposition 11.6 (Independent toral coordinates). *Suppose a component is identified with the full product torus $(\mathbb{R}/2\pi\mathbb{Z})^m$, surjectively in independent coordinates, and a mean on it equals*

$$a_0 + \sum_{l=1}^m (a_l \cos \theta_l + b_l \sin \theta_l).$$

Its contribution to (11.1) is $\frac{1}{2} \sum_l \sqrt{a_l^2 + b_l^2}$.

Proof. Each summand has the symmetric range with amplitude $\sqrt{a_l^2 + b_l^2}$, and all extrema are simultaneously attainable in the full product. Divide the resulting oscillation by four. $\square$

On a proper subtorus, extrema must instead be taken subject to its relations. Ambient coordinates do not certify independence. Similarly, invisible directions do not create an obstruction: for $H = \{\text{diag}(R_\theta, 1)\}$, seed and query e_3 give constant mean one, whereas seed ρe_1 and query e_1 have threshold $\rho/2$.

Proposition 11.7 (Compact similarity). *The classification applies to a finite alphabet in $GL(D, \mathbb{R})$ whose generated group, including inverses, is uniformly bounded, with the executable return hypothesis retained. Conjugating to an invariant Euclidean metric changes neither output probabilities nor stochastic widths.*

Proof. Uniform bounds on matrices and inverses make the closure a compact group. The matrix $Q = \int_H h^T h d\mu(h)$ is positive definite and satisfies $A^T Q A = Q$. Hence $Q^{1/2} A Q^{-1/2}$ is orthogonal. Transport the continuous output maps through this conjugacy; for a linear mean transport x_s to $Q^{1/2} x_s$ and q_j to $Q^{-1/2} q_j$. Exact identity products and their word lengths are preserved. $\square$

11.1. A curved categorical image with distinct component centers.

Proposition 11.8. *Let $H = (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z}$, with an identity letter, an irrational rotation in the first factor, and a toggle of the second factor. Put*

$$q_0 = (\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{6}), \quad q_1 = (\tfrac{1}{6}, \tfrac{1}{3}, \tfrac{1}{2}), \quad t = \tfrac{1}{12},$$

and prescribe the three-outcome law

$$F(\theta, c)_l = q_{c,l} + t \cos(\theta - 2\pi l/3), \quad l = 0, 1, 2.$$

Each component image is a proper closed curve in the interior of the simplex, with unique constrained total-variation center q_c and radius t . The union of both component images has radius

1/4. Thus the sharp clocked threshold is 1/12, and the stationary minimum at that threshold is exactly two labels.

Proof. The three cosines sum to zero and every coordinate is at least 1/12, so F is a legal law in the simplex interior. The first two independent trigonometric coordinates identify a nondegenerate ellipse in its affine plane. For a zero-sum vector $z \in \mathbb{R}^3$, one has $\frac{1}{2}\|z\|_1 = \max_l |z_l|$. Each coordinate of the curve ranges over $[q_{c,l} - t, q_{c,l} + t]$. Hence for every center $y \in \Delta_3$,

$$\sup_{\theta} \text{TV}(F(\theta, c), y) = t + \max_l |q_{c,l} - y_l|.$$

This proves uniqueness of q_c and radius t . For the union, the least possible value of $\max_{c,l} |q_{c,l} - y_l|$ is 1/6, forced by the two endpoints in the first coordinate and attained by $y = (1/3, 1/3, 1/3)$. Its radius is therefore $t + 1/6 = 1/4$. A one-label stationary machine has a constant decoder and cannot reach error t ; two component labels attain it. The clocked threshold follows from Theorem 7.2. $\square$

Remark 11.9. The total-variation diameter of a probability simplex is one, so half its diameter is 1/2; the three-vertex example above has radius 2/3. In the unitary corollary, $U(d)$ is connected. These facts explain respectively why coordinate midpoints do not solve the categorical problem and why the full-unitary component is the whole group.

12. MATCHING WIDTH LAWS THROUGHOUT THE SUBCRITICAL INTERVAL

We now distinguish the dimension of the command arithmetic from the dimension of the physical orbit. Let $s \geq 1$ and let the alphabet contain the rotations $R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_s}$ and I . For the matching upper bound the alphabet is exactly these letters, optionally with the reflection $F = \text{diag}(1, -1)$. Use signed coordinate seeds of radius ρ and coordinate queries. Extra letters do not affect the lower bounds, but are not silently included in the upper construction.

For a real number x , write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The dual Diophantine condition is

$$(12.1) \quad \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|n\|_1^{-\tau} \quad (0 \neq n \in \mathbb{Z}^s), \quad b > 0, \quad \tau \geq s.$$

In particular $1, \alpha_1, \dots, \alpha_s$ are rationally independent. When $\tau = s$ this is the dual badly approximable condition. All implicit constants below may depend on $\alpha, b, s, \rho, \varepsilon$, but not on N or a stochastic realization.

Theorem 12.1 (Full-subcritical matching law). *Suppose (12.1) holds with $\tau = s$, and let $0 < \rho < 1$. For every fixed $0 \leq \varepsilon < \rho/2$,*

$$(12.2) \quad W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}).$$

The upper bound is achieved by an exact horizon-dependent realization. At $\varepsilon \geq \rho/2$, one command label suffices. If $\tau \geq s$ is arbitrary, the lower bound $W_{N,\varepsilon} = \Omega(N^{s/(2\tau+1)})$ still holds throughout the same subcritical interval.

The exponent for one badly approximable rotation is 1/3, not logarithmic. The statement includes zero error when the signal has strict amplitude slack $\rho < 1$. The threshold theorem continues to include $\rho = 1$, but the exact polynomial upper bound here is not asserted at that endpoint. The earlier arithmetic argument obtained this scale in a smaller noise range. The proof below supplies the missing full-noise lower estimate and states the exact upper realization in full.

Proposition 12.2 (An explicit family in every command dimension). *Let ξ be a real algebraic integer of degree $s + 1$. Then $\alpha = (\xi, \xi^2, \dots, \xi^s)$ satisfies (12.1) with $\tau = s$ and some effectively computable $b > 0$.*

Proof. For $0 \neq n \in \mathbb{Z}^s$ put $H = \|n\|_1$, and choose $n_0 \in \mathbb{Z}$ so that $z = n_0 + \sum_{l=1}^s n_l \xi^l$ has modulus $\|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}$. The degree assumption gives $z \neq 0$. Write $\xi_1 = \xi, \dots, \xi_{s+1}$ for the conjugates and put $M_j = \max_{1 \leq l \leq s} |\xi_j|^l$. Then $|n_0| \leq HM_1 + 1/2$ and

$$\left| n_0 + \sum_{l=1}^s n_l \xi_j^l \right| \leq H(M_1 + M_j + 1/2) \quad (j \geq 2).$$

The norm of the nonzero algebraic integer z is a nonzero integer. Thus

$$|z| \geq H^{-s} \prod_{j=2}^{s+1} (M_1 + M_j + 1/2)^{-1}.$$

Choose a positive rational b below this computable positive constant and $1/2$. This proves the claim. It concerns algebraic *angles*, not algebraic entries of their rotation matrices. $\square$

12.1. Harmonic localization. For the harmonic calculation assume that the alphabet contains I, R_θ and that one seed-query pair has mean $\rho \cos \phi$ after accumulated angle ϕ , where $0 < \rho \leq 1$. Other seeds, queries, or commands impose additional constraints and do not invalidate the lower bound. Let $\lambda = e^{i\theta}$ have infinite order. Put $\delta = 2\varepsilon < \rho$, and choose an integer

$$(12.3) \quad r \geq 1, \quad r > \frac{\delta}{\rho - \delta}, \quad m = r + 1.$$

For example $r = \lfloor \delta/(\rho - \delta) \rfloor + 1$ works. Define

$$(12.4) \quad \Delta_r = 2 \left(\frac{\rho r}{r+1} - \delta \right) > 0, \quad A_r = \frac{2(1 + \rho + \delta)(2r + 1)m}{\Delta_r} > 1.$$

Lemma 12.3 (Explicit harmonic localization). *The nonnegative trigonometric polynomials*

$$p_{\pm}(\phi) = \frac{(1 \pm \cos \phi)^r}{2^{-r} \binom{2r}{r}}$$

have Haar mean one and weighted means

$$\int p_{\pm}(\phi) \rho \cos \phi \frac{d\phi}{2\pi} = \pm \frac{\rho r}{r+1}.$$

For the moving feature vector

$$Z_m(\phi) = (\cos \phi, \sin \phi, \dots, \cos m\phi, \sin m\phi),$$

any terminal realization with wordwise mean error at most δ obeys

$$(12.5) \quad \mathbb{E} \|\mathbb{E}[Z_m(\phi) \mid S]\| \geq \frac{\Delta_r}{2(1 + \rho + \delta)(2r + 1)}.$$

Proof. The identity $1 + \cos \phi = 2 \cos^2(\phi/2)$, or the constant Laurent coefficient, gives

$$C_r := \int (1 + \cos \phi)^r \frac{d\phi}{2\pi} = 2^{-r} \binom{2r}{r}, \quad C_{r+1}/C_r = \frac{2r+1}{r+1}.$$

Hence the weighted cosine mean is $C_{r+1}/C_r - 1 = r/(r+1)$. Translation by π gives the negative case.

We use the unweighted real coefficient vector in the basis $1, \cos \phi, \sin \phi, \dots$, without orthonormal Fourier rescaling. The absolute sum of the real cosine coefficients of either $p_{\pm}$, including the constant coefficient, is

$$B_r = \frac{4^r}{\binom{2r}{r}} \leq 2r + 1.$$

For p_+ its coefficients are nonnegative and their sum is $p_+(0)$; for p_- they differ only by alternating signs. The inequality follows because the central binomial coefficient is the largest of the $2r + 1$ coefficients summing to 4^r . Multiplication by $\rho \cos \phi$ has coefficient absolute sum at most ρB_r , since $\cos u \cos v = (\cos(u + v) + \cos(u - v))/2$. Thus the Euclidean norms of the moving coefficient vectors of $p_\pm$ and $p_\pm \rho \cos \phi$ are at most $2r + 1$ and $\rho(2r + 1)$, respectively. For completeness, let $c = \mathbb{E}D$, $Q = \mathbb{E}\|\mathbb{E}[Z_m | S]\|$. For either weight, the conditional coefficient identity gives $|c - \int p_\pm \rho \cos \phi| \leq \delta + (1 + \rho + \delta)(2r + 1)Q$. Subtracting the two inequalities gives (12.5). Correctness is averaged word by word before using this identity. The starting feature norm is $\sqrt{m} \leq m$; A_r deliberately uses the latter bound to obtain a rational closed form. $\square$

The integer r in (12.3) is the localization order; it is unrelated to the number s of rotation letters. For fixed subcritical error it is finite and independent of N and k .

Theorem 12.4 (A full-noise arithmetic occupation bound). *Assume (12.1) and $0 \leq \varepsilon < \rho/2$. Fix the localization parameters r, m, A_r in (12.3)–(12.4). For an integer $k \geq 1$, put*

$$L = \lceil (2k)^{1/s} \rceil, \quad B = s(L - 1).$$

Every error- ε realization of peak at most k satisfies

$$(12.6) \quad N < B \left(1 + b^{-2}(mB)^{2\tau} \log A_r \right).$$

With no restriction on other widths, the number of cuts of width at most k is at most

$$(12.7) \quad \left\lceil B - 1 + Bb^{-2}(mB)^{2\tau} \log A_r \right\rceil.$$

Proof. Use s consecutive groups of $L - 1$ command positions. In group i , choose J_i uniformly from $\{0, \dots, L - 1\}$, apply J_i copies of the i th rotation, and fill the other positions with identities. This defines an external law on length- B words with $L^s \geq 2k$ equally likely products. On the harmonic feature use

$$R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt}).$$

For two distinct count vectors J, J' and any frequency $1 \leq l \leq m$, condition (12.1) and $\|l(J - J')\|_1 \leq mB$ give

$$|e^{2\pi i l(J - J') \cdot \alpha} - 1| = 2|\sin(\pi l(J - J') \cdot \alpha)| \geq 4b(mB)^{-\tau} =: \sigma.$$

Here $2\sin(\pi x) \geq 4x$ for $0 \leq x \leq 1/2$. Also $b \leq 1/2$, $m \geq 2$, and $B \geq 1$, so $\sigma \leq 1$.

For a unit vector $u = (u_1, \dots, u_m)$ in the frequency blocks, the squared distance between the two represented orbit points is

$$\sum_{l=1}^m \|u_l\|^2 |e^{2\pi i l(J - J') \cdot \alpha} - 1|^2 \geq \sigma^2.$$

An open ball of radius $\sigma/2$ about any unit center contains at most one point. At least half the L^s points therefore have distance at least $\sigma/2$ from all of any k centers. Since the inner-product defect between unit vectors equals half their squared distance, the finite word law has defect at least

$$d_k = \sigma^2/16 = b^2(mB)^{-2\tau}.$$

Apply conditional-centroid contraction to successive blocks and Lemma 12.3 to the terminal query. The starting norm is $\sqrt{m} \leq m$. Thus

$$\left\lfloor \frac{N}{B} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq b^{-2}(mB)^{2\tau} \log A_r,$$

which proves (12.6). The greedy endpoint count, with identity gaps and no prefix, gives (12.7). Internal registers remain unrestricted. Finally $B < s(2k)^{1/s}$, so (12.6) implies the asserted $\Omega(N^{s/(2\tau+1)})$ lower bound. $\square$

Remark 12.5. Testing a coordinate unit vector in (12.1) gives $b \leq 1/2$. In the block construction, $L = \lceil (2k)^{1/s} \rceil \geq 2$, hence $B = s(L-1) \geq 1$. The integer r is the localization order and s is the number of rotations. For the algebraic-angle example, $1, \xi, \dots, \xi^s$ are linearly independent over $\mathbb{Q}$ by the degree $s+1$ hypothesis; this ensures that the algebraic integer used in its norm bound is nonzero.

12.2. An exact polygon realization. Write $\eta_\alpha(q) = \max_i \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}$. The following elementary construction explains why a common denominator, rather than the number of numerical orbit points, controls the upper width.

Proposition 12.6 (Exact enclosure upper bound). *Let $0 < \rho < 1$, $a = (1 + \rho)/2$, and $\gamma = \log(a/\rho) > 0$. Choose an integer $q_0 \geq 4$ with $\cos(\pi/q_0) \geq a$. If $q \geq q_0$ and*

$$(12.8) \quad 4\pi^2 N \eta_\alpha(q)/q^2 \leq \gamma,$$

then the indicated rotation alphabet, with or without F , has an exact horizon- N realization with q command labels.

Proof. Let P_q be the regular polygon with vertices $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$, $0 \leq j < q$. For each i choose a nearest integer p_i to $q\alpha_i$ and put

$$t = \pi/q, \quad h_i = 2\pi|\alpha_i - p_i/q| \leq t, \quad \Lambda_i = \frac{\cos(t - h_i)}{\cos t}.$$

Writing $n_j = (\cos((2j+1)t), \sin((2j+1)t))$, its facet description is

$$P_q = \{x : \langle n_j, x \rangle \leq \cos t \ (0 \leq j < q)\}.$$

The facet inequalities of the regular polygon give $R_{2\pi\alpha_i}P_q \subset \Lambda_i P_q$: the nearest facet normal to a rotated vertex has angular difference $t - h_i$. Further,

$$\log \Lambda_i \leq \log(1 + \tan(t)h_i) \leq \tan(t)h_i \leq 4\pi^2 \eta_\alpha(q)/q^2,$$

since $\cos h + \tan(t) \sin h \leq 1 + \tan(t)h$ and $\tan t \leq 2t$ for $0 \leq t \leq \pi/4$. Let $\Lambda = \max(1, \Lambda_1, \dots, \Lambda_s)$. The standard reflection $F = \text{diag}(1, -1)$ sends v_j to $v_{-j \pmod q}$ and preserves the chosen vertex set. The identity and reflection therefore preserve P_q , so every command U satisfies $\Lambda^{-1}UP_q \subset P_q$.

For each command and each vertex choose a probability row with barycenter $\Lambda^{-1}Uv_j$ in P_q . These are actual common command rows, not independent prefix/suffix factorizations. At most three successors suffice by planar convexity. Initialize the seed ρu_s , $u_s \in \{\pm e_1, \pm e_2\}$, with a row whose barycenter is au_s . This is possible since $a \leq \cos(\pi/q)$, the polygon's inradius. After a length- N word the mean vertex is

$$a\Lambda^{-N}U_w u_s.$$

By (12.8), $\rho\Lambda^N \leq a$. Hence the terminal columns

$$d_j(v) = \frac{\rho\Lambda^N}{a} e_j^T v$$

belong to $[-1, 1]$ and return exactly $\rho e_j^T U_w u_s$. The rows can be stationary within a horizon; the terminal scaling and chosen denominator may depend on N . No one finite all-horizon exact machine is asserted. $\square$

Lemma 12.7 (Comparable simultaneous denominators). *Assume (12.1). For every integer $Q \geq 2$ there is q with $1 \leq q \leq Q^s$, $\eta_\alpha(q) \leq 1/Q$, and*

$$q \geq cQ^{s/(\tau+1-s)}, \quad c = \left(\frac{b}{s^{\tau+1}} \right)^{s/(\tau+1-s)}.$$

In particular, for $\tau = s$ one has $cQ^s \leq q \leq Q^s$.

Proof. Partition the torus into Q^s half-open cubes and apply the pigeonhole principle to $0, \alpha, \dots, Q^s \alpha$. Their difference gives $q \leq Q^s$ and integers p_i with $|q\alpha_i - p_i| \leq 1/Q$.

Put $M = \lfloor q^{1/s} \rfloor$. The $(M+1)^s > q$ vectors in $\{0, \dots, M\}^s$ cannot all have distinct residues $n \cdot p$ modulo q , even when some nearest integers p_i are negative. Subtract two to obtain a nonzero vector n with $q \mid n \cdot p$ and $H := \|n\|_1 \leq sq^{1/s}$. Therefore

$$bH^{-\tau} \leq \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad q^{(\tau+1-s)/s} \geq bQ/s^{\tau+1}.$$

The displayed bound follows. This argument supplies the needed quantitative transference directly; no unspecified denominator subsequence is being used. $\square$

Proof of Theorem 12.1. The lower estimate follows from Theorem 12.4. For $\tau = s$, take the constants a, γ, q_0, c above and an integer

$$Q \geq \max \left(2, (q_0/c)^{1/s}, \left(\frac{4\pi^2 N}{c^2 \gamma} \right)^{1/(2s+1)} \right).$$

Choosing the least such integer, Lemma 12.7 gives $q \leq Q^s = O(N^{s/(2s+1)})$, $q \geq q_0$, and

$$4\pi^2 N \eta_\alpha(q)/q^2 \leq 4\pi^2 N/(c^2 Q^{2s+1}) \leq \gamma.$$

Proposition 12.6 supplies the exact upper bound. At and above $\rho/2$ use the fair decoder from Corollary 11.5. This proves all claims. $\square$

The command count s is an arithmetic parameter, not a replacement for the dimension of the physical circle. The constants deteriorate as $\varepsilon \uparrow \rho/2$ through the localization degree, and as $\rho \uparrow 1$ through the enclosure slack. Both dependences are explicit in the finite inequalities. The theorem makes no uniform claim at these limiting parameters.

Corollary 12.8 (Unit amplitude at positive error). *For the exact rotation alphabet of Theorem 12.1, with or without its specified reflection, suppose the angle vector is dual badly approximable. At $\rho = 1$, for every fixed $0 < \varepsilon < 1/2$,*

$$W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}).$$

Proof. The harmonic localization and orbit-packing lower bound in Theorem 12.4 permits $\rho = 1$ and every fixed $\varepsilon < 1/2$. For the upper bound choose $0 < \eta < 2\varepsilon$ and put $\rho' = 1 - \eta \in (0, 1)$, decreasing η if necessary. Apply the exact polygon construction and comparable-denominator lemma to ρ' . On every seed, word, and query its mean differs from the unit-amplitude target by at most η , so the binary total-variation error is at most $\eta/2 < \varepsilon$. Its width has the required order. This is a positive-error upper bound, not an exact realization at unit amplitude. $\square$

The case $\rho = 1$, $\varepsilon = 0$ is not decided by this corollary. In particular, taking $\eta = 0$ would destroy the strictly positive enclosure slack required by Proposition 12.6.

13. SPHERICAL EXPERIMENTS WITH A SPECTRAL GAP

The next result concerns a full vector orbit, not an abelian factor of a nonabelian system. Write σ for normalized area on S^{d-1} , where $d \geq 2$. Let a finite orthogonal alphabet contain the identity, and let μ be a probability law on its letters. Assume that its averaging operator

$$Pf(x) = \mathbb{E}_{R \sim \mu} f(Rx)$$

satisfies

$$(13.1) \quad \|P^n f\|_{L^2(\sigma)} \leq \lambda^n \|f\|_{L^2(\sigma)} \quad \left(\int f d\sigma = 0 \right), \quad 0 < \lambda < 1.$$

The physical target is $F(g) = \rho g x_0$ for a fixed $x_0 \in S^{d-1}$, with output set the closed Euclidean unit ball and error measured in Euclidean norm.

Theorem 13.1 (A spherical width law up to a logarithm). *Assume (13.1), $0 < \rho < 1$, and $0 \leq \varepsilon < \rho$. There are constants $c, C > 0$ such that*

$$(13.2) \quad c \left(\frac{N}{\log(N+2)} \right)^{(d-1)/2} \leq W_{N,\varepsilon} \leq C(N+1)^{(d-1)/2}.$$

The upper realization is exact. Every error- ε realization has at most

$$Ck^{2/(d-1)} \log(k+2)$$

positive cuts of width at most k , with other widths unrestricted. The constants may depend on $d, \lambda, \rho, \varepsilon$, but not on the horizon or the realization. At unit amplitude the same bounds hold for every fixed $0 < \varepsilon < 1$; exact unit amplitude is not included in the upper assertion.

Lemma 13.2 (Spherical quantization and executable contraction). *Under (13.1), there are $c_d > 0$ and integers $B_k \leq C_{d,\lambda} \log(k+2)$ such that*

$$\mathbb{E}_{w \sim \mu^{B_k}} \max_{i \leq k} \langle v_i, u_w u \rangle \leq 1 - c_d k^{-2/(d-1)}$$

for all unit vectors $u, v_1, \dots, v_k$. The defect may be decreased so that it lies in $(0, 1)$. When $d = 3$ one may take defect $1/(2k)$ and

$$(13.3) \quad B_k = \left\lceil \frac{\log(64k^2)}{-\log \lambda} \right\rceil.$$

Proof. Normalized spherical cap measure obeys

$$c_d h^{d-1} \leq \sigma\{x : \|x - u\|_2 < h\} \leq C_d h^{d-1} \quad (0 < h \leq 1),$$

uniformly in u . This follows by writing area in polar angle as a constant times $\sin^{d-2} \theta d\theta$ and using $h = 2 \sin(\theta/2)$; for $d = 2$ the same formula is the arc-length formula. For $f(x) = \max_i \langle v_i, x \rangle$, choose $h = ck^{-1/(d-1)} \leq 1$ so that the union of the k radius- h caps has area at most $1/2$. Outside them, $1 - f(x) \geq h^2/2$. Thus

$$(13.4) \quad \int f d\sigma \leq 1 - a_d k^{-2/(d-1)}.$$

The function f is 1-Lipschitz for chordal distance and bounded by one in absolute value.

Let K_δ average over the chordal cap of radius $\delta \leq 1$. It commutes with all rotations, fixes constants, and satisfies $\|K_\delta f - f\|_\infty \leq \delta$. The cap density at a center has L^2 norm at most $C_d \delta^{-(d-1)/2}$. Hence (13.1) and Cauchy-Schwarz imply, uniformly in the center,

$$\left| P^B f(u) - \int f d\sigma \right| \leq \delta + C'_d \delta^{-(d-1)/2} \lambda^B.$$

Take δ to be a fixed sufficiently small multiple of $k^{-2/(d-1)}$ and then $B = O_{d,\lambda}(\log(k+2))$ so that the right side is at most half the defect in (13.4). The law of the B successive letters is a finite executable word law, proving the general assertion.

For $d = 3$, a chordal cap of radius $h \leq 2$ has normalized area $h^2/4$. The union bound gives $\Pr(1 - f < t) \leq kt/2$, so integration for $0 \leq t \leq 2/k$ gives $\int (1 - f) d\sigma \geq 1/k$. A radius- δ cap density has L^2 norm $2/\delta$, and $\|f - \int f\|_2 \leq 2$. Thus the mixing error is at most $\delta + 4\lambda^B/\delta$. Choose $\delta = 1/(4k)$ and (13.3); the error is at most $1/(2k)$. $\square$

Proof of Theorem 13.1. Let $Z = u_w x_0$ and let $D(S)$ be the terminal decoder. For every deterministic word, correctness and $\|Z\|_2 = 1$ give

$$\langle \mathbb{E}[D(S) \mid w], Z \rangle \geq \rho - \varepsilon.$$

Under any external word law this implies

$$(13.5) \quad \rho - \varepsilon \leq \mathbb{E} \langle D(S), Z \rangle \leq \mathbb{E} \|\mathbb{E}[Z \mid S]\|_2,$$

since the decoder lies in the unit ball. Lemma 7.5, applied with the executable law of Lemma 13.2, contracts the last conditional norm by $1 - a_k$ at each selected narrow endpoint, where $a_k = c_d k^{-2/(d-1)} \in (0, 1)$. Initially the norm is one. Fixed identity-letter gaps cannot increase it, even though their stochastic rows may process the register. After J selected endpoints, (13.5) gives

$$J[-\log(1 - a_k)] \leq \log \frac{1}{\rho - \varepsilon}.$$

Greedy spacing by B_k as in the occupation proof discards at most $B_k - 1$ initial cuts and charges at most B_k eligible cuts to each selected endpoint. Therefore the occupation count is at most

$$(13.6) \quad B_k \left(1 + a_k^{-1} \log \frac{1}{\rho - \varepsilon} \right).$$

If every register has width at most k , this bounds N . Inverting the bound gives the lower order in (13.2). Explicitly, when $k \leq (N+1)^{d-1}$ its logarithm is $O_d(\log(N+2))$, and otherwise the asserted lower bound is automatic.

For the upper bound, choose a maximal δ -separated set of vertices on S^{d-1} containing x_0 , with $0 < \delta \leq 1/2$. The disjoint caps of radius $\delta/2$ show that it has at most $C_d \delta^{-(d-1)}$ vertices. Maximality makes it a δ -net. Its convex hull Q satisfies

$$(13.7) \quad r\mathbb{B}^d \subset Q \subset \mathbb{B}^d, \quad r = 1 - \delta^2/2,$$

because for every unit direction u some vertex v has $\langle u, v \rangle \geq 1 - \delta^2/2$. The inclusion follows from the supporting half-space characterization of a closed convex body. For each command R and vertex v_i , the point rRv_i belongs to Q . Choose barycentric coordinates of that point as the i th row of one common stochastic command matrix. The identity command is treated in the same way, with the same factor r .

Initialize at the vertex x_0 and decode vertex v_i as $(\rho/r^N)v_i$. Induction on the row means gives output exactly $\rho u_w x_0$ on every length- N word. The decoder is legal if $r^N \geq \rho$. Choose

$$\delta^2 = \min \left(\frac{1}{4}, \frac{-\log \rho}{N+1} \right).$$

Since $\log(1 - \delta^2/2) \geq -\delta^2$, the required inequality holds. The number of vertices is $O_{d,\rho}((N+1)^{(d-1)/2})$. Both rows and decoder may depend on the horizon, as permitted by the clocked model.

For $\rho = 1$ and $\varepsilon > 0$, the same lower proof applies. An exact upper realization at amplitude $\rho' = 1 - \varepsilon/2$ differs from the unit-amplitude target by $\varepsilon/2$ in Euclidean norm, proving the stated upper bound. There is no assertion of exactness at unit amplitude. $\square$

Corollary 13.3 (A rational noncommuting alphabet on $\mathrm{SO}(3)$). *Let the alphabet consist of the identity, the matrices*

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/5 & -4/5 \\ 0 & 4/5 & 3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and their inverses. For $F(g) = \rho g e_3$, with $0 < \rho < 1$ and $0 \leq \varepsilon < \rho$, the bounds are

$$c \frac{N}{\log(N+2)} \leq W_{N,\varepsilon} \leq C(N+1).$$

At error $\varepsilon \geq \rho$ a single zero decoder suffices. Every subcritical realization has at most $Ck \log(k+2)$ positive cuts of width at most k .

Proof. The angle θ of both rotations has $2 \cos \theta = 6/5$. It is not a rational multiple of 2π : otherwise $2 \cos \theta$ would be a rational algebraic integer, hence an integer. The closures of the two cyclic groups are the full circle subgroups about their distinct axes. Their Lie algebras and bracket span $\mathfrak{so}(3)$, so the generated compact closure is $\mathrm{SO}(3)$. The matrices do not commute.

Under the double cover $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$, their unit-quaternion lifts are $(2+i)/\sqrt{5}$ and $(2+k)/\sqrt{5}$. They have algebraic matrix entries and generate a dense subgroup of $\mathrm{SU}(2)$: its closed image is all of $\mathrm{SO}(3)$, so its Lie algebra has dimension three. The spectral-gap theorem of Bourgain–Gamburd [16] applies. Adding the identity with positive mass makes the symmetric averaging operator have norm strictly less than one on mean-zero L^2 . The spherical representation is a subrepresentation of the regular representation, so the uniform law on the five displayed letters satisfies (13.1) on S^2 . Apply Theorem 13.1. The radius of the full sphere ρS^2 inside the unit ball is ρ , attained at zero. $\square$

The spectral gap in this corollary is an imported theorem, not a numerical assertion extracted from finite orbit checks. This retained block estimate has a logarithmic loss. Theorem 15.5 supplies the matching linear lower order under the same spectral-gap hypothesis. It does not compute a numerical gap for this particular alphabet; Theorem 27.2 gives a separate alphabet with explicit constants.

14. A UNIFORM ORBIT CRITERION FOR QUANTITATIVE WIDTH

The spherical estimate has an extension in which the dimension of the ambient feature space is not the relevant exponent. Two geometric quantities enter: a uniform small-ball exponent for every possible conditional-centroid direction, and the covering dimension of the particular target orbit. They need not agree. Keeping this distinction is essential for matrix-valued experiments.

Let G be a compact connected Lie group with normalized Haar measure m_G and a bi-invariant Riemannian distance. Let $R : G \rightarrow O(E)$ be a continuous orthogonal representation on a finite-dimensional real Euclidean space. Fix $z_0 \in E$ of norm one, put

$$\mathcal{O} = R(G)z_0, \quad Q = \mathrm{conv}(\mathcal{O}), \quad q = \dim \mathcal{O},$$

and assume that $\mathcal{O}$ spans E and that E has no nonzero fixed vector. Thus $q \geq 1$. A finite alphabet in G contains the identity and carries a probability law μ for which

$$(14.1) \quad \|P^n f\|_2 \leq \lambda^n \|f\|_2 \quad \left(\int_G f \, dm_G = 0 \right), \quad Pf(g) = \mathbb{E}_{a \sim \mu} f(u_a g), \quad 0 < \lambda < 1.$$

Letters outside the support of μ are allowed. All command products and stochastic rows retain the conventions of Definition 7.1.

The uniform geometric hypothesis is that, for some $p > 0$, $A < \infty$ and $h_0 > 0$,

$$(14.2) \quad \sup_{u,v \in S(E)} m_G \{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

It is imposed on all unit vectors u , not only on $\mathcal{O}$. A conditional mean of points in $\mathcal{O}$, after normalization, need not belong to $\mathcal{O}$.

Lemma 14.1 (From orbit small balls to executable contraction). *Under (14.1)–(14.2), constants $c, C > 0$ and integers $B_k \leq C \log(k+2)$ can be chosen so that*

$$(14.3) \quad \mathbb{E}_{w \sim \mu^{B_k}} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - ck^{-2/p}$$

for every $u, v_1, \dots, v_k \in S(E)$. The constant c can be decreased so that the defect lies in $(0, 1)$ for every $k \geq 1$.

Proof. Choose $h = c_0 k^{-1/p} \leq \min(h_0, 1)$ with $Ac_0^p \leq 1/2$. Outside the union of the k open balls in (14.2),

$$1 - \max_i \langle v_i, R(g)u \rangle \geq h^2/2.$$

The union has measure at most $1/2$. Consequently the Haar mean of $f(g) = \max_i \langle v_i, R(g)u \rangle$ is at most $1 - a_k$, where $a_k = c_0^2 k^{-2/p}/4$. Repeated centers are allowed. All such functions are bounded by one in absolute value and have a common Lipschitz constant L : differentiating the finite-dimensional representation bounds the derivative uniformly on unit vectors.

Write $D = \dim G$, distinct from the cap exponent p and orbit dimension q . The ball calculation uses the fixed bi-invariant Riemannian metric on G . In exponential coordinates at the identity, Haar measure has a smooth positive density. Hence a metric ball of radius $\delta \leq \delta_0$ has measure at least $c_G \delta^D$. Average $P^B f$ over this ball. Left translations are isometries, so $P^B f$ has Lipschitz constant at most L . Cauchy–Schwarz and (14.1) give

$$|P^B f(1) - \bar{f}| \leq L\delta + 2c_G^{-1/2} \delta^{-D/2} \lambda^B.$$

Take δ to be a sufficiently small fixed multiple of a_k , also bounded by δ_0 . Then choose $B_k \geq 1$ to make the second term at most $a_k/4$. The first term is at most $a_k/4$ as well. Since $\log(a_k^{-1}) = O(\log(k+2))$, this requires $B_k \leq C \log(k+2)$. Products under μ^{B_k} are genuine words of one common length. Their reverse chronological order has the same law because the letters are independent with the same law. Thus $P^{B_k} f(1)$ is the expectation in (14.3). This proves the lemma with defect $a_k/2$. $\square$

Lemma 14.2 (Quadratic inner approximation of an orbit hull). *There are constants $C_1, C_2, \delta_0 > 0$ such that, for $0 < \delta \leq \delta_0$, points $z_1, \dots, z_K \in \mathcal{O}$, including z_0 , can be chosen with*

$$(14.4) \quad K \leq C_1 \delta^{-q}, \quad (1 - C_2 \delta^2)Q \subseteq \text{conv}\{z_1, \dots, z_K\} \subseteq Q, \quad 0 < 1 - C_2 \delta^2 < 1.$$

Proof. The orbit is a compact smooth embedded homogeneous manifold. Finitely many coordinate charts with bounded derivatives and inverse derivatives give an intrinsic δ -net of size at most $C\delta^{-q}$; adjoining z_0 changes only C . Use the induced Riemannian metric on the orbit. If the functional $x \mapsto \langle b, x \rangle$ attains its maximum at $x_b \in \mathcal{O}$, its derivative along the orbit vanishes there. The second derivative along unit-speed orbit geodesics has absolute value at most $C\|b\|$, uniformly in their starting points and directions, by compactness of the unit tangent bundle and smoothness of the embedding. A net point within intrinsic distance δ of x_b therefore satisfies

$$\langle b, z_i \rangle \geq h_Q(b) - C\|b\|\delta^2, \quad h_Q(b) = \max_{x \in Q} \langle b, x \rangle.$$

Haar averaging gives $\int R(g)z_0 dm_G = 0$. Moreover 0 lies in the interior of Q in E . Otherwise a nonzero supporting functional would be nonnegative on the orbit and have Haar mean zero; continuity and full Haar support would make it vanish on the orbit, contrary to its spanning E . Thus $r\mathbb{B}_E \subset Q$ for some $r > 0$, and $h_Q(b) \geq r\|b\|$. The support inequality becomes $\max_i \langle b, z_i \rangle \geq (1 - C\delta^2/r)h_Q(b)$. The support-function criterion for inclusion of compact convex sets proves (14.4). Decrease δ_0 to obtain the last bounds. $\square$

Theorem 14.3 (Geometric width and occupation). *Assume (14.1)–(14.2). Take one seed and one query with target $F(g) = \rho R(g)z_0$, output set Q , and Euclidean error. For $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$, put $\zeta = \rho - \varepsilon \in (0, 1]$. Every error- ε realization satisfies*

$$(14.5) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \log(k+2) \left(1 + k^{2/p} \log \frac{1}{\zeta}\right),$$

where C depends on the geometric and spectral-gap data but not on ρ, ε, N, k or the realization. In particular, at fixed parameters,

$$(14.6) \quad W_{N,\varepsilon} \geq c \left(\frac{N}{\log(N+2)} \right)^{p/2}.$$

For $0 < \rho < 1$ there is an exact common-row realization with

$$(14.7) \quad W_{N,0} \leq C \left(1 + \frac{N+1}{-\log \rho} \right)^{q/2}.$$

For $\rho = 1$ and fixed $0 < \varepsilon < 1$, the upper order is $O_\varepsilon((N+1)^{q/2})$. When $p = q$, the lower and upper exponents agree, with the displayed logarithmic loss in the lower bound. The constrained error threshold is exactly ρ , attained by the constant zero decoder.

Proof. Put $Z = R(u_w)z_0$ and let $D(S) \in Q \subseteq \mathbb{B}_E$ be the decoder. For each deterministic word, the error bound gives $\langle \mathbb{E}[D(S) \mid w], Z \rangle \geq \rho - \varepsilon$. Under any external word law, conditioning on the register yields

$$\zeta \leq \mathbb{E} \langle D(S), Z \rangle \leq \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

The conditional-centroid Lemma 7.5 applies to the law of Lemma 14.1. The uniformity over every direction in (14.2) is precisely what justifies this application. It contracts the conditional norm by $1 - a$ at a selected endpoint of width at most k , where $a = ck^{-2/p} \in (0, 1)$. The initial norm is one. Fixed identity-letter fillers cannot increase it, regardless of their rows or intervening widths. After J selected endpoints,

$$J[-\log(1-a)] \leq \log(1/\zeta).$$

Greedily select narrow cuts at separation at least B_k , omitting the first $B_k - 1$ cuts. Each selected endpoint accounts for at most B_k cuts. The count is at most $B_k(1 + a^{-1} \log(1/\zeta))$, proving (14.5), also when $\zeta = 1$. In that case even one selected endpoint is impossible. More precisely one may take the integer bound

$$L_k = B_k \left(1 + \left\lceil c^{-1} k^{2/p} \log(1/\zeta) \right\rceil \right).$$

For a full width bound k , either $k \geq N^{p/2}$, or $\log(k+2) \leq C_p \log(N+2)$. In the latter case the occupation inequality gives $N \leq C_\zeta \log(N+2) k^{2/p}$. These alternatives prove (14.6).

For the upper bound take the net in Lemma 14.2, and write $r = 1 - C_2 \delta^2$. The orbit hull is invariant under every command. Hence $rR(u_a)z_i$ is in the convex hull of the net points. For each of the finitely many command-vertex pairs, choose barycentric coordinates independently; no measurable selection is involved. Use them as row i of a stochastic matrix T_a , the same at every epoch within this horizon. Initialize at z_0 . The conditional row means then evolve as $r^t R(u_w)z_0$. Decode label i by $(\rho/r^N)z_i$, which belongs to Q if $r^N \geq \rho$, since $0 \in Q$. Choosing

$$\delta^2 = \min \left(\delta_0^2, \frac{1}{2C_2}, \frac{-\log \rho}{2C_2(N+1)} \right)$$

gives this inequality by $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$. The net size gives (14.7). At unit amplitude and positive error use the exact construction for amplitude $1 - \varepsilon/2$; its discrepancy from the desired target is $\varepsilon/2$.

Finally, Haar mean zero and $\|R(g)z_0\| = 1$ imply

$$\int_G \|\rho R(g)z_0 - y\|^2 dm_G = \rho^2 + \|y\|^2.$$

Every constrained center therefore has worst error at least ρ , while $y = 0$ attains it. Below it (14.6) rules out bounded width. None of the constructions requires transition matrices to satisfy the physical group relations, and extra finitely many commands do not change the upper construction. $\square$

The geometric ingredients in Lemma 14.2 are the usual local geometry of compact homogeneous spaces; related covering estimates are developed in [15]. The additional realization statement is the common stochastic-row construction and its combination with a contraction uniform over conditional-centroid directions. A small-ball estimate restricted to the target orbit alone does not imply Theorem 14.3.

Remark 14.4 (Dependence of constants). The cap and executable-gap constants depend only on the fixed group with its bi-invariant metric, the representation, the uniform cap data (A, p, h_0) , and the declared L^2 gap λ . The hull constants also depend on the target orbit, its second derivatives, and the inradius of Q in its span E . The occupation bound further depends on $\zeta = \rho - \varepsilon$. None depends on the horizon, hidden transition rows, or masses of individual labels. The orbit is the smooth embedded quotient G/G_{z_0} and q is its real dimension. The logarithmic factor is the loss of this mixing-block argument. Theorem 15.5 removes it by a different entropy–transport argument; all statements and proofs of the block estimate remain valid.

15. SPECTRAL ENTROPY AND SHARP OCCUPATION BOUNDS

The pointwise mixing block in Lemma 14.1 costs a logarithm at each selected cut. We now replace that block argument by an entropy estimate along the selected cuts themselves. The auxiliary entropy is attached to a smoothed law of *centroid directions*; it is not the Shannon entropy of the hidden register. The normalization by centroid mass will be essential.

Let E have real dimension $m \geq 2$, let $\Sigma = \{u \in E : \|u\| = 1\}$, and let σ be normalized round measure on Σ . Write d_Σ for its geodesic distance and W_2 for the corresponding quadratic Wasserstein distance. All logarithms in this section are natural. Let $G \subseteq O(E)$ be compact, with normalized Haar measure m_G , and let λ be a probability on the command letters. On $L^2(\Sigma, \sigma)$ define

$$Pf(x) = \sum_a \lambda_a f(R(u_a)^{-1}x), \quad \Pi f(x) = \int_G f(R(g)^{-1}x) dm_G(g).$$

Here R denotes the given action; writing $G \subseteq O(E)$ identifies its image when convenient. The projection Π need not have one-dimensional range. Our spectral hypothesis is

$$(15.1) \quad \|Pf - \Pi f\|_2 \leq \kappa \|f - \Pi f\|_2, \quad 0 \leq \kappa < 1.$$

This is a bound on the whole function space, not merely on the finite-dimensional representation E . The group bound (14.1) implies (15.1), with the same κ : decompose the unitary representation on $L^2(\Sigma)$ into irreducibles; each nontrivial irreducible occurs in the regular representation, where the assumed bound holds (use the adjoint operator if needed for the inverse convention). The trivial summands are exactly the range of Π [13].

Remark 15.1 (The full spectral hypothesis). A contraction on E alone does not imply (15.1). For finitely many circle rotations, simultaneous rational approximation provides arbitrarily high characters on which the averaging eigenvalue is arbitrarily close to one, although its first harmonic may contract. Such an alphabet does not satisfy the full spectral hypothesis. This is why the present theorem does not replace the different Diophantine width laws by a spurious spherical exponent.

15.1. A compactly supported entropy potential. For $0 < h \leq 1$ set

$$(15.2) \quad K_h(u, x) = Z_h^{-1} \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2, \quad Z_h = \int_{\Sigma} \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2 d\sigma(x).$$

The normalizing constant is independent of u . If ν is a probability on Σ , write $K_h\nu(x) = \int K_h(u, x) d\nu(u)$. For a density f put

$$\mathcal{H}(f) = \int f \log f d\sigma, \quad \mathcal{I}(f) = \int_{\{f>0\}} \frac{\|\nabla f\|^2}{f} d\sigma, \quad 0 \log 0 = 0.$$

Lemma 15.2 (Entropy continuity for spherical mixtures). *There are constants $D_m \geq 1$ and $B_m < \infty$, depending only on m , such that every finitely supported probability ν satisfies*

$$(15.3) \quad 0 \leq \mathcal{H}(K_h\nu) \leq L_h := \log(D_m h^{-(m-1)}), \quad \mathcal{I}(K_h\nu) \leq B_m h^{-2}.$$

For any two finitely supported probabilities ν, η ,

$$(15.4) \quad |\mathcal{H}(K_h\nu) - \mathcal{H}(K_h\eta)| \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

For $m = 3$ one can take $D_3 = 12$ and $B_3 = 24$.

Proof. Put $s = \|x - u\|^2$. Polar coordinates give a density proportional to $s^{(m-3)/2}(1 - s/4)^{(m-3)/2}$ on $0 < s < 4$. On $0 < s \leq 1$ its second factor is bounded above and below by positive dimension-dependent constants. Substitution $s = h^2 v$ therefore gives $Z_h \asymp_m h^{m-1}$. Also $\|\nabla s\|^2 = 4s - s^2$, so

$$\mathcal{I}(K_h(u, \cdot)) = \frac{4}{Z_h h^4} \int_{s < h^2} (4s - s^2) d\sigma \leq B_m h^{-2}.$$

Increasing D_m gives $K_h\nu \leq D_m h^{-(m-1)}$. Jensen's inequality gives the lower entropy bound, and the pointwise upper bound gives the upper one. Cauchy-Schwarz in a finite mixture shows that its Fisher information is at most the mixture of the individual Fisher informations.

Choose an optimal finite coupling (π_{ij}) of ν and η . Join its paired centers by shortest constant-speed spherical geodesics. Each path is $u_{ij}(t) = \exp(tA_{ij})u_i$, where A_{ij} is skew-symmetric, acts on a two-dimensional plane, and has operator norm $d_{\Sigma}(u_i, v_j)$. An antipodal pair is treated by choosing any such plane. Define

$$f_t = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), \cdot), \quad J_t(x) = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), x) A_{ij} x.$$

Rotation invariance gives $\partial_t f_t + \operatorname{div} J_t = 0$. Pointwise Cauchy-Schwarz and integration yield

$$\int_{\{f_t>0\}} \frac{\|J_t\|^2}{f_t} d\sigma \leq \sum_{i,j} \pi_{ij} d_{\Sigma}(u_i, v_j)^2 = W_2(\nu, \eta)^2.$$

The preceding Fisher bound holds at every t . To justify entropy differentiation even where f_t vanishes, replace f_t by $(f_t + \delta)/(1 + \delta)$ and J_t by $J_t/(1 + \delta)$. The kernel is C^1 , the sphere is compact, and integration by parts followed by Cauchy-Schwarz gives

$$\left| \frac{d}{dt} \mathcal{H} \left(\frac{f_t + \delta}{1 + \delta} \right) \right| \leq \frac{1}{1 + \delta} \sqrt{\mathcal{I}(f_t)} \left(\int \frac{\|J_t\|^2}{f_t} d\sigma \right)^{1/2} \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

Here quotients at common zeros are zero; the displayed regularized identity can equivalently be obtained by smoothing the C^1 kernel. Integrate in t and let $\delta \downarrow 0$. Uniform boundedness and continuity of $x \log x$ at zero justify the limit and prove (15.4).

On $\Sigma = S^2$, the variable s has constant density $1/4$. Direct integration gives

$$Z_h = h^2/12, \quad \mathcal{I}(K_h(u, \cdot)) = 24/h^2 - 4.$$

These imply the asserted explicit constants. $\square$

The relation between entropy continuity and quadratic transport is classical; compare Polyanskiy–Wu [27, Proposition 1] for regular Euclidean densities. The proof above is included because its densities have compact support and zeros, and because the ambient sphere and the finite-mixture path, rather than Euclidean regularity, are used here.

Lemma 15.3 (Spectral entropy loss on a thin support). *Suppose (15.1) holds. Let f be a bounded probability density supported in a measurable set $A \subseteq \Sigma$ for which*

$$\Pi \mathbf{1}_A \leq \theta < 1.$$

Then

$$(15.5) \quad \mathcal{H}(f) - \mathcal{H}(Pf) \geq (1 - \kappa^2)(1 - \theta).$$

Proof. Orbital Cauchy–Schwarz gives $(\Pi\sqrt{f})^2 \leq (\Pi\mathbf{1}_A)(\Pi f)$ and hence $\|\Pi\sqrt{f}\|_2^2 \leq \theta$. The invariant and orthogonal components are preserved by P . Since $\|\sqrt{f}\|_2 = 1$,

$$\|P\sqrt{f}\|_2^2 \leq \|\Pi\sqrt{f}\|_2^2 + \kappa^2\|(I - \Pi)\sqrt{f}\|_2^2 \leq 1 - (1 - \kappa^2)(1 - \theta).$$

For nonnegative numbers z_a with $z = \sum_a \lambda_a z_a$, the relative-entropy/Hellinger inequality gives

$$\sum_a \lambda_a z_a \log z_a - z \log z \geq \sum_a \lambda_a (\sqrt{z_a} - \sqrt{z})^2.$$

For completeness this follows by summing $t \log(t/s) - t + s \geq (\sqrt{t} - \sqrt{s})^2$; continuity handles zeros. Apply it pointwise to $z_a = f(R(u_a)^{-1}x)$ and integrate. Invariance of σ and Cauchy–Schwarz give

$$\mathcal{H}(f) - \mathcal{H}(Pf) \geq 2 \left(1 - \int \sqrt{Pf} P\sqrt{f} d\sigma \right) \geq 2(1 - \|P\sqrt{f}\|_2).$$

The inequality $2(1 - \sqrt{1-d}) \geq d$ for $0 \leq d \leq 1$ completes the proof. No logarithmic Sobolev inequality or pointwise mixing estimate has been assumed. $\square$

15.2. Transport paid for by loss of centroid mass. For a unit physical random vector X and a finite register S , put

$$(15.6) \quad c_s = \mathbb{E}[X \mid S = s], \quad \alpha = \sum_s \Pr(S = s) \|c_s\|, \quad \nu = \frac{1}{\alpha} \sum_{c_s \neq 0} \Pr(S = s) \|c_s\| \delta_{c_s / \|c_s\|}.$$

Only positive-mass labels contribute. The directional law is defined when $\alpha > 0$; its weights are not, in general, the probabilities of the hidden labels.

Lemma 15.4 (Centroid-loss transport). *Apply an independent letter $a \sim \lambda$, followed by any fixed physical identity-product word, using the actual stochastic rows of the machine. Let α', ν' be the quantities at the endpoint and suppose $\alpha' \geq \zeta > 0$. If $\lambda * \nu$ denotes the mixture of the rotated laws $R(u_a)_* \nu$, then $\delta = \alpha - \alpha' \geq 0$ and*

$$(15.7) \quad W_2(\lambda * \nu, \nu')^2 \leq \frac{3\pi^2}{\zeta} \delta.$$

There is no bound on any intervening register.

Proof. Write $T_a(s, i)$ for the actual composite row, including the fixed filler. Conditional independence of the row from the unrecorded past is precisely the machine definition. Thus

$$p'_i c'_i = \sum_{s,a} p_s \lambda_a T_a(s, i) R(u_a) c_s.$$

Set $w_{sai} = p_s \lambda_a T_a(s, i) \|c_s\|$, $q_i = \sum_{s,a} w_{sai}$, and $m_i = p'_i \|c'_i\|$. For $m_i > 0$ put $v_i = c'_i / \|c'_i\|$; otherwise choose an arbitrary unit vector. With $u_{sa} = R(u_a) c_s / \|c_s\|$ for nonzero terms,

$$(15.8) \quad \sum_{s,a} w_{sai} \|u_{sa} - v_i\|^2 = 2(q_i - m_i).$$

In particular $m_i \leq q_i$, $\sum_i q_i = \alpha$, $\sum_i m_i = \alpha'$, and $\delta \geq 0$.

Normalize the flows by α . They couple $\lambda * \nu$ to $\hat{\nu} = \sum_i (q_i / \alpha) \delta_{v_i}$. Chordal and geodesic distance satisfy $d_\Sigma(u, v) \leq (\pi/2) \|u - v\|$, so this coupling has geodesic quadratic cost at most $\pi^2 \delta / (2\alpha)$. If $\delta > 0$, the excess masses give a probability χ with

$$\hat{\nu} = (\alpha' / \alpha) \nu' + (\delta / \alpha) \chi.$$

Keeping the common part and transporting the remaining part across the sphere costs at most $\pi^2 \delta / \alpha$. The triangle inequality for W_2 and $(a + b)^2 \leq 2a^2 + 2b^2$ give $W_2(\lambda * \nu, \nu')^2 \leq 3\pi^2 \delta / \alpha$. Since $\alpha \geq \alpha' \geq \zeta$, this proves the result. The case $\delta = 0$ follows from the same couplings without an excess measure. Zero centroids and zero-probability labels require no division. $\square$

15.3. The sharp lower law. Assume, for some $A \geq 1$, $p > 0$, and $h_0 > 0$, the all-direction cap bound

$$(15.9) \quad \sup_{u,v \in \Sigma} m_G \{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

This positive-power cap bound excludes every fixed unit vector. Hence $E^G = \{0\}$ and the Haar mean of each orbit is zero; in particular $0 \in \text{conv } R(G)z_0$. The constants are fixed with the representation. For $k \geq 1$ define

$$(15.10) \quad c_0 = \min\{1, h_0, (2A)^{-1/p}\}, \quad h_k = c_0 k^{-1/p}, \quad \gamma = (1 - \kappa^2)/2, \quad L_k^{\text{ent}} = \log(D_m h_k^{-(m-1)}).$$

Theorem 15.5 (Occupation without logarithmic loss). *Let the command alphabet contain a physical identity letter, and assume (15.1) and (15.9). Fix a unit seed z_0 , let $Q = \text{conv } R(G)z_0$, and suppose the target is $\rho R(u_w)z_0$, with $0 < \rho \leq 1$ and legal decoder values in Q . Error is Euclidean error in E . For $0 \leq \varepsilon < \rho$ write $\zeta = \rho - \varepsilon$. Every horizon- N realization, including one redesigned at that horizon, satisfies*

$$(15.11) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq \left\lceil 1 + \frac{2L_k^{\text{ent}}}{\gamma} + \frac{3\pi^2 B_m}{\gamma^2 h_k^2} \frac{1 - \zeta}{\zeta} \right\rceil.$$

In particular, for fixed subcritical parameters there are constants $C_\zeta, c_\zeta > 0$ such that

$$(15.12) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C_\zeta k^{2/p}, \quad W_{N,\varepsilon} \geq c_\zeta N^{p/2}.$$

The intervening widths are unrestricted. The constants depend only on $m, A, p, h_0, \kappa, \zeta$ and the fixed round normalization.

Proof. If there are $J \geq 2$ positive cuts of width at most k , list them as $t_1 < \dots < t_J$. Use a fixed prefix up to t_1 . In each subsequent gap put one independent letter with law λ , then identity letters until the next selected cut. Finish by a fixed suffix. This is an external distribution on actual words of length N , chosen from the advertised width profile, not from the machine's private state. It does not require the machine's rows on an identity letter to be identities.

Let X be the current unit physical vector. Write α_j, ν_j for (15.6) at t_j . Wordwise correctness and $\|D_i\| \leq 1$ imply at the terminal cut

$$\zeta \leq \mathbb{E} \langle X_N, D(S_N) \rangle = \sum_i \Pr(S_N = i) \langle \mathbb{E}[X_N \mid S_N = i], D_i \rangle \leq \alpha_N.$$

The first inequality averages the conditional *mean* error on each deterministic word. It makes no assertion about a single sampled label's error. Conditional Jensen, or the flow calculation above, shows that centroid mass cannot increase along any gap or fixed suffix. Thus $1 \geq \alpha_j \geq \zeta$, and

$$\delta_j := \alpha_j - \alpha_{j+1} \geq 0, \quad \sum_{j < J} \delta_j \leq 1 - \zeta.$$

Fix $h = h_k$ and put $f_j = K_h \nu_j$. Each f_j is supported on at most k open chordal caps of radius h . The cap assumption gives $\Pi \mathbf{1}_{\{f_j > 0\}} \leq A k h^p \leq 1/2$. Lemma 15.3 therefore gives $\mathcal{H}(P f_j) \leq \mathcal{H}(f_j) - \gamma$. Equivariance of the kernel says $P f_j = K_h(\lambda * \nu_j)$. Lemmas 15.2 and 15.4 imply

$$\mathcal{H}(f_{j+1}) \leq \mathcal{H}(f_j) - \gamma + \frac{\pi \sqrt{3 B_m}}{h \sqrt{\zeta}} \sqrt{\delta_j}.$$

Writing $M = J - 1$, summing, using $0 \leq \mathcal{H}(f_j) \leq L_h$, and applying Cauchy–Schwarz only over these M gaps, gives

$$(15.13) \quad \gamma M \leq L_h + \frac{\pi \sqrt{3 B_m}}{h} \sqrt{M \frac{1 - \zeta}{\zeta}}.$$

Young's inequality bounds the second term by $\gamma M/2 + 3\pi^2 B_m(1 - \zeta)/(2\gamma h^2 \zeta)$. This proves (15.11); $J \leq 1$ is immediate. The logarithm in $L_h = O(1 + \log k)$ is additive, not multiplied by $k^{2/p}$. Since $1 + \log k = O_p(k^{2/p})$, the first bound in (15.12) follows. If the whole profile has width at most k , its N positive cuts are all counted, proving the second. $\square$

Remark 15.6 (Endpoint information and synchronization). The unrelaxed inequality (15.13) also holds. When $\zeta = 1$, it gives $J \leq 1 + L_k^{\text{ent}}/\gamma$, consistent with exponential exact width for extreme outputs. For fixed positive error it gives the polynomial law, but does not classify every joint limit $N \rightarrow \infty$, $\varepsilon \downarrow 0$.

If the identity letter is absent but physical identity-product words exist at every length at least g , greedily retain narrow cuts at spacing at least $g + 1$. Each retained cut accounts for at most $g + 1$ original narrow cuts. One independent command followed by a return of the remaining length fits each retained gap. The prefix and suffix may be any fixed words of their required lengths. The same proof multiplies the occupation bound by at most $g + 1$. Approximate returns with accuracy-dependent thresholds are not included in this quantitative extension without additional estimates.

Corollary 15.7 (Matching power laws). *In the setting of Theorem 14.3, replace its converse by (15.12). In particular, if the all-direction exponent p equals the real dimension q of the target orbit, then*

$$W_{N,\varepsilon} = \Theta((N + 1)^{q/2})$$

for every fixed $0 < \rho < 1$, $0 \leq \varepsilon < \rho$, and also for $\rho = 1$, $0 < \varepsilon < 1$. For a general fixed-point-free compact connected Lie representation one may take $p = p_*$ from Theorem 24.1; the lower and upper exponents are then $p_*/2$ and $q/2$, respectively, with no logarithmic loss in the lower bound.

Proof. Apply Theorem 15.5. The upper common-row construction in Theorem 14.3 is exact for $\rho < 1$ and has $O((N + 1)^{q/2})$ labels. At $\rho = 1$ and positive error use its amplitude-slack version. The two bounds match when $p = q$. Theorem 24.1 supplies the remaining assertion. $\square$

The endpoint $\rho = 1, \varepsilon = 0$ is not an assertion of the matching polynomial corollary. No exact leading constant or asymptotic ratio is claimed. The spectral constant is effective only when a certified value is supplied; a fully numerical example is given below.

Remark 15.8 (Return-free extension). Theorem 16.2 below removes the identity-letter and return assumptions from the occupation bound. It uses covariance under the actual deterministic physical fillers, rather than treating approximate returns as exact. The present synchronized theorem is its special case; all smoothing and transport estimates remain unchanged.

16. ENTROPY TRANSPORT WITHOUT A RETURN CONDITION

The entropy potential is invariant under a fixed physical rotation. This allows arbitrary executable words between the spectral steps; they need not return the physical system to its initial state. We first make the covariance explicit and then retain the dependence on the error instead of absorbing it into a fixed-error constant.

Keep the sphere $\Sigma = S(E)$, round probability measure, kernel K_h , and constants D_m, B_m from Lemma 15.2. For an orthogonal map B , write $B_*\nu$ for transport of a measure and $U_B f(x) = f(B^{-1}x)$ for transport of a density. Directly from the radial kernel and invariance of round measure,

$$(16.1) \quad K_h(B_*\nu) = U_B(K_h\nu), \quad H(U_B f) = H(f), \quad W_2(B_*\nu, B_*\eta) = W_2(\nu, \eta).$$

No commutation between B and the averaging operator is asserted.

Lemma 16.1 (Transport after a deterministic physical drift). *In Lemma 15.4, replace the identity-product filler by any fixed executable word, with physical orthogonal action B . If the resulting centroid masses satisfy $\alpha' \geq \zeta > 0$, then $\alpha' \leq \alpha$ and*

$$(16.2) \quad W_2(B_*(\lambda * \nu), \nu')^2 \leq 3\pi^2 \frac{\alpha - \alpha'}{\zeta}.$$

The filler may have any length and its internal registers may have arbitrary widths. A fixed physical word, with no random spectral step, also cannot increase centroid mass.

Proof. Let $T_a(s, i)$ be the actual composite row for the random letter followed by this fixed word. Conditional independence at the incoming register and orthogonality give the exact identity

$$p'_i c'_i = \sum_{s,a} p_s \lambda_a T_a(s, i) B R(u_a) c_s.$$

Use the weights $w_{s,a,i} = p_s \lambda_a T_a(s, i) \|c_s\|$ and unit vectors $BR(u_a) c_s / \|c_s\|$ in the proof of Lemma 15.4. Their weighted sum is the displayed centroid, so its squared-distance identity and both couplings are unchanged. They give (16.2). Terms with zero mass or zero centroid are omitted, exactly as in that lemma. For a fixed word, conditional Jensen and the isometry B give $\sum_i p'_i \|c'_i\| \leq \sum_s p_s \|B c_s\| = \alpha$. $\square$

Let G be a compact connected Lie group with an orthogonal representation $R : G \rightarrow O(E)$, $m = \dim E \geq 2$. Fix a unit vector z_0 whose orbit spans E , and suppose $E^G = \{0\}$. Put $Q = \text{conv}(R(G)z_0)$. The finite nonempty command alphabet need not contain the identity or any exact return. Assume that a probability law λ on the *actual letters* satisfies the full action bound

$$(16.3) \quad \|Pf - \Pi f\|_2 \leq \kappa \|f - \Pi f\|_2, \quad Pf(x) = \sum_a \lambda_a f(R(u_a)^{-1}x), \quad \kappa < 1.$$

Here $\Pi f(x) = \int_G f(R(g)^{-1}x) dg$ projects onto all invariant functions, not merely the constants. Suppose also that

$$(16.4) \quad \sup_{u,v \in \Sigma} \int_G \mathbf{1}_{\{\|R(g)u-v\| < h\}} dg \leq Ah^p \quad (0 < h \leq h_0),$$

where $p > 0$. These are global action and all-direction hypotheses; contraction on E alone is not a substitute for (16.3).

Theorem 16.2 (Return-free sharp occupation and error budget). *Under (16.3)–(16.4), consider the target $F(g) = \rho R(g)z_0$, legal decoders in Q , and Euclidean error $0 \leq \varepsilon < \rho \leq 1$. Set*

$$\zeta = \rho - \varepsilon, \quad c_0 = \min\{1, h_0, (2A)^{-1/p}\}, \quad h = c_0 k^{-1/p}, \quad \gamma = (1 - \kappa^2)/2, \quad L_h = \log(D_m h^{-(m-1)}).$$

For any error- ε clocked realization, let $J = \#\{1 \leq t \leq N : |S_t| \leq k\}$. If $J \geq 1$ and $M = J - 1$, then

$$(16.5) \quad \gamma M \leq L_h + \frac{\pi \sqrt{3B_m}}{h} \sqrt{M \frac{1 - \zeta}{\zeta}}.$$

Consequently, as a real bound before taking an integer ceiling,

$$(16.6) \quad J \leq 1 + \frac{2L_h}{\gamma} + \frac{3\pi^2 B_m}{\gamma^2 h^2} \frac{1 - \zeta}{\zeta}.$$

No condition is imposed on return words or on intervening register widths. Constants other than ζ depend only on m, A, p, h_0, κ and the fixed spherical metric. In particular, for fixed $\zeta > 0$, $J = O_\zeta(k^{2/p})$ and $W_{N,\varepsilon} = \Omega_\zeta(N^{p/2})$.

Proof. If $J \leq 1$, the asserted real bound is immediate. Otherwise list the narrow cuts as $t_1 < \dots < t_J$. Choose an arbitrary deterministic prefix of length t_1 . In each gap put first one independent letter of law λ , and then a fixed word of length $t_{j+1} - t_j - 1$. Such a word exists at every length because the alphabet is nonempty: repeat any one of its letters. Finish with any fixed suffix of length $N - t_J$. This external word law depends on the advertised width profile, not on the machine's private state. Write B_j for the physical action of the fixed filler in gap j . The physical state at every cut remains one unit orbit point; no equality of the B_j to the identity is needed.

Let α_j, ν_j be its conditional-centroid mass and directional law at cut t_j . Terminal correctness gives $\alpha_N \geq \zeta$ by exactly the inner-product argument in Theorem 15.5. This uses wordwise correctness of the mean decoder, not correctness of each label. Lemma 16.1 and the fixed-suffix inequality imply

$$1 \geq \alpha_1 \geq \dots \geq \alpha_J \geq \zeta, \quad \sum_{j=1}^M (\alpha_j - \alpha_{j+1}) \leq 1 - \zeta.$$

There are at most k directions at each selected cut. Thus $f_j = K_h \nu_j$ is supported on at most k open chordal caps, and (16.4) gives $\Pi \mathbf{1}_{\text{supp}_+ f_j} \leq k A h^p \leq 1/2$. The kernel vanishes at each cap boundary. Lemma 15.3 and (16.1) therefore give

$$H(U_{B_j} P f_j) = H(P f_j) \leq H(f_j) - \gamma.$$

By covariance, $U_{B_j} P f_j = K_h(B_{j*}(\lambda * \nu_j))$. Entropy continuity and Lemma 16.1, with $\delta_j = \alpha_j - \alpha_{j+1}$, yield

$$H(f_{j+1}) \leq H(f_j) - \gamma + \frac{\pi \sqrt{3B_m}}{h \sqrt{\zeta}} \sqrt{\delta_j}.$$

Sum, use $0 \leq H(f_j) \leq L_h$, and apply Cauchy–Schwarz to the M nonnegative losses. This proves (16.5). The inequality $ab \leq (a^2 + b^2)/2$, with $a = \sqrt{\gamma M}$, gives (16.6). For a full peak bound k all N positive cuts are counted. Since $\log(k+1) = O(k^{2/p})$, the last conclusions follow. The constants are independent of the filler lengths, their actions, and all actual hidden tables. $\square$

Remark 16.3 (What has, and has not, been removed). The return hypothesis has disappeared from the quantitative converse, not from the general terminal stationarization theorem. The full action gap remains indispensable to this argument. A single irrational rotation letter has a unitary Koopman operator and cannot satisfy (16.3); its one-label clocked realization is therefore no counterexample. No entropy is assigned to the hidden register here: H is relative entropy of

an auxiliary smoothed directional law against normalized round measure. The classical entropy–transport principle [27] supplies neither the Markov row identity nor the width quantifiers. The logarithmic initial range L_h is retained in (16.5); it is not paid again at every narrow cut.

Corollary 16.4 (A fixed executable spectral block). *The same conclusions hold if (16.3) is supplied by a probability law on executable words of one fixed length $b \geq 1$, rather than individual letters. In (16.6) multiply the right side by b .*

Proof. Greedily select narrow cuts at separation at least b . Each selected cut accounts for at most b original cuts, including the last incomplete group of cuts. Apply the proof above with one independent spectral word at the start of every selected gap and an arbitrary fixed filler for its remaining length. There is no restriction before the first selected cut or after the last. The alphabet of spectral blocks is finite even when their internal hidden widths are unbounded. $\square$

Theorem 16.5 (A joint amplitude–accuracy law). *Assume the preceding geometric and action-gap hypotheses. Let $q = \dim R(G)z_0$, and write $W_N(\rho, \varepsilon)$ to display the amplitude dependence. Fix $\zeta_0 > 0$. Uniformly for $0 \leq \varepsilon < \rho \leq 1$ with $\rho - \varepsilon \geq \zeta_0$ and $\tau = 1 - \rho + \varepsilon > 0$, constants C_i depending only on the fixed data and ζ_0 give*

$$(16.7) \quad N \leq C_0 + C_1 \log k + C_2 \tau k^{2/p}, \quad k = W_N(\rho, \varepsilon),$$

$$(16.8) \quad W_N(\rho, \varepsilon) \leq C_3 \left(1 + \frac{N+1}{\tau}\right)^{q/2}.$$

If $p = q$ and a parameter sequence satisfies $\rho_N - \varepsilon_N \geq \zeta_0$ and $\log(1/\tau_N) = o(N)$, then

$$(16.9) \quad W_N(\rho_N, \varepsilon_N) = \Theta\left((N/\tau_N)^{p/2}\right) \quad (N \rightarrow \infty).$$

In particular, at pure amplitude and any positive subexponentially vanishing Euclidean tolerance, the matched law is $\Theta((N/\varepsilon_N)^{p/2})$. It includes polynomial and stretched-exponential accuracy sequences.

Proof. In (16.6), $1 - \zeta = \tau$ and $\zeta^{-1} \leq \zeta_0^{-1}$. Also $L_h = \log(D_m c_0^{-(m-1)}) + (m-1) \log k/p$. Counting all positive cuts proves (16.7).

For the upper bound set $\beta = \rho - \varepsilon/2$. Then $0 < \beta < 1$, $1 - \beta \geq \tau/2$, and the target change has norm at most $\varepsilon/2$. The proof of the upper construction in Theorem 14.3 uses only the invariant orbit hull and Lemma 14.2; it does not use its identity-letter or spectral assumptions. Applied at amplitude β , it gives an exact common-row machine with at most $C(1 + (N+1)/(-\log \beta))^{q/2}$ labels. The inequality $-\log \beta \geq 1 - \beta \geq \tau/2$ proves (16.8). When $\varepsilon = 0$, this is the same exact construction at $\beta = \rho < 1$; no limiting construction at $\beta = 1$ is used.

Let k_N be the minimum width. The uniform upper bound gives $\log k_N = O(1 + \log(N+1) + \log(1/\tau_N)) = o(N)$. Consequently the first two terms of (16.7) sum to at most $N/2$ for all sufficiently large N . It follows that $k_N \geq (N/(2C_2\tau_N))^{p/2}$. The matching upper bound when $p = q$ proves (16.9). The comparison constants depend only on the fixed data and ζ_0 ; the starting horizon can depend on the parameter sequence. $\square$

The least-orbit theorem supplies $p = p_*$ for each fixed-point-free compact connected Lie representation. Thus the joint law applies when $p_* = q$. The point $\tau = 0$ is precisely pure amplitude and zero error; there (16.5) instead gives $\gamma(J-1) \leq L_h$. Neither the joint theorem nor its proof determines optimal constants or the entire exponentially small transition window.

17. POSITIVE MATRIX REALIZATIONS IN EVERY DIMENSION

Let $d \geq 2$ and let $\mathcal{H}_d^0$ be the real Euclidean space of traceless Hermitian $d \times d$ matrices, with scalar product $\langle A, B \rangle = \text{tr}(AB)$ and Frobenius norm $\|\cdot\|_F$. Let

$$\mathcal{D}_d = \{D = D^* \succeq 0 : \text{tr } D = 1\}, \quad c_d = I/d, \quad r_d = \sqrt{1 - 1/d}.$$

This section concerns numerical outputs in $\mathcal{D}_d$. The requirement that every decoder value be positive with trace one is part of the model; it is not imposed merely on the final average.

Lemma 17.1 (Small balls on the complex Grassmannian). *Let P be a fixed rank- m orthogonal projection in $\mathbb{C}^d$, with $1 \leq m < d$. For Haar $U \in \text{SU}(d)$ there is $C_{d,m} < \infty$ such that*

$$\Pr\{\|UPU^* - Q\|_F < h\} \leq C_{d,m} h^{2m(d-m)} \quad (0 < h \leq 1)$$

for every rank- m projection Q . For rank one, writing $\text{dist}_{\text{pr}}(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, one has $\text{dist}_{\text{pr}}(P, Q) = \|P - Q\|_F / \sqrt{2}$ and exactly

$$(17.1) \quad \Pr\{\text{dist}_{\text{pr}}(UPU^*, Q) < h\} = h^{2(d-1)} \quad (0 \leq h \leq 1).$$

Proof. Conjugation is transitive, so fix $Q = \text{diag}(I_m, 0)$. A projection sufficiently close to Q has range the graph of a unique matrix $Z \in \mathbb{C}^{(d-m) \times m}$, with

$$P(Z) = \begin{pmatrix} I \\ Z \end{pmatrix} (I + Z^* Z)^{-1} \begin{pmatrix} I & Z^* \end{pmatrix}.$$

This notation means the product of the three displayed matrices. The map and its inverse are smooth near zero, and its derivative at zero is $Z \mapsto \begin{pmatrix} 0 & Z^* \\ Z & 0 \end{pmatrix}$, of Frobenius norm $\sqrt{2}\|Z\|_F$. The invariant probability measure is normalized Riemannian volume on this compact homogeneous manifold; in this chart its density is smooth and bounded on a sufficiently small closed neighborhood. The inverse chart puts a Frobenius h -ball inside a Euclidean Ch -ball in real dimension $2m(d-m)$. This proves the bound for small h ; increasing the constant covers the remaining interval. Transitivity makes the constant independent of Q .

For rank one choose $Q = e_1 e_1^*$. A uniform unit vector can be obtained by normalizing independent standard complex Gaussian coordinates. If X is the squared modulus of the first coordinate before normalization and Y is the sum of the other squared moduli, then X is exponential of mean one and Y is an independent sum of $d-1$ such variables. For $0 < z < 1$,

$$\Pr\left\{\frac{X}{X+Y} > 1-z\right\} = \mathbb{E} \exp\left(-\frac{1-z}{z} Y\right) = z^{d-1}.$$

Put $z = h^2$. The endpoint cases follow by continuity. This proves (17.1), and also shows that dist_{pr} is the Frobenius chordal distance divided by $\sqrt{2}$. $\square$

Lemma 17.2 (A uniform conjugacy-orbit exponent). *There is $C_d < \infty$ such that*

$$(17.2) \quad \sup_{\substack{A, B \in \mathcal{H}_d^0 \\ \|A\|_F = \|B\|_F = 1}} m_G\{U : \|UAU^* - B\|_F < h\} \leq C_d h^{2(d-1)} \quad (0 < h \leq 1).$$

The exponent cannot be increased uniformly over A and B .

Proof. Write the eigenvalues of A as $a_1 \geq \dots \geq a_d$. Since their sum is zero and their squared sum is one, $a_1 - a_d \geq d^{-1/2}$. Some adjacent gap therefore satisfies

$$(17.3) \quad a_m - a_{m+1} \geq \gamma_d := \frac{1}{\sqrt{d}(d-1)}.$$

Let P_A be the spectral projection onto the first m eigenvectors. It is well defined despite possible multiplicities within either block.

Two conjugates A', A'' of A , with eigenbases x_i, y_j in this order, obey the exact identity

$$(17.4) \quad \|A' - A''\|_F^2 = \sum_{i,j} (a_i - a_j)^2 |\langle x_i, y_j \rangle|^2.$$

Expanding both traces proves it. The cross-block terms and (17.3) imply

$$(17.5) \quad \|P_{A'} - P_{A''}\|_F \leq \gamma_d^{-1} \|A' - A''\|_F,$$

since the squared norm on the left is exactly the sum of the two cross-block overlap sums.

If the set in (17.2) is nonempty, choose one member U_0 . Every other member has $\|UAU^* - U_0AU_0^*\|_F < 2h$. Thus its rank- m spectral projection lies within $2h/\gamma_d$ of $U_0P_AU_0^*$. The Haar law of UP_AU^* is the invariant Grassmannian law. Lemma 17.1 bounds the measure by $C_{d,m}(2h/\gamma_d)^{2m(d-m)}$ for $h \leq \gamma_d/2$. Since $m(d-m) \geq d-1$, and there are finitely many m , this gives (17.2) with a constant independent of all eigenvalue multiplicities. Larger h are covered by increasing C_d .

For sharpness take $A = B = (P - c_d)/r_d$ with P rank one. By (17.1), for sufficiently small h the probability is

$$(r_d^2 h^2 / 2)^{d-1}.$$

It cannot be bounded by a constant times $h^{p'}$ as $h \downarrow 0$ for any $p' > 2(d-1)$. $\square$

The spectral split, rather than an assumption that all centroids remain rank one, supplies the uniformity needed by the stochastic contraction. The conjugation representation has ambient dimension $d^2 - 1$, but its worst small-ball exponent is $2(d-1)$.

Theorem 17.3 (Unitary matrix-orbit width). *Let a finite alphabet in $SU(d)$ contain the identity and support a probability law satisfying (14.1). Fix a rank-one projection P_0 . For $0 < a \leq 1$, set*

$$(17.6) \quad F_a(U) = c_d + a(UP_0U^* - c_d), \quad Y = \mathcal{D}_d,$$

with Frobenius error. Its exact bounded-width threshold is ar_d , attained by the one-label decoder c_d . For $0 \leq \varepsilon < ar_d$, put $\zeta = a - \varepsilon/r_d$. Every realization satisfies

$$(17.7) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \left(1 + \log(k+1) + \frac{1-\zeta}{\zeta} k^{1/(d-1)} \right).$$

For fixed $0 < a < 1$ and $0 \leq \varepsilon < ar_d$,

$$(17.8) \quad cN^{d-1} \leq W_{N,\varepsilon} \leq C(N+1)^{d-1},$$

and the upper realization can be exact. The same orders hold for $a = 1$ and every fixed $0 < \varepsilon < r_d$.

More explicitly, for $a < 1$ and $\varepsilon = 0$ set $a' = a$; for $\varepsilon > 0$ set $a' = a - \varepsilon/(2r_d)$. Whenever $a' < 1$, an error at most $\varepsilon/2$ realization has

$$(17.9) \quad K \leq C_d \left(1 + \frac{N+1}{-\log a'} \right)^{d-1}$$

labels. In the zero-error case this realization is exact. In (17.7) the constant depends only on d and the spectral-gap data, not on the amplitude or tolerance. Extra finitely many unitary commands are allowed in both bounds.

Proof. Center at c_d and divide all outputs by r_d . The resulting legal set is

$$Q = \text{conv}\{(P - c_d)/r_d : P \text{ rank one}\} \subseteq \mathbb{B}_{\mathcal{H}_d^0}.$$

The last inclusion follows from $\text{tr } D^2 \leq 1$ for $D \in \mathcal{D}_d$. The orbit spans $\mathcal{H}_d^0$: if a traceless Hermitian A has $\text{tr}(AP) = 0$ for every rank-one P , then $v^*Av = 0$ for every v , hence $A = 0$. Conjugation-fixed matrices are scalar, so there are no fixed vectors in this traceless space. The orbit is complex projective space, of real dimension $2(d-1)$. Lemma 17.2 and Theorem 15.5, with $p = q = 2(d-1)$,

prove the lower and occupation bounds, as well as the radius statement after multiplying the norm by r_d .

We give a direct inner approximation to make positivity of the upper rows transparent. Choose a maximal δ -separated family of rank-one projections P_i for dist_{pr} , containing P_0 . Its $\delta/2$ balls are disjoint, so (17.1) gives

$$K \leq (2/\delta)^{2(d-1)}.$$

For any traceless Hermitian A , put $M = \lambda_{\max}(A)$. When $A \neq 0$, $M > 0$ and $\lambda_{\min}(A) \geq -(d-1)M$. Approximate a top eigenline by P_i at distance at most δ . Spectral decomposition then gives

$$(17.10) \quad \text{tr}(AP_i) \geq M - (M - \lambda_{\min}(A))\delta^2 \geq (1 - d\delta^2)M.$$

The trivial $A = 0$ case is automatic. Comparing support functions in $\mathcal{H}_d^0$, with $K_0 = \mathcal{D}_d - c_d$, yields

$$(1 - d\delta^2)K_0 \subseteq \text{conv}\{P_i - c_d\} \subseteq K_0.$$

Put $r = 1 - d\delta^2 > 0$. For each actual command U_b choose one row of barycentric coordinates satisfying

$$(17.11) \quad \sum_j T_b(i, j)(P_j - c_d) = rU_b(P_i - c_d)U_b^*.$$

Rows are nonnegative and sum to one by construction. Initialize at P_0 and decode label i as

$$D_i = c_d + (a'/r^N)(P_i - c_d).$$

This is in $\mathcal{D}_d$ when $r^N \geq a'$: it is a convex combination of c_d and P_i . Choosing

$$\delta^2 = \min\left(\frac{1}{4d}, \frac{-\log a'}{2d(N+1)}\right)$$

ensures this inequality. Induction in (17.11) gives mean output exactly $F_{a'}(u_w)$. The target discrepancy is $|a - a'|r_d = \varepsilon/2$ when $\varepsilon > 0$. The cardinal bound proves (17.9). The case $a = 1, \varepsilon = 0$ is deliberately excluded from this dilation construction and is treated exactly in the next section. $\square$

The family contains the qubit sphere as $d = 2$, but for $d \geq 3$ the pure-state orbit is a proper curved subset of the unit sphere in $\mathcal{H}_d^0$. Thus (17.8) does not follow by substituting the ambient dimension into the spherical theorem. The entropy argument removes the logarithmic loss. No optimal leading constant, exact asymptotic ratio, or dimension-uniform constant is claimed.

Corollary 17.4 (Algebraic generating alphabets). *Theorem 17.3 applies to the identity together with any finite symmetric algebraic-entry set generating a dense subgroup of $\text{SU}(d)$, equipped with a lazy symmetric law. Such a finite alphabet exists for every $d \geq 2$.*

Proof. The spectral gap is the imported theorem of Bourgain–Gamburd [16]. Laziness makes its one-sided gap an absolute L^2 contraction on mean-zero functions. Existence can be seen by placing two algebraic irrational-angle $\text{SU}(2)$ rotations in every adjacent pair of coordinate directions. For example, use the quaternion lifts in the explicit adjacent-block construction below within each pair. Each pair generates its full $\text{SU}(2)$ subgroup. Their Lie algebras contain the adjacent off-diagonal matrix units and their brackets, which generate $\mathfrak{su}(d)$. The connected compact closure is consequently $\text{SU}(d)$. All entries remain algebraic. The spectral-gap constant is not computed by the finite regression checks. $\square$

17.1. A finite whole-law encoding. Let $q_0 = d^2 - 1$ and choose a Frobenius-orthonormal basis $H_1, \dots, H_{q_0}$ of $\mathcal{H}_d^0$. Since $\|H_l\|_{\text{op}} \leq 1$, the $2q_0$ matrices

$$E_{l,\pm} = \frac{I \pm H_l/2}{2q_0}$$

are positive and sum to I . Put $\Phi(D)_{l,\pm} = \text{tr}(E_{l,\pm}D)$ and $Y_\Phi = \Phi(\mathcal{D}_d)$. Every entry is at least $1/(4q_0)$. The map is injective, because

$$(17.12) \quad D - c_d = \sum_l 2q_0(\Phi(D)_{l,+} - \Phi(D)_{l,-})H_l.$$

On the vector space parallel to the affine span of Y_Φ , use the norm of the reconstructed matrix on the right. Then Φ is an affine isometry, and replacing each legal matrix decoder by its image preserves every clocked or stationary width exactly. This is one whole categorical law with a compact convex legality constraint, not separate binary event tolerances. For ordinary total variation the explicit norm comparison is

$$\frac{\|D - D'\|_{\text{F}}}{4q_0} \leq \text{TV}(\Phi(D), \Phi(D')) \leq \frac{\|D - D'\|_{\text{F}}}{4\sqrt{q_0}}.$$

It follows directly by comparing the ℓ^1 and ℓ^2 norms of the basis coefficients. Enlarging the legal output set from Y_Φ to the entire simplex is a different realization problem. In particular the extreme-output argument below cannot be transferred to that enlarged set: its target vectors are interior points of the simplex.

Remark 17.5 (An explicit algebraic alphabet in every dimension). In each adjacent coordinate plane $(j, j+1)$ embed the two matrices

$$\frac{1}{5} \begin{pmatrix} -3+4i & 0 \\ 0 & -3-4i \end{pmatrix}, \quad \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad 1 \leq j < d,$$

and act identically on all other coordinates. Include their inverses and the identity. Each pair has infinite-order eigenvalues, since its trace $-6/5$ is not an integer, and its two torus closures generate the adjacent $SU(2)$. The adjacent skew-Hermitian matrix units and their brackets span $\mathfrak{su}(d)$. The closed generated subgroup is therefore $SU(d)$. Bourgain–Gamburd is applied to this actual finite algebraic set, not merely to the existence of an unspecified set. The resulting spectral-gap constant is not numerically evaluated here.

The Frobenius chordal distance between rank-one projections is $\sqrt{2}$ times the projective distance used in the net construction. In the conjugacy-cap comparison, choosing one member of a nonempty target ball avoids imposing any spectral condition on its arbitrary center. All adjacent-gap and cap constants here depend on d . The amplitude $a' = a - \varepsilon/(2r_d)$ in the positive-error upper bound deliberately spends only half the available error.

18. EXTREME OUTPUTS AND THE EXACT ENDPOINT

The strict amplitude slack in the polytope construction is not a technical convenience that can simply be removed. At an extreme output, convex averaging has no room to hide a different physical state. This observation gives an exact finite-horizon formula, independently of spectral gap or synchronization.

Theorem 18.1 (The extreme-orbit formula). *Let a group act by affine bijections on a compact convex set Y . Let $x_0 \in Y$ have an orbit consisting of extreme points of Y , and let a finite nonempty command alphabet act through this group. There is one seed and one terminal query, with target $u_w x_0$. Put*

$$\mathcal{O}_t = \{u_w x_0 : |w| = t\}, \quad n_t = |\mathcal{O}_t|.$$

Every exact clocked stochastic realization at horizon N satisfies

$$(18.1) \quad |S_t| \geq n_t \quad (0 \leq t \leq N).$$

One deterministic realization attains all these inequalities simultaneously. Consequently

$$(18.2) \quad W_{N,0} = n_N.$$

Neither an identity letter nor an exact return is required.

Proof. Fix t and one suffix v of length $N - t$. Such a suffix exists because the alphabet is nonempty; at $t = N$ use the empty word. Let $D_v(i) \in Y$ be the conditional expected terminal decoder after this fixed suffix, when the cut- t label is i . Convexity guarantees that it lies in Y . For every prefix w of length t , exact correctness says

$$u_v u_w x_0 = \sum_i p_w(i) D_v(i),$$

where p_w is the actual distribution at that cut. Extremality forces $D_v(i) = u_v u_w x_0$ whenever $p_w(i) > 0$. Since the action of u_v is injective, prefixes with different physical outputs have disjoint nonempty supports in S_t . This proves (18.1), including the terminal cut.

Conversely, at cut t store the current point of $\mathcal{O}_t$. A command sends this point to its image in $\mathcal{O}_{t+1}$; the terminal decoder is the stored point. The initialization is deterministic. Finally, any one fixed command gives an injection $\mathcal{O}_t \rightarrow \mathcal{O}_{t+1}$, so n_t is nondecreasing. The peak of this realization is n_N , proving (18.2). $\square$

The same conclusion holds for an orbit on a Euclidean unit sphere with legal output set its convex hull: every orbit point is extreme by strict convexity of the ambient Euclidean ball. For density matrices, a rank-one projection is extreme. Indeed, if $P = \sum_i p_i D_i$ with $D_i \geq 0$ and $\text{tr } D_i = 1$, then $\text{tr}((I - P)D_i) = 0$ for every positive weight. Positivity forces D_i to be supported on the range of P , and hence $D_i = P$.

The cut argument fixes one suffix, whose affine action is injective. The cardinalities are nondecreasing because the nonempty alphabet contains a letter whose injective action maps each reachable set into the next one. No sampled-answer floor is included in this width.

19. EFFECTIVE ALGEBRAIC-CIRCLE BOUNDS

For algebraic unit-circle eigenvalues of infinite order, an explicit separation estimate replaces the Diophantine hypothesis of the preceding section. The parameters r, m, A_r below are those of (12.3)–(12.4); in this section r denotes the localization order, not the number of command rotations. The alphabet contains I, R_θ , and one interface mean is $\rho \cos \phi$.

Lemma 19.1 (Separated powers give a block defect). *Suppose $|\lambda^n - 1| \geq \sigma$ for every integer $1 \leq n \leq 2mk$, where $0 < \sigma \leq 1$. Write $R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt})$ for the representation acting on Z_m . On this representation, the uniform law on the $2k$ words with products R_θ^j , $0 \leq j < 2k$, each padded to length $2k$, has defect at least $\sigma^2/16$ in (7.4).*

Proof. Write a unit vector as $u = (u_1, \dots, u_m)$ in its two-dimensional frequency blocks. For distinct indices $i, j \in \{0, \dots, 2k - 1\}$,

$$\|R_m(i\theta)u - R_m(j\theta)u\|^2 = \sum_{l=1}^m \|u_l\|^2 |\lambda^{l(i-j)} - 1|^2 \geq \sigma^2.$$

A ball of radius strictly less than $\sigma/2$ about any of k centers contains at most one of these $2k$ points. Thus at least k points have distance at least $\sigma/2$ from all centers (the same conclusion follows by taking the open balls of radius $\sigma/2$). For unit centers the defect is one half of the squared distance to the closest center. Averaging gives at least $(1/2)(\sigma^2/8) = \sigma^2/16$. $\square$

Theorem 19.2 (Full-noise algebraic-circle bound). *Assume λ is algebraic and not a root of unity. Let its primitive irreducible integer polynomial be*

$$P(z) = az^d + a_{d-1}z^{d-1} + \cdots + a_0, \quad a > 0.$$

Set $H_P = \max(a, |a_{d-1}|, \dots, |a_0|)$ and

$$B_P = a(1 + H_P/a)^{d-1} > 1.$$

With r, m, A_r as in (12.3)–(12.4), every error- ε realization of peak at most k satisfies

$$(19.1) \quad N < 2k \left(1 + 16 \cdot 2^{2(d-1)} B_P^{4mk} \log A_r \right).$$

Its number of command cuts $t \in \{1, \dots, N\}$ with width at most k , even if other widths are unrestricted, is at most

$$(19.2) \quad 2k - 1 + 32k \cdot 2^{2(d-1)} B_P^{4mk} \log A_r.$$

Consequently, for fixed algebraic data, ρ , and $\varepsilon < \rho/2$,

$$(19.3) \quad W_{N,\varepsilon} \geq \frac{\log N - O(\log \log N)}{4m \log B_P}.$$

All constants in the finite inequalities are explicit. If $\lambda = (a_1 + ib_1)/c$ with integers $a_1, b_1, c > 0$, $a_1^2 + b_1^2 = c^2$, and λ not a root of unity, then (19.1)–(19.3) hold with the factor $2^{2(d-1)}$ deleted and B_P replaced by c .

Proof. List the conjugates with $\lambda_1 = \lambda$. They are distinct by irreducibility in characteristic zero. None is a root of unity: otherwise irreducibility would make P a cyclotomic polynomial up to its primitive sign. The resultant

$$\text{Res}(P, z^n - 1) = a^n \prod_{j=1}^d (\lambda_j^n - 1)$$

is therefore a nonzero integer. Each conjugate has modulus at most $R_P = 1 + H_P/a$. For completeness, if $|z| > R_P$, then

$$\sum_{j=0}^{d-1} |a_j| |z|^j < \frac{H_P |z|^d}{|z| - 1} < a |z|^d,$$

so $P(z) \neq 0$. Using $|z^n - 1| \leq 2 \max(1, |z|)^n$ for each other conjugate, the resultant bound gives

$$(19.4) \quad |\lambda^n - 1| \geq 2^{-(d-1)} \left(a \prod_{j=2}^d \max(1, |\lambda_j|) \right)^{-n} \geq 2^{-(d-1)} B_P^{-n}.$$

For $1 \leq n \leq 2mk$, take $\sigma = 2^{-(d-1)} B_P^{-2mk}$ in Lemma 19.1. The block defect is at least

$$d_k = \frac{2^{-2(d-1)} B_P^{-4mk}}{16}.$$

Repeated conditional contraction and Lemma 12.3 imply

$$\left\lfloor \frac{N}{2k} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq 16 \cdot 2^{2(d-1)} B_P^{4mk} \log A_r.$$

This proves (19.1). The greedy occupation argument in the proof of Theorem 7.2, with $b = 0$ and block length $2k$, proves (19.2). Taking logarithms gives (19.3): fix $C > 1/(4m \log B_P)$. If $k \geq C \log N$ the claim is immediate; otherwise $\log k \leq \log C + \log \log N$, which can be absorbed in the logarithm of (19.1). The implicit constant may depend on P, ρ, ε , not on N .

In the rational-circle case, $(a_1 + ib_1)^n - c^n$ is a nonzero Gaussian integer, of modulus at least one. Thus $|\lambda^n - 1| \geq c^{-n}$ directly. Repeat the proof with $\sigma = c^{-2mk}$. Here $c > 1$ automatically. A nonminimal supplied denominator is still valid but weakens the bound. $\square$

The threshold is sharp; these width rates are not asserted sharp. The factor m records the increased localization degree as the error approaches the boundary. At smaller errors, the retained coordinate-calibration estimates can have better constants. One should use the larger of applicable lower bounds, rather than claiming numerical improvement at every parameter value.

Proposition 19.3 (An explicit nonrational algebraic example). *Let $t = (1 - \sqrt{13})/2$ and*

$$\lambda = \frac{t + i\sqrt{4 - t^2}}{2}.$$

Then $|\lambda| = 1$, λ is not a root of unity, and one can take $d = 4$ and $B_P = 8$ in Theorem 19.2.

Proof. The number t lies strictly between -2 and 2 and solves $t^2 - t - 3 = 0$. Substitution of $t = z + z^{-1}$ gives

$$P(z) = z^4 - z^3 - z^2 - z + 1.$$

Its reduction modulo two is $z^4 + z^3 + z^2 + z + 1$. It has no linear factor over $\mathbb{F}_2$ and is not divisible by the only monic irreducible quadratic $z^2 + z + 1$; hence it is irreducible. Gauss's lemma proves irreducibility over $\mathbb{Q}$. The conjugate value $(1 + \sqrt{13})/2 > 2$ gives a positive real conjugate of λ greater than one. A root of unity cannot have such a conjugate. Finally $a = H_P = 1$ and $d = 4$, so $B_P = 2^3 = 8$. $\square$

For a reproducible rational calibration, take $\rho = 1/10$ and $\varepsilon = 1/25$. Then $\delta = 2/25$, $r = 5$, $m = 6$, $\Delta_r = 1/150$, and $A_r = 23364$. Thus the rational-circle alphabet above obeys

$$N < 2k(1 + 16 \cdot 5^{24k} \log 23364).$$

This is a concrete bound inside the interval between the former small-error obstruction and the sharp threshold $1/20$. The exact coefficient and resultant checks accompanying the article verify the displayed arithmetic, not the universal theorem by finite experimentation.

20. FINITE ENCODINGS AND BOUNDARY QUOTIENTS

The general continuous theorem is not an algorithm from arbitrary real data. We now give an explicit finite input class for which its conclusions are effective. A rational orthogonal input consists of a finite alphabet in $O(D, \mathbb{Q})$ containing I , finite seed and query sets, and rational polynomials $F_{s,j,l}$ in the matrix entries specifying categorical probabilities. The promise $F_{s,j}(H) \subset \Delta_{M_j}$ can itself be checked once the compact closure is computed. Real algebraic output numbers are encoded by polynomial equations and isolating data.

We use two established algorithmic inputs: computation of a finite defining description of the Zariski closure of a finitely generated rational matrix group [21, 23], and real quantifier elimination with effective semialgebraic connected components [19]. The former is a terminating closure algorithm, not merely an enumeration of valid polynomial identities with an unknown stopping point. Compact orthogonal groups are real algebraic, so the real points of the computed closure in $O(D)$ give the Euclidean closure H [20]. The complex algebraic components must not be substituted for the real connected components used here.

Theorem 20.1 (Effective rational classification). *For a rational orthogonal input, the following are computable:*

- (1) *a semialgebraic description of every connected component of H , the finite group C , and the component of each command;*

- (2) the critical total-variation error ε_c as an exact real algebraic number, and an attaining stationary component-permutation machine with algebraic decoders;
- (3) for each algebraic $0 \leq \varepsilon < \varepsilon_c$ and integer $k \geq 1$, an integer occupation constant L_k valid for every error- ε clocked realization;
- (4) for every fixed N and algebraic $\varepsilon \geq 0$, the least width $W_{N,\varepsilon}$ and an algebraic realization attaining it.

These are terminating computability statements. No polynomial running-time bound, practical implementation of general closure/quantifier elimination, or finite-random-bit cost bound is asserted.

Proof. Compute H using the cited closure algorithm, then its real connected components and algebraic sample points. The component containing I is K . Products of sample points and semialgebraic membership tests determine the component multiplication table and the command images. In particular K is now an effectively given compact semialgebraic group.

For a component c and a query, the condition that q be a radius- t center is the real first-order formula

$$(20.1) \quad q \in \Delta_{M_j}, \quad \forall h \in c : \sum_{l=1}^{M_j} |F_{s,j,l}(h) - q_l| \leq 2t.$$

Absolute values have a finite semialgebraic description. Quantifier elimination gives the set of admissible (q, t) . Its least t exists by compactness and is real algebraic; algebraic sampling supplies an attaining q . Maximize over the finite interface and components. This proves (1)–(2), including equality at the critical error.

We detail effectivity of the occupation proof, since computing the threshold alone would not supply it. Choose a component with radius greater than the given ε , and enumerate positive words until one lies in that component. Its rational product gives the prefix used in the proof of Theorem 7.2. The resulting map $f(h) = F_{s,j}(hu_v)$ is polynomial, so it can be used directly in place of G , with no approximation error.

For any finite tensor degree, all required polynomial functions are coefficients of the explicitly given tensor representation. Its fixed subspace is

$$\{z : \forall h \in K, (R(h) - I)z = 0\}.$$

This linear subspace is effectively semialgebraic. Real algebraic sampling, successively outside the span of the vectors already chosen, gives an algebraic basis in finitely many steps. Its Gram matrix gives the orthogonal projection P exactly. Thus every polynomial Haar moment is computable algebraically from $P\Phi(e)$; numerical integration of Haar measure is unnecessary.

Enumerate finite lists of squares of rational polynomials on K , discarding zero Haar integrals and normalizing the others. For a list p_i , compute $b_i = \int p_i f d\mu$ and its constrained simplex radius by quantifier elimination. Rational polynomials are uniformly dense among real polynomial restrictions on a compact matrix group; the localization proof therefore guarantees that some enumerated list has radius strictly greater than ε . Strict algebraic comparison detects that list. Fix a positive rational Δ below the radius difference and a positive rational upper bound L for all coefficient constants in Proposition 8.2. These give the certified residual $Q \geq \Delta/L$.

Next enumerate finite rational probability laws on executable words of a common length whose products lie in K , together with rational $d \in (0, 1)$. The assertion (7.4), quantified over the finitely many unit vectors in the algebraic moving space, is semialgebraic: each maximum of finitely many inner products can be expressed by a finite disjunction. It is therefore decidable. Lemma 7.4 and sufficiently small rational perturbations of its atom weights ensure that this search terminates. Padding by the identity letter makes every finite choice executable at one common length.

The resulting B, d, L, Δ, Q_0 determine (8.6). To avoid exact comparisons involving logarithms, take a rational $A' \geq \max(1, LQ_0/\Delta)$ and use

$$\frac{\log A}{-\log(1-d)} \leq \frac{A'}{d}.$$

An integer above $b + B - 1 + BA'/d$ is a computable occupation bound. This proves (3).

For (4), fix a candidate width k and use k labels at every cut, padding smaller registers with unused labels. Stochastic rows, initialization, and categorical decoder probabilities are finitely many variables in compact simplexes. For each of the finitely many length- N words, the output vector is polynomial in these variables, and the target vector is rational. The total-variation inequalities are semialgebraic. Quantifier elimination therefore decides feasibility and supplies an algebraic point when feasible. A deterministic register storing the seed and entire prefix is a finite exact realization, so increasing k must terminate. This proves the least-width claim for each finite horizon, without claiming a finite-horizon cutoff for all stationary or nonuniform realization questions. $\square$

The distinction between an effective theorem and an implemented solver matters. The accompanying finite checks verify displayed exact examples and proof constants; they do not execute the general group-closure and quantifier-elimination construction. The finite-bit compiler for rational finite groups remains in the complete predecessor supplement with its separate hypotheses and costs. It is not used to reprice arbitrary algebraic decoders as cost-free finite-bit procedures.

20.1. The least component-quotient realization. The upper bound $|\mathcal{S}||C|$ can often be reduced. Here is an exact finite characterization of that reduction, with its scope stated explicitly. Put $X = \mathcal{S} \times C$. A *component-quotient realization* has a deterministic state of the form $\pi(s, c)$, where $\pi : X \rightarrow Q$ is a surjection; its command update is induced by $(s, c) \mapsto (s, [u_a]c)$. It is stationary and has no additional state recording a representative word within a component.

Call a partition $\mathcal{B}$ of X equivariant if

$$(s, c) \sim (s', c') \implies (s, dc) \sim (s', dc') \quad (d \in C).$$

For a block $B \in \mathcal{B}$ and query j , set

$$A_{B,j} = \bigcup_{(s,c) \in B} F_{s,j}(c).$$

Theorem 20.2 (Finite quotient minimum). *The least number of command labels in an error- ε component-quotient realization is*

$$(20.2) \quad \min \left\{ |\mathcal{B}| : \mathcal{B} \text{ equivariant and } \text{rad}_{Y_j}(A_{B,j}) \leq \varepsilon \text{ for every } B, j \right\},$$

with value $+\infty$ if there is no admissible partition. For rational orthogonal categorical inputs this minimum and an attaining quotient are computable. If $\varepsilon_c = 0$ and $\varepsilon = 0$, the coarsest admissible partition is the future-output equivalence

$$(s, c) \sim (s', c') \iff F_{s,j}(dc) = F_{s',j}(dc') \quad \text{for every } d \in C, j,$$

where componentwise constant maps are identified with their values.

Proof. An equivariant partition induces well-defined command permutations on its blocks. Choosing a minimizing center for each $A_{B,j}$ gives exactly the stated error. Conversely, the fibers of π in any component-quotient realization form an equivariant partition. Its decoder must approximate every executable product in each corresponding component. Such products are dense in that component, so continuity extends the error bound to $A_{B,j}$. Its decoder is therefore a permitted center. This proves (20.2). The command generators suffice to check equivariance, since their images generate the finite group C .

There are finitely many partitions of X . For polynomial categorical data their radius tests are of the form (20.1), with a finite disjunction over the relevant seed/component pairs, so the minimum is computable. In the exact componentwise constant case, equality of all future outputs is an equivariant equivalence relation. Every admissible partition refines it, and its own blocks have singleton output images for each query. It is therefore the unique coarsest admissible partition. $\square$

The minimization in (20.2) is a deterministic quotient minimum, distinct from the unrestricted stationary minimum in Theorem 5.4 and from the unrestricted nonuniform minimum at the boundary. Randomized encodings can identify numerical behaviors without identifying component pairs. What is classified here is exactly the finite deterministic quotient construction that attains the radius threshold. Theorem 20.1(4) separately computes the unrestricted minimum at each fixed horizon.

20.2. Stationary minimization by permutation closures.

Theorem 20.3 (Effective unrestricted stationary minimum). *For rational orthogonal commands generating H , rational polynomial categorical outputs, and algebraic $\varepsilon \geq 0$, one can compute M_ε and an attaining algebraic machine when it is finite. An identity letter is not required. At each proposed width k , at most $(k!)^{|A|}$ augmented orthogonal group closures and finite real quantifier-elimination problems suffice.*

Proof. First compute H , its real connected components, and their constrained radii as in Theorem 20.1(1)–(2); those steps do not use an identity letter. The group $H \subset O(D)$ has finitely many components. Theorem 5.5 shows that $M_\varepsilon = \infty$ below their maximum radius. Otherwise $M_\varepsilon \leq |\mathcal{S}||H/H^\circ|$.

For k between one and this upper bound enumerate all command-permutation tuples σ . The matrices

$$\text{diag}(u_a^{-1}, \Pi_a) \in O(D + k, \mathbb{Q})$$

generate a compact closure which is exactly the joint group $\mathcal{L}_\sigma$ of Theorem 5.4. Compute it by the same algebraic closure procedure. In (5.2), the unknown seed rows and decoder rows are constrained to simplexes. The mean output is bilinear in these rows, the group variable is restricted by computed defining equations whose real zero set inside the augmented orthogonal group is exactly $\mathcal{L}_\sigma$, not merely by some equations vanishing on it, and $h^{-1} = h^T$ in the physical block. Consequently all target coordinates are polynomial in the universally quantified matrix entries. Total variation is semialgebraic. Real quantifier elimination decides the resulting existential–universal formula and supplies algebraic rows whenever it is feasible. The first feasible k is the minimum by purification. The finite upper bound proves termination of the entire search. $\square$

This is a structural reduction of stationary minimization, not merely an enumeration of arbitrary stochastic transition parameters. It does not equate stationary and finite-horizon minima. Theorem 4.2 identifies their limit under its additional return hypothesis. The two-cycle/three-cycle example in Proposition 5.6 already prevents replacing the permutation-feasibility test by deterministic component partitions.

20.3. Algorithmic cost and arithmetic representation. There are three distinct layers of effectivity. First, the group-closure procedure produces explicit defining equations; real-component decomposition and algebraic sampling then identify the component table. Second, radius computation and the stationary permutation tests are finite quantifier-elimination tasks on those equations. At fixed k , the stationary seed and output variables number $k(|\mathcal{S}| + \sum_j M_j)$ before normalization equalities are eliminated; the group variables can be taken among the $(D + k)^2$ augmented matrix entries. Thus the displayed finite search is independent of a horizon parameter. Its factorial number of discrete cases and the cost of the real-algebraic subproblems should not be

mistaken for an efficient algorithm. No bound for the full closure-to-machine process in terms of only the original bit length is extracted here.

Third, occupation certificates in Theorem 20.1(3) add searches over localizer degrees and executable word laws. Their termination follows from strict analytic margins; the present proof supplies no a priori useful bound on the successful degree or word length. Fixed-horizon minimization, by contrast, has finitely many word constraints at each N but their number is exponential in N before repetitions are identified. These stages should be priced separately rather than hidden behind the word “effective.”

All algebraic Haar moments in the localizer search are represented in a real algebraic number field with isolating data, using the fixed-subspace projector and its algebraic Gram inverse. Signs and strict comparisons are exact real-algebraic comparisons; a quadrature tolerance is not a certificate. Every rational command word has a rational matrix product and hence a rational target vector for rational polynomial readouts. The orthogonal closure fact used above is precisely that the Euclidean closure of a subgroup of $O(D)$ is the real zero set, inside $O(D)$, of its vanishing polynomial ideal, as used in [20]; replacing real connected components by complex algebraic ones would give the wrong quotient.

21. FINITE DESCRIPTIONS AND DECISION COMPLEXITY

The qualitative reduction is insensitive to the encoding of real numbers. Complexity statements are not. We first specify a finite input model and then give an explicit finite-precision counterpart to label width.

Theorem 21.1 (Finite-group stationary minimization is $\exists\mathbb{R}$ -complete). *The input consists of a finite group G given by its full multiplication table, finite seed and query sets, rational categorical vectors $F_{s,j}(g)$ for every $g \in G$, a nonnegative rational tolerance ε , and an integer $k \geq 1$ in binary. Deciding whether $M_\varepsilon \leq k$ is complete, under polynomial-time many-one reductions, for the existential theory of the reals. Hardness already holds for the trivial group, one query, and zero error. In particular the rational orthogonal polynomial-interface stationary problem is $\exists\mathbb{R}$ -hard even with constant readouts and trivial dynamics.*

Proof. Let $n = |G|$. If $k \geq n|\mathcal{S}|$, a deterministic seed-group register realizes every target exactly, so the answer is yes. Otherwise k is bounded by a polynomial in the explicit input length, even though its encoding is binary.

For membership, Theorem 6.1 permits permutation matrices P_g forming a representation of G . Introduce real variables for their entries, constrain each entry x by $x(x-1) = 0$, impose every row and column sum equal to one, and impose

$$P_1 = I, \quad P_g P_h = P_{gh} \quad (g, h \in G).$$

Introduce simplex variables α_s and categorical decoder rows $d_j(i)$. For every s, j, g and output coordinate c , use a nonnegative auxiliary variable $z_{s,j,g,c}$ with

$$z_{s,j,g,c} \geq \pm((\alpha_s P_{g^{-1}} d_j)_c - F_{s,j}(g)_c), \quad \sum_c z_{s,j,g,c} \leq 2\varepsilon.$$

This is an existential real formula with rational coefficients, degree at most three, and polynomial size. Padding a representation by fixed unused labels handles an optimum strictly smaller than k . Theorem 6.1 proves equivalence with the original unrestricted stationary problem. There is no group-closure computation in this explicit finite-group input model.

For hardness, let A be a rational nonnegative matrix and let k be a proposed nonnegative rank. Delete zero rows; the all-zero case is decided directly. Normalize each remaining row by its positive row sum to obtain a rational row-stochastic matrix F . This transformation is polynomial-time and preserves nonnegative rank. Interpret the rows as seeds and the columns as the categories of

one query on the trivial group. A zero-error realization gives a stochastic factorization $F = \alpha d$ of inner dimension at most k , so the nonnegative rank is at most k .

Conversely, from any nonnegative factorization $F = UV$ delete a zero row of V together with its column of U , and let $s_i = \sum_c V_{ic} > 0$. Put $d_{ic} = V_{ic}/s_i$ and $\alpha_{bi} = U_{bi}s_i$. Then d is row-stochastic and $\sum_i \alpha_{bi} = \sum_c F_{bc} = 1$, so α is row-stochastic. Identity command rows yield an exact stationary realization. Thus M_0 equals the real nonnegative rank of F . Shitov's universality theorem, specifically its complexity consequence [24, Theorem 2], makes this decision problem $\exists\mathbb{R}$ -complete. The last assertion follows by using the one-dimensional identity matrix as the only physical command and constant rational polynomial readouts. $\square$

The upper bound is for an explicitly tabulated finite group. The hardness transfers to rational matrix input, but a polynomial-size existential description of an arbitrary compact matrix closure has not been supplied. Theorem 20.3 remains the separate termination result for that broader input class. The complexity obstruction above already occurs in randomized initialization and decoding; it is not caused solely by closure algorithms.

21.1. A finite-table resource counterpart. For a categorical interface, a dyadic table of precision b has each probability in $2^{-b}\mathbb{Z}$. Besides peak label width k , charge its binary table length, all retained working registers, and the number of fair random bits used. A horizon- N clocked implementation can store initialization, N command tables, and terminal decoders. Its probability-table length is

$$O\left(b\left(|S|k + N|\mathcal{A}|k^2 + k\sum_j M_j\right)\right).$$

A uniform implementation is one finite algorithm that produces or accesses these tables from the encoded input, horizon, and accuracy. This requirement is additional to the original nonuniform model. An arbitrary real table, without an effective encoding, is not such an algorithm.

Lemma 21.2 (Dyadic rounding of a probability row). *A probability row p with m entries has a dyadic row $\hat{p}$ of precision b such that*

$$\text{TV}(p, \hat{p}) \leq (m-1)2^{-b}.$$

One exact draw from $\hat{p}$ uses at most b independent fair bits.

Proof. Round the first $m-1$ entries down to multiples of 2^{-b} and put the remaining mass in the last entry. This is a probability row. Its total variation error equals the mass transferred to the last entry, which is at most $(m-1)2^{-b}$. For sampling, draw a uniform integer in $\{0, \dots, 2^b - 1\}$ using b bits and compare it with the integer cumulative row sums. Degenerate rows can use fewer bits. $\square$

Theorem 21.3 (Precision and random-bit budgets). *A categorical horizon- N realization of peak width k and error at most ε admits a dyadic realization on the same registers whose error is at most*

$$(21.1) \quad \varepsilon + ((N+1)(k-1) + M_* - 1)2^{-b}, \quad M_* = \max_j M_j.$$

Sampling initialization, all transitions, and one terminal answer requires at most $(N+2)b$ fair bits.

For a stationary compact-group realization of width k , the corresponding error bound can instead be made independent of N :

$$(21.2) \quad \varepsilon + (k + M_* - 2)2^{-b}.$$

The stationary dyadic realization has at most k labels, permutation command tables, table length

$$O\left(|\mathcal{A}|k \log(k+1) + b\left(|\mathcal{S}|k + k \sum_j M_j\right)\right),$$

and uses at most $2b$ fair bits in total for one experiment of any length. Given exact algebraic encodings of the selected initialization and decoder rows, the rounding and sampling are effective finite procedures.

Proof. Apply Lemma 21.2 to every row. Couple the initial draws maximally and then, as long as the two labels agree, maximally couple their next transition draws. The probability of any disagreement is at most the sum of the initialization and N row errors, hence at most $(N+1)(k-1)2^{-b}$. Conditioned on matching terminal labels, the terminal categorical rows have total variation error at most $(M_* - 1)2^{-b}$. The coupling inequality gives (21.1), uniformly over words and queries. The sample count is the number of draws times b .

For the stationary claim, first use Theorem 5.3. Its transition matrices are permutations and are already dyadic at every precision. Round only initialization and terminal categorical rows. The initialization discrepancy cannot increase under a permutation, so the same coupling gives (21.2) with no factor N . There are no random transition draws, leaving only initialization and one terminal answer. A permutation is stored as k integer images, giving the displayed description bound. Exact comparison of algebraic numbers with dyadic rationals computes the necessary floors and handles equality, after which integer cumulative sampling is immediate. $\square$

For a prescribed extra error $\eta > 0$, choosing b with the added term in (21.1) or (21.2) at most η gives an explicit precision budget. The persistent label uses $\lceil \log_2 k \rceil$ bits; an implementation of the sampler also retains a b -bit random integer and bounded indexing registers while sampling. These temporary registers are not disguised as free persistent randomness. The description and random-bit bounds do not assert a polynomial time bound for constructing an algebraic optimum. They also do not turn an exact irrational realization into an exact finite-bit one: the extra tolerance is essential in that case.

22. POSITIVE FINITE-PRECISION MATRIX DECODERS

The label-width bounds leave real table descriptions uncharged. There is nevertheless a finite-description counterpart for each supplied encoded realization. Entrywise rounding of a density matrix is unsafe: it need not preserve positivity. A Gram-factor construction avoids this problem.

Lemma 22.1 (Rational positive rounding). *Let $D \in \mathcal{D}_d$ and let $L = D^{1/2}$. Approximate the real and imaginary parts of every entry of L by dyadic multiples of 2^{-b} , each within 2^{-b} , to obtain Q . Set $\delta = \sqrt{2}d2^{-b}$. If $\delta \leq 1/4$, then*

$$(22.1) \quad \hat{D} = \frac{QQ^*}{\text{tr}(QQ^*)} \in \mathcal{D}_d \cap M_d(\mathbb{Q} + i\mathbb{Q}), \quad \|\hat{D} - D\|_F \leq 8\sqrt{2}d2^{-b}.$$

The construction is effective when D is given by exact algebraic entries. Its rational output has $O(d^2(b + \log(d+1)))$ bits for fixed encoding conventions.

Proof. Write $Q = L + E$. Then $\|E\|_F \leq \delta$ and $\|L\|_F^2 = \text{tr } D = 1$. Hence $Q \neq 0$, $\text{tr}(QQ^*) \geq (1-\delta)^2$, and

$$\|QQ^* - D\|_F \leq (2+\delta)\delta, \quad |\text{tr}(QQ^*) - 1| \leq (2+\delta)\delta.$$

The first estimate follows from the two cross terms and EE^* , and the second from $\|Q\|_F^2 - \|L\|_F^2$. Since $\|D\|_F \leq 1$, division by $\text{tr}(QQ^*)$ gives an error at most $2(2+\delta)\delta/(1-\delta)^2 \leq 8\delta$. Positivity and trace one are exact, not numerical tolerances. After multiplying Q by 2^b , the normalization is a ratio of integer Gram entries to a positive integer sum of squares. Their bit sizes are

$O(b + \log(d + 1))$. The positive square root of an algebraic positive matrix has algebraic entries and can be specified and approximated by real-algebraic arithmetic. This proves the effectivity assertion without an efficiency claim for obtaining D . $\square$

Theorem 22.2 (A supplied realization with finite tables and fair bits). *Suppose an error- ε horizon- N realization for a matrix interface uses at most k labels, and its initialization, command rows, and decoders are supplied with exact algebraic encodings. Let $\sqrt{2}d2^{-b} \leq 1/4$. There is a rational finite-table implementation, with the same persistent label widths, whose Frobenius error is at most*

$$(22.2) \quad \varepsilon + \sqrt{2}((N + 1)(k - 1) + 8d)2^{-b}.$$

Each hidden draw uses b independent fair bits, so at most $(N + 1)b$ such bits suffice. For a supplied stationary permutation realization the additional error is at most

$$(22.3) \quad \sqrt{2}(k - 1 + 8d)2^{-b},$$

and only b fair bits are needed for initialization, independently of the horizon. The numerical matrix decoder itself is not a quantum preparation or a sampled terminal answer register.

Proof. Apply Lemma 21.2 to each initialization and command row after padding to k labels. Each row changes by at most $(k - 1)2^{-b}$ in total variation. Induction, or a maximal coupling at every actual draw, bounds the terminal distribution discrepancy by $(N + 1)(k - 1)2^{-b}$; this upper bound may harmlessly exceed one. Replace each decoder by the legal rational matrix in Lemma 22.1. The Frobenius diameter of $\mathcal{D}_d$ is $\sqrt{2}$, so changing the distribution incurs at most $\sqrt{2}$ times its total-variation discrepancy. The independent decoder error is at most $8\sqrt{2}d2^{-b}$. This proves (22.2). A dyadic probability row of denominator 2^b is sampled by one uniform b -bit integer and its cumulative integer table.

For a permutation realization the command rows already have exact integer entries and need not be rounded or sampled. Only the initialization distribution and decoder are changed. The same estimate therefore gives (22.3) and the horizon-independent fair-bit bound. $\square$

The row tables and their cumulative sums require $O((N|\mathcal{A}|k^2 + |\mathcal{S}|k)b)$ bits in the general case, in addition to $O(kd^2(b + \log(d + 1)))$ bits for one matrix decoder. Temporary random integers and table indices require their own $O(b + \log(k + 1))$ workspace; this workspace is not hidden in the persistent-label invariant. Multiple queries multiply the decoder storage by their number. To implement a sampled measurement one must additionally specify and charge its sampler. The theorem compiles a supplied encoded realization; it neither computes a width minimizer nor assigns finite encodings to arbitrary unspecified real tables. Applying it after the exact inner-polytope construction uses an explicit extra error budget, not an exact finite-precision claim at an extreme zero-error endpoint.

Here an exact algebraic entry is specified by defining polynomials and real isolating intervals (for both parts of a complex entry), with compatible field operations. The positive square root is the unique positive semidefinite solution of $L^2 = D$, and its isolating data can be obtained by real-algebraic algorithms. These representation and output bit-length statements are not running-time bounds for finding the original realization or its square root. The constructed rational qubit compiler is a separate, more explicit algorithm.

23. RELATION TO NUMERICAL AUTOMATA AND POSITIVE REALIZATION

The input/output object and the order of quantifiers are decisive in comparing realization theorems. Classical probabilistic automata assign numerical acceptance probabilities to words, while stochastic-language results also study threshold recognition [5, 6]. A numerical equivalence

of two fixed machines is not the assertion that for every horizon there exists a new bounded-width approximate machine. Conversely, language equivalence need not preserve the numerical probabilities.

For weighted automata over a field, the finite-Hankel-rank characterization gives finite linear representations and identifies their minimum dimension [7]. It does not enforce nonnegative rows, stochastic normalization, or simplex-valued decoders. Our common-row contraction concerns these constraints simultaneously. The invariant observable quotient in Theorem 11.4 uses the same reachable/invisible linear reduction; its additional conclusion is a finite-group criterion for exact positive realization with the full horizon-dependent quantifier.

The classical positive-realization problem asks for a positive system representing specified input/output data, and distinguishes positive dimension from unrestricted linear dimension [3]. Invariant cones and polyhedral sections are part of that theory, not new notions introduced here. Proposition 12.6 is an explicit common invariant-enclosure construction. The new matched statement pairs it with a full-subcritical lower estimate against all clocked stochastic machines. Neither one cutwise nonnegative factorization nor a finite-rank Hankel matrix supplies that lower estimate.

Brodsky and Pippenger identify bounded-error measure-once quantum languages as group languages [9, Theorem 3.3]. Their probabilistic simulation result preserves a cutpoint language with a potentially changed cutpoint [9, Theorem 3.6]; it is not uniform additive preservation of every acceptance probability. Our categorical unitary corollary fixes an entire terminal measurement law in total variation and allows horizon-dependent stochastic approximants. The distinction is already visible in the d -outcome threshold $1 - 1/d$.

The closure theorem and strict-threshold decision method of Blondel, Jeandel, Koiran, and Portier [20], and the effective Zariski-closure algorithms [21, 23], are algorithmic premises of Section 20. Computing the closure is not a new claim of this article. The additional outputs are a constrained minimax noise threshold, an attaining categorical decoder, and a terminating construction of an all-machine occupation constant. Recent Lie-algebra-based closure computations [22] give an alternative representation of the algebraic closure, but their complex connected components still require real-component analysis for the present application.

Peter–Weyl approximation, Haar averaging, and the representative-function algebra are classical compact-group and almost-periodic-function machinery [13]. Uniform approximation by a finite sum of matrix coefficients gives a finite *linear* representation of an output function. It does not give positive stochastic rows or the repeated information loss quantified by (7.3). Similarly, Steinberg’s minimal compact topological automaton and enveloping-monoid results [12, Theorems 2.2 and 2.6] concern recognition of a language. His compact topological-monoid criterion [12, Proposition 2.8] uses clopen recognition, not a continuous simplex-valued response with a prescribed error norm. The finite-quotient radius formula is specific to groups. The stationarization theorem itself now holds for irreversible topological monoids with eventual approximate returns (Theorem 4.2).

Distribution-based bisimulation metrics provide robust comparisons for specified probabilistic automata [11]. They do not by themselves impose a minimum over all horizon-specific stochastic realizations of an external group experiment. The finite future-output equivalence in Theorem 20.2 is deliberately a component-quotient statement, rather than a claim that approximate equivalence always has a unique minimal positive stochastic realization.

Comparison object	Error or equivalence	Machine quantifier
Compact stochastic groups [1]	Idempotent and permutation structure; no external error specification	One matrix group or semigroup kernel
Advice regular languages [8]	Neutral-letter recognition	One-scan advice automata; Boolean output
Weighted series [7]	Exact scalar coefficients; linear rank	One linear representation
Group-language simulation [9]	Cutpoint language; not additive probability error	One automaton
Positive realization [3]	Exact specified numerical response	One positive system
Topological recognition [12]	Clopen language acceptance	One recognizing action
Bisimulation metrics [11]	Distributional comparison of given systems	Two specified automata
Theorem 4.2	Same closed numerical error and label bound	Arbitrarily many narrow cuts; one extracted stationary machine
Theorem 7.2	Worst-word norm error; TV for a whole categorical law	A new clocked stochastic machine at every horizon
Theorem 12.1	Binary TV, every fixed $\varepsilon < \rho/2$	Same nonuniform lower model; exact upper construction

The contribution combines stationary purification and its exact finite-quotient criterion with a sharp constrained clocked radius, phasewise arbitrary-width occupation, and, for the specified Diophantine family, a matching full-subcritical growth law. The elementary radius center, the arithmetic pigeonhole argument, the closure algorithm, and the harmonic-analysis ingredients are not asserted individually new. The accompanying theorem-level audit records the closest comparisons and their changed hypotheses; it is not a claim of exhaustive priority clearance.

23.1. The classical matrix-semigroup boundary. Flor’s description of nonnegative matrix groups [1, Theorems 2 and 3] identifies their idempotent geometry and the finiteness of bounded groups; his account explicitly credits Brown and Schwarz for earlier compact and stochastic cases. We do not claim this matrix structure as new. Theorem 5.3 uses a minimal-rank idempotent in a joint semigroup, not just in the matrix projection: its physical coordinate must be the group identity. Sandwiching each command then preserves all uniform error inequalities by closure, even when the original stochastic rows violate every nontrivial physical group relation. The resulting initialization may be random, a point essential to exact state minimization.

The stationary theorem has no return hypothesis. The clocked theorem has a different quantifier and needs synchronization; the empty-return example and the phase-dependent example show why the distinction cannot be removed by terminology. The exact-return-length reduction is an additive-semigroup argument, whereas the profinite zero-error converse uses finite Hankel rank and continuous finite-dimensional representations. The new stationarization theorem removes the reversibility requirement for the equality of limiting width and stationary width, but does not supply a finite-quotient radius formula for an arbitrary irreversible monoid.

The finite-quotient existence theorem does not say that $[H : U]$ is the optimal number of stochastic labels. The actual minimum is the permutation-feasibility invariant in Theorem 5.4; the finite cyclic example separates them. For infinitely many components, the infimum of finite-quotient radii can fail to be attained, even at zero. Finally, the exactly matched quantitative law retains its specified Diophantine alphabet. Theorem 13.1 gives the retained block estimate, strengthened to a matching power law by Theorem 15.5, for spherical representations with a

spectral gap; Corollary 13.3 gives a concrete rational noncommuting instance. The exact unit-amplitude endpoint is not covered by these upper constructions. These distinctions delimit the proved statements without weakening any of the retained conclusions.

23.2. What the reductions add. The algebraic ingredients of Lemmas 5.1 and 5.2 are the classical minimum-rank kernel mechanism, the group corner of a compact matrix semigroup, and the recurrent simplex of a stochastic idempotent. Flor’s Theorem 2 describes nonnegative idempotents; his Theorem 3 and preceding Lemma 3 give the permutation structure of bounded nonnegative groups [1]. The paper explicitly attributes earlier cases to Brown and Schwarz. The present proof adds the physical coordinate to the closure, chooses the idempotent over its identity, and tracks the closed error balls through sandwiching. Theorem 6.1 further averages the finite kernel above the physical identity and descends to an actual continuous H -action. These are specified realization reductions using classical structure, not a new classification of matrix-semigroup kernels.

For deterministic Boolean recognition, neutral-letter removal of advice is an established phenomenon. Fijalkow–Paperman [8, Theorem 10] prove the Crane Beach property for advice regular languages by padding distinguishing words and suffixes, and their account relates one-scan programs to earlier Ramsey arguments. Their Myhill–Nerode argument in fact preserves the deterministic distinguishability bound K ; preservation of a state bound alone is not claimed as new here. A stochastic k -label register has a continuum of state distributions, so this discrete class count does not apply. Theorem 4.2 instead colors approximate tuples of stochastic composite kernels, extracts one common tuple, and absorbs a final filler matrix into the decoder. It preserves the same numerical tolerance at equality and the same k , and bounds all narrow-cut occurrences with unrestricted intermediate widths. Eventual approximate returns further replace a literal neutral letter. The finite Ramsey theorem itself is classical [25].

The spherical lower estimate uses a classical spectral gap, cap measure, and smoothing; its upper estimate uses a polytope enclosing a contracted sphere. The contribution is their realization-theoretic combination with the all-profile conditional-centroid bound. The rational $SO(3)$ example invokes [16] as a deep external input. Similarly, the lower-complexity input in Theorem 21.1 is Shitov’s nonnegative-factorization universality theorem [24]; the stochastic normalization and finite-group existential upper reduction are given explicitly. Neither external theorem is claimed as part of our proof technology’s novelty.

23.3. Companion results. The companion supplement gives additional results on compatible reachable sections, one-surplus geometry, stable sections, finite-horizon certificates, word distortion, two-state duality, finite-bit compilation, and global residual hierarchies. The scalar two-extreme argument and earlier quantitative constants are also retained there. The present main proofs are self-contained apart from the classical results explicitly cited; none of these companion derivations is used to conceal a missing step in a principal theorem.

23.4. Orbit geometry, positive matrix outputs, and recognition. The homogeneous-space covering estimates and the smooth orbit geometry used in Section 14 are classical; see, for example, Szarek [15]. The spectral-gap input is Bourgain–Gamburd [16]. Dense algebraic free generators in Section 18 come from Breuillard–Gelder [17, Theorem 1.1]; the later repair and supplementary proof details are recorded in [18]. None of these results is claimed as a new group-theoretic theorem here. The realization argument requires, in addition, a contraction uniform over every conditional-centroid direction, one common positive row for each command and label, and a terminal decoder in the original legal matrix set.

Quantum finite automata already exhibit strong state-complexity separations in language recognition; Ambainis–Freivalds [10] is a relevant early comparison. Theorem A.2 instead asks a classical stochastic machine to reproduce one entire legal density-matrix vector at every input

word, permits horizon-specific redesign and a free clock, and compares exact and fixed-error width for the same interface. Its exact lower bound uses extreme points of the decoder set; it does not apply unchanged to a single acceptance probability or to an enlarged unconstrained output simplex. A finite tomography map preserves width only when its legal image and reconstructed norm are retained. This comparison identifies different quantifiers and resources, not an exhaustive priority claim about all quantum or probabilistic automata formalisms.

The spectral-entropy argument removes the logarithm from the nonabelian lower bound. Equality of lower and upper powers requires equality of least-orbit and target-orbit dimensions; no universally matching law for every representation is asserted. General online outputs, adaptive feedback, and unrestricted compact semigroup radius formulas require additional arguments. The same-width stationarization result for topological monoids remains available under its actual approximate-return hypothesis; it does not provide that missing radius formula.

24. THE LEAST ORBIT DIMENSION AND UNIFORM SMALL BALLS

The geometric hypothesis needed for a stochastic converse concerns every conditional-centroid direction, not only the prescribed target orbit. For a fixed finite-dimensional orthogonal representation of a compact connected Lie group, this hypothesis follows from a rank invariant. Throughout this section Haar measure is normalized, and the Lie algebra has a fixed invariant inner product. No constants uniform in the dimension or in the representation are asserted.

Theorem 24.1 (Uniform caps from the least orbit dimension). *Let G be a compact connected Lie group and $R : G \rightarrow O(E)$ a continuous finite-dimensional representation with $E^G = \{0\}$ and $E \neq \{0\}$. Write $D = \dim G$ and choose a basis $X_1, \dots, X_D$ of its Lie algebra, using the same symbols for their infinitesimal actions on E . Set*

$$(24.1) \quad p_* = \min_{\|u\|=1} \text{rank}[X_1 u \cdots X_D u] = \min_{\|u\|=1} \dim R(G)u.$$

Then $p_* \geq 1$, and there is $C < \infty$ such that

$$(24.2) \quad \sup_{\|u\|=1, v \in E} \mu\{g : \|R(g)u - v\| < h\} \leq Ch^{p_*} \quad (0 < h \leq 1).$$

The exponent is optimal as a uniform exponent: it cannot be replaced by any larger exponent with a constant independent of u, v, h .

Proof. The tangent space to the orbit at u is the span of the $X_i u$. An orbit of dimension zero is a singleton, since G is connected; its point would be fixed. This proves $p_* \geq 1$. Put $r = p_*$ and fix orthonormal coordinates on E .

For each unit u , some r -row, r -column minor of $[X_1 u \cdots X_D u]$ is nonzero. The maximum of the absolute values of these finitely many minors is a continuous positive function on the unit sphere. Its minimum is therefore positive. The entries are uniformly bounded. Consequently there is $\sigma > 0$ such that for every unit u one can choose r input indices and r output coordinates whose derivative matrix A_u has least singular value at least σ .

For each of the finitely many permutations of the chosen Lie algebra basis consider the product chart

$$\phi(x, y) = \exp(x_1 X_{i_1}) \cdots \exp(x_r X_{i_r}) \exp(y_1 X_{i_{r+1}}) \cdots \exp(y_{D-r} X_{i_D}).$$

All these charts are diffeomorphisms on one sufficiently small cube about zero. Their images contain a common identity neighborhood W , and their Haar densities are bounded above by a common constant. By shrinking the cube once more, uniform continuity of the derivatives on the unit sphere and the finite list of charts ensures the following. For every unit u , choose the minor above and let $\pi_u : E \rightarrow \mathbb{R}^r$ be its coordinate projection. For all (x, y) in the cube,

$$(24.3) \quad \|\partial_x(\pi_u R(\phi(x, y))u) - A_u\| \leq \sigma/2.$$

For fixed y , the map $f_{u,y}(x) = \pi_u R(\phi(x, y))u$ is injective. Indeed integration of the derivative on the segment between x and x' gives

$$\|f_{u,y}(x) - f_{u,y}(x')\| \geq \|A_u(x - x')\| - \frac{1}{2}\sigma\|x - x'\| \geq \frac{1}{2}\sigma\|x - x'\|.$$

Every singular value of its derivative is at least $\sigma/2$. The change-of-variables formula therefore bounds the r -dimensional volume of the inverse image of a ball of radius h by $\omega_r(2/\sigma)^r h^r$. A ball in E projects into a ball of the same radius in these coordinates. Integrating in y and using the bounded Haar density proves the required estimate on this chart, uniformly in u and in the center v .

Choose finitely many right translates Wg_j covering G . On such a translate apply the preceding argument to the unit vector $R(g_j)u$. For its selected minor, use the corresponding permuted chart; that chart contains W . Haar invariance and a finite sum give (24.2) for sufficiently small h . Enlarging C covers the remaining $h \leq 1$. This proof does not require a spectral decomposition of the arbitrary center v .

Finally choose u_* whose orbit has dimension r . The orbit is the smooth embedded compact homogeneous space G/G_{u_*} . The pushforward of Haar measure is its invariant probability measure, with positive smooth density in local coordinates. At u_* , sufficiently small ambient chordal balls have measure bounded below by ch^r for some $c > 0$. Thus no larger uniform exponent is possible. $\square$

Proposition 24.2 (An algebraic rank certificate). *Suppose the infinitesimal matrices X_i are supplied with exact real algebraic entries, and are known to specify the representation in Theorem 24.1. Then p_* is computable by real quantifier elimination. A positive algebraic lower bound for the largest $p_* \times p_*$ minor on the unit sphere is computable as well. If a unit seed z_0 is algebraic, its orbit dimension q is the rank of $[X_1 z_0 \cdots X_D z_0]$.*

Proof. For each integer r ask whether a unit vector exists on which all $(r+1) \times (r+1)$ minors vanish. These are finitely many polynomial conditions over real algebraic numbers; the first affirmative answer is p_* . With $r = p_*$, the sum of squares of all r -minors has a positive minimum on the unit sphere, by the first paragraph of the preceding proof. Quantifier elimination computes that minimum as a real algebraic number with isolating data. If there are M minors, their largest absolute value is at least $\sqrt{\gamma/M}$, where γ is that minimum. Algebraic rank at the supplied seed gives the last assertion. $\square$

The proposition computes the rank invariant from *supplied* infinitesimal matrices. It neither certifies that arbitrary input matrices integrate to a compact group nor computes a random-walk spectral gap.

Corollary 24.3 (Uniform width exponents without a separate cap hypothesis). *In the orbit realization model of Theorems 14.3 and 15.5, assume the declared spectral gap, but replace its small-ball hypothesis by $E^G = \{0\}$. Let p_* be (24.1), and let q be the real dimension of the unit target orbit spanning E . For fixed $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$,*

$$W_{N,\varepsilon} \geq cN^{p_*/2}.$$

For $0 < \rho < 1$ the exact upper bound is $C(N+1)^{q/2}$; for $\rho = 1$ it holds at each fixed positive error. Every fixed width has the occupation bound of Theorems 14.3 and 15.5. Error is measured in the Euclidean norm defining the legal orbit hull.

Proof. Apply Theorem 24.1 to obtain its uniform cap estimate with exponent p_* . The lower and upper arguments of Theorems 14.3 and 15.5 then apply without alteration. $\square$

The group dimension D , least orbit dimension p_* , and target orbit dimension q need not coincide. For the standard sphere representation $p_* = q = d-1$. For conjugation on traceless Hermitian $d \times d$

matrices, an orbit with eigenvalue multiplicities $m_1, \dots, m_b$ has dimension $d^2 - \sum_i m_i^2$. A nonzero traceless matrix has $b \geq 2$, and the largest possible sum is $(d-1)^2 + 1$. Thus $p_* = 2(d-1)$, attained by a centered rank-one projection. The explicit Grassmannian estimates below also provide this exponent; they are retained as a direct geometric proof with stated dimension-dependent constants.

The converse now uses Theorem 15.5, rather than paying the pointwise-mixing block length at each narrow cut. Thus no logarithm separates the bounds when $p_* = q$. A supplied algebraic infinitesimal representation makes p_* computable; this statement does not compute a spectral gap or assert dimension-uniform constants.

25. A FIXED RATIONAL EXPERIMENT WITH AN EXPONENTIAL EXACT PROFILE

We now give fixed, explicit matrices and a rational pure seed. The freeness and trivial-stabilizer assertions have elementary finite-field proofs; no generic choice of generators or transcendental seed is used. The spectral-gap input is needed only for the noisy lower bound.

Lemma 25.1 (A quaternion free triple). *In the unit quaternions put*

$$a = (1 + 2\mathbf{i})/\sqrt{5}, \quad b = (1 + 2\mathbf{j})/\sqrt{5}, \quad c = (1 + 2\mathbf{k})/\sqrt{5}.$$

Every nonempty freely reduced word in $a^{\pm 1}, b^{\pm 1}, c^{\pm 1}$ is nonscalar. In particular these elements freely generate a group of rank three, also after passage to the projective unitary group.

Proof. Use the Hamilton relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$. Modulo five these relations are respected by

$$\mathbf{i} \mapsto \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{j} \mapsto \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}, \quad \mathbf{k} \mapsto \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

The six numerators $1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}$ therefore have respective images

$$(25.1) \quad \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \\ \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

They have rank one and their image lines are the six distinct points of $\mathbb{P}^1(\mathbb{F}_5)$. The kernel line of each matrix is the image line of the matrix with the opposite sign. Hence two consecutive matrices have zero product exactly when their letters are inverses. Writing each rank-one matrix as a column times a row shows that a product of them is nonzero whenever no adjacent pair is inverse.

A reduced word of length $n \geq 1$ is $5^{-n/2}Q$, where Q is an integral Hamilton quaternion of norm 5^n . Its image modulo five is nonzero by the preceding observation. Were Q scalar, its integer scalar entry t would satisfy $t^2 = 5^n$. For odd n this is impossible; for even $n \geq 2$ it forces $t = \pm 5^{n/2}$, making the image zero modulo five. This contradiction proves the assertion. Since the scalar unit quaternions are ± 1 , it proves both freeness statements. $\square$

Use the faithful complex representation

$$\rho(t + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \begin{pmatrix} t + xi & y + zi \\ -y + zi & t - xi \end{pmatrix}.$$

Define the following matrices, all of whose real and imaginary parts are rational:

$$(25.2) \quad U = \frac{1}{5} \begin{pmatrix} -3 + 4i & 0 \\ 0 & -3 - 4i \end{pmatrix}, \quad V = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad P_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Here $U = \rho(a^2)$, $V = \rho(b^2)$, and P_0 is a rank-one density matrix. The alphabet is exactly

$$(25.3) \quad \mathcal{A} = \{I, U, U^*, V, V^*\}.$$

Theorem 25.2 (Explicit rational exact-noisy separation). *For the fixed input (25.2)–(25.3), let the terminal target be $F(g) = gP_0g^*$ and require each decoder value to be a legal density matrix in $\mathcal{D}_2$. Error is Frobenius error. At every horizon $N \geq 0$ the exact minimum command-width profile is*

$$(25.4) \quad |S_t|_{\min} = 2 \cdot 3^t - 1 \quad (0 \leq t \leq N), \quad W_{N,0} = 2 \cdot 3^N - 1.$$

In fact the same profile is necessary, and is attained by the exact machine, throughout the closed range

$$(25.5) \quad 0 \leq \varepsilon \leq 625^{-N}/16.$$

For each fixed $0 < \varepsilon < 1/\sqrt{2}$, constants $c_\varepsilon, C_\varepsilon > 0$ satisfy

$$(25.6) \quad c_\varepsilon N \leq W_{N,\varepsilon} \leq C_\varepsilon(N+1).$$

At $\varepsilon \geq 1/\sqrt{2}$ one command label suffices. The upper bound in (25.6) has a uniform rational-table and fair-bit construction, specified in Theorem 26.2.

Proof. First, $\langle U, V \rangle$ is free of rank two: a nontrivial reduced word in $a^{\pm 2}, b^{\pm 2}$ expands to a nontrivial reduced word in $a^{\pm 1}, b^{\pm 1}$, and Lemma 25.1 applies. If a member g of this subgroup fixes P_0 , then it commutes with P_0 . It consequently commutes with

$$\rho(c) = (I + 2i(2P_0 - I))/\sqrt{5}.$$

A nonempty reduced word w in a, b cannot commute with c : wcw^{-1} is reduced, has length $2|w|+1$, and cannot equal c in the free group. Thus the projective stabilizer of P_0 in $\langle U, V \rangle$ is trivial.

The subgroup is dense in $SU(2)$. Both U and V have trace $-6/5$. If either had root-of-unity eigenvalues, its trace would be a rational algebraic integer and hence an integer, a contradiction. Its powers are therefore dense in its one-dimensional torus. The Lie algebra of the closed generated subgroup contains the $\mathbf{i}$ and $\mathbf{j}$ directions and their bracket, the $\mathbf{k}$ direction. It is all of $\mathfrak{su}(2)$; connectedness of $SU(2)$ implies density.

By freeness and the trivial stabilizer, the reachable pure states are in bijection with the reduced word ball of radius t . Identity padding makes this precisely the set reachable at length t . Its size is $1 + 4 \sum_{j=0}^{t-1} 3^j = 2 \cdot 3^t - 1$. The extreme-orbit profile theorem (Theorem 18.1) gives (25.4). Its proof uses one fixed suffix at each cut, not a query-dependent suffix.

We prove the stronger assertion (25.5) directly. Write $D = (I + x \cdot \sigma)/2$, with $x \in B_3$, for the Bloch coordinates of a density matrix. Then

$$(25.7) \quad \|D - D'\|_F = \|x - x'\|/\sqrt{2}.$$

The rotations induced by U, V, U^*, V^* have entries in $25^{-1}\mathbb{Z}$. At cut t , different reachable unit Bloch vectors have coordinates in $25^{-t}\mathbb{Z}$, and hence mutual Euclidean distance at least 25^{-t} . Fix any suffix of length $N - t$. Its action is an isometry, so the same separation holds for the corresponding terminal targets. For each cut label let $z_i \in B_3$ be its conditional terminal decoder mean after this fixed suffix. For a prefix with pure target n , error at most ε gives

$$\left\| \sum_i p_i z_i - n \right\| \leq \sqrt{2}\varepsilon, \quad \sum_i p_i \|z_i - n\|^2 \leq 2 \left(1 - \left\langle \sum_i p_i z_i, n \right\rangle \right) \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is therefore within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (25.5), twice this radius is at most $\sqrt{\sqrt{2}/2} 25^{-N} < 25^{-t}$. These closed balls are disjoint for different prefixes. They require distinct labels and prove the pointwise lower bound at every cut. Deterministic exact storage gives the simultaneous upper bounds.

For the positive-error converse, apply the spectral-gap theorem of Bourgain–Gamburd [16, Theorem 1] to the *displayed* algebraic generators U, V . The density just proved verifies its

hypothesis. The symmetric walk becomes an absolute L^2 contraction off the constants after making it lazy, for example with mass $1/2$ at I and $1/8$ at each other letter. The legal density-matrix width theorem in dimension two gives the lower bound in (25.6). The rational construction proved next gives its upper bound without using a spectral gap. The center $I/2$ is at distance $1/\sqrt{2}$ from every pure state, which gives the final threshold assertion; the lower bound excludes bounded width at any smaller fixed error. $\square$

This is a finite-input separation, not merely an existence statement in an arbitrary-real advice model. Every length- N target has real and imaginary entries with denominator dividing $2 \cdot 25^N$, hence bit length $O(N + 1)$. Neither the exponential exact lower bound nor its robust extension uses a spectral-gap theorem. The spectral-gap constant in the noisy converse remains non-effective in the present proof; no numerical value is inferred from finite word tests. The logarithmic interval in (25.6) is stated explicitly rather than suppressed.

26. A UNIFORM RATIONAL QUBIT COMPILER

The preceding example also admits a constructive upper realization. The compiler below constructs its rows rather than assuming that an optimal realization has already been supplied. Its persistent labels and finite table resources are accounted for separately.

Lemma 26.1 (A rational spherical inner hull). *For $m \geq 2$ an integer, let $\mathcal{V}_m$ consist of the distinct points*

$$(26.1) \quad v_{j,\ell,\eta} = \eta \frac{(2jm, 2\ell m, m^2 - j^2 - \ell^2)}{m^2 + j^2 + \ell^2}, \quad -m \leq j, \ell \leq m, \quad \eta \in \{-1, 1\}.$$

Put $r = 1 - m^{-2}$ and add the vertex 0 as a distinguished label. Then

$$rB_3 \subseteq \text{conv}(\mathcal{V}_m) \subseteq B_3, \quad K := 1 + |\mathcal{V}_m| \leq 1 + 2(2m + 1)^2 \leq 13m^2.$$

The points have $O(\log(m+1))$ -bit rational coordinates. For any supplied rational rotation $A \in SO(3)$ and any label v , the point rAv has a rational barycentric expression using 0 and at most three members of $\mathcal{V}_m$. Such an expression can be found by finitely enumerating triples and solving rational 3×3 systems.

Proof. The stereographic parametrization $\psi(s, t) = (2s, 2t, 1 - s^2 - t^2)/(1 + s^2 + t^2)$ has differential norm at most two. Every point in the closed northern hemisphere has a preimage in the unit disk, hence in $[-1, 1]^2$. Rounding its coordinates to the nearest multiples of $1/m$ gives a point of (26.1) within distance $\sqrt{2}/m$. Negation treats the southern hemisphere. For every unit n there is thus $v \in \mathcal{V}_m$ with $\langle n, v \rangle = 1 - \|n - v\|^2/2 \geq 1 - m^{-2}$. The support-function criterion proves the inner inclusion. The stated cardinality bound follows since $m \geq 2$; duplicates are discarded.

The polytope contains a neighborhood of zero. The ray through a nonzero rAv reaches its boundary in a face. Triangulate a facet containing that face using its vertices. A triangle containing the ray endpoint, together with zero, gives the desired expression, allowing zero coefficients on lower-dimensional faces. Its three nonzero vertices are linearly independent: their affine plane is a supporting plane not through zero. Enumerate all such triples, solve for their three coefficients, and retain any solution with nonnegative coefficients and sum at most one. The remaining mass is assigned to zero. Cramer's rule gives rational coefficients; for $v = 0$ use the point mass at zero. All arithmetic is finite, and no selection theorem is needed. $\square$

Theorem 26.2 (Constructed finite-input upper realizations). *Let the alphabet consist of a fixed finite list of rational Bloch rotations, and let the seed be the first coordinate unit vector. These*

include the explicit experiment (25.2). For a rational $0 < \beta < 1$ and horizon $N \geq 0$, choose

$$(26.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{N}{1-\beta}} \right\rceil \right), \quad r = 1 - m^{-2}.$$

There is an exactly correct rational stochastic realization of the matrix target $F_\beta(g) = I/2 + \beta(gP_0g^* - I/2)$ on at most $13m^2$ command labels. Each command row has at most four nonzero entries and is independent of the command epoch within the chosen horizon.

For any rational $0 < \varepsilon < 1/\sqrt{2}$, the pure target F_1 admits an error- ε realization on $O(1 + N/\varepsilon)$ labels with rational legal matrix decoders and dyadic rows. It can be sampled with at most Nb fair bits, where it suffices to take

$$(26.3) \quad b = \left\lceil \log_2 \frac{24(N+1)}{\varepsilon} \right\rceil.$$

Both constructions are uniform algorithms polynomial in N , $(1-\beta)^{-1}$ or ε^{-1} , and the input bit lengths. This is not a polynomial bound in the binary length of N or in the logarithm of the inverse accuracy.

Proof. The net includes the first coordinate unit vector, so initialization is deterministic. For each letter A and each net label v_i , select the rational row in Lemma 26.1 whose barycentric mean is rAv_i . The zero label stays at zero. Induction shows that the mean Bloch vector of the label at time t is $r^t A_{a_t} \cdots A_{a_1} e_1$. At label v_i use the matrix decoder

$$(26.4) \quad D_i = \frac{1}{2} \left(I + \frac{\beta}{r^N} v_i \cdot \sigma \right).$$

Bernoulli's inequality and (26.2) give $r^N \geq 1 - N/m^2 \geq \beta$. Thus every decoder is positive semidefinite of trace one. Its expectation is exactly F_β , for every length- N word; the same rows work for all those words. Formula (26.4) also gives $I/2$ at the zero label.

For the pure target choose $\eta = \min(\varepsilon, 1/2)$ and $\beta = 1 - \eta$. The amplitude loss contributes Frobenius error $\eta/\sqrt{2} \leq \varepsilon/\sqrt{2}$. Keep the rational decoders and round each sparse probability row to denominator 2^b without enlarging its support: round down the first at most three masses and give the last mass the remainder. Its total-variation error is at most $3 \cdot 2^{-b}$. Stochastic contraction bounds the terminal distribution discrepancy by $3N2^{-b}$. The Frobenius diameter of $\mathcal{D}_2$ is $\sqrt{2}$; therefore the additional error is at most $3\sqrt{2}N2^{-b} \leq \varepsilon/4$ under (26.3). Since $1/\sqrt{2} + 1/4 < 1$, the total is strictly less than ε . One uniform b -bit integer samples each row. There is no initialization draw and no sampled quantum output in this count.

For completeness we give finite construction costs. Enumerating the net uses $O(m^2)$ rational points. For each of $|\mathcal{A}|K$ rows, enumerating at most $\binom{K-1}{3}$ triples and solving fixed-size systems uses polynomially many rational operations. Cramer's rule shows that the exact row coefficients have bit length $O(\log(m+1) + B_{\mathcal{A}})$, where $B_{\mathcal{A}}$ bounds the supplied rotation-entry bit lengths. The denominator r^N has bit length $O(N \log(m+1))$. Expanded decoders therefore have $O(B_\beta + (N+1) \log(m+1))$ bits each. A naive full enumeration needs $O(|\mathcal{A}|K^4)$ fixed-size rational solves, each of polynomial bit cost. This proves the algorithmic assertion as well as the bit bounds. $\square$

Sparse row tables require $O(|\mathcal{A}|K(B_{\mathcal{A}} + \log(K+1) + \log(m+1)))$ bits before rounding, and $O(|\mathcal{A}|K(b + \log(K+1)))$ bits after rounding. Expanded matrix decoders require $O(K(B_\beta + (N+1) \log(m+1)))$ bits. The persistent label uses $\lceil \log_2 K \rceil$ bits; the horizon counter, temporary sampler, and arithmetic working storage are separately charged. Unlike a statement only about output bit length, the preceding proof gives a terminating polynomial table-construction algorithm for this specific family. It does not solve general minimum-width realization or certify a spectral gap. The full legal density-matrix constraint is maintained at every rounding step.

Proposition 26.3 (Supplied rational unit seeds). *Theorem 26.2 remains valid for any supplied rational unit Bloch seed, with the label bound replaced by $13m^2 + 1$.*

Proof. Add the seed to the rational unit net if it is absent. The inner-ball inclusion is unchanged, every contracted rotated vertex still has an exact rational sparse row, and initialization is now deterministic at the added seed. The same induction and terminal rescaling apply. $\square$

The implementation accepts the supplied finite rotation list and unit seed as exact rational data. Orthogonality, determinant one and unit seed norm are checked by rational identities before table allocation.

27. A RATIONAL EXPERIMENT WITH EXPLICIT SPECTRAL CONSTANTS

The original rational pair in Theorem 25.2 now has a sharp linear fixed-error law, but its spectral constant is still supplied existentially by Bourgain–Gamburd. We give a second finite input for which every constant in a useful linear lower bound is numerical. The deep input here is the classical Lubotzky–Phillips–Sarnak theorem, not a new spectral-gap calculation.

Let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices in their usual order, and put $D(x) = (I + \sum_j x_j \sigma_j)/2$ for $x \in \mathbb{B}_3$. Thus $D(\mathbb{B}_3) = \mathcal{D}_2$ and $\|D(x) - D(y)\|_F = \|x - y\|/\sqrt{2}$. Consider the following *rational Bloch rotations*:

$$(27.1) \quad A_1 = \frac{1}{5} \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & -4 \\ 0 & 4 & -3 \end{pmatrix}, \quad A_2 = \frac{1}{5} \begin{pmatrix} -3 & 0 & 4 \\ 0 & 5 & 0 \\ -4 & 0 & -3 \end{pmatrix}, \quad A_3 = \frac{1}{5} \begin{pmatrix} -3 & -4 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Take the alphabet

$$\mathcal{A}_{\text{LPS}} = \{I, A_1, A_1^\top, A_2, A_2^\top, A_3, A_3^\top\}$$

and the rational pure seed e_1 , so $D(e_1) = P_0$ as in (25.2). The induced unitary lifts can be taken as $(I - 2i\sigma_j)/\sqrt{5}$. They are algebraic, not rational matrices; the rational input in (27.1) describes the physical Bloch action directly. This is a different alphabet from the rational unitary pair of Theorem 25.2.

Lemma 27.1 (The classical six-rotation spectral bound). *Let P_{LPS} average uniformly over the six nonidentity letters in $\mathcal{A}_{\text{LPS}}$. On normalized S^2 ,*

$$(27.2) \quad \|P_{\text{LPS}}\|_{L_0^2(S^2) \rightarrow L_0^2(S^2)} = \frac{\sqrt{5}}{3}.$$

Their generated subgroup is dense in $SO(3)$.

Proof. The matrices in (27.1) are the adjoint actions of the unit Hamilton quaternions $(1 + 2\mathbf{i})/\sqrt{5}, (1 + 2\mathbf{j})/\sqrt{5}, (1 + 2\mathbf{k})/\sqrt{5}$. Together with their inverses they are exactly the norm-five set in the Lubotzky–Phillips–Sarnak theorem [28, Theorems 1.3 and 1.5]; see the precise restatement [29, Theorem 1.1], which gives $2\sqrt{p}/(p+1)$ for this set at $p = 5$. This proves (27.2). The cited theorem uses the theory of automorphic forms and Deligne’s theorem; no finite matrix enumeration replaces it.

Density can also be checked directly. Each rotation has cosine $-3/5$. If its angle were a rational multiple of 2π , twice this cosine would be a rational algebraic integer, contrary to $-6/5 \notin \mathbb{Z}$. Its powers are dense in its circle subgroup. The three coordinate-axis circle subgroups generate $SO(3)$, as follows either from their infinitesimal generators and brackets or from Euler rotations. Hence the closed generated subgroup is $SO(3)$. $\square$

Theorem 27.2 (An effective linear noisy law and an exponential endpoint). *For the fixed input (27.1), the terminal target is $D(A_w e_1)$, every decoder belongs to $\mathcal{D}_2$, and error is Frobenius error. For $0 < \varepsilon < 1/\sqrt{2}$ put $\zeta = 1 - \sqrt{2}\varepsilon$. Every admissible realization satisfies*

$$(27.3) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq 1 + 6 \log(12k) + 648\pi^2 \frac{1-\zeta}{\zeta} k \leq \frac{6500}{\zeta} k.$$

In particular,

$$(27.4) \quad \frac{\zeta N}{6500} \leq W_{N,\varepsilon} \leq 52 + \frac{104N}{\varepsilon}.$$

For rational ε in this interval the upper machine has uniformly constructed rational legal decoders and dyadic stochastic rows, each supported on at most four labels. It uses at most Nb fair bits for hidden transitions, where $b = \lceil \log_2(24(N+1)/\varepsilon) \rceil$.

At every horizon $N \geq 0$, the simultaneous exact minimum profile is

$$(27.5) \quad |S_t|_{\min} = 5^t \quad (0 \leq t \leq N).$$

The same profile remains necessary and is attained throughout

$$(27.6) \quad 0 \leq \varepsilon \leq 25^{-N}/16.$$

For $\varepsilon \geq 1/\sqrt{2}$, one label suffices.

Proof. For a spherical chordal cap of radius $h \leq 1$, normalized area is exactly $h^2/4$. Consequently the support of a mixture of k kernels K_h occupies at most $kh^2/4$ of each group orbit. Take $h = k^{-1/2}$. In Lemma 15.3 we may take

$$\theta = 1/4, \quad \kappa^2 = 5/9, \quad \gamma = (1 - \kappa^2)(1 - \theta) = 1/3.$$

Lemma 15.2 gives $L_h = \log(12k)$ and $B_3 = 24$. The proof of Theorem 15.5, with these constants and the normalization $\zeta = 1 - \sqrt{2}\varepsilon$, gives the first inequality in (27.3) as a real bound before taking an integer ceiling. For $k \geq 1$, $\log(12k) \leq 3k$, and $\pi^2 < 10$. Thus its right side is at most $19k + 6480(1 - \zeta)k/\zeta \leq 6500k/\zeta$. Counting all positive cuts proves the lower bound in (27.4).

For the upper bound apply the rational compiler of Theorem 26.2 to the displayed seven-letter list and the first-axis seed. At rational tolerance take $\beta = 1 - \varepsilon/2$ and $m = \max\{2, \lceil \sqrt{2N/\varepsilon} \rceil\}$. It gives an exact rational realization of $D(\beta A_w e_1)$ on at most $13m^2 \leq 52 + 52N/\varepsilon$ labels, with sparse common rows. The target change has Frobenius norm $\varepsilon/(2\sqrt{2})$. The stated dyadic precision adds at most $3\sqrt{2}N2^{-b} \leq \sqrt{2}\varepsilon/8$, so the total is less than ε . This proves the looser advertised numerical upper bound and the fair-bit statement. For a real tolerance choose a rational $\tilde{\varepsilon} \in (\varepsilon/2, \varepsilon)$ and use the same construction at $\tilde{\varepsilon}$; the bound $52 + 104N/\varepsilon$ follows. The algorithmic assertion uses rational input, not an oracle for an unencoded real number.

We next prove the exact profile independently of the spectral bound. Lemma 25.1 shows that the three norm-five quaternion generators freely generate modulo scalars, hence their adjoint rotations form a free group on a, b, c . The stabilizer of e_1 in this free group is exactly $\langle a \rangle$. Indeed, a rotation fixing e_1 commutes with the rotation a about that axis. In a free group the centralizer of this generator is its cyclic subgroup: write a reduced word outside $\langle a \rangle$ as $a^r v a^s$, where v is nonempty and begins and ends with neither a nor a^{-1} . Its conjugate of a reduces to $a^r v a v^{-1} a^{-r}$, which is not a . The converse, that every power of a fixes e_1 , is immediate.

The orbit is therefore indexed by the right cosets of $\langle a \rangle$. Each coset has a unique reduced representative with no terminal $a^{\pm 1}$, obtained by removing the maximal terminal power of a . It has least word length in that coset. At length $\ell \geq 1$ there are $4 \cdot 5^{\ell-1}$ such representatives. Identity padding gives exactly the ball of these cosets at command length t , whose cardinality is $1 + 4 \sum_{\ell=1}^t 5^{\ell-1} = 5^t$. Theorem 18.1 applies because pure states are extreme in $\mathcal{D}_2$, proving (27.5) at every cut; deterministic orbit storage attains the entire profile at once.

For robustness, distinct cut- t Bloch targets have coordinates in $5^{-t}\mathbb{Z}^3$ and hence distance at least 5^{-t} . One fixed suffix of length $N - t$ preserves these distances. If $z_i \in \mathbb{B}_3$ is the conditional terminal decoder mean from cut label i , wordwise error at a pure target n implies

$$\sum_i p_i \|z_i - n\|^2 \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (27.6), twice this radius is strictly less than $5^{-N} \leq 5^{-t}$. The corresponding closed balls for distinct prefix orbit points are disjoint, and so require distinct labels. The exact construction attains this bound. No fixed positive tolerance is asserted to lie in (27.6) for all horizons. Finally $D(0) = I/2$ is at Frobenius distance $1/\sqrt{2}$ from every pure state, proving the one-label assertion. $\square$

Remark 27.3 (What is effective). The input, orbit counts, robust tolerance, spectral norm, constants in (27.3)–(27.4), and rational compiler are explicit. The constant 6500 is conservative; it is not an optimal leading constant. The spectral norm is imported from [28, 29], not inferred from finite harmonic tests. For the earlier five-letter rational-unitary example, Theorem 15.5 removes the logarithm but does not supply a numerical Bourgain–Gamburd constant. Neither statement claims a complete two-parameter phase diagram as the error tends to zero. The finite-table and fair-bit resources refer to a classical hidden label and a numerical matrix output, not quantum-state preparation.

28. THE SIX-COMMAND SPHERE AND VARIABLE ACCURACY

The full spectral measure in Lemma 27.1 already lives on six nonidentity commands. The following result uses exactly those six commands for the experiment itself, with no idle instruction.

Theorem 28.1 (An explicit no-idle experiment). *Let A_1, A_2, A_3 be the rational matrices in (27.1). Take exactly*

$$\mathcal{A}_6 = \{A_1, A_1^\top, A_2, A_2^\top, A_3, A_3^\top\}, \quad x_0 = e_1, \quad F(w) = D(A_w e_1).$$

Every decoder must lie in $\mathcal{D}_2$, and error is Frobenius error. For $0 < \varepsilon < 1/\sqrt{2}$, put $\zeta = 1 - \sqrt{2}\varepsilon$ and $\theta = (1 - \zeta)/\zeta$. Then every admissible width profile satisfies

$$(28.1) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq 1 + 6 \log(12k) + 648\pi^2 \theta k.$$

For $N \geq 2$, the minimum peak width obeys the explicit joint bounds

$$(28.2) \quad \min \left\{ \frac{e^{(N-1)/12}}{12}, \frac{N-1}{1296\pi^2\theta} \right\} \leq W_{N,\varepsilon} \leq \min\{5^N, 52 + 104N/\varepsilon\}.$$

At every horizon the exact simultaneous cut profile is 5^t , $0 \leq t \leq N$, and the same profile is necessary and attained for $0 \leq \varepsilon \leq 25^{-N}/16$. At $\varepsilon \geq 1/\sqrt{2}$, one label suffices. For any fixed $\varepsilon_0 < 1/\sqrt{2}$ and sequence $0 < \varepsilon_N \leq \varepsilon_0$ with $\log(1/\varepsilon_N) = o(N)$,

$$(28.3) \quad W_{N,\varepsilon_N} = \Theta(N/\varepsilon_N).$$

The constants in this last comparison can be taken independent of the sequence; its starting horizon can depend on the sequence.

Proof. The classical full spherical norm is $\sqrt{5}/3$ for the uniform law on exactly $\mathcal{A}_6$ [29, Theorem 1.1]. Use the drift proof of Theorem 16.2, taking $h = k^{-1/2}$ and the sharper sphere constants $\gamma = 1/3$, $L_h = \log(12k)$, $B_3 = 24$ as in Theorem 27.2. This gives (28.1), with arbitrary deterministic commands in the gaps. It imports no uncomputed spectral constant.

For a full width bound k , at least one of $6 \log(12k)$ and $648\pi^2 \theta k$ is at least $(N-1)/2$. Inverting the corresponding inequality gives the lower bound in (28.2). The sparse rational construction

in Theorem 27.2 applies unchanged to the six-letter sublist; no row for the identity is needed. It gives the second upper bound, with the same rational-decoder and dyadic-row guarantees for rational tolerances.

For the exact profile, recall the independently proved free-group and stabilizer calculation in Theorem 27.2. The orbit is the right-coset space of $\langle a \rangle$ in the free group on a, b, c , where $a = A_1$. Every coset has a unique shortest representative w with no terminal $a^{\pm 1}$. If its length is $\ell \leq t$, the group word $wa^{t-\ell}$ has length t and the same action on e_1 . In chronological execution this means applying the a 's first; they fix the seed. There is no cancellation when w is nonempty, and the empty representative is padded by a^t . Thus even without an identity letter, the length- t reachable orbit is exactly the radius- t coset ball. Its size is 5^t . The extreme-orbit theorem gives the exact profile and deterministic orbit storage gives the upper bound 5^N .

The denominator-five lattice and fixed-suffix isometry argument of Theorem 27.2 does not require identity padding once the orbit count is established. It gives the same robust interval. The center $I/2$ gives the final constant-width assertion.

Finally $\theta \leq C_{\varepsilon_0} \varepsilon_N$, while $\log(N/\varepsilon_N) = o(N)$. The exponential alternative in the minimum on the left of (28.2) eventually exceeds a constant times N/ε_N . The other alternative already has that lower order. The upper bound proves (28.3). $\square$

Remark 28.2 (Three separate accuracy regimes). The exact count and its robust interval, the explicit two-parameter bounds (28.2), and the subexponential-accuracy asymptotic (28.3) are different statements. For example, $\varepsilon_N = N^{-a}$ gives width $\Theta(N^{a+1})$, and $\varepsilon_N = e^{-N^b}$, $0 < b < 1$, gives $\Theta(Ne^{N^b})$. Neither assertion extrapolates the exact plateau to a fixed positive error or resolves its full crossover at $\log(1/\varepsilon_N) \asymp N$. The constant 6500 remains a safe fixed-error simplification of (28.1), not an optimal leading constant. The seven-letter experiment and the older rational-unitary pair remain separate retained examples.

29. NONBACKTRACKING ENTROPY AND EXPONENTIAL ACCURACY

The estimates for subexponential inverse accuracy do not determine the width when $\log(1/\varepsilon)$ is comparable to the horizon. In this section a Ramanujan bound on a symmetric spherical action determines that exponential rate. The averaging polynomials and the relative-entropy affinity inequality are classical. The point is to retain their logarithmic strength in the centroid transport budget.

29.1. The logarithmic entropy defect. We continue to use the spherical kernel, entropy and invariant projection from Section 15. In the next lemma the projection may have more than one-dimensional range.

Lemma 29.1 (Affinity form of the entropy defect). *Let T be a finite convex combination of measure-preserving rotations and suppose $T\Pi = \Pi T = \Pi$ and $\|T(I - \Pi)\|_{2 \rightarrow 2} \leq \kappa < 1$. If a bounded probability density f is supported in A and $\Pi \mathbf{1}_A \leq \theta < 1$, then*

$$(29.1) \quad \mathcal{H}(f) - \mathcal{H}(Tf) \geq -\log(\kappa^2 + (1 - \kappa^2)\theta).$$

Proof. Write $Tf = \sum_j a_j f_j = g$, discarding zero coefficients. All f_j have integral one and the same entropy as f . The standard order-1/2 Rényi bound [33, Theorem 3], with its elementary Jensen proof, is

$$D(u\|v) = -2 \int u \log \sqrt{v/u} d\sigma \geq -2 \log \int \sqrt{uv} d\sigma.$$

Approximation or the extended-value convention handles zeros. Here $g \geq a_j f_j$, so each divergence needed below is finite. Convexity of $-\log$, followed by Cauchy–Schwarz, gives

$$\begin{aligned} \mathcal{H}(f) - \mathcal{H}(g) &= \sum_j a_j D(f_j \| g) \\ &\geq -2 \log \sum_j a_j \int \sqrt{f_j g} d\sigma \geq -2 \log \|T\sqrt{f}\|_2. \end{aligned}$$

Orbital Cauchy–Schwarz, exactly as in Lemma 15.3, gives $\|\Pi\sqrt{f}\|_2^2 \leq \theta$. Orthogonality of the invariant and orthogonal parts then gives $\|T\sqrt{f}\|_2^2 \leq \kappa^2 + (1 - \kappa^2)\theta$. This proves the claim. In contrast with its linearized version, (29.1) can grow with the length of an averaging block. $\square$

29.2. A spherical Ramanujan hypothesis. Fix $m \geq 2$, $p = m - 1$, a unit seed $z_0 \in \mathbb{R}^m$, and an alphabet of $2r$ indexed commands, $r \geq 2$, consisting of orthogonal maps $A_1, A_1^{-1}, \dots, A_r, A_r^{-1}$. Put $q = 2r - 1$. The legal output set is the closed unit ball $\mathbb{B}_m$, the target is $A_w z_0$, and error is Euclidean. Assume that the uniform command average P on $L^2(S^{m-1}, \sigma)$ satisfies

$$(29.2) \quad \|P|_{L_0^2}\|_{2 \rightarrow 2} \leq \frac{2\sqrt{q}}{q+1}.$$

This is a hypothesis on the *whole* spherical action, not just on $\mathbb{R}^m$. It implies that its compact group closure is transitive on S^{m-1} : otherwise averaging a continuous function distinguishing two compact orbits would give a nonconstant invariant function. In particular, the Haar orbit law is round measure and satisfies

$$(29.3) \quad \sigma\{x : \|x - u\| < h\} \leq A_m h^p \quad (0 < h \leq 1), \quad A_m \geq 1.$$

The harmless choice $A_m \geq 1$ will simplify the formulas. No identity letter or return assumption is made. The indexed inverse relation is used in counting formal words; no freeness assumption is needed for the following upper spectral bound. Reversing a formal nonbacktracking word preserves its law, so the chronological product convention gives the same averaging operator. The polynomial formula below does not replace a positive word average by a signed command kernel.

Let P_b average uniformly over the $(q+1)q^{b-1}$ formal words of length b having no adjacent inverse pair, for $b \geq 1$. Define

$$(29.4) \quad a_q = \frac{q-1}{q+1}, \quad \eta_b = (b + a_q)^2 q^{-b}.$$

Lemma 29.2 (Classical radial-polynomial estimate). *Under (29.2), $\|P_b|_{L_0^2}\|^2 \leq \eta_b$. In particular $\eta_b \leq 1/4$ for every $b \geq 4$ and $q \geq 3$.*

Proof. Let $A = (q+1)P$, and let S_b be the unnormalized formal word sum in the action. Cancellation of the first inverse pair gives

$$S_0 = I, \quad S_1 = A, \quad S_2 = A^2 - (q+1)I, \quad S_b = AS_{b-1} - qS_{b-2} \quad (b \geq 3).$$

The exceptional coefficient $q+1$ at length two is essential. With U_j the Chebyshev polynomial of the second kind and $U_{-1} = 0$, induction yields, for $b \geq 1$,

$$S_b = q^{b/2} \left[U_b \left(\frac{A}{2\sqrt{q}} \right) - q^{-1} U_{b-2} \left(\frac{A}{2\sqrt{q}} \right) \right],$$

where the second term is zero when $b = 1$. The operator A is self-adjoint and its spectrum on L_0^2 is in $[-2\sqrt{q}, 2\sqrt{q}]$. The elementary identity $U_j(\cos t) = \sum_{\ell=0}^j e^{i(j-2\ell)t}$ implies $|U_j(x)| \leq j+1$ on $[-1, 1]$. Dividing by $(q+1)q^{b-1}$ gives the bound $(b + a_q)q^{-b/2}$, including $b = 1$. For $b \geq 4$, the square of this bound is strictly decreasing in b : its successive ratio is at most $(1 + 1/4)^2/3 < 1$. At

$b = 4, q = 3$ its square is $(4 + 1/2)^2/81 = 1/4$, while for $q \geq 5$ it is at most $25/625 < 1/4$. This proves the non-strict bound. $\square$

The use of radial polynomials is not a new spectral theorem. For the LPS rotations, Pinochet Lobos–Pittet prove the sharper exact norm $(1 + a_q b)q^{-b/2}$ [29, Theorem 1.2]. We use the slightly larger bound (29.4) because the displayed polynomial argument proves it directly from (29.2).

29.3. A finite block inequality for arbitrary width profiles. Keep D_m, B_m from Lemma 15.2; here the entropy range exponent $m - 1$ equals the cap exponent p . For an error $0 \leq e < 1$ put $\xi = e/(1 - e)$.

Theorem 29.3 (Block entropy budget). *Let $t_0 < \dots < t_M$ be command cuts, each of width at most k , with $t_{j+1} - t_j \geq b$, where $b \geq 4$. Cut zero is allowed in this intermediate statement. Every error- e realization satisfies*

$$(29.5) \quad Md_b \leq \log \frac{D_m A_m k}{\eta_b} + C_m \left(\frac{A_m k}{\eta_b} \right)^{1/p} \sqrt{M\xi}, \quad d_b = -\log(2\eta_b - \eta_b^2), \quad C_m = \pi\sqrt{3B_m}.$$

All registers between these cuts are unrestricted. For a peak-width- k realization, one may take $M = \lfloor N/b \rfloor$. If instead J positive cuts have width at most k , one may take $M = \lceil J/b \rceil - 1$ whenever this number is positive.

Proof. Choose $h = (\eta_b/(A_m k))^{1/p} \leq 1$. At each selected cut the smoothed directional law is supported in at most k caps; its round support measure is at most η_b . By Lemmas 29.1 and 29.2, applying an independent P_b word lowers its entropy by at least d_b .

Use a deterministic prefix to t_0 . In each gap apply a fresh independent formal nonbacktracking word of length b , then any fixed word filling the rest of the gap. Use a fixed suffix after t_M . These are executable words of the original horizon, irrespective of possible cancellations between different blocks. Conditional on the incoming hidden label and the whole block word, the actual composite row is a stochastic kernel. The proof of Lemma 16.1 applies verbatim to this finite word law, not just to a one-letter law: its weighted flow identity sums over the word in place of the letter. Consequently the entropy correction at gap j is at most

$$\frac{\pi\sqrt{3B_m}}{h} \sqrt{\frac{\alpha_j - \alpha_{j+1}}{1 - e}}.$$

The deterministic filler rotates the potential and costs no entropy; no commutation with P_b is required. Wordwise terminal mean correctness and legal decoder norm at most one give $\alpha_j \geq 1 - e$. At the first selected cut the physical vector is deterministic, so $\alpha_0 = 1$, even with randomized initialization. Thus the losses sum to at most e . Telescoping the entropy, using its range $[0, \log(D_m h^{-p})]$, and then Cauchy–Schwarz proves (29.5).

For a peak bound take $t_j = jb$ and fill the final suffix. For J positive narrow cuts, greedily keep the first and then the next at distance at least b . Each selected cut accounts for at most b integer cuts, so there are at least $\lceil J/b \rceil$ selected cuts. Use the first $\lceil J/b \rceil$ of them. This proves the last claim without including cut zero in J . $\square$

Corollary 29.4 (Explicit inversion). *Put $H = Md_b - \log(D_m A_m / \eta_b)$. If $H > 0$, then for any $0 < \lambda < 1$,*

$$(29.6) \quad k \geq \min \left\{ \exp((1 - \lambda)H), \quad \frac{\eta_b}{A_m} \left(\frac{\lambda H}{C_m \sqrt{M\xi}} \right)^p \right\}.$$

When $e = 0$, the second term is interpreted as infinity and the stronger bound $k \geq \exp H$ holds.

Proof. Subtract $\log(D_m A_m / \eta_b)$ in (29.5). Either $\log k \geq (1 - \lambda)H$, or its remaining transport term is greater than λH . Solving the latter inequality proves the result. When $\xi = 0$ that term vanishes. $\square$

29.4. Uniform crossover on the exponential scale.

Theorem 29.5 (Spherical exponential-accuracy law). *Under (29.2), fix $0 < e_0 < 1$. There are constants $C, C' > 0$, depending only on the alphabet, m , and e_0 , such that for all $N \geq 1$ and $0 \leq e \leq e_0$,*

$$(29.7) \quad \exp[-C\sqrt{N \log(N+2)}] \min\{q^N, ((N+1)/e)^{p/2}\} \leq W_{N,e} \leq C' \min\{q^N, ((N+1)/e)^{p/2}\}.$$

The convention is $1/0 = \infty$. Moreover, if an error- e realization has $J \geq 1$ positive cuts of width at most k , then

$$(29.8) \quad k \geq \exp[-C\sqrt{J \log(J+2)}] \min\{q^J, ((J+1)/e)^{p/2}\}.$$

In particular, if $0 \leq e_N \leq e_0$ and $N^{-1} \log(1/e_N) \rightarrow a \in [0, \infty]$, then

$$(29.9) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,e_N}}{N} = \min\{\log q, pa/2\}.$$

Proof. We prove the lower estimate using a parameter n equal to N for peak width and to J for the occupation statement. For sufficiently large n set

$$b = \left\lceil \sqrt{n \log(n+2)} \right\rceil, \quad \lambda = n^{-1/2}.$$

Then $b \geq 4$, $b = o(n)$, and the respective choices of M in Theorem 29.3 satisfy $n/b - 1 \leq M \leq n/b$. By (29.4),

$$-\log \eta_b = b \log q - 2 \log(b + a_q), \quad d_b \geq b \log q - 2 \log(b + a_q) - \log 2.$$

It follows, with constants independent of e, n, k , that

$$\begin{aligned} H &\geq n \log q - C_1(b + (n/b) \log(b+1) + 1) \\ &\geq n \log q - C_2 \sqrt{n \log(n+2)}, \quad H \geq \tfrac{1}{2} n \log q \end{aligned}$$

for all sufficiently large n . The first term in (29.6) is therefore at least $q^n \exp[-C_3 \sqrt{n \log(n+2)}]$. For $e > 0$, use $\xi \leq e/(1 - e_0)$, $M \leq n/b$, $\eta_b \geq q^{-b}$ and $\lambda H \geq \frac{1}{2} \sqrt{n} \log q$. The second term in that formula is at least

$$c q^{-b} b^{p/2} / e^{p/2} \geq ((n+1)/e)^{p/2} \exp[-C_4 \sqrt{n \log(n+2)}].$$

The last inequality absorbs only constants and powers of n , besides q^{-b} . If $e = 0$, apply the last sentence of Corollary 29.4. Increasing C handles the finitely many smaller n , since the minimum in either statement is at most q^n and every register has at least one label.

For the upper bound, formal cancellation leaves a reduced word of length at most t after t commands. Thus at most $1 + (q+1) \sum_{j=0}^{t-1} q^j \leq C_q q^t$ physical products need be stored. Deterministic exact storage gives the first bound; it does not require physical freeness. For $e > 0$, the spherical inner-hull construction in Theorem 14.3 realizes the smaller amplitude $1 - e/2$ exactly with at most $C_{\text{net}}(1 + (N+1)/e)^{p/2}$ labels. Its bias is $e/2$ and its decoders remain in $\mathbb{B}_m$. The same conclusion follows directly by taking a spherical δ -net with $\delta^2 \asymp e/(N+1)$, placing each contracted rotated vertex in its inner hull, and rescaling the terminal decoder within the available amplitude slack. Transitivity established above makes the target orbit the whole sphere. Taking the better of the two machines proves the upper estimate. Finally divide logarithms in (29.7) by N ; the loss term is $o(N)$, proving (29.9), including $a = \infty$. $\square$

Remark 29.6 (What the crossover determines). The estimate is uniform in the error and determines the entire exponential rate, including the saturation rate. Its factor $\exp(O(\sqrt{N \log(N+2)}))$ is not asserted optimal. It does not give uniform multiplicative Θ bounds through the transition,

optimal leading constants, or the largest interval of exact equality with the orbit count. The subexponential-accuracy matching theorem 16.5 remains stronger in its own regime.

29.5. The fixed rational six-rotation experiment.

Corollary 29.7 (Explicit rational crossover). *For the six nonidentity rational Bloch commands of Theorem 28.1, seed e_1 , legal decoders $\mathcal{D}_2$ and Frobenius error, fix $0 < \varepsilon_0 < 1/\sqrt{2}$. Uniformly for $N \geq 1$ and $0 \leq \varepsilon \leq \varepsilon_0$,*

$$(29.10) \quad \exp[-C\sqrt{N \log(N+2)}] \min\{5^N, (N+1)/\varepsilon\} \leq W_{N,\varepsilon} \leq \min\{5^N, 52 + 104N/\varepsilon\}.$$

Consequently $\log(1/\varepsilon_N)/N \rightarrow a$ implies $\log W_{N,\varepsilon_N}/N \rightarrow \min\{a, \log 5\}$. At every cut the exact minimum is still 5^t for $\varepsilon \leq 25^{-N}/16$. No claim of exact saturation at $a = \log 5$ is made.

Proof. The affine Bloch identification multiplies Frobenius error by $\sqrt{2}$. Thus apply Theorem 29.5 with $m = 3$, $p = 2$, $q = 5$, and $e = \sqrt{2}\varepsilon$. The full-action Ramanujan hypothesis is precisely the imported LPS norm of Lemma 27.1; the nonbacktracking estimates above are for this same six-command alphabet. Constant factors of $\sqrt{2}$ are absorbed in C . The exact and rational-compiler upper bounds and the robust pointwise statement are Theorem 28.1. $\square$

For finite lower certificates on this experiment one may use its exact spherical cap area, rather than $A_3 \geq 1$. If $b \geq 4$, $M = \lfloor N/b \rfloor$, $\eta_b = (b + 2/3)^2 5^{-b}$, and $\xi = \sqrt{2}\varepsilon/(1 - \sqrt{2}\varepsilon)$, take $h = 2\sqrt{\eta_b/k} \leq 1$. The same proof yields

$$(29.11) \quad Md_b \leq \log(3k/\eta_b) + 3\pi\sqrt{2M\xi k/\eta_b}.$$

A strict violation excludes width k . This is a sufficient exclusion test, not a decision procedure for exact minimum width. The supplied `accuracy_profile.py` checks such violations with rational logarithm intervals and rational upper bounds for $\sqrt{2}$ and π ; it never treats a finite enumeration of words or harmonics as a proof of the imported spectral norm.

30. UNIFORM ONE-PASS COMPUTATION WITH CHARGED BIT-SPACE

The preceding label-width invariant permits arbitrary real transition tables. A lower bound for that invariant also constrains finite-bit algorithms, but an upper bound for it need not be implementable at a comparable cost. We now give a separate uniform result in which all integer storage and arithmetic are charged. Throughout this section the physical alphabet is exactly $\mathcal{A}_6$ in Theorem 28.1; in particular, no idle command is added. All logarithms are natural except those explicitly denoted by $\log_2$.

30.1. Computational convention and the space theorem. A numerical streaming algorithm has a fixed finite program. It reads parameters $N \geq 2$ and $L \geq 2$, followed by N commands, in one pass. A consumed command cannot be consulted again. No input-dependent information may be stored in a readable output tape, an external clock, or an uncharged address. After the last command it returns a rational matrix $\hat{F}(w) \in \mathcal{D}_2$. For randomized algorithms every returned matrix is required to be legal, and our correctness condition is the weaker, mean condition

$$(30.1) \quad \sup_{w \in \mathcal{A}_6^N} \|\mathbb{E}\hat{F}(w) - D(A_w e_1)\|_F \leq 2^{-L}.$$

Internal computation terminates almost surely between successive commands and at the output stage. Randomness is fresh; any retained random seed is part of the configuration.

Writable bit-space includes the data registers, temporary integer storage, retained parameter information, work-tape addresses and control information. Equivalently, one may charge a binary encoding of the complete private configuration. To relate these conventions, a fixed program with s work bits and a fixed number of work heads has at most $C(s+1)^{d_{2cs}}$ configurations, for

constants depending only on its machine convention. The prescribed position of the one-pass input head at the t th command boundary contains no input-dependent information. A private address that does contain such information must be counted. The parameters are not an advice table; they contain only the two integers N, L .

Theorem 30.1 (Uniform streaming space). *There is one deterministic numerical streaming algorithm for the six rational rotations such that, for every $N \geq 2$ and $L \geq 2$, every output is legal, the pointwise error is at most 2^{-L} , and the writable bit-space is*

$$(30.2) \quad O(\min\{N, L + \log_2(N + 1)\}).$$

It uses no real advice, no random sampling, and no oracle arithmetic. Writing $b_ = \min\{N, L + \lceil \log_2 N \rceil + 2\}$, its command updates can be performed in $O(b_*)$ bit operations each with $O(b_*)$ scratch space. Parameter scanning is charged separately by its input length; a binary rational output uses $O(b_*)$ bits.*

Conversely, there is an absolute $c > 0$ such that every one-pass algorithm satisfying (30.1) and having at most 2^B complete private configurations at each command boundary satisfies

$$(30.3) \quad B \geq c \min\{N, L + \log_2(N + 1)\}.$$

There is no running-time restriction in this converse. It applies even to arbitrary horizon-dependent stochastic updates. Consequently every fixed-program randomized bit algorithm has the same lower order of writable space, with machine-dependent constants. Deterministic and randomized uniform space therefore have the common order $\Theta(\min\{N, L + \log_2(N + 1)\})$ for this numerical task.

The spectral input for the converse is the full spherical Koopman norm $\sqrt{5}/3$ of the six-letter average, not the norm of its action on $\mathbb{R}^3$. The upper algorithm uses only rational orthogonality and does not need a spectral assumption. The theorem is a consequence of the entropy occupation law together with an explicit finite-arithmetic construction, not a new proof of the LPS spectral theorem.

30.2. From complete configurations to stochastic width.

Lemma 30.2 (Boundary reduction). *If a one-pass algorithm has at most k private configurations at every command boundary and satisfies mean error ε , its boundary laws define a clocked numerical realization of peak width at most k and the same error.*

Proof. Take the configurations at boundary t as S_t . Include every piece of retained information, whether or not it is called a data register. Starting from a boundary configuration and a supplied next command, integrate over all internal computation and fresh randomness up to the next boundary. Almost-sure termination gives a stochastic row of total mass one. The row depends only on that configuration and command, with the fixed parameters and boundary index permitted as public parameters. Consumed input and old random bits are inaccessible unless recorded in the configuration. This gives precisely the Markov rows of Definition 7.1.

At the terminal boundary, replace any randomized output computation by its conditional mean matrix. This mean is in $\mathcal{D}_2$ by convexity and compactness, and is independent of the discarded input conditional on the complete configuration. Thus the numerical decoder is legal and has exactly the mean output in (30.1). No running-time or exact-sampling assumption is introduced in this reduction. $\square$

Proposition 30.3 (A finite bit-accuracy obstruction). *For the six-command experiment, suppose a correct realization has at most 2^B complete configurations at every positive cut and error $0 \leq \varepsilon < 1/\sqrt{2}$. Put*

$$a_B = 1 + 6 \log(12 \cdot 2^B), \quad c_B = 648\pi^2 2^B.$$

Then

$$(30.4) \quad N \leq a_B + c_B \frac{\sqrt{2}\varepsilon}{1 - \sqrt{2}\varepsilon}.$$

In particular, if $d = N - a_B > 0$, then

$$(30.5) \quad \varepsilon \geq \frac{d}{\sqrt{2}(c_B + d)}.$$

At zero error the exact profile gives the stronger bound $B \geq N \log_2 5$.

Proof. Apply (28.1) to all N positive cuts with $k = 2^B$ and $\zeta = 1 - \sqrt{2}\varepsilon$. This gives (30.4), including its limiting zero-error case. Solving $d \leq c_B \sqrt{2}\varepsilon / (1 - \sqrt{2}\varepsilon)$ gives (30.5). For the last assertion use the exact terminal width 5^N in Theorem 28.1 instead of the weaker entropy estimate. $\square$

Lower bound in Theorem 30.1. Let k be the maximum number of boundary configurations. Since $\varepsilon = 2^{-L} \leq 1/4$, Proposition 30.3, or (28.1) with k , implies, for absolute positive constants C_0, C_1, C_2 ,

$$(30.6) \quad N \leq C_0 + C_1 \log k + C_2 \varepsilon k.$$

For $N \geq 2C_0$, either $\log k \geq N/(4C_1)$ or $k \geq N/(4C_2\varepsilon)$. Taking logarithms and enlarging an absolute additive constant yields

$$\log_2 k \geq c_0 \min\{N, \log_2(N/\varepsilon)\} - C_3.$$

The additive constant can be absorbed uniformly. Indeed $k \geq 2$ for all $N \geq 2$ in the stated error range: the all- A_1 word ends at e_1 , whereas $N - 1$ copies of A_1 followed by A_2 end at $A_2 e_1 = (-3/5, 0, -4/5)$. Their density matrices have Frobenius distance $\sqrt{8/5} > 1/2$. A single constant mean decoder cannot be within $1/4$ of both. If $\min\{N, \log_2(N/\varepsilon)\}$ is below a fixed large constant, $\log_2 k \geq 1$ suffices; above that constant it absorbs C_3 in the preceding estimate. The finitely bounded range $N < 2C_0$ is handled in the same way. Since $k \leq 2^B$, this proves (30.3). Finally the fixed-program configuration count stated above gives $\log_2 k = O(s + 1)$, so the same conclusion holds for work-tape space. This argument neither assumes uniform transition tables nor assumes that an individual sampled decoder approximates the physical target. $\square$

30.3. A legality-preserving integer algorithm. For $b \geq 0$, define coordinatewise truncation toward zero by

$$T_b(x)_i = 2^{-b} \operatorname{sgn}(x_i) \lfloor 2^b |x_i| \rfloor.$$

Unlike rounding to the nearest grid point, this operation preserves the closed Euclidean unit ball.

Lemma 30.4 (Directed rounding). *For every $x \in \mathbb{R}^d$,*

$$\|T_b(x)\| \leq \|x\|, \quad \|T_b(x) - x\| \leq \sqrt{d} 2^{-b}.$$

If $R_1, \dots, R_N$ are orthogonal, $x_t = R_t x_{t-1}$ with $\|x_0\| \leq 1$, and $v_0 = x_0$ is on the 2^{-b} grid, then $v_t = T_b(R_t v_{t-1})$ satisfies

$$(30.7) \quad \|v_t\| \leq 1, \quad \|v_t - x_t\| \leq t\sqrt{d} 2^{-b}.$$

Proof. Each absolute coordinate decreases under truncation and changes by less than 2^{-b} . Squaring and summing gives the first two estimates. Orthogonality and the triangle inequality give $\|v_t - x_t\| \leq \sqrt{d} 2^{-b} + \|v_{t-1} - x_{t-1}\|$. Induction proves both assertions in (30.7). $\square$

Write $A_a = M_a/5$, where M_a is one of the six fixed integer matrices from (27.1) and their transposes, multiplied by five. Set

$$(30.8) \quad b = L + \lceil \log_2 N \rceil + 2.$$

When $b < N$, initialize $n = (2^b, 0, 0) \in \mathbb{Z}^3$. On command a , compute

$$(30.9) \quad z = M_a n, \quad n'_i = \operatorname{sgn}(z_i) \lfloor |z_i|/5 \rfloor, \quad n \leftarrow n'.$$

The current numerical vector is $v = n/2^b$. The update is exactly $T_b(A_a v)$; in particular it does not use an approximate matrix. The common denominator is kept fixed.

When $b \geq N$, instead initialize $n = e_1$ and $d = 1$ and use the exact updates

$$(30.10) \quad n \leftarrow M_a n, \quad d \leftarrow 5d.$$

After t commands, $n/d = A_{a_t} \cdots A_{a_1} e_1$ and $d = 5^t$. No history of the commands is retained in either mode.

Upper bound in Theorem 30.1. In the fixed-grid mode, Lemma 30.4 gives legality at every step and

$$\|D(v_N) - D(x_N)\|_F = \frac{\|v_N - x_N\|}{\sqrt{2}} \leq \sqrt{3/2} N 2^{-b} < 2^{-L}.$$

The exact mode is legal and has zero error. In both modes the output is a rational density matrix. For example, if $v = n/d$, one may output

$$(30.11) \quad D(v) = \frac{1}{2d} \begin{pmatrix} d + n_3 & n_1 - in_2 \\ n_1 + in_2 & d - n_3 \end{pmatrix}.$$

The guarantee $\|n\| \leq d$ implies positive semidefiniteness. There is no square root, eigenvalue computation, projection oracle, or physical quantum preparation in the algorithm.

In the fixed-grid mode each coordinate needs $b + O(1)$ bits, including its sign. Every entry of M_a has absolute value at most five and each row has absolute sum at most seven. Thus the temporary integer z_i and the new coordinate also require only $b + O(1)$ bits. A constant number of buffers suffices. Addition, multiplication by these fixed integers, and division of a nonnegative integer by five are executable by binary digit scans with $O(b)$ bit operations and $O(b)$ storage. Truncation of a negative coordinate uses its sign and absolute value; rounding toward negative infinity would not give this algorithm.

In the exact mode $\|n\| = d = 5^t$, so all numerators, the denominator, and intermediate results have $O(t + 1)$ bits. Their updates use $O(t + 1)$ bit operations and storage. Taking $b < N$ for the first mode and $b \geq N$ for the second yields (30.2) and the asserted command time. Formula (30.11) can be written as signed binary integers over one common denominator in $O(b_*)$ time and space. Conversion to a decimal presentation is not needed for the bit-operation bound.

For completeness, parameter handling does not require storing a very large requested precision. Read N and store it in binary. While reading L , maintain its value capped at N by the recurrence $\ell \leftarrow \min\{N, 2\ell + \delta\}$ for each binary digit δ . If the cap is reached, the exact mode is already determined. Otherwise $L < N$ is stored exactly, and (30.8) is computed with $O(\log N)$ auxiliary bits. This also works with any fixed input radix. Reading the entire header takes its actual input length; a straightforward implementation uses $O(\log N)$ work bits and $O(\log N)$ bit operations per header digit. A counter validating that exactly N commands were read costs $O(\log N)$ additional bits, which are within the asserted bound. No command-time external clock is needed. $\square$

Corollary 30.5 (The zero-error and arbitrary-tolerance regimes). *Exact one-pass numerical realization of this fixed rational experiment has bit-space $\Theta(N)$. For every real $0 < \varepsilon \leq 1/4$, a sufficient binary accuracy parameter $L = \lceil \log_2(1/\varepsilon) \rceil$ gives optimal space order*

$$\Theta(\min\{N, \log(N + 1) + \log(1/\varepsilon)\}).$$

Thus the order is logarithmic at fixed positive error and linear once $\log(1/\varepsilon)$ is of order N or larger.

Proof. Use the exact profile 5^N for the first lower bound and (30.10) for its upper bound. For positive tolerance, the lower proof applies directly with that tolerance in (30.6). The chosen L differs from $\log_2(1/\varepsilon)$ by less than one and supplies the upper bound. $\square$

30.4. Rational orthogonal actions. The computational consequence is not restricted to the numerical LPS example. Legal unit-ball outputs allow the same argument for any fixed rational orthogonal alphabet satisfying the two geometric hypotheses used by the entropy theorem.

Corollary 30.6 (Uniform space for rational orthogonal actions). *Let $\mathcal{A} \subset O(\mathbb{Q}^m)$ be a fixed finite alphabet, let G be its compact generated group, and let $x_0 \in \mathbb{Q}^m$ be a unit vector. Assume the full action gap (16.3) and the all-direction cap bound (16.4) with a positive exponent p . The dimension, alphabet, rational denominators, seed, spectral constant and cap constants are fixed independently of N, L . Require numerical decoder values to belong to the closed unit ball of $\mathbb{R}^m$. In the one-pass model, with Euclidean mean error at most 2^{-L} , the optimal uniform bit-space order is*

$$\Theta(\min\{N, L + \log_2(N + 1)\})$$

for $L \geq 2$ and N above a constant depending only on the fixed input. There is no identity or return hypothesis. The upper bound is deterministic with pointwise error. Exact computation has order N .

Proof. The lower proof of Theorem 16.2 uses the legal output set only through $\|D_i\| \leq 1$ in the terminal centroid estimate. It therefore applies unchanged to unit-ball decoders, with $\rho = 1$ and $\zeta = 1 - \varepsilon$. Its finite budget gives

$$N \leq C_0 + C_1 \log k + C_2 \varepsilon k^{2/p} \quad (0 \leq \varepsilon \leq 1/4).$$

The same dichotomy as in (30.6) gives $\log k \geq c \min\{N, \log(N/\varepsilon)\} - C'$. For sufficiently large N this lower bound absorbs its additive constant uniformly, since $\log(N/\varepsilon) \geq \log(4N)$. At zero error the finite budget directly gives $\log k \geq cN - C'$. The complete-configuration reduction then applies.

For the upper bound write $A_a = M_a/D$ and $x_0 = n_0/d_0$ with fixed integer data and positive common denominators. At grid precision b initialize $v_0 = T_b(x_0)$ and perform the integer update $n'_i = \text{sgn}((M_a n)_i) \lfloor |(M_a n)_i|/D \rfloor$. The initial error contributes one additional rounding term, so Lemma 30.4 gives $\|v_N - x_N\| \leq (N + 1)\sqrt{m} 2^{-b}$ and legal outputs at all prefixes. Choose $b = L + \lceil \log_2(N + 1) \rceil + \lceil \log_2 \sqrt{m} \rceil + 2$. Alternatively retain the exact vector with denominator $d_0 D^t$. Since the dimension and integer coefficients are fixed, these modes use respectively $O(b)$ and $O(N + 1)$ bits, including scratch and parameter storage. Choose the smaller mode. The capped parameter scan and fixed-integer digit operations from the preceding proof apply without change. No orbit-covering algorithm or construction of a real transition table is used. $\square$

This corollary requires unit-ball decoders. It does not assert that coordinatewise truncation preserves an arbitrary orbit hull or the positive-semidefinite cone in higher matrix dimension. The Bloch ball is exactly the legal qubit output set, which is why the preceding explicit result gives legal density matrices.

30.5. Relation to finite automata and positive realization. The passage from a configuration to a stochastic label in Lemma 30.2 is elementary, as is directed finite-precision arithmetic. The quantitative content is that the entropy obstruction continues to force optimal uniform bit-space even when arbitrary randomization and unlimited internal running time are allowed. This is not a reinterpretation of free real-table lookup as an implemented algorithm. The upper program is independently specified. The logarithm of width records fewer asymptotic details than width

itself: the present matching space order does not remove the subexponential multiplicative loss in Theorem 29.5.

The invariant-cone characterization of positive realization, stated by Benvenuti–Farina [3, Theorem 2], concerns exact stationary nonnegative realizations of a rational transfer function. Our lower bound instead concerns all-word, horizon-dependent approximation by common stochastic rows. The affine Bloch recursion has constant real linear dimension, but its negative matrix entries do not make it a positive finite-label realization. A finite-bit implementation of that recursion must pay for its accuracy.

Ambainis–Watrous [35, Section 2 and Theorem 1] allow a two-way input head and a fixed-dimensional quantum register in a language recognition problem. Panduranga Rao–Vinay [36, Theorem 1] construct complex-weighted two-way automata for one-sided language recognition. Their transitions are not nonnegative stochastic rows with a legal density-matrix decoder. Neither comparison identifies algebraic amplitudes with finitely many classical configurations. The model here has a one-pass classical input, no physical quantum register, and a numerical output required for every command word. We do not claim a language-recognition separation or an extension to repeated quantum measurements.

31. POSITIVE FINITE-PRECISION MATRIX STATES

Coordinate truncation preserves a Euclidean ball, but not the cone of positive matrices. We give a replacement whose output is a density matrix in every dimension. The correction is explicit; no spectral projection or membership oracle is used in an update. Write

$$\mathcal{D}_d = \{\rho \in \mathbb{C}^{d \times d} : \rho = \rho^*, \rho \geq 0, \text{tr } \rho = 1\}, \quad \|X\|_1 = \text{tr } \sqrt{X^* X}.$$

All dimensions and all physical input data below are fixed independently of the horizon and the precision. Trace norm is not divided by two.

31.1. A fixed-denominator positive rounding map. Fix an integer $B \geq 1$. For $\rho \in \mathcal{D}_d$, form a Hermitian matrix $Q_B(\rho)$ as follows. Truncate the first $d - 1$ diagonal entries and the real and imaginary parts of every strictly upper triangular entry toward zero on the grid $B^{-1}\mathbb{Z}$. Reflect the upper triangle by conjugation, and set the last diagonal entry equal to one minus the sum of the other diagonal entries. Thus $\text{tr } Q_B(\rho) = 1$. Define

$$(31.1) \quad \mathcal{R}_B(\rho) = \frac{BQ_B(\rho) + 2dI}{B + 2d^2}, \quad \delta_B = \frac{6d^2}{B}.$$

Lemma 31.1 (Positive rounding with fixed denominator). *For every $\rho \in \mathcal{D}_d$, the matrix $\mathcal{R}_B(\rho)$ belongs to $\mathcal{D}_d$ and*

$$(31.2) \quad \|\mathcal{R}_B(\rho) - \rho\|_1 \leq \delta_B.$$

Its numerator is a Gaussian-integer Hermitian matrix of trace $B + 2d^2$. If $\rho = P/z$ with $P \in \mathbb{Z}[i]^{d \times d}$, $P \geq 0$ and $z = \text{tr } P > 0$, its numerator is computable by integer multiplication, addition and signed division, using $O_d(\log(B + 1) + \log(z + 1))$ bits of scratch space.

Proof. Set $E = Q_B(\rho) - \rho$. The first $d - 1$ diagonal errors have absolute value less than B^{-1} ; the last is at most $(d - 1)B^{-1}$. Each off-diagonal real or imaginary error has absolute value less than B^{-1} . Consequently

$$\|E\|_F^2 \leq \frac{(d - 1) + (d - 1)^2 + 2d(d - 1)}{B^2} = \frac{3d(d - 1)}{B^2}.$$

In particular $\|E\|_{\text{op}} \leq 2d/B$ and $\|E\|_1 \leq d\|E\|_{\text{op}} \leq 2d^2/B$. Since $\rho \geq 0$, the matrix $Q_B(\rho) + (2d/B)I$ is positive. Its trace is $1 + 2d^2/B$, proving positivity and trace one in (31.1). With

$$\eta = 2d/B,$$

$$\mathcal{R}_B(\rho) - \rho = \frac{E + \eta(I - d\rho)}{1 + d\eta}.$$

The estimate $\|I - d\rho\|_1 \leq 2d$ proves (31.2). The bounds deliberately leave slack.

For rational input, the independent entries of $BQ_B(\rho)$ are $\text{sgn}(a)[B|a|/z]$, where a is the corresponding integer part of P . The last diagonal entry is B minus the other diagonal numerators. Adding $2d$ to each diagonal gives the displayed common denominator. Positivity of P gives $|P_{ij}| \leq \sqrt{P_{ii}P_{jj}} \leq z$. Thus a fixed number of integer buffers of the asserted length suffices. Binary long division uses linear storage and quadratic bit operations in that length. $\square$

The positivity correction cannot simply be omitted. For $\rho = \frac{1}{13} \begin{pmatrix} 4 & 6 \\ 6 & 9 \end{pmatrix}$ and $B = 3$, trace-preserving entry truncation gives $Q_B(\rho) = \frac{1}{3} \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$, whose determinant is negative.

The map is not a quantum operation and is not asserted to be affine. It is a numerical update on a stored matrix. Its positivity and its uniform error are the only properties needed below.

31.2. The trace-norm estimate for an instrument. An instrument with outcomes in a finite set $\mathcal{Y}$ is a family of completely positive maps $(\Phi_y)_{y \in \mathcal{Y}}$ whose sum is trace preserving. The elementary contraction below is standard quantum-information theory [37, Corollary 3.40]; a proof is included to specify the norm and the treatment of zero outcomes.

Lemma 31.2 (Contraction and positive branch repair). *For every Hermitian X ,*

$$(31.3) \quad \sum_y \|\Phi_y(X)\|_1 \leq \|X\|_1.$$

For positive X , put

$$\overline{\mathcal{R}}_B(X) = \begin{cases} (\text{tr } X) \mathcal{R}_B(X / \text{tr } X), & \text{tr } X > 0, \\ 0, & X = 0. \end{cases}$$

Then $\text{tr } \overline{\mathcal{R}}_B(X) = \text{tr } X$ and

$$(31.4) \quad \|X - \overline{\mathcal{R}}_B(X)\|_1 \leq \delta_B \text{tr } X.$$

Proof. Use the Jordan decomposition $X = X_+ - X_-$, with $X_{\pm} \geq 0$ and $\text{tr } X_+ + \text{tr } X_- = \|X\|_1$. Positivity and the triangle inequality give

$$\sum_y \|\Phi_y(X)\|_1 \leq \sum_y \text{tr } \Phi_y(X_+ + X_-) = \text{tr}(X_+ + X_-).$$

Equation (31.4) is (31.2) multiplied by $\text{tr } X$. A positive matrix of trace zero is zero, so no normalization of a zero-probability branch is required. $\square$

Neither conclusion requires a lower bound on a nonzero outcome probability. Complete positivity is needed for the physical interpretation of the maps; positivity and preservation of the total trace already prove the displayed contraction.

Remark 31.3 (Diagonal coordinates before positive repair). For a density input, the specific Q_B above also has its final diagonal entry in $[0, 1]$: the other diagonal entries are nonnegative and their sum is at most $\sum_{i < d} \rho_{ii} \leq 1$. This scalar observation does not imply positivity of Q_B ; the rank-one example above has an indefinite rounded matrix despite nonnegative diagonal entries. The buffer repairs the matrix, not a missing scalar diagonal bound. The constants in the repair error are convenient upper bounds and are not optimized.

32. ONE-PASS SIMULATION OF DISTURBING QUANTUM INSTRUMENTS

We now allow genuine state disturbance. This is a constructive finite-horizon statement, not a finite-state classification. It is separate from the classification by repeatable nondisturbing probes.

32.1. Fixed physical data and operational error. Fix $d \geq 2$, a finite command set $\mathcal{A}$, finite outcome sets, and an instrument $(\Phi_{a,y})_y$ for each $a \in \mathcal{A}$. Assume a fixed Gaussian-rational Kraus description

$$(32.1) \quad \Phi_{a,y}(X) = q^{-2} \sum_r M_{a,y,r} X M_{a,y,r}^*, \quad M_{a,y,r} \in \mathbb{Z}[i]^{d \times d}, \quad \sum_{y,r} M_{a,y,r}^* M_{a,y,r} = q^2 I$$

with one fixed positive integer q . The initial density matrix is P_0/z_0 , where P_0 is Gaussian integral and positive, and $z_0 = \text{tr } P_0 > 0$. Rationality here refers to this supplied finite description; no assertion that every rational superoperator has a Gaussian-rational Kraus decomposition is needed.

At each step an external classical controller chooses a command as a function of the public history, possibly using its own randomness. The instrument emits y and updates the physical state accordingly. The simulator receives the same kind of command and emits one outcome immediately. It cannot reread consumed commands or emitted outcomes; any information retained privately is charged. A common controller is used when comparing the two experiments. Its memory is not simulator memory, and no access to an unrecorded external history is granted to the simulator.

For a complete control–outcome history h at time t , let A_h be the true subnormalized state, of trace equal to the probability of h . Let $\hat{A}_h$ be the analogous simulator matrix. Define

$$(32.2) \quad \mathfrak{D}_t = \sum_h \|A_h - \hat{A}_h\|_1.$$

This is the trace norm of the difference of the block-diagonal classical–quantum states, not an average conditioned on a rare observed history. Histories of probability zero contribute the zero matrix. Classical transcript total variation is at most $\mathfrak{D}_t/2$. The same bound holds after any common, history-dependent terminal measurement. The compared interface consists of the classical transcript and its numerical matrix blocks. We do not claim total variation of binary matrix encodings, nor simulation of an arbitrary external entangled quantum memory.

Theorem 32.1 (Charged simulation of rational instruments). *For every fixed input (32.1) there is one randomized classical one-pass program which, for all $N \geq 2$ and $L \geq 2$, simulates every N -step adaptive control policy and satisfies*

$$(32.3) \quad \sup_{\text{policies}} \mathfrak{D}_t \leq 2^{-L} \quad (0 \leq t \leq N).$$

Its stored and returned numerical state is always in $\mathcal{D}_d$. Writing

$$s_* = \min\{N, L + \lceil \log_2(N+1) \rceil\},$$

its writable space is $O_{\text{input}}(s_)$, equivalently $O(d^2 s_*)$ with constants allowed to depend on the fixed Kraus data. Command processing uses $O_{\text{input}}(s_*^2)$ expected bit operations and $O_{\text{input}}(s_*)$ expected fair bits per step. Every step terminates almost surely, with a worst-case space bound even during rejection sampling. No exact-probability sampling oracle is used.*

Exact simulation of the entire adaptive experiment, including its conditional numerical matrix states, is possible in $O_{\text{input}}(N)$ writable bits. No positivity bound on outcome probabilities, reversibility, physical return, or spectral gap is assumed. For one-outcome instruments the program is deterministic.

The finite program and physical data are fixed before N, L are supplied. Parameter scanning is charged by its actual input length. The theorem concerns a classical numerical simulation, not preparation of a quantum state. It gives no uniform error bound for normalized states conditional on arbitrary rare histories.

32.2. An integer realization of every step. For approximate simulation choose

$$(32.4) \quad b = L + \lceil \log_2(6d^2(N+1)) \rceil, \quad B = 2^b, \quad m = B + 2d^2.$$

Store $\mathcal{R}_B(P_0/z_0)$ as P/m with an integral Hermitian numerator P . Given command a , form

$$(32.5) \quad S_y = \sum_r M_{a,y,r} P M_{a,y,r}^*, \quad s_y = \text{tr } S_y, \quad T = \sum_y s_y = q^2 m.$$

Emit y with probability s_y/T . Only positive s_y can be emitted. Replace the state by $\mathcal{R}_B(S_y/s_y)$, using the divisions in Lemma 31.1. Its denominator is again exactly m .

These rational probabilities have an explicit fair-bit implementation. Let $\ell = \lceil \log_2 T \rceil$, draw ℓ independent fair bits as an integer $J \in \{0, \dots, 2^\ell - 1\}$, and retry if $J \geq T$. If $T = 1$ take $J = 0$ without drawing a bit. Accept the unique outcome whose cumulative interval of integer lengths $(s_y)_y$ contains J . Acceptance probability is greater than $1/2$, except that for a power of two it is one. Thus the expected number of bits is at most 2ℓ . Discarded trials are not stored, and their count is not needed. Zero-length intervals can never be selected.

There is also an exact mode. Start from P_0 and, on each command, form $(S_y)_y$ by (32.5), now with $T = q^2 \text{tr } P$. Sample the same integer intervals and set $P \leftarrow S_y$. The numerical state is $P/\text{tr } P$. This update is precisely the normalized physical branch, while the numerator itself is never divided. After t commands,

$$(32.6) \quad 1 \leq \text{tr } P \leq z_0 q^{2t}.$$

It is an integer because all input matrices are Gaussian integral. The upper bound follows from $\text{tr } S_y \leq \sum_y \text{tr } S_y = q^2 \text{tr } P$. All entries have modulus at most $\text{tr } P$ by positivity.

32.3. Uniform control of the whole path.

Proof of Theorem 32.1. In approximate mode, each stored state is legal by Lemma 31.1, and every sampling row has total mass one by (32.1). For an arbitrary common controller, write $\pi(a | h)$ for its next command law. The true and approximate subnormalized updates are

$$A_{hay} = \pi(a | h) \Phi_{a,y}(A_h), \quad \hat{A}_{hay} = \pi(a | h) \bar{\mathcal{R}}_B(\Phi_{a,y}(\hat{A}_h)).$$

This expression agrees with the actual emitted probabilities and normalized successor matrices. For each history, the triangle inequality and Lemma 31.2 imply

$$\sum_{a,y} \|A_{hay} - \hat{A}_{hay}\|_1 \leq \|A_h - \hat{A}_h\|_1 + \delta_B \text{tr } \hat{A}_h.$$

Summing over histories and using $\sum_h \text{tr } \hat{A}_h = 1$ gives $\mathfrak{D}_{t+1} \leq \mathfrak{D}_t + \delta_B$. Initial rounding contributes $\mathfrak{D}_0 \leq \delta_B$. Hence

$$\mathfrak{D}_t \leq (t+1)\delta_B \leq (N+1)6d^2/B \leq 2^{-L}.$$

This proof works uniformly for every controller. A controller with private randomness may be conditioned on its seed, followed by integration; the displayed bound is independent of that seed. Taking traces gives the classical total-variation assertion. Appending a terminal instrument and applying (31.3) gives the assertion about terminal measurements. No division by a small target probability has appeared.

In grid mode, the numerator P and the denominator m have $O_{\text{input}}(b+1)$ bits per entry. The fixed integral Kraus products, the outcome traces, the cumulative intervals, and the integer J have the same order. The numerators in the signed divisions have at most $2b + O_{\text{input}}(1)$ bits. A constant number, depending on the fixed physical data, of matrix buffers suffices. Elementary long division gives quadratic bit time and linear scratch space. Rejection sampling uses at most $O_{\text{input}}(b+1)$ space in every trial, and has the stated expected bit and time bounds.

Exact mode reproduces every branch probability and conditional state exactly. Equation (32.6), together with the fixed Kraus coefficients, bounds all stored and temporary integers by $O_{\text{input}}(t+1)$ bits. Thus the exact simulation uses linear space and has the stated conservative quadratic time bound per step.

Use grid mode when $b < N$ and exact mode otherwise. The choice is made before allocating B : store N , scan L with its value capped at N , and compute the threshold with binary integer lengths. A precision string of arbitrary length is consumed without being stored. A counter verifying the number of commands uses $O(\log N)$ bits. The fixed-radix numerical matrix output has $O_{\text{input}}(s_*)$ bits and requires no readable transcript tape. These observations complete the charged-space proof. $\square$

Corollary 32.2 (Rational channels with legal matrix outputs). *A fixed finite family of Gaussian-rational Kraus channels and a fixed rational initial density matrix admit one deterministic one-pass algorithm with pointwise trace-norm error at most 2^{-L} at every prefix, legal outputs, and space $O_{\text{input}}(\min\{N, L + \log(N+1)\})$. Exact output uses $O_{\text{input}}(N)$ bits.*

Proof. There is one outcome per command, so its probability is one and no random bits are needed. The single block in (32.2) is the difference of the two numerical matrices. Apply Theorem 32.1. $\square$

The contraction and hybrid argument are classical. Their use here is to give a fully specified positive integer simulator with bounded space, exact rational emissions and no rare-outcome assumption. This constructive theorem does not replace the repeatable-probe hypothesis in the finite-response-quotient classification.

33. SHARP SPACE WITH HIGHER-DIMENSIONAL DENSITY-MATRIX OUTPUTS

Positive rounding removes the restriction of the uniform upper construction to a Bloch ball. We retain the spectral hypotheses in the converse; none is needed by the algorithm.

Theorem 33.1 (Sharp matrix-output space). *Fix a finite family of Gaussian-rational unitary matrices in $U(d)$ and a Gaussian-rational pure initial state ρ_0 . Let G be the compact closure of their conjugation action on the real Euclidean space*

$$E = \{H \in \mathbb{C}^{d \times d} : H = H^*, \text{tr } H = 0\}, \quad \|H\| = \|H\|_F.$$

Assume a probability law supported on the given commands has a full Koopman action gap on the unit sphere of E , off all invariant functions, and an all-direction cap estimate (16.4) of exponent $p > 0$. The dimension, matrices, seed, rational denominators, cap constants and spectral constant are fixed independently of N, L .

For a single fixed-program classical one-pass simulator, assume almost-sure termination between commands and at final output, and require a legal numerical output $\hat{\rho}(w) \in \mathcal{D}_d$ on every run and only the mean condition

$$(33.1) \quad \sup_{|w|=N} \|\mathbb{E}\hat{\rho}(w) - U_w \rho_0 U_w^*\|_F \leq 2^{-L}.$$

For every such fixed program, a matching space lower bound holds for $L \geq 2$ and all sufficiently large N , with constants and threshold allowed to depend on that program and the fixed physical data. A single uniform algorithm attains the upper bound. In this fixed-program asymptotic sense, the optimal writable bit-space has order

$$(33.2) \quad \Theta_{\text{input}}(\min\{N, L + \log_2(N+1)\}).$$

The lower bound permits arbitrary randomization and unlimited internal running time. A deterministic integer program attains the upper bound with pointwise trace-norm error. Exact computation has order N . No idle or return assumption is made.

Proof. Put $c = I/d$ and $r_d = \sqrt{(d-1)/d}$. For every density matrix D ,

$$\|D - c\|_F^2 = \text{tr}(D^2) - 1/d \leq r_d^2;$$

equality holds for a pure state. Thus the physical target $X_w = (U_w \rho_0 U_w^* - c)/r_d$ is unit, and every normalized legal decoder $(D - c)/r_d$ has norm at most one.

The complete-configuration reduction of Lemma 30.2 works with $\mathcal{D}_d$ in place of $\mathcal{D}_2$: almost-sure termination gives stochastic rows, and conditional means of legal outputs remain legal by compact convexity. The induced rows may depend on the public boundary index, which only enlarges the lower-bound comparison class. The proof of Theorem 16.2 uses the decoder set only through the unit-norm bound. With error ε/r_d and $0 \leq \varepsilon \leq 1/4$, its finite budget therefore gives

$$(33.3) \quad N \leq C_0 + C_1 \log k + C_2 \varepsilon k^{2/p}$$

for a simulator with at most k complete private configurations at every positive boundary. Constants depend only on the fixed data. Indeed the residual mass is $1 - \varepsilon/r_d$, bounded away from zero uniformly in this error range.

For $N \geq 2C_0$, either $\log k \geq N/(4C_1)$ or $k^{2/p} \geq N/(4C_2\varepsilon)$. Taking logarithms yields

$$\log_2 k \geq c \min\{N, L + \log_2(N+1)\} - C'.$$

Since the minimum tends to infinity with N , uniformly for $L \geq 2$, the additive constant is absorbed above a fixed physical-data-dependent horizon. For a fixed finite program, the complete-configuration count is at most $C(s+1)^{a2^{bs}}$ when s work bits are used, with fixed machine-dependent constants. After absorbing the fixed finite-control factor C , this proves the work-space lower bound above a program-dependent horizon. No bound uniform over arbitrarily large uncharged finite programs is asserted. At zero error (33.3) gives $\log k \geq cN - C'$.

Regard each unitary as a one-outcome rational Kraus instrument. Corollary 32.2 supplies the deterministic trace-norm upper bound and the exact fallback. Trace norm dominates Frobenius norm. The algorithm does not use the spectral or cap constants, nor construct a stochastic orbit-hull table. $\square$

Corollary 33.2 (An explicit family in every fixed dimension). *For every fixed $d \geq 2$, take the adjacent-coordinate rational unitaries displayed in Section 17, their inverses and the identity, and take $\rho_0 = e_1 e_1^*$. Legal numerical density matrix simulation satisfies (33.2), with dimension-dependent constants.*

Proof. The density argument in that section shows that the generated compact group is $SU(d)$. All entries are algebraic. The Bourgain–Gamburd theorem [16], alternatively the Benoist–de Saxcé extension [38, Theorem 1.2], gives a regular-representation spectral gap for this fixed dense algebraic set. Symmetrization and laziness give a contraction norm strictly less than one. The identity is already an allowed command, so this is an averaging measure on actual letters. Peter–Weyl decomposition transfers that norm to the action off its invariant functions. For conjugation on traceless Hermitian matrices, the fixed space is zero. The least orbit dimension calculation in Theorem 24.1 gives $p_* = 2(d-1)$ and its uniform cap estimate. Theorem 33.1 applies. $\square$

This corollary does not supply a numerical spectral constant uniformly in d . It uses an explicit rational alphabet, not an unspecified existence argument. The construction for a given alphabet is effective without knowing its spectral gap; that gap is a premise of the lower proof only.

Corollary 33.3 (Sharpness for an instrument family containing an expanding unitary subsystem). *Suppose the fixed rational instrument family of Theorem 32.1 contains the unitary commands and pure seed of Theorem 33.1. Among simulators that additionally return a legal numerical matrix and satisfy the classical–quantum error requirement (32.3), the optimal space has order (33.2).*

Proof. The instrument simulator gives the upper bound. For the lower bound, restrict the controller to a deterministic word of unitary commands. Their outcome sets are singletons, so the error requirement implies (33.1). Apply the preceding theorem. This is not a lower bound for transcript-only simulators without a legal numerical matrix interface. $\square$

34. STRICT CUTPOINTS AND NUMERICAL SIMULATION

State cost for recognition and numerical precision are different invariants. Recent results make it necessary to state the distinction at theorem level, rather than relying on the words “classical simulation.”

Chen–Wu [39, Definition 2.1 and Theorems 3.2, 4.4] define simulation by equality of strict-cutpoint languages, allowing the simulator’s cutpoint to change. Their August 2026 version gives the exact worst-case cost $n^2 + 1$ for n -state one-way general quantum automata, $n \geq 2$. In [40, Theorems 11, 14 and Corollary 15] they give order cq^2 for $c, q \geq 2$ in a model with c classical states and a q -dimensional quantum register, and order n^2 for the measure-once model. These are relevant positive results. The first paper’s current title is *The State Cost of Classical Simulation of One-Way General Quantum Finite Automata*; its version and exact bound, rather than the earlier title or only its quadratic order, are used here.

Their stochasticization preserves a sign, not the numerical acceptance function. In particular, [39, Proposition 3.1] contains an attenuation factor $\delta\eta^{|w|} > 0$ multiplying a zero-cutpoint linear output. It can tend to zero exponentially without changing the recognized language. The following elementary construction isolates this point.

Proposition 34.1 (Attenuation preserves a strict cutpoint). *Let a probabilistic automaton have acceptance function $f(w)$, and fix $0 < \beta < 1$ and $0 \leq \lambda < 1$. Adding two absorbing states gives an automaton with acceptance function*

$$(34.1) \quad f_\beta(w) = \lambda + \beta^{|w|}(f(w) - \lambda).$$

It recognizes the same strict-cutpoint language at λ . If some length- N word has $|f(w) - \lambda| \geq c > 0$, its numerical error on that word is at least $(1 - \beta^N)c$.

Proof. On every letter, each old state follows its original stochastic row with total probability β . The remaining mass goes to an absorbing accepting state with probability $(1 - \beta)\lambda$ and to an absorbing rejecting state with probability $(1 - \beta)(1 - \lambda)$. The old terminal functional is unchanged. After t letters the surviving old distribution is β^t times the original distribution, while the absorbed accepting mass is $\lambda(1 - \beta^t)$. This proves (34.1). The factor $\beta^{|w|}$ is positive, proving language equality. Subtracting $f(w)$ proves the last assertion. $\square$

A legal density-matrix approximation gives a numerical guarantee for every supplied effect: if $0 \leq E \leq I$ and $\rho, \sigma \in \mathcal{D}_d$, then

$$|\operatorname{tr} E(\rho - \sigma)| \leq \frac{1}{2} \|\rho - \sigma\|_1.$$

Indeed the positive and negative parts of $\rho - \sigma$ have equal trace, and an effect takes a value between zero and that trace on either part. Therefore an approximation preserves a threshold decision whenever its numerical error is strictly smaller than the distance from the cutpoint. Without a margin, numerical approximation alone does not give exact language equivalence.

The prepare–test and sign-rank methods in [40] concern finite prefix–suffix sign patterns. Such patterns are unchanged by positive attenuation. Our entropy converse instead requires a calibrated residual mean and a legal convex decoder; attenuation can destroy that calibration. Conversely, its width lower bound cannot be transferred to unrestricted cutpoint-language simulation. Constant real linear dimension, fixed PFA state count and finite writable bit-space are not identified with one another. No quantum language separation, resolution of the Chen–Wu state-cost problem,

or unconditional lower bound for arbitrary noisy channels is claimed. The new positive process construction is an upper theorem for a different, explicitly specified numerical and transcript interface.

35. RATIONAL INSTRUMENTS STABLE UNDER EXTERNAL REFERENCES

A rounding operation on density matrices need not be affine. In particular, the map of Lemma 31.1 cannot be tensored with the identity on an unknown reference system. We therefore round a *description of the instrument*, once and independently of its input, rather than round the state on which it acts. The resulting maps are completely positive and jointly trace preserving. All constants below are explicit in the input and output dimensions and the number of outcomes.

35.1. An integral repair of Choi data. Write M_d for the complex matrices of order d . An instrument from M_d to M_n with m outcomes is a family of completely positive maps Φ_y whose sum is trace preserving. We use the input-first, *unnormalized* Choi convention

$$(35.1) \quad J_y = \sum_{i,j=1}^d |i\rangle\langle j| \otimes \Phi_y(|i\rangle\langle j|), \quad J_y \geq 0, \quad \sum_{y=1}^m \text{Tr}_n J_y = I_d.$$

Here Tr_n traces the output factor; in particular $\sum_y \text{tr} J_y = d$, not one. These are the classical Choi criteria [37, Theorems 2.22 and 2.26]. The channel with the outcome retained is $\mathcal{F}(X) = \bigoplus_y \Phi_y(X)$. We write $\|\cdot\|_\diamond$ for the completely bounded trace norm.

Lemma 35.1 (Choi trace norm and reference stability). *For any linear map $\Lambda : M_d \rightarrow M_n$ with unnormalized Choi matrix $J(\Lambda)$,*

$$d^{-1} \|J(\Lambda)\|_1 \leq \|\Lambda\|_\diamond \leq \|J(\Lambda)\|_1.$$

Consequently the diamond distance of two instruments, with the outcome retained, is at most the sum of the trace norms of their Choi-block differences.

Proof. Applying $\text{id}_d \otimes \Lambda$ to the normalized maximally entangled state gives the lower bound. For the upper bound, let u, v be unit vectors in $\mathbb{C}^r \otimes \mathbb{C}^d$. With $\Omega = \sum_i e_i \otimes e_i$, write $u = (A \otimes I_d)\Omega$ and $v = (B \otimes I_d)\Omega$. The matrices $A, B : \mathbb{C}^d \rightarrow \mathbb{C}^r$ have Hilbert–Schmidt norm one and therefore operator norm at most one. Thus

$$(\text{id}_r \otimes \Lambda)(|u\rangle\langle v|) = (A \otimes I_n)J(\Lambda)(B^* \otimes I_n)$$

has trace norm at most $\|J(\Lambda)\|_1$. The trace-norm unit ball is the convex hull of rank-one partial isometries (including zero as a convex combination of opposite such matrices). A singular-value decomposition of an arbitrary input matrix therefore proves the same induced trace-norm bound, independently of r . Taking the supremum over r proves the assertion. For an instrument the Choi matrix is, up to a fixed permutation of factors, the direct sum of its blocks. $\square$

Let B be a positive integer. The supplied approximations T_y are Hermitian matrices with entries in $B^{-1}\mathbb{Z}[i]$. Assume that every independent real diagonal coordinate, and every independent real and imaginary off-diagonal coordinate, differs from that of J_y by at most B^{-1} . The T_y need not be positive or satisfy the marginal constraint. Put

$$(35.2) \quad R = I_d - \sum_y \text{Tr}_n T_y, \quad Q_y = T_y + \mathbf{1}_{\{y=1\}} R \otimes |e_1\rangle\langle e_1|,$$

$$(35.3) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad \tilde{J}_y = \frac{BQ_y + cI_{dn}}{D}.$$

The choice of the first outcome and first output coordinate is merely a fixed convention. Any fixed outcome and output basis coordinate give the same legality and uniform error bound. If specified

zero outcomes must remain exactly zero, remove them before applying the theorem and restore them afterwards; m then counts only the repaired outcomes.

Theorem 35.2 (Reference-stable rational instrument rounding). *For every $d, n, m, B \geq 1$, the matrices in (35.3) are Choi matrices of an instrument. Their numerators are positive semidefinite matrices with Gaussian-integer entries, their common denominator is D , and*

$$(35.4) \quad \|\mathcal{F} - \tilde{\mathcal{F}}\|_{\diamond} \leq \sum_y \|J_y - \tilde{J}_y\|_1 \leq \frac{3mndc}{B + mnc} \leq \frac{K(d, n, m)}{B}, \quad K(d, n, m) = 6mn^2d^2(1 + mn).$$

The construction uses integer additions, multiplication by specified integers, and the supplied grid entries. It uses no spectral factorization, positivity projection, or rational Kraus factorization. Its complete output description has

$$O(md^2n^2 \log(B + mnc + 1))$$

bits, with an absolute implicit constant for the fixed encoding.

Proof. Set $E_y = T_y - J_y$. Counting the real coordinates gives $\|E_y\|_{\text{op}} \leq \|E_y\|_F \leq 2nd/B$. For a Hermitian E , the inequalities $-\|E\|_{\text{op}}I \leq E \leq \|E\|_{\text{op}}I$ imply $\|\text{Tr}_n E\|_{\text{op}} \leq n\|E\|_{\text{op}}$. It follows that

$$\|R\|_{\text{op}} \leq 2mn^2d/B, \quad \|Q_y - J_y\|_{\text{op}} \leq c/B.$$

Since $J_y \geq 0$, the numerator $BQ_y + cI_{dn}$ is positive. All its entries are Gaussian integers. The partial trace constraint is exact, because

$$\sum_y \text{Tr}_n(BQ_y + cI_{dn}) = (B + mnc)I_d.$$

The Choi criteria therefore give a completely positive instrument, including when some of the original outcomes are zero.

Let $F_y = Q_y - J_y$. The identity

$$D(\tilde{J}_y - J_y) = BF_y + cI_{dn} - mncJ_y$$

and $\|F_y\|_1 \leq dnc/B$ give

$$D \sum_y \|\tilde{J}_y - J_y\|_1 \leq mndc + mndc + mnc \sum_y \text{tr } J_y = 3mndc.$$

Lemma 35.1 proves the norm assertion. Finally each diagonal entry of a positive numerator is at most D , by the summed partial trace constraint. Positivity bounds every off-diagonal modulus by the geometric mean of two diagonal entries, hence also by D . Encoding $m(dn)^2$ entries and the common denominator proves the bit bound. All intermediate entries have polynomially bounded bit length in the stated parameters and the supplied data. $\square$

For exact rational input, directed truncation supplies T_y by integer division. For uncertain real input, the hypothesis is a certified error bound on the *final grid coordinates*; an uncertified floating estimate is not a premise of the theorem. Although B may be a power of two, $D = B + mnc$ generally is not. The output is rational, not in general dyadic. The construction may replace a zero outcome by a small nonzero one; its mass is charged in (35.4). Known zero outcomes can instead be removed before rounding and restored as zero blocks afterwards, with m counting the remaining outcomes.

Corollary 35.3 (A finite-description instrument net). *For fixed d, n, m and $0 < \delta \leq 1$, the preceding construction yields a finite diamond- δ net of legal instruments. Each member has an integer description of length*

$$O(md^2n^2 \log(2 + K(d, n, m)/\delta + mnc)).$$

This is a deterministic description net, not a statistically learned net. A particular member is computable in polynomial bit time from rational input data and the grid precision; enumeration of the entire net is not required.

Proof. Every Choi diagonal is at most one by (35.1), and positivity bounds every entry modulus by one. There are finitely many possible directed grid truncations at $B = \lceil K/\delta \rceil$. Apply the theorem to those arising from valid instruments. Computation of one truncation and its repair uses a polynomial number of elementary integer operations on polynomial-length operands. $\square$

The CP/TP constraints and adding a scalar identity buffer are not new principles. Projected least-squares process tomography already repairs Choi estimates and uses identity mixing to finish a TP-preserving positivity correction [42, Section 4]. Here the particular integer formula supplies an explicit denominator and a deterministic diamond-error budget from coordinate errors. No nearest-projection, statistical sample-complexity, or rank-optimality claim is made.

35.2. Adaptive use and stopped observations. A tester may retain an arbitrary finite-dimensional quantum memory, start with a state entangled across that memory and the instrument input, and insert arbitrary channels and finite-outcome measurements between uses. The command port is classical: its value may be selected from the public history and measured tester memory. Coherent access to a superposition of commands is not an interface assumed here. The *same* tester is used in the two compared experiments.

Theorem 35.4 (Adaptive reference-stable precision budget). *Suppose that at use t , uniformly over permitted commands and public histories, $\|\mathcal{F}_{t,a} - \tilde{\mathcal{F}}_{t,a}\|_{\diamond} \leq \delta_t$. For every common tester with a classical command port (not coherent superpositions of commands), every initial state, and every public stopping rule $\tau \leq N$, the final states, with the stopped history and tester memory retained, satisfy*

$$(35.5) \quad \|\Omega_{\tau} - \tilde{\Omega}_{\tau}\|_1 \leq \sum_{t=1}^N \delta_t.$$

The bound is independent of the tester-memory dimension. In particular, rounding each command by Theorem 35.2 with $B \geq NK(d, n, m)2^L$ gives stopped joint-state error at most 2^{-L} and transcript total-variation error at most 2^{-L-1} .

Proof. Replace the instruments one use at a time. Just before the replaced use, both members of a neighboring hybrid pair have the same joint state. Decomposing a classical command/history register into blocks, the diamond bound applies to each positive block and is multiplied by its trace. These traces sum to one. The error introduced at this use is therefore at most δ_t , including any quantum reference. All subsequent operations in the hybrid pair are the same channels; complete trace-norm contraction bounds their final difference by the same quantity. Summing the N hybrid differences proves (35.5). For a stopping rule retain an absorbing halt flag, use the identity on halted blocks, and finish all experiments at use N . A public history-dependent stopping rule is then part of the same tester. The same argument applies, without conditioning on a possibly rare stopping history. Taking traces and dividing by two gives the total-variation assertion. $\square$

The hybrid principle is classical; see [37, Section 3.3, equation (3.306)] and the operational strategy norms of [43]. The conclusion here is a uniform finite-description approximation by *genuine instruments*. It is not a classical device that accepts an unknown physical entangled state and prepares its output. A numerical simulation of a specified joint state must charge the joint matrix dimension. Nor does the nonlinear state repair in the preceding section become a channel by this theorem.

Proposition 35.5 (Posterior control at a stopping time). *Let $A_h = p_h \rho_h$ and $\hat{A}_h = q_h \hat{\rho}_h$ be the positive blocks at a common bounded stopping time, with both total traces one. At zero simulated mass assign any density matrix to $\hat{\rho}_h$. If $\sum_h \|A_h - \hat{A}_h\|_1 \leq \delta$, then*

$$(35.6) \quad \sum_h p_h \|\rho_h - \hat{\rho}_h\|_1 \leq 2\delta, \quad \Pr_p\{\|\rho_h - \hat{\rho}_h\|_1 > \eta\} \leq \min\{1, 2\delta/\eta\}.$$

The tail estimate holds for every $\eta > 0$. For any event of true probability $\alpha > 0$, the two pooled conditional states have distance at most $\min\{2, 2\delta/\alpha\}$, with an arbitrary conditional state when the simulated event has probability zero.

Proof. Writing $e_h = \|A_h - \hat{A}_h\|_1$, trace contraction gives $|p_h - q_h| \leq e_h$. For positive p_h ,

$$p_h \|\rho_h - \hat{\rho}_h\|_1 \leq e_h + |p_h - q_h| \leq 2e_h.$$

The inequality remains valid if $q_h = 0$. Sum and apply Markov's inequality. Applying the same calculation to the sums over an event gives the final assertion; two density matrices have distance at most two. Zero true mass makes no contribution. $\square$

The proposition also applies to the numerical history metric of Theorem 32.1. For its stopped version, freeze every halted block and apply the same contraction and mass-weighted repair argument only to live blocks. At each use their total mass is at most one, so the bound remains $(N+1)\delta_B$. Thus the theorem provides a high-probability posterior guarantee at a chosen stopping time without a probability floor or a union bound over all times. It does not assert simultaneous accuracy of every rare normalized posterior. This distinction is relevant when comparing finite-bit history approximation with numerical quantum filtering [41, Section II, equations (10)–(13)].

36. UNIFORM STREAMING FROM RATIONAL CHOI DESCRIPTIONS

The preceding instrument description is not supplied with a Kraus factorization over the Gaussian rationals. Computing such a factorization is unnecessary for classical numerical trajectories. We now allow the matrix dimension and the entire rational instrument to be input to one program, rather than fixed constants of that program.

36.1. Description and exact update. Fix an encoding of finite lists of signed binary integers. A valid description consists of $d \geq 2$, a positive semidefinite initial matrix with Gaussian-integer entries P_0 with trace $z_0 > 0$, and a finite command list. For each command a it gives a positive integer D_a and positive semidefinite matrices with Gaussian-integer entries $C_{a,y}$ of order d^2 such that

$$(36.1) \quad \sum_y \text{Tr}_d C_{a,y} = D_a I_d.$$

A zero outcome is simply a zero Choi matrix. The associated instrument has input-first, unnormalized Choi blocks $C_{a,y}/D_a$ and initial state P_0/z_0 . Let S be the total encoded length, including dimensions, delimiters, command identifiers and all matrix entries. The entire description is retained and charged. In particular the longest identifier is at most S bits and bounds the parser's temporary token store; a free external read-only description would define a different space model. For a stored numerator P define

$$(36.2) \quad (S_{a,y})_{\alpha\beta} = \sum_{i,j=1}^d (C_{a,y})_{i\alpha,j\beta} P_{ij}.$$

The indices in this formula are important: no conjugation or transposition of P_{ij} is inserted.

Lemma 36.1 (Integral Choi trajectory update). *In the input-first, unnormalized convention $J(\Phi) = \sum_{i,j} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|)$, for positive semidefinite P with Gaussian-integer entries the matrices in (36.2) are positive semidefinite matrices with Gaussian-integer entries and*

$$\sum_y \text{tr } S_{a,y} = D_a \text{tr } P.$$

The conditional outcome probabilities are $\text{tr } S_{a,y} / (D_a \text{tr } P)$, and the next normalized state is $S_{a,y} / \text{tr } S_{a,y}$ on a positive-mass outcome.

Proof. The Choi convention gives $S_{a,y} = D_a \Phi_{a,y}(P)$, so positivity follows from complete positivity, and integrality follows from the entry formula. Taking traces and using (36.1) gives the identity. The scalar factors D_a and any accumulated previous factors cancel from both probabilities and normalized states. A positive matrix of trace zero is zero and is never sampled. $\square$

Put $D_* = \max_a D_a$ and, using logarithms to base two in this section,

$$(36.3) \quad \begin{aligned} H &= \lceil \log_2(D_* + 1) \rceil, & \ell_0 &= \lceil \log_2(z_0 + 1) \rceil, \\ Q &= \ell_0 + NH, & b &= L + \lceil \log_2(6d^2(N + 1)) \rceil, & u &= \min\{Q, b\}. \end{aligned}$$

The description, once validated, may be kept read-only; charging its entire length S is included in the bound below.

Lemma 36.2 (Reduced Schur entries and bit length). *Let A be a Hermitian matrix of order q with Gaussian-integer entries, whose real and imaginary parts have absolute value at most 2^h . In exact Schur elimination, let I be the ordered indices of the positive pivots already eliminated. Every remaining entry is*

$$(36.4) \quad S_{ij} = \frac{\det A[I \parallel i, I \parallel j]}{\det A[I, I]},$$

where $I \parallel i$ appends i in that order, and the empty principal determinant is one. Reduced numerators and denominators have $O(q(h + \log(q + 1)))$ bits. Hermitian positive semidefiniteness is therefore decidable by this elimination in polynomial bit time and space in the explicit input length. No optimized polynomial exponent is asserted.

Proof. When the leading pivot t is positive, block congruence reduces

$$\begin{pmatrix} t & b^* \\ b & C \end{pmatrix} \geq 0 \iff C - bb^*/t \geq 0.$$

A negative pivot rejects. At a zero diagonal, each 2×2 principal minor forces the corresponding row and column to vanish; otherwise reject, and if they vanish delete that row and column. Positive pivots make $A[I, I]$ positive definite. The bordered determinant identity

$$\det \begin{pmatrix} A[I, I] & A[I, j] \\ A[i, I] & A_{ij} \end{pmatrix} = \det A[I, I] (A_{ij} - A[i, I] A[I, I]^{-1} A[I, j])$$

proves (36.4), including after zero rows are skipped. The specified ordering prevents a spurious permutation sign.

A minor of order $k \leq q$ has determinant modulus at most $k^{k/2}(\sqrt{2}2^h)^k$, by the product of its row norms. Its real and imaginary parts are integers with $O(k(h + \log(k + 1)))$ bits. The principal denominator is a positive integer. Cancelling common factors cannot enlarge these lengths. Each arithmetic operation on reduced Gaussian-rational entries thus has polynomial operand size. There are polynomially many operations and at most q^2 stored entries; ordinary multiplication, division and the Euclidean algorithm give the stated bit bounds. For rational rather than integral input, first clear a common positive denominator. Its bit length is at most the sum of the encoded denominator lengths, so this preliminary step also has polynomial cost. $\square$

Theorem 36.3 (Variable-description numerical simulation). *There is a single classical randomized program with the following property. Given any valid description above and $N, L \geq 2$, it consumes an adaptive command stream once, emits each outcome immediately, and retains a legal numerical density matrix. Against every common classical adaptive controller, its subnormalized history error at every prefix, and at every common public stopping time bounded by N , is at most 2^{-L} .*

Description validation has polynomial bit time and polynomial working space in S . After validation, the maximum writable space, including temporary arithmetic and the stored description, is

$$(36.5) \quad O(S + d^2[H + \log(d + 1) + u] + \log(N + 1)),$$

with an absolute implicit constant for the fixed machine encoding. The total peak space is the maximum of this bound and the stated polynomial preprocessing bound. Expected bit time per command is $\text{poly}(S)(1 + u)^2$ and expected fresh fair-bit consumption is $O(H + u + \log(d + 1))$. The space bound holds on every rejection trial; running time is not bounded on every such execution. The theorem requires neither a spectral gap nor a Kraus factorization.

Proof. In grid mode use $B = 2^b$ and retain the integer numerator of $R_B(\rho)$ from Lemma 31.1; its trace is always $B + 2d^2$. At a command compute the branch integers in (36.2), sample an outcome from their exact integer traces, and repair the chosen normalized branch back to the same state denominator. The sampler uses fresh binary words of length $\lceil \log_2(D_a \text{tr } P) \rceil$, rejecting words outside the required integer interval. Its acceptance probability is greater than one half, except for the power-of-two case where it is one. Rejected words are discarded, not recorded. Branch weights can be computed sequentially; a second scan finds and recomputes the selected branch. Thus the number of retained matrices does not grow with the number of outcomes.

The proof of Theorem 32.1 uses positivity and (36.1), not rational Kraus entries. It therefore applies verbatim at the level of its stated block recursion: the initial error is at most $6d^2/B$, and each subsequent use adds at most $6d^2/B$, weighted by total simulated branch mass. It gives

$$\sum_h \|A_h - \hat{A}_h\|_1 \leq (t + 1)6d^2/B \leq 2^{-L} \quad (t \leq N).$$

The outcome probabilities used here are computed from the approximate stored state, not from the unknown exact state. The block recursion accounts for that difference. Freezing halted branches proves the same stopped bound as in Proposition 35.5.

In exact mode retain the selected unnormalized matrix without repair. This numerator represents a path block up to omitted common scalar denominators. Those factors cancel in every conditional branch probability and normalized output; they are not reconstructed or stored merely to restore a redundant normalization. Positivity and the trace identity imply

$$1 \leq \text{tr } P_t \leq z_0 D_*^t < 2^{\ell_0 + tH}.$$

Every entry modulus is bounded by the trace. This mode has zero approximation error. Choose it when $Q \leq b$, and choose grid mode otherwise. The switch is an order bound, not an exact minimization of small-instance memory.

For space accounting, each Choi numerator entry has modulus at most D_a , by positivity and (36.1). Each summand in (36.2) is a product of integers of lengths at most H and $u + O(\log(d + 1))$. An accumulator over d^2 summands needs at most $H + u + O(\log(d + 1))$ bits. A constant number of $d \times d$ matrices, traces, indices, division buffers and one rejection word suffice. Initial grid rounding reads the initial entries sequentially; its extra operand length ℓ_0 is at most the charged input length S . Counters and address arithmetic give (36.5). Classical multiplication and long division on w -bit integers take $O(w^2)$ bit operations and $O(w)$ scratch. Scanning the explicit outcome list a fixed number of times gives the claimed polynomial factor in S ; powers of H are

absorbed there. The sampling bound follows from the rejection interval length and the expected number of trials, which is below two. A single nonzero branch requires no random bits.

Validation uses the reduced-rational Schur procedure of Lemma 36.2, not the unreduced fixed-dimensional scaled recursion. Its ratios-of-minors formula proves polynomial operand lengths. Hermitian symmetry, the exact partial trace identities and positive initial trace are checked separately. No pivot permutation is needed: a zero leading diagonal of a positive semidefinite matrix forces its entire row and column to vanish. This establishes a separate polynomial preprocessing bound, not a validator matching the subsequent streaming-space bound.

Parameters are parsed before commands. The precision L may be read with saturation at Q : once that threshold is reached, exact mode is selected and its exact value is unnecessary. The time to read all parameter digits is charged as preprocessing in their actual length; no equally long precision integer is retained. A binary counter checks the command count. The numerical output consists of signed binary integer entries and their common denominator; writing it requires no additional matrix-sized precision store. $\square$

Corollary 36.4 (Known joint systems and two error budgets). *Suppose the entire joint system to be simulated, including a specified reference and all specified controller operations, has a rational Choi description on dimension d_{joint} . Theorem 36.3 applies with that dimension and the full description length charged. If its instruments were first compiled by Theorem 35.2, the reference-stable description error and the numerical history error add. In particular, allocating $\varepsilon/2$ to each gives total history error at most ε .*

Proof. Use the joint Choi maps and joint initial state as input to the uniform program. Theorem 35.4 bounds the first replacement, and Theorem 36.3 bounds the numerical simulation of the replaced experiment. The triangle inequality gives the sum. An unspecified quantum tester remains covered by the *instrument-description* comparison only; a classical numerical simulation of it is not thereby supplied at no cost. $\square$

Theorem 36.3 is an upper bound with variable data. The lower bounds of Theorem 33.1 retain their own fixed-program quantifiers, numerical matrix interface, and full-action spectral and cap hypotheses. No dimension-uniform spectral constant, no sharp lower bound for arbitrary noisy transcript-only interfaces, and no classification of disturbing instruments follows from the upper construction. Conversely, a rational Choi input is enough to execute the construction even when a Gaussian-rational Kraus list is not supplied. This is a representation-level enlargement of the constructive theorem, not an alteration of the inherited lower model.

37. MEASUREMENT-DESCRIPTION OVERVIEW

A reusable description specifies one quantum instrument whose behaviour remains close to the supplied target through every allowed interaction. This problem depends not only on how many real parameters the target has, but on how repeated use resolves its different directions. We study the transition from noisy to extremal readout, together with singular conditional preparations and the geometry of reusable codes.

For two memoryless instruments of the same interface, define

$$d_N(\Phi, \Psi) = \sup_{\mathbf{T}} \|\mathbf{T}[\Phi] - \mathbf{T}[\Psi]\|_1.$$

The same causal tester $\mathbf{T}$ is used in both hypotheses. It may have initial reference entanglement, arbitrary finite quantum memory, feedback and a public stopping time bounded by N . The output includes its retained classical and quantum systems. Thus the distance is unhalved and has maximum two; event probabilities differ by at most $d_N/2$. Each device invocation uses the same memoryless instrument. Covering centres are legal instruments, not arbitrary arrays.

The unbiased-slice result concerns the ordered binary measurements $E_x^\pm = (I \pm x \cdot \sigma)/2$, $|x| \leq 1$. Theorem 49.1 supplies one finite-use metric comparison through both the erased and projective boundaries. It treats visibility as part of the target, rather than a public constant. The proof combines an exact radial binomial reduction with the retained same-visibility angular theorem. A monotonicity argument prevents cancellation between radial and angular perturbations; a simple triangle estimate alone would not establish the lower bound.

Theorem 49.3 turns this comparison into a geometric criterion on any compact Ahlfors-regular identifiable subset of the ball. Its weighted volume counts the number of local resolution cells. If the measure of the depth- t boundary layer has order t^α , Corollary 49.4 distinguishes $\alpha > k/2$, $\alpha = k/2$ and $\alpha < k/2$. Thus the horizon exponent depends on boundary contact, not just parameter dimension. For the disk the critical equality gives an additional $\log N$ in the codebook size; the ball has boundary-dominated order N^2 . The rational construction in Theorem 49.6 realizes both orders with a single charged index and no irrational normalization of the supplied vector. These are small-error covering theorems with arbitrary legal lower centres and family-valued upper centres. The biased-body extension is given in Theorems 50.1 and 51.1; the regular-subfamily criterion here retains its unbiased-slice hypotheses.

Single-shot discrimination of white-noise mixtures, including distinct noise levels, is explicitly treated by Sedláč-Ziman [57, Section VI, Example 4]. The optimized single-shot projective measurement distance is identified by Puchała et al. [58, Theorem 1]. We do not claim those reductions, classical simulation, or the metrological noise crossover as new. The new conclusions are the finite-use joint metric, weighted family-covering criterion, contact law and exact rational description.

We next recall the earlier fixed-visibility and instrument results, whose complete proofs are retained. The fixed-visibility family consists of the labelled qubit readouts

$$E_{\theta,\lambda}^\pm = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)).$$

The visibility $\lambda \in [0, 1]$ is public and may vary with N . Theorem 48.1 gives, with absolute constants,

$$d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \asymp \min\{1, K_{N,\lambda} h(\theta, \phi)\}, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, (1 - \lambda^2)^{-1}\}}.$$

Consequently the optimal supplied description has length $\log_2(1 + K_{N,\lambda}/\delta) + O(1)$ for $0 < \delta \leq 2^{-10}$, uniformly in N, λ, δ . At zero visibility the family is exactly a singleton. At unit visibility its readout resolution is N^{-1} . Between these endpoints, $N(1 - \lambda^2)$ determines the transition. In particular, a noise sequence $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ approaching zero gives horizon coefficient $(1 + \min\{\alpha, 1\})/2$ at fixed small error. This is a statement about the operational covering metric, not a universal parameter-count rule at a boundary.

The upper estimate represents each pair by common projective endpoints with two classical programme laws and retains their exact finite-use fidelity. It controls reference assistance and all adaptive testers. The lower estimate uses blocks of size comparable to $\min\{N, (1 - \lambda^2)^{-1}\}$, or shorter blocks when angular separation requires them. An explicit Bernoulli product estimate then gives a finite-distance witness. Noise-dependent loss of Heisenberg scaling and classical channel simulation are established metrological methods [47]; their use here is credited. A Fisher-information bound alone would not imply the displayed global pairwise metric law, arbitrary-centre covering, or its rational realization.

Two rational semicircle charts yield exact positive effects and a finite grid decision using only integer arithmetic. The lower cover allows arbitrary legal memoryless centres. Appending labelled conditional states of ranks at most r_+, r_- gives Corollary 48.6: the length acquires the term $(V/2) \log_2 N + V \log_2(1/\delta)$, where $V = \sum_{y=\pm} [r_y(2n - r_y) - 1]$. The maximally mixed input isolates those state coordinates uniformly in visibility, while discarding their quantum outputs isolates

readout. The rational upper code preserves each conditional rank. No arbitrary centre retraction or full large-error theorem is inferred for this fixed-visibility family.

The previously established observable projective family is

$$\mathcal{F}_{P,\sigma,(x,y)}(X) = \text{tr}(P_x X) \sigma_{xy}.$$

Here the P_x form an ordered rank-one projective measurement, both labels (x, y) are retained, and each conditional row of positive blocks is trace normalized. The rank bounds r_{xy} allow changing supports and zero blocks. Put $b = d^2 - d$ and $V = \sum_x (\sum_y r_{xy} (2n - r_{xy}) - 1)$. Theorem 46.1 proves

$$\log_2 \mathcal{R}_N(\delta) = (b + V/2) \log_2 N + (b + V) \log_2(1/\delta) + O(1), \quad 0 < \delta \leq 1/32.$$

The remainder is independent of $N \geq 1$ and δ , at fixed public dimensions and ranks. The lower bound allows arbitrary legal memoryless centres; the upper code returns rational family members without increasing conditional ranks. This is an observable-readout theorem, not a dimension count for all entanglement-breaking channels. Corollary 46.5 removes the explicit x flag when fixed orthogonal output supports recover it. Equal overlapping unlabelled rows, by contrast, can erase every basis parameter.

The basis scale follows from the established multiple-query theorem for von Neumann measurements [56], not from the intrinsic dimension of the flag manifold alone. Readout differences remain visible when the conditional outputs are discarded. Conditional-state differences at a common readout are visible by a basis-state input. These two pair-dependent witnesses justify a product packing at scales δ/N and $\delta/\sqrt{N}$. For the upper code, partial-pivot elimination selects bounded rational coordinates directly from the ordered projections. Normalized algebraic eigenvectors and exhaustive unitary-net search are unnecessary. Positive row-factor codes and a common programme comparison then control all adaptive testers.

The inherited Theorem 47.1, identifies the covering problem exactly when one common tester extracts the programme from one use throughout its ambient state domain. State/channel comparisons under environment parametrization and seizing are classical [53, 54, 55]. The additional centre argument retracts every legal memoryless centre through the same seizer and processor. Thus finite-use family covering numbers equal their state counterparts, not merely a binary asymptotic exponent. The inherited rank-state law then gives the full range $0 < \delta \leq \delta_* < 2$ for fixed cap; Corollary 47.2 illustrates it with Pauli channels. The retraction may increase Choi rank. Rank and zero preservation belong to the upper target encoder, not to arbitrary-centre retraction.

The preceding results are placed alongside their necessary preliminary geometry. The fixed-readout input-dependent theorem 45.1 has coefficient $V/2$ in $\log N$ and the full fixed submaximal-error range. The common-estimator theorem 44.3 explains that range by disjoint-event multiplicity, rather than pairwise separation beyond half the diameter. The coherent/preparation tensor theorem 43.1 has projective-unitary dimension $u = d^2 - 1$ and coefficient $u + v/2$; its small-error range is unchanged. These earlier theorems, the intrinsic Choi codec and the singular preparation code are retained with complete proofs. In particular the new readout quotient removes all column phases, not merely one global phase. Revision genealogy is recorded outside the article.

The encoder receives matrix data. Fixed dimensions, rank bounds, horizon, tolerance, error cap and, in the noisy-readout theorem, visibility are public. Finite chart headers, pivot masks and coordinate digits are charged in the reusable payload; expanded matrices, encoder workspace and quantum programme hardware are different resources. Exact rational implementation applies to rational targets; the real-target theorem constructs rational centres in exact real arithmetic. These supplied-matrix description theorems do not perform device learning. Sections 54 and 55 separately construct common learners from calls to an unknown device. A reusable description does not by itself specify physical hardware implementing the described channel.

38. INTRINSIC DESCRIPTION ENTROPY OF INSTRUMENTS

We now determine the high-resolution description exponent, rather than only an upper bound on the number of stored matrix entries. The objects being described remain genuine quantum instruments. A binary description of such an object is not a physical classical implementation of it.

Fix positive integers d, n, m , put $q = dn$, and use the input-first, unnormalized Choi convention of Section 35. Let

$$\mathcal{C} = \mathcal{C}_{d,n,m} = \{(J_y)_{y=1}^m : J_y \in \text{Herm}(q), J_y \geq 0, \sum_y \text{Tr}_n J_y = I_d\}.$$

The corresponding outcome-retaining channel is denoted by $\mathcal{F}_J$. On tuples of Hermitian matrices write

$$|X|_2 = \left(\sum_y \|X_y\|_F^2 \right)^{1/2}, \quad \|X\|_\diamond = \|\mathcal{F}_X\|_\diamond.$$

Here $\mathcal{F}_X$ is the linear map defined by the Choi blocks, whether or not they are positive. Distances in this section are unhalved norms. The instrument diamond norm means exactly the diamond norm of this flagged direct-sum channel, not the sum of the individual outcome-map diamond norms. All m labeled outcome slots are part of the body. A legal η -description code is a finite set of codewords with one member of $\mathcal{C}$ assigned to each codeword, such that every target is within diamond distance η of a decoded member. Each codeword selects one fixed memoryless instrument, not a target-dependent random mixture with uncharged mixing probabilities. A rational code requires the decoded matrices to have rational real and imaginary parts. Dimensions, the outcome names and the requested tolerance are fixed public parameters in the code-length statements. Their encodings are additional data when they vary.

Lemma 38.1 (Affine dimension and an interior ball). *The affine hull of $\mathcal{C}$ has real dimension*

$$(38.1) \quad s = mq^2 - d^2 = d^2(mn^2 - 1).$$

Let $V = \{X : \sum_y \text{Tr}_n X_y = 0\}$ and $J_y^\circ = I_q/(mn)$. For $X \in V$,

$$(38.2) \quad |X|_2 \leq \sum_y \|X_y\|_1 \leq d\|X\|_\diamond, \quad \|X\|_\diamond \leq \sqrt{mq} |X|_2.$$

The $\|\cdot\|_\diamond$ -ball in V of radius $r_0 = (dmn)^{-1}$, translated by J° , is contained in $\mathcal{C}$. Moreover $\mathcal{C} - J^\circ$ lies in the diamond ball of radius two.

Proof. The map from $\text{Herm}(q)^m$ to $\text{Herm}(d)$ given by summing output partial traces is surjective: an arbitrary $H \in \text{Herm}(d)$ is the image of the tuple with first block $H \otimes |e_1\rangle\langle e_1|$ and other blocks zero. Its kernel has codimension d^2 . All blocks of J° are positive definite, so small perturbations in that kernel remain legal. This proves (38.1) and that V is the translation space of the full affine hull, not merely of a containing affine space.

The first two inequalities follow from Lemma 35.1 and the direct-sum Choi matrix. The last follows from that lemma, $\|X_y\|_1 \leq \sqrt{q}\|X_y\|_F$, and Cauchy–Schwarz over y . If $\|X\|_\diamond \leq r_0$, every block has operator norm at most $|X|_2 \leq 1/(mn)$, which proves positivity of $J^\circ + X$. The affine constraint is unchanged. Finally a channel has diamond norm one, so the distance between any two outcome-retaining channels is at most two. This also follows from complete trace-norm contractivity of channels, as in [37, Corollary 3.40]. $\square$

The lower Choi estimate from Section 35 thus has a specific converse use: it transfers a diamond ball into the positive Choi body. Neither this step nor the dimension count uses a spectral gap.

Theorem 38.2 (Sharp high-resolution description exponent). *Suppose $s > 0$. Let $\mathcal{N}_\diamond(\eta)$ and $\mathcal{N}_\diamond^\mathbb{Q}(\eta)$ be the smallest cardinalities of legal and rational legal η -description codes for $\mathcal{C}$. For $0 < \eta \leq 1$,*

$$(38.3) \quad \max\{1, (r_0/\eta)^s\} \leq \mathcal{N}_\diamond(\eta) \leq \mathcal{N}_\diamond^\mathbb{Q}(\eta) \leq (1 + 8/\eta)^s.$$

In particular, for fixed d, n, m , both optimal fixed-length binary code lengths are

$$(38.4) \quad s \log_2(1/\eta) + O_{d,n,m}(1) \quad (\eta \downarrow 0).$$

When $s = 0$, $m = n = 1$, the unique instrument is the trace map and the code length is zero.

Proof. Equip V with Lebesgue measure and the diamond norm restricted to V . By Lemma 38.1, the target set contains a ball of radius r_0 . A union of M balls of radius η covering this ball has volume at most $M\eta^s$ times the unit-ball volume. Hence $M \geq (r_0/\eta)^s$. Centres anywhere in the affine hull give the same lower bound.

For the upper bound, take a maximal set of points of $\mathcal{C}$ whose mutual distances are greater than $\eta/2$. Balls of radius $\eta/4$ about them are disjoint up to null boundaries and lie in the ball of radius $2 + \eta/4$ about J° . The number of points is therefore at most $(1 + 8/\eta)^s$, and maximality makes them an $\eta/2$ -net. Rational legal instruments are dense, by the explicit construction in Theorem 38.3 below. Replace each centre by a rational legal point within $\eta/2$. This yields a rational η -net of the same cardinality. This dependence is not circular: the construction below is proved directly from the Choi constraints.

A codebook with M words can be indexed by $\lceil \log_2 M \rceil$ bits, and a code of at most b fixed-length bits has at most 2^b words. Taking logarithms in (38.3) proves (38.4). The singleton case follows directly from the constraint $J_1 = I_d$. $\square$

This is an elementary volumetric theorem for the precise legal Choi body and norm. The volume comparison is not a new general covering principle. The coefficient s is sharp in the high-resolution limit with dimensions fixed; the estimate does not assert optimal constants in a simultaneous dimension–accuracy limit.

38.1. An intrinsic rational coordinate code. Fix an outcome y_* and an output basis index α_* . The independent real coordinates of a Hermitian matrix are its real diagonal entries and the real and imaginary parts of its strictly upper-triangular entries. Omit all d^2 such coordinates belonging to the submatrix

$$((J_{y_*})_{i\alpha_*, j\alpha_*})_{1 \leq i, j \leq d}.$$

There are exactly s remaining real coordinates, in a fixed order. Every coordinate of a legal target has absolute value at most one: positivity and the marginal identity bound each diagonal by one, and $|J_{uv}|^2 \leq J_{uu}J_{vv}$ bounds the off-diagonal entries.

For $B \in \mathbb{N}$, truncate the s retained coordinates toward zero on the grid $B^{-1}\mathbb{Z}$, producing digits $z_1, \dots, z_s \in [-B, B] \cap \mathbb{Z}$. Fill the omitted submatrix by the exact marginal identity:

$$(38.5) \quad (Q_{y_*})_{i\alpha_*, j\alpha_*} = \delta_{ij} - \sum_{(y, \alpha) \neq (y_*, \alpha_*)} (Q_y)_{i\alpha, j\alpha}.$$

All summands on the right were retained. Lower-triangular entries are Hermitian copies, not additional digits.

Theorem 38.3 (Exact intrinsic codec). *Let*

$$(38.6) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad K = 3mndc.$$

The matrices

$$(38.7) \quad \hat{J}_y = (BQ_y + cI_q)/D$$

are positive semidefinite with Gaussian-integer numerators, satisfy $\sum_y \text{Tr}_n \hat{J}_y = I_d$, and obey

$$(38.8) \quad \|\mathcal{F}_J - \mathcal{F}_{\hat{J}}\|_{\diamond} \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/D \leq K/B.$$

They can be encoded by one integer in $[0, (2B+1)^s)$, using at most $\lceil \log_2((2B+1)^s) \rceil$ bits. Thus $B = \lceil K/\eta \rceil$ achieves the leading coefficient s in (38.4) by a direct coordinate code, without enumerating a net. For rational input, encoding, decoding and validity testing have polynomial bit complexity in the explicit input and output lengths. This is not a workspace bound for a trajectory simulator.

Proof. Non-omitted independent-coordinate errors are at most $1/B$. An omitted coordinate is reconstructed from at most $mn - 1$ errors, so its error is at most $(mn - 1)/B$. In particular every independent real error is at most mn/B . Counting the Hermitian copies gives the safe estimate

$$\|Q_y - J_y\|_{\text{op}} \leq \|Q_y - J_y\|_F \leq 2qmn/B \leq c/B.$$

The affine constraint is exact by (38.5). Therefore $BQ_y + cI_q \geq 0$ and its summed output partial trace is $(B + mnc)I_d$. With $E_y = Q_y - J_y$,

$$D(\hat{J}_y - J_y) = BE_y + cI_q - mncJ_y.$$

The sum of the trace norms of each of the three terms is bounded by $mndc$, using $\sum_y \text{tr } J_y = d$ for the last term. Lemma 35.1 proves (38.8).

Set $M = 2B + 1$. The integer with base- M digits $z_j + B$ determines all retained coordinates, and (38.5) then determines the omitted coordinates. A fixed number of digits is used, including leading zeros. Only integers whose reconstructed, buffered blocks are positive are valid codewords. They can be recognized by the exact Schur test in Lemma 36.2. Every code produced from a legal target is valid by the preceding proof. If a total decoder is required, assign each invalid word the canonical rational instrument J° ; unused binary strings can be treated in the same way. The valid codebook already covers all targets. In the singleton case the construction decodes the trace map from the empty string.

For rational data the coordinate digits are obtained by exact signed integer division. Matrix filling, buffering and base conversion are finite integer operations with polynomial operand lengths. Exact PSD validation has polynomial cost by Lemma 36.2. No eigenvector computation, rational Kraus factorization, random draw, statistical sample or floating-point decision is involved. For an arbitrary real target, the coordinate construction proves existence and density; it is not an algorithm for recovering unknown real input from finitely many unverified measurements. $\square$

Remark 38.4 (Known zero outcomes and public headers). If a specified set of outcomes is known to be zero, remove it before coding and restore it afterwards. With $m' \geq 1$ active outcomes the intrinsic dimension is $d^2(m'n^2 - 1)$, and m' replaces m in the constants. If the active set is not a fixed public parameter, its mask must also be encoded. The generic construction does not preserve zeros without this operation. The choice of (y_*, α_*) affects neither the dimension nor the bounds. Headers for dimensions, B , outcomes and the chosen coordinate convention are not part of a fixed-parameter codeword; a self-contained file carries and charges them separately.

39. DESCRIPTION ENTROPY UNDER REPEATED ADAPTIVE USE

The number of calls changes the precision needed in a reusable instrument description. On a strictly positive Choi body the relevant scale is $N^{-1/2}$, not the N^{-1} scale supplied by a general one-slot hybrid. We prove a finite, multiparameter coding statement against arbitrary common adaptive quantum testers.

For two instruments $J, L \in \mathcal{C}_{d,n,m}$, define

$$(39.1) \quad d_N(J, L) = \sup_{\mathsf{T}} \|\Omega_J^{\mathsf{T}} - \Omega_L^{\mathsf{T}}\|_1.$$

Here T is the same causal tester in both experiments, uses the same memoryless instrument at most N times, and may have arbitrary finite-dimensional quantum memory, initially entangled references, intermediate quantum channels and measurements, and classical feedback. Fresh ancillary systems of arbitrary finite dimension are allowed: each tester has finite-dimensional registers at every slot, but no common dimension bound is imposed in the supremum. Outcomes and final memory are retained. A public stopping time bounded by N is allowed. A changed input dimension can be handled by a tester channel between calls. There is no coherent superposition of command ports, no tester informed of the hidden choice J versus L , and no stopping based on inaccessible simulator state. Taking suprema of trace distances gives the triangle inequality. In particular $d_N \leq 2$. Public stopping is included by an absorbing flag, as in Theorem 35.4.

For a fixed a with $0 < a < 1/(mn)$ let

$$\mathcal{C}_a = \{J \in \mathcal{C} : J_y \geq aI_{dn} \text{ for every } y\}.$$

This is a Choi eigenvalue margin, not a spectral gap for an action or a mixing hypothesis on trajectories. Its affine dimension is the same $s = d^2(mn^2 - 1)$ as before.

Lemma 39.1 (Two-point classical programme bound). *Suppose $J_y, L_y \geq \lambda I_{dn}$ for every outcome and $r = \|J - L\|_2 \leq \lambda$, where $\lambda > 0$. Then, for every $N \geq 1$,*

$$(39.2) \quad d_N(J, L) \leq \min\{2, 2\sqrt{N}r/\lambda\}.$$

The same bound covers arbitrary initially entangled references and bounded public stopping in (39.1).

Proof. There is nothing to prove for $r = 0$. Put $K = (J + L)/2$ and $U = (J - L)/r$. Since $U \in V$ and each $\|U_y\|_{\text{op}} \leq 1$,

$$K^+ = K + \lambda U, \quad K^- = K - \lambda U$$

are legal instruments. With

$$p_{\pm} = \frac{1}{2} \pm \frac{r}{4\lambda}, \quad q_{\pm} = \frac{1}{2} \mp \frac{r}{4\lambda},$$

we have $J = p_+K^+ + p_-K^-$ and $L = q_+K^+ + q_-K^-$. All four probabilities are at least $1/4$. The elementary inequality $\log x \leq x - 1$ implies

$$D(p\|q) \leq \sum_{b \in \{+, -\}} \frac{(p_b - q_b)^2}{q_b} \leq \frac{2r^2}{\lambda^2}.$$

Choose the programme signs independently in advance for all N slots. Conditional on a sign string, a fixed tester acts on exactly the same sequence of genuine quantum instruments in the two experiments. It therefore defines one channel from the classical sign string to the final quantum-classical state. This remains true with adaptive feedback: conditioning fixes the instruments, not the tester's outcomes. If the tester stops early, the unused signs are ignored. Contractivity and the classical product identity give

$$\|\Omega_J^{\mathsf{T}} - \Omega_L^{\mathsf{T}}\|_1 \leq 2 \text{TV}(p^{\otimes N}, q^{\otimes N}) \leq \sqrt{2ND(p\|q)} \leq 2\sqrt{N}r/\lambda.$$

The middle bound is the usual Pinsker inequality. To recall an elementary proof, coarse-grain two distributions to the set where the first is larger. The log-sum inequality reduces relative entropy to binary relative entropy. Its second derivative in the first argument is $1/[x(1-x)] \geq 4$, so integration about its minimum proves $D \geq 2 \text{TV}^2$. Taking the supremum over testers proves the assertion. $\square$

The classical-programme idea is an established method in quantum metrology; Demkowicz-Dobrzański, Kołodyński and Guţă [47, Equations (4)–(6), Methods] use parameter-dependent mixtures of fixed channels to bound distinguishability. The lemma above gives the finite two-point trace-norm estimate needed here, including common adaptive testers and public stopping. It is not a claim that the mixture method or the square-root scale is a new metrological principle. The operations $K^\pm$ remain quantum operations, not classical physical substitutes for unknown input.

Lemma 39.2 (Repeated Choi tests). *For any two instruments $J, L \in \mathcal{C}$, with $t = |J - L|_2$,*

$$(39.3) \quad d_N(J, L) \geq 2 - 4 \exp\left(-\frac{Nt^2}{8d^2}\right).$$

A negative right-hand side is simply a vacuous lower bound.

Proof. At each call supply half of a normalized maximally entangled state and retain its other half and the outcome. The resulting density matrices are $A = d^{-1} \oplus_y J_y$ and $B = d^{-1} \oplus_y L_y$. They have trace one. The projector on the positive part of $A - B$ gives a binary measurement whose probability gap is

$$g = \frac{1}{2} \|A - B\|_1 \geq t/(2d).$$

The same binary measurement is made on each of the N independent outputs. With a threshold halfway between the two means, each wrong-side probability is at most $\exp(-Ng^2/2)$. Indeed for a Bernoulli variable Z , the second derivative of $\log \mathbb{E}e^{u(Z - \mathbb{E}Z)}$ is the variance of a tilted Bernoulli variable and is at most $1/4$. The value and first derivative at zero vanish; integration and Chernoff's inequality give $\Pr\{N^{-1} \sum Z_i - \mathbb{E}Z \geq v\} \leq e^{-2Nv^2}$, and likewise for the other tail. Set $v = g/2$. The total variation distance between the two product experiments is at least $1 - 2e^{-Ng^2/2}$; the unhalved trace norm of their classical outputs is twice this. This particular nonadaptive tester is allowed in (39.1), and the result follows. The tester may depend on the pair J, L , as is permitted when lower-bounding a metric; it is the same tester for the two members of that pair. $\square$

Theorem 39.3 (Sharp horizon exponent for interior descriptions). *Fix d, n, m with $s > 0$, a margin $0 < a < 1/(mn)$ and a tolerance $0 < \delta < 1$. Let $\mathcal{M}_N(\delta; a)$ be the minimum number of legal decoded instruments whose d_N -balls of radius δ cover $\mathcal{C}_a$. A decoded instrument need not itself have margin a . Write $\mathcal{M}_N^{\mathbb{Q}}$ when rational decoded instruments are required. Set*

$$R = \frac{1}{mn} - a, \quad c_\delta = \sqrt{8d^2 \log \frac{4}{1-\delta}}, \quad B_N = \left\lceil \frac{K}{a} \left(2 + \frac{4\lceil \sqrt{N} \rceil}{\delta} \right) \right\rceil,$$

where K is from (38.6). Then

$$(39.4) \quad \max\{1, (R\sqrt{N}/c_\delta)^s\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s.$$

Consequently the minimum fixed-length description satisfies

$$(39.5) \quad \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) = \frac{s}{2} \log_2 N + O_{d,n,m,a,\delta}(1),$$

and the same formula holds without the rationality restriction. The upper bound is achieved by the exact intrinsic codec applied once at grid B_N ; its decoded instrument is reused at every call.

Proof. The tuple-Frobenius ball in V of radius R about J° is contained in $\mathcal{C}_a$. Choose a maximal set of points in that ball with pairwise distances greater than $c_\delta/\sqrt{N}$. Balls of that radius about the selected points cover the original ball. Euclidean volume comparison therefore gives at least $(R\sqrt{N}/c_\delta)^s$ points. By Lemma 39.2, their pairwise d_N distances are greater than

$$2 - 4e^{-c_\delta^2/(8d^2)} = 1 + \delta > 2\delta.$$

One d_N -ball of radius δ cannot cover two of these points, regardless of the location of its legal centre. This proves the lower bound.

For the upper bound use Theorem 38.3. It gives

$$r = |J - \hat{J}|_2 \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/B_N \leq \min\{a/2, a\delta/(4\sqrt{N})\}.$$

Every block of $\hat{J}$ is consequently at least $(a/2)I$, and both J and $\hat{J}$ satisfy Lemma 39.1 with $\lambda = a/2$ and $r \leq \lambda$. Its conclusion gives $d_N(J, \hat{J}) \leq 4\sqrt{N}r/a \leq \delta$. There are at most $(2B_N + 1)^s$ possible digit strings. All codewords arising from targets in $\mathcal{C}_a$ decode legally. The remaining codewords may be rejected or assigned J° ; this does not affect the cover. Since $B_N = O_{d,n,m,a,\delta}(\sqrt{N})$, taking logarithms in (39.4) proves (39.5). $\square$

Remark 39.4 (What is charged). The lower bound concerns the number of possible reusable mathematical instrument descriptions, not the mutable memory of a simulator that receives and stores the entire instrument as input. It applies to all full-dimensional strictly positive instrument families above, without an expanding unitary subsystem. It therefore complements, rather than replaces, the hypothesis-sensitive streaming-space lower bounds. For variable a, δ, d, n, m , the displayed finite inequalities remain valid; (39.5) holds with those parameters fixed. The estimates of this section alone assert neither an optimal uniform dimension constant nor a complete joint N - δ asymptotic. The latter is supplied, for fixed dimensions and margin, by Theorem 40.3. The rational implementation takes rational a, δ and an asserted margin, which it can verify exactly. No samples of an unknown channel are supplied or required.

Proposition 39.5 (The boundary is quantitatively different). *There is no constant C such that $d_N(J, L) \leq C\sqrt{N}|J - L|_2$ holds for every pair of qubit channels and every N . In particular the strict margin in Theorem 39.3 cannot be removed from its proof by setting $a = 0$.*

Proof. Let $U_\theta = \text{diag}(1, e^{i\theta})$ and compare its unitary channel to the identity channel. For their unnormalized Choi matrices,

$$|J(U_\theta) - J(I)|_2 = 2\sqrt{2}|\sin(\theta/2)| \leq \sqrt{2}|\theta|.$$

At $\theta = \pi/N$, sequentially applying the unknown channel N times to $|+\rangle$ gives respectively $|-\rangle$ and $|+\rangle$. The final trace distance is two. The proposed upper bound would tend to zero. The general additive diamond hybrid remains valid at this boundary; the proposition asserts no global covering exponent for boundary or rank-constrained instrument classes. $\square$

39.1. Description, learning and physical simulation. A reusable rational description is different from a protocol that replaces quantum communication by classical messages. Naik, Zartab, Gisin and Banik [44, Theorems 2 and 5] show a finite-classical-message obstruction for the perfect qubit channel in a reference-sensitive joint-measurement model, and finite-resource possibility for noisy depolarizing channels. Their receiver retains a quantum system; the task is physical simulation. Here the decoded objects and the vertices in Lemma 39.1 are genuine quantum instruments. Our numerical algorithms instead start from specified matrices. Neither construction supplies a classical physical device accepting unknown entangled input.

Unknown-channel learning charges channel queries used to infer a classical description. Mele and Bittel [45, Theorem I.1] prove dimension-optimal query complexity at fixed diamond accuracy; Oufkir and Girardi [46] give further accuracy-dependent query lower bounds. The input here is already specified, and the lower bound in (39.4) counts descriptions, not the queries needed to select one from unknown data. The elementary packing and testing ingredients are shared methods, not a priority claim that those learning theorems are consequences of this code.

40. JOINT HORIZON AND ACCURACY IN DESCRIPTION LENGTH

The exponential test in Lemma 39.2 resolves pairs at fixed error, but its lower bound is vacuous when the two experiments are very close. A bounded statistic gives the local estimate needed to retain the accuracy parameter. Throughout this section $N \geq 1$ and the adaptive distance d_N is the unhalved final-state trace distance maximized over one common tester, as in (39.1). The unknown device is the same memoryless instrument at every call; commands are classical, and stopping is public and bounded by N .

Lemma 40.1 (A local binary test). *For $0 \leq p, q \leq 1$,*

$$(40.1) \quad \text{TV}(\text{Bern}(p)^{\otimes N}, \text{Bern}(q)^{\otimes N}) \geq \frac{3}{32} \min\{\sqrt{N}|p - q|, 1\}.$$

The estimate is uniform up to the endpoints of the probability interval.

Proof. Assume $p < q$ and first suppose $q - p \leq N^{-1/2}$. Put $b = N(p + q)/2$ and let $f : \mathbb{R} \rightarrow [0, 1]$ be the nondecreasing function which is affine with slope $(8\sqrt{N})^{-1}$ on $[b - 4\sqrt{N}, b + 4\sqrt{N}]$, zero to its left, and one to its right. For $S \sim \text{Bin}(N, t)$, differentiating the finite binomial polynomial gives the exact identity

$$\frac{d}{dt} \mathbb{E}_t f(S) = N \mathbb{E}_t \{f(T + 1) - f(T)\}, \quad T \sim \text{Bin}(N - 1, t).$$

It holds on $(0, 1)$ and extends continuously to both endpoints. For $t \in [p, q]$, the distance from $\mathbb{E}_t T$ to b is at most $\sqrt{N}/2 + 1$. Also $\text{Var}_t T \leq (N - 1)/4$. Chebyshev's inequality gives

$$\Pr_t\{|T - \mathbb{E}_t T| \leq \sqrt{N}\} \geq 3/4.$$

On this event, both T and $T + 1$ belong to the affine interval: their distance to b is at most $3\sqrt{N}/2 + 2 \leq 4\sqrt{N}$. Since f is nondecreasing everywhere,

$$\frac{d}{dt} \mathbb{E}_t f(S) \geq \frac{3}{32} \sqrt{N}.$$

Integration from p to q , and the characterization of total variation by $[0, 1]$ -valued statistics, prove the assertion in this case. The statistic depends only on the count, which is a function of the full binary string.

If $q - p > N^{-1/2}$, apply the same construction to p and $q_0 = p + N^{-1/2} < q$. Couple Bernoulli samples by common uniform random variables. Monotonicity of f then gives $\mathbb{E}_q f(S) \geq \mathbb{E}_{q_0} f(S)$, and the preceding lower bound is $3/32$. The cases $p = q$ and $q < p$ follow by equality and symmetry. $\square$

The proof uses only binomial differentiation and a bounded statistic; it requires neither a normal approximation nor a lower variance bound. In particular its constant does not deteriorate for a rare outcome.

Corollary 40.2 (Local repeated Choi separation). *For the instrument body $\mathcal{C}_{d,n,m}$, let $t = |J - L|_2$. Then*

$$(40.2) \quad d_N(J, L) \geq \frac{3}{16} \min \left\{ \frac{\sqrt{N}}{2d} t, 1 \right\}.$$

Proof. Use the same normalized Choi inputs as in Lemma 39.2. The positive spectral projection of $d^{-1} \oplus_y (J_y - L_y)$ defines a binary test with probability gap at least $t/(2d)$. Repeating this test independently and applying Lemma 40.1 gives (40.2); classical trace distance is twice total variation. This is one admissible nonadaptive tester. For each pair it is the same tester in both experiments, and it is allowed in the supremum defining d_N . $\square$

Theorem 40.3 (The joint interior coding law). *Fix d, n, m with $s = d^2(mn^2 - 1) > 0$ and $0 < a < 1/(mn)$. For all $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(40.3) \quad \max \left\{ 1, \left(\frac{R\sqrt{N}}{32d\delta} \right)^s \right\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s,$$

where $R = 1/(mn) - a$ and B_N is the explicit grid in Theorem 39.3. Consequently the optimal legal fixed-length description, with or without rational decoded matrices, has length

$$(40.4) \quad \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O_{d,n,m,a}(1),$$

uniformly on this whole range of N and δ . The decoded instrument is reused unchanged, and no spectral-gap hypothesis occurs.

Proof. The tuple-Frobenius ball of radius R about J° in V is contained in $\mathcal{C}_a$. Take a maximal set in this ball whose pairwise Euclidean distances are greater than $h = 32d\delta/\sqrt{N}$. Maximality and s -dimensional volume give at least $(R/h)^s$ points. Corollary 40.2 separates each pair by more than 2δ in d_N : before the minimum saturates its value is greater than 3δ , and after saturation it is $3/16 > 2\delta$. Thus no legal centre of a d_N -ball of radius δ can cover two packing points, even if that centre has no positive eigenvalue margin. This proves the lower bound.

The intrinsic rational codec and its classical-programme estimate in Theorem 39.3 give the upper bound with the same B_N . Since $N \geq 1$ and $\delta \leq 1/32$, there is a constant $C = C(d, n, m, a)$ such that $2B_N + 1 \leq C\sqrt{N}/\delta$. The lower bound and this inequality give (40.4), including rounding a logarithm up to a fixed-length binary code. When the volumetric lower expression is less than one, its logarithm is negative; the nonnegative code length still gives the same uniform additive estimate. Neither bound charges the queries of a learning procedure. $\square$

The earlier fixed-error theorem is retained. The new assertion is the uniform additive remainder in (40.4), which is independent of both the horizon and the accuracy. It does not assert sharp dimension-dependent constants, nor is it a lower bound for the mutable workspace of the variable-description trajectory algorithm.

41. BOUNDARY-UNIFORM CODING OF PREPARATION INSTRUMENTS

Strict positivity is essential to the two-point programme argument, as Proposition 39.5 shows. Nevertheless it is not necessary for a complete coding law in an irreversible family. We consider all outcome-retaining instruments that erase their input and prepare a specified quantum-classical state. This family contains zero outcomes and arbitrary rank-deficient output blocks.

For $m, n \geq 1$ put

$$\mathcal{S}_{m,n} = \{(\sigma_y)_{y=1}^m : \sigma_y \in \text{Herm}(n), \sigma_y \geq 0, \sum_y \text{tr } \sigma_y = 1\}.$$

Write $\sigma = \bigoplus_y \sigma_y$. For any input dimension $d \geq 1$, the associated instrument is

$$(41.1) \quad \mathcal{R}_{\sigma,y}(X) = \text{tr}(X)\sigma_y, \quad J_y(\mathcal{R}_\sigma) = I_d \otimes \sigma_y.$$

The output label y is retained. Its conditional quantum state is $\sigma_y/\text{tr } \sigma_y$ when that trace is positive. This is a specified mathematical quantum instrument; a description of it is not a numerical algorithm acting on an unknown physical input.

Lemma 41.1 (Reduction to a product-state experiment). *For arbitrary $\sigma, \tau \in \mathcal{S}_{m,n}$,*

$$(41.2) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_\tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

Here the tester may have arbitrary quantum memory, initial reference entanglement, feedback and a public stopping time at most N .

Proof. Supply N independent program states σ to one fixed processor. At the next device call the processor discards the incoming device system and returns the next program state, including its classical outcome. Between calls it performs the prescribed tester operations. This reproduces the experiment exactly, even when the incoming system is entangled with the tester memory: the incoming system is traced out in (41.1). Unused program states are ignored after a public stop. The processor is the same channel for σ and τ , so trace-norm contraction gives the upper bound. Conversely, a tester can make exactly N calls and retain every outcome and quantum output. Its final state is the displayed product, which proves equality. The processor is a quantum channel, not a claimed finite-classical-message physical simulator. $\square$

41.1. Triangular factors and a rational code. Every $\sigma \in \mathcal{S}_{m,n}$ has triangular factors L_y with $\sigma_y = L_y L_y^*$ and nonnegative real diagonal entries. The usual no-pivot Cholesky procedure also covers singular matrices: a zero diagonal pivot forces the entire residual row and column to vanish and contributes a zero factor column. A block of rank r therefore has exactly r positive pivots. All factors together satisfy $\sum_y \|L_y\|_F^2 = 1$.

Fix public bounds $0 \leq r_y \leq n$, not all zero, and write

$$(41.3) \quad D = \sum_{y=1}^m r_y(2n - r_y), \quad v = D - 1, \quad \mathcal{S}_r = \{\sigma \in \mathcal{S}_{m,n} : \text{rank } \sigma_y \leq r_y\}.$$

If the positive pivot positions of L_y are $T_y \subseteq \{1, \dots, n\}$, record only the real diagonal and the real and imaginary coordinates strictly below each such pivot. Their total number is

$$h = \sum_y \sum_{j \in T_y} \{1 + 2(n - j)\} \leq D.$$

The resulting real coordinate vector $x \in \mathbb{R}^h$ has norm one. The inequality follows by placing each block's pivots as early as possible. It holds also when the actual ranks are smaller than the public bounds.

Choose a coordinate j of maximal absolute value, put $a = |x_j|$, and set $w = x/a$. For an integer $B \geq 1$, take

$$(41.4) \quad z_j = B \operatorname{sgn}(x_j), \quad z_i = \operatorname{sgn}(x_i) \lfloor B|w_i| \rfloor \quad (i \neq j).$$

Insert these integers in the recorded factor positions to obtain Gaussian-integer triangular matrices Z_y ; all unrecorded columns are zero. Define

$$(41.5) \quad T = \sum_y \|Z_y\|_F^2, \quad \hat{\sigma}_y = Z_y Z_y^* / T.$$

Since $|z_j| = B$, the denominator satisfies $B^2 \leq T \leq hB^2$.

Theorem 41.2 (A boundary-uniform rational factor code). *The output (41.5) lies in $\mathcal{S}_r$, preserves every zero target outcome, and does not increase the rank of any individual block. For all $N \geq 1$,*

$$(41.6) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_{\hat{\sigma}}) \leq \min \left\{ 2, \frac{4\sqrt{N(h-1)}}{B} \right\} \leq \min \left\{ 2, \frac{4\sqrt{Nv}}{B} \right\}.$$

At fixed m, n, r, B a fixed-length code needs at most

$$(41.7) \quad mn + \lceil \log_2(2D(2B+1)^v) \rceil$$

bits. For rational target blocks, its encoder and decoder use exact rational and integer arithmetic, including integer square roots; no algebraic-number oracle, positive eigenvalue margin, or Kraus factorization is required.

Proof. The product $Z_y Z_y^*$ is positive semidefinite, and normalization makes the sum of its traces exactly one. Its number of possibly nonzero columns is no larger than that of L_y , so its rank does not exceed the target rank. When an outcome is zero it has no recorded columns and remains exactly zero.

Put $z' = z/B$. Its j th coordinate agrees with that of w ; every other coordinate differs by less than $1/B$. Both $\|w\|$ and $\|z'\|$ are at least one. The elementary normalization estimate gives

$$(41.8) \quad \left\| x - \frac{z}{\|z\|} \right\| \leq \frac{2\|w - z'\|}{\|w\|} \leq \frac{2\sqrt{h-1}}{B}.$$

Use the normalized vectors

$$|L\rangle = \sum_{y,i,j} (L_y)_{ij} |y, i\rangle |y, j\rangle, \quad |\widehat{L}\rangle = T^{-1/2} \sum_{y,i,j} (Z_y)_{ij} |y, i\rangle |y, j\rangle.$$

Their reduced states on the first factor are σ and $\widehat{\sigma}$. Their vector distance is the left side of (41.8). For any unit vectors u, v , diagonalization on their two-dimensional span gives

$$\| |u\rangle\langle u|^{\otimes N} - |v\rangle\langle v|^{\otimes N} \|_1 = 2\sqrt{1 - |\langle u, v \rangle|^{2N}} \leq 2\sqrt{N} \|u - v\|.$$

Indeed $1 - t^N \leq N(1 - t)$ on $[0, 1]$ and $1 - |\langle u, v \rangle|^2 \leq 2(1 - \Re\langle u, v \rangle)$. Taking partial traces and using Lemma 41.1 proves (41.6). These are standard purification and product-fidelity estimates; see [37, Section 3.2]. Their short proof specifies the normalization used here.

The pivot sets use mn bits. Given them, there are at most $2h$ anchor/sign choices and $(2B+1)^{h-1}$ remaining digit strings. Pad their index to the maximum length in (41.7). An invalid header may be rejected, or sent by a total mathematical decoder to a fixed legal state. Valid factor words always decode positively, even if they did not arise from this encoder.

For rational input, maintain the exact rational Schur residual A . At a positive pivot $u = A_{jj}$, the factor diagonal is $\sqrt{u}$ and a lower entry is $A_{ij}/\sqrt{u}$. Thus the square of each of its real coordinates is rational, respectively u , or $(\Re A_{ij})^2/u$, or $(\Im A_{ij})^2/u$; its sign is known. Updating the residual uses rational arithmetic only. Choose the largest squared coordinate $b_j > 0$ by exact comparisons. If another coordinate has squared value b_i , then

$$\lfloor B|w_i| \rfloor = \text{isqrt} \left(\left\lfloor B^2 b_i / b_j \right\rfloor \right).$$

Consequently no square root of a rational number must be represented. The ratios-of-minors bounds in Lemma 36.2 give polynomial operand lengths during this preprocessing. Integer square root, products and mixed-radix coding have polynomial bit complexity in the explicit input and output lengths. Encoding workspace and the expanded decoded matrix are not identified with the compressed payload. $\square$

41.2. Sharp description laws on the boundary. Let $\mathcal{P}_N(\delta; \mathbf{r})$ be the minimum number of legal preparation instruments whose d_N -balls of radius δ cover $\mathcal{S}_{\mathbf{r}}$; the decoded centre may have any output rank. Add a superscript $\mathbb{Q}$ when rational centres are required. For an upper construction our centres will obey the same rank bounds.

Theorem 41.3 (Rank-stratified joint coding law). *Fix $m, n, \mathbf{r}$ as in (41.3). If $v > 0$, there exist constants $c, C > 0$, depending only on these fixed data, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(41.9) \quad \max\{1, (c\sqrt{N}/\delta)^v\} \leq \mathcal{P}_N(\delta; \mathbf{r}) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C(\sqrt{N}/\delta)^v.$$

The optimal legal and rational fixed-length descriptions therefore have length

$$(41.10) \quad \frac{v}{2} \log_2 N + v \log_2(1/\delta) + O_{m,n,\mathbf{r}}(1)$$

uniformly in N and δ . The statement has no spectral-gap or positive-eigenvalue-margin hypothesis. It includes every rank-deficient boundary allowed by the public bounds. If $v = 0$, the family is a singleton and its optimal description length is zero.

Proof. For the upper bound take $B = \lceil 4\sqrt{Nv}/\delta \rceil$ in Theorem 41.2; for rational implementation it suffices to replace $\sqrt{Nv}$ by its integer ceiling. Its error is at most δ , and its codebook cardinality is at most $2^{mn}2D(2B+1)^v$. This has the asserted upper order uniformly on the stated parameter range.

For completeness we construct a full-dimensional local patch for the lower bound. For every block with $r_y > 0$, take an n by r_y lower-trapezoidal factor: entries above the first r_y diagonal positions vanish, and these diagonal entries are positive real numbers. These factors have D independent real coordinates. Impose $\sum_y \|L_y\|_F^2 = 1$ and work near the point with all these diagonal entries equal to $(\sum_y r_y)^{-1/2}$ and all other entries zero. An open patch of this unit sphere is a smooth v -dimensional manifold. The map

$$(L_y)_y \mapsto (L_y L_y^*)_y$$

is one-to-one there and has a smooth inverse on its image. To see the inverse explicitly, the leading r_y by r_y principal block is positive definite and determines its unique positive-diagonal Cholesky factor A_y . The bottom-left block is $B_y A_y^*$, so it determines B_y by multiplication by $(A_y^*)^{-1}$. These operations are smooth near the chosen point. When $r_y = 0$, the entire block is identically zero and contributes no coordinate.

Choose a closed coordinate ball of sufficiently small positive radius inside this patch. The chart and its inverse have bounded derivatives on a slightly larger neighborhood; hence distances between parameters in the ball are bounded above and below by constant multiples of the tuple-Frobenius distance between their states. More explicitly one may bound the derivative of the inverse on a convex neighborhood of the chosen positive principal blocks, where the same Cholesky formulas extend smoothly. A maximal Euclidean packing of this parameter ball at scale $A\delta/\sqrt{N}$ therefore has at least $(c\sqrt{N}/\delta)^v$ points whose state distances exceed $32\delta/\sqrt{N}$, after increasing the fixed A .

The binary argument of Corollary 40.2, applied directly to the trace-one block states (input dimension one), separates each pair by more than 2δ after N copies. By Lemma 41.1 the same separation holds in d_N . One radius- δ ball, regardless of the rank of its legal centre, cannot contain two packing points. This proves the lower bound and then (41.10). Finally $D = 1$ requires $n = 1$ and exactly one bound $r_y = 1$; all other bounds vanish. The sole permitted state has unit mass on that outcome, as asserted. $\square$

Corollary 41.4 (All ranks, with an explicit lower constant). *For the full family $\mathcal{S}_{m,n}$ put $v = mn^2 - 1 > 0$. Then*

$$(41.11) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{64mn\delta} \right)^v \right\} \leq \mathcal{P}_N(\delta) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta) \leq 2^{mn} 2mn^2 (2B+1)^v, \quad B = \left\lceil \frac{4\sqrt{Nv}}{\delta} \right\rceil.$$

In particular a general n -dimensional prepared state has coefficient $n^2 - 1$, while the class of pure prepared states has coefficient $2n - 2$ in (41.10). A classical m -outcome law has coefficient $m - 1$, including probabilities equal to zero.

Proof. The upper bound is Theorem 41.2 with every $r_y = n$. For the lower bound, the affine space of block states has dimension $mn^2 - 1$. Its tuple-Frobenius ball of radius $1/(2mn)$ about the blocks $I_n/(mn)$ lies in $\mathcal{S}_{m,n}$. Repeat the packing proof of Theorem 40.3 with input dimension one. For pure states take $m = 1, r_1 = 1$ in Theorem 41.3; for classical laws take $n = 1$. $\square$

Remark 41.5 (Operational and priority distinctions). Input erasure is decisive in (41.2). No analogous product-state reduction has been proved here for an arbitrary disturbing instrument;

coherent accumulation for unitary channels remains governed by Proposition 39.5. The present theorems classify reusable description lengths in the stated preparation families, not general simulator workspace. Constants remain dimension dependent.

Quantum population compression starts with physical copies of an unknown state and compresses their quantum information. Yang, Bai, Chiribella and Hayashi [48] obtain a half-parameter-count logarithmic memory law in that different task and show a limitation on purely classical compression of noncommuting families. Here the encoder is given a classical matrix description, and each codeword specifies a state or preparation instrument. It need not infer that word from unknown copies. The common half-dimension scale, purification inequalities and finite-dimensional covering methods are classical antecedents, not claimed inventions. The contribution established here is a uniform two-parameter legal-description law with a rational, zero-outcome-preserving, rank-nonincreasing encoder and matching lower bounds throughout the declared boundary families.

42. RETRACTION OF COVERING CENTRES FOR PREPARATION TARGETS

The centres in a covering problem need not belong to the family being covered. For preparation targets this distinction can be removed exactly, without a small-error or rank assumption. Throughout this section a centre is one memoryless instrument, reused at every call; it is not a device with an additional persistent environment.

Theorem 42.1 (Preparation-centre retraction). *Let $\mathcal{C} = (\mathcal{C}_y)_{y=1}^m$ be any instrument from M_d to M_n , and set*

$$(42.1) \quad \tau_y = \mathcal{C}_y(I_d/d) = d^{-1} \text{Tr}_{\text{in}} J_y(\mathcal{C}), \quad \mathfrak{RC} = \mathcal{R}_\tau.$$

Then $\mathfrak{R}$ maps legal instruments to preparation instruments, fixes every preparation instrument, and preserves rational Choi data. For every $N \geq 1$ and every $\sigma \in \mathcal{S}_{m,n}$,

$$(42.2) \quad d_N(\mathcal{R}_\sigma, \mathfrak{RC}) \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Here d_N is the unhalved trace distance optimized over the common adaptive testers of Lemma 41.1. Consequently, for any subset of preparation targets and any radius $\delta > 0$, the minimum covering cardinality is unchanged when arbitrary legal memoryless centres are restricted to preparation centres. The assertion also holds when centres are required to have rational Choi entries.

Proof. Complete positivity gives $\tau_y \geq 0$, and trace preservation gives $\sum_y \text{tr} \tau_y = 1$. The formula is rational and is the identity on the preparation family, so $\mathfrak{R}^2 = \mathfrak{R}$. Feed the state I_d/d into each of N calls to $\mathcal{C}$, use a fresh input at the next call, and retain all outputs and outcome labels. The resulting state is $\tau^{\otimes N}$, since the centre is memoryless. The same tester applied to $\mathcal{R}_\sigma$ returns $\sigma^{\otimes N}$. Therefore

$$\|\sigma^{\otimes N} - \tau^{\otimes N}\|_1 \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Lemma 41.1 identifies the left side with the left side of (42.2). Retracting every centre in any cover preserves coverage and cannot increase cardinality. The reverse inequality follows because preparation centres are among the legal centres. The same argument preserves rationality. $\square$

The retraction does not preserve a target rank bound: for example the identity channel retracts to the maximally mixed preparation. Thus the centre restriction in Theorem 42.1 is to the full preparation family, not necessarily to the target rank stratum. The rank-constrained rational upper centres of Theorem 41.2 remain a separate, stronger construction. The theorem concerns the geometry of mathematical descriptions. Its maximally mixed test is a quantum experiment, not a finite-classical-message simulation of unknown inputs.

The reduction mechanism for input-erasing channels has a direct antecedent in Cooney–Mosonyi–Wilde [50]. Their discussion of two replacer channels reduces channel divergences to state divergences; their principal results concern asymptotic discrimination and strong-converse exponents

for a channel versus a replacer. Lemma 41.1 uses the same input-erasure mechanism in an exact finite-use trace-norm identity, whereas Theorem 42.1 concerns the allowed centres of an entire covering problem. No exact nonadaptive optimality for an arbitrary non-replacer channel against a replacer is inferred from that identity. The processor in the product identity is quantum, and input erasure, rather than entanglement breaking alone, is its essential hypothesis.

43. COHERENT AND PREPARATION DIRECTIONS AT A SINGULAR BOUNDARY

The preparation theorem has a square-root horizon scale. Reversible quantum information can instead accumulate a coherent discrepancy at a linear scale. We now exhibit both scales in a single non-input-erasing family, and determine their combined description cost.

Fix $d \geq 2$, $n, m \geq 1$ and public rank bounds $0 \leq r_y \leq n$, not all zero. For $U \in U(d)$ and $\sigma \in \mathcal{S}_r$, define an instrument from M_d to M_{dn} by

$$(43.1) \quad \mathcal{F}_{U,\sigma,y}(X) = UXU^* \otimes \sigma_y.$$

Its outcome register is retained. Global phase is immaterial, so the unitary parameter lies in $PU(d)$. Put

$$u = d^2 - 1, \quad v = \sum_y r_y(2n - r_y) - 1.$$

The Choi matrix of each outcome is, up to a fixed tensor permutation, $|U\rangle\rangle\langle\langle U| \otimes \sigma_y$. Its rank is $\text{rank } \sigma_y$, not $d \text{rank } \sigma_y$ as in the input-erasing family. These instruments transmit the data system unitarily, including its entanglement with a reference. Their preparation supports may change and may lie on any of the permitted singular strata. Nevertheless, the outcome probabilities do not depend on the input. This is a specified boundary family, not the whole body of disturbing instruments.

Let $\mathcal{C}_N(\delta; \mathbf{r})$ denote its covering number in d_N , with centres allowed to be any legal memoryless instrument on the same input, output and outcome spaces. Let $\mathcal{C}_N^{\mathbb{Q}}$ impose the additional condition that the centres have Gaussian-rational Choi entries. A target is supplied as U and its preparation blocks; no unknown-device learning interface is used.

Theorem 43.1 (Joint coding at a coherent–preparation boundary). *There are $0 < c \leq C < \infty$, depending only on the fixed dimensions and rank bounds, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(43.2) \quad c \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v \leq \mathcal{C}_N(\delta; \mathbf{r}) \leq \mathcal{C}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v.$$

The metric is the unhalved final-state trace norm, maximized over common adaptive quantum testers using the same memoryless instrument at most N times, with arbitrary finite references, quantum memory, feedback and bounded public stopping. In particular, the optimal fixed-length reusable description has length

$$(43.3) \quad (u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O_{d,n,m,\mathbf{r}}(1).$$

The upper centres can be chosen within the family (43.1), with rational unitary matrices and rational preparation blocks. They do not increase a target block rank and preserve its zero outcomes. A finite exact encoder exists for supplied rational unitaries and rational preparation blocks. Description length is not its workspace or running time, and decoding describes a quantum instrument, not a classical physical processor for unknown entangled inputs.

43.1. The two metric scales. For $U, V \in U(d)$ write

$$(43.4) \quad \chi(U, V) = \left(1 - \frac{|\text{tr}(U^*V)|^2}{d^2}\right)^{1/2}, \quad \Delta_N(\sigma, \tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

The first expression is the chordal distance between normalized Choi rays; it is a metric after quotienting global phase.

Lemma 43.2 (Anisotropic adaptive comparison). *For the instruments (43.1),*

$$(43.5) \quad \max\{\min\{N\chi(U, V)/\sqrt{2}, 1\}, \Delta_N(\sigma, \tau)\} \leq d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}),$$

$$(43.6) \quad d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}) \leq \min\{2, 2dN\chi(U, V) + \Delta_N(\sigma, \tau)\}.$$

All lower testers are permitted to depend on the pair being compared.

Proof. Discarding the data-system outputs gives the product-state lower bound. For the unitary bound, discard the preparation systems and outcome labels, and apply the same known operation U^* to the data output after each call. The two resulting data operations are I and $W = U^*V$. If $e^{i\phi_j}$ are the eigenvalues of W , then

$$\chi(U, V)^2 = \frac{2}{d^2} \sum_{i,j} \sin^2((\phi_i - \phi_j)/2).$$

There is an eigenvalue pair whose principal half-angle $\theta \in [0, \pi/2]$ satisfies $\sin \theta \geq \chi(U, V)/\sqrt{2}$. Prepare the equal superposition of its orthogonal eigenvectors. After k calls its two possible pure states have trace distance $2|\sin(k\theta)|$. If $N\theta \leq \pi/2$, take $k = N$ and use $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. The result is at least $N \sin \theta$. If $N\theta > \pi/2$, take $k = \lceil \pi/(4\theta) \rceil \leq N$. Then $\pi/4 \leq k\theta \leq 3\pi/4$, giving distance at least $\sqrt{2}$. The case $\theta = 0$ is immediate. This proves the first minimum in (43.5); the two lower witnesses need not coincide.

For the upper bound first change U to V , keeping σ fixed. The unnormalized Choi difference of the unitary channels has trace norm $2d\chi(U, V)$ and bounds their diamond distance. Appending a fixed preparation state does not increase that norm. A slot-by-slot hybrid and trace contraction for the common tester therefore cost at most $2dN\chi(U, V)$. Next hold V fixed and change σ to τ . All calls to V and all tester operations form one common quantum processor consuming N preparation states; unused copies are discarded at a public stop. This costs at most $\Delta_N(\sigma, \tau)$, not the sum of N one-copy trace distances. Triangle inequality gives (43.6). $\square$

Finite-use amplification of unitary differences is classical; see Acín [51] and Duan–Feng–Ying [52]. The elementary estimate above is included to specify its uniform constant, its pair-dependent controls and its compatibility with bounded stopping. Neither the existence of unitary discrimination nor the real dimension $d^2 - 1$ is claimed as a new ingredient.

43.2. A finite rational atlas for unitary channels. For $\ell \geq 1$, a signed stereographic chart on S^ℓ has the form

$$(43.7) \quad S_{j,\xi}(t)_j = \xi \frac{1 - \|t\|^2}{1 + \|t\|^2}, \quad S_{j,\xi}(t)_i = \xi \frac{2t_i}{1 + \|t\|^2} \quad (i \neq j), \quad \xi \in \{-1, 1\}.$$

For a target $q \in S^\ell$, choose j of largest absolute coordinate, $\xi = \text{sgn}(q_j)$ and $t_i = \xi q_i / (1 + |q_j|)$. Then $|t_i| \leq 1$ and $S_{j,\xi}(t) = q$. Direct expansion gives

$$(43.8) \quad \|S_{j,\xi}(t) - S_{j,\xi}(t')\| = \frac{2\|t - t'\|}{\sqrt{(1 + \|t\|^2)(1 + \|t'\|^2)}}.$$

Rounding each coordinate towards zero at denominator B thus gives a rational unit vector at distance at most $2\sqrt{\ell}/B$. Every decoded chart word is on the sphere, regardless of whether it is the canonical encoding of a target.

Lemma 43.3 (Rational unitary code). *Put $L = d(d-1)/2$ and $K_d = 3L + 2(d-1)$. There is an explicitly enumerable set $\mathcal{U}_B \subset U(d, \mathbb{Q}(i))$ satisfying*

$$(43.9) \quad |\mathcal{U}_B| \leq 6^L 4^{d-1} (2B+1)^{d^2-1}, \quad \sup_U \min_{V \in \mathcal{U}_B} \chi(U, V) \leq K_d/B.$$

For rational U , membership in any prescribed rational squared-error ball $\chi(U, V)^2 \leq b$ is decidable by rational arithmetic alone.

Proof. The usual complex Givens elimination, in a fixed order of index pairs, expresses every unitary as a product of L embedded matrices

$$\begin{pmatrix} c & -\bar{s} \\ s & c \end{pmatrix}, \quad c \in \mathbb{R}, \quad c^2 + |s|^2 = 1,$$

and a diagonal unitary. For completeness, a column pair (a, b) is sent to one nonzero coordinate using $c = |a|/(|a|^2 + |b|^2)^{1/2}$ and $s = cb/a$ when $a \neq 0$; the cases $a = 0$ or $(a, b) = 0$ are obtained by $c = 0$ or by the identity. Applying the adjoint rotation eliminates the second coordinate. Successive eliminations give an upper triangular unitary, necessarily diagonal. Removing the phase of its first diagonal entry leaves $d-1$ free phases. All factors are taken in the corresponding fixed reversed order when reconstructing U .

Each Givens matrix has an S^2 parameter $(c, \Re s, \Im s)$; its operator-norm change equals the Euclidean change of that vector. Each remaining phase has an S^1 parameter. Apply (43.7) at grid B to these $L + d - 1$ factors. There are 6 chart headers for each S^2 and 4 for each S^1 , with $2L + d - 1 = d^2 - 1$ grid digits altogether. Telescoping a product of unitaries gives, up to a global phase, operator error at most

$$(2\sqrt{2}L + 2(d-1))/B \leq K_d/B.$$

For any fixed phase z of modulus one, $\chi(U, V)^2 \leq d^{-1} \|U - zV\|_F^2 \leq \|U - zV\|_{\text{op}}^2$. This proves (43.9). The formula (43.4) has rational square for rational U, V , proving the last assertion. Enumeration of these finite words followed by the squared-distance test is an exact encoder whenever the radius is at least K_d/B . This does not assert a fast nearest-centre algorithm. $\square$

43.3. The joint covering theorem.

Proof of Theorem 43.1. For the upper bound use

$$B_U = \lceil 4dK_d N / \delta \rceil, \quad B_P = \max\{1, \lceil 8\sqrt{Nv} / \delta \rceil\}.$$

Lemma 43.3 makes the first term in (43.6) at most $\delta/2$. The factor code in Theorem 41.2 makes the second term at most $\delta/2$. Its finite pivot headers and anchor code give at most $C_{n,m,r}(2B_P + 1)^v$ legal images. If $v = 0$ the preparation family is a singleton and no such digits are needed. Multiplication of the two cardinalities proves the upper bound and all legality, rank and zero-outcome assertions. For rational data, search the finite unitary list using the rational test $\chi(U, V)^2 \leq \delta^2/(16d^2N^2)$; the preceding estimate guarantees a result. The preparation encoder is already exact. Decoded Choi entries are rational products of the two rational outputs. The finite search and the expanded matrices are not charged to the reusable payload.

For the converse, a neighborhood of the identity in $PU(d)$ contains a quantitatively bi-Lipschitz u -dimensional Euclidean patch for the metric χ . Here is a direct verification. Use traceless Hermitian coordinates H near zero and the ray of e^{iH} . The differential of its normalized Choi projection at zero vanishes only when H is scalar, hence only when $H = 0$. On a sufficiently small convex coordinate ball its differential is uniformly closer than half its least singular value to that at zero. Integration along a line segment gives both Lipschitz inequalities. Since the projection Frobenius distance is $\sqrt{2}\chi$, the asserted metric comparison follows. Volumetric packing therefore

gives at least $c_d(N/\delta)^u$ unitaries separated by more than $4\delta/N$, with a smaller fixed constant covering the coarse-scale cases as well.

The maximal-rank preparation patch used in Theorem 41.3 has real dimension v and is bi-Lipschitz in tuple Frobenius norm. Pack it at separation greater than $32\delta/\sqrt{N}$, obtaining at least $c_{n,m,r}(\sqrt{N}/\delta)^v$ points. The local binary bound in Corollary 40.2, applied to the preparation state (input dimension one), separates their N -copy states by more than 2δ . Indeed the bound at that scale is $(3/16)\min\{16\delta, 1\}$, which is 3δ before saturation and at least $3/16 > 2\delta$ afterwards. For $v = 0$ use its single point.

Take the Cartesian product of these packings. Two elements with different unitaries are separated by (43.5), since $\min\{2\sqrt{2}\delta, 1\} > 2\delta$. Two elements with the same unitary and different preparation blocks are separated by the other lower witness. Thus a ball of radius δ about any legal centre can contain at most one product point. This proves the lower bound even for centres outside the family. Taking base-two logarithms and rounding a fixed-length word length upward changes the answer by at most a fixed constant, proving (43.3). $\square$

The coefficient $u + v/2$ reflects two different reuse scales, rather than half of the total parameter dimension $u + v$. No spectral-gap assumption or positive eigenvalue margin is used. The constants are nevertheless fixed-dimensional. Nor is this a classification of every tangent direction of an arbitrary instrument: coupling an input-dependent measurement to its disturbance is not part of (43.1).

43.4. An explicit qubit codec. For $q = (q_0, q_1, q_2, q_3) \in S^3$ put

$$(43.10) \quad U(q) = \begin{pmatrix} q_0 + iq_3 & q_2 + iq_1 \\ -q_2 + iq_1 & q_0 - iq_3 \end{pmatrix}.$$

The points q and $-q$ describe the same channel. Choose a largest absolute coordinate, flip that common sign to make it positive, and use (43.7) with $\ell = 3$ and $\xi = 1$. Its header has four possibilities, and its body has exactly three signed digits.

Corollary 43.4 (Exact rational coherent-preparation description). *For a supplied rational unit quaternion and rational preparation blocks, the preceding three-digit chart, at $B_U = \lceil 16N/\delta \rceil$, combined with the preparation codec at tolerance $\delta/2$, gives a legal rational instrument within d_N error δ . Its fixed-length description has*

$$(3 + v/2) \log_2 N + (3 + v) \log_2(1/\delta) + O_{n,m,r}(1)$$

bits, with no increase of a block rank and exact preservation of zero outcomes. It uses only integer and rational arithmetic, including the integer square roots in the preparation code. Its output is a mathematical instrument description, not physical classical simulation.

Proof. For quaternion matrices, $\|U(q) - U(q')\|_{\text{op}} = \|q - q'\|$. The unitary diamond error is at most twice this quantity. Thus the chart's coherent contribution is at most $4\sqrt{3}N/B_U < \delta/2$. Its squared upper certificate is $48N^2/B_U^2$. The preparation contribution is at most $\delta/2$. Triangle inequality proves the claim; one may certify its square by $2(e_U^2 + e_P^2)$, not by $e_U^2 + e_P^2$. The chart returns an exactly unit quaternion, and the preparation decoder returns positive blocks of total trace one. Formula (43.1) therefore gives exact complete positivity and trace preservation, including on singular inputs. The chart has at most $4(2B_U + 1)^3$ words, which gives the stated length after adding the inherited preparation payload. $\square$

The reference encoder replays both deterministic encoders when verifying a supplied target. Bare decoding checks only syntax, rational legality and the declared construction budget. It cannot certify proximity to an unspecified target. In particular, reference stability concerns the quantum instrument defined by the decoded Choi matrices, not access to a classical processor with equivalent action on an unknown reference-entangled state.

44. COVERINGS THROUGHOUT THE SUBMAXIMAL ERROR RANGE

A pairwise separation argument at radius δ requires distances larger than 2δ . Since our adaptive distance is an unhalved trace norm, this argument cannot by itself reach $\delta \geq 1$. We instead bound the number of targets that one centre can cover. The same measurement, rather than a different binary test for each pair, will be used in this part of the converse. Dimensions are fixed throughout. For a target family $\mathcal{F}$, write $\mathcal{M}_N(\delta; \mathcal{F})$ for its least covering cardinality in d_N with arbitrary legal memoryless centres, and add a superscript $\mathbb{Q}$ when the decoded Choi blocks must be rational.

Lemma 44.1 (Multiplicity of a covering centre). *Let $J_1, \dots, J_K$ be targets and let one common tester with at most N calls have disjoint classical events $A_1, \dots, A_K$ such that $\Pr_{J_i}(A_i) \geq 1 - \eta$. If $0 \leq \eta < 1$ and $0 \leq \delta < 2(1 - \eta)$, every covering of these targets by M legal centres at adaptive radius δ satisfies*

$$(44.1) \quad M \geq (1 - \eta - \delta/2)K.$$

A centre need not belong to the target family. Indeed the conclusion still holds for a centre specified by a legal N -slot process with private memory, provided its distance to each assigned target controls this same tester.

Proof. Let Q be the tester's output distribution at a centre. For each target assigned to that centre, trace contraction under the terminal measurement gives $Q(A_i) \geq 1 - \eta - \delta/2$. The division by two is required because classical total variation is half the unhalved trace norm. Disjointness gives at most $(1 - \eta - \delta/2)^{-1}$ assigned targets. Sum this bound over centres. No product structure, unbiasedness or independence is required of Q . $\square$

Lemma 44.2 (One common finite-dimensional estimator). *Put $b = m(dn)^2$. On $\mathcal{C}_{d,n,m}$ there is, for every $N \geq 1$, one nonadaptive tester and a Hermitian tuple estimator $\hat{J}$ satisfying*

$$(44.2) \quad \sup_{J \in \mathcal{C}_{d,n,m}} \mathbb{E}_J |\hat{J} - J|_2^2 \leq \frac{C_0}{N}, \quad C_0 = 2d^2b^2.$$

The estimator need not be positive or trace preserving. The tester and estimator are independent of the unknown target and of any later choice of a packing set.

Proof. One normalized maximally entangled input produces the flagged Choi state $\Omega_J = d^{-1} \bigoplus_y J_y$. Choose a Hilbert–Schmidt orthonormal Hermitian basis $H_1, \dots, H_b$ of the block-diagonal space. Every H_j has operator norm at most one. Thus $(I + H_j)/2, (I - H_j)/2$ is a binary POVM, and its signed outcome has mean $\text{tr}(H_j \Omega_J)$ and variance at most one. For $N \geq b$, use $k = \lfloor N/b \rfloor$ independent Choi inputs for each of the b measurements. There are at most N calls. Let $\hat{a}_j$ be the corresponding empirical signed mean and set $\hat{\Omega} = \sum_j \hat{a}_j H_j$. Orthonormality gives

$$\mathbb{E}_J \|\hat{\Omega} - \Omega_J\|_F^2 = \sum_j \text{Var}(\hat{a}_j) \leq b/k \leq 2b^2/N.$$

Set $\hat{J} = d\hat{\Omega}$, interpreted as a tuple. For $N < b$ use $\hat{J} = 0$ and make no calls. Then $|J|_2^2 = d^2 \text{tr} \Omega_J^2 \leq d^2 \leq C_0/N$. This proves every asserted range, without a rank or eigenvalue-margin condition. The binary POVMs are block diagonal in the public outcome, so coherent access to different classical flags is not needed. $\square$

This elementary informationally complete estimation argument is a standard finite-dimensional tomography construction; compare the process estimation setting of [42]. Here it is used only to build a common lower-bound witness. It does not constitute a new optimal-learning claim, and the description encoder below is still supplied its target.

Theorem 44.3 (A full-range covering lower bound). *Let $\mathcal{F} \subseteq \mathcal{C}_{d,n,m}$ contain the image of a nonempty a -dimensional Euclidean ball under a bi-Lipschitz map for tuple Frobenius distance, with $a > 0$. For every fixed $0 < \delta_* < 2$ there is $c > 0$, depending on this patch, the fixed dimensions and δ_* , such that*

$$(44.3) \quad \mathcal{M}_N(\delta; \mathcal{F}) \geq cN^{a/2} \quad (N \geq 1, 0 < \delta \leq \delta_*).$$

Centres may be arbitrary legal memoryless instruments. The same lower bound holds under the relaxation to legal N -slot centres described in Lemma 44.1.

Proof. Set $\eta = (2 - \delta_*)/4 > 0$ and $r_N = (C_0/(\eta N))^{1/2}$. The previous lemma and Markov's inequality give $\Pr_J\{|\tilde{J} - J|_2 \leq r_N\} \geq 1 - \eta$ for every target. A volumetric packing of the fixed patch supplies at least $c_1 r_N^{-a}$ points at pairwise tuple distance greater than $2r_N$. The constant can be decreased to cover coarse scales: a singleton packing always exists. The estimator events consisting of the closed r_N -balls at these points are disjoint. Lemma 44.1 applies because $1 - \eta - \delta/2 \geq 1 - \eta - \delta_*/2 = \eta$. It gives $\mathcal{M}_N \geq \eta c_1 r_N^{-a}$, as required. In particular one centre may cover several packing points when the error is large; the proof bounds that multiplicity rather than asserting it is one. $\square$

Corollary 44.4 (Full-range interior and preparation laws). *Fix $0 < \delta_* < 2$. In Theorem 40.3, with fixed d, n, m, a and $s = d^2(mn^2 - 1) > 0$, both real and rational legal-centre covering numbers satisfy*

$$(44.4) \quad \log_2 \mathcal{M}_N(\delta; a) = \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) + O(1) = \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The constants may depend on δ_* and the positive margin a . For the preparation family in Theorem 41.3, the same extension holds with s replaced by $v = \sum_y r_y(2n - r_y) - 1$, with fixed dimensions and public output-rank bounds. Its constants require no positive eigenvalue margin. If $v = 0$ the family is a singleton and needs no payload.*

Proof. For $0 < \delta \leq 1/32$ these are the inherited joint laws. The interior ball and maximal-rank preparation chart used in their proofs are bi-Lipschitz patches of dimensions s and v , respectively. Theorem 44.3 therefore supplies the lower term $(s/2) \log_2 N + O(1)$, or $(v/2) \log_2 N + O(1)$, on the remaining range $1/32 < \delta \leq \delta_*$. On that range $\log_2(1/\delta)$ is bounded above and below by constants depending only on δ_* . For the upper bound use the old code at accuracy $1/32$, which is also accurate to δ . This proves a uniform additive remainder for both kinds of centres. If $\delta_* \leq 1/32$ there is no remaining range. $\square$

The cap $\delta_* < 2$ is substantive: at unhalved error two every nonempty family is covered by a single legal centre. Nothing above is uniform as $\delta_* \uparrow 2$, or along a varying error sequence converging to two. The exact zero-error endpoint is also separate from the positive-error coding law. The older coherent/preparation theorem retains its own $\delta \leq 1/32$ range; the present common tomography argument gives the half-dimension scale, not its coherent N^{-1} scale.

45. INPUT-DEPENDENT BOUNDARY INSTRUMENTS

Fix an orthonormal basis $e_1, \dots, e_d$ of the input and the associated complete dephasing map $\mathcal{D}(X) = \sum_x \langle e_x, X e_x \rangle |e_x\rangle \langle e_x|$. For positive semidefinite $n \times n$ matrices σ_{xy} satisfying $\sum_y \text{tr } \sigma_{xy} = 1$ for every x , define

$$(45.1) \quad \Phi_{\sigma,y}(X) = \sum_{x=1}^d \langle e_x, X e_x \rangle \sigma_{xy}, \quad 1 \leq y \leq m.$$

These instruments measure a fixed input basis and then prepare an outcome-dependent state. Their public outcome probabilities $\sum_x \langle e_x, X e_x \rangle \text{tr } \sigma_{xy}$ may depend on the input. Measurement

destroys input coherences; the family is not in general input erasing, and it is not the coherent tensor family of Theorem 43.1.

The Choi blocks and their ranks are

$$(45.2) \quad J_y = \bigoplus_{x=1}^d \sigma_{xy}, \quad \text{rank } J_y = \sum_x \text{rank } \sigma_{xy}.$$

In particular the marginal constraint is exactly one unit-trace condition per input row. Fix public integers $0 \leq r_{xy} \leq n$, with $\sum_y r_{xy} > 0$ in every row, and let $\mathcal{F}_r$ be the family with $\text{rank } \sigma_{xy} \leq r_{xy}$. Put

$$(45.3) \quad v_x = \sum_y r_{xy}(2n - r_{xy}) - 1, \quad V = \sum_x v_x.$$

The ranks here are conditional output-state ranks, not independent rank bounds on arbitrary Choi blocks. The input basis, dimensions and rank bounds are fixed, not learned from an unknown device.

Theorem 45.1 (Rank-boundary coding with input-dependent outcomes). *Fix the preceding data and $0 < \delta_* < 2$. If $V > 0$, the least covering cardinality by arbitrary legal memoryless centres and the least cardinality with rational decoded Choi blocks both obey*

$$(45.4) \quad \log_2 \mathcal{M}_N(\delta; \mathcal{F}_r) = \frac{V}{2} \log_2 N + V \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The error is the unhalved final-state trace distance over arbitrary common quantum testers with finite references, quantum memory, feedback and public stopping bounded by N . A rational upper code has decoded centres in (45.1); it never increases any conditional block rank and preserves every zero block exactly. Its encoder on rational target data uses integer and rational arithmetic only. If $V = 0$ the family is a known singleton.*

The remainder depends on the fixed dimensions, rank bounds and δ_ , but not on N , δ or a smallest positive output eigenvalue. Payload length is distinct from encoder workspace and expanded decoder storage. The decoded object is a mathematical quantum instrument, not a physical classical simulator on an unknown reference-entangled input.*

45.1. Centres and the adaptive metric. Write $\Omega_x = \bigoplus_y \sigma_{xy}$ for the flagged state of one input row. A useful general observation first identifies the allowed centres.

Proposition 45.2 (Idempotent input retraction). *We use $\Phi\Pi = \Phi \circ \Pi$: the rightmost map acts on the input first. Let Π be a fixed completely positive trace-preserving idempotent input map and let every target satisfy $\Phi\Pi = \Phi$. Then*

$$(45.5) \quad d_N(\Phi, K\Pi) \leq d_N(\Phi, K)$$

for every legal memoryless centre K and every N . Thus covering numbers with all legal centres equal those with centres fixed by right composition with Π , at every radius. For $\Pi = \mathcal{D}$, the retracted centre has rows $\tau_{xy} = K_y(|e_x\rangle\langle e_x|)$ and is of the form (45.1). In the prescribed coordinate basis this operation preserves rationality. It need not preserve the target's rank bounds.

Proof. For any tester of Φ against $K\Pi$, insert Π on its input immediately before each use of the unknown device. This is a legal common tester of Φ against K : in the first hypothesis $\Phi\Pi = \Phi$, and in the second it implements $K\Pi$. References, feedback and stopping are unaffected. Taking the supremum proves (45.5). Idempotence says that $K\Pi$ is fixed by right composition with Π . Map every covering centre this way; cardinality cannot increase. The reverse covering inequality follows from inclusion of allowed centre classes. For dephasing, the displayed rows are positive and have unit total trace, since K is an instrument. They are diagonal input blocks of the original Choi matrices. $\square$

This is a data-processing retraction, not a rank-preserving projection. It generalizes the preparation-centre observation of Theorem 42.1, whose fixed input map discards its input and supplies one chosen state. In the converse of Theorem 45.1, no rank promise is imposed on centres.

Lemma 45.3 (Row witnesses and a common programme). *For two instruments of (45.1),*

$$(45.6) \quad \max_x \|\Omega_x^{\otimes N} - \Lambda_x^{\otimes N}\|_1 \leq d_N(\Phi_\sigma, \Phi_\tau) \leq \left\| \bigotimes_x \Omega_x^{\otimes N} - \bigotimes_x \Lambda_x^{\otimes N} \right\|_1.$$

If unit purifications a_x, b_x of the corresponding row states are chosen and $\eta_x = \|a_x - b_x\|$, then

$$(45.7) \quad d_N(\Phi_\sigma, \Phi_\tau) \leq 2\sqrt{N \sum_x \eta_x^2}.$$

For one use there is the separate exact identity

$$(45.8) \quad d_1 = \max_x \|\Omega_x - \Lambda_x\|_1.$$

Proof. For the first inequality repeatedly feed $|e_x\rangle\langle e_x|$ and retain all flagged outputs. For the second, prepare, for each of the N slots, one copy of every row state. A common processor measures the actual input in the prescribed basis and uses only the programme state with that outcome x . The remaining row states in that slot are discarded. If the input is entangled with the tester's memory, its conditional memory operator is retained by this measurement; appending the selected row state gives exactly (45.1) on that joint input. The unobserved input outcome is internal to the processor, not new feedback granted to the tester. All later operations and the discard of unused slots at a stop form one common CPTP processor. Trace contraction proves the upper bound.

Purify all these programme states by the chosen vectors. For unit vectors, $1 - |\langle a_x, b_x \rangle|^2 \leq \eta_x^2$. The elementary product inequality $1 - \prod t_j \leq \sum (1 - t_j)$ for $t_j \in [0, 1]$ gives pure-programme trace distance squared at most $4N \sum_x \eta_x^2$. Partial trace and the common processor give (45.7). Finally, for an arbitrary one-use input with reference, the difference is $\sum_x \xi_R^x \otimes (\Omega_x - \Lambda_x)$, where $\xi_R^x \geq 0$ and $\sum_x \text{tr} \xi_R^x = 1$. The triangle inequality gives the one-use upper maximum, and a basis input attains it. $\square$

The product programme consumes up to dN row states in the proof, whereas the actual tester makes at most N instrument calls. It is only a comparison device, not an equality of these resource counts. Neither inequality in (45.6) is asserted to be an equality for general N . In particular the absence of an asymptotic adaptive advantage in a binary discrimination exponent does not imply a finite-use metric equality; compare [49, Sections II and V].

45.2. Exact rational decoding and the covering proof.

Lemma 45.4 (A common-grid row code). *For rational target data, apply the singular factor encoder of Theorem 41.2 separately in every input row at a common integer grid $B \geq 1$. If h_x is that row's actual real factor-coordinate count, the decoded instrument satisfies*

$$(45.9) \quad d_N(\Phi, \tilde{\Phi})^2 \leq \frac{16N \sum_x (h_x - 1)}{B^2} \leq \frac{16NV}{B^2}.$$

There are at most $C_{d,n,m,r}(2B+1)^V$ legal decoder images. The conditional block ranks do not increase; consequently no outcome Choi rank increases either. All zero conditional blocks remain zero. The expanded common denominator is at most $B^{2d} \prod_x h_x$ and thus has $O_{d,n,m}(1 + \log B)$ bits.

Proof. The inherited anchor construction gives a unit factor purification in row x within distance $2\sqrt{h_x - 1}/B$ of the target's chosen unit factor vector. Apply (45.7). Trace normalization is performed separately in each row, so (45.2) is exactly trace preserving. The inherited zero-pivot proof, rational squared coordinates and integer-square-root ratio calculation apply row by row. Their finite masks and anchor headers have fixed cardinality; the row body has at most v_x signed grid coordinates. Multiplying these counts gives the asserted exponent V . Only valid decoded words count. The block-rank and zero statements hold row by row and pass to the direct sum in (45.2). Row x has normalization denominator $T_x \leq h_x B^2$. A common denominator is the least common multiple of the T_x , bounded by their product. This proves the final assertion. $\square$

Proof of Theorem 45.1. For the upper bound take $B = \max\{1, \lceil 4\sqrt{NV}/\delta \rceil\}$. The last lemma gives error at most δ and at most $C(2B + 1)^V$ centres, hence the upper description law for $\delta \leq \delta_* < 2$. Real targets admit the same factor rounding construction; the decoded matrices are still rational. For supplied rational targets, no irrational factor is stored: the exact inherited square-comparison algorithm determines the digits. Thus the real-family rational-centre upper bound does not restrict the targets to a countable subclass.

For the lower bound choose, in every row with $v_x > 0$, the gauge-fixed maximal-rank factor patch used in Theorem 41.3. Its inverse is the local leading-block Cholesky map. Rows with $v_x = 0$ are fixed. A Cartesian product of sufficiently small closed coordinate balls contains a V -dimensional ball. The map to the tuple (σ_{xy}) , and hence to J , is bi-Lipschitz; direct sum is an isometry for tuple Frobenius norm. This supplies one fixed V -dimensional patch with no claim of a global factor chart. For $\delta \leq 1/32$ pack it at separation greater than $32d\delta/\sqrt{N}$. Corollary 40.2 gives distances larger than 2δ , so even an unrestricted legal centre covers at most one packing point. This gives $c(\sqrt{N}/\delta)^V$ centres. For $1/32 < \delta \leq \delta_*$ use Theorem 44.3 on the same patch and the boundedness of $\log(1/\delta)$ on that interval. These lower bounds apply also to rational centres, which form a smaller class. Taking logarithms, with the usual ceiling for fixed-length words, completes the proof. If $V = 0$, each nonempty row consists of a specified one-dimensional unit block; its location is fixed by the public rank bounds. Thus the whole family is a known singleton. $\square$

For example, with scalar output $n = 1$ and all rank bounds equal to one, (45.4) has $V = d(m - 1)$: it is a boundary coding law for the rows of a classical stochastic matrix. With one quantum output and full row ranks, $V = d(n^2 - 1)$; with one pure output per row, $V = d(2n - 2)$. Mixed and pure conditional states need not commute. These dimensions are standard rank-stratum dimensions; the theorem identifies their joint coding scale under the declared adaptive metric. The measured input basis is fixed. No law for varying measurement bases, unknown instruments, or all irreversible quantum dynamics follows from this result.

46. VARYING PROJECTIVE READOUT AND CONDITIONAL PREPARATIONS

The fixed input basis can be allowed to vary when its outcome remains observable. This changes the coding scale: destruction of the measured quantum input does not make the readout coordinate a preparation coordinate. The distinguishing experiment may use entangled inputs.

Let $\mathfrak{F}_d$ be the space of ordered orthogonal rank-one projections $P = (P_1, \dots, P_d)$ on $\mathbb{C}^d$ with $\sum_x P_x = I_d$. It is the flag manifold $U(d)/\mathbb{T}^d$, of real dimension $b = d^2 - d$. The column phases are immaterial; permutation of the ordered outcomes is not a gauge. For each x let $\sigma_x = (\sigma_{xy})_{y=1}^m \in \mathcal{S}_{m,n}$ and set

$$(46.1) \quad \mathcal{F}_{P,\sigma,(x,y)}(X) = \text{tr}(P_x X) \sigma_{xy}, \quad J_{xy} = P_x^\top \otimes \sigma_{xy}.$$

Both labels (x, y) are retained. The transpose in this input-first Choi formula is essential, including for complex projections. Fix public bounds $0 \leq r_{xy} \leq n$, with at least one positive bound in each

row, and put

$$v_x = \sum_y r_{xy}(2n - r_{xy}) - 1, \quad V = \sum_x v_x.$$

The family $\mathfrak{R}_r$ consists of (46.1) with these rank bounds. Let $\mathcal{R}_N(\delta)$ be its covering number for d_N , allowing arbitrary legal memoryless instrument centres of the same interface; $\mathcal{R}_N^{\mathbb{Q}}(\delta)$ requires rational centres. Public dimensions, outcome labels, ranks, horizon and tolerance are not payload.

Theorem 46.1 (Mixed-scale coding with observable readout). *For fixed d, n, m, r there are constants $c, C > 0$ such that for $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(46.2) \quad c(N/\delta)^b(\sqrt{N}/\delta)^V \leq \mathcal{R}_N(\delta) \leq \mathcal{R}_N^{\mathbb{Q}}(\delta) \leq C(N/\delta)^b(\sqrt{N}/\delta)^V.$$

Consequently the optimal reusable description length is

$$(46.3) \quad (b + V/2) \log_2 N + (b + V) \log_2(1/\delta) + O(1),$$

with remainder uniform in N and δ . The upper centres belong to the displayed family, preserve its zero conditional blocks and do not increase any conditional rank. For supplied rational projections and rational row states there is a finite exact rational encoder, without choosing algebraic unit-vector representatives or searching an exhaustive unitary net. Real targets admit the same rational-centre cover by the exact-real construction. If $b + V = 0$, the family is a known singleton and needs zero payload.

The error range here is not the full submaximal range of Theorem 45.1. The readout part of the lower proof uses pairwise separation; it is not an application of the full-error half-dimensional estimator theorem.

46.1. The two distinguishing scales. Write $M_P(X) = (\text{tr}(P_x X))_x$ and $q(P, Q)^2 = \sum_x \|P_x - Q_x\|_F^2$.

Lemma 46.2 (Readout modulus). *If $z = d_1(M_P, M_Q)$, then*

$$(46.4) \quad \sqrt{2/d} q(P, Q) \leq z \leq \sqrt{d} q(P, Q).$$

For $z < 2$ one has

$$(46.5) \quad d_N(M_P, M_Q) = 2 \sin(\min\{N \arcsin(z/2), \pi/2\}).$$

For $z = 2$ the distance is two at every $N \geq 1$. In particular it is at least $(2/\pi) \min\{Nz, 1\}$ and at most Nz .

Proof. The difference of two rank-one projections has nonzero eigenvalues $\pm s$, so its Frobenius norm is $\sqrt{2}s$. An input eigenvector maximizing one event difference gives total unhalved outcome distance at least $2 \max_x \|P_x - Q_x\|_{\infty}$. This proves the lower bound. For an arbitrary reference input state ρ , duality gives

$$\|\text{tr}_A(((P_x - Q_x) \otimes I)\rho)\|_1 \leq \|P_x - Q_x\|_{\infty}.$$

Indeed a Hermitian contraction H on the reference tests this block by $\text{tr}(((P_x - Q_x) \otimes H)\rho)$. Sum over the classical blocks and apply Cauchy–Schwarz to obtain the upper bound in (46.4).

We use here the exact measurement-discrimination theorem of Puchała–Pawela–Krawiec–Kukulski–Oszmaniec [56, Corollary 1 and Theorem 2]. If U represents the relative ordered basis, their optimized eigenvalue-arc length $\Upsilon(U)$ gives distance $2 \sin(N\Upsilon(U)/2)$ until $N\Upsilon(U) \geq \pi$, when the distance is two; their Theorem 2 allows general adaptive networks. At one use, $z < 2$ implies $\Upsilon(U) = 2 \arcsin(z/2)$. The already-perfect case is treated separately and need not be inverted. This proves (46.5). A publicly stopped tester is contained in a fixed N -slot network by performing irrelevant calls after the stop and discarding their outputs. Conversely, parallel testers are admissible here. The elementary bound $\sin t \geq 2t/\pi$ for $0 \leq t \leq \pi/2$ gives the stated lower

modulus; the upper bound also follows directly by a hybrid argument. The exact multiple-use theorem is an imported result, not an inference from a finite-dimensional spectral gap. $\square$

Lemma 46.3 (Measurement and programme comparisons). *For arbitrary row states,*

$$d_N(\mathcal{F}_{P,\sigma}, \mathcal{F}_{Q,\tau}) \geq d_N(\mathbf{M}_P, \mathbf{M}_Q).$$

When $P = Q$, the distance is at least $\max_x \|\Omega_x^{\otimes N} - \Lambda_x^{\otimes N}\|_1$, where $\Omega_x = \bigoplus_y \sigma_{xy}$ and $\Lambda_x = \bigoplus_y \tau_{xy}$. For purifications whose rowwise vector differences have lengths η_x , one has

$$(46.6) \quad d_N(\mathcal{F}_{P,\sigma}, \mathcal{F}_{Q,\tau}) \leq Nd_1(\mathbf{M}_P, \mathbf{M}_Q) + 2\sqrt{N \sum_x \eta_x^2}.$$

Proof. Discarding the quantum outputs and y gives the measurement channel, so every measurement tester is a tester of the displayed instruments. For a common flag, feeding a unit vector in $\text{ran } P_x$ at every call gives the x th product-state witness. For the upper bound first change P to Q while keeping σ fixed. Appending the conditional row state is a common postprocessing of the readout, and a slotwise hybrid gives the first term. Then change the row states with Q fixed. At each slot a common processor measures Q , selects one of the supplied row programme states, retains x , and discards the unused rows. The product-purification argument of Lemma 45.3 gives the second term, also with references, feedback and stopping. This is the standard common-environment method [53, Proposition 2]; retaining the readout label changes neither its positivity nor its contraction argument. The upper comparison is not asserted to equal either lower comparison. $\square$

46.2. A bounded rational atlas without column phases. There is a direct flag codec, rather than an exhaustive search over unitaries. We give its quantitative bound because it also controls singular rational descriptions whose unit vectors are not rational.

Lemma 46.4 (Pivoted triangular flag charts). *Every $P \in \mathfrak{F}_d$ is obtained from an ordered basis of columns of $\Pi^{-1}L$, where Π is a permutation matrix and L is unit lower triangular with complex entries of modulus at most one below the diagonal. Its ordered rank-one projections are the successive differences of projections onto the first j columns. Thus $d!$ charts, each with $b = d^2 - d$ real coordinates, suffice. Given rational P_x , both the chart and its coordinates can be found rationally.*

Round each real and imaginary coordinate of L towards zero at grid $B \geq 1$ and let $\hat{P}$ be the decoded flag. Put

$$K_d = (2d)^{d-1}, \quad C_d = 4d^2 K_d.$$

Then every decoded flag is exactly legal and rational, and

$$(46.7) \quad d_N(\mathbf{M}_P, \mathbf{M}_{\hat{P}}) \leq C_d N/B.$$

There are at most $d!(2B+1)^b$ codewords. The expanded projection entries have $O_d(1 + \log B)$ bits.

Proof. Choose a nonzero column $v_x = P_x e_{k_x}$, taking the first positive diagonal position. These columns are rational whenever the input projections are rational. The columns are nonzero and mutually orthogonal; hence $A = [v_1 \ \cdots \ v_d]$ is invertible. Gaussian elimination with partial row pivoting gives $\Pi A = LR$, with L unit lower triangular and R upper triangular and invertible. At every pivot choose a residual entry of largest modulus, breaking ties by row order. Each multiplier below it has modulus at most one. Subsequent row interchanges interchange the already recorded multipliers as well. All comparisons of squared moduli and all arithmetic are rational for rational A . Right multiplication by an invertible upper triangular matrix preserves each initial column span. Thus the first j columns of $\Pi^{-1}L$ span $\text{ran}(P_1 + \cdots + P_j)$. This proves the chart assertion without choosing square roots.

Any unit lower triangular matrix in the coordinate cube $|\Re L_{ij}|, |\Im L_{ij}| \leq 1$ is invertible, even if it was not produced by elimination. If Z_j is its first j columns, put

$$E_j = Z_j(Z_j^* Z_j)^{-1} Z_j^*, \quad E_0 = 0.$$

The differences $E_j - E_{j-1}$ are mutually orthogonal rank-one projections with sum I . Permute their coordinates back by Π . The same formulas decode the rounded rational matrix exactly.

Throughout the cube, $\|L\|_2 \leq 2d$ and $\det L = 1$, so its least singular value is at least K_d^{-1} . Restriction to any first j columns preserves this lower bound. For two such matrices L, L' , write E_j, E'_j for their column-space projections. Using $(I - E'_j)Z'_j = 0$ and $E_j = Z_j Z_j^\dagger$ gives

$$\|(I - E'_j)E_j\|_F \leq K_d \|Z_j - Z'_j\|_F.$$

The analogous bound with the primes interchanged and the identity $E_j - E'_j = (I - E'_j)E_j - E'_j(I - E_j)$ imply $\|E_j - E'_j\|_F \leq 2K_d \|L - L'\|_F$. Consequently each rank-one difference is at most $4K_d \|L - L'\|_F$. The one-use block estimate used above bounds the measurement distance by the sum of these norms. Rounding changes L by at most $\sqrt{b}/B \leq d/B$, proving the one-use bound C_d/B and then (46.7) by hybridization.

Record the permutation and the b signed digits in $[-B, B]$. No phases or rational unit-vector normalizations are encoded. All inverse Gram matrices in the displayed decoding formula have fixed size and rational entries of degree bounded in terms of d in the digits and B . The determinant is nonzero. Clearing denominators and the adjugate formula bound every numerator and denominator by $B^{O_d(1)}$ times a fixed constant. This proves the bit-length claim. $\square$

Partial pivoting is used here only to select a bounded chart. The decoder itself needs no pivot test on the target and always returns a flag for any valid chart word. A target-bound verifier nevertheless checks the entire canonical encoding, not just legality.

46.3. Proof of the covering law.

Proof of Theorem 46.1. For the upper bound, use the flag code at $B_f = \max\{1, \lceil 2C_d N / \delta \rceil\}$ and the conditional factor code at $B_s = \max\{1, \lceil 8\sqrt{NV} / \delta \rceil\}$. The two terms of (46.6) are each at most $\delta/2$. The factor code counts every actual pivot mask, anchor and sign. Its total cardinality is at most $C(2B_s + 1)^V$. Multiply by $d!(2B_f + 1)^b$. On the stated range this proves the upper bound in (46.2). Each factor row is trace normalized and rank nonincreasing. Formula (46.1) then gives positive Choi blocks with joint partial trace I_d ; their ranks equal those of the conditional states. Rationality and zero preservation follow exactly. Real target projectors admit the same chart selection and rounding; only the algorithmic assertion about rational targets uses finite rational comparisons.

For the lower bound, choose a small closed coordinate box around a flag in $\mathfrak{F}_d$ on which q is bi-Lipschitz to a Euclidean metric in b dimensions. Such a chart can also be checked directly from $L = I + T$: to first order the off-diagonal entries of the ordered projections recover every real and imaginary entry of strictly lower triangular T . The derivative is injective, and shrinking to a compact box gives fixed forward and inverse Lipschitz constants. Grid-pack it at q -separation greater than $A_d \delta / N$, where A_d is fixed large enough that Lemma 46.2 gives distance strictly greater than 2δ . Indeed its constant lower plateau $2/\pi$ exceeds 2δ on this range. The packing has at least $c(N/\delta)^b$ points after reducing c to include the coarse-grid cases.

In every nontrivial row choose the compact maximal-rank factor patch from Theorem 41.3. The product row patch has V real dimensions. Pack it in tuple Frobenius distance at more than $A_{d,n,m,r} \delta / \sqrt{N}$. Some row of two distinct grid points then differs by at least the tuple distance divided by $\sqrt{d}$. The local product-Choi test (Corollary 40.2, with input dimension one) implies

that its N -copy states are separated by more than 2δ . The row packing has cardinality at least $c(\sqrt{N}/\delta)^V$.

Take the Cartesian product of these two finite packings. If two points have different flags, discarding the conditional outputs gives the separating measurement witness regardless of their rows. If their flags agree, the product-state witness in a differing row separates them. These are legitimate pair-dependent testers, not incompatible tests assumed to hold in one common experiment. Hence all distinct product points have d_N -distance greater than 2δ . By the triangle inequality, a ball centred at any legal memoryless instrument covers at most one of them. Multiplying cardinalities proves the lower bound. A fixed-dimensional ceiling in the word length changes only the additive constant. Finally, $b + V = 0$ forces $d = 1$ and a singleton rank-state family, proving the last assertion. $\square$

Corollary 46.5 (Readout recovered from orthogonal output supports). *Suppose the label x in (46.1) is not output, but the x th row is supported in a fixed public subspace H_x , with the H_x mutually orthogonal. Then the same coding law holds with n replaced by $n_x = \dim H_x$ in each row dimension. For public rational coordinate embeddings the upper code is rational.*

Proof. Measuring the fixed output projections onto H_x recovers x without disturbing the target's conditional state. This operation and forgetting x are fixed CPTP postprocessings inverse on the family. Inserting them at every slot proves equality of the two metrics on target pairs, with references and adaptive feedback. The preceding upper code and product packing therefore apply. The lower packing still allows arbitrary legal centres. We do not infer a target-independent input retraction for varying flags, or a rank bound on the postprocessing of an arbitrary centre. $\square$

Remark 46.6 (What can disappear when readout is unobservable). If the label x is discarded and all conditional rows equal one state tuple σ , then $\sum_x \text{tr}(P_x X) \sigma_y = \text{tr}(X) \sigma_y$ for every flag. Thus the entire basis coordinate disappears exactly. This example separates an observable readout theorem from an unjustified parameter count for arbitrary unlabelled overlapping output rows. The two proved families above do have the asserted mixed scale; neither statement classifies a general coupled coherent/support-changing instrument boundary.

47. UNIFORMLY SEIZABLE PROGRAMMES AND UNRESTRICTED CENTRES

Programme-state comparisons become exact when a single physical procedure can recover the programme from one use, uniformly over the whole family. The state/channel discrimination principle is classical [54, Definition 36 and Theorem 37]; see also the channel-box formulation [55, Section III.A]. We state its covering consequence with the centre classes explicit.

Let $\mathcal{S}$ be the full state space of a fixed finite direct-sum matrix algebra. Let $\mathcal{A}(\omega)$ be the instrument obtained by feeding a state $\omega \in \mathcal{S}$ into a fixed CPTP processor along with its input. Let $\mathcal{B}$ be a fixed one-use tester which returns a state in $\mathcal{S}$ from any legal instrument of the same interface. Assume

$$(47.1) \quad \mathcal{B}(\mathcal{A}(\omega)) = \omega \quad (\omega \in \mathcal{S}).$$

The tester and processor do not depend on the target, the pair of hypotheses, the tolerance or the covering centre. They may have a fixed finite quantum memory. In particular a common programme representation alone is not this hypothesis.

Theorem 47.1 (Exact transfer of family covering numbers). *Under (47.1), for every $N \geq 1$,*

$$(47.2) \quad d_N(\mathcal{A}(\omega), \mathcal{A}(\tau)) = \|\omega^{\otimes N} - \tau^{\otimes N}\|_1.$$

The map $\mathcal{R} = \mathcal{A}\mathcal{B}$ on legal memoryless instrument centres is an idempotent one-use superchannel, and

$$(47.3) \quad d_N(\mathcal{A}(\omega), \mathcal{R}(K)) \leq d_N(\mathcal{A}(\omega), K).$$

For every subset $S \subseteq \mathcal{S}$ and every radius δ , its N -copy state covering number with arbitrary centres in $\mathcal{S}$ equals the d_N covering number of $\mathcal{A}(S)$ with arbitrary legal memoryless instrument centres.

In particular, for the rank-bounded state family $\mathcal{S}_r \subseteq \mathcal{S}_{m,n}$ with $v = \sum_y r_y(2n - r_y) - 1$, the latter logarithmic covering number is

$$(47.4) \quad (v/2) \log_2 N + v \log_2(1/\delta) + O(1)$$

for every fixed $0 < \delta_* < 2$, uniformly over $N \geq 1$, $0 < \delta \leq \delta_*$. If $\mathcal{A}$ sends rational programme states to rational Choi matrices, the same asymptotic law holds with rational centres and is attained by the inherited rational factor code followed by $\mathcal{A}$. The code preserves programme block ranks and zeros. No Choi-rank preservation of $\mathcal{R}$ is asserted.

Proof. Give a common processor N independent copies of the programme, and use one fresh copy at each device call. All tester operations, including discarding the unused programmes after a stop, compose into one common channel. Data processing proves the upper bound in (47.2). Repeating the one-use seizing procedure N times and retaining its states gives the reverse bound. This is the established environment-seizing reduction with its finite-use normalization made explicit.

To implement $\mathcal{R}(K)$ on an external input, retain that input in a quantum memory, use one fresh invocation of K inside $\mathcal{B}$, and feed the resulting programme together with the retained input to $\mathcal{A}$. This is a fixed one-use pre- and postprocessing, hence a superchannel. Equation (47.1) implies $\mathcal{R}^2 = \mathcal{R}$ and $\mathcal{R}(\mathcal{A}(\omega)) = \mathcal{A}(\omega)$. Replacing each slot of an arbitrary tester against $\mathcal{A}(\omega)$ and $\mathcal{R}(K)$ by this construction produces a common tester against $\mathcal{A}(\omega)$ and K with identical two output states. It uses no more device calls and is compatible with reference entanglement and stopping. Taking suprema proves (47.3).

A state cover $\{\tau_i\}$ gives an instrument cover $\{\mathcal{A}(\tau_i)\}$ by (47.2). Conversely, map each arbitrary instrument centre K_i to $\tau_i = \mathcal{B}(K_i)$. Equations (47.3) and (47.2) show that every target covered by K_i is covered by τ_i at the same radius in the state metric. This proves exact equality of covering cardinalities, not just a two-hypothesis equivalence. The centres τ_i need not obey the rank promises on the targets; this is why (47.1) is imposed on the whole ambient programme space.

Apply the preparation-state law of Theorem 41.3, the preparation-centre retraction Theorem 42.1, and its full-error extension Corollary 44.4. They give (47.4) for arbitrary programme centres. Rationality-preserving $\mathcal{A}$ maps the rational rank-preserving upper code into legal rational instrument centres, while the unrestricted lower bound applies to this smaller class. If $v = 0$, zero payload suffices for the known singleton. The constants can depend on the fixed dimensions, rank bounds and error cap; they are not uniform as $\delta_* \uparrow 2$. $\square$

Corollary 47.2 (Pauli-channel descriptions). *Let $\sigma_0 = I, \sigma_1 = X, \sigma_2 = Y, \sigma_3 = Z$ be the Pauli matrices and let $\Phi_p(\rho) = \sum_{j=0}^3 p_j \sigma_j \rho \sigma_j^*$. Fix a nonempty public allowed support of cardinality k for p , allowing zero probabilities on that support. With arbitrary legal qubit-channel centres, its optimal reusable description length is*

$$\frac{k-1}{2} \log_2 N + (k-1) \log_2(1/\delta) + O(1), \quad N \geq 1, \quad 0 < \delta \leq \delta_* < 2.$$

Rational upper descriptions preserve zero probabilities and never increase the Pauli-channel Choi rank of the target.

Proof. The classical programme p controls which Pauli matrix is applied. One maximally entangled input and a Bell-basis measurement recover p exactly. Both operations are fixed over the full probability simplex, so Theorem 47.1 applies with $n = 1$ and one-dimensional blocks at the allowed positions. The orthogonal Bell Choi vectors show that the Choi rank equals the number of positive probabilities. The inherited scalar factor encoder keeps zeros zero and has rational output. This

is a coding consequence of the standard Pauli/Bell seizing mechanism, not a new discrimination protocol. $\square$

For contrast, $U = (I + iX + iY + iZ)/2$ is unitary, so its channel has Choi rank one. Its Bell probabilities are all $1/4$, and $\mathcal{AB}$ sends it to the uniform Pauli mixture of Choi rank four. This exact example explains why arbitrary-centre retraction and rank-preserving target encoding are different claims. The processor in a programme-state comparison is quantum. Description length in (47.4) does not count its Hilbert-space dimension or give an unknown-channel learning or classical physical simulation theorem.

48. A NOISE-UNIFORM READOUT TRANSITION

The dependence on the readout basis can change scale when noise approaches an extremal measurement. We determine that transition in a fixed one-dimensional family, without a positive spectral margin uniform in the noise. The proof is a finite-distance argument; no conclusion is inferred merely from a Fisher-information calculation.

Let σ_1, σ_2 be the first two Pauli matrices. For $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and a public visibility $0 \leq \lambda \leq 1$, put

$$(48.1) \quad E_{\theta,\lambda}^{\pm} = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)), \quad \mathcal{M}_{\theta,\lambda}^{\pm}(X) = \text{tr}(E_{\theta,\lambda}^{\pm} X).$$

Both outcome labels are retained. The output quantum system is one-dimensional. In particular, θ and $\theta + \pi$ are not identified by an outcome relabelling. Let $h(\theta, \phi) \in [0, \pi]$ be circular distance and set

$$(48.2) \quad \eta = 1 - \lambda^2, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, \eta^{-1}\}}, \quad N \geq 1,$$

where $\eta^{-1} = +\infty$ at $\lambda = 1$. Thus $K_{N,1} = N$ and $K_{N,0} = 0$. The visibility may depend arbitrarily on N and the requested error; it is not a target coordinate charged in this theorem.

Write $\mathcal{C}_{N,\lambda}(\delta)$ for the least number of legal memoryless binary instruments of this interface whose d_N -balls of radius δ cover all targets in (48.1). Centres need not belong to the displayed family. Define $\mathcal{C}_{N,\lambda}^{\text{rat}}(\delta)$ by restricting centres to family members with rational $\cos \theta, \sin \theta$; rational visibility then gives rational effects. A constant in the following statement is independent of all three parameters N, λ, δ .

Theorem 48.1 (Noise-uniform finite-use coding). *For all $N \geq 1$, $0 \leq \lambda \leq 1$, and all pairs of angles,*

$$(48.3) \quad \frac{1}{64} \min\{1, K_{N,\lambda} h(\theta, \phi)\} \leq d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min\{2, K_{N,\lambda} h(\theta, \phi)\}.$$

There are absolute $c, C > 0$ such that, for $0 < \delta \leq 2^{-10}$,

$$(48.4) \quad c \left(1 + \frac{K_{N,\lambda}}{\delta}\right) \leq \mathcal{C}_{N,\lambda}(\delta) \leq \mathcal{C}_{N,\lambda}^{\text{rat}}(\delta) \leq C \left(1 + \frac{K_{N,\lambda}}{\delta}\right).$$

Consequently the optimum fixed-length reusable payload is

$$(48.5) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta}\right) + O(1).$$

The upper construction is an exact rational algorithm on rational visibility, tolerance and unit-circle input coordinates. At $\lambda = 0$ the family is a singleton and its optimum payload is zero.

The constants and the additive remainder are uniform up to the erased and projective endpoints. In contrast, they do not assert an optimal leading constant, a large-error coherent covering law, or a coding law in which an unknown visibility is also encoded.

48.1. A two-point programme with its noise dependence retained.

Lemma 48.2 (An adaptive upper modulus). *Let $h = h(\theta, \phi)$ and $a = \lambda \sin(h/2)$. If $\eta > 0$, then*

$$(48.6) \quad d_N(\mathcal{M}_{\theta, \lambda}, \mathcal{M}_{\phi, \lambda}) \leq \min \left\{ 2, 2Na, 2\sqrt{1 - \left(\frac{\eta}{\eta + a^2} \right)^N} \right\}.$$

For $\eta = 0$ the bound $\min\{2, 2Na\}$ remains valid.

Proof. The difference of the two positive-outcome effects has operator norm a . On a bipartite input state, the two output blocks of the difference are B and $-B$, where $B = \text{tr}_A[(E_{\theta, \lambda}^+ - E_{\phi, \lambda}^+) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B\|_1 \leq a$. An eigenstate of the effect difference attains equality. Hence $d_1 = 2a$, including reference assistance. Replacing the calls one at a time and using trace-norm contraction proves $d_N \leq 2Na$ for every common adaptive tester.

Choose angle representatives with difference h and let u, v be the unit vectors along their midpoint and its oriented perpendicular in the equatorial plane. The two Bloch vectors are $m \pm av$, where $m = \lambda \cos(h/2)u$. Put $b = (1 - \|m\|^2)^{1/2} = (\eta + a^2)^{1/2}$. The two Bloch vectors $m \pm bv$ have norm one, so they define two legal binary projective instruments, say $\mathcal{A}_+$ and $\mathcal{A}_-$. The target pair is exactly the two mixtures of these same instruments with programme laws

$$p_{\pm} = ((1 \pm t)/2, (1 \mp t)/2), \quad t = a/b.$$

This construction depends on the pair, which is permissible in a metric upper bound. It is not one fixed processor for the whole circle.

For a fixed tester, first supply N independent programme bits and, at each requested call, apply the indicated $\mathcal{A}_+$ or $\mathcal{A}_-$. All subsequent operations, including the reference, feedback and a stopping rule, are a common channel of these bits. Unused bits are discarded. The root fidelity of p_+, p_- is $\sqrt{1 - t^2}$. Purifying the classical laws and using contraction gives

$$\|p_+^{\otimes N} - p_-^{\otimes N}\|_1 \leq 2\sqrt{1 - (1 - t^2)^N}.$$

This also follows directly from the classical fidelity inequality. Substitution proves (48.6). Finally $1 - (1 - t^2)^N \leq Nt^2$ and $a \leq \lambda h/2$ imply

$$d_N \leq \min\{N\lambda h, \sqrt{N/\eta} \lambda h, 2\}.$$

The limiting projective bound is supplied by the hybrid estimate, not by division by zero. $\square$

Classical simulation by common extremal channels and the effect of noise on the Heisenberg scale are established methods [47]. Environment/programme-state processing under adaptive use is also an established framework [53, 54, 55]. The point needed here is the exact two-point expression (48.6) and its uniform conversion to a family cover, rather than a new claim about the origin of those methods.

48.2. Finite entangled blocks and a binary product estimate.

Lemma 48.3 (Symmetric Bernoulli products). *Let $P_{\pm}$ be Bernoulli laws with success probabilities $(1 \pm a)/2$, where $0 \leq a \leq 1$. For every integer $m \geq 1$,*

$$(48.7) \quad \|P_+^{\otimes m} - P_-^{\otimes m}\|_1 \geq \frac{1}{2} \min\{1, a\sqrt{m}\}.$$

Proof. Retain the largest odd number $r = 2j + 1 \leq m$ of coordinates, so $r \geq m/2$. For their majority event write $g_r(a) = P_+(\text{majority}) - P_-(\text{majority})$. Differentiating the finite binomial sum and cancelling consecutive terms gives

$$g'_r(a) = r \binom{2j}{j} 4^{-j} (1 - a^2)^j.$$

For $j \geq 1$, the bound $\binom{2j}{j}4^{-j} \geq 1/(2\sqrt{j})$ follows by induction: the ratio of consecutive left sides is $(2j+1)/(2j+2) \geq \sqrt{j/(j+1)}$. For $0 \leq a \leq r^{-1/2}$, Bernoulli's inequality gives $(1-a^2)^j \geq 1-j/r \geq 1/2$. Thus $g'_r(a) \geq \sqrt{r}/(2\sqrt{2})$ on that interval. The case $r = 1$ is immediate. Integration from zero and monotonicity yield $g_r(a) \geq (2\sqrt{2})^{-1} \min\{1, a\sqrt{r}\}$. Unhalved distance is at least twice this event difference; using $r \geq m/2$ proves the stated bound. $\square$

Lemma 48.4 (A noise-uniform pairwise witness). *The lower inequality in (48.3) is witnessed by nonadaptive entangled blocks, each of at most N qubits, followed by classical parity and majority processing.*

Proof. A block of k inputs is prepared in $(|0\rangle^{\otimes k} + e^{i\alpha}|1\rangle^{\otimes k})/\sqrt{2}$. Multiplying the k output signs of (48.1) has mean $\lambda^k \cos(k\theta - \alpha)$. This identity follows by applying $\bigotimes_{i=1}^k (E^+ - E^-)$ to the two off-diagonal terms of the block density matrix. For the two hypotheses choose the one common phase $\alpha = k(\theta + \phi)/2 + \pi/2$. Their parity laws then have opposite biases of magnitude

$$(48.8) \quad a_k = \lambda^k |\sin(kh/2)|.$$

Repeating $m = \lfloor N/k \rfloor$ independent blocks and ignoring remaining calls is an admissible tester. Its unhalved output distance is at least $\frac{1}{2} \min\{1, \sqrt{m} a_k\}$ by Lemma 48.3. The phase and block size may depend on the pair, but never on the unknown hypothesis or a private outcome.

Assume $\lambda > 0$ and put $T = \min\{N, \eta^{-1}\}$, $k_0 = \lfloor T \rfloor$. Then $T/2 \leq k_0 \leq T$. For every $1 \leq k \leq k_0$,

$$(48.9) \quad \lambda^k \geq \lambda/2.$$

Indeed, when $0 < \eta < 1$, the inequality $\log(1 - \eta) \geq -\eta/(1 - \eta)$ and $k - 1 \leq \eta^{-1} - 1$ give $(k - 1) \log \lambda \geq -1/2$. At $\eta = 0$ the assertion is immediate.

If $h \leq 1/k_0$, choose $k = k_0$. Since $\sin(kh/2) \geq kh/\pi$ and $m \geq N/(2k)$, we obtain

$$\sqrt{m} a_k \geq \frac{\lambda \sqrt{Nk_0}}{2\pi\sqrt{2}} h \geq \frac{K_{N,\lambda}}{4\pi} h.$$

This gives a lower constant $1/(8\pi)$. If $k_0 = 1$, use $k = 1$ for all $h \leq \pi$ instead. Now $T < 2$ unless $T = 1$, and $\sqrt{N}\lambda \sin(h/2) \geq K_{N,\lambda}h/(\pi\sqrt{2})$; this is stronger than required.

It remains that $k_0 \geq 2$ and $h > 1/k_0$. Here $\lambda \geq 1/\sqrt{2}$. Choose $k = \max\{1, \lfloor 1/h \rfloor\}$. Then $k \leq k_0$ and $1/2 \leq kh \leq \pi$. Equations (48.8) and (48.9) give $a_k \geq 1/(4\pi\sqrt{2})$. Even one such block has distance at least $1/(8\pi\sqrt{2}) > 1/64$. The three cases prove the claim. For $\lambda = 0$ both sides of the desired inequality vanish. $\square$

48.3. Covering and exact rational realization.

Proof of Theorem 48.1. The metric assertion follows from the preceding lemmas. For a lower cover choose points on the circle with circular separation at least $256\delta/K_{N,\lambda}$ whenever that quantity is at most π . An equally spaced packing has at least a fixed constant times $K_{N,\lambda}/\delta$ points. Every two points have d_N -distance at least 4δ , because $256\delta \leq 1/4$. No radius- δ ball, even about an arbitrary legal instrument, can contain two such points. If the required angular separation exceeds π , the trivial lower bound one suffices, with a smaller absolute constant. This proves the first inequality in (48.4).

For the upper bound use the two rational semicircle charts

$$(48.10) \quad z_s(t) = s \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right), \quad s \in \{-1, 1\}, \quad -1 \leq t \leq 1.$$

Each chart has angular speed $2/(1+t^2) \leq 2$. Round t to the nearest j/B , $-B \leq j \leq B$, with a fixed rule at ties. The circular error is at most $1/B$. Thus any integer $B \geq \max\{1, K_{N,\lambda}/\delta\}$ yields

a cover with at most $4B + 2$ centres. For rational inputs, an entirely rational choice is

$$(48.11) \quad B = \max \left\{ 1, \min \left\{ \left\lceil \frac{4N\lambda}{\delta} \right\rceil, \left\lceil \sqrt{\frac{16N\lambda^2}{\eta\delta^2}} \right\rceil \right\} \right\},$$

omitting the second entry when $\eta = 0$. Integer squaring and rational comparison compute the ceiling square root exactly. Both entries are ceilings of the corresponding real bounds, so their minimum is $\lceil 4K_{N,\lambda}/\delta \rceil$. For $z = (x, y)$ on the rational unit circle choose $s = 1$ when $x \geq 0$ and $s = -1$ otherwise, and set $t = sy/(1 + sx)$. Its denominator is at least one, so it is rational and belongs to $[-1, 1]$.

The chart digit (s, j) determines a legal effect pair with eigenvalues $(1 \pm \lambda)/2$. No approximate positivity test, irrational normalization or search over real angles is needed. A fixed-length index over $4B + 2$ slots includes the chart sign. Its length gives (48.5). Redundant chart endpoints cause at most a constant overhead. At $\lambda = 0$ dispense with the chart and emit the single fair, input-erasing measurement. $\square$

Corollary 48.5 (A continuous family of horizon exponents). *Fix $0 < \delta \leq 2^{-10}$. If $\lambda_N \rightarrow 1$ and $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ for $\alpha \geq 0$, then*

$$\frac{\log_2 \mathcal{C}_{N,\lambda_N}(\delta)}{\log_2 N} \rightarrow \frac{1 + \min\{\alpha, 1\}}{2}.$$

If $\lambda_N = N^{-\beta+o(1)} \rightarrow 0$, with $\beta \geq 0$, the limit is $\max\{1/2 - \beta, 0\}$. The projective and erased endpoints have coefficients one and zero, respectively.

Proof. Substitute the two regimes in (48.2) and use (48.4). The term $1 + K/\delta$ handles a vanishing or subpolynomial scale without taking the logarithm of zero. $\square$

Corollary 48.6 (Noisy readout with conditional quantum states). *Fix n and ranks $1 \leq r_+, r_- \leq n$. Replace the scalar output in (48.1) by*

$$\mathcal{F}_{\theta,\lambda,\rho}^y(X) = \text{tr}(E_{\theta,\lambda}^y X) \rho_y, \quad \text{tr } \rho_y = 1, \quad \text{rank } \rho_y \leq r_y,$$

retaining the outcome y . Put $V = \sum_{y \in \{+, -\}} [r_y(2n - r_y) - 1]$. For some fixed $\delta_0 > 0$, the optimum covering length is

$$(48.12) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta} \right) + \frac{V}{2} \log_2 N + V \log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The constants are uniform in $N \geq 1$ and $\lambda \in [0, 1]$, at fixed n, r_+, r_- . For rational visibility, upper centres can be rational and never increase either conditional state rank; lower centres may be arbitrary legal memoryless instruments of the same interface.

Proof. Discarding the quantum output leaves $\mathcal{M}_{\theta,\lambda}$, so the readout lower bound is independent of the conditional states. Conversely, using $I/2$ independently at each call produces exactly $\Omega_\rho^{\otimes N}$, where $\Omega_\rho = \frac{1}{2}\rho_+ \oplus \frac{1}{2}\rho_-$, independently of both θ and λ . Choose compact smooth patches of the two state rank strata. Their product has dimension V and is locally bi-Lipschitz in the Hilbert–Schmidt metric on Ω_ρ . The local binary-test proof of Corollary 40.2, applied to these states, supplies a packing of size $c(\sqrt{N}/\delta)^V$ at output distance greater than 2δ for all sufficiently small fixed δ_0 . Cross it with the angular packing used above. Two differing angles are separated by the readout witness regardless of their states; pairs with the same angle are separated by the maximally mixed-input witness. Triangle inequality therefore proves the product lower bound for arbitrary centres.

For the upper cover first encode the angle to error at most $\delta/2$. For fixed conditional states their preparation is a common operation following the readout, so it cannot enlarge this error, including within an adaptive tester. Encode each trace-one state separately by Theorem 41.2,

with preparation N -copy error at most $\delta/4$. At each call a common processor can consume one copy of each row state and select the state indexed by the measurement outcome. Providing all $2N$ row programmes in advance controls every tester. Triangle inequality on the two tensor products bounds their contribution by $\delta/2$. The product of the two rank-aware factor codebooks has size $C(\sqrt{N}/\delta)^V$. Rational state factors and the rational angle charts give rational Choi blocks $(E_{\theta,\lambda}^y)^\top \otimes \rho_y$ when λ is rational. Each encoded state is trace one and rank nonincreasing, which proves the asserted legality and payload bound. $\square$

Remark 48.7 (The retained flag and the scope of the transition). For $n = 1$, forgetting y yields the trace map for every θ and λ ; its covering number is one. For conditional states, forgetting the flag likewise erases the readout whenever $\rho_+ = \rho_-$. Thus the observable interface is essential. At $0 < \lambda < 1$ the measurement Choi blocks have full rank, whereas they have rank one at $\lambda = 1$. Their approach to that boundary is controlled explicitly by $N(1 - \lambda^2)$; a fixed-interior theorem alone would leave its constants undetermined. The transition is not a general classification of all coupled Choi tangent directions. Nor does a pairwise GHZ witness supply a single globally successful estimator or extend this coherent covering theorem to all fixed radii below two. Those are different statements from (48.4).

Remark 48.8 (Rational-circle convention). The rational upper-centre notation in this section specifies rational circle coordinates. Decoded effects are fully rational when the public visibility is rational; for irrational visibility no such matrix rationality follows. Section 49 gives a different joint code for rational Cartesian targets, in which visibility is encoded and every decoded matrix is rational.

49. COUPLED GEOMETRY OF THE BINARY MEASUREMENT BALL

We now encode the visibility as well as the direction. More generally, we give a local geometric criterion for arbitrary regular subfamilies of the unbiased binary measurement ball. The parameter space in this criterion is the identifiable image of the family, not a redundant choice of angular coordinates.

Let $\mathbb{B} = \{x \in \mathbb{R}^3 : |x| \leq 1\}$. With ordered outcomes put

$$(49.1) \quad E_x^\pm = \frac{1}{2}(I \pm x \cdot \sigma), \quad \mathcal{M}_x^\pm(X) = \text{tr}(E_x^\pm X), \quad x \in \mathbb{B}.$$

Here $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ is the Pauli triple. We use d_N with its unhalved trace-norm convention and all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping. The channel outputs the ordered classical outcome and no residual quantum system. Define

$$(49.2) \quad a_N(r) = \sqrt{\frac{N}{1 - r^2 + N^{-1}}}, \quad w_N(x) = a_N(|x|), \quad N \geq 1.$$

The cutoff N^{-1} is part of a finite-use estimate. It is not a positive-noise assumption on the target.

Theorem 49.1 (Finite-use metric on the measurement ball). *For $x, y \in \mathbb{B}$, let $s = \min(|x|, |y|)$ and $D_N(x, y) = a_N(s)|x - y|$. Then*

$$(49.3) \quad \frac{1}{256} \min\{1, D_N(x, y)\} \leq d_N(\mathcal{M}_x, \mathcal{M}_y) \leq \min\{2, 6D_N(x, y)\}.$$

In particular $d_1(\mathcal{M}_x, \mathcal{M}_y) = |x - y|$. The constants in (49.3) are independent of the visibility, direction, horizon and pair separation. The lower bound is attained up to these constants by pair-dependent nonadaptive tests; it does not assert the existence of one estimator for the whole ball. The comparison constants are not optimized.

This supplies a finite-distance local comparison even at erasure and at projective effects. The comparison is not obtained by integrating a Fisher metric through a singular boundary. We first prove its radial part, for which there is an exact classical reduction.

Lemma 49.2 (Binomial distance with a support cutoff). *For $0 \leq s \leq r \leq 1$, a unit u , and $P_t = \text{Bern}((1-t)/2)$,*

$$(49.4) \quad d_N(\mathcal{M}_{ru}, \mathcal{M}_{su}) = \|P_r^{\otimes N} - P_s^{\otimes N}\|_1.$$

Writing $R = a_N(s)(r-s)$, this distance lies between $\frac{1}{64} \min\{1, R\}$ and $\min\{2, 3R\}$.

Proof. First perform the projective measurement along u and then flip its output with probability $(1-t)/2$. For any fixed adaptive tester, all its outputs are a common channel of N independent flip bits. Supply all bits in advance and discard any unused suffix after public stopping. This proves the upper inequality in (49.4). Repeated input of the $+u$ eigenstate gives exactly these bits and proves equality.

Put $p = (1-r)/2$, $q = (1-s)/2$, and $\Delta = q - p$. If $q = 0$, the ordering forces $s = r = 1$, so the two projective measurements coincide. More generally, the case $\Delta = 0$ is immediate. Otherwise $0 < q \leq 1/2$. Let

$$k = \min\{N, \lfloor (2q)^{-1} \rfloor\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, q^{-1}\}$ and $m \geq N/(2k)$. In a block of k trials the no-success event has probabilities $e_p = (1-p)^k$ and $e_q = (1-q)^k$. Their difference obeys

$$e_p - e_q \geq k\Delta(1-q)^{k-1} \geq k\Delta/2.$$

The second inequality follows from $(k-1)q \leq 1/2$ and Bernoulli's inequality. A common randomized processing maps a Bernoulli parameter e to $b + ce$, where

$$c = \frac{1}{\max\{e_p + e_q, 2 - e_p - e_q\}} \geq \frac{1}{2}, \quad b = \frac{1}{2}(1 - c(e_p + e_q)).$$

Indeed $b \geq 0$, $b + c \leq 1$, and the processed pair has parameters $(1 \pm c(e_p - e_q))/2$. Apply Lemma 48.3 to the m independent processed blocks. Contraction under processing gives

$$\|P_r^{\otimes N} - P_s^{\otimes N}\|_1 \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, q^{-1})}\}.$$

Since $1 - s^2 = 2q(1+s)$, we have $a_N(s) \leq \sqrt{N \min(N, q^{-1})}$. Also $\Delta = (r-s)/2$. The asserted lower constant $1/64$ follows.

For the upper bound a one-bit coupling gives $N(r-s)$. Let $F = \sqrt{pq} + \sqrt{(1-p)(1-q)}$. Directly,

$$1 - F = \frac{1}{2}[(\sqrt{p} - \sqrt{q})^2 + (\sqrt{1-p} - \sqrt{1-q})^2] \leq \frac{\Delta^2}{2q(1-q)} = \frac{(r-s)^2}{2(1-s^2)}.$$

The no-success statistic above is a lower-bound proof device; its optimality is not asserted. The product-fidelity inequality gives the second bound $2\sqrt{N}(r-s)/\sqrt{1-s^2}$ when $s < 1$. For $z = N(1-s^2) \geq 0$,

$$\min\{1, 2/\sqrt{z}\}\sqrt{z+1} \leq \sqrt{5} < 3.$$

Combining the two upper estimates proves the claim, without division by zero at the identical endpoint $r = s = 1$. $\square$

Proof of Theorem 49.1. Assume $x = ru$, $y = sv$, $r \geq s > 0$, and write $h \in [0, \pi]$ for the angle between u, v . Any two directions lie on a great circle; common unitary conjugation of all input systems preserves d_N . Thus the fixed-visibility theorem 48.1 applies with $K = K_{N,s}$. Notice

$$(49.5) \quad sa_N(s) \leq K \leq \sqrt{2} sa_N(s).$$

Let $d = d_N(\mathcal{M}_{ru}, \mathcal{M}_{sv})$ and $d_r = d_N(\mathcal{M}_{ru}, \mathcal{M}_{su})$. Inputting the $+u$ eigenstate gives Bernoulli parameters $p = (1 - r)/2$ and $q' = (1 - s \cos h)/2 \geq q = (1 - s)/2$. For fixed p , the total variation of binomial products increases with the larger parameter. To see this without a differentiability assumption, the binomial likelihood ratio is monotone in the success count; its optimal event is a lower count interval. Each such event has probability decreasing in the other parameter. Endpoint cases follow by continuity. Consequently (49.4) gives $d_r \leq d$. The triangle inequality now yields

$$d_N(\mathcal{M}_{su}, \mathcal{M}_{sv}) \leq d_r + d \leq 2d.$$

Set $A = a_N(s)(r - s)$ and $B = sa_N(s)h$. The radial lemma and the fixed-visibility lower bound imply

$$d \geq \frac{1}{64} \min\{1, A\}, \quad d \geq \frac{1}{128} \min\{1, B\}.$$

Hence $d \geq \frac{1}{256} \min\{1, A + B\}$. Since $|ru - sv| \leq (r - s) + sh$, this proves the required lower bound. For the upper bound the radial and angular estimates give $d \leq 3A + \sqrt{2}B$. Moreover $r - s \leq |ru - sv|$ and $sh \leq (\pi/2)|ru - sv|$, by $|ru - sv|^2 = (r - s)^2 + 4rs \sin^2(h/2)$. Thus $d \leq (3 + \pi/\sqrt{2})D_N < 6D_N$. When $s = 0$, the radial lemma proves both bounds, independently of the undefined direction at the origin. The one-use identity follows, as in Lemma 48.2, from the eigenvalues $\pm|x - y|/2$ of $E_x^+ - E_y^+$ and reference trace-norm duality. $\square$

49.1. A local volume criterion and boundary contact. For compact $X \subset \mathbb{B}$ let $\mathcal{C}_N(X, \delta)$ allow arbitrary legal memoryless centres of the binary qubit interface, and let $\mathcal{C}_N^X(X, \delta)$ require centres in $\{\mathcal{M}_x : x \in X\}$. The measure in the next theorem is ordinary Euclidean local geometry on the identifiable set X .

Theorem 49.3 (Weighted local covering criterion). *Suppose a finite Borel measure μ on a nonempty compact $X \subset \mathbb{B}$ satisfies, for some $k > 0$, $r_0 > 0$ and $c_0, C_0 > 0$,*

$$(49.6) \quad c_0 t^k \leq \mu(X \cap B(x, t)) \leq C_0 t^k \quad (x \in X, 0 < t \leq r_0).$$

There are $c_X, C_X, \delta_X > 0$, depending only on these regularity constants, such that, for every $N \geq 1$ and $0 < \delta \leq \delta_X$,

$$(49.7) \quad c_X \delta^{-k} \int_X w_N(x)^k d\mu(x) \leq \mathcal{C}_N(X, \delta) \leq \mathcal{C}_N^X(X, \delta) \leq C_X \delta^{-k} \int_X w_N(x)^k d\mu(x).$$

No map retracting arbitrary centres onto X is assumed. In a parametrized family the set X is its image under (49.1); indistinguishable parameter values are not charged twice.

Proof. Put $e_N(x) = 1 - |x|^2 + N^{-1}$, so $e_N \geq N^{-1}$ and $|e_N(x) - e_N(y)| \leq 2|x - y|$. If $|x - y| \leq t/w_N(x)$ and $t \leq 1/4$, then

$$|e_N(x) - e_N(y)| \leq 2t\sqrt{e_N(x)/N} \leq 2te_N(x).$$

Thus the weights are comparable on these small Euclidean balls. Conversely, if $d_N(\mathcal{M}_x, \mathcal{M}_y) \leq t < 1/256$, Theorem 49.1 implies $|x - y| \leq 256t/a_N(s)$, $s = \min(|x|, |y|)$. Applying the same estimate first at the point of smaller norm shows comparability to $w_N(x)$ when, for example, $t \leq 2^{-14}$. Consequently absolute $c, C > 0$ satisfy the two local inclusions

$$X \cap B(x, ct/w_N(x)) \subseteq B_{d_N}(x, t) \cap X \subseteq X \cap B(x, Ct/w_N(x)).$$

Reduce t by a fixed multiple of r_0 if necessary. Since $w_N(x) \geq 1/\sqrt{2}$, all Euclidean radii used here lie below r_0 . The measure ν_N is auxiliary geometric volume, rather than a statistical prior or a hidden-register entropy. The weighted measure $d\nu_N = w_N^k d\mu$ of either operational ball is bounded above and below by fixed multiples of t^k , by (49.6) and weight comparability.

A maximal δ -separated set in the compact metric space (X, d_N) covers X by radius- δ balls. Its radius- $\delta/3$ balls are disjoint, proving the upper bound in (49.7). A maximal 3δ -separated set covers X by radius- 3δ balls, so it has at least $c\nu_N(X)\delta^{-k}$ points. Any radius- δ ball about

any legal centre contains at most one of these points, by the triangle inequality. This gives the lower bound for unrestricted centres. Compactness and the one-use identity ensure that these are genuine metrics on X ; the finite hybrid bound gives the required continuity. $\square$

Corollary 49.4 (Boundary-contact law). *In Theorem 49.3, put $F(t) = \mu\{x \in X : 1 - |x|^2 \leq t\}$. Suppose $F(0) = 0$ and $ct^\alpha \leq F(t) \leq Ct^\alpha$ for $0 < t \leq t_0$, with $\alpha > 0$. Uniformly on the same small-error range,*

$$(49.8) \quad \mathcal{C}_N(X, \delta) \asymp_X \delta^{-k} \begin{cases} N^{k/2}, & \alpha > k/2, \\ N^{k/2} \log(N+2), & \alpha = k/2, \\ N^{k-\alpha}, & 0 < \alpha < k/2. \end{cases}$$

Here $F(0) = \mu(X \cap S^2)$ is boundary mass of the target measure, not ambient surface measure. If instead $F(0) > 0$, the law is $\asymp_X \delta^{-k} N^k$.

Proof. Write $b = k/2$ and $\tau = N^{-1}$. Stieltjes integration by parts gives

$$\int_{[0, t_0]} (t + \tau)^{-b} dF(t) = (t_0 + \tau)^{-b} F(t_0) + b \int_0^{t_0} \frac{F(t)}{(t + \tau)^{b+1}} dt$$

when $F(0) = 0$. Substitute the two bounds on F and split the last integral at τ . It is bounded, logarithmic, or comparable to $\tau^{\alpha-b}$ according as $\alpha > b$, $\alpha = b$, or $\alpha < b$. The part beyond t_0 is bounded, and multiplication by N^b proves (49.8). Bounded N is absorbed into the constants. An atom at depth zero contributes exactly $N^k F(0)$ to the weighted volume; $w_N \leq N$ supplies the matching upper bound. $\square$

For example, in a regular k -dimensional chart with a boundary subset of codimension ℓ , a two-sided relation between the support deficit $1 - |x|^2$ and the m th power of transverse distance yields $\alpha = \ell/m$, provided the transverse volume has order t^ℓ . This is a geometric condition to check, not an automatic regularity assertion. It distinguishes ordinary interior accumulation, critical logarithmic enhancement and boundary-dominated coherent accumulation.

Corollary 49.5 (Joint visibility and direction). *Let $\mathbb{B}_d$ be the closed unit ball in a fixed d -dimensional linear subspace of $\mathbb{R}^3$, $d = 1, 2, 3$. For an absolute small-error range,*

$$(49.9) \quad \mathcal{C}_N(\mathbb{B}_d, \delta) \asymp \begin{cases} \sqrt{N} \delta^{-1}, & d = 1, \\ N \log(N+2) \delta^{-2}, & d = 2, \\ N^2 \delta^{-3}, & d = 3. \end{cases}$$

For the coordinate disk and full ball, rational target data admit exact rational family-centred codes attaining these orders. In particular, the optimal fixed-length payloads for $d = 2, 3$ are respectively

$$\log_2 N + \log_2 \log(N+2) + 2 \log_2(1/\delta) + O(1), \quad 2 \log_2 N + 3 \log_2(1/\delta) + O(1).$$

Visibility is a charged target coordinate in these statements.

Proof. Lebesgue measure on $\mathbb{B}_d$ satisfies (49.6) with $k = d$, and its boundary-depth distribution has exponent $\alpha = 1$. For the disk $(k, \alpha) = (2, 1)$ is critical: its weighted volume contains $N \int_0^1 (t + N^{-1})^{-1} dt \asymp N \log(N+2)$. For the ball $(k, \alpha) = (3, 1)$, the boundary layer instead dominates and gives N^2 . These factors are geometric and are unchanged in order by replacing the Ahlfors measure by an equivalent one, with constants depending on the regularity data. Apply Corollary 49.4. The exact rational upper construction follows from Theorem 49.6 below. In a coordinate line the direct scalar grid in the same construction gives the stated one-dimensional order. $\square$

49.2. Exact joint coding without irrational normalization. We give a rational construction for $d = 2, 3$ that takes the Cartesian Bloch vector as input. Its norm need not be rational. Fix public $N \geq 1$ and rational $0 < \delta \leq 1/4$. Set

$$(49.10) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad t_i = i/B, \quad r_i = \frac{2t_i}{1+t_i^2}, \quad A_i = \max \left\{ 1, \left\lceil \sqrt{64(d-1)K_{N,r_i}^2/\delta^2} \right\rceil \right\} \quad (i > 0).$$

The radial layer $i = 0$ has one codeword. At $i > 0$ use $2d$ signed stereographic charts and the grid $\{-A_i, \dots, A_i\}^{d-1}/A_i$. For chart axis j and sign s , the map is

$$(49.11) \quad u_j = s \frac{1 - |z|^2}{1 + |z|^2}, \quad u_\ell = \frac{2z_\ell}{1 + |z|^2} \quad (\ell \neq j).$$

Every decoded centre is $r_i u \in \mathbb{B}_d$ with rational entries. The exact capacity, including possibly redundant chart words, is

$$(49.12) \quad L_{N,\delta,d} = 1 + \sum_{i=1}^B 2d(2A_i + 1)^{d-1}.$$

A single fixed-length integer index in this union of layers charges the visibility digit as well as the direction digits.

Theorem 49.6 (Rational joint codec). *For rational $x \in \mathbb{B}_d$, $d = 2, 3$, an exact rational algorithm returns an index in (49.12). Its decoded measurement is within δ in d_N , and every index decodes to a legal rational measurement. There are absolute constants such that*

$$L_{N,\delta,2} \leq CN \log(N+2)\delta^{-2}, \quad L_{N,\delta,3} \leq CN^2\delta^{-3}.$$

The stored payload is $\lceil \log_2 L_{N,\delta,d} \rceil$ bits. The JSON envelope, encoder workspace, public d, N, δ and expanded matrices are not this payload. No input visibility is public, and no floating-point operation decides a chart or grid digit.

Proof. Write $r = |x|$, $\beta = \arcsin r$, $t = \tan(\beta/2) \in [0, 1]$. Choose a nearest i/B to t , resolving ties towards zero. Then $|\beta - 2\arctan(i/B)| \leq 1/B$. The radial flip programmes have root fidelity $\cos((\beta - \beta_i)/2)$, and hence their N -copy distance is at most $\sqrt{N}|\beta - \beta_i| \leq \delta/8$. When $i = 0$, decode the unique erased measurement and stop. Otherwise choose a maximal-absolute-coordinate axis of x , with its sign. The exact chart parameters of x/r are

$$z_\ell = \frac{x_\ell}{r + |x_j|}, \quad \ell \neq j.$$

They lie in $[-1, 1]$. Round each to a nearest grid point at denominator A_i . The differential norm of (49.11), measured in spherical length, is $2/(1 + |z|^2) \leq 2$. Thus the angular error is at most $\sqrt{d-1}/A_i$ and its operational cost is at most $\delta/8$, by the fixed-visibility upper bound and (49.10). The triangle inequality leaves at least three quarters of the requested budget unused.

All comparisons above can be made in the rationals. If $Q = \sum x_j^2$, comparison of t with a rational $v \in [0, 1]$ is comparison of Q with $[2v/(1+v^2)]^2$. For a direction digit compare $b/(\sqrt{Q} + a)$ with rational $v \geq 0$, where $a = |x_j|$ and $b = |x_\ell|$. First test the sign of $b - va$. If it is negative, the comparison is negative; otherwise compare $(b - va)^2$ and v^2Q . The sign test is essential before squaring. Binary integer searches therefore determine every nearest grid digit, including ties, without computing an irrational normalization. Integer ceiling square roots suffice for both grids.

It remains to bound the union capacity. Here $B \asymp \sqrt{N}/\delta$. For $i \leq B/2$, $K_{N,r_i} \leq C\sqrt{N}$, which contributes at most $CN^{d/2}\delta^{-d}$. For $i > B/2$, put $j = B - i$. Since $1 - r_i^2 = (1 - t_i^2)^2/(1 + t_i^2)^2$, we have

$$K_{N,r_i} \leq C \min\{N, \sqrt{N}B/(j+1)\},$$

with an adjusted absolute constant also at $j = 0$. Split the sum at $j + 1 = B/\sqrt{N} \asymp \delta^{-1}$. For $d = 2$, the remaining sum is harmonic and contributes at most $CN\delta^{-2}\log(N+2)$. For $d = 3$, the sum of $(j+1)^{-2}$ contributes at most $CN^2\delta^{-3}$. The near-boundary layers have these same or smaller orders. The added constants in A_i and the single erased word are absorbed into these bounds. This proves the capacity and payload assertions. Counting layer sizes and ranking or unranking the integer grid are finite exact algorithms, with cost polynomial in the numerical radial grid and operand lengths; no polynomial bound in the encoded length of N or $1/\delta$ is asserted. $\square$

Remark 49.7 (Priority and interface). Sedlák–Ziman [57, Section VI, Example 4, equation (37)] already treat distinct white-noise levels in single-shot qubit measurement discrimination. Puchała–Pawela–Krawiec–Kukulski [58, Theorem 1] identify the single-shot diamond distance of projective measurements with an optimized unitary-channel distance. Neither result is claimed anew here. Our conclusion is the uniform finite-use metric comparison (49.3), its weighted covering criterion, and the resulting contact and exact rational joint-description laws. The classical programme and entangled-block mechanisms retain their earlier attribution. The geometric criterion covers regular identifiable subsets of (49.1), not arbitrary biased POVMs or general disturbing instruments. The supplied vector is encoded; no unknown measurement is learned. Pairwise testing, a common estimator, a global environment seizer and a physical classical simulation of unknown quantum inputs are different tasks.

50. THE FULL BINARY EFFECT BODY

The spectral boundary of a biased measurement has two distinct endpoints. A vanishing eigenvalue of one effect need not give a coherent angular scale: the other outcome can still destroy coherence. We therefore retain both endpoint coordinates and determine their joint finite-use geometry.

Let

$$(50.1) \quad \mathcal{E} = \left\{ (a, x) \in \mathbb{R} \times \mathbb{R}^3 : 0 \leq a - |x| \leq a + |x| \leq 2 \right\}, \quad E_{a,x} = \frac{1}{2}(aI + x \cdot \sigma).$$

We identify a point (a, x) of $\mathcal{E}$ with its effect $E_{a,x}$. Write $\mathcal{M}_{a,x}$ for the measurement with ordered effects $E_{a,x}, I - E_{a,x}$ and no residual quantum output. Its maximum and minimum eigenvalues, spectral gap, and bias are

$$p = \frac{1}{2}(a + |x|), \quad q = \frac{1}{2}(a - |x|), \quad r = p - q = |x|, \quad b = p + q - 1.$$

Thus $0 \leq q \leq p \leq 1$ and $|b| + r \leq 1$. When $r > 0$ put $u = x/r$, so that $E_{a,x} = pP_u + q(I - P_u)$, where $P_u = (I + u \cdot \sigma)/2$. Directions are not identified under outcome exchange.

For an integer $N \geq 1$ set $\tau = N^{-1}$ and define

$$(50.2) \quad T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + \tau}}, \quad 0 \leq s, t \leq 1,$$

$$V(p, q) = p(1-p) + q(1-q) = \frac{1}{2}(1 - b^2 - r^2),$$

$$(50.3) \quad K_N(p, q) = r \sqrt{\frac{N}{V(p, q) + \tau}}.$$

For two effects with data (p, q, u) and (p', q', v) let $h \in [0, \pi]$ be the angle between u, v when both spectral gaps are positive. If either gap is zero, set the following angular term equal to zero, without choosing a direction at the repeated eigenvalue:

$$(50.4) \quad H_N(E, F) = T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\} h.$$

Theorem 50.1 (Finite-use geometry of binary qubit effects). *For every $E, F \in \mathcal{E}$ and $N \geq 1$,*

$$(50.5) \quad \frac{1}{8192} \min\{1, H_N(E, F)\} \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 3H_N(E, F)\}.$$

The distance permits all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping, and uses the unhalved trace norm. The constants are independent of both spectra, both directions, and the horizon. In particular,

$$(50.6) \quad d_1(\mathcal{M}_{a,x}, \mathcal{M}_{a',x'}) = |a - a'| + |x - x'|.$$

The lower comparison is witnessed by pair-dependent nonadaptive tests. It does not require a common estimator of all four target coordinates.

The denominator in (50.3) involves the two outcome variances. Replacing it by the distance to the nearest spectral face would give an incorrect angular scale on a face such as $q = 0$, $0 < p < 1$. The proof below supplies both a scalar-overlap dilation upper bound and explicit finite-distance lower witnesses.

50.1. The spectral endpoints. Put

$$B_N(s, t) = \|\text{Bern}(s)^{\otimes N} - \text{Bern}(t)^{\otimes N}\|_1.$$

Lemma 50.2 (A Bernoulli estimate at both support endpoints). *For $s, t \in [0, 1]$,*

$$(50.7) \quad \frac{1}{64} \min\{1, T_N(s, t)\} \leq B_N(s, t) \leq \min\{2, 3T_N(s, t)\}.$$

For a fixed one of the two parameters, B_N increases as the other parameter moves away from it on either side.

Proof. Assume $s < t$, put $\Delta = t - s$, and write $v = \max\{s(1-s), t(1-t)\}$. The identical pair is immediate. A coupling of individual bits gives $B_N \leq 2N\Delta$. The root fidelity $F = \sqrt{st} + \sqrt{(1-s)(1-t)}$ satisfies

$$\begin{aligned} 1 - F &= \frac{\Delta^2}{2} \left(\frac{1}{(\sqrt{s} + \sqrt{t})^2} + \frac{1}{(\sqrt{1-s} + \sqrt{1-t})^2} \right) \\ &\leq \frac{\Delta^2(1+\Delta)}{2t(1-s)} \leq \frac{\Delta^2}{2v}, \end{aligned}$$

whenever $v > 0$. For the last inequality, both endpoint variances are at most $t(1-s)/(1+\Delta)$: after cancellation the required inequalities are $s(1+\Delta) \leq t$ and $(1-t)(1+\Delta) \leq 1-s$. They follow respectively from $\Delta(1-s) \geq 0$ and $t\Delta \geq 0$. The product-fidelity inequality gives $B_N \leq 2\sqrt{N}\Delta/\sqrt{v}$. For $z = Nv$, the inequality $\min\{2, 2/\sqrt{z}\}\sqrt{z} + 1 \leq 2\sqrt{2} < 3$ combines the two upper bounds. At $v = 0$ use the coupling bound.

For the lower bound, complement both bits if necessary to have $s + t \leq 1$. If $t > 3/4$, then $\Delta \geq 2t - 1 > 1/2$ and $B_N \geq B_1 = 2\Delta > 1$, which suffices. Otherwise $0 < t \leq 3/4$. Set

$$k = \min \left\{ N, \max \left\{ 1, \left\lfloor \frac{1}{2t} \right\rfloor \right\} \right\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, t^{-1}\}$, $m \geq N/(2k)$, and $(k-1)t \leq 1/2$. The event of no success in a block of k trials has probabilities $e_s = (1-s)^k$ and $e_t = (1-t)^k$, with

$$e_s - e_t \geq k\Delta(1-t)^{k-1} \geq k\Delta/2.$$

There is a common stochastic processing of this binary event whose success probability is $d + ce$, where

$$c = \frac{1}{\max\{e_s + e_t, 2 - e_s - e_t\}} \geq \frac{1}{2}, \quad d = \frac{1}{2}(1 - c(e_s + e_t)).$$

Indeed $d \geq 0$, $d+c \leq 1$, and the processed pair has parameters $(1 \pm c(e_s - e_t))/2$. Apply Lemma 48.3 to m independent blocks and discard the unused calls. Contraction under this processing gives

$$B_N(s, t) \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, t^{-1})}\}.$$

Since $v \geq t(1-t) \geq t/4$, $T_N(s, t) \leq 2\Delta\sqrt{N \min(N, t^{-1})}$. This proves the stated lower constant.

For monotonicity, the likelihood ratio of the two binomial count laws is monotone in the count. For a fixed larger parameter their total variation is the maximum, over upper count intervals, of the larger-parameter probability minus the smaller-parameter probability. The latter probabilities decrease when the smaller parameter decreases. Fixing the smaller parameter gives the analogous statement with lower count intervals. Endpoint cases follow by continuity of the finite sums. $\square$

Lemma 50.3 (Spectral projection without cancellation). *For arbitrary directions and spectra,*

$$(50.8) \quad B_N(p, p') \leq d_N(\mathcal{M}_E, \mathcal{M}_F), \quad B_N(q, q') \leq d_N(\mathcal{M}_E, \mathcal{M}_F).$$

If the two directions agree, then

$$(50.9) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p',q',u}) \leq B_N(p, p') + B_N(q, q').$$

Consequently, with $d = d_N(\mathcal{M}_E, \mathcal{M}_F)$,

$$(50.10) \quad \begin{aligned} d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) &\leq 3d, \\ d_N(\mathcal{M}_{p',q',u}, \mathcal{M}_{p',q',v}) &\leq 3d. \end{aligned}$$

Proof. Suppose first $p \geq p'$. Repeatedly input a maximum-eigenvalue eigenstate of E . Its success probabilities under the two hypotheses are p and some $t \leq p'$. The monotonicity in Lemma 50.2 gives $B_N(p, p') \leq B_N(p, t) \leq d$. If $p < p'$, interchange the hypotheses. For $q \leq q'$, use a minimum-eigenvalue eigenstate of E ; the other success probability is at least q' . Reverse the hypotheses if $q > q'$. These are separate admissible witnesses; no single input is asserted to witness both endpoint changes.

For aligned directions, first make the common projective measurement $\{P_u, I - P_u\}$ and, according to its outcome, generate a bit with success probability p or q . This implements exactly the measurement channel, including on inputs entangled with a reference: the reference blocks are the corresponding partial matrix elements weighted by the displayed success probabilities. Supply in advance N independent p -bits and N independent q -bits; at a call consume the next unused bit from the selected array. Every fixed adaptive tester is a common channel of these two arrays. The triangle inequality for tensor products therefore bounds its output distance by $B_N(p, p') + B_N(q, q')$. This proves (50.9) with arbitrary references, feedback, and stopping; all unqueried row bits are discarded. Combining it with (50.8) bounds every aligned spectral change by $2d$. The triangle inequality, aligning at u or at v , then proves (50.10). $\square$

50.2. Angular geometry with both eigenvalues retained.

Theorem 50.4 (Angular transition for a biased measurement). *For fixed $0 \leq q \leq p \leq 1$, $N \geq 1$, and directions at angle $h \in [0, \pi]$,*

$$(50.11) \quad \frac{1}{512} \min\{1, K_N(p, q)h\} \leq d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) \leq \min\{2, \sqrt{2}K_N(p, q)h\}.$$

More precisely, put

$$A = \sqrt{p(1-p)} + \sqrt{q(1-q)}, \quad z = (p-q) \sin(h/2).$$

When $A > 0$,

$$(50.12) \quad d_N \leq \min \left\{ 2, 2Nz, 2\sqrt{1 - \left(\frac{A^2}{A^2 + z^2} \right)^N} \right\}.$$

When $A = 0$, the bound $\min\{2, 2Nz\}$ remains valid.

Proof. The one-use distance is $2z$: the effect difference is traceless with operator norm z , and the same reference trace-norm duality as in Lemma 48.2 applies. Replacing the calls one at a time gives the adaptive bound $2Nz$.

We next construct a dilation with scalar cross overlap. Common unitary conjugation puts the two directions in one real meridian. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_t = \exp(tJ), \quad t = h/2, \quad E = \text{diag}(p, q), \quad F = R_t E R_t^\top.$$

A measurement dilation is

$$W_E \psi = |+\rangle_C |+\rangle_L \sqrt{E} \psi + |-\rangle_C |-\rangle_L \sqrt{I-E} \psi.$$

The square-root vectors belong to a further two-dimensional environment. Tracing both environments leaves the ordered classical output and implements the measurement even with a reference. Put

$$f = \sqrt{pq} + \sqrt{(1-p)(1-q)}.$$

For any ϕ , apply the common rotation $R_{\phi-t}$ to the two-dimensional environment of the dilation W_F . The cross overlap is

$$\begin{aligned} W_E^\dagger (I \otimes R_{\phi-t}) W_F &= (\sqrt{E} R_\phi \sqrt{E} + \sqrt{I-E} R_\phi \sqrt{I-E}) R_{-t} \\ &= (\cos \phi I + f \sin \phi J) R_{-t}. \end{aligned}$$

Here the rotation acts as the identity on the outcome registers. When $f > 0$, choose

$$D = \sqrt{f^2 \cos^2 t + \sin^2 t}, \quad \cos \phi = \frac{f \cos t}{D}, \quad \sin \phi = \frac{\sin t}{D}.$$

Then the displayed overlap is cI , where $c = f/D \in [0, 1]$. This construction also covers $t = 0$ and $t = \pi/2$. To express it in the spectral coordinates, put $T = \sqrt{p(1-q)} + \sqrt{q(1-p)}$. Direct multiplication gives

$$fT = A, \quad (1-f^2)T^2 = (p-q)^2.$$

If $A > 0$ then $f, T > 0$, and therefore

$$(50.13) \quad c^2 = \frac{f^2}{f^2 \cos^2 t + \sin^2 t} = \frac{A^2}{A^2 + (p-q)^2 \sin^2 t}.$$

Fix any common adaptive tester. Purify its initial state and all intermediate operations, and retain the measurement environments in this proof. The environments are never accessed by the tester. At each call the scalar identity for the cross overlap multiplies the inner product of the two purifications by c , independently of the current input and reference. The common intervening isometries preserve the inner product. A tester that stops early is padded to N calls on dummy inputs, with their outcomes ignored in its final output. Thus two purifications of that final output have inner product c^N . Trace-norm contraction and the pure-state formula give $d_N \leq 2\sqrt{1-c^{2N}}$, proving (50.12). This is a pairwise dilation comparison; it does not require a processor for the entire effect body or an unknown-input environment-seizing procedure.

Since $V \leq A^2 \leq 2V$ and $1 - (1+y)^{-N} \leq Ny$ for $y \geq 0$, the two nontrivial upper bounds imply

$$d_N \leq \min\{Nrh, \sqrt{N}rh/\sqrt{V}\} \leq \sqrt{2}K_N(p, q)h$$

when $V > 0$. The final inequality follows by putting $z = NV$ and using $\min\{1, z^{-1/2}\}\sqrt{z+1} \leq \sqrt{2}$. If $V = 0$, the only possibilities are scalar deterministic effects or a projective measurement, and the hybrid bound suffices.

For the lower bound, two explicit witnesses suffice. First suppose $r > 0$ and $h > 0$. Input a pure state in the direction $u - v$. Its success probabilities are

$$\frac{1 + b + r \sin(h/2)}{2}, \quad \frac{1 + b - r \sin(h/2)}{2}.$$

Their difference is $r \sin(h/2)$. The larger of their Bernoulli variances is at most $1 - b^2$: after complementing both output laws we may assume $a = 1 + b \leq 1$; both success probabilities lie in $[0, a]$, so their variances are at most $a \leq a(2 - a) = 1 - b^2$. Lemma 50.2, together with $\sin(h/2) \geq h/\pi$, now gives

$$(50.14) \quad d_N \geq \frac{1}{64\pi} \min \left\{ 1, rh \sqrt{\frac{N}{1 - b^2 + \tau}} \right\}.$$

The repeated pure input is a nonadaptive test.

For the second witness, choose a half-turn of the input qubit about an axis normal to the plane of u, v . It sends both Bloch directions to their negatives. With probability $1/2$ use the original measurement, and with probability $1/2$ apply this half-turn before the call and swap the two recorded outcomes afterwards. Discard the random choice. The resulting measurement has effects $(I \pm ru \cdot \sigma)/2$ or $(I \pm rv \cdot \sigma)/2$ under the respective hypotheses: the bias has been removed and the visibility is r . The same random preprocessing and postprocessing are used under both hypotheses and on every call. Any test for this unbiased pair is consequently an admissible test for the original pair. Theorem 48.1 gives

$$(50.15) \quad d_N \geq \frac{1}{64} \min \left\{ 1, rh \sqrt{\frac{N}{1 - r^2 + \tau}} \right\}.$$

The witness is the finite entangled-block test of Lemma 48.4, with this common processing. If the directions coincide or are antipodal, the required half-turn is chosen about any perpendicular axis.

Finally, feasibility $|b| + r \leq 1$ implies

$$\min\{1 - b^2, 1 - r^2\} \leq 2(1 - b^2 - r^2) = 4V.$$

For example, write $x = \max\{|b|, r\}$ and $y = \min\{|b|, r\}$; then $x \leq 1 - y$, $0 \leq y \leq 1/2$, and $x^2 + 2y^2 \leq (1 - y)^2 + 2y^2 \leq 1$, which is the asserted inequality. Hence the larger of the two coefficients in (50.14) and (50.15) is at least $K_N(p, q)/2$. Their stronger lower bound is at least $(128\pi)^{-1} \min\{1, K_N(p, q)h\}$, which implies (50.11). If $r = 0$ or $h = 0$, the two channels coincide and the assertion is immediate. $\square$

Proof of Theorem 50.1. For positive gaps use a path whose angular leg has the spectrum with the smaller angular coefficient. Lemma 50.3, followed by Lemma 50.2 and Theorem 50.4, gives

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 3T_N(p, p') + 3T_N(q, q') + \sqrt{2} \min\{K_N(p, q), K_N(p', q')\}h \leq 3H_N(E, F).$$

If one gap vanishes, that scalar effect may be aligned with the other direction at no cost, so the same estimate holds with angular term zero.

For the lower bound, put $A = T_N(p, p')$, $B = T_N(q, q')$, and $C = \min\{K_N(p, q), K_N(p', q')\}h$. The spectral projection lemma gives $d_N \geq 64^{-1} \min\{1, A\}$ and $d_N \geq 64^{-1} \min\{1, B\}$. Its angular projections and Theorem 50.4 give $d_N \geq 1536^{-1} \min\{1, C\}$. The largest of these three bounds is at least $4608^{-1} \min\{1, A + B + C\}$, which proves the weaker displayed constant $1/8192$. At a scalar effect only the two spectral bounds are needed. To check the assertion about lower witnesses, let d_N^{na} be the supremum over nonadaptive tests. The two spectral lower bounds hold with d_N^{na} , and the aligned spectral upper bound is consequently at most $2d_N^{\text{na}}$. Apply the triangle inequality to the output of a nonadaptive angular witness on the three-channel path. Its actual-pair leg is at

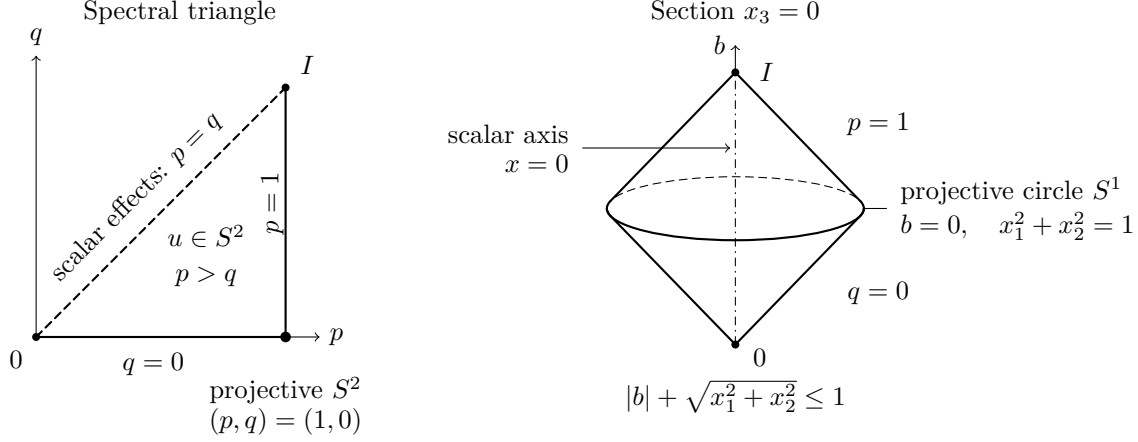


FIGURE 1. The four-dimensional effect body $|b| + |x| \leq 1$. Each point of the spectral triangle with $p > q$ carries a direction $u \in S^2$; on the scalar diagonal all directions represent the same effect. The boundary pieces $q = 0$ and $p = 1$ meet at the projective stratum $(p, q) = (1, 0)$, which carries the full sphere S^2 . The right panel is a projection of the three-dimensional section $x_3 = 0$; its equatorial circle is the intersection of that sphere with the section.

most d_N^{na} , and its aligned spectral leg is at most $2d_N^{\text{na}}$. The same lower comparison thus holds with d_N^{na} in place of d_N .

It remains to verify the one-use identity. For a reference-assisted input state ρ , the difference has output blocks B_ρ and $-B_\rho$, where $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An extremal eigenstate of $E - F$ attains equality. Thus $d_1 = 2\|E - F\|_{\text{op}}$; the two eigenvalues of $E - F$ are $((a - a') \pm |x - x'|)/2$. This proves (50.6). $\square$

Remark 50.5 (Boundary faces and repeated eigenvalues). On $p = q = t$, the measurement is the input-independent coin $\text{Bern}(t)$ and all directions represent the same effect. Its exact finite-use distance along this scalar interval is $B_N(t, t')$. On the face $q = 0$, $0 < p < 1$, the angular coefficient is

$$K_N(p, 0) = p \sqrt{\frac{N}{p(1-p) + N^{-1}}}.$$

For a fixed p in this interval it has square-root-horizon order, despite the zero eigenvalue. At the projective edge $(p, q) = (1, 0)$ it equals N . At the two deterministic apices $p = q = 0$ and $p = q = 1$, the angular coefficient vanishes while the spectral moduli retain their finite-use support cutoff. These cases are part of the same theorem, without a spectral margin imposed on the target.

51. THE DESCRIPTION ENTROPY OF ALL BINARY QUBIT EFFECTS

Let

$$\mathfrak{E} = \{E = E^* \in \mathbb{C}^{2 \times 2} : 0 \leq E \leq I\}.$$

An effect E specifies the ordered measurement with effects $E, I - E$. Write $\mathcal{C}_N(\mathfrak{E}, \delta)$ for its covering number under d_N , allowing arbitrary legal memoryless centres of this interface, and write $\mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta)$ when the centres have rational Cartesian coordinates. Both eigenvalues and the eigenspaces are charged. The spectral representation $E = pP_u + q(I - P_u)$, $p \geq q$, represents a scalar effect only once, independently of u .

Theorem 51.1 (The full effect body). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every integer $N \geq 1$ and $0 < \delta \leq \delta_0$,*

$$(51.1) \quad cN^2 \log(N+2) \delta^{-4} \leq \mathcal{C}_N(\mathfrak{E}, \delta) \leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta) \leq CN^2 \log(N+2) \delta^{-4}.$$

Consequently the optimum fixed-length reusable payload is

$$(51.2) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1).$$

The constants are uniform over the whole closed effect body, including scalar, deterministic and projective effects. For rational accuracy input, rational target data admit an exact rational encoder attaining this order.

The logarithm records accumulation near the projective stratum, where both outcome probabilities can approach their support endpoints. We prove the upper bound by a finite rational construction and the lower bound by separated families of projective-corner boxes.

51.1. A rational spectral and angular code. For $N \geq 1$ and rational $0 < \delta \leq 1/4$, put

$$(51.3) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad s_i = \begin{cases} \frac{i^2}{B^2 + i^2}, & 0 \leq i \leq B, \\ 1 - \frac{(2B-i)^2}{B^2 + (2B-i)^2}, & B < i \leq 2B. \end{cases}$$

Thus $0 = s_0 < s_1 < \dots < s_{2B} = 1$. For $0 \leq j < i \leq 2B$ define

$$(51.4) \quad K_{ij}^2 = \frac{N(s_i - s_j)^2}{s_i(1-s_i) + s_j(1-s_j) + N^{-1}}, \quad A_{ij} = \max \left\{ 1, \left\lceil \sqrt{256K_{ij}^2/\delta^2} \right\rceil \right\}.$$

Use the six signed stereographic charts

$$(51.5) \quad u_a = \epsilon \frac{1 - |z|^2}{1 + |z|^2}, \quad u_b = \frac{2z_b}{1 + |z|^2} \quad (b \neq a), \quad a \in \{1, 2, 3\}, \quad \epsilon \in \{-1, 1\},$$

with $z \in \{-A_{ij}, \dots, A_{ij}\}^2 / A_{ij}$ for the spectral pair (s_i, s_j) . Every chart word decodes to $s_i P_u + s_j(I - P_u)$. A diagonal pair $i = j$ has the single word $s_i I$. The exact number of words, including redundant chart words, is

$$(51.6) \quad L_{N,\delta} = (2B+1) + \sum_{0 \leq j < i \leq 2B} 6(2A_{ij} + 1)^2.$$

Theorem 51.2 (Exact rational realization). *The words in (51.6) form a radius- $\delta/2$ cover of $\mathfrak{E}$ by legal rational effects, and*

$$(51.7) \quad L_{N,\delta} \leq CN^2 \log(N+2) \delta^{-4}$$

for an absolute C . If

$$E = \frac{1}{2}((1+b)I + x \cdot \sigma), \quad b \in \mathbb{Q}, \quad x \in \mathbb{Q}^3, \quad |b| + |x| \leq 1,$$

then a covering word is computable by rational arithmetic and integer comparisons. Its payload is one fixed-length index in the set of (51.6).

Proof. Write $f(t) = t^2/(1+t^2)$, $0 \leq t \leq 1$. For a probability in $[0, 1/2]$, round its inverse coordinate t to the nearest multiple of $1/B$; for a probability in $[1/2, 1]$, use the reflected chart $1 - f(t)$. Choose the multiple of smaller magnitude at ties. This defines a nondecreasing quantizer on $[0, 1]$, so separately quantizing $p \geq q$ preserves their order.

The square-root vector associated with $f(t)$ is $(t, 1)/\sqrt{1+t^2}$; its derivative has norm $1/(1+t^2) \leq 1$. The reflected chart has the same bound. Hence the Euclidean square-root distance between each probability and its quantization is at most $1/(2B)$. If H is this distance, the product-fidelity

inequality gives unhalved distance at most $2\sqrt{N}H$ between the corresponding N -fold Bernoulli laws. The common two-coin programme in Lemma 50.3 therefore gives a spectral error at most

$$2\sqrt{N}/B \leq \delta/4.$$

This argument includes adaptive testers and entangled references.

If the quantized eigenvalues coincide, this completes the encoding. Otherwise choose a signed chart whose distinguished coordinate has maximal absolute value in the target direction. Its inverse coordinate belongs to $[-1, 1]^2$. Rounding its two coordinates to the grid of denominator A_{ij} gives angular error at most $2\sqrt{2}/A_{ij}$, since geodesic distance is bounded by the length of the mapped straight segment and the differential of stereographic projection has norm at most two. The angular upper bound in Theorem 50.4 gives error at most

$$\sqrt{2}K_{ij} \frac{2\sqrt{2}}{A_{ij}} \leq \delta/4.$$

The triangle inequality proves the covering assertion. Formula (51.5) preserves the unit sphere exactly, so every decoded effect is legal and rational.

We verify exact computability even when the input eigenvalues are irrational. Put $Q = |x|^2 \in \mathbb{Q}$. The eigenvalues are $(1 + b \pm \sqrt{Q})/2$. Each decision in the spectral rounding compares one of these numbers with a rational threshold $f((k + 1/2)/B)$ or its reflection. Such a comparison is the sign of a rational number plus or minus $\sqrt{Q}$; checking signs before squaring makes it an exact rational comparison. If $Q = 0$, the two quantized eigenvalues coincide. Otherwise choose the smallest index a maximizing $|x_a|$ and put $\epsilon = \text{sgn}(x_a)$. The inverse chart is

$$z_b = \frac{x_b}{\sqrt{Q} + |x_a|} \quad (b \neq a).$$

Its comparison with any rational rounding threshold again reduces to the sign of a rational number plus a rational multiple of $\sqrt{Q}$. The integers B, A_{ij} are computed by integer squaring and rational comparison. Thus no irrational normalization is required. The decoder uses only the displayed rational formulae.

It remains to count the words. Set $m_i = \min(i, 2B - i)$ and $U = B^2/N$. Directly from (51.3),

$$\frac{m_i^2}{4B^2} \leq s_i(1 - s_i) \leq \frac{m_i^2}{B^2}, \quad K_{ij}^2 \leq \frac{4NB^2}{m_i^2 + m_j^2 + U}.$$

Each value of m_i occurs at most twice. Since $(2A_{ij} + 1)^2 \leq C(1 + K_{ij}^2/\delta^2)$, we obtain

$$(51.8) \quad L_{N,\delta} \leq CB^2 + \frac{CNB^2}{\delta^2} \sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U}.$$

Here $U \geq 1$. The part with $\max(a, b) \leq \sqrt{U}$ is bounded by an absolute constant. Each successive dyadic square ring contributes at most another absolute constant. There are at most $C \log(N + 2)$ such rings, because $B/\sqrt{U} = \sqrt{N}$. Consequently

$$\sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U} \leq C \log(N + 2).$$

Finally $B^2 \leq CN/\delta^2$, so (51.8) proves (51.7). The support cutoff in the denominator is retained throughout the count, including the endpoint grid words. $\square$

51.2. Packing the projective corner.

Proof of Theorem 51.1. The upper bound follows from Theorem 51.2. For a real δ , choose a rational δ' with $\delta \leq \delta' \leq 2\delta$ and use its radius- $\delta'/2$ code; decrease δ_0 to at most $1/8$. For the lower bound first let $N \geq 16$. For every

$$t = 8^j/N \leq 1/16, \quad j \geq 0,$$

consider the effects with

$$(51.9) \quad p = 1 - \alpha, \quad q = \beta, \quad t \leq \alpha, \beta \leq 2t, \quad u = u(z), \quad z \in [-1/4, 1/4]^2,$$

where $u(z)$ is the chart in (51.5) with $a = 3, \epsilon = 1$. On this square, spherical distance is bounded above and below by absolute positive multiples of Euclidean distance in z . Throughout (51.9), the eigenvalue gap is at least $3/4$ and

$$p(1-p) \asymp q(1-q) \asymp t, \quad p(1-p) + q(1-q) + N^{-1} \asymp t.$$

The finite-distance comparison of Theorem 50.1 therefore supplies an absolute $c_1 > 0$ such that any two points in this box satisfy

$$(51.10) \quad d_N \geq c_1 \min \left\{ 1, \sqrt{N/t} |(\alpha, \beta, z) - (\alpha', \beta', z')|_\infty \right\}.$$

Choose an absolute sufficiently large A and then an absolute sufficiently small δ_0 . In each of the four coordinates take a grid of spacing $A\delta\sqrt{t/N}$. Since $Nt \geq 1$, this spacing is at most half the width of either spectral interval when $\delta \leq \delta_0$. The resulting points have pairwise distance at least 4δ by (51.10), and their number is at least

$$c \left(\frac{t}{\delta\sqrt{t/N}} \right)^2 \left(\frac{1}{\delta\sqrt{t/N}} \right)^2 = cN^2\delta^{-4}.$$

Points from different boxes are also uniformly separated. Indeed, if $t' \geq 8t$, then $\alpha' \geq t'$ and $\alpha \leq 2t \leq t'/4$. The spectral modulus of the corresponding upper eigenvalues is bounded below by an absolute positive multiple of $\sqrt{Nt'} \geq 1$. The spectral noncancellation bound in Theorem 50.1 gives $d_N \geq c_2 > 0$. Shrinking δ_0 ensures that this is at least 4δ . There are at least $c \log(N+2)$ permitted boxes. Their union is therefore a 4δ -packing of cardinality $cN^2 \log(N+2)\delta^{-4}$.

For $1 \leq N < 16$, use instead a fixed spectral rectangle $5/8 \leq p \leq 3/4$, $1/4 \leq q \leq 3/8$ and the same angular chart. Theorem 50.1 gives a uniform local Euclidean lower bound, and a grid of spacing $C\delta$ has at least $c\delta^{-4}$ points. The factor $N^2 \log(N+2)$ is bounded on this finite range. This proves the same lower estimate for all $N \geq 1$.

A radius- δ ball about any legal centre contains at most one point of a 4δ -packing, by the triangle inequality. Hence the packing proves the lower bound with the stated centre convention. Taking binary logarithms and rounding the index length proves (51.2). $\square$

51.3. The source of the logarithm. The spectral and angular weights also give the following integral form of the same accumulation. Put $v_p = p(1-p)$, $v_q = q(1-q)$ and $\tau = N^{-1}$. Then

$$(51.11) \quad \int_{0 \leq q \leq p \leq 1} \frac{N^2(p-q)^2 dp dq}{(v_p + v_q + \tau) \sqrt{(v_p + \tau)(v_q + \tau)}} \asymp N^2 \log(N+2).$$

To verify this, near $p = 1, q = 0$ put $\alpha = 1 - p$, $\beta = q$. After removing the factor N^2 , the integrand is comparable to

$$\frac{1}{(\alpha + \beta + \tau) \sqrt{(\alpha + \tau)(\beta + \tau)}}.$$

For $s = \alpha + \beta$, integration over $0 \leq \alpha \leq s$ is bounded above by an absolute constant times $1/(s + \tau)$. For $s \geq 2\tau$, restriction to $s/4 \leq \alpha \leq 3s/4$ gives the matching lower bound. Integration

in s proves logarithmic growth. Near the scalar endpoint $p = q = 0$, the factor $(p - q)^2$ is at most $(p + q)^2$, while $v_p + v_q \asymp p + q$; the remaining integral is uniformly bounded. Complementation gives the same conclusion near $p = q = 1$. Away from these endpoint neighbourhoods, $v_p + v_q$ is bounded below and the factors $v_p^{-1/2}, v_q^{-1/2}$ are integrable. Bounded N is absorbed into the constants. Thus the extra logarithm is contributed by the projective corner. The finite covering construction and packing above also include the collapsed scalar coordinates, where the spectral representation ceases to be a chart.

52. MATRIX GEOMETRY IN ARBITRARY INPUT DIMENSION

The binary effect body has a finite-use description that does not require eigenvector coordinates. Its regularization is the same at every spectral rank, and two-dimensional compressions supply lower witnesses for the full matrix problem. Throughout this section $d, N \geq 1$ are integers and

$$(52.1) \quad \mathfrak{E}_d = \{E \in \text{End}(\mathbb{C}^d) : E = E^*, 0 \preceq E \preceq I_d\}.$$

The measurement $\mathcal{M}_E$ has the ordered effects $E, I_d - E$ and no residual quantum output. The distance d_N has the same reference-assisted adaptive meaning as in Theorem 50.1. Write d_N^{na} for its restriction to nonadaptive experiments, allowing entanglement across the calls and a retained reference. These experiments may depend on the known pair of hypotheses.

Put $\tau = N^{-1}$ and $V_E = E(I_d - E)$. On the real Hilbert space of Hermitian matrices with Hilbert–Schmidt inner product, define

$$(52.2) \quad \mathcal{A}_E(H) = V_E H + H V_E + \tau H,$$

$$(52.3) \quad g_{N,E}(H)^2 = N \langle H, \mathcal{A}_E^{-1}(H) \rangle_{\text{HS}}.$$

The operator $\mathcal{A}_E$ is strictly positive, including at a singular effect. For a pair E, F set

$$(52.4) \quad M = \tfrac{1}{2}(E + F), \quad H = F - E, \quad Q_N(E, F) = g_{N,M}(H).$$

In any orthonormal eigenbasis of M , with eigenvalues m_i and $v_i = m_i(1 - m_i)$, this is

$$(52.5) \quad Q_N(E, F)^2 = N \sum_{i,j=1}^d \frac{|H_{ij}|^2}{v_i + v_j + N^{-1}}.$$

Definition (52.4) makes the expression independent of all choices at repeated eigenvalues. We use it as a comparison modulus; a triangle inequality for Q_N is not required.

The inverse Sylvester operator is familiar from the Bures metric g^{B} on the positive definite cone [67]. Its normalization gives the algebraic identity

$$g_{N,E}(H)^2 = 2N g_{V_E + \tau I_d/2}^{\text{B}}(H, H).$$

Here H remains the effect tangent; the differential of the variance map $E \mapsto V_E$ is $H - EH - HE$.

Theorem 52.1 (Finite-use geometry of the matrix effect body). *For every $E, F \in \mathfrak{E}_d$,*

$$(52.6) \quad \frac{1}{8192d} \min\{1, Q_N(E, F)\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The constants are uniform in the horizon and in both effects, without a spectral margin or a condition on eigenvalue multiplicities. The adaptive upper bound permits arbitrary retained references, common feedback, and bounded public stopping. Each lower witness uses inputs in one fixed subspace of dimension at most two. Moreover,

$$(52.7) \quad d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}.$$

The theorem includes every rank of projection and all of the intermediate singular strata. The variance $E(I_d - E)$, rather than the smallest eigenvalue of either outcome, determines the matrix weights. The factor N^{-1} makes the formula finite on the entire closed body. The proof first establishes a path inequality valid for every common adaptive tester, and then obtains the converse from one- and two-dimensional experiments.

52.1. A horizontal measurement dilation. Kraus-gauge bounds for channel estimation were developed in [62, 47, 63]. Their integration along channel paths appears in [65, Section III, equation (26)] and, with general adaptive bounds, in [66, Section IV.A]. We use an explicit horizontal lift for binary effects and control the finite-horizon regularization by a trace-norm remainder.

Lemma 52.2 (Horizontal derivative and regularized path bound). *Every continuously differentiable path $G : [0, 1] \rightarrow \mathfrak{E}_d$ satisfies*

$$(52.8) \quad d_N(\mathcal{M}_{G(0)}, \mathcal{M}_{G(1)}) \leq 4 \int_0^1 g_{N,G(t)}(G'(t)) dt.$$

Proof. We begin with an interior effect G , so that G and $I_d - G$ are invertible. Set $S = \sqrt{G(I_d - G)}$. For a Hermitian matrix K put $H_0 = SK + KS$. At the two factors $A_1 = \sqrt{G}$, $A_0 = \sqrt{I_d - G}$ prescribe derivatives

$$(52.9) \quad B_1 = \sqrt{I_d - G} K, \quad B_0 = -\sqrt{G} K.$$

They obey

$$(52.10) \quad \begin{aligned} A_1^* B_1 + B_1^* A_1 &= H_0, & A_0^* B_0 + B_0^* A_0 &= -H_0, \\ \sum_{y=0}^1 A_y^* B_y &= 0, & \sum_{y=0}^1 B_y^* B_y &= K^2. \end{aligned}$$

These tangent equations can be realized by genuine dilations. Let $G_1(s) = G + sH_0$, $G_0(s) = I_d - G - sH_0$ for s in a sufficiently small open interval, and put $R_y(s) = \sqrt{G_y(s)}$. The first line of (52.10) implies that

$$T_y = (B_y - R'_y(0))R_y(0)^{-1}$$

is anti-Hermitian. Indeed, multiplying $T_y + T_y^*$ on the left and right by $R_y(0)$ gives, with $R = R_y(0)$,

$$R(T_y + T_y^*)R = RB_y + B_y^*R - (RR'_y(0) + R'_y(0)R) = 0,$$

by differentiating $R_y(s)^2 = G_y(s)$. Thus $\tilde{A}_y(s) = \exp(sT_y)R_y(s)$ has derivative B_y at zero and satisfies $\tilde{A}_y(s)^* \tilde{A}_y(s) = G_y(s)$ exactly. A measurement isometry is

$$W(s)\psi = \sum_{y=0}^1 |y\rangle_C |y\rangle_L \tilde{A}_y(s)\psi.$$

Here C is the observed outcome, and L and the factor space of $\tilde{A}_y$ are environments. Tracing the environments implements $\mathcal{M}_{G(s)}$, also on inputs entangled with a reference. At zero, (52.10) gives

$$(52.11) \quad W^* \dot{W} = 0, \quad \dot{W}^* \dot{W} = K^2.$$

The horizontal Kraus condition has the antecedents [62, Theorems 4–5]. Its adaptive channel-extension form is $F_N \leq 4N\|\alpha\|$ when $\beta = 0$; see [63, equation (9)] and [64, equations (5)–(10)]. In the present lift $\alpha = K^2$ and $\beta = 0$. We give the cancellation explicitly for this binary measurement.

Fix a common tester with N slots. Purify its initial state and all intervening operations, and retain the measurement environments only for this comparison; neither the outcome copy nor the factor environment is accessible to the tester. The derivative of its final purification in the

direction H_0 is the sum of N terms, each having one differentiated occurrence of W . The terms are mutually orthogonal. For two different differentiated slots, cancel the common isometries after the later slot; the remaining overlap contains $W^* \dot{W} \otimes I = 0$. Each term has norm at most $\|\dot{W}\|_{\text{op}} = \|K\|_{\text{op}}$, independently of the reference and memory dimensions. Hence the derivative of the purification has norm at most $\sqrt{N}\|K\|_{\text{op}}$. The pure-state trace-norm derivative bound, followed by contraction under the partial trace, bounds the observed output derivative by

$$(52.12) \quad 2\sqrt{N}\|K\|_{\text{op}}.$$

A publicly stopped tester is padded to N slots by calls on fixed dummy inputs, whose outcomes and environments are both discarded. The same purification argument then applies to its original output. Classical feedback is included by purifying the common controlled operations.

Now let H be an arbitrary Hermitian tangent at G , and set $\kappa = N^{-1/2}$. There is a unique Hermitian solution of

$$(52.13) \quad SK + KS + \kappa K = H.$$

Write $H = H_0 + R$, with $H_0 = SK + KS$ and $R = \kappa K$. Both are legitimate tangents at the interior effect G . The derivative of a fixed tester's output is linear in the inserted channel tangent. The part arising from H_0 is bounded by (52.12). The part arising from R is bounded by $2N\|R\|_{\text{op}}$. To verify the latter bound, the one-call tangent has two opposite reference blocks

$$B_\rho = \text{tr}_A[(R \otimes I)\rho], \quad -B_\rho.$$

Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|R\|_{\text{op}}$ for every input state. Insert this tangent into each of the N slots in turn; all subsequent common channels contract the Hermitian trace norm. This proves the bound for its sum. Thus the actual tangent H has output derivative bounded by

$$(52.14) \quad 2\sqrt{N}\|K\|_{\text{op}} + 2N\|R\|_{\text{op}} = 4\sqrt{N}\|K\|_{\text{op}}.$$

The splitting is used only for the differential at an interior effect; it does not postulate a physical mixture realizing the entire path.

In an eigenbasis of G , with $v_i = g_i(1 - g_i)$, equation (52.13) reads

$$K_{ij} = \frac{H_{ij}}{\sqrt{v_i} + \sqrt{v_j} + \kappa}.$$

Since $(\sqrt{v_i} + \sqrt{v_j} + \kappa)^2 \geq v_i + v_j + \tau$,

$$(52.15) \quad \sqrt{N}\|K\|_{\text{op}} \leq \sqrt{N}\|K\|_{\text{HS}} \leq g_{N,G}(H).$$

The integration of local distinguishability has direct antecedents: [65, Section III, equation (26), PDF version 3] treats the parallel path bound, and [66, Section IV.A, equations (14)–(22)] applies adaptive metrological estimates. Here the regularized binary Sylvester split and its residual produce the explicit integrand. For an interior path, integrate the resulting output derivative bound for each fixed tester, and then take the supremum. This proves (52.8) without exchanging a supremum with a derivative.

For a path in the closed body, apply the result to $G_\epsilon(t) = (1 - 2\epsilon)G(t) + \epsilon I_d$, where $0 < \epsilon < 1/2$. At fixed N , the form $g_{N,G}(H)$ is continuous on the whole body and bounded above by $N\|H\|_{\text{HS}}$. The integrals therefore converge as $\epsilon \downarrow 0$. The hybrid inequality $d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 2N\|E - F\|_{\text{op}}$ gives continuity of the distances at both endpoints. Passing to the limit proves the asserted closed-body statement. $\square$

52.2. The midpoint comparison.

Proof of Theorem 52.1. For the upper bound use $G(t) = (1-t)E + tF$. The map $G \mapsto V_G = G - G^2$ is operator concave. For $0 < t \leq 1/2$, write $G(t) = (1-2t)E + 2tM$ and discard the positive multiple of V_E in the concavity inequality. Applying the same argument from the other endpoint gives

$$(52.16) \quad V_{G(t)} \succeq a(t)V_M, \quad a(t) = 2 \min\{t, 1-t\}.$$

For a positive matrix D and a Hermitian matrix X ,

$$\langle X, (L_D + R_D)X \rangle_{\text{HS}} = 2 \operatorname{tr}(DX^2) \geq 0.$$

Thus left-plus-right multiplication preserves the stated order on the real Hilbert space of Hermitian matrices. Since $a(t) \leq 1$, $\mathcal{A}_{G(t)} \succeq a(t)\mathcal{A}_M$. Inverting positive operators reverses their order, whence

$$g_{N,G(t)}(H) \leq a(t)^{-1/2}Q_N(E, F).$$

The scalar factor has integral 2 on $[0, 1]$. Lemma 52.2 yields $d_N \leq 8Q_N$; the trace-norm bound $d_N \leq 2$ gives the stated upper estimate.

For the lower bound choose any orthonormal eigenbasis e_i of M and put

$$(52.17) \quad q_{ij} = |H_{ij}| \sqrt{\frac{N}{v_i + v_j + \tau}}, \quad Q_N^2 = \sum_{i,j} q_{ij}^2.$$

Repeated input e_i gives Bernoulli probabilities $s = m_i - H_{ii}/2$, $t = m_i + H_{ii}/2$. Their variances satisfy

$$\max\{s(1-s), t(1-t)\} \leq s(1-s) + t(1-t) \leq 2m_i(1-m_i).$$

Lemma 50.2 therefore gives

$$(52.18) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{64} \min\{1, q_{ii}\}.$$

For $i \neq j$, restrict each input to $\mathcal{S} = \operatorname{span}\{e_i, e_j\}$. The resulting channels have the compressed effects $E_{\mathcal{S}}, F_{\mathcal{S}}$, including on inputs entangled with a retained reference. Their midpoint is $M_{\mathcal{S}} = \operatorname{diag}(m_i, m_j)$. Write their spectral data as (p, q, u) and (p', q', v) in the notation of Section 50, with gaps r, r' and angle h when both gaps are positive. Put

$$V_E^{\mathcal{S}} = p(1-p) + q(1-q), \quad V_F^{\mathcal{S}} = p'(1-p') + q'(1-q').$$

For the compressed matrices the exact identity

$$V_{M_{\mathcal{S}}} = \frac{1}{2}V_{E_{\mathcal{S}}} + \frac{1}{2}V_{F_{\mathcal{S}}} + \frac{1}{4}(F_{\mathcal{S}} - E_{\mathcal{S}})^2$$

shows, on taking traces, that

$$(52.19) \quad v_i + v_j + \tau \geq \frac{1}{2}(V_E^{\mathcal{S}} + \tau), \quad v_i + v_j + \tau \geq \frac{1}{2}(V_F^{\mathcal{S}} + \tau).$$

The scalar part of $F_{\mathcal{S}} - E_{\mathcal{S}}$ has zero off-diagonal entry. Its traceless part has operator norm $|ru - r'v|/2$. Decomposing this vector along the smaller radius gives

$$(52.20) \quad \begin{aligned} |H_{ij}| &\leq \frac{1}{2}|r - r'| + \min\{r, r'\} \sin(h/2) \\ &\leq \frac{1}{2}(|p - p'| + |q - q'| + \min\{r, r'\}h). \end{aligned}$$

When one radius is zero, the angular term is zero and the same inequality holds without choosing its direction. Each endpoint variance is bounded by the appropriate $V_E^{\mathcal{S}}$ or $V_F^{\mathcal{S}}$. Equations (52.19)–(52.20) therefore imply

$$(52.21) \quad q_{ij} \leq \frac{1}{\sqrt{2}} (T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\}h).$$

The nonadaptive assertion of Theorem 50.1 applied to this compressed pair yields

$$(52.22) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{8192} \min\{1, q_{ij}\}.$$

Every such witness is a legitimate experiment for the original effects by a fixed isometric embedding of $\mathcal{S}$. Since $\max_{i,j} q_{ij} \geq Q_N/d$, taking the strongest of (52.18) and (52.22) proves the lower bound. For $d = 1$ only the diagonal argument is needed.

Finally, a one-use reference-assisted difference has output blocks B_ρ and $-B_\rho$, with $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Trace-norm duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An eigenstate of $E - F$ at an extremal eigenvalue attains equality. This proves (52.7). $\square$

Corollary 52.3 (Uniform comparison with nonadaptive experiments). *For all $E, F \in \mathfrak{E}_d$ and all $N \geq 1$,*

$$(52.23) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 65536d d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

Here nonadaptive experiments may entangle all inputs and retain a reference; they use no feedback between calls. The dimensional constant comes from the maximum of d^2 weighted entries and is not optimized.

Proof. If $Q_N \leq 1$, divide the upper estimate $8Q_N$ by the lower estimate $Q_N/(8192d)$. If $Q_N \geq 1$, the upper estimate is 2 and the lower estimate is $1/(8192d)$, which gives a smaller constant. The identical pair needs no division. $\square$

52.3. All spectral ranks and local forms.

Lemma 52.4 (Spectral noncancellation in every rank). *Let $\lambda_1(E) \geq \dots \geq \lambda_d(E)$ and $\lambda_1(F) \geq \dots \geq \lambda_d(F)$ be the ordered spectra. For every $1 \leq k \leq d$,*

$$(52.24) \quad B_N(\lambda_k(E), \lambda_k(F)) \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

For effects diagonal in the same ordered orthonormal basis,

$$(52.25) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \sum_{k=1}^d B_N(\lambda_k(E), \lambda_k(F)).$$

Proof. Suppose $\lambda_k(E) \geq \lambda_k(F)$. The span of the first k eigenvectors of E intersects the orthogonal complement of the first $k - 1$ eigenvectors of F in a nonzero subspace. A unit vector in that intersection has success probabilities $s \geq \lambda_k(E) \geq \lambda_k(F) \geq t$ under the two hypotheses. The monotonicity in Lemma 50.2 implies $B_N(s, t) \geq B_N(\lambda_k(E), \lambda_k(F))$. Repeated use of this vector proves the first assertion; reverse the hypotheses for the opposite ordering. Multiplicities cause no change in the dimension argument.

For the upper bound, first measure the common eigenbasis and then produce a bit whose success probability is its corresponding eigenvalue. Supply N independent bits in each of the d rows before the experiment. Every common adaptive tester is a common channel of these row arrays, including its reference blocks and all publicly stopped histories. The tensor-product triangle inequality gives (52.25); unused row bits are discarded. $\square$

The following local comparison will permit volume estimates at every boundary stratum. It concerns matrices themselves and does not require a coordinate chart for eigenvectors.

Lemma 52.5 (Uniform local comparison of the variance forms). *Set $W_G = G(I_d - G) + \tau I_d/2$. For a Hermitian H , put $q = g_{N,G}(H)$. Then*

$$(52.26) \quad \|W_G^{-1/2} H W_G^{-1/2}\|_{\text{HS}} \leq 2q.$$

For either sign for which $G \pm H/2 \in \mathfrak{E}_d$, one has

$$(52.27) \quad \|W_G^{-1/2} (W_{G \pm H/2} - W_G) W_G^{-1/2}\|_{\text{op}} \leq q + \frac{3}{4}q^2.$$

In particular, for $q \leq 1/4$,

$$(52.28) \quad \frac{1}{2}W_G \preceq W_{G \pm H/2} \preceq 2W_G.$$

Thus, if either $Q_N(E, F) \leq 1/4$ or $g_{N,E}(F - E) \leq 1/4$, the midpoint and the fixed-endpoint quadratic forms are comparable within factors 2.

Proof. In an eigenbasis of W_G , with eigenvalues $w_i \geq \tau/2$,

$$\frac{w_i + w_j}{Nw_iw_j} \leq 4.$$

Multiplying this inequality by $N|H_{ij}|^2/(w_i + w_j)$ and summing proves (52.26). Put $Z = W_G^{-1/2}HW_G^{-1/2}$. The exact expansion

$$W_{G \pm H/2} - W_G = \pm \frac{1}{2}(H - GH - HG) - \frac{1}{4}H^2$$

has normalized linear part $\pm((I_d - 2G)Z + Z(I_d - 2G))/4$, since G commutes with W_G . Its operator norm is at most $\|Z\|_{\text{op}}/2 \leq q$. The normalized quadratic part is $-ZW_GZ/4$. Since $\tau \leq 1$, one has $\|W_G\|_{\text{op}} \leq 3/4$, and its norm is at most $3q^2/4$. This proves (52.27) and hence (52.28). The same order holds for left plus right multiplication, and inversion gives the quadratic-form comparison. Apply it first at $G = M$ or at $G = E$, respectively, to obtain the final assertion. $\square$

53. DESCRIPTION ENTROPY IN ARBITRARY MATRIX DIMENSION

The intrinsic comparison of Theorem 52.1 permits the full effect body to be treated without choosing eigenvector coordinates near repeated spectra. Throughout this section the input dimension $d \geq 1$ is fixed. Constants may depend on d , but not on the integer horizon $N \geq 1$ or on the accuracy parameter.

Write $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ for the covering number under d_N , with centres in the same legal memoryless ordered binary interface, and write $\mathcal{C}_N^{\text{rat}}$ when the real and imaginary parts of every matrix entry of each centre are rational.

Theorem 53.1 (Entropy of the full effect body). *For every integer $d \geq 1$ there are constants $c_d, C_d, \delta_d > 0$ such that, for all integers $N \geq 1$ and all $0 < \delta \leq \delta_d$,*

$$(53.1) \quad \begin{aligned} c_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2} &\leq \mathcal{C}_N(\mathfrak{E}_d, \delta) \\ &\leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}_d, \delta) \\ &\leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}. \end{aligned}$$

Consequently the optimum fixed-length reusable description has length

$$(53.2) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The estimates include all support and spectral multiplicities. The rational-centre assertion here follows from a covering argument. Theorem 56.1 gives a deterministic finite exact encoder and decoder attaining the same order in every dimension. Theorem 51.2 retains its separate direct construction in dimension two. Computational cost is accounted for separately from the number of transmitted bits.

For $d = 1$, the result is the Bernoulli covering law $N^{1/2}\delta^{-1}$. For $d = 2$, it recovers Theorem 51.1. The next two cases have orders $N^{9/2} \log(N+2)\delta^{-9}$ and $N^8 [\log(N+2)]^2 \delta^{-16}$. The logarithmic power counts independent nested approaches of paired eigenvalues to the two support endpoints. We first establish a uniform measure estimate for actual operational balls, and then evaluate the total measure.

53.1. Operational balls and a regularized volume form. Set $n = d^2$, $\tau = N^{-1}$ and

$$W_E = E(I - E) + \frac{\tau}{2}I.$$

As in Section 52, let

$$(53.3) \quad g_{N,E}(H)^2 = N \langle H, (L_{W_E} + R_{W_E})^{-1} H \rangle_{\text{HS}}$$

on the real Hilbert space of Hermitian matrices. Let $G_{N,E}$ denote this positive quadratic form in Hilbert–Schmidt orthonormal coordinates, and define

$$(53.4) \quad \rho_N(E) = \sqrt{\det G_{N,E}}, \quad d\mu_N(E) = \rho_N(E) dE,$$

The real orthonormal basis consists of e_{ii} and, for $i < j$, $(e_{ij} + e_{ji})/\sqrt{2}$ and $i(e_{ij} - e_{ji})/\sqrt{2}$; dE is Lebesgue measure in these coordinates. The cutoff makes ρ_N positive and continuous on the whole closed body for each N .

Lemma 53.2 (Uniform volume of operational balls). *For fixed d there are $a_d, A_d, r_d > 0$ such that*

$$(53.5) \quad a_d r^{d^2} \leq \mu_N(B_{d_N}(E, r)) \leq A_d r^{d^2}$$

for every $N \geq 1$, every $E \in \mathfrak{E}_d$, and every $0 < r \leq r_d$. Balls are taken inside $\mathfrak{E}_d$.

Proof. We make explicit the two local consequences of Lemma 52.5 that will be used. If H is Hermitian and $q = g_{N,E}(H)$, diagonalizing W_E gives

$$(53.6) \quad \|W_E^{-1/2} H W_E^{-1/2}\|_{\text{HS}}^2 \leq 4q^2,$$

since its eigenvalues w_i are at least $\tau/2$ and

$$\frac{w_i + w_j}{N w_i w_j} = \tau(w_i^{-1} + w_j^{-1}) \leq 4.$$

For any legal $E + tH$, direct expansion yields

$$W_{E+tH} - W_E = \frac{t}{2}((I - 2E)H + H(I - 2E)) - t^2 H^2.$$

The matrices E and W_E commute, $\|I - 2E\|_{\text{op}} \leq 1$, and $\|W_E\|_{\text{op}} \leq 3/4$. Thus the norm of the normalized difference is at most $2|t|q + 3t^2q^2$. For $|t| \leq 1$ and q below an absolute constant, it follows that

$$(53.7) \quad \frac{1}{2}W_E \preceq W_{E+tH} \preceq 2W_E.$$

Loewner comparison passes to $L_W + R_W$ and, by inversion, to the quadratic forms. It also gives

$$(53.8) \quad 2^{-n/2} \rho_N(E) \leq \rho_N(E + tH) \leq 2^{n/2} \rho_N(E).$$

Apply this comparison from an endpoint to the midpoint and from the midpoint to either endpoint. Theorem 52.1 then provides constants $A, B, r_0 > 0$, depending only on d , such that

$$(53.9) \quad d_N(E, F) \leq r \implies g_{N,E}(F - E) \leq Ar,$$

$$(53.10) \quad g_{N,E}(F - E) \leq Br \implies d_N(E, F) \leq r,$$

for legal E, F and $0 < r \leq r_0$. Decrease r_0 so that Ar_0 lies in the small-form range of (53.7).

Let ω_n be the Euclidean volume of the unit ball in $\mathbb{R}^n$. The ellipsoid $g_{N,E}(H) \leq u$ has Lebesgue volume $\omega_n u^n / \rho_N(E)$. Equations (53.9) and (53.8) therefore imply

$$\mu_N(B_{d_N}(E, r)) \leq 2^{n/2} \omega_n (Ar)^n.$$

For the lower bound we construct a full legal ellipsoid, including when E lies on a support boundary. Choose a small constant $a > 0$ depending only on d , set $s = ar\tau$, and move towards the scalar interior point:

$$(53.11) \quad E_s = (1 - 2s)E + sI.$$

Take r_0 sufficiently small that $s \leq 1/4$. Then $sI \preceq E_s \preceq (1 - s)I$. Since the displacement commutes with E , its form is bounded by

$$g_{N,E}(E_s - E)^2 = N \sum_i \frac{s^2(1 - 2\lambda_i)^2}{2\lambda_i(1 - \lambda_i) + \tau} \leq d \frac{s^2}{\tau^2} = da^2r^2.$$

Choose $a\sqrt{d} \leq B/4$. By (53.10), $d_N(E, E_s) \leq r/4$.

Let $X = X^*$ satisfy $g_{N,E_s}(X) \leq br$, with $b > 0$ to be chosen. Write ℓ_i for the eigenvalues of E_s and $w_i = \ell_i(1 - \ell_i) + \tau/2$. Since $\ell_i \geq s$ and $w_i \leq \ell_i + \tau/2$, we have

$$\begin{aligned} \|E_s^{-1/2} X E_s^{-1/2}\|_{\text{HS}}^2 &\leq g_{N,E_s}(X)^2 \max_{i,j} \frac{w_i + w_j}{N\ell_i\ell_j} \\ &\leq b^2 r^2 \left(\frac{2\tau}{s} + \frac{\tau^2}{s^2} \right) = b^2 \left(\frac{2r}{a} + \frac{1}{a^2} \right). \end{aligned}$$

The same bound holds with $I - E_s$ in place of E_s , because $1 - \ell_i \geq s$ and $w_i \leq 1 - \ell_i + \tau/2$. Taking $b \leq a/4$ and $ar_0 \leq 1$ makes both operator norms at most $1/2$. Consequently every point of the entire ellipsoid

$$\mathcal{A}_s = \{E_s + X : g_{N,E_s}(X) \leq br\}$$

is a legal effect. Choose also $b \leq B/4$. Equation (53.10) and the triangle inequality give $\mathcal{A}_s \subset B_{d_N}(E, r)$. The density comparison about E_s and the exact ellipsoid volume now imply

$$\mu_N(B_{d_N}(E, r)) \geq 2^{-n/2} \omega_n (br)^n.$$

All choices depend only on d . This proves the assertion on every boundary stratum as well as in the interior. $\square$

53.2. The total regularized volume.

Proposition 53.3 (Nested endpoint accumulation). *For fixed $d \geq 1$, uniformly over $N \geq 1$,*

$$(53.12) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} [\log(N+2)]^{[d/2]}.$$

Proof. In a spectral basis, the quadratic form has weight $N/(2w_i)$ on the i th diagonal direction and weight $N/(w_i + w_j)$ on each of the two Hilbert–Schmidt orthonormal real directions belonging to $i < j$. Hence

$$(53.13) \quad \rho_N(E) = 2^{-d/2} N^{d^2/2} \prod_i w_i^{-1/2} \prod_{i < j} (w_i + w_j)^{-1}.$$

We use the classical spectral change of variables [68]. For a simple spectrum, differentiation of $E = U \text{diag}(\lambda) U^*$ gives

$$U^*(dE)U = d \text{diag}(\lambda) + [U^* dU, \text{diag}(\lambda)].$$

Each complex off-diagonal pair therefore contributes $(\lambda_i - \lambda_j)^2$ to the real Jacobian. Integration over the compact unitary orbit and over permutations supplies a positive constant depending only on d . The repeated-spectrum set is Lebesgue null. This spectral change of variables, together with (53.13), yields

$$(53.14) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} I_d(\tau),$$

where

$$(53.15) \quad I_d(\tau) = \int_{[0,1]^d} \prod_i (v_i + \tau)^{-1/2} \prod_{i < j} \frac{(\lambda_i - \lambda_j)^2}{v_i + v_j + \tau} d\lambda, \quad v_i = \lambda_i(1 - \lambda_i).$$

Replacing $v_i + \tau/2$ by $v_i + \tau$ in the single-coordinate factors changes the integral by a factor between fixed positive constants depending only on d .

We prove

$$(53.16) \quad I_d(\tau) \asymp_d [\log(2 + \tau^{-1})]^{[d/2]}, \quad 0 < \tau \leq 1.$$

For the upper bound, partition each coordinate into $[0, 1/4]$, $[1/4, 3/4]$, and $[3/4, 1]$. Factors involving an eigenvalue in the middle interval are bounded above by constants, so these coordinates can be integrated out. For each of the remaining $l \leq d$ coordinates put $x_i = \lambda_i$ near zero and $x_i = 1 - \lambda_i$ near one. On a sector $0 \leq x_1 \leq \dots \leq x_l \leq 1/4$, let $\epsilon_i \in \{1, -1\}$ record the endpoint and put

$$S_j = \sum_{i=1}^j \epsilon_i, \quad S_0 = 0, \quad y_j = x_j + \tau.$$

The individual factor is at most $Cy_j^{-1/2}$. For $i < j$, a same-endpoint pair factor is at most Cy_j , whereas an opposite-endpoint factor is at most C/y_j . The sector integrand is consequently bounded by

$$(53.17) \quad C_d \prod_{j=1}^l y_j^{a_j-1}, \quad a_j = \frac{1}{2} + \epsilon_j S_{j-1} = \frac{S_j^2 - S_{j-1}^2}{2}.$$

Its cumulative exponents satisfy

$$(53.18) \quad A_j = \sum_{i=1}^j a_i = \frac{S_j^2}{2} \geq 0.$$

Set $T = 1/4 + \tau$ and $L = \log(T/\tau)$. Substitute $y_j = T \exp(-z_j)$, where $z_1 \geq \dots \geq z_l \geq 0$, and then $u_j = z_j - z_{j+1}$ for $j < l$, $u_l = z_l$. The sector integral is bounded by

$$(53.19) \quad C_d \int_{u_j \geq 0, \sum_j u_j \leq L} \exp\left(-\sum_j A_j u_j\right) du.$$

The omitted factor T^{A_l} is bounded above uniformly in $0 < \tau \leq 1$. Each $A_j > 0$ is at least $1/2$, so its variable has an integrable exponential tail. Each vanishing A_j contributes at most one factor $1 + L$. A vanishing prefix sum requires an even j , and there are at most $[l/2]$ such indices. Summing the finitely many sectors proves the upper estimate in (53.16). The case $l = 0$ contributes only a bounded constant.

For the lower bound let $m = [d/2]$. Choose scales

$$(53.20) \quad \tau \leq t_1, \quad 8t_j \leq t_{j+1} \quad (j < m), \quad t_m \leq 1/32,$$

and restrict $2m$ eigenvalues in pairs by

$$(53.21) \quad \lambda_j^- = a_j \in [t_j, 2t_j], \quad \lambda_j^+ = 1 - b_j, \quad b_j \in [t_j, 2t_j], \quad 1 \leq j \leq m.$$

If d is odd, restrict the remaining eigenvalue to $[3/8, 5/8]$; all its factors are bounded above and below by positive constants. For one pair the two individual spectral factors and its opposite-endpoint factor have product comparable to t_j^{-2} . For two pairs at scales $t_i \leq t_j/8$, the two same-endpoint factors are comparable to t_j and the two opposite-endpoint factors to t_j^{-1} . Their product is therefore bounded above and below by fixed constants. Thus the integrand on the box is comparable to $\prod_j t_j^{-2}$ and its volume to $\prod_j t_j^2$. Each box contributes at least $c_d > 0$.

Take the scales from $\tau, 8\tau, 8^2\tau, \dots$ up to $1/32$. The boxes for distinct increasing m -tuples are disjoint. Their number is a binomial coefficient bounded below by $c_d[\log(\tau^{-1})]^m$ once $\log(\tau^{-1})$ is sufficiently large depending on d . In the remaining bounded range, a fixed box of distinct eigenvalues strictly inside $(0, 1)$ has a positive uniform integral and supplies the lower estimate. For $m = 0$ this interior argument suffices on the full range. This proves (53.16) and, by (53.14), the proposition. $\square$

The balanced-prefix identity (53.18) accounts for all nested endpoint scales. A simultaneous approach of all eigenvalues to the endpoints detects only one possible critical scaling; the separated pairs in (53.21) show why the logarithmic multiplicity can be larger.

53.3. Covering, rational centres, and description length.

Proof of Theorem 53.1. Choose δ_d smaller than the radius in Lemma 53.2. A cover by L balls of radius δ satisfies

$$\mu_N(\mathfrak{E}_d) \leq LA_d \delta^{d^2},$$

which gives the required lower bound. Conversely, take a maximal δ -separated subset of $\mathfrak{E}_d$. Its balls of radius $\delta/3$ are pairwise disjoint, and the lower bound in (53.5) bounds its cardinality by

$$C_d \mu_N(\mathfrak{E}_d) \delta^{-d^2}.$$

It is a radius- δ cover by maximality. Such finite maximal sets exist: the effect body is compact, d_N is continuous by the hybrid bound, and $d_1(E, F) = 2\|E - F\|_{\text{op}}$ separates distinct effects. Proposition 53.3 now gives both estimates for legal centres.

Rational legal effects are dense. Indeed, first move towards $I/2$ by (53.11), and then approximate all independent real matrix coordinates by rational numbers while remaining strictly between 0 and I . For fixed N , continuity in the operational distance follows again from the hybrid bound. Replace every centre of a radius- $\delta/2$ cover by a rational legal effect at distance less than $\delta/2$. The resulting radius- δ cover has the same cardinality. This proves the rational-centre upper bound.

A fixed-length payload indexes the centres of a cover, and conversely each legal reusable program with a fixed payload supplies a centre. Taking base-two logarithms of (53.1) and rounding the index length gives (53.2). $\square$

54. LEARNING THE COMPLETE BINARY QUBIT EFFECT BODY

We now pass from supplied descriptions to an unknown device. Throughout this section the input dimension is two. Write

$$E = \frac{1}{2}((1+b)I + ru \cdot \sigma), \quad 0 \leq r \leq 1, \quad |b| + r \leq 1,$$

with the direction absent when $r = 0$. The bias, contrast, and direction are all unknown. The outputs are ordered and classical, and each call consumes its input without a residual quantum output. A learning procedure may prepare entangled input blocks, retain the unused input qubits, and use classical feedback between blocks. It has no access to a measurement environment. Its choices depend only on the public parameters and the observed record.

The integer N below is the horizon of a future comparison, measured by d_N . The number M of training calls is a separate resource. The output is a classical description of a legal effect. For a unit vector w set $P_w = (I + w \cdot \sigma)/2$ and write $\mathcal{M}_{p,q,w}$ for the measurement associated with $pP_w + q(I - P_w)$, where $p = (1 + b + r)/2$ and $q = (1 + b - r)/2$. We use the spectral quantities

$$V = p(1 - p) + q(1 - q), \quad K_N(p, q) = r\sqrt{N/(V + N^{-1})}.$$

Theorem 54.1 (Common learning at the finite-use scale). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common learning procedure returns an effect $\hat{E} \in \mathfrak{E}_2$ with*

$$(54.1) \quad \sup_{E \in \mathfrak{E}_2} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record it uses at most

$$(54.2) \quad M \leq CN\delta^{-2} \log(1/\eta)$$

calls, and every entangled input block has length at most N . Conversely, every common procedure satisfying (54.1), with at most M calls on every record, obeys

$$(54.3) \quad M \geq cN\delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. The constants are uniform over the closed effect body, including scalar, deterministic, and projective effects.

The budget in (54.2) is a deterministic upper bound for each completed record, including unsuccessful records. It counts calls during training; N specifies only the subsequent loss. The construction uses independent input blocks, retaining at most $N - 1$ unused input qubits after the next qubit has been supplied, and no quantum system between blocks. Its preparation model permits ideal trusted finite-state preparation and fresh fair random signs. Preparation cost, classical running time, and classical workspace are separate resources. The reusable payload is charged separately in Corollary 54.6. The proof below specifies a written statistical algorithm; the accompanying finite identity checks do not constitute a software implementation of its risk guarantee.

The proof constructs one decision tree for the entire parameter body. Its experiments are not selected from a known pair of hypotheses. The angular part uses multiscale phase estimation, a method with antecedents in [71, 72]. We give the phase and confidence bounds explicitly. A data-dependent amplitude threshold selects the block length without prior calibration of the spectrum.

54.1. Bernoulli errors and estimated axes. For $s, t \in [0, 1]$ put

$$(54.4) \quad \mathfrak{h}(s, t)^2 = (\sqrt{s} - \sqrt{t})^2 + (\sqrt{1-s} - \sqrt{1-t})^2.$$

Write $B_N(s, t)$ for the unhalved distance between the two N -fold Bernoulli product laws. We use the following elementary estimates at both support endpoints.

Lemma 54.2 (Endpoint estimation). *For every $s, t \in [0, 1]$,*

$$(54.5) \quad B_N(s, t) \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For each fixed $A > 0$ there is C_A such that, if $0 < \lambda \leq 1$ and

$$(54.6) \quad |s - t| \leq A(\sqrt{t(1-t)\lambda} + \lambda),$$

then $\mathfrak{h}(s, t) \leq C_A\sqrt{\lambda}$. If two estimates of an ordered pair (p, q) each have $\mathfrak{h}$ error at most ε , sorting the estimates preserves this bound up to an absolute constant.

Proof. The root fidelity is $1 - \mathfrak{h}^2/2$. Tensorization and the trace-fidelity inequality give

$$B_N(s, t) \leq 2\sqrt{1 - (1 - \mathfrak{h}(s, t)^2/2)^{2N}} \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For the first square root in (54.4), if $t \geq \lambda$ divide the difference by $\sqrt{t}$ in (54.6). If $t < \lambda$, both t and s are at most $C_A \lambda$. Apply the same argument to their complements. For sorting, set $\phi(t) = \arcsin \sqrt{t}$. Then

$$\mathfrak{h}(s, t) = 2 \sin(|\phi(s) - \phi(t)|/2).$$

The two distances are uniformly comparable on $[0, \pi/2]$. Sorting is nonexpansive for the sup norm against an already ordered pair of real numbers. Apply this fact in the ϕ coordinate. $\square$

If $\hat{t}$ is the empirical frequency in n Bernoulli trials, Bernstein's inequality gives, with failure at most $2e^{-L}$,

$$(54.7) \quad |\hat{t} - t| \leq \sqrt{2t(1-t)L/n} + \frac{2L}{3n}.$$

Thus Lemma 54.2 applies with $\lambda = L/n$, including at $t = 0, 1$.

For a unit estimate w of u , write

$$(54.8) \quad h = \angle(u, w), \quad \gamma = \frac{r}{2}(1 - u \cdot w).$$

The input states on the axes $+w$ and $-w$ have respective success probabilities

$$(54.9) \quad s_+ = p - \gamma, \quad s_- = q + \gamma, \quad p = \frac{1}{2}(1 + b + r), \quad q = \frac{1}{2}(1 + b - r).$$

If $\hat{x}$ estimates $x = ru$ with Euclidean error at most e , normalizing $\hat{x}$ gives

$$(54.10) \quad rh \leq \pi \min(r, e), \quad \gamma \leq \min(r, e^2/r) \quad (r > 0).$$

Indeed $\|u - w\| \leq 2e/r$, $h \leq (\pi/2)\|u - w\|$, and $\gamma = r\|u - w\|^2/4$. If $\hat{x} = 0$, choose a fixed direction; then $e \geq r$ gives the same bounds. At $r = 0$ both quantities vanish.

Lemma 54.3 (Product learning away from unit contrast). *Suppose $r \leq 7/8$. There is a product-input procedure, not requiring b, r, u as input, which uses eight groups of n calls and returns a legal effect F such that, for $1 \leq L \leq n$,*

$$(54.11) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq C(\sqrt{NL/n} + NL/n)$$

outside an event of probability at most Ce^{-L} .

Proof. First use the positive and negative eigenstates of each of the three Pauli axes, with n calls per state. Encode an observed outcome as $Z \in \{-1, 1\}$. The two means on axis j are $b + x_j$ and $b - x_j$. Estimate x_j by half their empirical difference. Put

$$w_0 = 1 - b^2, \quad \lambda = L/n.$$

The two variances on axis j sum to $2(w_0 - x_j^2)$. Bernstein's inequality and a union bound therefore give

$$(54.12) \quad e = \|\hat{x} - x\| \leq 3(\sqrt{w_0 \lambda} + \lambda).$$

More explicitly, the coordinate error is at most $\sqrt{2w_0 \lambda} + 4\lambda/3$; the sign observation has range length two. Taking the Euclidean norm gives (54.12), since $\sqrt{6} < 3$ and $4\sqrt{3}/3 < 3$.

Normalize to a direction w . Feasibility and $r \leq 7/8$ imply

$$(54.13) \quad w_0 \geq r(2 - r), \quad V = \frac{1}{2}(w_0 - r^2) \geq w_0/9.$$

Use (54.10) and Theorem 50.4. The square-root term in (54.12) contributes at most $C\sqrt{N\lambda}$ to $K_N(p, q)h$, and its additive term contributes at most $CN\lambda$. Consequently,

$$(54.14) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w}) \leq C(\sqrt{N\lambda} + N\lambda).$$

This calculation does not require a lower bound on r or w_0 .

It remains to control both endpoint probabilities in (54.9). For either $t = p$ or $t = q$ put $v_t = t(1 - t)$. Since

$$|p(1 - p) - q(1 - q)| = r|b| \leq r, \quad w_0 = 2V + r^2 \leq 4v_t + 3r,$$

we claim

$$(54.15) \quad \gamma \leq 9\sqrt{v_t\lambda} + 72\lambda.$$

Equation (54.12) gives $e^2 \leq 18(w_0\lambda + \lambda^2)$. If $r \leq \lambda$, use $\gamma \leq r$. If $r > \lambda$, use

$$\gamma \leq 72v_t\lambda/r + 72\lambda, \quad \gamma \leq r,$$

and $\min(r, 72v_t\lambda/r) \leq \sqrt{72v_t\lambda}$. This proves (54.15).

Make two fresh groups of n calls, along $+w$ and $-w$. Conditional on the previous record, these are Bernoulli trials with the parameters in (54.9). Their variances differ from v_t by at most γ . Equations (54.7) and (54.15) imply

$$|\hat{t}_{\text{raw}} - t| \leq C(\sqrt{v_t\lambda} + \lambda).$$

For the additional term use $\sqrt{\gamma\lambda} \leq C(\sqrt{v_t\lambda} + \lambda)$, which follows from $\sqrt{AB} \leq (A + B)/2$ with $A = \sqrt{v_t\lambda}$ and $B = \lambda$. Lemma 54.2 gives root-distance error $C\sqrt{\lambda}$ for both endpoints. Sort the two empirical parameters to obtain $\hat{p} \geq \hat{q}$ and return $F = \hat{p}P_w + \hat{q}(I - P_w)$. The aligned programme bound (50.9) and (54.5) give a spectral contribution at most $C\sqrt{N\lambda}$. Combine with (54.14). All concentration bounds for the fresh groups hold conditionally, so their failure probabilities may be added to those of the first six groups. $\square$

54.2. A common entangled experiment with unknown bias.

Lemma 54.4 (Two-plane phase observation). *Fix an orthonormal frame (e_1, e_2, w) known to the experimenter. For each $i = 1, 2$ and integer $m \geq 1$, there is an m -call experiment with an observation $Y \in \{-1, 1\}$ whose mean, for a chosen phase φ , is*

$$(54.16) \quad \mathbb{E}_E Y = \Re(e^{i\varphi} Z_i), \quad Z_i = [x \cdot w + ix \cdot e_i]^m, \quad x = ru.$$

Two phases estimate the complex number Z_i . For $0 < \zeta \leq 1$ and $0 < \epsilon \leq 1/2$, four settings, using the two planes, estimate both complex numbers with error at most ζ and failure at most ϵ , with at most $C\zeta^{-2} \log(C/\epsilon)$ independent blocks per setting.

Fix $a = 1/64$. If $r > 0$ and $h = \angle(u, w) \leq 2a/m$, then

$$(54.17) \quad .99r^m \leq |Z_i| \leq r^m, \quad \text{Arg } Z_i = m \arctan \frac{u \cdot e_i}{u \cdot w},$$

where the principal argument is used. If the amplitudes are bounded below by a fixed positive constant and ζ is sufficiently small, these observations produce a direction estimate w' with

$$(54.18) \quad \angle(u, w') \leq C\zeta/m.$$

Proof. Choose a computational basis along the axis perpendicular to the plane spanned by w, e_i , with phase convention such that the off-diagonal matrix element of $O = 2E - I = bI + ru \cdot \sigma$ is $r(u \cdot w + iu \cdot e_i)$. Prepare

$$|G_{\varphi, s}\rangle = \frac{|0\rangle^{\otimes m} + se^{i\varphi}|1\rangle^{\otimes m}}{\sqrt{2}},$$

where s is a fresh fair sign for this block. Conditional on the entire preceding record, the two signs have probability $1/2$ each; the random draw is independent of all earlier preparation randomness. Pass its qubits through the measurement and record $Y = s \prod_{j=1}^m Z_j$, where each output is encoded as $Z_j \in \{-1, 1\}$. Memorylessness makes the joint outcome law the product POVM on the entangled input; the qubits may be supplied sequentially while the unused ones are retained.

The expansion of $\langle G_{\varphi,s} | O^{\otimes m} | G_{\varphi,s} \rangle$ has diagonal term

$$\frac{1}{2}((b + ru \cdot f_i)^m + (b - ru \cdot f_i)^m),$$

where f_i is the chosen perpendicular axis. It is independent of s and disappears after multiplication by s and averaging. The off-diagonal terms give (54.16). At $x = 0$ the mean is zero and no direction is needed. In particular the experiment removes the unknown bias exactly. The phases $0, \pi/2$ give the real part and the negative imaginary part. More precisely, at each of the four settings use

$$(54.19) \quad B(\zeta, \epsilon) = \left\lceil 4\zeta^{-2} \log(8/\epsilon) \right\rceil$$

fresh independent blocks. Hoeffding's inequality bounds the error of each real sample mean by $\zeta/\sqrt{2}$ outside an event of probability $2\exp(-B\zeta^2/4)$. The union bound over the four means is at most ϵ , and each complex error is at most ζ . This bound holds conditional on every preceding record: the frame and the setting are then fixed, and the memoryless device acts on fresh blocks. Each estimate batch uses its prescribed number of blocks without stopping within the batch.

For the phase conclusions assume $r > 0$ and put $\theta_i = \arctan((u \cdot e_i)/(u \cdot w))$. Then $|m\theta_i| \leq 2a$ and

$$|Z_i| = r^m (1 - (u \cdot f_i)^2)^{m/2} \geq r^m \cos(h)^m \geq r^m (1 - 2a^2/m).$$

This proves (54.17), with no phase ambiguity. For the estimates set

$$\hat{\theta}_i = \text{Arg}(\hat{Z}_i)/m, \quad w' = \frac{w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2}{\|w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2\|}.$$

The exact formula with θ_i recovers u . For $|\hat{Z}_i - Z_i| \leq |Z_i|/2$,

$$|\text{Arg}(\hat{Z}_i) - \text{Arg}(Z_i)| \leq 2|\hat{Z}_i - Z_i|/|Z_i|.$$

On $|\theta_i| \leq 1/4$ the displayed map from two angles to a unit vector is Lipschitz; a bound of 16 for angular distance against the largest coordinate error follows from differentiation of $\tan$ and normalization. This proves (54.18). Our fixed cone lies strictly inside the principal-argument interval, and the argument uses no unwrapping threshold from a separate phase-estimation theorem. $\square$

We make the reconstruction rule total. Fix a coordinate basis of $\mathbb{R}^3$ and a tie-breaking order. Given a unit vector w , choose the first coordinate vector minimizing its absolute inner product with w , normalize its projection onto $w^\perp$ to obtain e_1 , and put $e_2 = w \times e_1$. The projection has norm at least $\sqrt{2/3}$, so the frame is defined at every w . For complex estimates z_1, z_2 use the phase formula in the preceding proof only when both pass the guard

$$(54.20) \quad \Re z_i > 0, \quad 8|\Im z_i| \leq \Re z_i \quad (i = 1, 2).$$

Write $\mathcal{R}_m(w; z_1, z_2) = (\text{ok}, w')$ for the resulting update, and set $\mathcal{R}_m = (\text{fail}, w)$ otherwise. A passed guard gives $|\text{Arg } z_i| \leq \arctan(1/8) < 1/4$; the vector to be normalized has inner product one with w and cannot vanish. Zero complex estimates, the branch cut, and an out-of-cone phase therefore have prescribed outcomes before an argument or tangent is evaluated.

Lemma 54.5 (Selecting a block length from the observations). *Suppose $r \geq 5/8$ and an initial known direction w satisfies $\angle(u, w) \leq a = 1/64$. A common adaptive procedure selects a length m_* and a direction w_* , for $N \geq 1$ and $0 < \eta \leq 1/8$, using at most $CN \log(C/\eta)$ calls on every record. With failure at most $\eta/16$,*

$$(54.21) \quad \frac{1}{8} \min\{N, (1-r)^{-1}\} \leq m_* \leq N,$$

$$(54.22) \quad r^{m_*} \geq 1/5, \quad \angle(u, w_*) \leq a/m_*.$$

The reciprocal at $r = 1$ is interpreted as infinity. New observations at m_*, w_* have both complex amplitudes at least $1/6$.

Proof. Fix $\kappa = a/1024$, and let H be the largest power of two not exceeding N . At a tested length m , estimate both complex numbers in Lemma 54.4 to error at most κ with $B(\kappa, \epsilon_m)$ blocks per setting, where

$$(54.23) \quad \epsilon_m = \frac{\eta m}{32N}.$$

The following procedure stops at the first rejection; it never tests a larger length after that rejection. Here w_0 is the input direction and $\perp$ denotes an empty register.

SELECTBLOCK(N, η, w_0)

1. Set $m \leftarrow 1$, $w \leftarrow w_0$, and $A \leftarrow \perp$.
2. Obtain z_1, z_2 from the four fresh settings at length m and frame w , with budgets (54.23).
3. If $\min_i |z_i| < 1/4$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
4. Set $(s, v) \leftarrow \mathcal{R}_m(w; z_1, z_2)$. If $s = \text{fail}$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
5. Set $A \leftarrow (m, v)$. If $m = H$, return A .
6. Set $w \leftarrow v$, $m \leftarrow 2m$, and return to step 2.

Equality in the amplitude threshold is accepted, and equality in the second phase guard is allowed. All registers, decisions, and fallbacks are functions of the observed record.

We prove the assertions on the event that the successive complex estimates obey their bounds. Before a length m is tried the angular error is at most $2a/m$: this holds initially, and is preserved by each accepted update. If the amplitude test passes, each true amplitude $\rho = |Z_i|$ is at least $1/4 - \kappa > 1/5$. Since $|\text{Arg } Z_i| \leq 2a$,

$$\Re z_i \geq \rho(1 - 2a^2) - \kappa > 8(2a\rho + \kappa) \geq 8|\Im z_i|.$$

The strict inequality follows directly from $a = 1/64$, $\kappa = a/1024$, and $\rho \geq 1/4 - \kappa$. Thus the guard cannot reject such a stage on this event. The argument bound and the Lipschitz constant in Lemma 54.4 give an updated angular error at most

$$\frac{32\kappa}{(1/4 - \kappa)m} \leq \frac{a}{4m}.$$

Furthermore $r^m \geq 1/4 - \kappa$, whence $m(1 - r) \leq 2$.

On rejection, the amplitude test must have failed, and (54.17) gives $r^m < (1/4 + \kappa)/.99 < 1/3$. Since $r \geq 5/8$,

$$-\log r \leq (1 - r)/r \leq 2(1 - r), \quad m(1 - r) \geq (\log 3)/2.$$

The procedure returns the preceding accepted length $m/2$, which is therefore at least $(\log 3)/(4(1 - r))$. Length one is always accepted on the good event, since its true amplitudes exceed $.99(5/8)$. If the cap is reached, the selected length is $H \geq N/2$. These facts prove (54.21) and (54.22). At the updated direction the amplitude is at least $r^{m_*} \cos(a/m_*)^{m_*} > 1/6$.

For completeness, let $\mathcal{F}_m$ contain the record immediately before a possible stage m , let $T_m \in \mathcal{F}_m$ be the event that this stage is reached, and let G_m be its complex-error event. On every reached record the concentration bound gives $\mathbb{P}_E(G_m^c \mid \mathcal{F}_m) \leq \epsilon_m$. Consequently

$$(54.24) \quad \mathbb{P}_E\left(\bigcup_m (T_m \cap G_m^c)\right) \leq \sum_m \mathbb{E}_E[\mathbf{1}_{T_m} \mathbb{P}_E(G_m^c \mid \mathcal{F}_m)] \leq \sum_{m=1,2,4,\dots,H} \epsilon_m \leq \eta/16.$$

The sample-mean bound holds for every preceding record, even one whose frame has left the cone; the cone is needed only to turn those sample means into directional information. Independence

between stage-failure events is not assumed. By Lemma 54.4, the number of calls is bounded by an absolute constant times

$$(54.25) \quad \sum_{m=1,2,4,\dots,H} m \log \frac{CN}{\eta m} \leq C' N \log(C'/\eta).$$

To see this, write $m = H/2^k$ and use the convergent sums of 2^{-k} and $k2^{-k}$, together with $1 \leq N/H < 2$. The public cap and these budgets bound the cost on every record, including records on which an estimate fails. Rejection can only shorten the sum. $\square$

54.3. The learning bound and its converse. All statistical batches use fresh inputs and fresh randomization. Table 2 allocates their error probabilities. Each conditional bound is uniform over the preceding record. A row belonging to a branch not taken contributes no failure event.

TABLE 2. Confidence allocation for one execution of the learner.

Batch	Failure allowance	Conditioning and scope
Coarse Pauli vector	$\eta/8$	Unconditional; all six input groups jointly.
Product-branch Pauli vector	$\eta/8$	Given the coarse record; six fresh input groups jointly.
Dyadic stage m	$\eta m/(32N)$	Given the record before that stage; all reached stages together have allowance at most $\eta/16$.
Fine two-plane estimate	$\eta/8$	Given the complete search record, including the selected m_*, w_* .
Fresh endpoint groups	$\eta/16$ each	Given the final estimated axis; the two allowances sum to $\eta/8$, in either branch.

The conservative sum of all rows is $9\eta/16 < \eta$. The conditional union calculation (54.24) applies to every row, so this sum requires no independence between the events attached to different batches.

Proof of Theorem 54.1. We choose constants in the following order. Fix the local cone $a = 1/64$ and $\kappa = a/1024$ as above. Choose a sufficiently small absolute constant $c_* > 0$ in the final phase tolerance below. Then choose the absolute multipliers in the Bernoulli sample counts sufficiently large. Finally reduce δ_0 , if needed, so that $\delta_0 \leq 1/4096$ and $c_*\delta_0 \leq a/100$. No optimized value of these constants is used. Set

$$(54.26) \quad L = \log(C_2/\eta), \quad n_0 = \lceil C_0 L \rceil, \quad n = \lceil C_1 N \delta^{-2} L \rceil,$$

where C_0, C_1, C_2 are absolute constants large enough for the concentration bounds and loss estimates below. The common procedure is the following.

- (1) Make the six coarse Pauli experiments, with n_0 calls per input, and form their half-difference vector $\hat{x}_0$.
- (2) If $\|\hat{x}_0\| < 3/4$, make six fresh Pauli groups of n calls per input and form $\hat{x}$. Set $w_f = \hat{x}/\|\hat{x}\|$, using the fixed third coordinate direction if $\hat{x} = 0$, and proceed to step 4.
- (3) Otherwise set $w_0 = \hat{x}_0/\|\hat{x}_0\|$ and run $\text{SELECTBLOCK}(N, \eta, w_0)$. At its returned (m_*, w_*) , obtain two fresh complex estimates with tolerance $\zeta_* = c_*\delta\sqrt{m_*/N}$ and failure allowance $\eta/8$. Apply $\mathcal{R}_{m_*}$ to them; take its updated direction as w_f on success and retain $w_f = w_*$ on failure.
- (4) Make n fresh calls along each of $+w_f$ and $-w_f$. If the empirical success frequencies are s_+, s_- , return

$$\hat{E} = \max(s_+, s_-)P_{w_f} + \min(s_+, s_-)(I - P_{w_f}).$$

The empirical frequencies already lie in $[0, 1]$. Ties in the coarse comparison use step 3; ties between endpoint frequencies give a scalar effect. Together with the frame, phase, and first-rejection conventions above, these rules define an output for every record, with no retry or additional calibration step.

We verify the risk. With failure at most $\eta/8$, the coarse half-difference estimate satisfies

$$\|\hat{x}_0 - x\| \leq \varepsilon_0, \quad \varepsilon_0 \leq a/100.$$

On this event the two decisions have the explicit overlap

$$\begin{aligned} \|\hat{x}_0\| < 3/4 &\implies r < 3/4 + 1/6400 < 7/8, \\ \|\hat{x}_0\| \geq 3/4 &\implies r \geq 3/4 - 1/6400 > 5/8. \end{aligned}$$

For the product branch, Lemma 54.3 gives error at most $\delta/2$ with the sample sizes in (54.26), after increasing C_1 . Its first six groups have joint allowance $\eta/8$ conditional on the coarse record. After their axis is fixed, each endpoint group has allowance $\eta/16$. These are the product-branch and endpoint rows of Table 2, and all samples here are fresh relative to the coarse test.

For the entangled branch, the normalized initial estimate has angular error at most $\pi\varepsilon_0/r \leq 8\pi a/500 < a$ by (54.10). Apply Lemma 54.5. At the selected length and updated direction, make fresh two-plane observations with complex error tolerance

$$(54.27) \quad \zeta_* = c_*\delta\sqrt{m_*/N},$$

and failure allowance $\eta/8$, conditional on the full selection record. On the preceding good event, the two true amplitudes are at least $1/6$ and their phases have absolute value at most a . On the current complex-error event, $\zeta_* \leq a/100$ and the inequalities $|\sin t| \leq |t|$ and $\cos t \geq 1 - t^2/2$ show that both phase guards pass. The resulting direction w_f satisfies

$$(54.28) \quad h_f = \angle(u, w_f) \leq Cc_*\delta/\sqrt{Nm_*}.$$

The calls used in this step are at most

$$(54.29) \quad Cm_*\zeta_*^{-2} \log(C/\eta) \leq C'N\delta^{-2} \log(C/\eta).$$

Put $\nu = 1 - r$. Since $|b| \leq \nu$ and $r \geq 5/8$,

$$r\nu \leq V = \frac{1}{2}(1 - r^2 - b^2) \leq \nu.$$

Equation (54.21) implies

$$(54.30) \quad m_*(V + N^{-1}) \geq c_0 > 0$$

for an absolute c_0 . Thus (54.28) and Theorem 50.4 give

$$(54.31) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w_f}) \leq Cc_*\delta.$$

The endpoint misalignment is also bounded:

$$(54.32) \quad \gamma = \frac{r}{2}(1 - u \cdot w_f) \leq rh_f^2/4 \leq Cc_*^2\delta^2/N.$$

The final fresh groups have the public size n in (54.26). Conditional on the record fixing w_f , their success probabilities are exactly $p - \gamma$ and $q + \gamma$, and each group obeys Bernstein's bound with allowance $\eta/16$. Conditional Bernstein bounds, Lemma 54.2, and $\mathfrak{h}(s, t)^2 \leq 2|s - t|$ give

$$(54.33) \quad \begin{aligned} \mathfrak{h}(\hat{p}_{\text{raw}}, p), \mathfrak{h}(\hat{q}_{\text{raw}}, q) &\leq C\sqrt{\log(C/\eta)/n} + C\sqrt{\gamma} \\ &\leq c'\delta/\sqrt{N}. \end{aligned}$$

The constant c' can be made as small as required by the preceding choices of c_* and the sample multiplier. Sort the estimates and return $\hat{E} = \hat{p}P_{w_f} + \hat{q}(I - P_{w_f})$. Equations (50.9) and (54.5)

make the spectral loss at most $\delta/4$; choose c_* so that (54.31) contributes at most $\delta/4$. Table 2 now gives the stated risk.

Finally we check the budget for every record. The product branch uses exactly $6n_0 + 8n$ calls. The entangled branch uses at most

$$(54.34) \quad 6n_0 + 4 \sum_{m=1,2,4,\dots,H} mB(\kappa, \epsilon_m) + 4m_*B(\zeta_*, \eta/8) + 2n.$$

This is a pathwise upper bound: the sum includes even the stages not reached after the first rejection. Its dyadic term is $O(N \log(C/\eta))$ by (54.25), including the ceilings in (54.19). For every returned $1 \leq m_* \leq N$, even on a failed record,

$$4m_*B(\zeta_*, \eta/8) \leq 4N + 16c_*^{-2}N\delta^{-2} \log(64/\eta).$$

Here $m_*\zeta_*^{-2} = c_*^{-2}N\delta^{-2}$ exactly. Since $N \geq 1$, $\delta \leq \delta_0$, and $\eta \leq 1/8$, both branches satisfy (54.2). The prescribed fallbacks always leave a unit direction and a legal effect, and every block has at most $H \leq N$ inputs. This proves the upper bound with the same worst-record convention as the converse.

For the converse consider the scalar effects

$$(54.35) \quad E_0 = \tfrac{1}{2}I, \quad E_1 = (\tfrac{1}{2} + \Delta)I, \quad \Delta = 256\delta/\sqrt{N}.$$

Since $\delta \leq 1/4096$, both parameters lie in $[1/2, 9/16]$. The finite Bernoulli lower bound in Lemma 50.2 yields

$$(54.36) \quad \begin{aligned} d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_1}) &= B_N(1/2, 1/2 + \Delta) \\ &\geq \tfrac{1}{64} \min\{1, \Delta\sqrt{N/(5/4)}\} > 2\delta. \end{aligned}$$

For these devices every call gives a fresh Bernoulli bit independent of the input. On an entangled input, each conditional reference block is the same reduced state multiplied by the scalar outcome probability. Thus any adaptive learning record, including its final classical estimate, is a common stochastic processing of M independent Bernoulli bits and parameter-independent resources. Publicly stopped records may be padded. The relative entropy before processing is at most

$$(54.37) \quad MD(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) = -\tfrac{M}{2} \log(1 - 4\Delta^2) \leq 3M\Delta^2.$$

The two radius- δ success balls are disjoint by (54.36). Testing whether the estimated effect belongs to the first ball therefore gives two errors at most η . For classical laws P, Q , Jensen's inequality gives root fidelity at least $e^{-D(P\parallel Q)/2}$. The trace-fidelity inequality then implies that the sum of testing errors is at least $\frac{1}{2}e^{-D(P\parallel Q)}$. Consequently

$$2\eta \geq \tfrac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this proves (54.3). □

Corollary 54.6 (A learned joint description). *For rational δ in the range of Theorem 54.1, the learner may return one word of the rational joint code, with the same risk and call-count bounds. Its payload length is*

$$(54.38) \quad 2\log_2 N + \log_2 \log(N + 2) + 4\log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The decoding centre is a legal rational effect. This is a statement about device calls and payload length; classical arithmetic, workspace, and state-preparation costs are separate resources.

Proof. Use the learner with loss budget $\delta/2$. Its output is a finite computable expression determined by a finite record. In particular, the empirical frequencies are rational, and the guarded phase

roots and normalized frames specify algebraic directions. Choose a positive rational $\varepsilon \leq \delta/(100N)$ and rational approximations (b_0, x_0) to its bias and Bloch vector with

$$|b_0 - \widehat{b}| + \|x_0 - \widehat{x}\| \leq \varepsilon.$$

Set $(b_1, x_1) = (b_0, x_0)/(1 + 2\varepsilon)$. Then

$$|b_1| + \|x_1\| \leq \frac{1 + \varepsilon}{1 + 2\varepsilon} \leq 1,$$

and the exact one-use formula (50.6) bounds the distance from the unrounded hypothesis by 3ε . The N -use hybrid inequality adds at most $3N\varepsilon$. Apply Theorem 51.2 to this rational estimated effect with public accuracy $\delta/2$. Its certified radius is $\delta/4$, so the total loss is less than δ . The capacity estimate in that theorem, at a constant rescaling of accuracy, gives (54.38). This postprocessing uses no additional device calls and receives no exact description of the unknown target. $\square$

The loss in Theorem 54.1 is a future adaptive distance. One-use measurement tomography is studied in [45, 69]; combining the stated one-use accuracy bounds only with the hybrid inequality gives a quadratic dependence on the future horizon. The argument here uses finite-use geometry and data-dependent entangled blocks to obtain the linear dependence in (54.2), with a matching scalar converse. Learning a stored programme from which a measurement is later retrieved, as in [70], has a different output resource. The present theorem gives a classical estimate over the entire ordered binary qubit body. Section 55 uses this common qubit procedure after a separate calibration to obtain the corresponding learning law on the full matrix effect body.

55. COMMON LEARNING IN ARBITRARY INPUT DIMENSION

The two-dimensional learner extends to the full effect body through a calibration step. The calibration must resolve the two support endpoints simultaneously. A bound in operator norm alone does not provide this information: after compression, a poorly aligned subspace can introduce noise at a scale larger than N^{-1} . The procedure below learns a basis in which this additional noise is bounded by the finite-horizon cutoff.

Throughout this section, $\mathcal{M}_E$ is the ordered binary measurement associated with $E \in \mathfrak{E}_d$. Training may use entangled input blocks and classical feedback, but it has no access to the measurement environment. Its output is a classical description of a legal effect. The future horizon N and the number of training calls remain separate resources.

Theorem 55.1 (Common learning on the matrix effect body). *There are absolute constants $c, C, \delta_0 > 0$ with the following property. For every $d, N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common procedure returns a legal effect $\widehat{E} \in \mathfrak{E}_d$ such that*

$$(55.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\widehat{E}}) > \delta\} \leq \eta.$$

On every record its number of calls satisfies

$$(55.2) \quad M \leq Cd^4 N \delta^{-2} \log(d/\eta),$$

and every entangled input block has length at most N . Conversely, any common procedure satisfying (55.1) with at most M calls on every record satisfies

$$(55.3) \quad M \geq cN \delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. In particular, for each fixed d , the minimax number of calls is $\Theta_d(N \delta^{-2} \log(1/\eta))$ on the complete closed effect body. The displayed dependence on d in the upper bound is not asserted to be optimal.

We first give the calibration, including its finite budgets. All matrix orders below are Loewner orders. The use of a basis selected from earlier observations is always followed by fresh samples.

Lemma 55.2 (Simultaneous endpoint calibration). *Let $\tau = N^{-1}$ and $0 < \epsilon \leq 1/8$. There is a common product-input procedure returning $A \in \mathfrak{E}_d$ with at most $Cd^4N \log(d/\epsilon)$ calls on every record. With probability at least $1 - \epsilon$, it satisfies*

$$(55.4) \quad \frac{1}{2}(A + \tau I) \preceq E + \tau I \preceq 2(A + \tau I),$$

$$(55.5) \quad \frac{1}{2}(I - A + \tau I) \preceq I - E + \tau I \preceq 2(I - A + \tau I),$$

$$(55.6) \quad \|E - A\|_{\text{op}} \leq \sqrt{\tau}.$$

The input states depend only on the public parameters and the preceding record; no spectral data about E are supplied.

Proof. Set $J = \lceil \log_2 N \rceil$, $H = 2^J$, and $t_j = 2^{-j}$ for $0 \leq j \leq J$. Thus $N \leq H < 2N$ unless $N = 1$, and $\tau/2 < t_J \leq \tau$. Start from $A_0 = I/2$. If $J = 0$, return A_0 : all three conclusions hold without samples.

We use

$$h_d = d + 4 \binom{d}{2} = 2d^2 - d, \quad K_d = (320d)^2, \quad \alpha = 1/16.$$

At stage $j \geq 1$, assign failure allowance and sample size

$$(55.7) \quad \epsilon_j = \frac{\epsilon 2^j}{4H}, \quad L_j = \log \frac{2h_d}{\epsilon_j}, \quad n_j = \left\lceil K_d t_j^{-1} L_j \right\rceil.$$

Choose an orthonormal eigenbasis $e_1, \dots, e_d$ of the current legal estimate A_{j-1} . The choice is deterministic and measurable, including at multiple eigenvalues. More explicitly, fix a public input basis. All calibration matrices have algebraic entries in that basis: the sample frequencies are rational, and spectral projections, clipping at rational endpoints, and normalization preserve algebraic entries. Order the distinct eigenvalues, project the public basis onto each eigenspace, and apply Gram–Schmidt in that fixed order, discarding exactly zero vectors. Exact algebraic arithmetic therefore supplies a finite choice at every stage and requires no oracle for arbitrary real spectral data. Make n_j independent calls on each of the h_d states

$$(55.8) \quad e_i, \quad \frac{e_i + e_j}{\sqrt{2}}, \quad \frac{e_i - e_j}{\sqrt{2}}, \quad \frac{e_i + ie_j}{\sqrt{2}}, \quad \frac{e_i - ie_j}{\sqrt{2}} \quad (i < j).$$

Here and below a unit vector denotes its pure input state. Estimate each diagonal entry of E by its empirical outcome frequency. Estimate the real part of each off-diagonal entry by half the frequency difference between the positive and negative real superpositions. Its imaginary part is half the frequency on $(e_i - ie_j)/\sqrt{2}$ minus half the frequency on $(e_i + ie_j)/\sqrt{2}$. These formulas define a Hermitian matrix B_j .

To define the procedure on every record, clip the eigenvalues of B_j to $[-\alpha t_j, 1 + \alpha t_j]$, obtaining $\tilde{B}_j$, and set

$$(55.9) \quad A_j = \frac{\tilde{B}_j + \alpha t_j I}{1 + 2\alpha t_j}.$$

The returned matrix is always a legal effect. In particular, a failed concentration event cannot affect the subsequent call budget or require an additional experiment.

We prove the invariant

$$(55.10) \quad \frac{1}{2}(A_j + t_j I) \preceq E + t_j I \preceq 2(A_j + t_j I), \quad \frac{1}{2}(I - A_j + t_j I) \preceq I - E + t_j I \preceq 2(I - A_j + t_j I)$$

on the successive good events. It holds for $j = 0$ because $0 \preceq E \preceq I$.

Fix a preceding good record and abbreviate $t = t_j$, $A = A_{j-1}$, $B = B_j$, $D = B - E$, and $q = L_j/n_j$. All expectations and probabilities in the next calculation are conditional on that record. Bernstein's inequality and a union bound over the h_d groups show, with failure at most $2h_d e^{-L_j} = \epsilon_j$, that every empirical frequency has error at most

$$\sqrt{2p(1-p)q} + 2q/3.$$

For a pair of opposite superposition states, the sum of the outcome probabilities is $E_{ii} + E_{jj}$. Cauchy–Schwarz in each half-difference therefore gives the following two simultaneous bounds, also valid for $i = j$ after enlarging the constants:

$$(55.11) \quad |D_{ij}| \leq \sqrt{2(E_{ii} + E_{jj})q} + 2q, \quad |D_{ij}| \leq \sqrt{2(2 - E_{ii} - E_{jj})q} + 2q.$$

For an off-diagonal complex entry, apply the real estimate to the two quadratures and take their Euclidean norm. The factor $\sqrt{2}$ in (55.11) includes this operation.

Write a_i for the eigenvalues of A . The preceding invariant, whose regularization is $2t$, gives

$$E_{ii} \leq 2a_i + 2t, \quad 1 - E_{ii} \leq 2(1 - a_i) + 2t, \quad q \leq t/K_d.$$

Since

$$t(a_i + a_j + 2t) \leq 2(a_i + t)(a_j + t), \quad t \leq \sqrt{(a_i + t)(a_j + t)},$$

the first bound in (55.11) yields

$$(55.12) \quad \frac{|D_{ij}|}{\sqrt{(a_i + t)(a_j + t)}} \leq \frac{2\sqrt{2}}{\sqrt{K_d}} + \frac{2}{K_d} \leq \frac{5}{\sqrt{K_d}}.$$

Replacing a_i by $1 - a_i$ gives the same estimate for the complement. The Hilbert–Schmidt norm bounds the operator norm, so our choice of K_d implies

$$(55.13) \quad \begin{aligned} -\frac{1}{64}(A + tI) &\preceq D \preceq \frac{1}{64}(A + tI), \\ -\frac{1}{64}(I - A + tI) &\preceq D \preceq \frac{1}{64}(I - A + tI). \end{aligned}$$

The preceding invariant and $t_{j-1} = 2t$ also imply

$$A + tI \preceq 4(E + tI), \quad I - A + tI \preceq 4(I - E + tI).$$

Consequently

$$(55.14) \quad -\alpha(E + tI) \preceq D \preceq \alpha(E + tI), \quad -\alpha(I - E + tI) \preceq D \preceq \alpha(I - E + tI).$$

These inequalities place the spectrum of B inside $[-\alpha t, 1 + \alpha t]$. Thus the clipping in (55.9) does nothing on this event. On it,

$$A_j - E = \frac{D + \alpha t(I - 2E)}{1 + 2\alpha t}.$$

Since $-I \preceq I - 2E \preceq I$, equations (55.14) imply

$$\begin{aligned} -2\alpha(E + tI) &\preceq A_j - E \preceq 2\alpha(E + tI), \\ -2\alpha(I - E + tI) &\preceq A_j - E \preceq 2\alpha(I - E + tI). \end{aligned}$$

Adding the corresponding regularized endpoint gives factors $1 - 2\alpha = 7/8$ and $1 + 2\alpha = 9/8$. This proves the weaker factor-two invariant (55.10) for stage j .

The unweighted entry estimate also gives the operator-norm bound needed later. Because $E_{ii} + E_{jj} \leq 2$, we have

$$\|D\|_{\text{op}} \leq 4d\sqrt{t/K_d} = \sqrt{t}/80.$$

It follows that

$$(55.15) \quad \|A_j - E\|_{\text{op}} \leq \sqrt{t}/80 + \alpha t \leq \sqrt{t}.$$

At the final stage, adding $(\tau - t_J)I$ preserves the factor-two orders, and (55.15) gives (55.6).

The conditional failure allowances sum to less than $\epsilon/2$, since $\sum_{j=1}^J 2^j < 2H$. This proves the asserted probability bound. Finally, the deterministic call count is $h_d \sum_j n_j$. Writing $2^j = H/2^k$ yields

$$\sum_{j=1}^J 2^j L_j \leq H \sum_{k=0}^{J-1} 2^{-k} \left(\log \frac{8h_d}{\epsilon} + k \log 2 \right) \leq CH \log(d/\epsilon).$$

The ceiling terms contribute at most $h_d J \leq h_d H$. As $h_d K_d \leq Cd^4$ and $H < 2N$ when $N > 1$, the stated bound holds on every record. In particular, the number of dyadic stages does not introduce a factor $\log N$. $\square$

Lemma 55.3 (Calibrated variance and compression). *Suppose $A, E \in \mathfrak{E}_d$ satisfy (55.4)–(55.6). Set $W_G = G(I - G) + \tau I/2$. Then*

$$(55.16) \quad \frac{1}{12} W_A \preceq W_E \preceq 12 W_A.$$

If P is any orthogonal projection commuting with A and $C = PEP|_{P\mathbb{C}^d}$, then, on $P\mathbb{C}^d$,

$$(55.17) \quad \frac{1}{12} P W_A P \preceq W_C \preceq 14 P W_A P.$$

All constants are independent of the spectrum, its multiplicities, and the rank of P .

Proof. For a legal effect G , define the harmonic mean without a factor of two by

$$R_G = ((G + \tau I)^{-1} + (I - G + \tau I)^{-1})^{-1}.$$

The two factors commute because both are functions of G , so

$$(55.18) \quad R_G = \frac{G(I - G) + (\tau + \tau^2)I}{1 + 2\tau}, \quad \frac{1}{3}(G(I - G) + \tau I) \preceq R_G \preceq G(I - G) + \tau I.$$

The inequalities use only $0 < \tau \leq 1$. Inversion reverses Loewner order. Apply it to both endpoint comparisons, add, and invert again to get $R_E \preceq R_A$ with factors two. Equation (55.18) therefore compares $E(I - E) + \tau I$ and $A(I - A) + \tau I$ with factors six. Replacing the regularization τI by $\tau I/2$ proves (55.16).

The compression identity is exact:

$$(55.19) \quad C(I - C) = PE(I - E)P + PE(I - P)EP.$$

Because P commutes with A , $PE(I - P) = P(E - A)(I - P)$. Thus

$$0 \preceq PE(I - P)EP \preceq \|E - A\|_{\text{op}}^2 P \preceq \tau P.$$

The first term in (55.19), after adding $\tau P/2$, is $P W_E P$. Apply (55.16) and use $P W_A P \succeq \tau P/2$ to absorb the final τP with a factor two. This gives (55.17). $\square$

Proof of Theorem 55.1. For $d = 1$, Bernoulli sampling and Lemma 54.2 give the upper bound directly. Assume $d \geq 2$ and put $s = \binom{d}{2}$. Apply Lemma 55.2 with failure allowance $\eta/2$. Conditional on its good event, fix an orthonormal eigenbasis of the returned A , with eigenvalues a_i , and write

$$w_i = a_i(1 - a_i) + \tau/2.$$

For each $i < j$, let P_{ij} project onto $\text{span}(e_i, e_j)$ and let C_{ij} be the corresponding compression of E . An input qubit, including its entanglement with other input qubits or a retained reference, can be embedded isometrically into this two-dimensional subspace. The resulting observed channel is exactly $\mathcal{M}_{C_{ij}}$.

Run the common qubit learner of Theorem 54.1 on each compression, using fresh records, accuracy

$$(55.20) \quad \epsilon = c_* \delta/d,$$

and failure allowance $\eta/(2s)$. The constant $c_* > 0$ will be chosen below. This returns legal 2×2 effects $\hat{C}_{ij}$. Every guarantee is conditional on the calibration record and is uniform over the resulting compression. A union bound therefore shows that, with total failure at most η , calibration succeeds and

$$(55.21) \quad d_N(\mathcal{M}_{C_{ij}}, \mathcal{M}_{\hat{C}_{ij}}) \leq \varepsilon \quad \text{for every } i < j.$$

We record explicitly how an operational error controls an endpoint form. Theorem 52.1 in dimension two and Lemma 52.5 imply that there are absolute $a_0, A_0 > 0$ such that, for legal $C, D \in \mathfrak{E}_2$,

$$(55.22) \quad d_N(\mathcal{M}_C, \mathcal{M}_D) \leq u \leq a_0 \implies g_{N,C}(D - C) \leq A_0 u.$$

Indeed, for $u < 1/16384$ the lower metric bound forces $Q_N(C, D) \leq 16384u$; decreasing a_0 makes this at most $1/4$, where the local comparison applies. Choose c_* small enough that $\varepsilon \leq a_0$ and lies within the qubit learner's accuracy range.

For a Hermitian matrix X on $\text{span}(e_i, e_j)$ put

$$\gamma_{ij}(X)^2 = N \left(\frac{|X_{ii}|^2}{2w_i} + \frac{|X_{jj}|^2}{2w_j} + \frac{2|X_{ij}|^2}{w_i + w_j} \right).$$

Equation (55.17), applied to left plus right multiplication and then inverted, gives

$$\gamma_{ij}(X) \leq \sqrt{14} g_{N,C_{ij}}(X).$$

Hence (55.21) and (55.22) imply

$$(55.23) \quad \gamma_{ij}(\hat{C}_{ij} - C_{ij}) \leq \sqrt{14} A_0 \varepsilon.$$

Assemble a Hermitian matrix B by taking each off-diagonal entry from its corresponding estimate $\hat{C}_{ij}$. For each diagonal entry choose once and for all one incident pair and use that pair's estimate. No consistency among different estimates of the same diagonal entry is assumed. In the diagonal coordinates of A , the form $g_{N,A}$ is the sum of the nonnegative coordinate terms displayed above. Each coordinate used in $B - E$ occurs in at least one of the pair forms, while unused duplicate diagonal terms can be discarded. Therefore

$$(55.24) \quad g_{N,A}(B - E)^2 \leq \sum_{i < j} \gamma_{ij}(\hat{C}_{ij} - C_{ij})^2 \leq 14sA_0^2\varepsilon^2.$$

The assembled matrix need not be legal. Define the estimate by minimizing its distance to the legal effect body in the known positive quadratic norm $g_{N,A}$. The minimum is attained and is unique. An exact minimizer F satisfies

$$g_{N,A}(F - E) \leq 2g_{N,A}(B - E),$$

because E is a feasible competitor. To obtain a finite procedure with a rational legal output, it suffices instead to choose $F \in \mathfrak{E}_d$ with rational entries such that

$$(55.25) \quad g_{N,A}(F - B) \leq \inf_{G \in \mathfrak{E}_d} g_{N,A}(G - B) + \varepsilon.$$

This changes the bound only to

$$(55.26) \quad g_{N,A}(F - E) \leq 2\sqrt{14s}A_0\varepsilon + \varepsilon \leq C_0d\varepsilon$$

for an absolute C_0 .

For completeness, (55.25) is an effective finite selection, not a query to the unknown effect. Since $W_A \succeq \tau I/2$, $g_{N,A}(X) \leq N\|X\|_{\text{HS}}$. A finite rational grid covering $\mathfrak{E}_d$ in Hilbert–Schmidt norm to $\varepsilon/(4N)$ therefore covers it in $g_{N,A}$ to $\varepsilon/4$. Such a legal grid is obtained by rounding after the inward move $G \mapsto (1 - 2r)G + rI$: entrywise rounding with Hilbert–Schmidt error at most $r/2$ preserves positivity of both outcomes, and the total rounding displacement is at most $C_d r$.

Concretely, choose positive rational $r \leq \varepsilon/[4N(\sqrt{d} + 1)]$ and a rational entry mesh at most $r/(2d)$. The inward displacement is at most $r\sqrt{d}$, and the rounding displacement is at most $r/2$, so their sum is at most $\varepsilon/(4N)$. The grid and its rational entries are defined in the fixed public input basis, independently of the random eigenbasis used for calibration. Enumerate this bounded grid, retain legal matrices using exact positivity tests, and evaluate its finitely many objective values to tolerance $\varepsilon/4$. Selecting the smallest computed value proves (55.25) with slack. The calibration matrices are algebraic, and the qubit subroutine's outputs are finite computable expressions in its record, so these evaluations require no oracle for the unknown effect. Only the observed matrices, the public parameters, and finite classical calculations are used. Arithmetic and state preparation are not included in the device-call bound.

Finally, (55.16) gives

$$g_{N,E}(F - E) \leq \sqrt{12} g_{N,A}(F - E) \leq \sqrt{12} C_0 d \varepsilon.$$

Choose the absolute c_* in (55.20) so that this is at most $\delta/(8\sqrt{2})$, and reduce δ_0 if necessary so that it is also at most $1/4$. The endpoint-to-midpoint comparison in Lemma 52.5 and the upper bound in Theorem 52.1 then give

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 8Q_N(E, F) \leq 8\sqrt{2} g_{N,E}(F - E) \leq \delta.$$

This proves the risk bound.

The calibration costs at most $Cd^4N \log(d/\eta)$ calls. There are $s \leq d^2/2$ qubit learners, each with accuracy $c_*\delta/d$ and confidence $\eta/(2s)$. Their combined call count is at most $Cd^4N\delta^{-2} \log(d/\eta)$ by Theorem 54.1. All budgets hold on every record, all compressed entangled blocks have length at most N , and postprocessing uses no additional calls.

The confidence and resource allocation can be recorded as follows; the constants in the call bounds are absolute.

Stage	Failure allowance	Calls on every record
Endpoint calibration	$\eta/2$	$Cd^4N \log(d/\eta)$
Each of $s = \binom{d}{2}$ pair learners	$\eta/(2s)$	$CN(d/\delta)^2 \log(d/\eta)$
Legal rational selection	0	0

The conditional guarantees of the pair learners allow their allowances to be added after calibration, despite the dependence of the selected basis on its record.

For the lower bound take

$$E_0 = \frac{1}{2}I_d, \quad E_1 = (\frac{1}{2} + \Delta)I_d, \quad \Delta = 256\delta/\sqrt{N}.$$

Choose $\delta_0 \leq 1/4096$. The Bernoulli separation in (54.36) is unchanged by the input dimension, and the two effects are separated by more than 2δ under d_N . Every use of either scalar device produces a fresh Bernoulli bit independent of the input. Its conditional reference state is the same reduced state multiplied by that scalar outcome probability. Thus an arbitrary adaptive training record with at most M calls, padded if necessary, is a common processing of M independent Bernoulli bits and parameter-independent resources. Its relative entropy is at most $3M\Delta^2$ by (54.37). The two disjoint success balls turn the estimated effect into a test with both errors at most η . The same fidelity inequality used in the proof of Theorem 54.1 gives

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2\delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this is (55.3). □

The common learner now has the same dimension and boundary scope as the metric and entropy theorems. Its calibration is a sequence of product experiments, while the compression learners provide the enhanced angular resolution when an estimated two-dimensional subspace is close

to opposite support endpoints. No pair of candidate hypotheses is supplied to either part of the procedure.

56. A FINITE RATIONAL CODE IN EVERY DIMENSION

The volume argument can be made into a single deterministic code whose decoder reconstructs its dictionary from the public parameters. The construction is finite and exact. Its purpose is to settle the constructive description problem; the enumeration below can be very large.

Theorem 56.1 (Deterministic rational matrix code). *Let $d, N \geq 1$ be integers and let $0 < \delta \leq 1$ be rational. There is a deterministic finite procedure, using rational arithmetic, that constructs an ordered list*

$$\mathcal{C}_{d,N,\delta} = (C_0, \dots, C_{M-1}) \subset \mathfrak{E}_d$$

of Gaussian-rational effects with the following properties. Every $E \in \mathfrak{E}_d$ has an assigned index $j(E)$ satisfying

$$(56.1) \quad d_N(\mathcal{M}_E, \mathcal{M}_{C_{j(E)}}) \leq \frac{11}{16}\delta.$$

For fixed d ,

$$(56.2) \quad M \leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2},$$

uniformly in N and δ . The encoder on a supplied Gaussian-rational effect and the decoder on its index both terminate using exact arithmetic. They require no matrix-dependent advice or preassigned list of centres. With (d, N, δ) public, the payload length is $\lceil \log_2 M \rceil$, and for $0 < \delta \leq \delta_d$ it is

$$(56.3) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

Here δ_d may be the small-error constant in Theorem 53.1. The procedure makes no assertion of polynomial running time in the payload length.

Proof. Set

$$(56.4) \quad K = \left\lceil \frac{32dN}{\delta} \right\rceil, \quad s = \frac{2d}{K}, \quad \eta = \frac{\delta}{16}.$$

Fix the order of independent real coordinates to be the d diagonal entries, followed by the real and imaginary parts of G_{ij} for $i < j$ in lexicographic order. Let $\mathcal{G}_{d,K}$ consist of the legal Hermitian matrices whose diagonal coordinates belong to $\{0, 1/K, \dots, 1\}$ and whose remaining coordinates belong to $\{-1, -1 + 1/K, \dots, 1\}$. List these matrices in the induced lexicographic order. This list is finite, and it contains zero.

Traverse the list once. Retain its first element. Retain a subsequent element G exactly when

$$(56.5) \quad Q_N(G, C)^2 > \eta^2 \quad \text{for every previously retained element } C.$$

The retained elements, in their order of appearance, form $\mathcal{C}_{d,N,\delta}$. Every grid element either is retained or has a retained predecessor C with $Q_N(G, C) \leq \eta$. No triangle inequality for Q_N is used.

We first verify that the whole procedure is exact. Legality of a Gaussian-rational Hermitian matrix is decidable by exact Schur elimination, applied to G and $I - G$. A negative pivot excludes positive semidefiniteness. A zero pivot in a positive semidefinite matrix forces the corresponding row and column of the remaining matrix to vanish; otherwise the test rejects. A positive pivot reduces the question to the exact Schur complement. These rules terminate after at most d pivots,

including on singular strata. For a legal rational pair, its midpoint A and variance $A(I - A)$ are Gaussian rational. The equation

$$A(I - A)X + XA(I - A) + N^{-1}X = G - C$$

has an invertible Gaussian-rational coefficient matrix. Solving it gives $Q_N(G, C)^2 = N \operatorname{tr}((G - C)X) \in \mathbb{Q}_{\geq 0}$. Consequently (56.5) is an exact rational comparison. It requires neither spectral coordinates nor a numerical eigenvalue threshold.

To encode E , first form the inward displacement

$$E_s = (1 - 2s)E + sI.$$

Since $s \leq 1/16$, its spectrum lies in $[s, 1 - s]$. Round each independent real coordinate of E_s to its nearest multiple of $1/K$, choosing the larger multiple at a tie, and call the resulting Hermitian matrix $G(E)$. A diagonal entry changes by at most $1/(2K)$ and the modulus of an off-diagonal entry changes by at most $\sqrt{2}/(2K)$. Thus

$$\|G(E) - E_s\|_{\text{op}} \leq \frac{d}{K}.$$

In particular,

$$(56.6) \quad \frac{d}{K}I \preceq G(E) \preceq \left(1 - \frac{d}{K}\right)I.$$

Every coordinate lies in the prescribed grid range, and hence $G(E) \in \mathcal{G}_{d,K}$. Moreover,

$$(56.7) \quad \|E - G(E)\|_{\text{op}} \leq s + \frac{d}{K} = \frac{3d}{K} \leq \frac{3\delta}{32N}.$$

Choose the first retained centre with $Q_N(G(E), C_j)^2 \leq \eta^2$. Such a centre exists by the completed greedy traversal. The hybrid inequality and Theorem 52.1 yield

$$\begin{aligned} d_N(\mathcal{M}_E, \mathcal{M}_{C_j}) &\leq 2N\|E - G(E)\|_{\text{op}} + 8Q_N(G(E), C_j) \\ &\leq \frac{3\delta}{16} + \frac{\delta}{2} = \frac{11\delta}{16}. \end{aligned}$$

This proves (56.1). If E is Gaussian rational, the prescribed rounding and every subsequent operation are exact finite rational calculations.

For two distinct retained centres, Theorem 52.1 gives

$$(56.8) \quad d_N(\mathcal{M}_C, \mathcal{M}_{C'}) \geq \frac{1}{8192d} \min\{1, Q_N(C, C')\} > \frac{\delta}{131072d},$$

where $\delta \leq 1$ was used. Let a_d, r_d be the constants of Lemma 53.2, and put

$$b_d = \min\left\{r_d, \frac{1}{393216d}\right\}, \quad r = b_d\delta.$$

The operational balls of radius r about distinct retained centres are disjoint, by (56.8) and the triangle inequality for d_N . Their measures are each at least $a_d(b_d\delta)^{d^2}$. Hence

$$Ma_d(b_d\delta)^{d^2} \leq \mu_N(\mathfrak{E}_d).$$

Proposition 53.3 proves (56.2).

Both parties reconstruct this same finite list from (d, N, δ) . The encoder transmits the binary representation of j in $\lceil \log_2 M \rceil$ bits. The decoder reconstructs the list and returns C_j for $j < M$; every unused word of the same length is assigned the legal effect $I/2$. The codeword contains no additional centre coordinates. Equation (56.2) gives the upper bound in (56.3); the covering lower bound of Theorem 53.1 gives the matching lower bound, since this list is a radius- δ cover by (56.1). The description of the fixed reconstruction algorithm contributes only a constant. As usual,

if the parameters are not public, their explicitly chosen header representation must be charged separately. $\square$

Real input and finite precision. The covering statement concerns the entire real effect body. To interpret its encoder computationally on a real input, a representation of that input must be supplied. A certified rational approximation of each independent real coordinate to error at most $1/(8K)$ is sufficient. Apply the inward displacement to this approximation and then use the same exact nearest-grid rounding. Relative to the true E_s , each independent coordinate then has error at most $5/(8K)$; each complex off-diagonal error is at most $5\sqrt{2}/(8K) < 1/K$. Therefore (56.6)–(56.7) and the full error bound remain valid. The exact supplied-matrix implementation takes Gaussian-rational input; no physical-device learning operation is implicit in that interface. Computational scope. The ambient grid contains at most

$$(56.9) \quad (K+1)^d(2K+1)^{d(d-1)}$$

candidate coordinate tuples. The literal algorithm performs exact legality tests and, for each legal tuple, comparisons with the retained list. Retaining the list uses $O(Md^2)$ rational entries, and the decoder repeats the same construction. These are finite bounds, but their dependence on the requested horizon and precision prevents an efficiency assertion in the payload length. The exact kernel `matrix_codec.py` implements this specified enumeration, greedy test, rounding, and replay. A resource cutoff is reported as an incomplete construction and never as a codebook. Regression checks of small complete cases and of matrix kernels certify those computations; they do not assert that a full large-dimensional dictionary has been enumerated.

Corollary 56.2 (Learning into an optimal-order description). *For every fixed $d \geq 1$, every integer $N \geq 1$, every rational $0 < \delta \leq \min\{1, \delta_0, \delta_d\}$, and every $0 < \eta \leq 1/8$, there is one learner, common to all $E \in \mathfrak{E}_d$, which returns a fixed-length word whose deterministic public decoder produces a legal effect C with*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

The training budget on every record is at most $Cd^4N\delta^{-2}\log(d/\eta)$, and each entangled block has at most N calls. The word has length

$$(56.10) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For fixed d , both the training order $N\delta^{-2}\log(1/\eta)$ and the payload order in (56.10) are optimal. Payload optimality is among fixed-length words with a fixed public decoder into legal memoryless ordered binary effects. Encoding after learning uses no additional calls to the unknown device.

Proof. Apply Theorem 55.1 with accuracy $\delta/2$. It returns a legal effect F with rational entries in the public input basis. Encode F using Theorem 56.1 with accuracy $\delta/2$. On the learning success event, the triangle inequality gives

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \frac{\delta}{2} + \frac{11\delta}{32} = \frac{27\delta}{32} < \delta.$$

The second operation is a deterministic computation on the recorded estimate. It uses no device calls and does not increase the failure probability. Replacing δ by $\delta/2$ changes the training constant and adds only $O_d(1)$ to the payload length.

The scalar subfamily converse in Theorem 55.1 applies to this learning task as well. For the payload converse, a B -bit word and fixed decoder have at most 2^B possible decoded effects. Every E has a strictly positive success probability, so at least one of these effects lies within distance δ of E . The decoded family is therefore a radius- δ cover of $\mathfrak{E}_d$. The lower bound in Theorem 53.1 gives (56.10) with the opposite inequality up to $O_d(1)$. $\square$

57. THE COST OF LIMITING INDEPENDENT QUANTUM BLOCKS

The training law changes when the number of calls in one independent quantum experiment is prescribed. The relevant restriction concerns access to the next queried input. Quantum systems retained from completed experiments may still be stored and measured together. This distinction allows a direct comparison with tomography based on independently prepared probe–reference pairs.

Definition 57.1 (Independent blocks with retained outputs). A training protocol has independent blocks of length at most b if it has the following form. At a block boundary it may apply an arbitrary common quantum instrument to all its retained systems. The resulting classical history h , including its classical random choices and the number of calls already charged, determines whether to stop, or else a length $1 \leq t(h) \leq b$ and a fresh $t(h)$ -use tester. Conditional on h , this tester starts in a parameter-independent state tensor-separated from all retained systems. It may use arbitrary references, entangled inputs, and adaptive control within its $t(h)$ calls.

From its preparation until its last call, the entire fresh tester, including its reference and unused inputs, is isolated from the retained systems. Those systems are available again at the next block boundary, when the new outputs may be joined to them and processed collectively. Thus classical feedback may select the next tester, while no retained quantum register can be coupled into a block that still has calls to make. A block length is declared before that block and charged in full; an internal early exit is padded within its declared length. The sum of declared lengths is at most M on every record. Stopping at block boundaries may depend on the full classical history. The final estimate may use an arbitrary joint measurement of all retained systems. Classical record spaces are standard Borel and quantum registers are separable.

The retained quantum systems belong to the tester; the unknown measurement still consumes its input and returns only its ordered binary classical label. The definition permits an unbounded amount of retained quantum storage and classical computation. It also permits all probes to be prepared in advance when their states factor into the prescribed independent blocks: their preparation can be deferred without changing the experiment. In particular, independent one-call Choi preparations followed by a collective measurement belong to the case $b = 1$. The distinction between classical input control and quantum memory on the output side has an antecedent in the channel-discrimination model of [49, Section II.2]. The familiar quadratic information gain within a block and additive gain across independent blocks also occur in the metrological bounds of [73, 74]. We give a finite-budget fidelity argument because a local information bound alone does not supply the confidence dependence needed here.

Write $M_b^*(d, N, \delta, \eta)$ for the least every-record call budget of a protocol in Definition 57.1 whose legal estimate satisfies

$$(57.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

The future distance d_N retains its unrestricted adaptive definition and its unhalved trace-norm normalization. The restriction on b applies only to training.

Theorem 57.2 (Sharp tradeoff for independent training blocks). *There are absolute constants $c, C, \delta_* > 0$ such that, for every $d \geq 2$, $N \geq 1$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$, and $0 < \eta \leq 1/8$,*

$$(57.2) \quad c \frac{N^2}{b\delta^2} \log \frac{1}{\eta} \leq M_b^*(d, N, \delta, \eta) \leq C d^4 \frac{N^2}{b\delta^2} \log \frac{d}{\eta}.$$

The upper bound is attained by one procedure common to the entire closed effect body. The lower bound allows classical feedback, arbitrary quantum storage and collective processing at block

boundaries, and every-record bounded stopping. Consequently, for each fixed $d \geq 2$,

$$(57.3) \quad M_b^*(d, N, \delta, \eta) = \Theta_d \left(\frac{N^2}{b\delta^2} \log \frac{1}{\eta} \right).$$

In dimension $d = 1$ the corresponding order is $\Theta(N\delta^{-2} \log(1/\eta))$, independently of b .

The endpoints $b = 1$ and $b = N$ differ by a factor of order N . The claim is uniform in the block cap, horizon, accuracy, and confidence; the upper factor d^4 is not asserted to be sharp in growing dimension.

Lemma 57.3 (Changing the future horizon). *For all effects E, F and integers $N, b \geq 1$,*

$$(57.4) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \left\lceil \frac{N}{b} \right\rceil d_b(\mathcal{M}_E, \mathcal{M}_F).$$

Proof. Fix an N -use tester and partition its calls into consecutive groups of at most b . Replace the device in one group at a time. For each replacement the common prefix prepares a reference-assisted input to a tester with at most b calls, including all memory carried from the prefix. The common suffix is a channel on that tester's output and cannot increase trace distance. Each replacement therefore costs at most d_b . Sum these costs and take the supremum over the original tester. Public stopping is included by padding to N calls with discarded dummy experiments. $\square$

Upper bound in Theorem 57.2. Put $q = \lceil N/b \rceil$. Apply Theorem 55.1 with future horizon b , accuracy δ/q , and failure allowance η . Its construction uses fresh independent blocks of length at most b , with classical records between them, so it satisfies Definition 57.1. Lemma 57.3 converts its success event into (57.1). On every record its call count is at most

$$Cd^4 b q^2 \delta^{-2} \log(d/\eta) \leq 4Cd^4 \frac{N^2}{b\delta^2} \log(d/\eta),$$

because $1 \leq b \leq N$. Choose δ_* no larger than the accuracy constant in Theorem 55.1. $\square$

For positive, possibly subnormalized operators define root fidelity by

$$\mathfrak{f}(A, B) = \|\sqrt{A}\sqrt{B}\|_1.$$

We use its tensor-product identity, its additivity on direct sums, and its monotonicity under a common channel. In particular, for a common instrument $(\mathcal{J}_z)_z$,

$$(57.5) \quad \sum_z \mathfrak{f}(\mathcal{J}_z(A), \mathcal{J}_z(B)) \geq \mathfrak{f}(A, B).$$

These facts hold for subnormalized operators by homogeneity. For states, the Bures angle $\arccos \mathfrak{f}$ is a contractive metric; see [37, 65].

Lemma 57.4 (Fidelity with adaptive choices of fresh blocks). *Fix a two-dimensional subspace and, for $\phi \in \mathbb{R}$, set*

$$(57.6) \quad E_\phi = \frac{1}{2}(I_2 + \cos \phi \sigma_1 + \sin \phi \sigma_2) \oplus \frac{1}{2}I_{d-2},$$

with the second summand absent when $d = 2$. If $b|\theta| \leq 1$, the final states Γ_0, Γ_θ of any protocol in Definition 57.1, applied to E_0, E_θ with every-record budget M , satisfy

$$(57.7) \quad \mathfrak{f}(\Gamma_0, \Gamma_\theta) \geq \exp(-Mb\theta^2/4).$$

Furthermore, if $N|\theta| \leq \pi$, then

$$(57.8) \quad d_N^{\text{na}}(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2 \sin(N|\theta|/2).$$

Proof. For qubits, (57.8) follows from [56, Corollary 1 and Theorem 2]. We give the calculation to specify the normalization and the fidelity estimate needed below.

Let $U_\theta = e^{-i\theta\sigma_3/2} \oplus I_{d-2}$. Then $E_\theta = U_\theta E_0 U_\theta^*$ and $\mathcal{M}_{E_\theta} = \mathcal{M}_{E_0} \circ \text{Ad}_{U_\theta^*}$. If $|\theta| \leq \pi$, every purified input ψ , including an arbitrary reference, satisfies

$$|\langle \psi, (U_\theta^* \otimes I)\psi \rangle| \geq \text{Re} \langle \psi, (U_\theta^* \otimes I)\psi \rangle \geq \cos(|\theta|/2).$$

The last inequality follows from the spectrum of $(U_\theta + U_\theta^*)/2$. Applying the common measurement and discarding a purification can only increase fidelity. Thus one call changes the output Bures angle by at most $|\theta|/2$, uniformly over its input and reference.

For a common adaptive tester, apply the triangle inequality before each query: the angle between the two current states is contracted by the same channel, and changing the channel on the second state adds at most $|\theta|/2$. The interposed controls are common channels and also contract the angle. Induction from the common initial state proves that every t -use tester has output states ρ_0, ρ_θ satisfying

$$(57.9) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq \cos(t|\theta|/2) \quad \text{when } t|\theta| \leq \pi.$$

This argument permits arbitrary adaptive quantum memory inside the tester. Its output includes all internal outcomes before any postselection. For $t \leq b$ and $b|\theta| \leq 1$, $\cos x \geq e^{-x^2}$ for $0 \leq x \leq 1/2$ gives

$$(57.10) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq e^{-t^2\theta^2/4} \geq e^{-at}, \quad a = b\theta^2/4.$$

We next retain the whole classical history, including all branch lengths and stopping decisions. At a boundary write the two classical-quantum states as direct sums of subnormalized retained states A_h, B_h . Their traces include the respective probabilities of h . Let $m(h)$ be the charged calls in that history, and consider

$$(57.11) \quad \Psi = \sum_h e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h).$$

Initially $\Psi = e^{-aM}$, since the initial resources are the same. A common boundary instrument preserves $m(h)$ along each of its children, so (57.5) shows that it cannot decrease Ψ . The history is kept until the end; in particular, records with different charged call counts are never merged in this calculation.

At a continuing history h , the prescribed fresh tester has the same specification under both hypotheses. Conditional on h , its output states ρ_0^h, ρ_θ^h are normalized and are tensor-separated from the old retained state. Immediately after this block the contribution of this history is therefore

$$\begin{aligned} & e^{-a(M-m(h)-t(h))} \mathfrak{f}(A_h \otimes \rho_0^h, B_h \otimes \rho_\theta^h) \\ & \geq e^{-a(M-m(h)-t(h))} e^{-at(h)} \mathfrak{f}(A_h, B_h) = e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h). \end{aligned}$$

The subsequent joint instrument is covered by (57.5). A stopped history is carried unchanged. Since each continuing block charges at least one call, all histories are terminal after at most M such generations. At that time all weights in (57.11) are at most one. Additivity of root fidelity on the terminal histories therefore gives

$$\mathfrak{f}\left(\bigoplus_h A_h, \bigoplus_h B_h\right) = \sum_h \mathfrak{f}(A_h, B_h) \geq \Psi \geq e^{-aM}.$$

Any final collective processing or discarding of history again increases fidelity. This proves (57.7) without asserting independence of the unconditional records or imposing a common support condition on their laws.

For general measurable classical outcomes, choose a measure dominating the two history laws and interpret A_h, B_h as their operator-valued densities. Direct-sum additivity becomes the corresponding integral identity, and the same proof uses integrals in (57.5) and (57.11). These identities follow equivalently by finite coarse-graining of the classical record and refinement. The charged call count has only the values $0, \dots, M$ and is retained in every such coarse-graining.

Finally, use the first two coordinates for a parallel N -input state

$$\frac{|0\rangle^{\otimes N} + e^{i\alpha}|1\rangle^{\otimes N}}{\sqrt{2}}, \quad \alpha = N\theta/2 + \pi/2.$$

The parity of the ordered output signs has mean $\cos(N\phi - \alpha)$ under E_ϕ , hence means $-\sin(N\theta/2)$ and $\sin(N\theta/2)$ under the two hypotheses. Its unhalved distance is $2|\sin(N\theta/2)|$. For $N|\theta| \leq \pi$, apply (57.9) with $t = N$ to any future adaptive tester and use $\|\rho - \sigma\|_1 \leq 2\sqrt{1 - \mathfrak{f}(\rho, \sigma)^2}$. The resulting upper bound is $2\sin(N|\theta|/2)$, equal to the parallel witness. This proves (57.8). $\square$

Lower bound in Theorem 57.2. Reduce δ_* further so that $\delta_* \leq 1/4$, and set $\theta = 4\delta/N$ in (57.6). Then $b|\theta| \leq 1$, while (57.8) gives

$$d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2\sin(2\delta) > 2\delta.$$

The two radius- δ success balls are disjoint. Thus any estimate satisfying (57.1) determines a test with both error probabilities α, β at most η : choose the first hypothesis precisely when the estimate lies in its radius- δ ball. The classical root fidelity of that test is

$$\sqrt{(1-\alpha)\beta} + \sqrt{\alpha(1-\beta)} \leq 2\sqrt{\eta}.$$

Fidelity is nondecreasing under the map from the training output to this test. Lemma 57.4 consequently implies

$$e^{-Mb\theta^2/4} \leq 2\sqrt{\eta}, \quad M \geq \frac{N^2}{8b\delta^2} \log \frac{1}{4\eta} \geq \frac{N^2}{24b\delta^2} \log \frac{1}{\eta},$$

where the last inequality uses $\eta \leq 1/8$. This proves the asserted lower bound in every $d \geq 2$.

For $d = 1$ every effect is scalar. Bernoulli endpoint estimation in Lemma 54.2 gives the upper order with independent one-call samples. The scalar converse in Theorem 55.1 gives the matching lower order even with unrestricted adaptive quantum training. Thus increasing b has no effect in dimension one. $\square$

Corollary 57.5 (The limitation of independent one-call acquisition). *Fix $d \geq 2$. Every learner that acquires its data using fresh one-call probes and fresh references, with classical feedback and arbitrary subsequent collective processing as in Definition 57.1, requires $\Omega_d(N^2\delta^{-2} \log(1/\eta))$ calls for (57.1). The complete closed effect body admits the same risk with $O_d(N\delta^{-2} \log(1/\eta))$ calls when independent blocks of length up to N are permitted.*

Proof. Use $b = 1$ and $b = N$ in Theorem 57.2. $\square$

This conclusion applies to the acquisition interfaces of [45, 69]. The channel learner in [45, Section III.2 and Algorithm I.2] prepares independent Choi copies before their collective processing. The measurement tomography procedure in [69, Section 3] queries separately prepared known inputs. Both are contained in $b = 1$ above. Any estimator retaining these interfaces and required to satisfy the present worst-case future- N loss therefore incurs the quadratic horizon order, including when its final estimator is collective. The block lower bound settles this interface comparison directly; it does not rely on substituting δ/N into a sufficient one-use accuracy guarantee.

Corollary 57.6 (Learning with bounded blocks into an optimal-order word). *For fixed $d \geq 2$, $1 \leq b \leq N$, rational $0 < \delta \leq \min\{1, \delta_*, \delta_d\}$, and $0 < \eta \leq 1/8$, there is a common learner in*

Definition 57.1 that returns a fixed-length word whose deterministic public decoder produces a legal effect C with

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

Its every-record training budget is at most $Cd^4N^2(b\delta^2)^{-1} \log(d/\eta)$ and its payload has length

$$(57.12) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For each fixed d , both the training order in this block model and the payload order among fixed public decoders into legal effects are optimal. Encoding uses no additional device calls.

Proof. Run the learner of Theorem 57.2 at accuracy $\delta/2$. The upper construction inherits the legal rational output of Theorem 55.1. Apply the exact codec of Theorem 56.1 at accuracy $\delta/2$. On the learning success event its decoded effect obeys

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \delta/2 + 11\delta/32 = 27\delta/32 < \delta.$$

The codec is a deterministic computation on the recorded rational estimate. Theorem 56.1 gives (57.12). The training converse is Theorem 57.2. For the payload converse, positive success probability for every E forces the decoded family to cover $\mathfrak{E}_d$ by radius- δ balls; apply Theorem 53.1. As before, reconstruction time, workspace, and state-preparation complexity are separate from the two resource counts. $\square$

58. DIMENSION AND ACCURACY IN THE UNIFORMLY NOISY INTERIOR

The block converse concerns projective boundary effects. A different obstruction is present in the full-dimensional interior: an effect close to a fair coin supplies little information per call, even when the learner retains coherent quantum memory. We quantify this obstruction before analysing a one-use estimator under the future loss. Entropies and relative entropies in this section use natural logarithms. The quantum information identities used below are in [37]; the particular conditional bound is proved here.

Lemma 58.1 (Information supplied by a weak binary measurement). *Let X have a finite alphabet. Conditional on $X = x$, an input register A and a retained register R have an arbitrary joint state ρ_{AR}^x . Apply the consuming binary measurement with effect $E_x = I/2 + H_x$, where $\|H_x\|_{\text{op}} \leq r < 1/2$, to A , and call its classical outcome Y . Then*

$$(58.1) \quad I(X : YR)_{\text{out}} \leq I(X : AR)_{\text{in}} + \log(1 + 4r^2) \leq I(X : AR)_{\text{in}} + 4r^2.$$

The input states need not be independent of x or of one another.

Proof. Fix x , abbreviate $\rho = \rho_R^x$, and let Δ be the partial trace of $(H_x \otimes I)\rho_{AR}^x$ over A . Positivity of $rI \pm H_x$ gives

$$-r\rho \preceq \Delta \preceq r\rho, \quad \omega_{YR}^x = |1\rangle\langle 1| \otimes (\rho/2 + \Delta) + |0\rangle\langle 0| \otimes (\rho/2 - \Delta).$$

On the support of ρ , write $\Delta = \rho^{1/2}K\rho^{1/2}$ with $\|K\|_{\text{op}} \leq r$. Set $\sigma = (I_Y/2) \otimes \rho$. The support inclusion $\text{supp } \omega^x \subseteq \text{supp } \sigma$ holds, and direct multiplication gives

$$\begin{aligned} \text{tr}[(\omega^x)^2 \sigma^{-1}] &= 1 + 4 \text{tr}(\Delta^2 \rho^{-1}) \\ &= 1 + 4 \text{tr}(\rho K^2) \leq 1 + 4r^2. \end{aligned}$$

All inverses in this calculation are support inverses. For density matrices a, b with the same support inclusion, $D(a||b) \leq \log \text{tr}(a^2 b^{-1})$ follows directly from scalar Jensen: in eigenbases of a and b , use the probability weights $a_i |\langle u_i, v_j \rangle|^2$ and apply the concavity of $\log$ to a_i/b_j . Consequently

$$D(\omega_{YR}^x || (I_Y/2) \otimes \rho_R^x) \leq \log(1 + 4r^2).$$

Average over x . Subtracting $D(\omega_{YR} \| (I_Y/2) \otimes \omega_R)$ from the average on the left gives exactly $I(X : Y | R)_{\text{out}}$. Relative entropy is nonnegative. The retained marginal is unchanged by the call, so data processing gives $I(X : R)_{\text{out}} = I(X : R)_{\text{in}} \leq I(X : AR)_{\text{in}}$. Adding these two inequalities proves the claim. For a separable retained register, apply finite-rank compression to R and pass to the limit. The bound is uniform, and mutual information with the finite classical register X is continuous under trace-norm convergence. $\square$

The same calculation applies to any finite classical-output measurement uniformly close to a scalar one. More precisely, if $p_y > 0$, $\sum_y p_y = 1$, and $(1-r)p_y I \preceq E_{x,y} \preceq (1+r)p_y I$, its per-call information increment is at most $\log(1+r^2)$. Indeed, the corresponding collision expression is $1 + \sum_y p_y^{-1} \text{tr}(\Delta_y^2 \rho^{-1}) \leq 1 + r^2$; the linear terms cancel because $\sum_y \Delta_y = 0$. We use only the binary case in what follows.

Theorem 58.2 (A joint dimension and future-loss lower bound). *There is an absolute $c > 0$ such that, for every $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$, a common learner with*

$$\sup_{I_d/4 \preceq E \preceq 3I_d/4} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\widehat{E}}) > \delta\} \leq \eta$$

requires at least

$$(58.2) \quad M \geq cd^2 N \delta^{-2}$$

calls in its worst record. The learner may retain arbitrary quantum memory, entangle its training calls, use coherent adaptive controls, and stop publicly. Its estimate may be any legal binary effect; it need not lie in the interior subfamily.

Proof. Put $r = 2048\delta/\sqrt{N} \leq 1/4$. The real vector space of Hermitian matrices has dimension $p = d^2$. A maximal finite $r/4$ -separated set $\{H_x\}_{x=1}^L$ in its operator-norm ball of radius r covers that ball by translates of the radius- $r/4$ ball. Comparing Lebesgue volumes gives $L \geq 4^p$. Such a finite maximal set exists by compactness. Set $E_x = I/2 + H_x$, so every E_x lies between $I/4$ and $3I/4$.

For $x \neq z$, take a unit eigenvector of $H_x - H_z$ achieving its operator norm and repeat this input N times. The two outcome laws are Bernoulli products with parameters $s, t \in [1/4, 3/4]$ and $|s - t| \geq r/4$. Since

$$T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + N^{-1}}} \geq \frac{1}{2} \sqrt{N} |s - t| \geq 256\delta,$$

Lemma 50.2 and $256\delta \leq 1$ give

$$(58.3) \quad d_N(\mathcal{M}_{E_x}, \mathcal{M}_{E_z}) \geq 4\delta.$$

The witness may depend on the pair; it is used only to establish separation, not as a common learning rule.

Choose X uniformly from this packing. The learner's initial resources are independent of X . Common intervening channels cannot increase their mutual information with X . Lemma 58.1 applies to each call even when its input and memory already depend on X through previous calls. Pad a stopped experiment by ignored dummy calls to its worst-record budget M . Its final quantum and classical registers therefore have mutual information at most $4Mr^2$ with X .

From the returned effect choose a nearest member of the finite packing, with a fixed tie rule. This is a statistical testing reduction; no computation of an adaptive optimum is required from the learner. The selection is measurable because the hybrid bound makes d_N continuous in each effect. The disjointness in (58.3) makes the chosen member correct whenever the learning loss is at most δ . Fano's inequality and data processing yield

$$4Mr^2 \geq (1 - \eta) \log L - \log 2 \geq \frac{3}{4} d^2 \log 2.$$

Substituting r proves (58.2). $\square$

Corollary 58.3 (Joint dimension and accuracy in one-use learning). *At failure probability at most $1/8$, the minimax number of calls needed to learn an arbitrary ordered binary effect to operator-norm error ϵ is*

$$\Theta(d^2\epsilon^{-2}), \quad d \geq 1, \quad 0 < \epsilon \leq 2^{-14},$$

with absolute constants. The lower bound permits arbitrary coherent adaptive training. The upper bound uses independent Choi probes and collective processing of their outputs.

Proof. For binary measurements $d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}$, including retained references; see [45, Lemmas III.7–III.8]. Theorem 58.2 with $N = 1$ and $\delta = 2\epsilon$ gives the lower bound. The upper bound is [45, Corollary III.9], at $\eta = 1/8$. Its stated condition $\eta > 4\exp(-4d^2)$ holds for every $d \geq 1$. Thus the lower argument couples dimension and accuracy, while the upper algorithm is the existing one-use estimator. $\square$

Lemma 58.4 (Dimension-free interior comparison). *If $I_d/8 \preceq E, F \preceq 7I_d/8$, then*

$$(58.4) \quad \frac{1}{128} \min\{1, \sqrt{N}\|E - F\|_{\text{op}}\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 4\sqrt{N}\|E - F\|_{\text{op}}\}.$$

The constants are independent of d and N .

Proof. The same repeated eigenvector input and Lemma 50.2 give the lower bound. For the upper bound take $G(t) = E + t(F - E)$ and $H = F - E$. Along this path $S(t) = \sqrt{G(t)(I - G(t))} \succeq I/4$. The unregularized Sylvester equation $SK + KS = H$ has solution

$$K = \int_0^\infty e^{-uS} H e^{-uS} du, \quad \|K\|_{\text{op}} \leq 2\|H\|_{\text{op}}.$$

The horizontal calculation (52.12) bounds the derivative of each fixed common tester's output by $2\sqrt{N}\|K\|_{\text{op}}$. Integrate on $[0, 1]$ and then take the supremum over testers. This proves the upper bound using the same horizontal Kraus and adaptive path methods as Lemma 52.2; see also [62, 63, 64, 66]. $\square$

Theorem 58.5 (Sharp learning on a fixed full-dimensional interior). *For the family $I_d/4 \preceq E \preceq 3I_d/4$, the minimax training cost for future- N loss δ and failure at most $1/8$ is*

$$(58.5) \quad \Theta(d^2 N \delta^{-2}), \quad d, N \geq 1, \quad 0 < \delta \leq 2^{-13}.$$

The constants are absolute. The upper procedure uses fresh one-call probes and permits collective final processing. The lower bound permits arbitrary coherent adaptive training. Thus the same order holds for every fresh-block cap $1 \leq b \leq N$.

Proof. Theorem 58.2 supplies the lower bound on this very family. For the upper bound apply the binary estimator of [45, Corollary III.9] at operator-norm accuracy $\epsilon = \delta/(8\sqrt{N})$ and failure $1/8$. It uses $O(d^2 N \delta^{-2})$ calls and returns a legal effect F . On its successful event $\|E - F\|_{\text{op}} \leq \epsilon \leq 1/8$, so both endpoints lie in $[I/8, 7I/8]$. Lemma 58.4 then gives $d_N(E, F) \leq \delta/2$. Independent input/reference pairs prepare the Choi copies; all coherent processing acts on completed outputs. This is a fresh one-call experiment in Definition 57.1. $\square$

This establishes a stronger future-loss analysis of an existing tomography estimator on a fixed interior. On the complete effect body, Corollary 57.5 supplies the different quadratic-horizon obstruction for the same acquisition class. The interior comparison is dimension-free, and the interior learning order is joint in d, N, δ ; neither assertion changes the dimension dependence or boundary range of the global comparison and covering theorems. In particular, for $d \geq 2$,

$1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$ and $0 < \eta \leq 1/8$, the full-body fresh-block learner has the two simultaneous lower bounds

$$M \geq c\delta^{-2} \left(d^2 N + \frac{N^2}{b} \log(1/\eta) \right),$$

after decreasing an absolute c , while its constructive upper bound remains $Cd^4(N^2/b)\delta^{-2} \log(d/\eta)$. The displayed d^4 upper factor is not matched by this lower bound.

59. JOINT DIMENSION BOUNDS FOR OPERATIONAL DESCRIPTIONS

The dimension-free comparison of Lemma 58.4 allows a description theorem whose constants are uniform in the matrix dimension. This complements, rather than replaces, the closed-body law of Theorem 53.1. Put

$$\mathfrak{I}_d = \{E = E^* : I_d/4 \preceq E \preceq 3I_d/4\}, \quad p = d^2.$$

Throughout this section, centres are legal ordered binary effects and the operational distance is the unhalved distance d_N of the same consuming, classical-output interface. In particular, the future tester may retain a reference, use coherent adaptive controls, and stop at a public bounded time. All logarithms in description lengths have base two.

Theorem 59.1 (Uniform interior operational entropy). *Let $d, N \geq 1$ and $0 < \delta \leq 2^{-13}$. The least cardinality $\mathcal{C}_N(\mathfrak{I}_d, \delta)$ of a d_N -cover of $\mathfrak{I}_d$, allowing centres anywhere in $\mathfrak{E}_d$, satisfies*

$$(59.1) \quad \left(\frac{\sqrt{N}}{2048\delta} \right)^p \leq \mathcal{C}_N(\mathfrak{I}_d, \delta) \leq \left(\frac{25\sqrt{N}}{\delta} \right)^p.$$

The upper bound is attained by a public finite exact rational construction when δ is rational. For real δ , a volumetric net gives the stated upper bound. Consequently the optimal deterministic fixed-length description into legal effects is

$$(59.2) \quad B_{\text{int}}^*(d, N, \delta) = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2),$$

where the remainder is uniform in all three parameters.

Proof. The real vector space of Hermitian matrices has dimension p , and $\mathfrak{I}_d$ is precisely the translate of its operator-norm ball of radius $1/4$. Set $s = 512\delta/\sqrt{N}$. A maximal subset $\{E_1, \dots, E_L\}$ of $\mathfrak{I}_d$ whose distinct points have operator distance strictly greater than s is finite and its closed s -balls cover $\mathfrak{I}_d$. Translation-invariant Lebesgue measure and homogeneity therefore give

$$Ls^p \geq (1/4)^p, \quad L \geq \left(\frac{\sqrt{N}}{2048\delta} \right)^p.$$

The unit-ball volume cancels; no dimension-dependent volume estimate is used. Lemma 58.4 separates any two selected effects in d_N by at least 4δ , since $512\delta \leq 1/16$. A closed operational ball of radius δ contains at most one selected effect, irrespective of its centre. This uses the triangle inequality for d_N , not a property of a comparison modulus.

For the upper bound choose a maximal operator-norm $\delta/(4\sqrt{N})$ -separated subset of $\mathfrak{I}_d$. It covers the latter at that operator radius and hence at operational radius δ . The disjoint open balls of half that radius lie inside the concentric ball of radius $1/4 + \delta/(8\sqrt{N})$. Their number is at most $(1 + 2\sqrt{N}/\delta)^p$, which is smaller than the asserted bound. The rational construction with a specified finite stopping rule is proved next. A fixed public B -bit decoder has at most 2^B outputs, and indexing a smallest finite cover suffices. Taking logarithms, with one bit for rounding to an integer, proves (59.2). $\square$

59.1. An exact public interior dictionary. Only the public parameters (d, N, δ) determine the dictionary. Let $\delta \in \mathbb{Q} \cap (0, 1]$ and set

$$(59.3) \quad k = \lceil \sqrt{N} \rceil, \quad K = \left\lceil \frac{128dk}{\delta} \right\rceil, \quad t = \frac{\delta}{16k}.$$

The integer k is computed by an integer square-root test. Enumerate Hermitian matrices with diagonal coordinates in $\{0, 1/K, \dots, 1\}$ and real and imaginary off-diagonal coordinates in $\{-1, -1 + 1/K, \dots, 1\}$. Order coordinates as in Theorem 56.1 and retain only the matrices in $[I_d/8, 7I_d/8]$. From this ordered finite stream construct a list $\mathcal{D}$ greedily, accepting G precisely when $\|G - C\|_{\text{op}} > t$ for every previously accepted C . For a Hermitian rational difference H , the complementary test is

$$(59.4) \quad \|H\|_{\text{op}} \leq t \iff tI_d - H \succeq 0 \text{ and } tI_d + H \succeq 0.$$

Thus exact rational positive-semidefinite tests settle the comparison, including equality, without computing eigenvalues. The Schur test of Theorem 56.1 applies to both inequalities.

Theorem 59.2 (Finite exact interior coding). *For every $d, N \geq 1$ and rational $0 < \delta \leq 1$, the preceding construction terminates and defines a common encoder and decoder. For a supplied Gaussian-rational $E \in \mathfrak{J}_d$, the decoded centre $C \in [I_d/8, 7I_d/8]$ satisfies*

$$(59.5) \quad d_N(E, C) \leq 5\delta/16.$$

The same conclusion holds for a real $E \in \mathfrak{J}_d$ supplied with a Gaussian-rational Hermitian approximation A certified to satisfy $\|A - E\|_{\text{op}} \leq d/K$. The payload is an index of length $\lceil \log_2 |\mathcal{D}| \rceil$, and

$$(59.6) \quad |\mathcal{D}| \leq (1 + 12k/\delta)^p \leq (25\sqrt{N}/\delta)^p.$$

All words of that length have legal decodings. No nonpublic matrix coordinates or target-dependent dictionary are free advice.

Proof. Round every real coordinate of A to the nearest multiple of $1/K$, with ties towards positive infinity, and preserve conjugate entries. Call the result G . In every row the diagonal rounding error is at most $1/(2K)$ and each off-diagonal error has modulus at most $1/(\sqrt{2}K)$. The Hermitian row-sum bound gives $\|G - A\|_{\text{op}} \leq d/K$. Therefore

$$\|G - E\|_{\text{op}} \leq 2d/K \leq \delta/(64k) \leq 1/64.$$

For rational E take $A = E$; the same, slightly looser, bound applies. Weyl's inequality puts G in $[I_d/8, 7I_d/8]$. Its diagonal and off-diagonal coordinates also lie in the enumerated ranges, so it occurs in the legal finite stream.

If G is retained, select $C = G$. Otherwise the strict greedy rule supplies an earlier retained centre with $\|G - C\|_{\text{op}} \leq t$; select the first such centre. Hence

$$\|E - C\|_{\text{op}} \leq \frac{\delta}{64k} + \frac{\delta}{16k} = \frac{5\delta}{64k}.$$

Both endpoints lie in the range of Lemma 58.4, which proves (59.5). Legality and both selection tests use finite exact rational arithmetic.

Distinct dictionary elements have operator distance greater than t , and every element lies within operator radius $3/8$ of $I_d/2$. The open operator balls of radius $t/2$ about them are disjoint and lie in the ball of radius $3/8 + t/2$. Cancelling the common unit-ball volume gives $(1 + 3/(4t))^p = (1 + 12k/\delta)^p$. Since $k \leq 2\sqrt{N}$ and $\sqrt{N}/\delta \geq 1$, the second inequality in (59.6) follows.

The decoder reconstructs the entire ordered list from the public parameters before interpreting its index. Unused fixed-length words return $I_d/2$. The list is nonempty because the rounded centre $I_d/2$ occurs. This proves termination and total legal decoding. A rational approximation

to a real input is supplied data under the displayed certificate, not an arbitrary exact-real oracle provided by the theorem. $\square$

The raw enumeration has at most $(K+1)^d(2K+1)^{d(d-1)}$ tuples. Exact comparisons against the retained list and the bit lengths of Schur-complement entries are additional reconstruction costs. They are not controlled by (59.2). The reference implementation rejects a resource-limited incomplete enumeration without emitting a codebook or payload. Thus the theorem is a finite exact construction, not a polynomial-time coding result. Volumetric packing and rational nets are standard mechanisms; the dimension-uniform operational comparison is what transfers them to the present future-use loss.

59.2. Randomized prefix descriptions. The fixed-length conclusion has a counterpart which permits shared randomness and counts expected transmitted length. This distinguishes an index-cardinality converse from a converse for a randomized communication protocol.

Definition 59.3 (One-way prefix description). The encoder is supplied $E \in \mathfrak{I}_d$. A public random variable R , independent of E , is available to both parties. Conditionally on each public seed, the possible finite binary messages W form a prefix-free set. Private randomization is allowed, and the decoder returns a legal effect C from (R, W) and its independent private randomness. No timing, stopping signal, nonpublic header, or other side channel is uncharged. The cost is $\sup_E \mathbb{E}_E |W|$ and the risk is $\sup_E \mathbb{P}_E \{d_N(E, C) > \delta\}$. The public parameters d, N, δ, η are not part of the message.

Theorem 59.4 (Randomized expected description length). *Let $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 \leq \eta \leq 1/8$. Write*

$$A_d(N, \delta) = d^2 \log_2(\sqrt{N}/\delta), \quad h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2(1 - \eta).$$

Here $0 \log_2 0 = 0$. The infimum $\mathcal{B}_\eta$ of the costs in Definition 59.3 with risk at most η satisfies

$$(59.7) \quad (1 - \eta)(A_d(N, \delta) - 11d^2) - h_2(\eta) \leq \mathcal{B}_\eta \leq (1 - \eta)(A_d(N, \delta) + d^2 \log_2 25 + 1).$$

In particular, $\mathcal{B}_\eta = (1 - \eta)A_d(N, \delta) + O(d^2)$, with absolute uniform constants. For rational δ the achievable nonempty messages are supplied by the finite exact interior dictionary.

Proof. Use the 4δ -separated packing of Theorem 59.1 and let X be uniform on its L labels, independently of R . A decoded effect within δ of the true packing point identifies its label uniquely. Choose a nearest label with a fixed tie rule otherwise. This produces $\hat{X}$ with error probability at most η .

For completeness, if J indicates a classification error, the chain rule gives

$$H(X | \hat{X}, R) \leq H(J) + \mathbb{P}(J = 1) \log_2(L - 1) \leq h_2(\eta) + \eta \log_2 L.$$

The final inequality uses $\eta \leq 1/8$ and $L \geq 4$. Since X and R are independent, data processing gives

$$(1 - \eta) \log_2 L - h_2(\eta) \leq I(X; \hat{X} | R) \leq I(X; W | R) \leq H(W | R).$$

The decoder's private randomness is independent conditionally on (R, W) and does not increase this information. For each public seed, Kraft's inequality and nonnegativity of relative entropy give $H(W | R = r) \leq \mathbb{E}(|W| | R = r)$; integrating proves the same conditional inequality. These statements also apply to countable messages of finite expected length by truncation; infinite cost already satisfies the lower bound. Worst-target expected length is at least its average over the packing. The packing cardinality in (59.1) now proves the first inequality. This is the usual Fano–Kraft argument, written here to include all the stated randomization and message conventions.

For achievability, use a public coin independent of the target. With probability η transmit the empty word and decode to $I_d/2$. With the complementary probability use the deterministic index code of Theorem 59.2, or an interior net for real δ . For each seed the permitted set is either the singleton empty word or the fixed-length index set, and hence is prefix-free. The decoder knows which case occurs from R , so there is no free transmitted abort flag. Failure can occur only on the public abort event. Equations (59.5) and (59.6), with integer length rounding, give the upper bound. The model permits the specified Bernoulli public coin; the zero-failure finite exact dictionary does not require it. $\square$

59.3. Learning and transmitting at the joint rates.

Theorem 59.5 (Jointly optimal learned interior descriptions). *In the ideal-control query model of Theorem 58.5, let $d, N \geq 1$ and let $0 < \delta \leq 2^{-13}$ be rational. There is one common learner on $\mathfrak{I}_d$ with a finite binary output and a fixed public deterministic decoder such that*

$$(59.8) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq 1/8, \quad M = O(d^2 N \delta^{-2}), \quad B = A_d(N, \delta) + O(d^2).$$

The constants are absolute. The acquisition uses independent one-call input/reference pairs, followed by collective processing of completed outputs. No device call is used for encoding or replay. Any learner with these error and decoder conventions has $M = \Omega(d^2 N \delta^{-2})$ and $B \geq A_d(N, \delta) - 11d^2$.

Proof. Apply the binary estimator of [45, Corollary III.9] at operator accuracy $\epsilon = \delta/(64k)$ and failure $1/8$. It returns a legal effect F with $\|F - E\|_{\text{op}} \leq \epsilon$ on its good event, using $O(d^2 k^2 \delta^{-2})$ calls. Let T be the spectral clipping of F to $[I/4, 3I/4]$. The good event and Weyl's inequality imply $\|T - F\|_{\text{op}} \leq \epsilon$, whence

$$\|T - E\|_{\text{op}} \leq 2\epsilon, \quad d_N(E, T) \leq 8\sqrt{N}\epsilon \leq \delta/8.$$

This argument does not assume that scalar clipping is operator-Lipschitz for a noncommuting pair.

Round the coordinates of T and encode their grid point by the finite dictionary of Theorem 59.2. Its proof gives $d_N(T, C) \leq 5\delta/16$ even though the displayed estimator may initially be real-valued. The resulting map from the original classical outcome to a dictionary index is measurable: spectral clipping is continuous, coordinate rounding is Borel with specified ties, and the final list is finite. Composing the final tomography POVM with this finite partition defines a common finite-output measurement. It needs neither an unknown-effect-dependent measurement nor an exact-real side channel to the decoder. Equivalently, certified finite coordinate approximations can be used with the rounding slack of the coding theorem when supplied by a classical implementation. On the good event the total operational error is at most $7\delta/16 < \delta$. Since $k^2 \leq 4N$, the call bound and the payload upper bound follow.

For the call converse, a decoded word is also an estimator, so Theorem 58.2 applies. For the length converse, positive success probability for every target implies that the finite range of the fixed deterministic decoder is a δ -cover. The first inequality of (59.1) applies even when those centres lie outside $\mathfrak{I}_d$. $\square$

The last theorem is a query and transmitted-length statement. It inherits the ideal trusted tomography operations of [45]; the measurable finite readout above does not assert an efficient gate synthesis for them. The separate finite-control theorem for our full-body learners remains unchanged. The interior description bounds neither remove the endpoint logarithms on the complete body nor optimize its d^4 learning upper bound. Rather, on the same full-dimensional interior, they identify both training and description costs with constants uniform in dimension, horizon and accuracy.

60. THE JOINT COST OF DIMENSION, HORIZON, AND CONFIDENCE

The fixed-confidence interior law has a sharp extension to arbitrary confidence. The dependence on dimension and on the confidence logarithm is additive after their respective dimensional factors are included. In this section the unknown measurement is still ordered, binary, consuming, and classical-output. Put $\mathfrak{I}_d = \{E : I_d/4 \preceq E \preceq 3I_d/4\}$ as in Section 59. Logarithms in call bounds are natural unless indicated otherwise.

Theorem 60.1 (Uniform interior learning at every confidence). *There are absolute $c, C > 0$ such that, for every $d, N \geq 1$, $0 < \delta \leq 2^{-13}$, and $0 < \eta \leq 1/8$, the minimax every-record call budget for*

$$(60.1) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta$$

is between

$$(60.2) \quad c \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right) \quad \text{and} \quad C \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right).$$

The lower bound permits arbitrary coherent adaptive training, references, quantum memory, and public bounded stopping. The estimate may be any legal effect. The upper bound uses independent one-call Choi preparations and collective processing of completed outputs. Consequently the same order, with absolute constants, holds for every fresh-block cap $1 \leq b \leq N$.

The proof combines the dimension lower bound of Theorem 58.2 with a separate confidence obstruction and an amplification of the estimator in [45]. The latter procedure is the tomography primitive throughout; the amplification uses its dimension-dependent tail guarantee.

Lemma 60.2 (Confidence requires observations in every coordinate). *In the parameter range of Theorem 60.1, every learner satisfying (60.1) has*

$$(60.3) \quad M \geq \frac{dN}{6 \cdot 512^2 \delta^2} \log \frac{1}{\eta}.$$

This bound already holds when the unknown effects are diagonal in a fixed known basis.

Proof. Set $\Delta = 512\delta/\sqrt{N} \leq 1/16$ and consider

$$E_0 = I/2, \quad E_i = I/2 + \Delta|i\rangle\langle i|, \quad 1 \leq i \leq d.$$

All these effects belong to $\mathfrak{I}_d$. For each pair (E_0, E_i) , repeating the input $|i\rangle$ gives Bernoulli parameters $1/2$ and $1/2 + \Delta$. The comparison quantity in Lemma 50.2 satisfies

$$T_N \geq \frac{1}{2} \sqrt{N} \Delta = 256\delta.$$

That lemma, together with $256\delta \leq 1$, therefore gives

$$(60.4) \quad d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_i}) \geq 4\delta.$$

We bound the information used to distinguish all these alternatives from the same null effect. For a diagonal effect with entries $p_1, \dots, p_d$, the reference-assisted output is unchanged if the input is first measured in the known basis, yielding $J = j$, and a Bernoulli bit of parameter p_j is then drawn. We retain the basis label in an extra register for the analysis. Discarding it gives the original device, so any original learner, including its final estimate, is recovered by ignoring these extra registers. This is the diagonal classicalisation used in [45, Lemma IV.19]; the following relative-entropy calculation also treats different quantum inputs under the two hypotheses.

For an input-reference state ρ_{AR} , write $\tau_j = \langle j | \rho_{AR} | j \rangle$. The revealed output is

$$\widetilde{\mathcal{M}}_p(\rho_{AR}) = \sum_{j=1}^d |j\rangle\langle j| \otimes ((1-p_j)|0\rangle\langle 0| + p_j|1\rangle\langle 1|) \otimes \tau_j.$$

Let ρ_0, ρ_i be arbitrary current input-reference states, where the reference includes all retained memory. The common channel $\rho \mapsto \sum_j |j\rangle\langle j| \otimes \tau_j$ cannot increase relative entropy. Appending the coin conditional on j adds its classical relative entropy with weight $\text{tr } \tau_j^0$. Thus the classical-quantum relative-entropy identity gives

$$(60.5) \quad \begin{aligned} D(\widetilde{\mathcal{M}}_0(\rho_0) \parallel \widetilde{\mathcal{M}}_i(\rho_i)) &\leq D(\rho_0 \parallel \rho_i) + \kappa_\Delta \text{tr } \tau_i^0, \\ \kappa_\Delta &= D(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) \\ &= -\frac{1}{2} \log(1 - 4\Delta^2) \leq 3\Delta^2. \end{aligned}$$

The subscript on $\widetilde{\mathcal{M}}$ now denotes the hypothesis. This calculation uses no equality or independence assumption for ρ_0, ρ_i . Starting from the common initial state, it also establishes finite relative entropy at every stage.

Keep each revealed label without feeding it into the original controls. All intervening operations, including classically controlled or coherent ones, are common channels. Pad a stopped record to its every-record cap M using ignored dummy calls. Let T_i count the revealed labels equal to i among these M calls, and let Ω_0, Ω_i denote the complete final states with the extra labels retained. Iterating (60.5) yields

$$D(\Omega_0 \parallel \Omega_i) \leq 3\Delta^2 \mathbb{E}_0 T_i, \quad \sum_{i=1}^d \mathbb{E}_0 T_i = M.$$

The same null experiment appears for every i , which is essential for summing these inequalities. General measurable records are included by the relative-entropy chain rule for classical-quantum states; no countability assumption on the learner's observations is required.

By (60.4), choose hypothesis 0 precisely when the returned effect lies in its closed radius- δ ball. This is a common binary test with both errors α_i, β_i at most η . The test is measurable by the hybrid continuity bound for d_N . Data processing to this test gives

$$\begin{aligned} D(\Omega_0 \parallel \Omega_i) &\geq D(\text{Bern}(1 - \alpha_i) \parallel \text{Bern}(\beta_i)) \\ &\geq (1 - \eta) \log(1/\eta) - \log 2 \geq \frac{1}{2} \log(1/\eta), \end{aligned}$$

where $\eta \leq 1/8$ was used. Summing over i and substituting Δ proves (60.3). $\square$

Lemma 60.3 (Amplification using a dimension-dependent seed). *For all $d \geq 1$, $0 < \epsilon \leq 1$, and $0 < \eta \leq 1/8$, there is a common estimator on $\mathfrak{E}_d$ with legal output F such that*

$$(60.6) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{\|F - E\|_{\text{op}} > \epsilon\} \leq \eta, \quad M \leq C\epsilon^{-2} \left(d^2 + d \log \frac{1}{\eta} \right).$$

The acquisition uses independent one-call input/reference pairs. The output is a Borel function of collective measurements on their completed outputs.

Proof. Corollary III.9 of [45] gives a legal binary-effect estimator with operator error u and failure q using $O(u^{-2}(d^2 + d \log(1/q)))$ calls when $q > 4e^{-4d^2}$. Choose the rational seed probability

$$q_d = 2^{-3d}.$$

It satisfies that strict condition for every $d \geq 1$, since $4d^2 - 3d \log 2 - \log 4 > 0$. A seed run with $u = \epsilon/3$ therefore costs at most $Cd^2\epsilon^{-2}$ calls.

Let k be the smallest positive odd integer with $k \geq 2d^{-1} \log_2(1/\eta)$. Make k independent seed runs, obtaining legal effects $F_1, \dots, F_k$. Select the first index j for which more than $k/2$ of the effects have operator distance at most $2\epsilon/3$ from F_j . If there is no such index, return F_1 . Norm continuity and the fixed tie rule make this selection Borel, and it always returns a legal effect.

If more than half the seed estimates are within $\epsilon/3$ of E , a selectable index exists. Every selectable index has a neighbour among these good estimates, and so its distance from E is at most ϵ . The probability of at least $k/2$ failed seeds is at most

$$2^k q_d^{k/2} = 2^{k-3dk/2} \leq 2^{-dk/2} \leq \eta.$$

This proves the risk bound. Since $k < 2d^{-1} \log_2(1/\eta) + 2$, the total call count has the order in (60.6). Each seed uses fresh Choi preparations; all subsequent joint processing acts on completed outputs. The amplification is a modification of the repetition option in [45, Remark III.4], with seed failure chosen at the dimension scale rather than fixed independently of d . $\square$

Proof of Theorem 60.1. Theorem 58.2 gives $M \geq c_1 d^2 N \delta^{-2}$ throughout the stated range. Lemma 60.2 supplies $M \geq c_2 d N \delta^{-2} \log(1/\eta)$. The maximum of these two bounds is at least half their sum after decreasing the absolute constant, which proves the lower bound in (60.2).

For the upper bound apply Lemma 60.3 at $\epsilon = \delta/(8\sqrt{N})$. On its good event, $\|F - E\|_{\text{op}} \leq \epsilon \leq 1/8$. Since $E \in \mathfrak{J}_d$, Weyl's inequality puts both endpoints in $[I/8, 7I/8]$. Lemma 58.4 gives $d_N(E, F) \leq 4\sqrt{N}\epsilon = \delta/2$. The call count is the required upper bound. The procedure is in the fresh one-call class, whereas the lower bound imposed no restriction on coherent training. This also proves the assertion for every fresh-block cap. $\square$

For $d = 1$, (60.2) reduces to $\Theta(N\delta^{-2} \log(1/\eta))$, the scalar order, because $\log(1/\eta) \geq \log 8$. For growing d , the confidence term becomes comparable to the dimension term when $\log(1/\eta)$ is of order d .

Corollary 60.4 (Binary operator-norm learning at every confidence). *For $d \geq 1$, $0 < \epsilon \leq 2^{-14}$, and $0 < \eta \leq 1/8$, the minimax every-record call budget for operator-norm error ϵ on the complete ordered binary effect body is*

$$(60.7) \quad \Theta\left(\epsilon^{-2} \left(d^2 + d \log \frac{1}{\eta}\right)\right),$$

with absolute constants. The lower bound permits arbitrary coherent adaptive training; the upper bound uses independent Choi preparations and collective completed-output processing.

Proof. The upper bound is Lemma 60.3. For the lower bound restrict to $\mathfrak{J}_d$ and use Theorem 60.1 with $N = 1$ and $\delta = 2\epsilon$, since $d_1(E, F) = 2\|E - F\|_{\text{op}}$. $\square$

On the complete effect body, the confidence obstruction also combines with the independent-block obstruction. For $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$, and $0 < \eta \leq 1/8$, Theorem 57.2 and the interior subfamily give

$$(60.8) \quad M_b^*(d, N, \delta, \eta) \geq c\delta^{-2} \left[d^2 N + \left(dN + \frac{N^2}{b} \right) \log \frac{1}{\eta} \right].$$

A second upper construction follows from Lemma 60.3 at operator accuracy $\epsilon = \delta/(2N)$. The hybrid inequality $d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$ converts its good event to the required future loss. It uses only fresh one-call preparations, so it is admissible for every block cap b . Combining its public call budget with the construction in Theorem 57.2 gives, in the same parameter range,

$$(60.9) \quad M_b^*(d, N, \delta, \eta) \leq C \frac{N^2}{\delta^2} \min \left\{ d^2 + d \log \frac{1}{\eta}, \frac{d^4}{b} \log \frac{d}{\eta} \right\}.$$

The learner may choose the construction with the smaller public every-record budget before making any calls. Both constructions are common to all unknown effects. The two-sided bounds (60.8)–(60.9) do not in general match when dimension and horizon grow together.

Corollary 60.5 (Jointly optimal learned descriptions at every confidence). *For $d, N \geq 1$, rational $0 < \delta \leq 2^{-13}$, and $0 < \eta \leq 1/8$, one common learner on $\mathfrak{I}_d$ returns a fixed-length binary word whose public deterministic decoder is legal and satisfies (60.1), with*

$$(60.10) \quad M = O\left(\frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta}\right)\right), \quad B = d^2 \log_2(\sqrt{N}/\delta) + O(d^2).$$

All constants are absolute. Acquisition uses independent one-call Choi preparations, and encoding uses no additional device calls. Every learner with these risk and decoder conventions requires the lower call order in (60.2) and $B \geq d^2 \log_2(\sqrt{N}/\delta) - 11d^2$.

Proof. Put $k_N = \lceil \sqrt{N} \rceil$. Apply Lemma 60.3 at $\epsilon = \delta/(64k_N)$ and failure η , and clip its legal output F spectrally to $[I/4, 3I/4]$, obtaining T . On the good event, Weyl's inequality gives $\|T - F\|_{\text{op}} \leq \epsilon$ and hence $\|T - E\|_{\text{op}} \leq 2\epsilon$. This uses only the eigenvalue displacements of F and does not assume operator-Lipschitz spectral clipping. The interior comparison yields $d_N(E, T) \leq \delta/8$.

Use the exact dictionary of Theorem 59.2 at the same (d, N, δ) . As in the proof of Theorem 59.5, spectral clipping and the specified coordinate rounding give a Borel finite-valued index map, which can be incorporated into the final collective measurement. The decoded centre satisfies $d_N(T, C) \leq 5\delta/16$, so its total error is at most $7\delta/16 < \delta$ on the original good event. Thus the finite output does not increase the failure probability or supply an exact-real side channel to the decoder.

The bound $k_N^2 \leq 4N$ and Lemma 60.3 give the call count. Theorem 59.2 gives the payload. Conversely, a decoded word is an estimator, so Theorem 60.1 applies to its calls. For each target, positive success probability forces some decoded effect into its radius- δ ball. The fixed decoder's range is therefore a cover, and Theorem 59.1 gives the stated bit lower bound. $\square$

These upper bounds use the ideal trusted tomography operations of [45]. The Borel finite readout is a statistical construction; it alone supplies no gate synthesis or reconstruction-time bound. The confidence parameter increases the required observations while the same public interior dictionary supplies the transmitted word.

61. LEARNING WITH FINITE TRUSTED CONTROL SPECIFICATIONS

The learning theorems count unknown-device calls. We now remove exact physical preparation from their hypotheses, while keeping their loss and call orders. A trusted preparation instruction has a finite classical description and a certified trace-norm error; a trusted control instruction has a finite description and a certified diamond-norm error. These are accuracy assumptions on the available controls. They assert neither a universal hardware implementation nor a bound on preparation time, gate count, or classical workspace. All trace norms in this section are unhalved.

Lemma 61.1 (An adaptive control-error budget). *Fix a finite classical decision tree of experiments, with bounded numbers of calls and trusted instructions on every record. Its unknown channel is unchanged. At a scheduled trusted event j , conditional on the preceding classical history h , replace a channel $\mathcal{C}_{j,h}$ by a channel $\tilde{\mathcal{C}}_{j,h}$ such that*

$$(61.1) \quad \|\tilde{\mathcal{C}}_{j,h} - \mathcal{C}_{j,h}\|_{\diamond} \leq e_j(h), \quad \sum_{j \text{ on } h_*} e_j(h_{<j}) \leq e \quad \text{for every complete record } h_*.$$

For a fresh state preparation, its trace-norm error suffices in place of the diamond norm: conditional on h , the new state is tensor-separated from every carried system in both experiments. The final classical-quantum states satisfy

$$(61.2) \quad \|\tilde{\rho}_{\text{out}} - \rho_{\text{out}}\|_1 \leq e.$$

In particular, the laws of any unchanged classical output have total variation distance at most $e/2$.

Proof. Include each observed outcome in the classical history and pad stopped branches by identities. Quantum registers of different sizes can be placed in a finite direct sum. Each event is then a channel on the complete classical–quantum state. A quantum instrument means its complete channel $\rho \mapsto \sum_x |x\rangle\langle x| \otimes \mathcal{C}_x(\rho)$: if errors are instead supplied separately for its completely positive branches, their sum is the event’s error charge.

At each event, insert the ideal preceding state between the actual and ideal updates. Contractivity of the actual channel and the diamond-norm bound give

$$D_j \leq D_{j-1} + \sum_h p_{j-1}(h) e_j(h),$$

where D_j is the unhalved trace-norm difference and p_{j-1} is the ideal history law. Unknown-channel calls and unchanged classical calculations have charge zero. Summing the inequality gives the ideal expectation of the pathwise sum in (61.1), which is at most e . A replacement preparation channel has diamond distance equal to the trace distance of the prepared states, including when another register is retained. Finally, classical readout and deterministic postprocessing are contractions, and classical total variation is one half of the trace norm. No continuity of the decision rules or of the final loss is used. $\square$

We check that the learners to which this lemma is applied have finite, computable control specifications. Take integer dimensions and horizons and rational positive accuracy and confidence targets. All absolute proof constants may be fixed once as sufficiently conservative rational numbers. For sample budgets, replace each $\log x$, $x > 1$ rational, by

$$\ell(x) = \min\{k \in \mathbb{N} : 2^k \geq x\}.$$

Here $\log x \leq \ell(x) \leq 1 + \log x / \log 2$. Thus the concentration bounds and call orders are preserved, while ceilings of the resulting rational sample budgets are decidable by integer arithmetic. This convention avoids requiring an exact comparison of a transcendental expression with an integer.

Calibration frequencies and all complex empirical means are rational. The ordered spectral projections, fixed-order Gram–Schmidt procedure, clipping, normalized frames, and tie rules of Sections 54–55 use algebraic arithmetic. The notation for phase recovery does not require a transcendental oracle. For a nonzero empirical z passing (54.20), put $q = z/|z|$ and select the root v of $v^m = q$ with largest real part. It is unique: the passed guard places $\text{Arg } q$ strictly between $-\pi$ and π , so its principal m th root is strictly closer in angle to 1 than every other root. All roots are algebraic, and comparison of their real parts is decidable. Consequently

$$\tan(\text{Arg}(z)/m) = \frac{\Im v}{\Re v}$$

is algebraic; its denominator is positive under the guard. Zero, amplitude, frame-selection, and phase-guard equalities are therefore decidable exactly. The finite rational legalization grid in Theorem 55.1 is in the public input basis; its positivity tests are exact, and its finitely many algebraic objectives can be evaluated with the prescribed slack. The same holds for the exact rational codec. Every record consequently has a finite, terminating classical computation and a legal output.

There are finitely many records at fixed public parameters: the device outputs and preparation signs have finite alphabets and their numbers are bounded on every path. This does not discretize the unknown effect or the retained quantum state. Within a block the state still lies in its full continuous quantum state space. Only the experiment’s classical instructions and observations have finite descriptions.

Finite descriptions can also specify controls with any required positive accuracy. If a computable unit vector $\psi \in \mathbb{C}^D$ is prescribed and $0 < \epsilon \leq 1$, compute a nonzero Gaussian rational vector z

with $\|z - \psi\|_2 \leq \epsilon/8$ and set $\phi = z/\sqrt{\sum_k |z_k|^2}$. Then

$$(61.3) \quad \| |\phi\rangle\langle\phi| - |\psi\rangle\langle\psi| \|_1 \leq 2\|\phi - \psi\|_2 \leq 4\|z - \psi\|_2 \leq \epsilon/2.$$

Coordinate accuracy $O(\epsilon/\sqrt{D})$ suffices, so z and its normalization rule have a description of $O(D \log(D/\epsilon))$ bits. The normalization is specified algebraically; rational normalized amplitudes are not assumed. A trusted preparation within $\epsilon/2$ of this finitely specified target has total error at most ϵ from the ideal state.

For completeness, finite specifications can preserve the exact channel constraints as well. Approximate a known algebraic Stinespring isometry V by a Gaussian rational matrix A with $\|A - V\|_{\text{op}} \leq r < 1/2$, and specify $W = A(A^*A)^{-1/2}$. Then $W^*W = I$ and $\|W - V\|_{\text{op}} \leq 2r$: every singular value of A differs from one by at most r . The channels obtained from these isometries differ in diamond norm by at most $4r$, by expansion of $W\rho W^* - V\rho V^*$ and partial-trace contractivity. Keeping a finite outcome flag gives the same construction for instruments. Their actual trusted realizations are charged separately in (61.1). Independent rounding of Kraus entries without restoring the channel constraints would not suffice.

Theorem 61.2 (Finite-control form of the learning guarantees). *The common matrix learner of Theorem 55.1 and the learner with training-block cap $1 \leq b \leq N$ of Theorem 57.2 admit the preceding finite trusted-control model in their stated ranges. They have the original success target*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta$$

with every-record call orders, respectively,

$$(61.4) \quad Cd^4N\delta^{-2}\log(d/\eta), \quad Cd^4\frac{N^2}{b}\delta^{-2}\log(d/\eta).$$

Their original block caps and exact legality of the returned classical effect are preserved. The same transfer applies when the classical output is the public codeword. In dimension one the order remains $CN\delta^{-2}\log(1/\eta)$ independently of b .

Proof. Use the corresponding ideal learner at the same accuracy δ and confidence $\eta/2$, with the finite sample-budget convention above. Fix a computable public upper bound M_* on its calls on every record. The explicit constructions use fresh product states or fresh GHZ states embedded into a known two-dimensional subspace. They can be implemented by one preparation event per block, hence at most M_* such events. Given any preceding record, compute that record's prescribed state to trace accuracy η/M_* , using (61.3) and the trusted realization allowance. Lemma 61.1 gives final output total variation at most $\eta/2$. For each fixed unknown E , apply it to the set of returned effects whose original loss exceeds δ . The ideal probability is at most $\eta/2$, so the actual probability is at most η . Discontinuous decisions, unsuccessful records, and exact equality branches are included in this comparison.

Classical postprocessing is identical in the two experiments; it does not round the returned effect or test a perturbed loss. Thus legality is exact on every actual record and no additional δ -dependent loss allowance is needed. Halving η and upper-rounding sample budgets change only the absolute constants in (61.4). $\square$

Each approximate preparation above supplies a state on the same $m \leq b$ probe registers as its ideal block; preparation workspace is discarded or reset before the experiment and no quantum system is passed to the next block. Approximation therefore does not enlarge entangled input width or introduce coherent coupling from old memory into fresh probes. In a realization with S additional scheduled control instructions, use a public bound on the total number of trusted events and charge every instruction in (61.1). Per-event norm precision is then of order $\eta/(M_* + S)$. For an m -probe state the direct coordinate description has $D = d^m$ and size $O(D \log(D(M_* + S)/\eta))$

bits. The precision per coordinate is logarithmic in $(M_* + S)/\eta$, with its dimension dependence explicitly charged; this is not a polynomial preparation bound. Real-valued target tolerances may be replaced by supplied rational targets between one half and the requested values, preserving all orders and the requested guarantees. No representation of an arbitrary real input, and no oracle for the unknown effect, is assumed.

62. EFFECTIVE SYNTHESIS OF AN OPTIMAL INTERIOR LEARNER

The finite readout in Theorem 59.5 does not by itself give computable entries for its measurement operators. We now construct a finite algebraic readout, and then a finite trusted-control realization, at the joint query and description orders. The construction uses exact quantifier elimination. The tomography estimator of [45] supplies an existence bound; we do not compile its continuous outcome measurement or assume that its coarse-grained operators have computable entries.

62.1. A decidable collective-readout problem. Fix integers $d, m \geq 1$. Prepare m independent maximally entangled probe-reference pairs and apply the unknown measurement to each probe. Put $E_1 = E$ and $E_0 = I_d - E$. The complete classical-quantum record is

$$(62.1) \quad \Gamma_m(E) = \bigoplus_{y \in \{0,1\}^m} R_y(E), \quad R_y(E) = d^{-m} \bigotimes_{\ell=1}^m E_{y_\ell}^\top.$$

The transpose is in the public input basis. The blocks are *subnormalized*: their traces are the probabilities of the classical strings y . The real and imaginary parts of every entry of $R_y(E)$ are polynomials with rational coefficients in the d^2 real coordinates of E . No division by an unknown outcome probability occurs.

A final L -outcome readout consists of a POVM $(A_{j,y})_{j=1}^L$ on the d^m retained reference dimensions, selected by the observed classical string y . Its outcome probabilities are

$$(62.2) \quad p_j(E; A) = \sum_y \text{tr}(A_{j,y} R_y(E)), \quad A_{j,y} \succeq 0, \quad \sum_{j=1}^L A_{j,y} = I_{d^m}.$$

This description uses the outcome bits classically. It entails no coherent access to those bits or to a measurement environment. Conversely, any collective POVM on (62.1) can be dephased in y without changing its probabilities, giving precisely such a family.

Lemma 62.1 (Effective algebraic readout selection). *Given d, m , finitely many Gaussian-rational Hermitian centres $C_1, \dots, C_L$, and rational $a > 0$, $0 < \alpha < 1$, it is decidable whether a readout (62.2) satisfies*

$$(62.3) \quad \sum_{j: \|E - C_j\|_{\text{op}} \leq a} p_j(E; A) \geq 1 - \alpha \quad \text{for every } E \in \mathfrak{I}_d.$$

When it is feasible, one finite deterministic computation returns such a family with algebraic entries, valid for every real $E \in \mathfrak{I}_d$.

Proof. Parameterize each Hermitian matrix by its real coordinates. Positivity is a finite system of polynomial inequalities: use all principal minors, or the equivalent real symmetric representation for a complex Hermitian matrix. Trace and tensor products in (62.2) yield real polynomials with rational coefficients. The same is true of the input constraint $I_d/4 \preceq E \preceq 3I_d/4$ and of

$$\text{Good}_j(E) : \quad aI_d - (E - C_j) \succeq 0, \quad aI_d + (E - C_j) \succeq 0.$$

For each subset $S \subseteq \{1, \dots, L\}$ impose the implication

$$\left(\bigwedge_{j \in S} \text{Good}_j(E) \right) \wedge \left(\bigwedge_{j \notin S} \neg \text{Good}_j(E) \right) \implies \sum_{j \in S} p_j(E; A) \geq 1 - \alpha.$$

The conjunction of these finitely many implications, universally quantified over $E \in \mathfrak{I}_d$, is exactly (62.3). Equality belongs to **Good**; its complement uses the corresponding strict inequalities. Hence boundary cases require no numerical tolerance.

Eliminate the E variables while retaining the readout entries as free variables, and include the positivity and normalization conditions in (62.2). Effective real quantifier elimination decides whether this rational semialgebraic feasible set is empty. If it is nonempty, the real-closed-field transfer principle gives a point with real algebraic coordinates. One explicit selection rule uses a fixed effective enumeration of algebraic tuples and tests the resulting quantifier-free formula by exact algebraic sign determination. It terminates on this nonempty set. These are standard effective real-algebraic operations; see [78, Theorems 2.77, 2.80 and Chapter 14]. The eliminated formula is equivalent over the reals, so the selected point satisfies the condition for all real effects, not merely algebraic ones. The unknown effect is a quantified variable, never supplied data or an oracle queried by this computation. $\square$

62.2. Finite controls at the joint optimal rates.

Theorem 62.2 (Effective optimal interior learning and description). *Let $d, N \geq 1$ be integers, and let $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$ be rational. There is a common learner on $\mathfrak{I}_d$ whose entire classical specification is computed from (d, N, δ, η) in finitely many steps. In the finite trusted-control model of Section 61, it returns a fixed-length word decoded by a public deterministic decoder into a legal rational effect C , with*

$$(62.4) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq \eta,$$

$$(62.5) \quad M \leq C_0 \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right),$$

$$(62.6) \quad B = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

All constants are absolute. The number M and the finite readout are fixed before any unknown-device call. Acquisition uses fresh one-call probe–reference pairs; only completed outputs undergo collective processing. Thus the construction belongs to every fresh-block class $1 \leq b \leq N$.

The call order matches the coherent adaptive lower bound of Theorem 60.1, and the description order matches Theorem 59.1. No bound on classical search time, algebraic-description length, quantum workspace, or elementary known-gate count is asserted.

Proof. Construct the finite rational dictionary $\mathcal{D} = (C_1, \dots, C_L)$ of Theorem 59.2 at parameters (d, N, δ) . Set

$$k = \lceil \sqrt{N} \rceil, \quad a = \frac{\delta}{8k}, \quad \alpha = \eta/2.$$

For $m = 1, 2, \dots$, apply Lemma 62.1 to these centres, a , and α . At the first feasible m , extract its algebraic readout and set $M = m$. Each infeasible candidate is rejected by a terminating decision procedure; there is no unbounded search for a witness at such a candidate.

We prove termination and the call bound without computing any operator of the imported tomography procedure. Apply Lemma 60.3 at operator accuracy $\epsilon = \delta/(64k)$ and failure $\eta/2$. Its independent Choi acquisition uses

$$m_0 \leq C_1 k^2 \delta^{-2} \left(d^2 + d \log \frac{2}{\eta} \right)$$

calls and returns a legal effect F . Clip F spectrally to $[I_d/4, 3I_d/4]$, obtaining T . On the estimation success event, Weyl's inequality gives $\|T - F\|_{\text{op}} \leq \epsilon$ and hence $\|T - E\|_{\text{op}} \leq 2\epsilon$. The rounding and

predecessor selection in Theorem 59.2 produce an index j with $\|T - C_j\|_{\text{op}} \leq 5\delta/(64k)$. Therefore

$$(62.7) \quad \|E - C_j\|_{\text{op}} \leq \frac{7\delta}{64k} < \frac{\delta}{8k} = a$$

on that event.

Clipping, coordinate rounding with its prescribed ties, and selection from a finite list are Borel operations. Composing them with the final tomography measurement defines a finite-outcome POVM on the m_0 Choi records. Its dephased form is a readout (62.2) satisfying (62.3). Only its existence is used here; its entries and any defining integrals are not inputs to the algebraic search. Thus the search stops with $M \leq m_0$. Since $k^2 \leq 4N$ and $\log(2/\eta) \leq C_2 \log(1/\eta)$, this proves (62.5).

The selected ideal readout has probability at least $1 - \eta/2$ of returning a centre at operator distance at most a from E . The target lies in $\mathcal{I}_d$ and every centre lies in $[I_d/8, 7I_d/8]$. Lemma 58.4 therefore gives

$$d_N(E, C_j) \leq 4\sqrt{N}a \leq \delta/2$$

on this event.

It remains to realize the computed readout with finite trusted controls. For each classical string y , its algebraic POVM has the algebraic isometry

$$V_y \psi = \sum_{j=1}^L |j\rangle_C |j\rangle_Z \sqrt{A_{j,y}} \psi.$$

Tracing out Z and the factor space gives the required classical readout. This is a finite table, including strings of probability zero under some targets. Algebraic positive square roots and the finite descriptions of Section 61 make each entry effectively approximable. In particular, the normalization of a rational approximation to an isometry preserves the exact channel constraints; a fixed final dephasing preserves the classical output type.

There are M fresh-pair preparation events and one final readout event. Require total unhalved error at each event at most

$$(62.8) \quad e_{\text{event}} = \frac{\eta}{M+1}.$$

For a preparation this is trace-norm error in the fresh joint pair; for the readout it is complete diamond-norm error, uniformly over all y and all retained reference states. Split each allowance equally between approximation by a finite rational specification and its actual trusted realization. Any additional elementary instructions are charged by the same total pathwise budget η . Lemma 61.1 then bounds the total variation between ideal and actual output indices by $\eta/2$. Applying it to the original bad-output event $d_N(E, C_j) > \delta$ proves (62.4). No continuity of a decision boundary or perturbation of the returned effect is assumed.

Finally transmit the dictionary index using $B = \lceil \log_2 L \rceil$ bits. The exact public decoder, including its legal values on unused words, is unchanged. Theorem 59.2 gives the upper order in (62.6). Since the dictionary covers $\mathcal{I}_d$, Theorem 59.1 gives its matching lower order. More generally, positive success probability at every target forces any fixed finite decoder's range to cover the target family. This proves the stated optimality claims. $\square$

The synthesis is an effective construction of a new collective learner at the existing tomography query order. It is not a claim of efficient compilation of the original tomography measurement. At the selected M , there are 2^M possible classical strings, each selecting an L -outcome measurement on dimension d^M . The algebraic search, its coefficients, the corresponding isometry descriptions, and their preparation work are separate resources from the transmitted index. The per-event norm tolerance in (62.8) introduces no additional δ/N requirement; dimensions and coordinate

precision still depend on δ through M and L . Fresh preparation remains tensor-separated from the stored references, so finite-control approximation preserves the one-call acquisition restriction.

63. RATIONAL COLLECTIVE READOUTS WITH FINITE RISK CERTIFICATES

The effective selection of Section 62 can be realized using rational arithmetic and finite enumeration. The readout operators themselves may be chosen Gaussian rational. Their uniform guarantee is certified by exact inequalities on an explicit finite net. The net certifies the statistical risk of a fixed readout; it does not discretize the unknown measurement in the learning model.

63.1. A finite certificate for a collective readout. Keep the public dictionary $\mathcal{D} = (C_1, \dots, C_L)$ of Theorem 59.2, constructed at (d, N, δ) . Put $k = \lceil \sqrt{N} \rceil$ and $a = \delta/(8k)$. For $m \geq 1$, let $D = d^m$ and retain the subnormalized Choi records $\Gamma_m(E)$ of (62.1). For rational $0 < \eta \leq 1/8$, define

$$(63.1) \quad r_m = \min \left\{ \frac{\delta}{32k}, \frac{\eta}{16m} \right\}, \quad K_m = \left\lceil \frac{4d}{r_m} \right\rceil, \quad Q_m = \left\lceil \frac{64L^2 D}{\eta} \right\rceil.$$

Let $\mathcal{G}_m$ be the ordered finite set of Gaussian-rational Hermitian matrices with denominator K_m in $\mathfrak{I}_d = [I_d/4, 3I_d/4]$. More explicitly, enumerate diagonal coordinates in $\{0, 1/K_m, \dots, 1\}$ and real and imaginary coordinates above the diagonal in $\{-1, -1 + 1/K_m, \dots, 1\}$, in the public order, and retain exactly those matrices in $\mathfrak{I}_d$. All tests use exact rational positive-semidefinite elimination. For $G \in \mathcal{G}_m$, set

$$J(G) = \{j : \|G - C_j\|_{\text{op}} \leq a\}.$$

These sets are decidable by the two positive-semidefinite tests in (59.4), including equality.

A rational readout is a family $(A_{j,y} : 1 \leq j \leq L, y \in \{0, 1\}^m)$ of Gaussian-rational Hermitian $D \times D$ matrices satisfying

$$A_{j,y} \succeq 0, \quad \sum_{j=1}^L A_{j,y} = I_D \quad \text{for every } y.$$

Its certificate consists of these exact legality conditions and the finite inequalities

$$(63.2) \quad \sum_{j \in J(G)} \sum_{y \in \{0, 1\}^m} \text{tr}(A_{j,y} R_y(G)) \geq 1 - \frac{\eta}{4} \quad (G \in \mathcal{G}_m).$$

Every term is rational. A verifier reconstructs the dictionary and net from the public parameters and checks every displayed inequality. A proper prefix of either enumeration does not constitute a certificate.

Theorem 63.1 (Uniform validity of the finite certificate). *Every readout satisfying (63.2) has, for all real $E \in \mathfrak{I}_d$,*

$$(63.3) \quad \mathbb{P}_E\{d_N(E, C_J) > 5\delta/8\} \leq 5\eta/16.$$

Here J is the readout index, and C_J is its unchanged public rational decoding. More generally, if a finite operator-norm r -net $\mathcal{G} \subseteq \mathfrak{I}_d$ satisfies the same finite inequalities with right side $1 - \alpha$ and good-label radius a , then

$$(63.4) \quad \sup_{E \in \mathfrak{I}_d} \mathbb{P}_E\{\|E - C_J\|_{\text{op}} > a + r\} \leq \alpha + mr.$$

The centres may be any fixed finite legal family for this last assertion.

Proof. We first verify that $\mathcal{G}_m$ is an operator-norm r_m -net. Write $K = K_m$, and for any $E \in \mathfrak{J}_d$ form

$$T = \left(1 - \frac{8d}{K}\right)E + \frac{4d}{K}I_d.$$

Since $8d/K \leq 2r_m < 1$, this is a convex displacement towards $I_d/2$. Its spectrum lies in $[1/4 + 2d/K, 3/4 - 2d/K]$ and $\|T - E\|_{\text{op}} \leq 2d/K$. Round each independent real coordinate to the nearest multiple of $1/K$, with the prescribed ties, and preserve Hermitian conjugacy. The row-sum estimate gives $\|G - T\|_{\text{op}} \leq d/K$. Consequently $G \in \mathfrak{J}_d$, its coordinates occur in the specified finite stream, and

$$(63.5) \quad \|G - E\|_{\text{op}} \leq 3d/K \leq 3r_m/4 \leq r_m.$$

For one Choi record,

$$\|\Gamma_1(E) - \Gamma_1(G)\|_1 = \frac{2}{d}\|E - G\|_1 \leq 2\|E - G\|_{\text{op}}.$$

Tensor-product telescoping between normalized states therefore yields

$$(63.6) \quad \|\Gamma_m(E) - \Gamma_m(G)\|_1 \leq 2m\|E - G\|_{\text{op}}.$$

Apply the same fixed readout under the two target effects. Their index laws have total variation at most $mr_m \leq \eta/16$. For the particular G in (63.5), every $j \in J(G)$ satisfies

$$\|E - C_j\|_{\text{op}} \leq a + r_m \leq \frac{5\delta}{32k}.$$

Both effects lie in $[I_d/8, 7I_d/8]$, so Lemma 58.4 gives $d_N(E, C_j) \leq 4\sqrt{N}(a + r_m) \leq 5\delta/8$. The event $\{J \in J(G)\}$ has probability at least $1 - \eta/4 - \eta/16$ under E , by (63.2). This proves (63.3). No continuity of the index-selection boundary is used: one fixed subset $J(G)$ is transferred between the two experiments. Replacing the numerical allowances by a, r, α in that same argument proves (63.4). $\square$

63.2. A prescribed rational denominator suffices.

Proposition 63.2 (Rational approximation of a finite classical readout). *Let $(B_{j,y})$ be any family of L -outcome POVMs on dimension D , with finitely many classical conditioning values y . For rational $0 < \eta \leq 1/8$, put $s = \eta/16$ and $Q = \lceil 64L^2D/\eta \rceil$. There is a family $(A_{j,y})$ of Gaussian-rational POVMs, with all real and imaginary coordinates in $Q^{-1}\mathbb{Z}$, such that, for every input classical-quantum state, its output-index law differs from that of $(B_{j,y})$ in total variation by at most $3\eta/32$. The rational family belongs to an explicitly finite enumeration determined solely by (D, L, Q) and the conditioning alphabet.*

Proof. When $L = 1$, take $A_{1,y} = I_D$; there is no approximation error. Suppose $L \geq 2$. First put

$$\overline{B}_{j,y} = (1 - s)B_{j,y} + \frac{s}{L}I_D.$$

These effects sum to I_D and are all at least $(s/L)I_D$. For each y , round the independent real coordinates of the first $L - 1$ effects to multiples of $1/Q$, preserving Hermitian conjugacy, and call the rounded matrices $A_{j,y}$. Set

$$A_{L,y} = I_D - \sum_{j < L} A_{j,y}.$$

Thus normalization holds exactly. With $F_{j,y} = A_{j,y} - \overline{B}_{j,y}$, the row-sum estimate gives

$$\|F_{j,y}\|_{\text{op}} \leq D/Q \quad (j < L), \quad \|F_{L,y}\|_{\text{op}} \leq (L - 1)D/Q.$$

Since

$$(L - 1)D/Q \leq \frac{\eta(L - 1)}{64L^2} < \frac{\eta}{16L} = \frac{s}{L},$$

all $A_{j,y}$ are positive. Together with their exact sum, this also implies $A_{j,y} \preceq I_D$. The rounded coordinates therefore lie in the finite ranges $[0, 1]$ on the diagonal and $[-1, 1]$ for each real or imaginary off-diagonal coordinate.

Mixing with the uniform outcome changes every event probability by at most s . Rounding changes it by at most

$$\sum_j \|F_{j,y}\|_{\text{op}} \leq 2(L-1)D/Q \leq 2LD/Q \leq \eta/32.$$

The same bound holds after averaging over any subnormalized classical conditioning blocks. Thus the total variation of the index laws is at most $s + \eta/32 = 3\eta/32$.

For clarity, no exact entries of B are required by the algorithm below. This rounding argument proves that a suitable rational tuple exists in a prescribed finite set. Enumerate the first $L-1$ Hermitian matrices in the indicated coordinate ranges, define the last by the residual formula, and retain only positive tuples. The resulting finite stream contains the family just constructed. $\square$

Theorem 63.3 (Certified effective learning at the joint rates). *Let $d, N \geq 1$, and let $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$ be rational. A finite deterministic computation from (d, N, δ, η) returns a query number M , a Gaussian-rational collective readout, and the finite risk certificate (63.2). In the finite trusted-control model, the resulting common learner on $\mathfrak{J}_d$ satisfies*

$$(63.7) \quad \sup_{E \in \mathfrak{J}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq \eta,$$

$$(63.8) \quad M \leq C_0 \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right),$$

$$(63.9) \quad B = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

All constants are absolute. The acquisition consists of fresh one-call probe-reference pairs, followed by collective processing of completed outputs. The call and description orders are the minimax orders of Theorems 60.1 and 59.1. No classical running-time, workspace, or elementary known-gate bound is asserted.

Proof. Construct the public dictionary once. For each $m = 1, 2, 4, \dots$, compute (63.1) and exhaust the finite rational readout stream of Proposition 63.2, with 2^m conditioning values and dimension $D = d^m$. For every candidate, check exact POVM legality and every inequality in (63.2). If one passes, return the first such candidate and set $M = m$. If none passes, the finite enumeration ends and the algorithm moves to $2m$. Thus it does not wait indefinitely at an infeasible query count. All arithmetic, tensor products, comparisons, and positive-semidefinite tests are over the Gaussian rationals.

It remains to show that some stage passes at the claimed call order. Lemma 60.3, used at accuracy $\epsilon = \delta/(128k)$ and failure $\eta/8$, supplies an independent-Choi estimator with legal output F and

$$m_0 \leq C_1 k^2 \delta^{-2} \left(d^2 + d \log \frac{8}{\eta} \right).$$

Clip F spectrally to $[I_d/4, 3I_d/4]$, obtaining T . On its good event, Weyl's inequality gives $\|T - E\|_{\text{op}} \leq 2\epsilon$. The dictionary rounding and predecessor selection give

$$\|T - C_j\|_{\text{op}} \leq \frac{5\delta}{64k}, \quad \|E - C_j\|_{\text{op}} \leq \frac{3\delta}{32k} < a.$$

These operations define a Borel finite partition of the original tomography outcomes. Hence there exists an L -outcome collective POVM, dephased in the classical acquisition strings, with failure at most $\eta/8$ for the operator-radius- a event uniformly on $\mathfrak{J}_d$.

For every $m \geq m_0$, this readout is available on m Choi records by ignoring the last $m - m_0$ records. Equivalently, its effects are tensored with the identity on the extra reference factors and do not depend on the extra outcome bits. Apply Proposition 63.2 at this m . At every $G \in \mathcal{G}_m$, the resulting rational candidate has failure at most

$$\frac{\eta}{8} + \frac{\eta}{16} + \frac{\eta}{32} = \frac{7\eta}{32} < \frac{\eta}{4}.$$

It therefore passes every finite certificate inequality. The first dyadic integer at least m_0 is at most $2m_0$; thus the outer search terminates with $M \leq 2m_0$. This proof uses existence of the tomography readout only. Its entries, continuous outcome integrals, and exact optimization outputs are never inputs to the rational search. Since $k^2 \leq 4N$ and $\log(8/\eta) = O(\log(1/\eta))$, the query bound follows.

By Theorem 63.1, the selected exact readout has original-loss failure at most $5\eta/16$, with successful loss at most $5\delta/8$. Each rational POVM has an effectively computable algebraic Stinespring isometry, obtained from the positive square roots of its effects as in Section 62. Use the normalized rational approximations of Section 61 to specify its trusted implementation. There are M fresh-pair preparation events and one complete classically controlled readout event. Give each event total unhalved error allowance $\eta/(M+1)$, split equally between finite target approximation and actual trusted realization. This final event includes the complete readout, its classical conditioning, and its final dephasing. If it is implemented as several separately charged instructions, subdivide its allowance among those instructions.

The adaptive control-error lemma bounds the total variation of the ideal and actual index laws by $\eta/2$. The exact public decoder is unchanged. Consequently

$$\mathbb{P}_E\{d_N(E, C) > \delta\} \leq \frac{5\eta}{16} + \frac{\eta}{2} = \frac{13\eta}{16} < \eta \quad (E \in \mathfrak{I}_d).$$

Approximate pair preparations remain tensor-separated from every stored reference, so the fresh one-call restriction is preserved. The dictionary index supplies the description order in (63.9); its lower order follows from the interior covering converse. The unrestricted adaptive learning converse applies to this learner as well. $\square$

The certificate consists entirely of finite exact rational checks. Its continuum implication is Theorem 63.1, whose analytic input is the tensor-product Lipschitz estimate and the dimension-uniform interior metric comparison. Exhaustive rational search replaces real quantifier elimination in this construction. The output arrays have $2^M L d^{2M}$ independent real matrix coordinates before normalization is used; their construction and verification are separate resources from the transmitted dictionary index. At stage m , at most $(2K_m + 1)^{d^2}$ coordinate tuples are inspected for the parameter net, and at most $(2Q_m + 1)^{(L-1)2^m D^2}$ tuples for the readout. At a fixed stage m , the readout array can be stored using $(L-1)2^m D^2 \lceil \log_2(2Q_m + 1) \rceil$ bits in its fixed coordinate order, together with the public parameters; the residual effects are reconstructed exactly. The stage m is included in a finite header or recovered by rerunning the prescribed deterministic search. These explicit finite bounds are not polynomial-time or payload bounds. A software check of a finite instance establishes that instance's certificate, while the stated lemmas establish its uniform interpretation and the termination bound for the complete construction.

64. FINITE-OUTCOME GEOMETRY AND EXACT DESCRIPTIONS

The interior geometry does not require two outcomes. We now allow any fixed finite alphabet, while retaining the input-consuming, ordered, classical-output interface. The effects below need not commute. The binary results on the complete effect body remain separate: no positive lower spectral bound is imposed in those results.

For integers $d \geq 1$ and $k \geq 2$, put

$$(64.1) \quad \mathfrak{P}_{d,k} = \left\{ \mathbf{E} = (E_1, \dots, E_k) : E_j = E_j^*, \quad \sum_{j=1}^k E_j = I_d, \quad \frac{I_d}{2k} \preceq E_j \preceq \frac{3I_d}{2k} \right\}.$$

An element specifies the channel $\mathcal{M}_{\mathbf{E}}(\rho) = \sum_j \text{tr}(E_j \rho) |j\rangle\langle j|$. It returns no residual quantum system. Write D_N for its N -use distance, with precisely the reference, feedback, bounded public stopping, and unhalved trace-norm conventions defining d_N . The superscript na denotes nonadaptive experiments, not a restriction to unentangled inputs. Set

$$r(\mathbf{E}, \mathbf{F}) = \max_j \|E_j - F_j\|_{\text{op}}, \quad p = (k-1)d^2.$$

The affine space $\sum_j E_j = I_d$ has real dimension p . In its translated vector space $\sum_j H_j = 0$, the set $\mathfrak{P}_{d,k}$ is exactly the closed r -ball of radius $R = 1/(2k)$ about $\mathbf{U} = (I_d/k, \dots, I_d/k)$.

Theorem 64.1 (A dimension-independent interior comparison). *For $d, N \geq 1$, $k \geq 2$, and $\mathbf{E}, \mathbf{F} \in \mathfrak{P}_{d,k}$,*

$$(64.2) \quad \frac{1}{128} \min\{1, \sqrt{N} r(\mathbf{E}, \mathbf{F})\} \leq D_N^{\text{na}}(\mathbf{E}, \mathbf{F}) \leq D_N(\mathbf{E}, \mathbf{F}) \leq \min\{2, 2k\sqrt{N} r(\mathbf{E}, \mathbf{F})\}.$$

The lower bound holds for arbitrary legal ordered k -outcome measurements, without the spectral bounds in (64.1). The upper constants are independent of d and N ; dependence on k is displayed and is not asserted sharp.

Proof. Let $H_j = F_j - E_j$ and $A_j(t) = E_j + tH_j$ for $0 \leq t \leq 1$. Each $A_j(t)$ is positive definite. Solve the matrix differential equation

$$(64.3) \quad \dot{X}_j(t) = \frac{1}{2} X_j(t) A_j(t)^{-1} H_j, \quad X_j(0) = \sqrt{E_j}.$$

Its coefficients are continuous. Differentiating $X_j^* X_j$ shows that it solves the linear equation

$$\dot{B}_j = \frac{1}{2} H_j A_j^{-1} B_j + \frac{1}{2} B_j A_j^{-1} H_j, \quad B_j(0) = E_j.$$

The function A_j solves this same equation, so uniqueness gives $X_j^* X_j = A_j$. Consequently

$$(64.4) \quad X_j^* \dot{X}_j = \frac{1}{2} H_j, \quad \dot{X}_j^* \dot{X}_j = \frac{1}{4} H_j A_j^{-1} H_j.$$

This argument neither diagonalizes the tuple nor assumes that A_j and H_j commute.

Use the Stinespring isometry

$$W_t \psi = \sum_{j=1}^k |j\rangle_Y |j\rangle_Z X_j(t) \psi.$$

Tracing out Z and the last factor gives $\mathcal{M}_{\mathbf{A}(t)}$. The environment in this dilation is used only in the proof and is not supplied by the device. Since $\sum_j H_j = 0$, (64.4) yields

$$(64.5) \quad W_t^* W_t = I_d, \quad W_t^* \dot{W}_t = 0, \quad \dot{W}_t^* \dot{W}_t = \frac{1}{4} \sum_j H_j A_j(t)^{-1} H_j.$$

For completeness, purify a common adaptive tester, retaining a fresh inaccessible environment for each call. Its controls are isometries independent of t . Differentiate the final purified state. It is a sum of at most N terms, with $\dot{W}_t$ replacing one occurrence of W_t in each term. In the inner product of two distinct terms, cancel the later common isometries. At the later differentiated slot the remaining contraction is $W_t^* \dot{W}_t$ or its adjoint, hence zero. Each diagonal term has norm at most $\|\dot{W}_t\|_{\text{op}}$. The derivative of the purified state therefore has norm at most $\sqrt{N} \|\dot{W}_t\|_{\text{op}}$. The derivative of its rank-one density operator has trace norm at most twice this number. Partial trace and all final processing contract trace norm. Bounded stopping is covered by a coherent

stopping flag and dummy calls on a fixed discarded input after stopping; the resulting final state of the original tester is a common postprocessing of this padded experiment.

Integrating the derivative bound gives the more informative inequality

$$(64.6) \quad \begin{aligned} D_N(\mathbf{E}, \mathbf{F}) &\leq \sqrt{N} \int_0^1 \left\| \sum_j H_j A_j(t)^{-1} H_j \right\|_{\text{op}}^{1/2} dt \\ &\leq \sqrt{2kN} \left\| \sum_j H_j^2 \right\|_{\text{op}}^{1/2} \leq \sqrt{2k} \sqrt{N} \max_j \|H_j\|_{\text{op}}. \end{aligned}$$

We used $A_j(t)^{-1} \preceq 2kI_d$ and the order-preserving map $B \mapsto H_j B H_j$; commutation is unnecessary. The weaker constant $2k$ in (64.2) is convenient below. The trace distance is always at most two.

For the lower bound, choose a label j and a unit eigenvector of $F_j - E_j$ attaining its operator norm. Send that same vector in every call and replace the observed label by the indicator of j . The two records are products of Bernoulli laws with parameters $s, t \in [0, 1]$ and $|s - t| = \|F_j - E_j\|_{\text{op}}$. Since $\max\{s(1-s), t(1-t)\} + N^{-1} \leq 5/4$, the quantity in (50.2) is at least $\frac{1}{2}\sqrt{N}|s-t|$. Lemma 50.2 gives the lower constant $1/128$. Maximize over j . No interior assumption was used in this part. $\square$

64.1. The affine norm ball and its entropy. Let $\mathcal{C}_{N,k}(\delta)$ be the least number of radius- δ D_N -balls covering $\mathfrak{P}_{d,k}$. Their centres may be arbitrary legal memoryless ordered k -outcome measurements.

Theorem 64.2 (Dimension-uniform finite-outcome entropy). *For $d, N \geq 1$, $k \geq 2$, and $0 < \delta < 1/128$,*

$$(64.7) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{256k\delta} \right)^p \right\} \leq \mathcal{C}_{N,k}(\delta) \leq \left(\frac{3\sqrt{N}}{\delta} \right)^p, \quad p = (k-1)d^2.$$

In particular, uniformly in d, N, δ for each fixed k ,

$$(64.8) \quad \log_2 \mathcal{C}_{N,k}(\delta) = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1)).$$

Proof. If $D_N(\mathbf{E}, \mathbf{F}) \leq \delta < 1/128$, the lower bound of Theorem 64.1 implies $r(\mathbf{E}, \mathbf{F}) \leq 128\delta/\sqrt{N}$. Lebesgue volume in the p -dimensional translated affine space gives the lower covering bound, because the target is its r -ball of radius $1/(2k)$. This argument also applies when the centre lies outside $\mathfrak{P}_{d,k}$.

Take a maximal ϵ -separated subset of that ball, where $\epsilon = \delta/(2k\sqrt{N})$. It is an r -net. The open r -balls of radius $\epsilon/2$ about its points are disjoint and lie in the ball of radius $R + \epsilon/2$. Their number is at most $(1 + 2R/\epsilon)^p = (1 + 2\sqrt{N}/\delta)^p$. By the upper comparison it is a D_N -cover at radius δ . This proves (64.7); taking logarithms gives (64.8), including the range where the displayed power in the lower bound is less than one. $\square$

64.2. A finite grid with exact normalization. An independent rounding of all k effects need not preserve their sum. The following grid preserves both normalization and the two spectral inequalities simultaneously.

Lemma 64.3 (Rational nets of the balanced body). *Let $0 < t \leq 1/(8k)$ be rational and choose*

$$(64.9) \quad K = \left\lceil \max \left\{ \frac{4(k-1)d}{t}, 8k(k-1)d \right\} \right\rceil.$$

Enumerate the first $k-1$ Hermitian matrices using diagonal coordinates $0, 1/K, \dots, 1$ and real and imaginary coordinates above the diagonal in $K^{-1}\mathbb{Z} \cap [-1, 1]$. Set the last matrix equal to I_d minus their sum and retain exactly the legal tuples in $\mathfrak{P}_{d,k}$. This finite ordered grid, denoted $\mathcal{G}(t)$,

is an r -net of radius t . All membership and pairwise norm-threshold tests are finite exact rational positive-semidefinite tests.

Proof. Put $e = (k - 1)d/K$ and $\gamma = 4ke \leq 1/2$. For a target tuple form $E'_j = (1 - \gamma)E_j + \gamma I_d/k$. Each E'_j has margin $2e$ from both spectral faces in (64.1), and $\|E'_j - E_j\|_{\text{op}} \leq 2e$. Round the independent real coordinates of the first $k - 1$ matrices to nearest multiples of $1/K$, with a fixed tie rule and Hermitian conjugacy. The row-sum bound gives operator error at most d/K for each rounded matrix. Define the last matrix by exact subtraction from I_d ; its error is at most $(k - 1)d/K = e$. Thus every rounded effect still has margin at least e , their sum is exactly I_d , and their distance from the target is at most $3e \leq 3t/4 < t$. The rounded coordinates occur in the stated enumeration.

Legality is checked on $E_j - I_d/(2k)$ and $3I_d/(2k) - E_j$, for every j . The relation $r(\mathbf{E}, \mathbf{F}) \leq s$ is decided by checking $sI_d \pm (E_j - F_j) \succeq 0$ for every j . Exact Schur elimination, including the zero-pivot case, decides these tests over the Gaussian rationals. Equality is included. $\square$

Theorem 64.4 (A public exact code). *For integers $d, N \geq 1$, $k \geq 2$, and rational $0 < \delta < 1/128$, there is a finite deterministic dictionary with centres in $\mathfrak{P}_{d,k}$ and covering error at most $\delta/2$. Its fixed-length index uses*

$$(64.10) \quad B = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k + 1))$$

bits, with absolute implied constants. A fixed public decoder into arbitrary legal k -outcome measurements cannot improve this order for worst-case radius δ . The construction is finite and exact; no polynomial running-time or workspace assertion is made.

Proof. Put $n = \lceil \sqrt{N} \rceil$ and $s = \delta/(8kn)$, and enumerate $\mathcal{G}(s)$ from Lemma 64.3. Retain a point precisely when its r -distance from every earlier retained point is strictly greater than s . The list is finite and nonempty. It is an r -cover of the grid at radius s ; together with the grid approximation it covers the body at radius $2s$. Its D_N -covering error is therefore at most $2k\sqrt{N}(2s) \leq \delta/2$.

Distinct retained points have r -distance greater than s . The volume argument of Theorem 64.2 gives dictionary size at most

$$(1 + 2R/s)^p = (1 + 8n/\delta)^p \leq (17\sqrt{N}/\delta)^p.$$

The lower bound follows from the same theorem because the dictionary covers at radius δ , even allowing arbitrary legal centres. Taking the ceiling of the logarithm changes the length by at most one.

Encoder and decoder reconstruct the same ordered dictionary from (d, k, N, δ) ; only its index is transmitted. Unused binary words decode to the fixed uniform tuple. For a supplied rational target, scan the list for the first centre within $2s$, using the exact tests in Lemma 64.3. For a real target supplied with certified rational approximations, search for a grid point at certified strict distance less than s and use its assigned predecessor. Such a point exists at distance at most $3s/4$ by the grid lemma, so progressively finer certified input approximations make the strict test terminate. An unrestricted oracle for arbitrary real coordinates is not part of the encoding model. The covering assertion itself concerns every real target, irrespective of its computational representation. $\square$

The grid has at most $[(K + 1)^d(2K + 1)^{d(d-1)}]^{k-1}$ candidates before the legality tests. Reconstructing this list and comparing its retained points can cost far more than the transmitted index. The estimate concerns the description of an unknown measurement, not free communication of a target-dependent dictionary.

65. JOINT LEARNING AND FINITE CERTIFICATES FOR A FINITE ALPHABET

We next construct one learner common to all unknown elements of $\mathfrak{P}_{d,k}$. Its training experiments do not use the pair-dependent eigenvectors in the geometric lower bound. The outcome count k is public, as are d, N, δ, η . The upper procedure uses independent one-call probe-reference pairs; coherent collective processing is confined to completed outputs. The converse permits arbitrary coherent adaptive training, with an every-record budget and bounded public stopping.

Theorem 65.1 (Joint finite-outcome learning and coding). *Let $d, N \geq 1$, $k \geq 2$, $0 < \delta \leq 2^{-24}$, and $0 < \eta \leq 1/8$. Write $M_{d,k}^*(N, \delta, \eta)$ for the minimum every-record call budget with worst-case failure at most η under D_N -error δ on $\mathfrak{P}_{d,k}$. There are absolute constants $c, C > 0$ such that*

$$(65.1) \quad c \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right) \leq M_{d,k}^*(N, \delta, \eta) \leq C \frac{k^3 N}{\delta^2} \left(d^2 + d \log \frac{k}{\eta} \right).$$

For each fixed k , this is a matching order jointly in d, N, δ, η , including $d = 1$:

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k \left(N \delta^{-2} [d^2 + d \log(1/\eta)] \right).$$

The displayed dependence on growing k is not claimed optimal.

For rational accuracy and confidence, the upper procedure has a finite deterministic specification, returns an index with a fixed public decoder into $\mathfrak{P}_{d,k}$, and simultaneously uses

$$(65.2) \quad B = \frac{(k-1)d^2}{2} \log_2 N + (k-1)d^2 \log_2(1/\delta) + O((k-1)d^2 \log(k+1))$$

bits. On its good event the decoded error is at most $5\delta/8$. The same call order is available with finite trusted controls. Classical search, readout storage, known-state preparation, and known-control realization are separate resources.

Proof. We first give the constructive upper bound for rational parameters. Set $n = \lceil \sqrt{N} \rceil$ and

$$(65.3) \quad \epsilon = \frac{\delta}{16kn}.$$

For each j consider the binary effect $F_j = I_d/4 + E_j/2$. It belongs to $[I_d/4, 3I_d/4]$. One call to $\mathcal{M}_E$ simulates a call to $\mathcal{M}_{F_j}$: conditional on the original label x , return bit one with probability $3/4$ when $x = j$ and with probability $1/4$ otherwise. This is an equality of channels, including their action on a retained reference, because the bit-one effect is

$$\frac{3}{4}E_j + \frac{1}{4} \sum_{x \neq j} E_x = \frac{1}{4}I_d + \frac{1}{2}E_j.$$

Only public classical randomness is added. Each simulated call uses exactly one call to the unknown device.

Apply Theorem 63.3 to each F_j at future horizon one, accuracy ϵ , and failure η/k , using fresh records for different j . Its output $\hat{F}_j$ is Gaussian rational and legal. The binary identity $d_1(F_j, \hat{F}_j) = 2\|F_j - \hat{F}_j\|_{\text{op}}$ implies that, with probability at least $1 - \eta$, all the Hermitian matrices $B_j = 2\hat{F}_j - I_d/2$ obey

$$(65.4) \quad \max_j \|B_j - E_j\|_{\text{op}} \leq \epsilon.$$

Only a union bound is needed; no independence assertion concerning selected quantum readout outcomes is used.

The matrices B_j need not sum to I_d . Reconstruct the finite legal grid $\mathcal{G}(\epsilon/2)$ and choose its first tuple $\mathbf{G}$ satisfying

$$\|G_j - B_j\|_{\text{op}} \leq 2\epsilon \quad \text{for every } j.$$

Each comparison is an exact rational positive-semidefinite test. On (65.4), such a tuple exists by Lemma 64.3; it has $r(\mathbf{G}, \mathbf{E}) \leq 3\epsilon$. On every other record, if the finite search finds no tuple, return $\mathbf{U}$. Thus every record has a legal output and the same predetermined training budget. The geometric upper bound gives

$$D_N(\mathbf{E}, \mathbf{G}) \leq 6k\sqrt{N}\epsilon \leq 3\delta/8.$$

Encode $\mathbf{G}$ with the dictionary of Theorem 64.4 at parameter $\delta/2$. Its additional error is at most $\delta/4$, so the decoded error is at most $5\delta/8$. It uses no further device calls and has length (65.2).

There are k binary learners, each using at most $C_0\epsilon^{-2}[d^2 + d\log(k/\eta)]$ calls. Since $n^2 \leq 4N$, this proves the upper bound. The inherited rational synthesis and finite-control theorem applies to the simulated binary channels as to any channels of the same form. The public $1/4, 3/4$ relabeling is exact classical control. The error guarantee used in (65.4) is the inherited actual finite-control guarantee, not an idealized readout subsequently assumed exact. The construction terminates in finite classical time, with no assertion that this time is polynomial. For nonrational error parameters, choose rational $\delta' \in [\delta/2, \delta]$ and $\eta' \in [\eta/2, \eta]$ to obtain the stated existence bound; these choices change only absolute constants.

For the lower bound let $m = \lfloor k/2 \rfloor$ and $q = 2m/k \geq 2/3$. Embed any binary interior effect $F \in [I_d/4, 3I_d/4]$ by the tuple

$$(65.5) \quad E_j(F) = \begin{cases} 2F/k, & 1 \leq j \leq m, \\ 2(I_d - F)/k, & m < j \leq 2m, \\ I_d/k, & j = k = 2m + 1. \end{cases}$$

This belongs to $\mathfrak{P}_{d,k}$. A query to this tuple is simulated using one binary F query: with probability q , retain the binary outcome and select a uniform label from its corresponding group of m labels; otherwise erase the bit and return the extra label. In the odd case the erased reference state is the sum of its two subnormalized binary states, as required by the effect I_d/k . Thus the simulation is valid for entangled inputs and adaptive controls, not merely for outcome probabilities on pure states. For even k there is no erasure.

Suppose a k -outcome learner returns any legal tuple $\hat{\mathbf{E}}$ within D_N -error δ . Coarse-graining the first m labels gives a binary effect $H = \sum_{j \leq m} \hat{E}_j$ satisfying $d_N(qF, H) \leq \delta$. This follows by executing the same coarse-graining inside any binary tester of the two induced channels. The Bernoulli lower bound in Theorem 64.1, applied to these two binary effects, implies

$$\|H - qF\|_{\text{op}} \leq 128\delta/\sqrt{N},$$

since $\delta < 1/128$. Spectrally clip H/q to $[I_d/4, 3I_d/4]$, obtaining $\hat{F}$. Weyl's inequality bounds the clipping displacement by $\|H/q - F\|_{\text{op}}$; the triangle inequality therefore gives

$$\|\hat{F} - F\|_{\text{op}} \leq \frac{256}{q} \frac{\delta}{\sqrt{N}} \leq \frac{384\delta}{\sqrt{N}}.$$

No operator-Lipschitz claim about spectral clipping is used. The last accuracy is at most 2^{-14} under the stated cap. The proof of Corollary 60.4 gives the binary operator-loss lower bound already on this interior, with arbitrary coherent adaptive training. Applying it to the simulation proves the left inequality of (65.1), including the contribution $d\log(1/\eta)$ and the scalar case. The same simulation preserves every-record budgets and public stopping. $\square$

The length in (65.2) has the optimal leading order also among learners with a fixed-length, fixed public decoder. Indeed, if failure is less than one at every target, at least one word must decode inside that target's loss ball. The decoder's image is therefore a covering of the entire target family. The lower bound in Theorem 64.2 applies regardless of the number of training

calls. This is separate from the statistical query lower bound. Shared-randomness expected-prefix conventions are different and are not part of this assertion.

65.1. Finite uniform-risk certificates. The certification mechanism also survives a finite outcome alphabet. For an ideal one-call maximally entangled acquisition, its normalized classical-quantum record is

$$(65.6) \quad \Omega(\mathbf{E}) = \frac{1}{d} \bigoplus_{j=1}^k E_j^\top.$$

The classical labels remain ordered. A collective readout after M independent acquisitions can be described by one POVM $(A_{c,x})_c$ on dimension d^M for each classical string $x \in \{1, \dots, k\}^M$. Here $A_{c,x} \succeq 0$ and $\sum_c A_{c,x} = I_{d^M}$. Its output label c is decoded by a fixed finite family $\mathbf{C}_c$ of legal k -outcome tuples.

Theorem 65.2 (Finite-alphabet certificate transfer). *Let $\mathcal{G} \subset \mathfrak{P}_{d,k}$ be a complete finite r_0 -net in the tuple norm r . Suppose a fixed readout satisfies*

$$(65.7) \quad \sum_{c: r(\mathbf{G}, \mathbf{C}_c) \leq a} \sum_{x \in \{1, \dots, k\}^M} \text{tr} \left[A_{c,x} d^{-M} \bigotimes_{i=1}^M G_{x_i}^\top \right] \geq 1 - \alpha \quad (\mathbf{G} \in \mathcal{G}).$$

Then every real $\mathbf{E} \in \mathfrak{P}_{d,k}$ satisfies

$$(65.8) \quad \mathbb{P}_{\mathbf{E}}\{r(\mathbf{E}, \mathbf{C}_J) > a + r_0\} \leq \alpha + \frac{Mkr_0}{2}.$$

If all decoding centres are in $\mathfrak{P}_{d,k}$, the same failure bound holds for D_N -error greater than $2k\sqrt{N}(a + r_0)$. For Gaussian-rational data, rational a, r_0, α , and the net of Lemma 64.3, the hypotheses admit a finite exact verification. A proper prefix of the net or of the classical string list is not a certificate.

Proof. The block-diagonal formula (65.6) gives

$$\|\Omega(\mathbf{E}) - \Omega(\mathbf{G})\|_1 = d^{-1} \sum_j \|E_j - G_j\|_1 \leq kr(\mathbf{E}, \mathbf{G}).$$

Tensor telescoping between normalized states gives

$$(65.9) \quad \|\Omega(\mathbf{E})^{\otimes M} - \Omega(\mathbf{G})^{\otimes M}\|_1 \leq Mkr(\mathbf{E}, \mathbf{G}).$$

Apply the same readout to both states. Its index laws differ in total variation by at most $Mkr_0/2$ when $r(\mathbf{E}, \mathbf{G}) \leq r_0$. Choose such a $\mathbf{G}$. The fixed set of labels $\{c : r(\mathbf{G}, \mathbf{C}_c) \leq a\}$ has probability at least $1 - \alpha$ at $\mathbf{G}$, by (65.7); every label in this set has $r(\mathbf{E}, \mathbf{C}_c) \leq a + r_0$. Transferring this one event proves (65.8). There is no assumption that the success-label sets vary continuously with the unknown target.

The geometric upper comparison proves the assertion about D_N . For exact verification, reconstruct the whole legal net and every classical string. All traces in (65.7) are Gaussian-rational expressions with rational real values. Check complete POVM positivity and normalization at every string, use $sI_d \pm (G_j - C_{c,j}) \succeq 0$ to determine the good labels including equality, and check each rational inequality. These are finite computations. Their soundness on the real parameter body is supplied by (65.9), not by a numerical sampling argument. $\square$

The certificate describes the ideal specified rational readout. When implemented controls have a separately certified output-law error ζ , the right side of (65.8) increases by at most ζ . Such an error must be budgeted, not silently identified with zero. The finite-control guarantee in Theorem 65.1 instead uses the already budgeted binary procedures. The exact certificate transfer and the actual-control learning guarantee thus have explicitly distinct hypotheses.

66. A COVARIANCE PRINCIPLE ON THE CLOSED MEASUREMENT BODY

Let $\mathcal{P}_{d,k}$ denote the complete body of ordered k -outcome measurements on $\mathbb{C}^d$: its elements are Hermitian tuples $E = (E_1, \dots, E_k)$ with $E_j \succeq 0$ and $\sum_j E_j = I_d$. Here $d \geq 1$, $k \geq 2$, and $N \geq 1$ are integers. Zero effects, changing ranks, and noncommuting supports are allowed throughout this section. The operational distance D_N uses unhalved trace norm, retained references, common adaptive control, and at most N calls with bounded public stopping. The device consumes its input and returns only its ordered classical label.

The normalized tangent space and its inner product are

$$\mathcal{T}_{d,k} = \{H_j = H_j^* : \sum_j H_j = 0\}, \quad \langle H, K \rangle = \sum_j \text{tr}(H_j K_j).$$

All operator inequalities below on tuple spaces refer to this real Hilbert structure, unless stated otherwise. For any Hermitian tuple K , not necessarily of zero sum, put

$$(66.1) \quad S_E(K) = \sum_j E_j K_j,$$

$$(66.2) \quad (\mathcal{C}_E K)_j = \frac{1}{2}(E_j K_j + K_j E_j - E_j S_E(K) - S_E(K)^* E_j).$$

The matrix $S_E(K)$ need not be Hermitian. Keeping its adjoint in (66.2) is essential. The operator $\mathcal{C}_E$ is polynomial in the effects, annihilates constant tuples, and has range in $\mathcal{T}_{d,k}$. In particular its restriction to $\mathcal{T}_{d,k}$ is well defined.

For $\tau = N^{-1}$ set

$$(66.3) \quad g_{N,E}(H)^2 = N \langle H, (\mathcal{C}_E + \tau \text{Id})^{-1} H \rangle,$$

$$(66.4) \quad \mathcal{Q}_N(E, F) = g_{N,(E+F)/2}(F - E).$$

The inverse acts on $\mathcal{T}_{d,k}$. Positivity proved below makes it nonsingular even on the spectral boundary. We do not identify $\mathcal{Q}_N$ with an operational distance or assume a triangle inequality for it.

Theorem 66.1 (Closed-body adaptive bound). *For every $E, F \in \mathcal{P}_{d,k}$,*

$$(66.5) \quad D_N(E, F) \leq \min\{2, 2\sqrt{k+1} \mathcal{Q}_N(E, F)\}.$$

More generally, every continuously differentiable path $E(t)$ in $\mathcal{P}_{d,k}$ satisfies

$$(66.6) \quad D_N(E(0), E(1)) \leq \sqrt{k+1} \int_0^1 g_{N,E(t)}(E'(t)) dt.$$

These statements are uniform over all ranks and support patterns. For $k = 2$, $E = (A, I_d - A)$ and $H = (B, -B)$, the restriction is exactly

$$(66.7) \quad (\mathcal{C}_E H)_1 = A(I_d - A)B + BA(I_d - A), \quad \mathcal{Q}_N(E, F)^2 = 2Q_N(A, F_1)^2,$$

where Q_N is the binary comparison modulus in (52.4).

Thus the binary variance operator has a normalized-tuple extension, not an independently regularized list of effects. The theorem gives a global upper certificate. The two-sided balanced-interior classification and its entropy theorem remain Theorems 64.1 and 64.2; a matching general lower bound for (66.4) is a separate question.

66.1. Positive covariance and horizontal decomposition.

Lemma 66.2 (Covariance identities). *On all Hermitian tuples, $\mathcal{C}_E$ is self-adjoint and positive semidefinite, and*

$$\begin{aligned} \langle K, \mathcal{C}_E K \rangle &= \sum_j \operatorname{tr}(E_j K_j^2) - \|S_E(K)\|_{\text{HS}}^2 \\ (66.8) \quad &= \sum_j \|E_j^{1/2}(K_j - S_E(K))\|_{\text{HS}}^2. \end{aligned}$$

For $E_t = (1-t)E + tF$, $0 \leq t \leq 1$,

$$(66.9) \quad \langle K, [\mathcal{C}_{E_t} - (1-t)\mathcal{C}_E - t\mathcal{C}_F]K \rangle = t(1-t)\|S_F(K) - S_E(K)\|_{\text{HS}}^2.$$

The kernel consists precisely of tuples satisfying $E_j^{1/2}(K_j - S_E(K)) = 0$ for every j . Moreover $\mathcal{C}_E = 0$ on $\mathcal{T}_{d,k}$ if and only if the E_j are mutually orthogonal projections, with zero projections permitted.

Proof. Write $A_j = E_j^{1/2}$ and let A be the vertical stack of the A_j . Then $A^*A = I_d$, so $P = I - AA^*$ is an orthogonal projection. On complex matrix tuples with real Hilbert–Schmidt inner product, define

$$(T_A B)_j = A_j^* B_j + B_j^* A_j.$$

Its real adjoint is $(T_A^* K)_j = 2A_j K_j$. Direct multiplication gives

$$(66.10) \quad \mathcal{C}_E = \frac{1}{4} T_A P T_A^*.$$

This proves self-adjointness, positivity, and (66.8). It also proves the kernel claim. Constant tuples are killed and the displayed formula has zero sum, so no extra projection onto $\mathcal{T}_{d,k}$ is implicit.

In the first line of (66.8), the first term is linear in E and $S_E(K)$ is linear in E . Expanding its squared norm gives (66.9).

If the restriction of $\mathcal{C}_E$ to $\mathcal{T}_{d,k}$ vanishes, subtracting the arithmetic mean from any tuple shows that its form vanishes on all tuples. Take $K_j = I_d$ at one index and zero elsewhere. Its form is $\operatorname{tr}(E_j - E_j^2) = 0$. Since $0 \preceq E_j \preceq I_d$, this forces $E_j^2 = E_j$. Their sum being I_d , they are mutually orthogonal: each positive term in $\sum_{i \neq j} E_j E_i E_j = 0$ must vanish. Conversely, for orthogonal projections the summands $E_j K_j$ have mutually orthogonal ranges, and $\|\sum_j E_j K_j\|_{\text{HS}}^2 = \sum_j \|E_j K_j\|_{\text{HS}}^2$. Their covariance form is zero. $\square$

Lemma 66.3 (Regularized horizontal energy). *For $H \in \mathcal{T}_{d,k}$ and $\tau > 0$,*

$$(66.11) \quad \langle H, (\mathcal{C}_E + \tau \operatorname{Id})^{-1} H \rangle = \min_{\substack{A^*B=0, R \in \mathcal{T}_{d,k} \\ T_A B + R = H}} \left(4\|B\|_{\text{HS}}^2 + \tau^{-1}\|R\|_{\text{HS}}^2 \right).$$

If $K = (\mathcal{C}_E + \tau \operatorname{Id})^{-1} H$, a minimizer is

$$(66.12) \quad B_j = \frac{1}{2} A_j (K_j - S_E(K)), \quad R = \tau K.$$

Proof. The constraint $A^*B = 0$ means $PB = B$. Thus $B = P T_A^* K / 4$ gives $T_A B = \mathcal{C}_E K$ and is feasible with $R = \tau K$. Its objective is $\langle K, \mathcal{C}_E K \rangle + \tau \|K\|^2 = \langle H, K \rangle$. For any feasible perturbation $(\Delta B, \Delta R)$, its linear objective term is

$$8\langle B, \Delta B \rangle + 2\tau^{-1}\langle R, \Delta R \rangle = 2\langle K, T_A \Delta B + \Delta R \rangle = 0.$$

The remaining terms are nonnegative. This proves minimality, including at singular E ; no inverse of an effect was used. $\square$

66.2. Adaptive differentiation and boundary passage.

Proof of Theorem 66.1. First suppose every effect at the point of differentiation is positive definite. The tangent $H_0 = T_A B$ in (66.12) has zero sum. For small real s , $E_j + s(H_0)_j$ is positive and the tuple remains normalized. If $R_j(s)$ is its positive square root, then

$$U_j = (B_j - R'_j(0))A_j^{-1}$$

is anti-Hermitian. Indeed, multiplication of $U_j + U_j^*$ on the left and right by A_j gives $A_j B_j + B_j^* A_j - A_j R'_j(0) - R'_j(0) A_j = 0$. The factors $\exp(sU_j)R_j(s)$ therefore realize the derivative B_j while preserving the effects exactly.

Use the measurement isometry with an observed label and an inaccessible duplicate label, together with the factor environment. At the point under consideration it obeys

$$W^* \dot{W} = \sum_j A_j^* B_j = 0, \quad \|\dot{W}\|_{\text{op}} \leq \|B\|_{\text{HS}}.$$

Purify any fixed common N -slot tester. In the inner product of terms differentiated at distinct slots, cancel the common later isometries; the later differentiated slot contains $W^* \dot{W}$ or its adjoint and vanishes. Thus the observed output derivative in the direction H_0 has trace norm at most $2\sqrt{N}\|B\|_{\text{HS}}$. This holds for every reference and memory dimension. The environments used for comparison are not returned to the tester.

The output derivative of the same tester is linear in the channel tangent. For the residual tuple R , one call on a joint state ρ has reference blocks $\text{tr}_A[(R_j \otimes I)\rho]$. Duality against Hermitian contractions bounds the trace norm of the j th block by $\|R_j\|_{\text{op}}$. Inserting the residual at each slot and using trace-norm contraction hence costs at most

$$N \sum_j \|R_j\|_{\text{op}} \leq N\sqrt{k}\|R\|_{\text{HS}}.$$

This is a differential decomposition, not a proposed physical mixture of measurements. With $\tau = N^{-1}$, Cauchy–Schwarz and (66.11) now give

$$\begin{aligned} 2\sqrt{N}\|B\|_{\text{HS}} + N\sqrt{k}\|R\|_{\text{HS}} &\leq \sqrt{k+1} \sqrt{4N\|B\|_{\text{HS}}^2 + N^2\|R\|_{\text{HS}}^2} \\ &= \sqrt{k+1} g_{N,E}(H). \end{aligned}$$

Integrate for the fixed tester, then take the supremum. Padding a publicly stopped experiment with discarded dummy calls makes its original stopped output a common postprocessing of an N -slot experiment, so the same bound applies.

For a boundary path, use $E^\epsilon(t) = (1 - \epsilon)E(t) + \epsilon U$, where $U_j = I_d/k$ and $0 < \epsilon < 1$. The covariance is polynomial and positive semidefinite, so its regularized inverse is continuous. Moreover $g_{N,E}(H) \leq N\|H\|_{\text{HS}}$, uniformly on the closed body. Dominated convergence applies to the integrals. The hybrid bound $D_N(E, F) \leq N \sum_j \|E_j - F_j\|_{\text{op}}$ gives convergence of the endpoint distances. This proves (66.6) on the entire body.

For the segment $E(t) = (1 - t)E + tF$, put $M = (E + F)/2$ and $H = F - E$. Concavity and positivity imply, for $0 < t \leq 1/2$,

$$\mathcal{C}_{E(t)} + \tau \text{Id} \succeq 2t(\mathcal{C}_M + \tau \text{Id}).$$

Inverse order bounds $g_{N,E(t)}(H)$ by $(2t)^{-1/2}g_{N,M}(H)$; use $2(1 - t)$ on the second half. The two scalar integrals sum to 2, proving (66.5). Finally, substituting $K = (B, -B)$ into (66.2) gives (66.7). The factor two in the quadratic form comes from the two equal Hilbert–Schmidt contributions. $\square$

66.3. Classical postprocessing.

Proposition 66.4 (Data processing of horizontal energy). *Let $T = (T_{aj})$ be a column-stochastic classical channel and $F_a = \sum_j T_{aj} E_j$. Write $(T^* L)_j = \sum_a T_{aj} L_a$. Then, on all Hermitian tuples,*

$$(66.13) \quad \langle L, \mathcal{C}_F L \rangle \geq \langle T^* L, \mathcal{C}_E(T^* L) \rangle.$$

Consequently the extended quadratic energy

$$\mathcal{I}_E(H) = \sup_K \{2\langle H, K \rangle - \langle K, \mathcal{C}_E K \rangle\}$$

satisfies $\mathcal{I}_F(TH) \leq \mathcal{I}_E(H)$. The value may be infinite at a boundary; when finite it is $\langle H, \mathcal{C}_E^\dagger H \rangle$.

Proof. The two S matrices coincide. The difference of the covariance forms is therefore

$$\sum_j \text{tr} \left[E_j \left(\sum_a T_{aj} L_a^2 - \left(\sum_a T_{aj} L_a \right)^2 \right) \right] \geq 0.$$

The bracket equals the positive sum $\frac{1}{2} \sum_{a,b} T_{aj} T_{bj} (L_a - L_b)^2$. Substitution in the variational definition and enlargement from $K = T^* L$ to all K prove the energy inequality. The pseudoinverse statement follows by diagonalizing the real self-adjoint covariance; a component along its kernel makes the supremum infinite. $\square$

66.4. A uniform noise crossover, sharp in horizon and noise.

Theorem 66.5 (Regularization of arbitrary support patterns). *For $0 \leq \epsilon \leq 1$ put $E^\epsilon = (1 - \epsilon)E + \epsilon U$. On $\mathcal{T}_{d,k}$,*

$$(66.14) \quad \mathcal{C}_{E^\epsilon} \succeq (1 - \epsilon)\mathcal{C}_E + \frac{\epsilon}{k} \text{Id}.$$

In particular, for arbitrary $E, F \in \mathcal{P}_{d,k}$,

$$(66.15) \quad D_N(E^\epsilon, F^\epsilon) \leq \min \left\{ 2, \frac{2\sqrt{k+1}(1-\epsilon)N}{\sqrt{1+N\epsilon/k}} \|F - E\|_{\text{HS}} \right\}.$$

The dependence on N and ϵ is attained, up to constants depending only on k , by a two-dimensional family. Namely, let $P = |0\rangle\langle 0|$, $P_\theta = |u_\theta\rangle\langle u_\theta|$, $u_\theta = \cos \theta |0\rangle + \sin \theta |1\rangle$, and

$$E = (P, I_2 - P, 0, \dots, 0), \quad F = (P_\theta, I_2 - P_\theta, 0, \dots, 0).$$

For $0 \leq \theta \leq \pi/2$, $0 \leq \epsilon \leq 1/2$, and $z = (1 - \epsilon)N \sin \theta / \sqrt{1 + N\epsilon}$,

$$(66.16) \quad \frac{\min\{1, z\}}{16384} \leq D_N^{\text{na}}(E^\epsilon, F^\epsilon) \leq D_N(E^\epsilon, F^\epsilon) \leq 4\sqrt{k(k+1)} \min\{1, z\}.$$

Proof. For $K \in \mathcal{T}_{d,k}$, $S_U(K) = 0$ and $\mathcal{C}_U K = K/k$. Concavity proves (66.14). Apply it at the midpoint and use Theorem 66.1 to obtain (66.15).

For the displayed family, keep labels one and two and send every other label to an independent fair bit. This common postprocessing produces the binary effects $B = \beta P + \epsilon I_2/2$ and $B_\theta = \beta P_\theta + \epsilon I_2/2$, where $\beta = 1 - \epsilon$. The equality holds on subnormalized reference states. It therefore preserves the nonadaptive lower witnesses as well as channel postprocessing. The midpoint eigenvalues are $(1 \pm \beta \cos \theta)/2$, and direct substitution in (52.5) gives

$$Q_N(B, B_\theta)^2 = \frac{4N^2 \beta^2 \sin^2 \theta}{2 + N(1 - \beta^2 \cos^2 \theta)} \geq \frac{4z^2}{2 + z^2}.$$

Here $1 - \beta^2 \leq 2\epsilon$ and $N \geq 1$ were used. It follows that $\min\{1, Q_N(B, B_\theta)\} \geq \min\{1, z\}$. The binary lower theorem in dimension two gives the first inequality of (66.16). For the upper, $\|F - E\|_{\text{HS}} = 2 \sin \theta$ and $(1 + N\epsilon)/(1 + N\epsilon/k) \leq k$. Combine (66.15) with $D_N \leq 2$. $\square$

The family passes from projections and zero effects at $\epsilon = 0$ to strictly positive effects at every $\epsilon > 0$. Thus a bound uniform over the closed body must accommodate both the N scale and the $\sqrt{N}/\epsilon$ scale. This conclusion does not assert a matching entropy law on all multi-outcome boundary strata.

66.5. Exact evaluation without effect square roots.

Proposition 66.6 (Polynomial-size rational certificate). *For Gaussian-rational $E, F \in \mathcal{P}_{d,k}$ and integer $N \geq 1$, $\mathcal{Q}_N(E, F)^2$ is rational and can be evaluated by exact linear algebra on a positive definite real system of dimension $p = (k - 1)d^2$. The bit complexity is polynomial in d, k , the total input bit length, and $\log N$. Legality is decidable by exact positive-semidefinite tests. No effect square root, ODE integration, rank decision by tolerance, or enumeration of a dictionary is required.*

Proof. Use the rational Hermitian matrix basis consisting of diagonal units and real and imaginary off-diagonal pairs. For each of the first $k - 1$ slots, insert one matrix basis element there and its negative in the last slot. These tuples B_a form a rational basis of $\mathcal{T}_{d,k}$; they need not be orthonormal. Form

$$G_{ab} = \langle B_a, B_b \rangle, \quad L_{ab} = \langle B_a, \mathcal{C}_{(E+F)/2} B_b \rangle, \quad b_a = \langle B_a, F - E \rangle.$$

All entries are rational, G is positive definite, and $L \succeq 0$. Solve $(L + N^{-1}G)x = b$. Then $\mathcal{Q}_N^2 = Nb^\top x$. Omitting the Gram matrix would compute a different regularization. Fraction-free elimination performs polynomially many rational arithmetic operations; clearing input denominators and applying determinant bounds gives polynomial bit length for all minors and for the output. The same reasoning applies to exact symmetric elimination for each effect's positivity. The complexity statement concerns this certificate evaluator, not the exhaustive public code or collective-readout synthesis. $\square$

67. SEARCH-FREE LEGALIZATION OF BALANCED ESTIMATES

The exact public dictionary and the statistical legalization step have different purposes. A dictionary exploits global packing to reduce transmitted length. Legalizing a collection of component estimates need not reconstruct such a dictionary.

Theorem 67.1 (Affine repair with a uniform error bound). *Let $B = (B_1, \dots, B_k)$ be Hermitian matrices and $a > 0$. Define*

$$(67.1) \quad A_j = B_j + \frac{I_d - \sum_i B_i}{k},$$

$$(67.2) \quad L_a(B)_j = \frac{A_j + 4aI_d}{1 + 4ka}.$$

Suppose $E \in \mathfrak{P}_{d,k}$, the balanced interior from (64.1), and $\max_j \|B_j - E_j\|_{\text{op}} \leq a$. Then $L_a(B) \in \mathfrak{P}_{d,k}$ and

$$(67.3) \quad \max_j \|L_a(B)_j - E_j\|_{\text{op}} \leq \frac{4a}{1 + 4ka} \leq 4a.$$

For rational input and rational a , this is a rational map. Testing its balanced legality and returning the uniform tuple upon failure defines a total legal procedure on every Hermitian input record. It uses $O(kd^2)$ rational arithmetic operations apart from $2k$ exact positive-semidefinite tests of order d , and polynomial bit complexity. There is no grid enumeration.

Proof. Let $\Delta_j = B_j - E_j$. Then $A_j - E_j = \Delta_j - k^{-1} \sum_i \Delta_i$, whose norm is at most $2a$, and $\sum_j A_j = I_d$. Consequently

$$(1/(2k) - 2a)I_d \preceq A_j \preceq (3/(2k) + 2a)I_d.$$

Adding $4aI_d$ and dividing by $1 + 4ka$ gives exactly the two balanced spectral inequalities. The sum of the repaired effects is I_d . Finally,

$$L_a(B)_j - E_j = \frac{A_j - E_j + 4ka(I_d/k - E_j)}{1 + 4ka}.$$

The balanced hypothesis gives $\|I_d/k - E_j\|_{\text{op}} \leq 1/(2k)$, proving (67.3). Rationality and the operation count follow from the formula. The exact legality test makes no good-event assumption; if it fails the uniform tuple is an explicitly specified legal output. On the good event it never fails. $\square$

Corollary 67.2 (A common learner with polynomial legalization). *Under the range and access conventions of Theorem 65.1, replace its tuple-grid legalization by the guarded map in Theorem 67.1. Taking*

$$a = \frac{\delta}{32k \lceil \sqrt{N} \rceil}$$

and allocating failure η/k to each fresh binary component learner gives, on every record,

$$M \leq Ck^3 N \delta^{-2} [d^2 + d \log(k/\eta)]$$

for an absolute C , with future-use error at most $\delta/4$ on a common event of probability at least $1 - \eta$. Encoding with the public dictionary at parameter $\delta/2$ gives error at most $\delta/2$ on the same event, without further calls, and retains the optimal fixed- k fixed-length description order. Only the legalization cost is made polynomial; neither dictionary construction nor general collective-readout synthesis is asserted to be efficient.

Proof. Use precisely the reference-preserving binary simulation and actual finite-control component learners from Theorem 65.1, at the displayed smaller accuracy. Its conditional good-event argument gives component error at most a with total failure at most η . Affine repair then gives tuple error at most $4a$. The balanced metric upper bounds the future-use error by $8k\sqrt{N}a \leq \delta/4$. The dictionary adds at most $\delta/4$. Fresh calls and all finite-control budgets are charged exactly as in the component theorem; decreasing the component accuracy by a factor two changes its call bound only by an absolute factor. The guarded map is total even off the good event, so the recordwise call bound and public stopping convention are unchanged. $\square$

68. MEASUREMENT DISCRIMINATION, BLOCK RESOURCES, AND LEARNING

The present binary interface retains the ordered classical outcome and an arbitrary external reference, but no postmeasurement quantum system from the device. This fixes the comparison with the measurement-discrimination literature. The projective theorem of Puchała and collaborators [56, Corollary 1 and Theorem 2] gives an exact multiple-use formula with ancillary access. It is used in the retained observable-readout theorem. The restricted ancillary interface in Fiurášek–Mičuda [59] leads to a different comparison of adaptive and parallel probes.

Krawiec–Pawela–Puchała [60, Section 4] construct a pair of nine-outcome qutrit rank-one POVMs that admit perfect adaptive discrimination with two calls and no perfect discrimination by any finite parallel experiment. Datta–Biswas–Saha–Augusiak [61, Theorems 3–5] study entanglement-assisted single-use discrimination, including perfect constructions and a qubit entanglement advantage. Their qubit construction has three outcomes. These are direct antecedents for general POVM discrimination with a retained reference. They do not supply a full binary-effect

family covering law. Our comparison (52.23) permits an adaptive advantage and controls its size uniformly in N for fixed input dimension.

The differential argument in Lemma 52.2 belongs to the established horizontal-gauge and channel-extension method [62, 47, 63, 64]. Yuan–Fung give the parallel path estimate [65, Section III, equation (26), PDF version 3]; Sieniawski–Demkowicz–Dobrzański integrate adaptive bounds [66, Section IV.A]. The binary Sylvester choice here gives the specific regularized variance $M(I - M)$, which can be integrated explicitly and matched by compressed nonadaptive tests. The inverse-Sylvester expressions of Bures geometry [67] and its spectral integration [68] are related methods. Our form acts on the effect displacement H with variance $V = M(I - M)$; it is not the ordinary Bures pullback under $M \mapsto V$, whose differential would be $H - MH - HM$. The operational-ball lemma and the nested endpoint integral are both required for Theorem 53.1.

68.1. One-use tomography and its acquisition resources. Mele–Bittel [45, Theorem I.1] give channel tomography with $O(d_{\text{in}}d_{\text{out}}k\epsilon^{-2})$ calls at fixed confidence. Their binary specialization [45, Lemmas III.7–III.8 and Corollary III.9] returns a legal effect, with operator error at most ϵ , using

$$M = 1024d^2\epsilon^{-2} + O(d\epsilon^{-2}\log(1/\eta)), \quad 4e^{-4d^2} < \eta < 1, \quad 0 < \epsilon < 1.$$

The construction prepares independent maximally entangled input/reference pairs, then processes the Choi copies collectively [45, Algorithm I.2, Figure 1 and Section III.2]. Consequently it belongs to the fresh-block class with $b = 1$. Calling all invocations one parallel block, as the source does, does not entangle the inputs of distinct invocations. Coherent purification and collective output tomography are allowed by our $b = 1$ class. For all $0 < \eta \leq 1/8$, the amplification in [45, Remark III.4, equation (165)] gives the sufficient budget $O(d^2\epsilon^{-2}\log(1/\eta))$ with the same acquisition restriction. Section 60 further derives $O(\epsilon^{-2}[d^2 + d\log(1/\eta)])$ for every $0 < \eta \leq 1/8$ from the same Corollary III.9, using a dimension-dependent base confidence and standard repetition. That extension is distinguished from the source’s printed restricted-confidence corollary and its stated amplification.

Zambrano–Ramos-Calderer–Kueng [69, Definition 1 and Theorem 2] use worst-input one-use outcome total variation. For L outcomes, their global-2-design protocol has sufficient budget

$$M \geq \frac{8(d^2 + \epsilon(d^2 + 1)/6)}{\epsilon^2} \log \frac{2^{L+1}9^{2d}}{\eta}.$$

Known inputs are probed separately; least squares and projection return a legal POVM. This is also $b = 1$, with no retained reference. For $L = 2$ their loss is $\|E - F\|_{\text{op}}$, whereas ours has $d_1(E, F) = 2\|E - F\|_{\text{op}}$. Their Theorem 9 gives a $\Omega((d^3 + d^2L)\epsilon^{-2})$ lower bound at constant confidence for nonadaptive single-copy measurements on known input states. This restriction stays attached to that lower bound.

68.2. What stronger future-loss analysis can establish. The hybrid inequality

$$d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$$

transfers both one-use guarantees to sufficient budgets of order $O_d(N^2\delta^{-2}\log(1/\eta))$. By itself this remains an upper-bound conversion. Theorem 57.2 and Corollary 57.5 now give the distinct converse: for every fixed $d \geq 2$, every fresh $b = 1$ acquisition procedure that learns the entire ordered binary effect body under d_N requires

$$M \geq cN^2\delta^{-2}\log(1/\eta).$$

The statement permits classical adaptive choices, stored reference outputs, their collective processing, and public bounded stopping. It therefore applies to the acquisitions in both cited tomography protocols, including any reconstruction from the same acquired systems. No reanalysis of those acquisitions can give the uniform full-body $O_d(N\delta^{-2}\log(1/\eta))$ guarantee.

There is nevertheless a useful positive reanalysis on a promised interior class. Theorem 58.5 applies Mele–Bittel’s binary estimator with $\epsilon = \delta/(8\sqrt{N})$ and $\eta = 1/8$ to

$$\frac{1}{4}I \preceq E \preceq \frac{3}{4}I.$$

The horizontal path estimate gives $d_N(E, F) \leq 4\sqrt{N}\|E - F\|_{\text{op}}$ when $\|E - F\|_{\text{op}} \leq 1/8$. Thus $O(d^2 N \delta^{-2})$ fresh single-call probes suffice at this confidence. Theorem 58.2 proves the matching $\Omega(d^2 N \delta^{-2})$ lower bound even for arbitrary coherent adaptive training. This is a joint dimension, horizon and accuracy law on the displayed interior class at constant confidence. The projective boundary supplies the additional full-body $b = 1$ obstruction. The two conclusions answer different quantifiers in the estimator-reanalysis question.

68.3. The full-body resource law and its antecedents. For $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$ and $0 < \eta \leq 1/8$, Theorem 57.2 states

$$c \frac{N^2}{b} \delta^{-2} \log(1/\eta) \leq M_b^*(d, N, \delta, \eta) \leq C d^4 \frac{N^2}{b} \delta^{-2} \log(d/\eta).$$

Here M_b^* is the worst-case call budget for a common classical estimate in the fresh-block model of Section 57. A block may use its own reference and adaptive quantum controls; only the classical record can determine the next fresh probe. Stored quantum outputs may be processed jointly but cannot become quantum inputs to a later block. The bound holds on every record. At $b = N$ it recovers the retained fixed-dimensional learning order; at $b = 1$ it proves the direct quadratic-horizon obstruction. The scalar case $d = 1$ instead has order $N \delta^{-2} \log(1/\eta)$, independently of b .

The all-confidence operator upper bound also improves the explicit full-body dimension dependence. Applying Lemma 60.3 at $\epsilon = \delta/(2N)$ and using $d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$ gives the independent-Choi budget $C N^2 \delta^{-2} [d^2 + d \log(1/\eta)]$. This $b = 1$ procedure is admissible under every larger block cap. Choosing between it and the retained block construction yields (60.9), namely

$$M_b^*(d, N, \delta, \eta) \leq C \frac{N^2}{\delta^2} \min \left\{ d^2 + d \log \frac{1}{\eta}, \frac{d^4}{b} \log \frac{d}{\eta} \right\},$$

for $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$ and $0 < \eta \leq 1/8$. The choice uses only public parameters. This is a sufficient-budget improvement from the credited estimator and the hybrid conversion; the lower and upper dimension dependences on the full body remain unmatched.

Result and acquisition	Loss and parameter range	Training-call order
Mele–Bittel plus Lemma 60.3; independent Choi probes, $b = 1$	Entire binary body; one-use operator error ϵ ; $0 < \eta \leq 1/8$	$O(\epsilon^{-2}[d^2 + d \log(1/\eta)])$; source estimator with a derived confidence extension
Zambrano et al., known individual probes; $b = 1$	Binary specialization of global designs; one-use operator error ϵ ; $0 < \eta < 1$	$O(d^2 \epsilon^{-2}[d + \log(1/\eta)])$
Theorem 60.1, independent Choi probes; $b = 1$	$\frac{1}{4}I \preceq E \preceq \frac{3}{4}I$; future d_N error δ ; $d, N \geq 1$, $0 < \eta \leq 1/8$	$\Theta(N \delta^{-2}[d^2 + d \log(1/\eta)])$; lower allows fully coherent training
Theorem 62.2; synthesized finite readout, $b = 1$	Same interior and future loss; rational δ, η in the stated range	$O(N \delta^{-2}[d^2 + d \log(1/\eta)])$; exact public dictionary and algebraic trusted controls
Theorem 57.2, fresh blocks of size $\leq b$	Entire binary body; future d_N error δ ; $d \geq 2$, $1 \leq b \leq N$, $0 < \eta \leq 1/8$	$\Theta_d((N^2/b) \delta^{-2} \log(1/\eta))$; improved explicit upper in (60.9)

All new learning rows have an absolute small-error cap. The full-body dimension dependence is not sharp; neither this table nor the block theorem proves a growing-dimensional minimax law for that body.

The metrological block scaling is established background. Hyllus et al. [73, Observation 1, equation (9)] and Tóth [74, Observation 3, equation (6)] prove, for b -producible qubit probes under linear phase encoding,

$$F_Q \leq \lfloor M/b \rfloor b^2 + (M - b\lfloor M/b \rfloor)^2 \leq Mb.$$

Our block parameter also permits sequential quantum controls inside a fresh block; it is not a general measure of multipartite entanglement in the final state. The finite-pair fidelity proof supplies the confidence factor without an unbiasedness assumption. Salek–Hayashi–Winter [49, Section II.2 and Section V.2] already formulate classical feed-forward with arbitrary quantum memory at the output and no quantum memory at the input, and analyze asymptotic discrimination exponents. Wilde et al. [54, Proposition 21 and Appendix B] give adaptive divergence converses and fidelity-to-error inequalities. Section 57 checks the finite-budget conditional fidelity argument, variable block lengths and stopping explicitly. The all-effect upper bound follows by blocking the future tester and invoking the retained common learner at the shorter horizon. The asserted increment is this common future-loss minimax statement and its application to specified tomography acquisitions; the Mb metrological scale is not new.

A different literature bounds how many already supplied state copies can be measured jointly [76, 77]. Those output measurement restrictions differ from the fresh-query restriction here, which allows an unrestricted final joint measurement. Mele–Bittel’s proposed blockwise Choi processing [45, Section I.5] concerns that other resource.

Recent channel-learning lower bounds also require careful domain comparison. Oufkir–Girardi [46] and Chen–Zhang–Yu [75, Theorems 1.1 and 4.2] treat fully coherent channel learning in one-use diamond loss. The latter gives $\Omega(rd_{\text{in}}d_{\text{out}}\epsilon^{-2})$ when $rd_{\text{out}} \geq 2d_{\text{in}}$ at constant confidence. A two-dimensional quantum output need not be classical, so this general-channel theorem is not imported as a binary-qc lower bound. Mele–Bittel’s actual binary converses are their projector packing [45, Proposition IV.17] and the separate diagonal-coin bound $\Omega(d\epsilon^{-2}\log(d/\eta))$ [45, Corollary IV.20]. The self-contained near-fair binary information argument in Lemma 58.1 yields the growing-dimension lower bound used here. Its relation to other channel hard-instance constructions remains a specific priority comparison, not a claim that such lower-bound methods are new.

68.4. All-confidence interior and operator learning. Theorem 60.1 extends the retained constant-confidence law to

$$M_{\text{int}}^*(d, N, \delta, \eta) = \Theta\left(N\delta^{-2}[d^2 + d\log(1/\eta)]\right)$$

with absolute constants, for all integers $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$. The upper uses independent Choi inputs and hence $b = 1$; the lower allows unrestricted coherent adaptive training. Corollary 60.4 gives the corresponding binary operator-learning order $\Theta(\epsilon^{-2}[d^2 + d\log(1/\eta)])$ for $0 < \epsilon \leq 2^{-14}$.

The upper bound continues to use the existing Mele–Bittel binary estimator. The base failure $q_d = 2^{-3d}$ satisfies the hypothesis $4e^{-4d^2} < q_d$ of their Corollary III.9. Running it to a fixed fraction of the desired accuracy and repeating independently a number of times equal to the smallest positive odd integer at least $2\log_2(1/\eta)/d$ permits a majority-cluster selection. The usual binomial estimate gives error at most η and the displayed additive $d^2 + d\log(1/\eta)$ budget. This is a confidence extension of an imported estimator, not a new tomography primitive or a claim that amplification is new.

For the confidence converse, the relevant direct antecedent is Mele–Bittel’s classicalisation of diagonal multiple-coin channels [45, Lemma IV.19 and Corollary IV.20]. Their corollary gives

$\Omega(d\epsilon^{-2}\log(d/\eta))$ for small operator error and $0 < \eta < 1/4$. Lemma 60.2 provides the explicit null-versus-coordinate argument under the present coherent adaptive interface and future-use loss. Revealing the sampled diagonal index isolates the hypothesis-dependent Bernoulli coin after a common basis-dephasing map. The classical–quantum relative-entropy calculation permits different current quantum inputs under the two hypotheses. Summing the resulting relative-entropy testing inequalities yields $\Omega(dN\delta^{-2}\log(1/\eta))$. Combining this with Theorem 58.2 uses the maximum of two lower bounds, and hence their sum up to a constant; it does not multiply unrelated hard-instance bounds. The claim is this joint operational law and the stated proofs, with the existing diagonal-coin argument credited at its use. No priority clearance is inferred from the source comparison.

68.5. Finite synthesis, efficient tomography, and exact coding. Theorem 59.5 uses measurable finite coarsening of an imported ideal measurement. A finite outcome set alone does not supply computable measurement entries. Section 62 addresses that separate question. Lemma 62.1 writes the risk of a finite collective readout on independent Choi records as a first-order formula over the reals with rational coefficients. The formula quantifies over every real effect in the promised interior, includes positivity and completeness of the conditional POVMs, and uses the fixed exact public dictionary. Real quantifier elimination decides feasibility; transfer between real closed fields supplies an algebraic point, which can be found by exact enumeration and sign tests [78, Theorems 2.77 and 2.80 and Chapter 14]. Searching the sample count in order terminates because the imported statistical upper bound supplies a feasible witness at the stated query order. No entries of the Mele–Bittel measurement need be computed in this argument.

The resulting Theorem 62.2 gives a finite algebraic readout, a finite trusted-control precision prescription, and the same query bound $O(N\delta^{-2}[d^2 + d\log(1/\eta)])$ for rational public errors. Its decoded word remains an exact legal rational effect, with

$$B = \frac{1}{2}d^2 \log_2 N + d^2 \log_2(1/\delta) + O(d^2).$$

The precision budget is imposed on the complete preparation and conditional readout instruments, including all classical branches. It does not require coherent access to the device’s classical outputs or environment. The theorem places no useful bound on classical preprocessing time, gate count, workspace or algebraic description length.

Finite and efficient quantum tomography have substantial prior constructions. Pelecanos–Spilecki–Tang–Wright [79, Theorems 1.1–1.3, Section 3.3 and Figures 8–9] give an efficient purification reduction and efficient pure- and mixed-state tomography; their Theorem 4.2 supplies discrete approximate-design circuits used in the pure-state protocol. Those results should not be described as merely formal infinite measurements. Conversely, an efficient state-tomography procedure does not automatically preserve the covariance and binary operator-error analysis used in the imported channel estimator. The present proof avoids that implementation question by synthesizing a different feasible measurement. Its assertion is finite computability with a universal future-loss certificate and the exact dictionary, not a first finite implementation or an efficient realization of the existing estimator.

68.6. Retained calibration and exact descriptions. Multiscale phase recovery is established [71, 72]. The qubit learner proves its own safe-angle induction, bias cancellation, unknown-noise block selection and conditional confidence allocation. The matrix learner calibrates both support endpoints, controls the variance created by data-selected compressions, and assembles a legal estimate. These supply one procedure for the entire closed binary effect body, including rank changes and repeated eigenvalues. Pair-dependent tests in the metric proof retain their different role.

The deterministic rational construction in Theorem 56.1 supplies a finite exact selection rule and optimal-order payload. Exhaustive reconstruction time, workspace and state preparation remain separate from call count and payload. No efficient high-dimensional encoder is inferred. The closed-body entropy constants remain fixed-dimensional and its matching entropy and payload orders have a small-error range. In contrast, Theorem 59.1 has explicit dimension-uniform constants on $[I/4, 3I/4]$. Its norm-ball volume argument and the Fano–Kraft proof of Theorem 59.4 use standard mechanisms. Their role is to derive the precise future-loss description statements, not to claim priority for volume comparison or information inequalities. The retained v79 learned-description theorem credits the Mele–Bittel estimator and retains its ideal trusted operations; it adds finite-output coarsening and a public dictionary without extra device calls. The additional finite synthesis theorem above supplies a separate computability argument for the strengthened confidence law. Multiple-outcome measurements, residual quantum outputs, sharp full-body growing-dimensional learning, and efficient exact reconstruction require additional arguments.

69. COMPARISON AND BOUNDARY GEOMETRY

69.1. Adaptive discrimination and description. Salek–Hayashi–Winter [49, Sections II–V] study asymptotic binary channel-discrimination exponents. Their results include equal adaptive and nonadaptive exponents for classical–quantum channels, and separations for some entanglement-breaking channels. Our object is instead a covering number of a supplied family in the finite-use metric d_N . Its lower packing may use a different binary test for each pair; it does not supply a universal measurement which identifies every codeword. The exact reduction in Lemma 41.1 uses input erasure, not merely entanglement breaking, and implies no general absence of an adaptive advantage for entanglement-breaking channels.

Cooney–Mosonyi–Wilde [50] is a direct antecedent for the input-erasure mechanism. Its two-replacer discussion and asymptotic channel-versus-replacer exponents are distinct from an exact finite-use covering problem. The retraction of Theorem 42.1 concerns admissible centres, not an improvement to those discrimination exponents. The coherent amplification used here likewise has the established unitary discrimination antecedents [51, 52].

The two-point programme method has a metrological antecedent in [47]. Likewise the half-parameter scale has antecedents in quantum population compression [48]; there the input consists of unknown physical copies and the memory interface may include quantum storage. Here the encoder is given classical matrix data and the decoder specifies an operation or state. These are different coding tasks, even when a logarithmic coefficient has the same form.

Mele–Bittel [45] and Oufkir–Girardi [46] investigate learning unknown channels in diamond distance. The present encoder receives its target and asserts no query or sample bound. The reference-sensitive physical classical-simulation problem of Naik–Zartab–Gisin–Banik [44] is different again: a rational description with a diamond guarantee remains a description of a quantum operation, not a finite-classical-message implementation accepting an unknown entangled input. No bound here is asserted for the hardware dimension of a universal programmable processor.

69.2. Ranks, tangent directions and remaining joint regimes. For a general instrument J , let P_y project onto the support of J_y . The one-sided tangent cone of the legal instrument body is

$$T_{\mathcal{C}}(J) = \overline{\{t(L - J) : t \geq 0, L \in \mathcal{C}\}}.$$

Every element H of this cone satisfies the marginal constraint and $P_y^\perp H_y P_y^\perp \geq 0$. These necessary conditions identify where positivity becomes active; by themselves they assert no reuse modulus. A boundary classification must examine the metric of repeated use along support-changing directions and along reversible coherent directions, rather than count affine coordinates alone. Proposition 39.5 shows that a coherent unitary phase can have scale N^{-1} , whereas the preparation family in Theorem 41.3 has an $N^{-1/2}$ coding scale even at rank-deficient outputs. In that

preparation family the Choi ranks are d rank σ_y , and the rank bounds in the theorem are output ranks. They are not arbitrary Choi-rank strata of all instruments.

Theorem 43.1 gives a joint law in the non-input-erasing tensor family $UXU^* \otimes \sigma_y$. It includes a full projective unitary manifold and support-changing preparation blocks, but no input-dependent measurement probabilities. Its mixed coefficient $u + v/2$ is established by two explicit adaptive moduli and matching product packings, not by a tangent-dimension count alone. These qualifications are essential: it is not a classification of all coherent/noisy couplings.

Theorems 40.3 and 41.3 settle the joint N, δ dependence in their specified fixed-dimensional families. They do not assert dimension-uniform constants, a uniform limit as the positive margin tends to zero, or a full tangent-cone classification for general disturbing instruments. The older exponential-accuracy width result retains its multiplicative gap in the separate label-width resource. None of these description laws is a mutable-workspace converse for general variable-input trajectory simulation.

69.3. Input dependence and a common full-range witness. The fixed-readout family of Theorem 45.1 is a classical–quantum channel with a fixed quantum input measurement. Its N -slot programme selects conditional row states and therefore handles input-dependent probabilities. This is within the channel families considered by Salek–Hayashi–Winter [49, Sections II and V] for asymptotic binary discrimination. Their exponent comparisons are direct antecedents, but are not a finite-use family covering number or a rational rank-aware code. We do not infer equality between the row-product upper bound and the adaptive distance, nor a finite-use prohibition on adaptive advantage. The outcome/state rows may be noncommuting and have zero blocks. The measured basis remains fixed, so a varying measurement basis and general coherent/noisy couplings are not parameterized by V .

The common-estimator argument in Section 44 uses standard informationally complete tomography [42] and an elementary event-counting inequality. Its point is to control covering multiplicity at error greater than one, where the earlier requirement of pairwise distance greater than 2δ is unavailable. This witness is deliberately common to all packing points; the local small-error proof still uses pair-dependent binary tests. Neither witness turns the supplied-description code into an unknown-instrument learning result. The full-range extension concerns half-dimensional coding scales and does not extend the coherent N^{-1} scale by fiat. The author-side comparisons here identify premises and conclusions; they do not constitute an independent exhaustive priority review.

69.4. Environment-state antecedents and the observable extension. Das–Wilde [53, Definition 3 and Proposition 2] formulate a common interaction for an environment-parametrized memory cell and reduce adaptive reading to processing its environment states. This is a direct antecedent of our all-row programme upper bound. It does not by itself provide one-use seizing of all those row states. Wilde–Berta–Hirche–Kaur [54, Definition 36 and Theorem 37] give finite-use discrimination equalities for environment-seizable channels. Wang–Wilde [55, Section III.A] formulate their channel-box equivalence under common transformations. We use these state-processing principles explicitly, and do not claim to introduce either environment parametrization or seizing. Theorem 47.1 records a family-uniform ambient left inverse and transfers arbitrary covering centres; the rank-aware rational code then supplies a finite-use family covering law. The Pauli/Bell protocol is a standard example of the premise, not a new source of discrimination advantage.

The imported theorem [56, Corollary 1 and Theorem 2] determines the adaptive multiple-query distance of ordered projective measurements. It supplies the linear- N readout modulus in Theorem 46.1. Our contribution in this use is the mixed family covering, direct rational projection chart and conditional rank-preserving code, not the measurement-discrimination formula. Its flag

dimension $d^2 - d$ differs from the projective-unitary dimension $d^2 - 1$ in the retained coherent tensor theorem.

Fiurášek–Mičuda [59, Sections I, III and IV] study two uses of projective qubit measurements, with the decision based on the two device outcomes and no additional ancillary measurement. Within that class, adaptive product probes improve fixed product probes; entangled probes with feed-forward allow perfect discrimination for sufficiently separated measurements. This is a direct antecedent for the finite-use adaptive interface. The parallel optimality theorem of [56, Theorem 2] allows ancillary systems and a final ancillary measurement, so the two comparisons have different admissible tests. Neither supplies a noisy-family metric comparison or a covering theorem.

Varying readout is treated when the readout remains observable, and uniformly seizable families have an exact covering reduction. A further boundary problem is to classify nonrecoverable varying readout with overlapping conditional outputs, or coupled reversible and support-changing directions lacking a common ambient seizing map. The exact equal-row collapse in Remark 46.6 shows that this requires an identifiability analysis before assigning any basis exponent. These observable-readout and seizing theorems and the earlier full-error half-dimensional laws do not assert a general tangent-cone classification or uniform constants in growing dimension.

69.5. Finite-distance noise transition and its antecedents. Demkowicz-Dobrzański, Kołodyński and Guţă [47, Eq. (4) and the dephasing example] use classical channel simulation to derive metrological bounds, including the noise factor $\sqrt{1 - \lambda^2}/\lambda$ for dephasing (their visibility notation is different). Thus neither the noisy Heisenberg-to-square-root transition nor common convex simulation is claimed as a new principle. Our Lemma 48.2 instead keeps an exact two-point programme and finite- N fidelity expression. A proof by finite entangled blocks gives the reverse modulus, without turning a local estimation bound into a global distance assertion.

Theorem 48.1 counts legal reusable descriptions at all public visibilities, including sequences approaching zero or one. Its constants are uniform in noise, horizon and small error. Circle charts give an exact rational upper code, and metric packing allows centres outside the equatorial family. The conditional-state corollary combines this noise-dependent direction with the retained rank-state geometry. These are the asserted covering and realization consequences; no independent exhaustive priority clearance is implied by identifying their ingredients. The pair-dependent programme is not the family-uniform ambient seizer required by Theorem 47.1. The programme methods of [53, 54, 55] remain direct antecedents, not inferred novelty of the present proof.

For a general coupled boundary, the remaining question is how a nonidentifiable readout or a non-product support change modifies these operational scales. The equal-row collapse remains an exact warning against counting a basis that cannot be observed. The fixed-visibility noise-uniform theorem supplies one explicit transverse approach to an extremal readout, not a classification of all such directions, nor a large-error consequence of pairwise packing.

69.6. Joint visibility and boundary contact. The single-shot white-noise example of Sedlák–Ziman [57] already permits different visibilities. The single-shot projective measurement theorem of [58] and the multiple-shot theorem used in Theorem 46.1 have different hypotheses and conclusions. In particular, neither the one-use formula nor the projective multiple-use formula is an exact noisy-ball formula. Theorem 49.1 is instead a two-sided comparison with absolute constants, proved by a radial support-cutoff argument and entangled angular blocks. Its covering consequence encodes visibility and direction together, keeps unrestricted legal centres in the converse, and identifies the additional logarithmic factor for the disk. The contact criterion depends on explicit regularity and boundary-depth hypotheses; it does not classify every instrument tangent cone.

69.7. Bias and the complete binary-qubit effect body. For

$$E = \frac{(1+b)I + x \cdot \sigma}{2}, \quad |b| + |x| \leq 1,$$

the spin flip satisfies $\Gamma(E) - (I - E) = bI$. Thus the symmetry underlying the white-noise example of [57] holds only on the unbiased section. Theorem 50.1 treats the entire ordered binary-qubit effect body: both eigenvalues and the eigenbasis may vary, and the axis disappears at scalar effects. Its proof treats the two spectral endpoints by finite Bernoulli cutoffs, controls the basis change at fixed spectrum, and combines the contributions by eigenvalue extremality and monotonicity. This extends the operational comparison beyond a fixed-trace ball without assigning a metric from the number of parameters.

The angular upper bound constructs pairwise dilations with scalar cross overlap. The lower bound combines a product Bernoulli test with common bias-removing processing followed by finite entangled blocks. Both estimates retain the two endpoint variances, so a single rank-deficient effect need not produce the projective angular scale. The dilation comparison is pair-dependent and supplies no family-uniform seizing map.

Theorem 51.1 gives the full-body covering order $N^2 \log(N+2)\delta^{-4}$ for $N \geq 1$ and $0 < \delta \leq \delta_0$, with an absolute $\delta_0 > 0$. It counts all four target coordinates and permits arbitrary legal centres in the converse. Near the projective corner, each geometric depth scale between N^{-1} and a fixed constant contributes order $N^2\delta^{-4}$; their accumulation produces the logarithm. Theorem 51.2 realizes the upper bound with rational centres and a single payload index. These conclusions use the anisotropic finite-use metric, including its scalar degeneracy and projective boundary. The preceding discrimination results remain the direct antecedents for their respective one-use, two-use and projective all-use premises. The new covering argument uses pair-dependent tests; it asserts no common estimator or unknown-device sample bound. The separate common experiment in Theorem 54.1 provides the latter for the complete qubit effect body.

70. A RATIONAL COMPILER FOR ADAPTIVE OBSERVATION PROCESSES

The common-row construction also yields a causal approximation, but its proof requires control of the full observation law. Correct terminal means alone are not enough. We give that additional argument here.

The physical state is $x \in B_3$, initialized at a supplied rational $x_0 \in S^2$. A finite rational rotation list $\{A_a\} \subset SO(3)$ acts by $x \mapsto A_a x$ and emits a null symbol. A finite probe list has rational affine probability laws

$$(70.1) \quad q_{j,y}(x) = b_{j,y} + \langle \ell_{j,y}, x \rangle, \quad \sum_y b_{j,y} = 1, \quad \sum_y \ell_{j,y} = 0, \quad b_{j,y} - \|\ell_{j,y}\| \geq \eta > 0.$$

A probe emits a fresh draw and leaves x unchanged. Controls may depend on the entire observed transcript. A simulator has finite retained labels and a nonnegative joint emission/transition matrix for each control and output; these matrices may depend on horizon and epoch. Let $\text{PW}_{N,\varepsilon}$ denote its minimum peak label count for one total-variation budget ε on the entire transcript, uniformly over all adaptive randomized policies. This paragraph defines a causal model independently of the terminal matrix-output model.

Put $A = \max_j \sum_y \|\ell_{j,y}\|^2$ and $C_0 = \max(3, \max_j (M_j - 1))$. The lower probability bound in (70.1) is used only for the relative-entropy upper estimate, not for the finite-testing converse below.

Theorem 70.1 (Adaptive rational and fair-bit realization). *For rational $0 < \varepsilon < 1$ and integer $N \geq 0$, choose*

$$(70.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{4AN^2}{\eta\varepsilon^2}} \right\rceil \right), \quad r = 1 - m^{-2}, \quad b = \max \left(0, \left\lceil \log_2 \frac{2C_0 \max(1, N)}{\varepsilon} \right\rceil \right).$$

There is an error- ε causal realization on at most $13m^2 + 1$ labels. Before probability rounding every rotation row is rational, has at most four nonzero entries, and has barycentric mean rA_av . Probe emissions leave the label unchanged. After rounding, all emission and transition probabilities are dyadic, with at most b fair bits per control and Nb bits in total. The tables are stationary within the chosen horizon.

An exact rational algorithm constructs these tables in time polynomial in the numerical parameters $N, \varepsilon^{-1}, A/\eta$, the explicit input size, and the input bit lengths. In particular, for fixed probe and rotation lists,

$$(70.3) \quad \text{PW}_{N,\varepsilon} = O(1 + N^2/\varepsilon^2).$$

The bit count now includes the actual finite probe outputs as well as hidden transitions; it does not describe quantum-state preparation.

Proof. Take the rational net V_m from Lemma 26.1, including zero, and add x_0 if needed. This adds at most one label and cannot shrink the convex hull. Initialize deterministically at x_0 . For every rotation and label v , the same exact triple search supplies a row with mean rA_av . At probe j , emit y with probability $q_{j,y}(v)$ and retain v . All points lie in B_3 .

Fix an arbitrary policy. On the same finite trajectory space of controls, outputs and hidden labels, introduce an auxiliary law P : hidden labels follow the just constructed rotation kernels, but every probe emits $q_{j,y}(X)$ from the true physical state X , independently of the hidden label conditional on the public history. Its public marginal is exactly the target experiment. Let Q be the constructed simulator law. The policy kernels, initial label and hidden rotation kernels are the same under P, Q ; only probe emissions differ.

Under P , if the current physical and hidden points are X, V , both X and every nonzero hidden vertex are unit vectors. At a rotation step, for $V \neq 0$,

$$\begin{aligned} \mathbb{E}[\|V' - A_a X\|^2 \mid X, V, a] &\leq 2 - 2r\langle V, X \rangle \\ &= r\|V - X\|^2 + 2(1 - r). \end{aligned}$$

For $V = 0$ choose its row to remain at zero; the same upper bound holds because the left side is one and the right side is $2 - r \geq 1$. The added seed is a unit vector, so these two cases exhaust the state set. This is why a general interior seed has not been silently added to the theorem. A probe changes neither point. Iteration, with the control selected adaptively, gives

$$(70.4) \quad \mathbb{E}_P\|V_t - X_t\|^2 \leq 2t/m^2.$$

The inequality is unconditional; no posterior error bound after an unlikely observation is assumed.

For probability vectors p, q with $q_y \geq \eta$, the inequality $\log z \leq z - 1$ gives

$$D(p\|q) \leq \sum_y (p_y - q_y)^2 / q_y.$$

Consequently the conditional divergence at a probe is at most $A\|X - V\|^2/\eta$. The finite-trajectory relative-entropy chain rule cancels the common policy and rotation kernels. Applying (70.4) at the probe times gives

$$(70.5) \quad D(P\|Q) \leq \frac{2AN^2}{\eta m^2}, \quad \text{TV}(P_{\text{pub}}, Q_{\text{pub}}) \leq \sqrt{D(P\|Q)/2} \leq \frac{N}{m} \sqrt{A/\eta} \leq \varepsilon/2.$$

Here marginalization can only decrease divergence. These elementary entropy facts apply to the joint classical trajectory, not to a quantum state. One may verify the total-variation inequality by partitioning into the set where $P \geq Q$ and its complement: the log-sum inequality reduces divergence to a Bernoulli divergence, whose second derivative in its first parameter is at least four. This yields $D(P\|Q) \geq 2\text{TV}(P, Q)^2$. The chain rule follows by factoring each trajectory

probability into its conditional kernels. All probe probabilities in Q are positive, so absolute continuity causes no exception. If $A = 0$, every probe is constant and the divergence is zero.

Round each rotation row and each probe row by rounding all but its last positive mass down to denominator 2^b and assigning the remainder to the last. A row supported on d points changes in total variation by at most $(d-1)2^{-b}$. Thus every one-step joint kernel changes by at most $C_0 2^{-b}$. Coupling the two label-transcript processes until their first discrepancy proves, for every adaptive policy, an additional path error at most $NC_0 2^{-b} \leq \varepsilon/2$. Combining this with (70.5) proves the asserted error. Each rounded row can be sampled by one uniform b -bit integer and cumulative integer masses; zero or deterministic rows may use fewer bits.

The net and an added rational unit seed are explicit. For every command-vertex pair the exact enumeration in Lemma 26.1 examines at most $\binom{k-1}{3}$ triples, where $k \leq 13m^2 + 1$. The total is $O(|\mathcal{A}|k^4)$ constant-size rational linear-algebra operations, in addition to the probe tables. All determinants, comparisons and row identities involve rational numbers with polynomially bounded bit lengths in the supplied data and $\log m$; dyadic rounding adds b bits. The computation of m and b can use integer comparisons, not floating-point decisions. This proves the stated uniform, numerical-parameter complexity. It is a construction, not a minimum-width algorithm, and its exhaustive fallback need not be practical for large horizons. $\square$

70.1. An effective process converse for the rational free experiment. Keep the rational rotations and seed of (25.2)–(25.3), now exposing the three nondisturbing binary probes

$$(70.6) \quad q_{j,+}(x) = (1 + \lambda x_j)/2, \quad q_{j,-}(x) = (1 - \lambda x_j)/2, \quad j = 1, 2, 3,$$

where $0 < \lambda < 1$ is rational. This defines a fixed rational classical experiment. The probing law is not an implementation of nondisturbing quantum measurement. Let PW refer to the causal model above, not to terminal density-matrix width W .

Theorem 70.2 (Effective repeated-observation lower bound). *Let $0 \leq \varepsilon < 1$ and $0 < \delta < 1 - \varepsilon$. For an integer $t \geq 0$ put*

$$(70.7) \quad L_t = \max \left(1, \left\lceil 18\lambda^{-2} 625^t \log(6/\delta) \right\rceil \right).$$

At every horizon $N \geq t + 3L_t$, any error- ε causal machine has

$$(70.8) \quad |S_t| \geq (1 - \varepsilon - \delta)(2 \cdot 3^t - 1).$$

If $\varepsilon + \delta < 1/2$, it has the stronger lower bound $|S_t| \geq 2 \cdot 3^t - 1$. More generally, the same bounds hold at every cut u with $t \leq u \leq N - 3L_t$ after padding the prefixes by probes. In particular, for each fixed $0 < \varepsilon < 1$, there are explicit positive constants $c_{\varepsilon,\lambda}, C_{\varepsilon,\lambda}$ such that, for all sufficiently large N ,

$$(70.9) \quad c_{\varepsilon,\lambda} N^{\log 3 / \log 625} \leq \text{PW}_{N,\varepsilon} \leq C_{\varepsilon,\lambda} (N + 1)^2.$$

The converse uses no spectral-gap theorem. Its exponent is not asserted to be sharp.

Proof. The rational freeness and stabilizer argument in Theorem 25.2 gives $2 \cdot 3^t - 1$ different reachable pure Bloch vectors at time t . Their coordinates lie in $25^{-t}\mathbb{Z}^3$. Hence each distinct pair differs in some coordinate by at least 25^{-t} , and the associated plus-probabilities in (70.6) differ by at least $\lambda 25^{-t}/2$. Use one common continuation: repeat each of the three probes L_t times. Hoeffding's elementary Bernoulli bound and a union bound give

$$\mathbb{P}_x \left\{ \max_j |\hat{q}_j - q_{j,+}(x)| \geq \lambda 25^{-t}/6 \right\} \leq 6 \exp(-L_t \lambda^2 625^{-t}/18) \leq \delta.$$

The empirical neighborhoods for the different targets are disjoint. At a cut label z , let Q_z be the law of this same continuation. Marginalizing prefix outputs and using total-variation correctness shows that the continuation law from each prefix gives its own neighborhood probability at least

$1 - \varepsilon - \delta$. Writing this law as $\sum_z \alpha_x(z) Q_z$ and summing over the disjoint neighborhoods yields (70.8). If the probability exceeds $1/2$, each neighborhood needs a different label assigning it probability greater than $1/2$, giving the stronger conclusion. The proof does not condition on the success of a previous observation. Padding by arbitrary fixed probes leaves the target states unchanged and proves the assertion at later cuts. Unused final controls may also be probes. A terminal cut without enough remaining observations is not claimed to satisfy this lower bound.

For the power lower bound take $\delta = (1 - \varepsilon)/2$ and $B = 54\lambda^{-2} \log(6/\delta) + 4$. Then $t + 3L_t \leq B625^t$. For $N \geq B$, choose $t = \lfloor \log_{625}(N/B) \rfloor$. Since $3^t \geq (N/B)^{\log 3 / \log 625} / 3$ and $2 \cdot 3^t - 1 \geq 3^t$, (70.8) gives, for example, $c_{\varepsilon, \lambda} = (1 - \varepsilon)B^{-\log 3 / \log 625} / 6$. The upper bound follows from Theorem 70.1 with $\eta = (1 - \lambda)/2$ and $A = \lambda^2/2$. For irrational requested ε , construct at any rational tolerance between $\varepsilon/2$ and ε . All constants can thus be bounded from the displayed data without computing a group spectral gap. $\square$

The tests in this theorem consume horizon. The full-state terminal exponential profile and the causal lower bound therefore express different requirements. Conversely, the causal upper construction controls one joint transcript law under adaptive control, not just its one-time marginals. These process statements do not by themselves sharpen the terminal spectral-gap converse. Its logarithmic loss is removed separately by Theorem 15.5.

APPENDIX A. AN EXISTENTIAL HIGHER-DIMENSIONAL ENDPOINT

A.1. A fixed nonabelian family with an exponential exact endpoint. The next example uses two established inputs, explicitly separated from the realization argument. Breuillard and Gelander prove that a dense subgroup of a connected semisimple real Lie group contains a dense free subgroup on two generators [17, Theorem 1.1]; their later correction supplies repairs and details for other parts of the original proof [18]. The spectral-gap input for dense algebraic unitary generators is [16]. No algorithm for finding free generators, or for certifying a spectral-gap constant, is asserted here.

Lemma A.1 (A line with trivial algebraic projective stabilizer). *Fix a real number τ transcendental over $\mathbb{Q}$ and put*

$$v_\tau = \frac{(1, \tau, \dots, \tau^{d-1})^\top}{(\sum_{j=0}^{d-1} \tau^{2j})^{1/2}}, \quad P_\tau = v_\tau v_\tau^*.$$

An invertible matrix with algebraic entries that preserves the line $\mathbb{C}v_\tau$ is scalar.

Proof. If $Mv_\tau = \lambda v_\tau$, the eigenvalue λ is algebraic because it is a root of the characteristic polynomial of the algebraic matrix M . Each row of $(M - \lambda I)(1, \tau, \dots, \tau^{d-1})^\top = 0$ is a polynomial identity in τ with algebraic coefficients. Transcendence over $\mathbb{Q}$ implies transcendence over the field of algebraic numbers: an algebraic relation over a finite number field would make τ algebraic over $\mathbb{Q}$ by transitivity of algebraic extensions. Every row polynomial therefore has all coefficients zero. Thus $M = \lambda I$. $\square$

One may take a fixed computable transcendental such as $\tau = \sum_{n=1}^{\infty} 10^{-n!}$. For completeness, its truncations have rational denominator $10^{m!}$ and strictly positive tail at most $2 \cdot 10^{-(m+1)!}$. For a nonzero integer polynomial A of degree D , a rational p/q not among its finitely many roots satisfies $|A(p/q)| \geq q^{-D}$. If $A(\tau) = 0$, a bounded derivative near τ would instead bound $|A(p/q)|$ by a constant times $q^{-(m+1)}$ along these truncations, a contradiction for large m . This also records precisely why the chosen seed is not part of the rational-input specialization.

Theorem A.2 (Exponential exact width and polynomial noisy width). *For every $d \geq 2$, with a transcendental seed and existentially selected generators, there is a fixed five-letter algebraic*

unitary alphabet

$$\{I, U, U^{-1}, V, V^{-1}\} \subset SU(d)$$

and a fixed rank-one seed P_τ as above such that the full matrix interface $F(g) = gP_\tau g^*$, with legal decoder set $\mathcal{D}_d$, has

$$(A.1) \quad W_{N,0} = 2 \cdot 3^N - 1 \quad (N \geq 0).$$

For each fixed $0 < \varepsilon < r_d = \sqrt{1 - 1/d}$, its Frobenius-error width instead satisfies

$$(A.2) \quad c_{d,\varepsilon} \left(\frac{N}{\log(N+2)} \right)^{d-1} \leq W_{N,\varepsilon} \leq C_{d,\varepsilon} (N+1)^{d-1}.$$

At $\varepsilon \geq r_d$ one label suffices. For the same alphabet and seed, every fixed amplitude $0 < a < 1$ admits the exact polynomial upper bound in (17.9) and the corresponding lower bound in (17.8).

Proof. The subgroup of $SU(d)$ consisting of matrices with algebraic entries is dense. One direct verification is to approximate a unitary matrix by a nonsingular matrix with entries in $\mathbb{Q} + i\mathbb{Q}$, apply Gram–Schmidt, and multiply its final column by the inverse determinant of the resulting unitary matrix. All entries produced are algebraic; continuity near the original unitary matrix gives density. Apply [17, Theorem 1.1] to this subgroup. It contains algebraic U, V freely generating a dense free group Γ of rank two. The scalar matrices in $SU(d)$ form a finite group. Since a free group is torsion-free, Γ contains no nonidentity scalar matrix. Lemma A.1 now shows that the orbit map $\Gamma \rightarrow \mathcal{D}_d$, $g \mapsto gP_\tau g^*$, is injective.

With the identity available for padding, the products at length N are exactly the elements of reduced free-word length at most N . There are four choices for the first letter of a nonempty reduced word and three for each subsequent letter. Thus the ball has cardinality

$$1 + 4 \sum_{j=1}^N 3^{j-1} = 2 \cdot 3^N - 1.$$

Apply Theorem 18.1 to obtain (A.1). In fact the entire pointwise minimum profile is $|S_t| = 2 \cdot 3^t - 1$.

Assign a symmetric positive law to the four nonidentity commands and positive mass to the identity. The algebraic dense-generator theorem [16], with laziness, gives (14.1). Consequently Theorem 17.3 applies to this same fixed alphabet and seed. It yields (A.2), the threshold r_d , and the asserted strict-amplitude conclusion. $\square$

This is a full legal-matrix-output statement. It does not resolve the zero-error unit-amplitude problem for a single scalar planar readout: that readout need not expose every orbit point as an extreme decoder value. Nor does it identify a sampled measurement outcome with a density matrix stored in a classical label. The distinction between a prescribed convex output set and a larger simplex is essential here.

REFERENCES

Stochastic semigroups and positive realization

- [1] P. Flor, On groups of non-negative matrices, *Compositio Mathematica* **21** (1969), no. 4, 376–382.
 - [2] Š. Schwarz, On the structure of the semigroup of stochastic matrices, *Publications of the Mathematical Institute of the Hungarian Academy of Sciences, Series A* **9** (1964), 297–311.
 - [3] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Transactions on Automatic Control* **49** (2004), no. 5, 651–664. DOI: 10.1109/TAC.2004.826715.
 - [4] A. Heller, On stochastic processes derived from Markov chains, *Annals of Mathematical Statistics* **36** (1965), no. 4, 1286–1291.
- Automata, weighted series, and recognition*
- [5] M. O. Rabin, Probabilistic automata, *Information and Control* **6** (1963), no. 3, 230–245.
 - [6] A. Paz, *Introduction to Probabilistic Automata*, Academic Press, New York, 1971.

- [7] J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Encyclopedia of Mathematics and its Applications 137, Cambridge University Press, Cambridge, 2011.
- [8] N. Fijalkow and C. Paperman, Monadic second-order logic with arbitrary monadic predicates, arXiv:1709.03117 (2017).
- [9] A. Brodsky and N. Pippenger, Characterizations of 1-way quantum finite automata, *SIAM Journal on Computing* **31** (2002), no. 5, 1456–1478.
- [10] A. Ambainis and R. Freivalds, 1-way quantum finite automata: strengths, weaknesses and generalizations, arXiv:quant-ph/9802062 (1998).
- [11] Y. Feng, L. Song, and L. Zhang, Distribution-based bisimulation and bisimulation metric in probabilistic automata, arXiv:1512.05027 (2015).
- [12] B. Steinberg, Topological dynamics and recognition of languages, arXiv:1306.1468 (2013).
Compact groups and orbit geometry
- [13] G. B. Folland, *A Course in Abstract Harmonic Analysis*, second ed., Chapman & Hall/CRC, Boca Raton, 2016.
- [14] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, second ed., Graduate Texts in Mathematics 222, Springer, Cham, 2015.
- [15] S. J. Szarek, Metric entropy of homogeneous spaces, *Banach Center Publications* **43** (1998), 395–410.
- [16] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *Journal of the European Mathematical Society* **14** (2012), no. 5, 1455–1511.
- [17] E. Breuillard and T. Gelander, On dense free subgroups of Lie groups, *Journal of Algebra* **261** (2003), no. 2, 448–467.
- [18] E. Breuillard and T. Gelander, On dense free subgroups of Lie groups—revisited, arXiv:2605.20568 (2026).
Algebraic computation and complexity
- [19] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second ed., Algorithms and Computation in Mathematics 10, Springer, Berlin, 2006.
- [20] V. D. Blondel, E. Jeandel, P. Koiran, and N. Portier, Decidable and undecidable problems about quantum automata, *SIAM Journal on Computing* **34** (2005), no. 6, 1464–1473.
- [21] H. Derksen, E. Jeandel, and P. Koiran, Quantum automata and algebraic groups, *Journal of Symbolic Computation* **39** (2005), nos. 3–4, 357–371.
- [22] W. A. de Graaf, Computing the Zariski closure of a finitely generated matrix group, arXiv:2505.02577v2 (2026).
- [23] K. Nosan, A. Pouly, S. Schmitz, M. Shirmohammadi, and J. Worrell, On the computation of the Zariski closure of finitely generated groups of matrices, arXiv:2106.01853v6 (2025).
- [24] Y. Shitov, A universality theorem for nonnegative matrix factorizations, arXiv:1606.09068v2 (2018).
- [25] F. P. Ramsey, On a problem of formal logic, *Proceedings of the London Mathematical Society* (2) **30** (1930), no. 1, 264–286.
- [26] S. Singh, M. R. James, and M. Rudary, *Predictive state representations: A new theory for modeling dynamical systems*, Proceedings of the Twentieth Conference on Uncertainty in Artificial Intelligence, AUA Press, 2004, 512–519. See Theorems 2 and 6; arXiv:1207.4167.
Entropy transport and explicit spherical spectral bounds
- [27] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Transactions on Information Theory* **62** (2016), no. 7, 3992–4002. DOI: 10.1109/TIT.2016.2562630.
- [28] A. Lubotzky, R. Phillips, and P. Sarnak, Hecke operators and distributing points on the sphere I, *Communications on Pure and Applied Mathematics* **39** (1986), suppl., S149–S186. DOI: 10.1002/cpa.3160390710.
- [29] A. Pinochet Lobos and C. Pittet, The exact convergence rate in the ergodic theorem of Lubotzky–Phillips–Sarnak, arXiv:1805.05261v2 (2019): Theorem 1.1 restates the one-step LPS bound; Theorem 1.2 gives the exact radial-average norms. Published in expanded form as *The exact convergence rate in the ergodic theorem of Lubotzky–Phillips–Sarnak and a universal lower bound on discrepancies*, *L’Enseignement Mathématique* **67** (2021), nos. 1–2, 63–94. DOI: 10.4171/LEM/1003.
Nonhomogeneous Markov chains and observation tails
- [30] D. Blackwell, Finite non-homogeneous chains, *Annals of Mathematics* (2) **46** (1945), no. 4, 594–599. DOI: 10.2307/1969199.
- [31] H. Cohn, On the tail σ -algebra of the finite inhomogeneous Markov chains, *Annals of Mathematical Statistics* **41** (1970), no. 6, 2175–2176. DOI: 10.1214/aoms/1177696725.
- [32] M. Grabchak and I. Sonin, A zero-one law for Markov chains, arXiv:2011.04063 (2020), Theorem 1 and Section 5.1; published in *Stochastics*, DOI: 10.1080/17442508.2021.1980569.
Affinity inequalities and arithmetic gate approximation
- [33] T. van Erven and P. Harremoës, Rényi divergence and Kullback–Leibler divergence, *IEEE Transactions on Information Theory* **60** (2014), no. 7, 3797–3820. DOI: 10.1109/TIT.2014.2320500. Theorem 3; arXiv:1206.2459.

- [34] O. Parzanchevski and P. Sarnak, Super-Golden-Gates for $PU(2)$, *Advances in Mathematics* **327** (2018), 869–901. DOI: 10.1016/j.aim.2017.06.022. Section 3; arXiv:1704.02106.
Positive realization and finite automata
- [35] A. Ambainis and J. Watrous, Two-way finite automata with quantum and classical states, *Theoretical Computer Science* **287** (2002), no. 1, 299–311. arXiv:cs/9911009, Section 2 and Theorem 1.
- [36] M. V. Panduranga Rao and V. Vinay, Quantum finite automata and weighted automata, arXiv:quant-ph/0701144 (2007), Theorem 1.
Quantum instruments and current simulation-cost comparisons
- [37] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. Theorems 2.22 and 2.26 (Choi criteria), Corollary 3.40 (trace-norm contraction), and Section 3.3, equation (3.306) (hybrid comparison).
- [38] Y. Benoist and N. de Saxcé, A spectral gap theorem in simple Lie groups, arXiv:1405.1808 (2014), Theorem 1.2.
- [39] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2 (27 August 2026).
- [40] Z. Chen and J. Wu, On the simulation cost of quantum finite automata, arXiv:2605.10682v1 (11 May 2026).
Quantum trajectories, process estimation and adaptive comparison
- [41] P. Rouchon and J. F. Ralph, Efficient quantum filtering for quantum feedback control, *Phys. Rev. A* **91** (2015), 012118; arXiv:1410.5345v2, 6 January 2015.
- [42] T. Surawy-Stepney, J. Kahn, R. Kueng and M. Guță, Projected least-squares quantum process tomography, *Quantum* **6** (2022), 844; arXiv:2107.01060v2, 18 October 2022.
- [43] G. Gutoski, On a measure of distance for quantum strategies, *J. Math. Phys.* **53** (2012), 032202; arXiv:1008.4636v4, 14 March 2012.
Description, statistical learning and physical simulation
- [44] S. G. Naik, M. Zartab, N. Gisin and M. Banik, No-Go Theorem for Generic Simulation of Qubit Channels with Finite Classical Resources, arXiv:2501.15807v2 (16 July 2025), Theorems 2 and 5.
- [45] A. A. Mele and L. Bittel, Optimal learning of quantum channels in diamond distance, arXiv:2512.10214v3 (2026), Theorem I.1, Lemmas III.7–III.8, Corollary III.9 and Sections III.1.1, IV.1.1.
- [46] A. Oufkir and F. Girardi, Improved Lower Bounds for Learning Quantum Channels in Diamond Distance, arXiv:2601.04180v4 (2026).
- [47] R. Demkowicz-Dobrzański, J. Kołodyński and M. Guță, The elusive Heisenberg limit in quantum-enhanced metrology, *Nature Communications* **3** (2012), 1063. DOI: 10.1038/ncomms2067; arXiv:1201.3940v2, equations (4)–(6) (classical simulation), equations (10)–(13) (channel extension), and the classical-simulation argument in Methods.
- [48] Y. Yang, G. Bai, G. Chiribella and M. Hayashi, Compression for quantum population coding, *IEEE Transactions on Information Theory* **64** (2018), no. 7, doi:10.1109/TIT.2017.2788407; arXiv:1701.03372v8.
- [49] F. Salek, M. Hayashi and A. Winter, Usefulness of adaptive strategies in asymptotic quantum channel discrimination, *Phys. Rev. A* **105** (2022), 022419; arXiv:2011.06569v3 (17 March 2024).
- [50] T. Cooney, M. Mosonyi and M. M. Wilde, Strong converse exponents for a quantum channel discrimination problem and quantum-feedback-assisted communication, arXiv:1408.3373v3 (26 February 2016), p. 5 and Sections 4.1–4.2.
- [51] A. Acín, Statistical distinguishability between unitary operations, *Phys. Rev. Lett.* **87** (2001), 177901. DOI: 10.1103/PhysRevLett.87.177901.
- [52] R. Duan, Y. Feng and M. Ying, Entanglement is not necessary for perfect discrimination between unitary operations, *Phys. Rev. Lett.* **98** (2007), 100503. DOI: 10.1103/PhysRevLett.98.100503.
- [53] S. Das and M. M. Wilde, Quantum reading capacity: General definition and bounds, *IEEE Trans. Inform. Theory* **65** (2019), 7566–7583; arXiv:1703.03706v3; doi:10.1109/TIT.2019.2929925.
- [54] M. M. Wilde, M. Berta, C. Hirche and E. Kaur, Amortized channel divergence for asymptotic quantum channel discrimination, *Lett. Math. Phys.* **110** (2020), 2277–2336; arXiv:1808.01498v2; doi:10.1007/s11005-020-01297-7.
- [55] X. Wang and M. M. Wilde, Resource theory of asymmetric distinguishability for quantum channels, *Phys. Rev. Research* **1** (2019), 033169; arXiv:1907.06306v2; doi:10.1103/PhysRevResearch.1.033169.
- [56] Z. Puchała, Ł. Paweła, A. Krawiec, R. Kukulski and M. Oszmaniec, Multiple-shot and unambiguous discrimination of von Neumann measurements, *Quantum* **5** (2021), 425; arXiv:1810.05122v4; doi:10.22331/q-2021-04-06-425.
- [57] M. Sedláč and M. Ziman, Optimal single-shot strategies for discrimination of quantum measurements, *Phys. Rev. A* **90** (2014), 052312; arXiv:1408.0934; doi:10.1103/PhysRevA.90.052312. Section VI, Example 4, equation (37).
- [58] Z. Puchała, Ł. Paweła, A. Krawiec and R. Kukulski, Strategies for optimal single-shot discrimination of quantum measurements, *Phys. Rev. A* **98** (2018), 042103; arXiv:1804.05856; doi:10.1103/PhysRevA.98.042103. Theorem 1.

- [59] J. Fiurášek and M. Mičuda, Optimal two-copy discrimination of quantum measurements, *Phys. Rev. A* **80** (2009), 042312; arXiv:0909.2940; doi:10.1103/PhysRevA.80.042312.
Measurement discrimination, channel geometry and common learning
- [60] A. Krawiec, Ł. Paweł and Z. Puchała, Discrimination of POVMs with rank-one effects, *Quantum Information Processing* **19** (2020), 428; arXiv:2002.05452v1; doi:10.1007/s11128-020-02883-3.
- [61] C. Datta, T. Biswas, D. Saha and R. Augusiak, Perfect discrimination of quantum measurements using entangled systems, *New Journal of Physics* **23** (2021), 043021; arXiv:2012.07069v2; doi:10.1088/1367-2630/abecaf.
- [62] A. Fujiwara and H. Imai, A fibre bundle over manifolds of quantum channels and its application to quantum statistics, *J. Phys. A: Math. Theor.* **41** (2008), 255304; doi:10.1088/1751-8113/41/25/255304. Theorems 4 and 5.
- [63] R. Demkowicz-Dobrzański and L. Maccone, Using entanglement against noise in quantum metrology, *Phys. Rev. Lett.* **113** (2014), 250801; arXiv:1407.2934v3; doi:10.1103/PhysRevLett.113.250801.
- [64] S. Kurdziałek, W. Górecki, F. Albarelli and R. Demkowicz-Dobrzański, Using adaptiveness and causal superpositions against noise in quantum metrology, *Phys. Rev. Lett.* **131** (2023), 090801; arXiv:2212.08106v2; doi:10.1103/PhysRevLett.131.090801.
- [65] H. Yuan and C.-H. F. Fung, Fidelity and Fisher information on quantum channels, arXiv:1506.00819v3 (2017), Section III, equation (26).
- [66] S. Sieniawski and R. Demkowicz-Dobrzański, Adaptive quantum channel discrimination using methods of quantum metrology, *New Journal of Physics* **28** (2026), 024502; arXiv:2510.15506v3; doi:10.1088/1367-2630/ae3edd.
- [67] J. Dittmann, Explicit formulae for the Bures metric, *J. Phys. A: Math. Gen.* **32** (1999), 2663–2670; doi:10.1088/0305-4470/32/14/007; arXiv:quant-ph/9808044, entitled *Note on explicit formulae for the Bures metric*.
- [68] H.-J. Sommers and K. Życzkowski, Bures volume of the set of mixed quantum states, *J. Phys. A: Math. Gen.* **36** (2003), 10083–10100; arXiv:quant-ph/0304041v3; doi:10.1088/0305-4470/36/39/308.
- [69] L. Zambrano, S. Ramos-Calderer and R. Kueng, Fast quantum measurement tomography with optimal error bounds, *Quantum* **10** (2026), 2162; arXiv:2507.04500v3; doi:10.22331/q-2026-07-15-2162.
- [70] A. Bisio, G. M. D’Ariano, P. Perinotti and M. Sedláč, Quantum learning algorithms for quantum measurements, *Physics Letters A* **375** (2011), 3425–3434; arXiv:1103.0480v2; doi:10.1016/j.physleta.2011.08.002.
- [71] B. L. Higgins, D. W. Berry, S. D. Bartlett, M. W. Mitchell, H. M. Wiseman and G. J. Pryde, Demonstrating Heisenberg-limited unambiguous phase estimation without adaptive measurements, *New Journal of Physics* **11** (2009), 073023; arXiv:0809.3308v4; doi:10.1088/1367-2630/11/7/073023.
- [72] S. Kimmel, G. H. Low and T. J. Yoder, Robust calibration of a universal single-qubit gate set via robust phase estimation, *Phys. Rev. A* **92** (2015), 062315; doi:10.1103/PhysRevA.92.062315; erratum, *Phys. Rev. A* **104** (2021), 069901, doi:10.1103/PhysRevA.104.069901; arXiv:1502.02677v3.
- [73] P. Hyllus, W. Laskowski, R. Krischek, C. Schwemmer, W. Wieczorek, H. Weinfurter, L. Pezzé and A. Smerzi, Fisher information and multiparticle entanglement, *Phys. Rev. A* **85** (2012), 022321; arXiv:1006.4366v3; doi:10.1103/PhysRevA.85.022321. Section III.A, Observation 1, equation (9).
- [74] G. Tóth, Multipartite entanglement and high-precision metrology, *Phys. Rev. A* **85** (2012), 022322; arXiv:1006.4368v3; doi:10.1103/PhysRevA.85.022322. Observation 3, equation (6).
- [75] K. Chen, Z. Zhang and N. Yu, Optimal lower bound for quantum channel tomography in away-from-boundary regime, arXiv:2601.10683v1 (15 January 2026), Theorems 1.1, 3.2 and 4.2.
- [76] S. Chen, J. Li and A. Liu, An optimal tradeoff between entanglement and copy complexity for state tomography, arXiv:2402.16353v1 (26 February 2024).
- [77] A. Nayak and X. Zhou, Optimal Low-Rank Quantum State Tomography with Bounded-Sample Joint Measurements, arXiv:2609.10514v1 (9 September 2026), Theorems 1.1–1.2.
- [78] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics, vol. 10, Springer, Berlin, Heidelberg, 2006; doi:10.1007/3-540-33099-2. Theorems 2.77 and 2.80 and Chapter 14.
- [79] A. Pelecanos, J. Spilecki, E. Tang and J. Wright, Mixed state tomography reduces to pure state tomography, arXiv:2511.15806v1 (19 November 2025). Theorems 1.1–1.3 and 4.2, Section 3.3, Figures 8–9.